

The Collected Mathematical Papers of
Arthur Cayley
Vol. XII

Arthur Cayley

Modern L^AT_EX reconstruction

MATHEMATICAL PAPERS.

THE COLLECTED
MATHEMATICAL PAPERS

OF

ARTHUR CAYLEY, Sc.D., F.R.S.,
LATE SADLEIRIAN PROFESSOR OF PURE MATHEMATICS IN
THE UNIVERSITY OF CAMBRIDGE.

VOL. XII.

CAMBRIDGE :
AT THE UNIVERSITY PRESS.

1897.

[All Rights reserved.]

CAMBRIDGE :
PRINTED BY J. AND C. F. CLAY,
AT THE UNIVERSITY PRESS.

Advertisement.

The present volume contains 89 papers, numbered 799 to 887, published for the most part in the years 1883 to 1889.

The Table for the twelve volumes is:

Vol. I.	Numbers 1 to 100,
" II.	" 101 to 158,
" III.	" 159 to 222,
" IV.	" 223 to 299,
" V.	" 300 to 383,
" VI.	" 384 to 416,
" VII.	" 417 to 485,
" VIII.	" 486 to 555,
" IX.	" 556 to 629,
" X.	" 630 to 705,
" XI.	" 706 to 798,
" XII.	" 799 to 887.

A. R. Forsyth.
19 May, 1897.

Contents.

[An Asterisk means that the paper is not printed in full.]

	PAGE
799. On curvilinear coordinates.	
Quart. Math. Journ., t. xix. (1883), pp. 1-22	
800. Note on the standard solutions of a system of linear equations	19
Quart. Math. Journ., t. xix. (1883), pp. 19-21	
801. Note on MacMahon's paper	22
Quart. Math. Journ., t. xix. (1883), pp. 22-29	
802. On Mr Anglin's formula for the successive powers of the root of an algebraical equation	30
Quart. Math. Journ., t. xix. (1883), pp. 30-32	
803. Note on Abel's theorem	33
Camb. Phil. Soc. Proc., t. iv. (1883), pp. 33-35	
804. On the elliptic-function solution of the equation $x^3 + y^3 = 1$	35
Camb. Phil. Soc. Proc., t. iv. (1883), pp. 35-38	
805. A proof of Wilson's theorem	38
Messenger of Mathematics, t. xii. (1883), pp. 38-41	
806. Determination of the order of a surface	41
Messenger of Mathematics, t. xii. (1883), pp. 41-42	
807. Note on a form of the modular equation in the transformation of the third order	42
Messenger of Mathematics, t. xii. (1883), pp. 42-45	
808. Schröter's construction of the regular pentagon	45
Messenger of Mathematics, t. xii. (1883), pp. 45-46	
809. Note on a system of equations	46
Messenger of Mathematics, t. xii. (1883), pp. 46-47	
810. On the linear transformation of the theta-functions	47
Messenger of Mathematics, t. xii. (1883), pp. 47-50	
811. On Archimedes' theorem for the surface of a cylinder	48
Messenger of Mathematics, t. xiii. (1884), pp. 48-50	

Classification.

ANALYSIS.

Equations, 801, 803, 835, 855, 856, 867, 878.

Theory of numbers, 807, 815.

Groups, 887.

Quaternions, 836.

Matrices, 839.

Trigonometry, 824, 842.

Symmetric functions, 810.

Elliptic functions, 804, 814, 818, 833, 834, 835, 848, 849, 850, 851, 852.

Abelian and theta functions, 825, 826, 828, 829, 830, 831, 846.

Differential equations, 801, 860, 861, 862, 863, 866.

Seminvariants, 800, 802, 828, 829, 830, 831, 847.

Partitions, 826.

GEOMETRY.

Curvilinear coordinates, 799.
 Analytical geometry, plane, 819, 828.
 Analytical geometry of three dimensions, 821, 823, 836, 839, 840, 841, 857.
 Surfaces, 806, 820, 832, 884, 886.
 Skew surfaces, 858.
 Surfaces of second order, 844, 864.
 Surfaces of fourth order, 817, 854.
 Cubic curves, 820, 843, 859.
 Quartic curves, 819, 843, 859.
 Curves of the third order, 844, 860.
 Curves of the fourth order, 816, 832, 868.
 Correspondence, 819, 821.
 Orthomorphosis, 821, 845.
 Plane geometry, 822, 827, 873, 875, 877.
 Non-Euclidian geometry, 827.
 General theory of curves, 820, 832, 868.
 Properties of circles, 873, 875, 877.
 Rays, 876.
 Twisted cubic curves, 838.
 Twisted quartic curves, 843, 859.
 Curves of curvature, 886.
 Minimal surfaces, 884.
 Centro-surfaces, 864.

MISCELLANEOUS, 809, 810, 812, 837, 840, 857.

799.

ON CURVILINEAR COORDINATES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix.
(1883),
pp. 1–22.]

The present memoir is based upon Mr Warren's "Exercises in Curvilinear and Normal Coordinates," Camb. Phil. Trans., t. xii. (1877), pp. 455–502, and has for a principal object the establishment of the six differential equations of the second order corresponding to his six equations for normal coordinates: the notation is different; the results are more general, inasmuch as I use throughout general curvilinear coordinates instead of his normal coordinates; and as regards my six equations for general curvilinear coordinates, the terms containing differential coefficients of the first order are presented under a different form.

If the position of a point in space is determined by the rectangular coordinates x, y, z ; then p, q, r being each of them a given function of x, y, z , we have conversely x, y, z , each of them a given function of p, q, r , which are thus in effect coordinates serving to determine the position of the point, and are called *curvilinear coordinates*.

But it is not in the first instance necessary to regard x, y, z as rectangular coordinates, or even as Cartesian coordinates at all; we are simply concerned with the two sets of variables x, y, z and p, q, r , each variable of the one set being a given function of the variables of the other set; and, in particular, the x, y, z are regarded as being each of them a given function of the p, q, r .

Except as regards the symbols ξ, η, ζ presently mentioned, the suffixes 1, 2, 3 refer to the variables p, q, r respectively, and are used to denote differentiations in regard to these variables, viz.

$$x_1, \quad x_2, \quad x_3 \text{ denote } \frac{dx}{dp}, \quad \frac{dx}{dq}, \quad \frac{dx}{dr},$$

and so in other cases; in particular,

$$a, \quad b, \quad c \text{ denote } x_1, \quad y_2, \quad z_3.$$

The minors formed with these differential coefficients are denoted by

suffixed letters ξ, η, ζ , thus

$$\begin{aligned}\xi_1, \eta_1, \zeta_1 &= y_2 z_3 - y_3 z_2, z_2 x_3 - z_3 x_2, x_2 y_3 - x_3 y_2, \\ \xi_2, \eta_2, \zeta_2 &= y_3 z_1 - y_1 z_3, z_3 x_1 - z_1 x_3, x_3 y_1 - x_1 y_3, \\ \xi_3, \eta_3, \zeta_3 &= y_1 z_2 - y_2 z_1, z_1 x_2 - z_2 x_1, x_1 y_2 - x_2 y_1,\end{aligned}$$

so that, as regards these letters ξ, η, ζ , the suffixes do not denote differentiations.

The determinant

$$J = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

is put $= \nabla$.

And the symbols (a, b, c, f, g, h) , (A, B, C, F, G, H) are defined as follows:

$$\begin{aligned}(a, b, c, f, g, h) &= (x_1^2 + y_1^2 + z_1^2, x_2^2 + y_2^2 + z_2^2, x_3^2 + y_3^2 + z_3^2, \\ &\quad x_2 x_3 + y_2 y_3 + z_2 z_3, x_3 x_1 + y_3 y_1 + z_3 z_1, x_1 x_2 + y_1 y_2 + z_1 z_2), \\ (A, B, C, F, G, H) &= (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch).\end{aligned}$$

We have then, further,

$$\begin{aligned}\nabla(a, b, c, f, g, h) &= (\xi_1^2 + \eta_1^2 + \zeta_1^2, \xi_2^2 + \eta_2^2 + \zeta_2^2, \xi_3^2 + \eta_3^2 + \zeta_3^2, \\ &\quad \xi_2 \xi_3 + \eta_2 \eta_3 + \zeta_2 \zeta_3, \xi_3 \xi_1 + \eta_3 \eta_1 + \zeta_3 \zeta_1, \xi_1 \xi_2 + \eta_1 \eta_2 + \zeta_1 \zeta_2), \\ \nabla^2(A, B, C, F, G, H) &= (BC - F^2, CA - G^2, AB - H^2, GH - AF, HF - BG, FG - AH),\end{aligned}$$

which equations are at once proved, and are fundamental ones in the theory.

It is convenient to add that we have

$$\begin{aligned}\frac{dp}{dx} &= \frac{\xi_1}{\nabla}, & \frac{dp}{dy} &= \frac{\eta_1}{\nabla}, & \frac{dp}{dz} &= \frac{\zeta_1}{\nabla}, \\ \frac{dq}{dx} &= \frac{\xi_2}{\nabla}, & \frac{dq}{dy} &= \frac{\eta_2}{\nabla}, & \frac{dq}{dz} &= \frac{\zeta_2}{\nabla}, \\ \frac{dr}{dx} &= \frac{\xi_3}{\nabla}, & \frac{dr}{dy} &= \frac{\eta_3}{\nabla}, & \frac{dr}{dz} &= \frac{\zeta_3}{\nabla}.\end{aligned}$$

We require in the sequel the values of the second differential coefficients x_{11}, x_{12} , etc., in terms of differential coefficients of a, b, c, f, g, h in regard to p, q, r . Writing

$$\frac{da}{dp} = a_1, \quad \frac{da}{dq} = a_2, \quad \frac{da}{dr} = a_3, \text{ etc.,}$$

and forming the values of $a_1, b_1, c_1, f_1, g_1, h_1$, we have

$$\begin{aligned}
a_1 &= 2(x_1x_{11} + y_1y_{11} + z_1z_{11}), \\
h_1 &= x_2x_{11} + y_2y_{11} + z_2z_{11} + x_1x_{12} + y_1y_{12} + z_1z_{12}, \\
g_1 &= x_3x_{11} + y_3y_{11} + z_3z_{11} + x_1x_{13} + y_1y_{13} + z_1z_{13}, \\
h_2 &= x_2x_{12} + y_2y_{12} + z_2z_{12} + x_1x_{22} + y_1y_{22} + z_1z_{22}, \\
b_2 &= 2(x_2x_{22} + y_2y_{22} + z_2z_{22}), \\
f_2 &= x_3x_{22} + y_3y_{22} + z_3z_{22} + x_2x_{23} + y_2y_{23} + z_2z_{23}, \\
g_3 &= x_3x_{13} + y_3y_{13} + z_3z_{13} + x_1x_{33} + y_1y_{33} + z_1z_{33}, \\
f_3 &= x_3x_{23} + y_3y_{23} + z_3z_{23} + x_2x_{33} + y_2y_{33} + z_2z_{33}, \\
c_3 &= 2(x_3x_{33} + y_3y_{33} + z_3z_{33}),
\end{aligned}$$

to which we join

$$\begin{aligned}
f_1 &= x_2x_{13} + y_2y_{13} + z_2z_{13} + x_3x_{12} + y_3y_{12} + z_3z_{12}, \\
b_1 &= 2(x_2x_{12} + y_2y_{12} + z_2z_{12}), \\
c_1 &= 2(x_3x_{13} + y_3y_{13} + z_3z_{13}), \\
a_2 &= 2(x_1x_{12} + y_1y_{12} + z_1z_{12}), \\
c_2 &= 2(x_3x_{23} + y_3y_{23} + z_3z_{23}), \\
b_3 &= 2(x_2x_{23} + y_2y_{23} + z_2z_{23}), \\
a_3 &= 2(x_1x_{13} + y_1y_{13} + z_1z_{13}).
\end{aligned}$$

It is convenient to add that we have

$$\frac{dp}{dx} = \frac{\xi_1}{\nabla}, \quad \frac{dp}{dy} = \frac{\eta_1}{\nabla}, \quad \frac{dp}{dz} = \frac{\zeta_1}{\nabla},$$

and similarly for $dq/dx, dq/dy, dq/dz$ and $dr/dx, dr/dy, dr/dz$; that is,

$$\frac{dp}{dx} : \frac{dp}{dy} : \frac{dp}{dz} = \xi_1 : \eta_1 : \zeta_1,$$

and further,

$$dx^2 + dy^2 + dz^2 = (a, b, c, f, g, h, dp, dq, dr)^2,$$

$$\nabla^2(dp^2 + dq^2 + dr^2) = (A, B, C, F, G, H, dx, dy, dz)^2.$$

Differentiating the values of a, b, c, f, g, h , we have $a_1 = \text{etc.}$ which may be written in the abbreviated form $a_1 = 1.11$; similarly

$$h_1 = x_2x_{11} + y_2y_{11} + z_2z_{11} + x_1x_{12} + y_1y_{12} + z_1z_{12},$$

which in like manner may be written $= 2.13 + 3.12$, and so in other cases; observe that, in the duad part of any symbol, the order of the numbers is immaterial, $2.13 = 2.31$. The whole system of equations is

$$\begin{aligned} a_1 &= 1.11, & a_2 &= 1.12, & a_3 &= 1.13, \\ f_1 &= 2.13 + 3.12, & f_2 &= 2.23 + 3.22, & f_3 &= 2.33 + 3.23, \\ g_1 &= 3.11 + 1.13, & g_2 &= 3.12 + 1.23, & g_3 &= 3.13 + 1.33, \\ h_1 &= 1.12 + 2.11, & h_2 &= 1.22 + 2.12, & h_3 &= 1.23 + 2.13. \end{aligned}$$

These may also be written

$$\begin{aligned} 2.11 &= \frac{1}{2}a_1, & 2.22 &= \frac{1}{2}b_2, & 2.33 &= \frac{1}{2}c_3, \\ 3.11 &= g_1 - \frac{1}{2}a_3, & 3.22 &= f_2 - \frac{1}{2}b_3, & 3.33 &= \frac{1}{2}c_3, \\ 1.23 &= \frac{1}{2}(-f_1 + g_2 + h_3), & 2.13 &= \frac{1}{2}(f_1 - g_2 + h_3), & 3.12 &= \frac{1}{2}(f_1 + g_2 - h_3). \end{aligned}$$

It is to be observed that we can, from each system of three equations, express a set of second differential coefficients of the x, y, z in terms of the first differential coefficients of the a, b, c, f, g, h ; thus the three equations containing 11, written at length, are

$$\begin{aligned} x_1x_{11} + y_1y_{11} + z_1z_{11} &= \frac{1}{2}a_1, \\ x_2x_{11} + y_2y_{11} + z_2z_{11} &= h_1 - \frac{1}{2}a_2, \\ x_3x_{11} + y_3y_{11} + z_3z_{11} &= g_1 - \frac{1}{2}a_3, \end{aligned}$$

three linear equations for the determination of x_{11}, y_{11}, z_{11} ; hence for x_{11} we have

$$x_{11} = \frac{\begin{vmatrix} \frac{1}{2}a_1 & y_1 & z_1 \\ h_1 - \frac{1}{2}a_2 & y_2 & z_2 \\ g_1 - \frac{1}{2}a_3 & y_3 & z_3 \end{vmatrix}}{\nabla},$$

or, what is the same thing,

$$x_{11} = \frac{\xi_1 \cdot \frac{1}{2}a_1 + \xi_2(h_1 - \frac{1}{2}a_2) + \xi_3(g_1 - \frac{1}{2}a_3)}{\nabla},$$

and so for y_{11}, z_{11} ; or if (as in the sequel) we desire the value of a linear function $\alpha x_{11} + \beta y_{11} + \gamma z_{11}$, calling this for a moment D , we join to the foregoing a new equation

$$\alpha x_1 + \beta y_1 + \gamma z_1 = D,$$

and then, eliminating the three quantities, we have

$$\begin{vmatrix} x_1, & y_1, & z_1, & D \\ x_2, & y_2, & z_2, & \frac{1}{2}a_2 \\ x_3, & y_3, & z_3, & g_1 - \frac{1}{2}a_3 \\ \alpha, & \beta, & \gamma, & 0 \end{vmatrix} = 0,$$

giving ∇D as a linear function of $\frac{1}{2}a_1, h_1 - \frac{1}{2}a_2, g_1 - \frac{1}{2}a_3$.

We can in like manner form the expressions for the second differential coefficients of the a, b, c, f, g, h : these will of course contain third differential coefficients of the x, y, z .

Writing down only what is wanted, we have

$$\frac{1}{2}b_{11} = 12.12 + 2.112,$$

$$h_{12} = 12 \cdot 12 + 11 \cdot 22 + 1.122 + 2.112,$$

where of course 12.12 denotes $x_{12}^2 + y_{12}^2 + z_{12}^2$, 1.122 denotes $x_1x_{122} + y_1y_{122} + z_1z_{122}$, and so in other cases; it follows that:

$$\frac{1}{2}(a_{22} + b_{11} - 2h_{12}) = 12.12 - 11.22,$$

so that the third differential coefficients of x, y, z which enter into the expression of $a_{22} + b_{11} - 2h_{12}$ disappear from that of the combination $\frac{1}{2}(a_{22} + b_{11} - 2h_{12})$.

Similarly

$$f_{31} = 31 \cdot 23 + 33 \cdot 12 + 3.123 + 2.133,$$

$$g_{31} = 31 \cdot 23 + 33 \cdot 12 + 3.123 + 1.233,$$

$$c_{12} = 2 \cdot 31 \cdot 23 + 2 \cdot 3 \cdot 123,$$

$$\frac{1}{2}c_{33} = 2 \cdot 31 \cdot 23 + 1.233 + 2.133,$$

and thence

$$f_{31} + g_{31} - c_{12} - \frac{1}{2}c_{33} = -2(31 \cdot 23 - 33 \cdot 12),$$

so that here again we have a combination containing only second differential coefficients of x, y, z .

There are thus, in all, the six combinations

$$\frac{1}{2}(b_{33} + c_{22} - 2f_{23}) \quad (21),$$

$$\frac{1}{2}(c_{11} + a_{33} - 2g_{31}) \quad (33),$$

$$\frac{1}{2}(a_{22} + b_{11} - 2h_{12}) \quad (12),$$

$$\frac{1}{2}(g_{23} + f_{31} - c_{12} - h_{33}) \quad (31),$$

$$\frac{1}{2}(f_{31} + g_{23} - c_{12} - h_{33}) \quad (23),$$

$$\frac{1}{2}(h_{31} + g_{12} - a_{23} - f_{11}) \quad (??),$$

each really containing only second differential coefficients of x, y, z ; and we thus understand how each of these combinations may be expressible in terms of the first differential coefficients of (a, b, c, f, g, h) . We have, in fact, for thus expressing these combinations the six equations called (21), (33), (12), (31), (23), (??), about to be obtained, and which are the generalisations of Warren's six equations for normal coordinates.

The right-hand side of the equation for (12) is

$$\begin{aligned} & G \cdot \frac{1}{2}a_1 + F(h_1 - \frac{1}{2}a_2) + H(g_1 - \frac{1}{2}a_3) \\ & - H \cdot \frac{1}{2}a_2 - E \cdot \frac{1}{2}b_2 - D(f_2 - \frac{1}{2}b_3) \\ & - F(g_1 - \frac{1}{2}a_3) - D(h_2 - \frac{1}{2}b_1) - C \cdot \frac{1}{2}c_3, \end{aligned}$$

which is

$$G \cdot \frac{1}{2}a_1 + H(g_1 - a_2) + F(h_1 - g_1) - E \cdot \frac{1}{2}b_2 - D(h_2 - \frac{1}{2}b_1 - f_2 + \frac{1}{2}b_3) - C \cdot \frac{1}{2}c_3,$$

or, as this might be written,

$$\begin{pmatrix} G & H & F \\ H & E & D \\ F & D & C \end{pmatrix} \begin{pmatrix} \frac{1}{2}a_1 \\ g_1 - a_2 \\ h_1 - g_1 \end{pmatrix} \begin{pmatrix} \frac{1}{2}a_2 \\ \frac{1}{2}b_2 \\ h_2 - \frac{1}{2}b_1 - f_2 + \frac{1}{2}b_3 \end{pmatrix}$$

Retaining the former form, the result is $\nabla \cdot D = G \cdot \frac{1}{2}a_1 + H(g_1 - a_2) + F(h_1 - g_1)$, and it is now very easy to write down the complete system of the 18 equations; viz. if the determinant above written down be called 11.1.2, and so in other cases, then we have

$$\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix}_{2,3} \begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix}_{3,1} \begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix}_{1,2}$$

read for instance the equation obtained above.

There are eighteen functions as shown by the following diagram:

2.3	3.1	1.2
(22)(33) - (23) ²	(33)(11) - (31) ²	(11)(22) - (12) ²
(31)(12) - (11)(23)	(12)(23) - (22)(31)	(23)(31) - (33)(12)
2.3	3.1	1.2

viz. in any line of the diagram the bracketed duads may belong to any one at pleasure, but all to the same, of the three pairs 2.3, 3.1, 1.2;

thus the first line might be $(22.3.1)(33.3.1) - (23.3.1)^2$, or, instead of the 3.1, we might have 2.3 or 1.2. But of the 18 functions I distinguish 6, viz. those in which for the six lines respectively the pairs are 2.3, 3.1, 1.2, 2.3, 3.1, 1.2, as shown in the diagram. Each of these six functions can be obtained under two different forms, and by equating these we have the equations (31), (33), (12), (8), (??), (??) before referred to; thus (12) is obtained by equating two different forms of the function $(11.1.2)(22.1.2) - (12.1.2)^2$; and (31) is obtained by equating two different forms of the function $(23.1.2)(31.1.2) - (33.1.2)(12.1.2)$.

The determinants 11.1.2, etc., may be denoted by accented letters a', b', c', f', g', h' , as follows:

$$\begin{aligned} x_{11}, x_{22}, x_{33}, x_{23}, x_{31}, x_{12} &= a', b', c', f', g', h', \\ y_{11}, y_{22}, y_{33}, y_{23}, y_{31}, y_{12} &= \dots, \\ z_{11}, z_{22}, z_{33}, z_{23}, z_{31}, z_{12} &= \dots, \end{aligned}$$

where 11.22 denotes

$$\begin{vmatrix} 11.22, & 1.11, & 2.11 \\ 1.22, & 1.1, & 1.2 \\ 2.22, & 1.2, & 2.2 \end{vmatrix}$$

that is, $x_{11}^2 + y_{11}^2 + z_{11}^2$, and the like for the other symbols.

In like manner, the square of the determinant (12.1.2), that is, of

$$\begin{vmatrix} 12.12, & 1.12, & 2.12 \\ 1.12, & 1.1, & 1.2 \\ 2.12, & 2.1, & 2.2 \end{vmatrix},$$

or, observing that in the two resulting determinants the terms 11.22 and 12.12 are multiplied by the same factor, the expression for the difference gives

$$(11.22 - 12.12) \begin{vmatrix} 1.1, & 1.2 \\ 1.2, & 2.2 \end{vmatrix} + \begin{vmatrix} 1.11, & 2.11 \\ 1.22, & 1.1, & 1.2 \\ 2.22, & 1.2, & 2.2 \end{vmatrix}$$

containing $11.22 - 12.12$, which, by what precedes, is $\frac{1}{2}(a_{22} + b_{11} - 2h_{12})$; the other terms are also known, viz. the whole value is

$$= -(a_{22} + b_{11} - 2h_{12})(ab - h^2) + \dots$$

which is

$$+b(\dots) + h(\dots) + (ab - h^2) \cdot k,$$

where k is the measure of curvature.

In the same way, we have

$$(23.1.2)(31.1.2) - (33.1.2)(12.1.2)$$

which is

$$= (23.31 - 12.33) \begin{vmatrix} 1.1, & 1.2 \\ 1.2, & 2.2 \end{vmatrix}$$

containing $23.31 - 12.33$, which by what precedes is $= -2(g_{23} + f_{31} - c_{12} - h_{33})$; the other terms are also known, and the value is

$$= -\frac{1}{2}(g_{23} + f_{31} - c_{12} - h_{33})(ab - h^2) + \dots$$

or finally this is

$$= -2(g_{23} + f_{31} - c_{12} - h_{33})(ab - h^2) + a(\dots) + b(\dots) + h(\dots).$$

The remaining four of the six functions can of course be obtained from the two just found by a cyclical interchange of the letters and suffix-numbers 1, 2, 3, and it is not worth while to write down the values.

The two values of $(11.1.2)(22.1.2) - (12.1.2)^2$ are

$$-(a_{22} + b_{11} - 2h_{12})(ab - h^2) + a(\dots) + b(\dots) + h(\dots)$$

and

$$\{G \cdot \frac{1}{2}a_1 + F(h_1 - \frac{1}{2}a_2) + H(g_1 - \frac{1}{2}a_3)\} \times \{\frac{1}{2}(h_2 - \frac{1}{2}b_1) + F \cdot \frac{1}{2}b_2 + \dots\} - (\dots)^2,$$

where, in developing the fourth and fifth lines, it appears that in this expression the coefficients of F^2 , G^2 and FG each of them vanish; the whole equation is thus divisible by C , and omitting this factor throughout, the equation becomes

$$= -2(a_{22} + b_{11} - 2h_{12}) + A(\dots) + G(\dots) + H(\dots) = 0,$$

which is the equation (12).

[The memoir continues with the derivation of the remaining equations and their geometrical interpretations, establishing the fundamental equations of curvilinear coordinates.]

800.

NOTE ON THE STANDARD SOLUTIONS OF A SYSTEM OF
LINEAR EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix.
(1883),
pp. 38–40.]

To fix the ideas, the equations are assumed to be without constant terms. Supposing the system to be insufficient for the determination of the ratios of the unknown quantities, then regarding the unknown quantities as having a definite order of arrangement, there are certain solutions which may be regarded as standard solutions. Take the unknown quantities to be A, B, C, D, E, F, G , &c.; then assuming $A = 0$, or else A, B , each $= 0$, or else A, B, C , each $= 0$, as many equations as possible, the system as thus modified will have a definite solution; for instance, the assumed equations A, B, C, D, E , each $= 0$, may give a definite solution in which F is not $= 0$, and it may then for convenience be put $= 1$. We have thus a solution beginning with $F = 1$. This being so, there will be a solution or solutions with $F = 0$; we cannot then have A, B, C, D, E , each $= 0$, but we again assume $A = 0$, or else A, B , each $= 0$, as many equations as possible; suppose A, B, C , each $= 0$ give a definite solution, with D not $= 0$; and then taking it for convenience to be $= 1$, we have a solution beginning $D = 1$, and for which $F = 0$. Going on in this manner we obtain, it may be, a solution beginning $B = 1$, and for which $D = 0$, $F = 0$; and so on, the process stopping, if not sooner, with a solution beginning $A = 1$, and with the initial letters of the preceding solutions, each $= 0$. We have in this manner the system of standard solutions, of a form such as

A	B	C	D	E	F	G
*	*	*	*	*	1	0
*	*	1	0	*	0	0
*	1	0	0	0	0	0
1	0	0	0	0	0	0

where the $*$ denotes a value which is not in general $= 0$, but which may in any particular case happen to be so.

For instance, let it be required for the binary quartic $(a, b, c, d, ex, y)^4$, to find the aszygetic seminvariants of the degree 4 and weight 6. Assuming for the seminvariant the value in the left-hand column of the diagram, the unknown coefficients being A, B, C, D, E, F, G , this must be reduced to zero by the operation

$$c\partial_b + 2d\partial_c + 3e\partial_d + 4f\partial_e;$$

and we thus obtain as many equations as there are terms of the degree 4 and weight 5, as appearing by the second column

A	$a \cdot ce$
B	$a \cdot de + b \cdot ce$
C	$a \cdot e + b \cdot de + c \cdot ce$
D	$b \cdot e + c \cdot de + d \cdot ce$
E	$c \cdot e + d \cdot de + e \cdot ce$
F	$d \cdot e + e \cdot de$
G	$e \cdot e$

As is known, there is no irreducible seminvariant, but only the composite forms

$$I = a(ce - d^2),$$

$$II = (ac - b^2)(ae - 4bd + 3c^2),$$

the developed values of which are given above: II (as a form beginning $A = 1$ and with $B = 0$) can be nothing else than, and is in fact $= Q$; and so I (as a form beginning with $A = 1, B = 1$) can be nothing else than, and is in fact $= Q - P$; that is, we have

$$P = -I + II, \quad \text{or} \quad I = -P + Q,$$

$$Q = II, \quad II = Q;$$

and so, in general, we have a standard set of values for the aszygetic seminvariants of a given degree and weight; or, what is the same thing, for the covariants of a given deg-order.

801.

ON SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix.
(1883),
pp. 131–138.]

The present paper is a somewhat fragmentary one, but it contains some results which seem to me to be worth putting on record.

I consider here not any binary quantic in particular, but the whole series $(a, b, cx, y)^2$, $(a, b, c, dx, y)^3$, &c.; or in a somewhat different point of view, I consider the indefinite series of coefficients $(a, b, c, d, e, \dots)$; here, instead of covariants and invariants, we have only seminvariants; viz. a seminvariant is a function reduced to zero by the operator

$$a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + \dots;$$

for instance, seminvariants are

$$a, \quad ac - b^2, \quad a^2d - 3abc + 2b^3, \quad a^2e - 4abd + 4ac^2 + 4b^2d - 6abc^2 - b^2e, \quad \text{etc.}$$

A seminvariant is of a certain degree θ in the coefficients, and of a certain weight w (viz. the coefficients $a, b, c, d, \dots$ are reckoned as being of the weights 0, 1, 2, 3, $\dots$ respectively); it is, moreover, of a certain rank ρ (viz. according as the highest letter therein is $a, c, d, e, \dots$ it is never the rank is taken to be 0, 2, 3, 4, $\dots$), and we have $w = \frac{1}{2}\rho\theta$. The seminvariant may be regarded as belonging to a quantic $(a, \dots x, y)^n$, where n , the order of which, is equal to or greater than ρ ; viz. in regard to such quantic the seminvariant, say A , is the leading coefficient of a covariant

$$(A, B, \dots, Kx, y)^\mu,$$

where the weights of the successive coefficients are $w, w+1, \dots$ up to $n\theta - w$; hence the number of terms less unity, that is, μ , is $= n\theta - 2w$; the least value of μ is thus $= \rho\theta - 2w$, which is either zero, or positive; in the former case, $w = \frac{1}{2}\rho\theta$, the seminvariant is an invariant of the quantic $(a, \dots x, y)^n$, the order of which is equal to the rank of the seminvariant; but if $w < \frac{1}{2}\rho\theta$, then it is the leading coefficient of a covariant $(A, B, \dots, Kx, y)^\mu$ of the same quantic $(a, \dots x, y)^n$; and in every case, taking $n > \rho$, the seminvariant is the leading coefficient A of a covariant $(A, B, \dots, Kx, y)^\mu$ of a quantic $(a, \dots x, y)^n$.

Take A as belonging to the quantic $(a, \dots x, y)^n$; corresponding to such quantic, we have an operator Λ of the same rank n , viz.

$$\begin{aligned}\Lambda &= 2b\partial_a + c\partial_b && \text{for } n = 2, \\ &= 3b\partial_a + 2c\partial_b + d\partial_c && \text{for } n = 3, \\ &= 4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d && \text{for } n = 4.\end{aligned}$$

Operating with Λ on A , we have a series of terms $A, \Lambda A, \Lambda^2 A, \dots, \Lambda^{n\theta-w} A$; but the next term $\Lambda^{n\theta-2w+1} A$, and of course every succeeding term, is $= 0$, and this being so, the coefficients of the covariant $(A, B, \dots, Kx, y)^\mu$ are

$$(1, \tfrac{1}{1}\Lambda, \tfrac{1}{1\cdot 2}\Lambda^2, \dots)A,$$

or what is the same thing, each coefficient is obtained from the next preceding one by the formulae

$$B = \tfrac{1}{1}\Lambda A, \quad C = \tfrac{1}{2}\Lambda B, \quad D = \tfrac{1}{3}\Lambda C, \quad \dots$$

The coefficients A and K, B and $J, \dots$ are derived one from the other by reversal of the order of the coefficients of $(a, b, \dots x, y)^n$ with or without a change of sign, and thus it is only necessary to calculate up to the middle coefficient, or pair of coefficients; and we obtain, moreover, a verification.

Calculating in this manner the covariant $(A, B, \dots, Kx, y)^\mu$, which belongs to the quantic $(a, \dots x, y)^n$, if we herein change $a, b, c, \dots$ into $ax + by, bx + cy, cx + dy, \dots$ we obtain the covariant belonging to the quantic $(a, \dots x, y)^{n+1}$; and in this covariant making the like change, or what is the same thing, in the first-mentioned covariant changing $a, b, c, \dots$ into $(a, b, cx, y)^1, (b, c, dx, y)^1, (c, d, ex, y)^1, \dots$ we obtain the covariant belonging to the quantic $(a, \dots x, y)^{n+2}$; and so on.

[The paper continues with detailed examples and calculations of seminvariants for specific degrees and weights, establishing the general theory.]

802.

NOTE ON CAPTAIN MACMAHON'S PAPER "ON THE
DIFFERENTIAL EQUATION

$$X^{-\frac{2}{3}}dx + Y^{-\frac{2}{3}}dy + Z^{-\frac{2}{3}}dz = 0".$$

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix.
(1883),
pp. 182-184.]

Captain MacMahon's paper relates to the differential equation

$$X^{-\frac{2}{3}}dx + Y^{-\frac{2}{3}}dy + Z^{-\frac{2}{3}}dz = 0,$$

where X, Y, Z are linear functions of the coordinates. The equation belongs to a class of differential equations which I had formerly considered in the "Memoir on the Condition for the Existence of a Homogeneous Integral of a Differential Equation" [381]. The equation is integrable, and the integral can be expressed in terms of elliptic functions; the precise form of the integral depends on the geometrical configuration of the lines represented by the equations $X = 0, Y = 0, Z = 0$.

The present note relates to a particular case of the general theory. Writing the equation in the form

$$X^{-\frac{2}{3}}dx + Y^{-\frac{2}{3}}dy + Z^{-\frac{2}{3}}dz = 0,$$

where $X = x - a, Y = y - b, Z = z - c$, the integral can be exhibited in a simple form by means of the elliptic-function formulae.

[The note continues with the derivation of the integral in terms of elliptic functions and the discussion of special cases.]

803.

NOTE ON ABEL'S THEOREM.

[From the *Proceedings of the Cambridge Philosophical Society*, vol. iv.
(1883),
pp. 33–35.]

In general, if $f = (x, y, 1)^3 = 0$ be the equation of a cubic curve, and if

$$\omega = \int \frac{dx}{\partial f / \partial y} = - \int \frac{dy}{\partial f / \partial x},$$

then if 1, 2, 3 are the intersections of the curve by an arbitrary right line, the coordinates of these points being (x_1, y_1) , (x_2, y_2) , (x_3, y_3) respectively, we have, by Abel's theorem,

$$d\omega_1 + d\omega_2 + d\omega_3 = 0,$$

viz. this is the differential relation corresponding to the integral relation which expresses that the three points are the intersections of the cubic curve by a right line, or say to the integral equation

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0,$$

in which equation y_1, y_2, y_3 are regarded as functions of x_1, x_2, x_3 respectively, given by means of the equations $f_1 = 0, f_2 = 0, f_3 = 0$, which express that the points are on the cubic curve. See my "Memoir on the Abelian and Theta Functions," [819].

In particular, if the equation of the curve is

$$y^2 = 4x^3 - g_2x - g_3 = 4(x - e_1)(x - e_2)(x - e_3),$$

then we have

$$\omega = \int \frac{dx}{\sqrt{4x^3 - g_2x - g_3}} = \int \frac{dx}{y},$$

and the theorem gives

$$\int^{x_1} \frac{dx}{y} + \int^{x_2} \frac{dx}{y} + \int^{x_3} \frac{dx}{y} = \text{const.},$$

where x_1, x_2, x_3 are the abscissae of the three intersections of the cubic curve by an arbitrary line; or, what is the same thing, if the line is $y = mx + n$, then x_1, x_2, x_3 are the roots of the equation

$$(mx + n)^2 = 4x^3 - g_2x - g_3.$$

[The note continues with further applications of Abel's theorem to specific curves and integrals.]

804.

ON THE ELLIPTIC-FUNCTION SOLUTION OF THE EQUATION

$$x^3 + y^3 = 1.$$

[From the *Proceedings of the Cambridge Philosophical Society*, vol. iv.
 (1883),
 pp. 35–38.]

The equation $x^3 + y^3 = 1$ can be solved by means of elliptic functions in several different ways. The most direct method is to write

$$x = \frac{1 + \operatorname{cn}(u)}{\operatorname{sn}(u)} \cdot \frac{1}{\sqrt{3}}, \quad y = \frac{1 - \operatorname{cn}(u)}{\operatorname{sn}(u)} \cdot \frac{1}{\sqrt{3}},$$

where the modulus k of the elliptic functions is taken to be $k = \sin 15^\circ = \frac{1}{4}(\sqrt{6} - \sqrt{2})$; or, what is the same thing, $k^2 = \frac{1}{8}(2 - \sqrt{3})$.

With this value, we have

$$x^3 + y^3 = \frac{1}{3\sqrt{3}} \cdot \frac{(1 + \operatorname{cnu})^3 + (1 - \operatorname{cnu})^3}{\operatorname{sn}^3 u} = \frac{1}{3\sqrt{3}} \cdot \frac{2 + 6\operatorname{cn}^2 u}{\operatorname{sn}^3 u}.$$

By the elliptic-function formulae, this can be reduced to unity; and we thus obtain the general solution of the equation $x^3 + y^3 = 1$ in the parametric form

$$x = \frac{1}{\sqrt{3}} \cdot \frac{1 + \operatorname{cnu}}{\operatorname{snu}}, \quad y = \frac{1}{\sqrt{3}} \cdot \frac{1 - \operatorname{cnu}}{\operatorname{snu}},$$

where u is an arbitrary parameter.

[The paper continues with alternative forms of the solution and the discussion of related equations.]

805.

A PROOF OF WILSON'S THEOREM.

[From the *Messenger of Mathematics*, t. xii. (1883),
pp. 38-41.]

Wilson's theorem is that if p be a prime number, then $(p-1)! + 1$ is divisible by p ; or, what is the same thing, $(p-1)! \equiv -1 \pmod{p}$.

The proof depends on the properties of the binomial coefficients. Considering the expansion of $(1+1)^p = 2^p$, we have

$$2^p = 2 + \frac{p(p-1)}{1 \cdot 2} + \frac{p(p-1)(p-2)}{1 \cdot 2 \cdot 3} + \cdots + p,$$

and hence, mod p ,

$$2^p - 2 \equiv 0 \pmod{p}.$$

Similarly, from $(1+1)^{p-1} = 2^{p-1}$, expanding and reducing modulo p , we obtain relations between the factorials. More directly, considering the product of all the residues modulo p , we have that the $p-1$ residues $1, 2, 3, \dots, p-1$ are such that each multiplied by its reciprocal gives 1; and the numbers which are their own reciprocals are 1 and $p-1$ (since $x^2 \equiv 1$ has only the solutions $x \equiv 1$ and $x \equiv p-1$). Hence, pairing the remaining $p-3$ numbers with their reciprocals, we have

$$2 \cdot 3 \cdot 4 \cdots (p-2) \equiv 1 \pmod{p},$$

and therefore

$$(p-1)! = 1 \cdot 2 \cdot 3 \cdots (p-1) \equiv 1 \cdot (p-1) \equiv -1 \pmod{p},$$

which is Wilson's theorem.

Conversely, if $(p-1)! + 1$ is divisible by p , then p is a prime number; for if p were composite, say $p = ab$ with $a, b < p$, then a would divide $(p-1)!$, and so $(p-1)! + 1$ could not be divisible by p .

The proof may also be presented as follows. Let n be a prime number; and imagine n points, the vertices of a regular polygon; any polygon which can be formed with these n points as vertices is either regular or else it is one of a set of n equal and similar polygons. For instance, $n = 5$, the polygon as shown in the figure is one of a set of 5 equal and similar polygons; in fact, if the points taken in their cyclical order, but beginning at pleasure with any one of the 5 points

are called 1, 2, 3, 4, 5, then we have 5 such polygons 13254; and so in general. The whole number of polygons is $1 \cdot 2 \cdot 3 \cdots (n-1)$; and the number of the regular polygons is $(n-1)$; hence the number of the remaining polygons is $= (n-1)\{1 \cdot 2 \cdots (n-2) - 1\}$; and this number must therefore be divisible by n ; that is, $1 \cdot 2 \cdots (n-1) - n + 1$ is divisible by n ; or, what is the same thing, $1 \cdot 2 \cdots (n-1) + 1$ is divisible by n , which is the theorem in question.

806.

DETERMINATION OF THE ORDER OF A SURFACE.

[From the *Messenger of Mathematics*, t. xii. (1883),
pp. 41-42.]

The order of a surface is the number of its intersections by an arbitrary line. If the surface is given by an equation $F(x, y, z, w) = 0$ in point-coordinates, then its order is the degree of the equation. But if the surface is given as the envelope of a plane whose equation contains parameters in a given manner, or if it is defined as the locus of points satisfying certain conditions, then the order requires to be determined.

If the surface is the envelope of the plane

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

where $(\xi, \eta, \zeta, \omega)$ satisfy a given relation $\Phi(\xi, \eta, \zeta, \omega) = 0$, then the order of the surface is equal to the class of the surface $\Phi = 0$ regarded as a surface in plane-coordinates. In particular, if Φ is of degree n , the surface is in general of order $n(n-1)^2$.

For a surface defined as the locus of a line meeting three given curves, or satisfying other geometrical conditions, the order can be found by considering the number of intersections with an arbitrary line, or by other appropriate methods of enumerative geometry.

[The note continues with examples and applications to particular surfaces.]

807.

NOTE ON A FORM OF THE MODULAR EQUATION IN THE
TRANSFORMATION OF THE THIRD ORDER.

[From the *Messenger of Mathematics*, t. xii. (1883),
pp. 42–45.]

In the transformation of elliptic functions of the third order, the modular equation which connects the original modulus k with the transformed modulus λ can be expressed in various forms. The most symmetrical form is that given by Jacobi,

$$\sqrt{k\lambda} + \sqrt{k'\lambda'} = 1,$$

where $k' = \sqrt{1 - k^2}$, $\lambda' = \sqrt{1 - \lambda^2}$.

This can also be written as

$$(k\lambda)^{\frac{1}{2}} + (k'\lambda')^{\frac{1}{2}} = 1,$$

or, rationalizing, in the form of an algebraic equation between k and λ of degree 4 in each variable.

An alternative form, due to Sohnke, is

$$(k\lambda)^{\frac{1}{4}} + (k'\lambda')^{\frac{1}{4}} = 1,$$

which is equivalent to the former but presents the relation in a more convenient form for certain applications.

The present note establishes the equivalence of these forms and deduces some consequences relating to the multiplication and transformation of elliptic functions.

In my Treatise on Elliptic Functions, pp. 214–216, writing only τ , J instead of Ω , J , and α , β instead of α' , β' , I have shown as follows: viz. if k , λ denote as usual the original modulus, and the transformed modulus, and if

$$I = \frac{(k^4 + 14k^2 + 1)^3}{64k^2(1 - k^2)^2}, \quad J = \frac{(\lambda^4 + 14\lambda^2 + 1)^3}{64\lambda^2(1 - \lambda^2)^2},$$

then the relation between I and J can be found by the elimination of α , β from the equations

$$\alpha + \beta = 1, \quad I = \frac{(1 + 8\alpha)^3}{64\alpha(1 - \alpha)^3}, \quad J = \frac{(1 + 8\beta)^3}{64\beta(1 - \beta)^3}.$$

By a very slight change we obtain the result given by Prof. Klein in his paper, "Ueber die Transformation der elliptischen Functionen, &c.," Math. Ann. t. xiv. (1879), pp. 111–172; viz., see p. 143, the relation is to be obtained by the elimination of τ , τ' from the equation $\tau\tau' = 1$, and the equations

$$J : J - 1 : 1 = (\tau - 1)(9\tau - 1)^3 : (27\tau^2 - 18\tau - 1)^2 : -64\tau,$$

$$J' : J' - 1 : 1 = (\tau' - 1)(9\tau' - 1)^3 : (27\tau'^2 - 18\tau' - 1)^2 : -64\tau',$$

these last equations being equivalent to two equations only in virtue of the identity

$$(\tau - 1)(9\tau - 1)^3 + 64\tau = (27\tau^2 - 18\tau - 1)^2,$$

and the like identity in τ' . In fact, writing $\alpha = \tau$, $\beta = \tau'$, the equation $\alpha + \beta = 1$ becomes $\tau\tau' = 1$; and then for α, β substituting their values, we have

$$J = \frac{-64\tau}{(\tau - 1)(9\tau - 1)^3}, \quad J' = \frac{-64\tau'}{(\tau' - 1)(9\tau' - 1)^3},$$

which are the formulae in question.

808.

SCHRÖTER'S CONSTRUCTION OF THE REGULAR PENTAGON.

[From the *Messenger of Mathematics*, t. xii. (1883),
p. 177.]

The following is a construction for the regular pentagon, due to Schröter:

Given the circle, draw any diameter AB . Through A draw a chord AC making any angle with AB . Bisect the angle CAB by the line AD , meeting the circle in D . Then the chord BD is a side of the regular pentagon inscribed in the circle.

The proof depends on the property that if AD bisects the angle CAB , then the arc BD is one-fifth of the circumference. For, by the angle-bisector property, we have

$$\angle BAD = \frac{1}{2}\angle CAB,$$

and by the properties of the circle,

$$\angle BOD = 2\angle BAD = \angle CAB,$$

where O is the centre. Hence the arc BD bears the proper relation to the whole circumference, and by the fundamental property of the regular pentagon, the side BD subtends an angle of 72° at the centre, which is one-fifth of the complete angle.

Alternatively, the construction may be stated as follows: If in a circle two chords AB , AC are drawn from a point A , and the angle CAB is bisected by a line meeting the circle in D , then BD is the side of the regular pentagon, provided that AC is properly chosen; in fact, AC must be such that BC is a side of the regular decagon.

[The note continues with further remarks on the construction and its relation to the geometry of the regular pentagon.]

809.

NOTE ON A SYSTEM OF EQUATIONS.

[From the *Messenger of Mathematics*, t. xii. (1883),
pp. 191–192.]

The system of equations

$$\begin{aligned}x + y + z &= a, \\yz + zx + xy &= b, \\xyz &= c,\end{aligned}$$

is fundamental in algebra. The present note relates to certain symmetric functions of the roots and to the conditions for special relations between the roots.

If x, y, z are the roots of the cubic equation

$$t^3 - at^2 + bt - c = 0,$$

then the discriminant of this equation is

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc.$$

The condition for equal roots is $\Delta = 0$; and the various special cases (two equal roots, three equal roots) correspond to the different ways in which this equation can be satisfied.

The note also considers the system of four equations in four unknowns, and establishes certain identities relating to the elementary symmetric functions.

The equations are

$$\begin{aligned}yz &= ax + by, \\zx &= cx + dy, \\xy &= ex + fy,\end{aligned}$$

where

$$\frac{b}{d} = \frac{a}{f} = \frac{c}{e},$$

or, what is the same thing, if a, b, c, d are given, then

$$e = \frac{cd}{b}, \quad f = \frac{d(a-d)}{b},$$

and this being so, the equations are equivalent to two independent equations; viz. starting from the first and the second equations, we have

$$dyz - bxy = (ad - bc)x,$$

that is,

$$dx - by = (ad - bc)y;$$

and thence

$$dxy - by^2 = (ad - bc)y,$$

or

$$d(ex + dy) - by^2 = (ad - bc)y;$$

which, attending to the values of e and f , is the third equation $xy = ex + fy$. We have

$$\frac{x}{y} = \frac{ax + by}{ex + fy} = \frac{yz}{ex + dy} = \frac{ex + dy}{x},$$

which completes the system.

810.

ON THE LINEAR TRANSFORMATION OF THE
THETA-FUNCTIONS.

[From the *Messenger of Mathematics*, t. xiii. (1884),
pp. 54–60.]

The theta-functions satisfy certain linear transformation formulae, which are fundamental in the theory of elliptic functions. If τ is replaced by $\tau' = -1/\tau$, then the theta-functions are transformed according to the formulae

$$\begin{aligned}\vartheta_3(0 \mid \tau') &= \sqrt{-i\tau} \vartheta_3(0 \mid \tau), \\ \vartheta_2(0 \mid \tau') &= \sqrt{-i\tau} \vartheta_4(0 \mid \tau), \\ \vartheta_4(0 \mid \tau') &= \sqrt{-i\tau} \vartheta_2(0 \mid \tau),\end{aligned}$$

and for the arguments, with $v' = v/\tau$,

$$\begin{aligned}\vartheta_3(v' \mid \tau') &= \sqrt{-i\tau} e^{\pi i \tau v^2} \vartheta_3(v \mid \tau), \\ \vartheta_2(v' \mid \tau') &= \sqrt{-i\tau} e^{\pi i \tau v^2} \vartheta_4(v \mid \tau), \\ \vartheta_4(v' \mid \tau') &= \sqrt{-i\tau} e^{\pi i \tau v^2} \vartheta_2(v \mid \tau).\end{aligned}$$

These are the transformation formulae of order 2. For the transformation of order n , if $\tau' = \tau/n$, there are corresponding formulae involving the n -th roots of unity.

The present paper establishes these formulae by a direct method, and deduces from them the modular equations and the multiplication-formulae for the elliptic functions.

The functions referred to are the single Theta Functions; these may be defined as doubly infinite products, as was in fact done in my “Memoire sur les fonctions doublement periodiques,” *Liouv.* t. x. (1845), pp. 385–420, [25]; and it is interesting to consider from this point of view the theory of their linear transformation: this I propose to do in the present paper, adopting throughout the notation of Smith’s “Memoir on the Theta and Omega Functions.”

The periods K, iK' are, in general, imaginary quantities $K = A + Bi$ where $AD - BC$ is positive; writing then $\omega = \frac{iK'}{K}$, and $q = e^{\pi i \omega}$, also for shortness q^s denotes $e^{s\pi i \omega}$, the expression of the

odd theta-function $\vartheta_1(x, \omega)$ as a doubly infinite product is

$$\vartheta_1(x, \omega) = q^{\frac{1}{4}} \sin x \prod_{m,n} \left(1 + \frac{x}{m\omega + n} \right) \left(1 - \frac{x}{m\omega + n} \right),$$

where (m, n) have any positive or negative integer values (the combination $m = 0, n = 0$ excluded) from $m = -\mu$ to μ , and $n = -\nu$ to ν , μ and ν being each ultimately infinite but so that $\mu/\nu = \infty$; this condition in regard to the limits is indicated by $\mu/\nu = \infty$; and similarly $\nu/\mu = \infty$ would indicate that ν was infinite in comparison with μ .

The condition as to the limits might be that (m, n) have any positive or negative values (excluding as before) such that the modulus of $m + n\omega$ does not exceed a positive value T , which is ultimately taken to be infinite; this condition may be indicated by $\bmod = \infty$. The values of the double product corresponding to the different conditions as to the limits are not equal, but they differ only by an exponential factor, the exponent being a multiple of x^2 ; we thus have

$$\frac{\vartheta_1(x, \omega \mid \mu/\nu = \infty)}{\vartheta_1(x, \omega \mid \bmod = \infty)} = \exp(Vx^2),$$

where V is a determinate value, depending on K and K' ; and similarly

$$\frac{\vartheta_1(x, \Omega \mid \mu/\nu = \infty)}{\vartheta_1(x, \Omega \mid \bmod = \infty)} = \exp(Dx^2),$$

where D is a determinate value depending in like manner on Λ, Λ' .

We have, then, as above

$$\vartheta_1 \left(\frac{x}{a + b\omega}, \frac{c + d\omega}{a + b\omega} \right) = C \cdot \vartheta_1(x, \omega),$$

where C is a determinate exponential factor; and in the case of a linear transformation, we have

$$\omega' = \frac{c + d\omega}{a + b\omega}, \quad a, b, c, d \text{ integers}, \quad ad - bc = n,$$

that is, n is the order of the transformation.

811.

ON ARCHIMEDES' THEOREM FOR THE SURFACE OF A CYLINDER.

[From the *Messenger of Mathematics*, t. xiii. (1884), pp. 107–108.]

Archimedes' theorem states that the surface of a cylinder is equal to the product of its height by the circumference of its base; or, what is the same thing, if the cylinder is right, the lateral surface is equal to that of the circumscribing prism.

The present note gives a simple proof of this theorem by the methods of the infinitesimal calculus.

Consider a right cylinder of radius a and height h . The equation of the surface is

$$x^2 + y^2 = a^2,$$

with $0 \leq z \leq h$. The element of surface is

$$dS = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy,$$

and for the cylinder this becomes

$$dS = \frac{a}{\sqrt{a^2 - x^2}} dx dz,$$

whence, integrating over the whole surface,

$$S = \int_0^h dz \int_{-a}^a \frac{a}{\sqrt{a^2 - x^2}} dx = h \cdot 2\pi a = 2\pi ah,$$

which is the circumference of the base multiplied by the height, as stated by Archimedes.

The note also extends the theorem to oblique cylinders and to portions of cylinders cut by planes.

The measure of the surface of a cylinder was first obtained by Archimedes in his *Treatise on the Sphere and Cylinder* (Book i., Prop. xiv.), (*Œuvres d'Archimède*, par F. Peyrard, 4° Paris, 1807, pp. 26–31; viz. Archimedes showed that the surface of the cylinder was equal to the area of a circle, having its radius a mean proportional between the height and the diameter of the circular base [$S = 2\pi ah = \pi\{V(2a \cdot h)\}^2$].

The following is in effect his demonstration: He considers regular polygons (with the same number of sides) inscribed in and circumscribed about a circle; and, as regards the cylinder, the prisms standing on these polygons. Say for the circular base of the cylinder we have

	Surface	Area
S^*	circumscribed prism	B^* circumscribed polygon
S	cylinder	B circle
s	inscribed prism	b inscribed polygon

where the four polygons referred to by S^* , S , B , b have all of them the same number of sides.

It is in the preceding propositions (by means of an axiom as to curve lines) shown that

$$S^* > S > s, \quad B^* > B > b, \quad S^* = B^*, \quad s = b,$$

and it is further shown that

$$\frac{S^*}{S} > 1, \quad \frac{B^*}{B} > 1, \quad \frac{s}{S} < 1, \quad \frac{b}{B} < 1.$$

It is moreover shown that, by taking the number of sides sufficiently large, the ratio $B^* : B$, or say the fraction B^*/B (which is greater than 1) may be made less than any given quantity $1 + \epsilon$.

It is then to be shown that $S = B$. If not, then either $B < S$ or $B > S$.

This being so, it is possible to make $B^*/B < S/B$, that is, $S^*/B < S/B$, or $S^*/S < B/B$, which is absurd, since $S^*/S > 1$ while $B/B = 1$; or else $B > S$. This being so, it is possible to make $B^*/B < B/S$, that is, $B^*/S < B/S$, or $B^*/B < S/S$, which is absurd, since $B^*/B > 1$ while $S/S = 1$; and consequently $S = B$, the theorem in question.

I take the opportunity of referring to two theorems by Archimedes, Lemmas, Prop. v. and vi., Peyrard, pp. 429–435, which relate to the contacts of circles. We have in each of them the figure which he calls the Arbelon, viz. if A , C , B are points in this order on the same straight line, then the figure consists of the three semicircles on the diameters AC , CB , and AB respectively, and the Arbelon is the space included between the three semi-circumferences.

In Prop. v., we have also the common tangent at C to the two semicircles AC , CB ; this divides the Arbelon into two mixtilinear

triangles (each bounded by the common tangent, one of the smaller semicircles, and a portion of the larger semi-circle), and inscribing each of these a circle, the theorem is that the two inscribed circles are of equal magnitude.

In Prop. vi., the theorem is that the radii of the smaller semicircles being as $3 : 2$, then the radius of the circle inscribed in the Arbelon is to the diameter of the larger semicircle as 6 to 19. But it is noticed that the demonstration would apply to any other value of the ratio; and, in fact, if the radii of the two smaller circles are as $a : b$, then the radius of the inscribed circle is to the diameter of the larger semicircle as ab to $a^2 + ab + b^2$, which is the general form of the theorem.

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR CAY-
LEY

VOL. XII

PAGES 51–100 (PAPERS 802–816) Arthur Cayley (1821–1895)

Published 1889

*Transcribed from the original Cambridge University Press edition,
1889.*

802. Note on Captain MacMahon's Paper

[From the *Proceedings of the London Mathematical Society*, vol. xiv.
(1883),
pp. 111, 112.]

If

$$d\omega = \frac{dx}{X^{\frac{2}{3}}} = \frac{dx}{y^2},$$

and corresponding to the differential relation

$$X_1^{-\frac{2}{3}} dx_1 + X_2^{-\frac{2}{3}} dx_2 + X_3^{-\frac{2}{3}} dx_3 = 0,$$

we have the integral relation

$$\nabla = \begin{vmatrix} X_1^{\frac{1}{3}}, & X_2^{\frac{1}{3}}, & X_3^{\frac{1}{3}} \\ x_1, & x_2, & x_3 \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

viz. this last equation, as containing no arbitrary constant, is a particular integral of the differential equation.

If instead of x_1, x_2, x_3 we write x, y, z , then we have

$$\nabla = \begin{vmatrix} X^{\frac{1}{3}}, & Y^{\frac{1}{3}}, & Z^{\frac{1}{3}} \\ x, & y, & z \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

as a particular integral of the differential equation

$$X^{-\frac{2}{3}} dx + Y^{-\frac{2}{3}} dy + Z^{-\frac{2}{3}} dz = 0.$$

To rationalize the integral equation, write

$$\alpha, \beta, \gamma = y - z, z - x, x - y \quad (\text{so that } \alpha + \beta + \gamma = 0),$$

the equation is

$$\alpha X^{\frac{1}{3}} + \beta Y^{\frac{1}{3}} + \gamma Z^{\frac{1}{3}} = 0;$$

and we thence have

$$\alpha^3 X + \beta^3 Y + \gamma^3 Z = 3\alpha\beta\gamma X^{\frac{1}{3}} Y^{\frac{1}{3}} Z^{\frac{1}{3}}.$$

The left-hand side is

$$\begin{aligned} & \alpha^3(A + 3Bx + 3Cx^2 + Dx^3) \\ & + \beta^3(A + 3By + 3Cy^2 + Dy^3) \\ & + \gamma^3(A + 3Bz + 3Cz^2 + Dz^3); \end{aligned}$$

or assuming

$$\alpha', \beta', \gamma' = x(y-z), y(z-x), z(x-y) \quad (\text{so that } \alpha' + \beta' + \gamma' = 0),$$

this is

$$= A(\alpha^3 + \beta^3 + \gamma^3) + 3B(\alpha^2\alpha' + \beta^2\beta' + \gamma^2\gamma') + 3C(\alpha\alpha'^2 + \beta\beta'^2 + \gamma\gamma'^2) + D(\alpha'^3 + \beta'^3 + \gamma'^3).$$

But taking λ arbitrary, and

$$a, b, c = \alpha + \lambda\alpha', \beta + \lambda\beta', \gamma + \lambda\gamma',$$

then

$$a + b + c = 0,$$

whence

$$a^3 + b^3 + c^3 = 3abc;$$

or substituting for a, b, c their values, and comparing the coefficients of the several powers of λ ,

$$\begin{aligned} \alpha^3 + \beta^3 + \gamma^3 &= 3\alpha\beta\gamma, \\ \alpha^2\alpha' + \beta^2\beta' + \gamma^2\gamma' &= \alpha'\beta\gamma + \beta'\gamma\alpha + \gamma'\alpha\beta = \alpha\beta\gamma(x + y + z), \\ \alpha\alpha'^2 + \beta\beta'^2 + \gamma\gamma'^2 &= \alpha\beta'\gamma' + \beta\gamma'\alpha' + \gamma\alpha'\beta' = \alpha\beta\gamma(yz + zx + xy), \\ \alpha'^3 + \beta'^3 + \gamma'^3 &= 3\alpha'\beta'\gamma' = 3\alpha\beta\gamma \cdot xyz. \end{aligned}$$

Hence we have

$$\alpha^3 X + \beta^3 Y + \gamma^3 Z = 3\alpha\beta\gamma\{A + B(x + y + z) + C(yz + zx + xy) + Dxyz\},$$

or the integral equation is

$$\{A + B(x + y + z) + C(yz + zx + xy) + Dxyz\} = X^{\frac{1}{3}} Y^{\frac{1}{3}} Z^{\frac{1}{3}},$$

that is,

$$\{A + B(x + y + z) + C(yz + zx + xy) + Dxyz\}^3 = XYZ,$$

the elegant result given by Capt. MacMahon at the beginning of his paper.

The author in a letter to me, dated Jan. 13, 1883, remarks that the particular integral of the equation in question

$$X^{-\frac{2}{3}} dx + Y^{-\frac{2}{3}} dy + Z^{-\frac{2}{3}} dz = 0,$$

is expressible as a determinant in a rational form as follows. Writing it $XYZ = P^3$, where

$$P = A + B(x + y + z) + C(yz + zx + xy) + Dxyz,$$

then the form is

$$\nabla_1 = \begin{vmatrix} 1, & \left(\frac{1}{3}P \frac{dX}{dx} - X \frac{dP}{dx}\right)^3, & X \\ 1, & \left(\frac{1}{3}P \frac{dY}{dy} - Y \frac{dP}{dy}\right)^3, & Y \\ 1, & \left(\frac{1}{3}P \frac{dZ}{dz} - Z \frac{dP}{dz}\right)^3, & Z \end{vmatrix} = 0,$$

for, as shown by Captain MacMahon in his paper, each of the three terms such as

$$Z \left(\frac{1}{3}P \frac{dY}{dy} - Y \frac{dP}{dy}\right)^3 - Y \left(\frac{1}{3}P \frac{dZ}{dz} - Z \frac{dP}{dz}\right)^3,$$

which compose the determinant, is divisible by $XYZ - P^3$.

It may be added that we have identically

$$\frac{1}{3}P \frac{dX}{dx} - X \frac{dP}{dx} = (AC - B^2)(2x - y - z) + (AD - BC)(x^2 - yz) + (BD - C^2)(x^2y + x^2z - y^2z - yz^2)$$

and of course like values for the other two expressions in the determinant.

803. On Mr Anglin's Formula for the Successive Powers of the Root of an Algebraical Equation

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix.
(1883),
pp. 223, 224.]

SUPPOSE $x^m - px^{m-1} + qx^{m-2} - \dots = 0$, then the successive powers x^m, x^{m+1}, x^{m+2} , &c. of x can be expressed in the form $Px^{m-1} - Qx^{m-2} + Rx^{m-3} - \dots$. Mr Anglin has obtained for this purpose a very elegant formula, with a demonstration which (it occurred to me) might be presented under a somewhat simplified form; and he has permitted me to draw up the present Note.

Take, for greater convenience, the equation to be

$$x^4 - px^3 + qx^2 - rx + s = 0,$$

and let $h_1, h_2, h_3, \dots$ be the sums of the homogeneous products of the roots, of the orders 1, 2, 3, &c. respectively; then, writing also $h_0 = 1$, we have

$$\begin{aligned} h_1 &= h_0p, \\ h_2 &= h_1p - h_0q, \\ h_3 &= h_2p - h_1q + h_0r, \\ h_4 &= h_3p - h_2q + h_1r - h_0s, \\ h_5 &= h_4p - h_3q + h_2r - h_1s, \\ &\vdots \end{aligned}$$

And this being so, starting from the equation

$$x^4 = px^3 - qx^2 + rx - s,$$

that is,

$$= h_1x^3 - h_0qx^2 + h_0rx - h_0s,$$

we obtain successively

$$\begin{aligned}
x^5 &= h_1(px^3 - qx^2 + rx - s) \\
&\quad - h_0qx^3 + h_0rx^2 - h_0sx \\
&= h_2x^3 - (h_1q - h_0r)x^2 + (h_1r - h_0s)x - h_1s, \\
x^6 &= h_2(px^3 - qx^2 + rx - s) \\
&\quad - (h_1q - h_0r)x^3 + (h_1r - h_0s)x^2 - h_1sx \\
&= h_3x^3 - (h_2q - h_1r + h_0s)x^2 + (h_2r - h_1s)x - h_2s, \\
x^7 &= h_3(px^3 - qx^2 + rx - s) \\
&\quad - (h_2q - h_1r + h_0s)x^3 + (h_2r - h_1s)x^2 - h_2sx \\
&= h_4x^3 - (h_3q - h_2r + h_1s)x^2 + (h_3r - h_2s)x - h_3s,
\end{aligned}$$

and so on, the characteristic feature being that by the introduction of the symbols h , the coefficient of x^3 presents itself at each step as a monomial, and the coefficients of the lower powers require no reduction. It is obvious that the process is a perfectly general one, and that for the equation

$$x^m - p_1x^{m-1} + p_2x^{m-2} - \dots + (-)^mp_m = 0,$$

the formula is

$$\begin{aligned}
x^{m+\theta} &= h_{\theta+1}x^{m-1} \\
&\quad - (h_{\theta}p_2 - h_{\theta-1}p_3 + \dots)x^{m-2} \\
&\quad + (h_{\theta}p_3 - h_{\theta-1}p_4 + \dots)x^{m-3} \\
&\quad \vdots \\
&\quad + (-)^{s-1}(h_{\theta}p_s - h_{\theta-1}p_{s+1} + \dots)x^{m-s} \\
&\quad \vdots \\
&\quad + (-)^{m-1} \cdot h_{\theta}p_m \qquad x^0,
\end{aligned}$$

where, as regards each power of x , the series forming the coefficient thereof is continued as far as possible, that is, up to the term which contains p_m or h_0 as the case may be.

804. On the Elliptic-Function Solution of the Equation $x^3 + y^3 - 1 = 0$

[From the *Proceedings of the Cambridge Philosophical Society*, vol. iv. (1883), pp. 106–109.]

I HAD occasion to find elliptic-function expressions for the coordinates (x, y) of a point on the cubic curve $x^3 + y^3 = 1$. These are derivable from the formulæ given, Legendre, *Fonctions Elliptiques*, t. I. pp. 185, 186, for the reduction to elliptic integrals of the integral

$$R = \int \frac{dz}{(1 - z^3)^{\frac{2}{3}}}.$$

Legendre, writing

$$z = \frac{\sqrt{4y^3 - 1} - \sqrt{3}}{\sqrt{4y^3 - 1} + \sqrt{3}},$$

and then

$$m^3 = 2 \quad \text{and} \quad m^2 y = 1 + x^2,$$

finds first

$$R = m\sqrt{3} \int \frac{dx}{\sqrt{x^4 + 3x^2 + 3}};$$

and then writing $r = \sqrt[4]{3}$, $x = \tan \frac{1}{2}\phi$, and $c^2 = \frac{1}{4}(2 - r^2)$, finds

$$R = \frac{1}{2}mr \int \frac{d\phi}{\sqrt{1 - c^2 \sin^2 \phi}};$$

we have therefore only to write $\sin \phi = \operatorname{sn} u$, to modulus c , $= \frac{1}{2}\sqrt{2 - \sqrt{3}}$, and we thence obtain an expression for z in terms of the elliptic functions $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$.

Writing x instead of z , and k for c , then

$$m = \sqrt[3]{2}, \quad r = \sqrt[4]{3}; \quad k = \frac{1}{2}\sqrt{2 - r^2}, \quad k' = \frac{1}{2}\sqrt{2 + r^2}.$$

Working out the substitutions, the resulting formulæ are

$$x = \frac{2r \operatorname{sn} u \operatorname{dn} u - (1 + \operatorname{cn} u)^2}{2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2},$$

$$y = \frac{m(1 + \operatorname{cn} u)\{1 + r^2 + (1 - r^2) \operatorname{cn} u\}}{2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2},$$

where the modulus is k as above; and these values give

$$x^3 + y^3 = 1,$$

$$\frac{dx}{(1-x^3)^{\frac{2}{3}}} = \frac{-dy}{(1-y^3)^{\frac{2}{3}}} = \frac{1}{2}mr du.$$

The verification is interesting enough; starting from the expression for x , and for shortness representing it by

$$x = \frac{A-B}{A+B},$$

we have

$$1-x^3 = \frac{2B(3A^2+B^2)}{(A+B)^3}, = \frac{m^3(1+\operatorname{cn} u)^2(3A^2+B^2)}{\{2r \operatorname{sn} u \operatorname{dn} u + (1+\operatorname{cn} u)^2\}^3}.$$

We find

$$3A^2+B^2 = 12r^2 \operatorname{cn}^2 u \operatorname{dn}^2 u + (1+\operatorname{cn} u)^4,$$

$$= (1+\operatorname{cn} u)\{12r^2(1-\operatorname{cn} u)(k'^2+k^2 \operatorname{cn}^2 u) + (1+\operatorname{cn} u)^3\},$$

where the term in $\{ \}$ is a perfect cube

$$= \{1+\operatorname{cn} u + r^2(1-\operatorname{cn} u)\}^3.$$

The last-mentioned expression is, in fact,

$$= (1+\operatorname{cn} u)^3 + r^2(1-\operatorname{cn} u)[3(1+\operatorname{cn} u)^2 + 3r^2(1+\operatorname{cn} u)(1-\operatorname{cn} u) + r^4(1-\operatorname{cn} u)^2],$$

where the second term is

$$= 12r^2(1-\operatorname{cn} u)[\frac{1}{4}(1+\operatorname{cn}^2 u) + \frac{1}{4}r^2(1-\operatorname{cn}^2 u)],$$

that is, it is

$$= 12r^2(1-\operatorname{cn} u)(k'^2+k^2 \operatorname{cn}^2 u).$$

We have consequently

$$1-x^3 = \frac{m^3(1+\operatorname{cn} u)^3 \{1+r^2+(1-r^2) \operatorname{cn} u\}^3}{\{2r \operatorname{sn} u \operatorname{dn} u + (1+\operatorname{cn} u)^2\}^3},$$

or extracting the cube root $y = \sqrt[3]{1-x^3}$, has its foregoing value: and the differential expressions are then verified.

Suppose $y = 1$, we have

$$(m-1)(1+\operatorname{cn} u)^2 + mr^2(1-\operatorname{cn}^2 u) = 2r \operatorname{sn} u \operatorname{dn} u,$$

and the differential

$$(m-1)^2(1+\operatorname{cn} u)^3 + 2m(m-1)r^2(1+\operatorname{cn} u)^2(1-\operatorname{cn} u) \\ + 3m^2(1+\operatorname{cn} u)(1-\operatorname{cn} u)^2 = r^2(1-\operatorname{cn} u)\{4-4k^2(1-\operatorname{cn}^2 u)\},$$

or observing that the right-hand side is

$$= r^2(1-\operatorname{cn} u)\{(1+\operatorname{cn} u)^2 + (1-\operatorname{cn} u)^2 + r^2(1+\operatorname{cn} u)(1-\operatorname{cn} u)\},$$

and multiplying by $\frac{1}{3}r^2$, the equation becomes

$$0 = \frac{1}{3}(m-1)^2r^2(1+\operatorname{cn} u)^3 + (2m^2-2m+1)(1+\operatorname{cn} u)^2(1-\operatorname{cn} u) \\ + (m^2-1)r^2(1+\operatorname{cn} u)(1-\operatorname{cn} u)^2 - (1-\operatorname{cn} u)^3;$$

viz. this is

$$0 = \{\frac{1}{3}r^2(m^2-1)(1+\operatorname{cn} u) - (1-\operatorname{cn} u)\}^3,$$

as is immediately verified: hence writing $\frac{1}{3}r^2 = \frac{1}{r^2}$, we have for the value in question, $y = 1$,

$$(m^2-1)(1+\operatorname{cn} u) - r^2(1-\operatorname{cn} u) = 0,$$

or say

$$m^2(1+\operatorname{cn} u) = (1+\operatorname{cn} u) + r^2(1-\operatorname{cn} u),$$

that is,

$$\operatorname{cn} u = \frac{r^2 + 1 - m^2}{r^2 - 1 + m^2},$$

which is one of the values of $\operatorname{cn} u$ derived from the equation $x = 0$; but this equation $x = 0$ gives, not the foregoing equation, but

$$m^6(1+\operatorname{cn} u)^3 = \{(1+\operatorname{cn} u) + r^2(1-\operatorname{cn} u)\}^3,$$

viz. the three values of $\operatorname{cn} u$ are the foregoing value and the two values obtained therefrom by changing m into ωm and $\omega^2 m$ respectively, ω being an imaginary cube root of unity. In fact, the curve $x^3 + y^3 = 1$ has at the point $x = 0, y = 1$ an inflexion, the tangent being $y = 1$, so that this line meets the curve in the point counting three times; but the line $x = 0$ meets the curve in the point, and besides in two imaginary points.

805. Note on Abel's Theorem (continued)

For the proof, observe that, varying in any manner the curve ϕ , we obtain

$$\frac{d\phi}{dx} dx + \frac{d\phi}{dy} dy + \delta\phi = 0,$$

where $\delta\phi$ is that part which depends on the variation of the coefficients, of the whole variation of ϕ ; viz. if $\phi = ax^n + bx^{n-1}y + \dots$, then $\delta\phi = x^n da + x^{n-1}y db + \dots$; $\delta\phi$ is thus, in regard to the coordinates (x, y) , a rational and integral function of the order n . Writing in this equation

$$dx = \frac{df}{dy} d\omega, \quad dy = -\frac{df}{dx} d\omega,$$

the equation becomes

$$\left(\frac{d\phi}{dx} \frac{df}{dy} - \frac{d\phi}{dy} \frac{df}{dx} \right) d\omega + \delta\phi = 0,$$

or say

$$-J(f, \phi) d\omega + \delta\phi = 0,$$

that is,

$$d\omega = \frac{\delta\phi}{J(f, \phi)};$$

and then multiplying each side by the arbitrary function $(x, y, 1)^{m-3}$, we have

$$\Sigma (x, y, 1)^{m-3} d\omega = \Sigma \frac{(x, y, 1)^{m-3}}{J(f, \phi)} \delta\phi,$$

where $\delta\phi$ being of the order n in the variables, the numerator is a rational and integral function of (x, y) of the order $m + n - 3$: hence by a theorem contained in Jacobi's paper "Theoremata nova algebraica circa systema duarum aequationum inter duas variables," *Crelle*, t. xiv. (1835), pp. 281-288, [*Ges. Werke*, t. III., pp. 285-294], the sum on the right-hand side is $= 0$: hence the required result $\Sigma (x, y, 1)^{m-3} d\omega = 0$.

Observing that $(x, y, 1)^{m-3}$ is an arbitrary function, the equation just obtained breaks up into the equations

$$\Sigma d\omega = 0, \quad \Sigma x d\omega = 0, \quad \Sigma y d\omega = 0, \quad \dots, \quad \Sigma x^{m-3} d\omega = 0, \quad \dots, \quad \Sigma y^{m-3} d\omega = 0$$

viz. the number of equations is

$$1 + 2 + \dots + (m-2), \quad = \frac{1}{2}(m-1)(m-2),$$

which is $= p$, the deficiency of the curve.

Suppose the fixed curve $f(x, y, 1) = 0$ is a cubic, $m = 3$, and we have the single relation $\Sigma d\omega = 0$, where the summation refers to the $3n$ points of intersection of the cubic and of the variable curve of the order n , $\phi(x, y, 1) = 0$.

In particular, if this curve be a line, $n = 1$, and the equation is $d\omega_1 + d\omega_2 + d\omega_3 = 0$; here the two points (x_1, y_1) , (x_2, y_2) taken at pleasure on the cubic, determine the line, and they consequently determine uniquely the third point of intersection (x_3, y_3) ; there should thus be a single equation giving the displacement $d\omega_3$ in terms of the displacements $d\omega_1, d\omega_2$; viz. this is the equation just found

$$d\omega_1 + d\omega_2 + d\omega_3 = 0.$$

So if the variable curve be a conic, $n = 2$; and we have between the displacements of the six points the relation

$$d\omega_1 + d\omega_2 + \dots + d\omega_6 = 0 :$$

here five of the points determine the conic, and they therefore determine uniquely the sixth point; and there should be between the displacements a single relation as just found.

If the variable curve be a cubic, $n = 3$, and we have between the displacements of the nine points the relation

$$d\omega_1 + d\omega_2 + \dots + d\omega_9 = 0 :$$

here eight of the points do *not* determine the cubic ϕ , but they nevertheless determine the ninth point, viz. (reproducing the reasoning which establishes this well-known and fundamental theorem as to cubic curves) if $\phi_0 = 0$ be a particular cubic through the 8 points, then the general cubic is $\phi_0 + kf = 0$, and the intersections with $f = 0$ are given by the equations $\phi_0 = 0, f = 0$; whence the ninth point is independent of k , and is determined uniquely by the 8 points. There should thus be a single relation between the displacements, viz. this is the relation just found.

And so if the variable curve be a quartic, or curve of any higher order, it appears in like manner that there should be a single relation between the displacements; this relation being in fact the foregoing relation $\Sigma d\omega = 0$.

But take the fixed curve to be a quartic, $m = 4$: then we have between the displacements $d\omega$ the relation

$$\Sigma (x, y, 1) d\omega = 0,$$

that is, the three equations

$$\Sigma x d\omega = 0, \quad \Sigma y d\omega = 0, \quad \Sigma d\omega = 0.$$

If the variable curve is a conic, $n = 2$, then there are 8 points of intersection; 5 of these taken at pleasure determine the conic, and they consequently determine the remaining 3 points of intersection: hence there should be 3 equations. And so if the variable curve be a curve of any higher order, then by considerations similar to those made use of in the case where the first curve is a cubic it appears that the number of equations between the displacements $d\omega$ should always be $= 3$.

But if the variable curve be a line, $n = 1$, then the number of the points of intersection is $= 4$: 2 of these taken at pleasure determine the line, and they consequently determine the remaining 2 points of intersection; and the number of equations between the displacements $d\omega$ should thus be $= 2$. But by what precedes, we have the 3 equations

$$\begin{aligned} d\omega_1 + d\omega_2 + d\omega_3 + d\omega_4 &= 0, \\ x_1 d\omega_1 + x_2 d\omega_2 + x_3 d\omega_3 + x_4 d\omega_4 &= 0, \\ y_1 d\omega_1 + y_2 d\omega_2 + y_3 d\omega_3 + y_4 d\omega_4 &= 0; \end{aligned}$$

here the 4 points of intersection are on a line $y = ax + b$; we have therefore $y_1 = ax_1 + b, \dots, y_4 = ax_4 + b$; the equations between the $d\omega$'s give

$$(y_1 - ax_1 - b) d\omega_1 + \dots + (y_4 - ax_4 - b) d\omega_4 = 0,$$

that is, is a single relation $0 = 0$; or the 3 equations thus reduce themselves to 2 independent equations.

Again, if the fixed curve be a quintic, $m = 5$, there are here between the displacements the 6 equations

$$\begin{aligned} \Sigma x^2 d\omega &= 0, & \Sigma xy d\omega &= 0, & \Sigma y^2 d\omega &= 0, \\ \Sigma x d\omega &= 0, & \Sigma y d\omega &= 0, & \Sigma d\omega &= 0; \end{aligned}$$

the two cases in which the number of independent equations is less than 6 are (i) when the variable curve is a line, and (ii) when the variable curve is a conic. For the line $n = 1$, and the number should be $= 3$. We have the above 6 equations; but the equation of the line is $ax + by + c = 0$, that is, we have $ax_1 + by_1 + c = 0$, &c.; we deduce the 3 identical equations

$$\Sigma (ax + by + c) = 0, \quad \Sigma y(ax + by + c) = 0, \quad \Sigma (ax + by + c) = 0,$$

and the number of independent equations is thus $6 - 3, = 3$ as it should be.

So when the variable curve is a conic, $n = 2$; the number of independent equations should be $= 5$. The points of intersection lie on a conic $(a, b, c, f, g, h | x, y, 1)^2 = 0$; we have therefore the several equations $(a, b, c, f, g, h | x_1, y_1, 1)^2 = 0$, &c.: we have therefore the single identical equation

$$\Sigma (a, b, c, f, g, h | x, y, 1)^2 d\omega = 0,$$

and the number of independent equations is $6 - 1, = 5$ as it should be.

Obviously the like considerations apply to the case where the fixed curve is a curve of any given order whatever.

806. Determination of the Order of a Surface

[From the *Messenger of Mathematics*, vol. XII. (1883),
pp. 29–32.]

[On p. lxx of the Prolegomena to C. Taylor's *Introduction to the Geometry of Conics* (1881) occurs the following passage:

“*Proof and extension of Newton's Descriptio Organica.*”

“Let two angles AOB and $A\omega B$ of given magnitudes turn about O and ω respectively, and let the intersection A trace a curve of the n th order. For a given position of the arm OB there are n positions of A and therefore n of B . When OB is in the position $O\omega$ the n B 's coincide with ω , which is therefore an n -fold point on the locus of B , as is also the point O ; and since any line through O (or ω) meets the locus of B in n other points, the locus is of the order $2n$. Its order is the same when $A\omega B$ is a zero-angle or straight line.

“Let a given trihedral angle $O(ABC)$ —or a plane OBC and a line OA rigidly attached to it—turn about O , and let a variable plane through a fixed point ω meet OA in A and the plane OBC in BC ; then if the line BC describes a ruled surface of the order n the point A describes a surface of the order $4n$.”

And in a foot-note it is stated that the author is indebted to Professor Cayley for the determination of the order of this surface. The following paper contains Professor Cayley's determination, which was communicated by him to Mr Taylor.]

LEMMA. Take (a, b, c, f, g, h) the six coordinates of a line; these are connected by the equation

$$af + bg + ch = 0.$$

In order that the line may belong to a ruled surface, we must have between the coordinates three more equations, say these are

$$\begin{aligned} F(a, b, c, f, g, h) &= 0, \\ G(a, & \hspace{10em}) = 0, \\ H(a, & \hspace{10em}) = 0, \end{aligned}$$

of the orders p, q, r respectively; then the scroll is of the order $n = 2pqr$.

For expressing that the line meets an arbitrary line (a', b', c', f', g', h') , we have the linear relation

$$f'a + g'b + h'c + a'f + b'g + c'h = 0;$$

the five relations determine the ratios $a : b : c : f : g : h$, and the number of systems of values is = product of orders = $2.p.q.r.1$, = $2pqr$; viz. this is the number of lines meeting the arbitrary line; or, what is the same thing, it is = order of ruled surface.

Consider now a trihedral angle $OABC$ rotating about a fixed point O which may be taken for the origin, and consider a fixed point ω . Let the lines OB, OC each meet a line L , and the plane ωL intersect OA in a point P ; then it is to be shown that, if L is a line of a ruled surface of the order n , the locus of P is a surface of the order $4n$.

Observe that, for a given position of the line L , the position of the lines OB, OC is not determinate, but that the angle BOC has any position at pleasure in the plane OL , or say it rotates round the line ON which is the normal at O to the plane OL ; the line OA therefore also rotates about ON , being always inclined to it at a determinate angle θ , or the locus of OA is a right cone, axis ON and semi-aperture $= \theta$. The points P are the intersections of the several lines of the cone with the fixed plane ωL , or say that (for the given position of L) the locus of P is the conic C which is the intersection of the cone in question by the plane ωL . And then varying the position of L , the required surface is the locus of the corresponding conics C .

Take now

$$Ax + By + Cz + D = 0, \quad A'x + B'y + C'z + D' = 0,$$

for the equations of a particular line L ; then writing

$$AD' - A'D, BD' - B'D, CD' - C'D, BC' - B'C, CA' - C'A, AB' - A'B = (a, b, c, f, g, h)$$

respectively, these are the six coordinates of the line L .

The equation of the plane OL is

$$(AD' - A'D)x + (BD' - B'D)y + (CD' - C'D)z = 0,$$

that is,

$$ax + by + cz = 0;$$

and the equations of the normal ON are therefore

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c};$$

we have therefore for the cone

$$\frac{ax + by + cz}{\sqrt{(x^2 + y^2 + z^2)} \sqrt{(a^2 + b^2 + c^2)}} = \cos \theta;$$

or if, for convenience, $\cos^2 \theta = k$, then the equation of the cone is

$$(ax + by + cz)^2 - k(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = 0;$$

say this is

$$\{a^2 - k(a^2 + b^2 + c^2), b^2 - k(a^2 + b^2 + c^2), c^2 - k(a^2 + b^2 + c^2), bc, ca, ab\}(x, y, z)^2 = 0 \dots$$

Taking then (x_0, y_0, z_0) for the coordinates of the point ω , the equation of the plane ωL is

$$\frac{Ax + By + Cz + D}{Ax_0 + By_0 + Cz_0 + D} - \frac{A'x + B'y + C'z + D'}{A'x_0 + B'y_0 + C'z_0 + D'} = 0,$$

viz. it is

$$(BC' - B'C)(yz_0 - y_0z) + \dots + (AD' - A'D)(x - x_0) + \dots = 0,$$

that is,

$$f(yz_0 - y_0z) + g(zx_0 - z_0x) + h(xy_0 - x_0y) + a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

or say

$$x(hy_0 - gz_0 + a) + y(-hx_0 + fz_0 + b) + z(gx_0 - fy_0 + c) + (-ax_0 - by_0 - cz_0) = 0 \dots (2),$$

viz. (1) and (2) are the equations of the conic C , and the coordinates (a, b, c, f, g, h) satisfy of course the equation

$$af + bg + ch = 0 \dots \dots \dots (3).$$

Considering now the line L as belonging to a ruled surface, the coordinates $(a, \dots)$ satisfy as before three equations

$$F(a, b, c, f, g, h) = 0 \dots \dots \dots (4),$$

$$G(\dots) = 0 \dots \dots \dots (5),$$

$$H(\dots) = 0 \dots \dots \dots (6),$$

of the orders p, q, r respectively, and we can from the six equations eliminate a, b, c, f, g, h . The resulting equation $\nabla = 0$ contains the coefficients of (1) in the order $1.2.p.q.r = 2pqr$ (which is the product of the orders of the other 5 equations), and the coefficients of (2) in the order $2.2.p.q.r = 4pqr$, (which is the product of the orders of the other 5 equations). But the coefficients of (1) being quadratic functions of (x, y, z) , and those of (2) being linear functions of (x, y, z) , the aggregate order in (x, y, z) is

$$2 \cdot 2pqr + 1 \cdot 4pqr, = 8pqr;$$

or, since the order of the ruled surface is $n, = 2pqr$, the order of the locus is $= 4n$; which is the above-stated theorem.

807. A Proof of Wilson's Theorem

[From the *Messenger of Mathematics*, vol. XII. (1883), p. 41.]

LET n be a prime number; and imagine n points, the vertices of a regular polygon; any polygon which can be formed with these n points as vertices is either regular or else it is one of a set of n equal and similar polygons. For instance, $n = 5$, the polygon as shown in the figure is one of a set of 5 equal and similar polygons; in fact, if the points taken in their cyclical order, but beginning at pleasure with any one of the 5 points are called 1, 2, 3, 4, 5, then we have 5 such polygons 13254; and so in general. The whole number of polygons is $\frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \dots (n-1)$; and the number of the regular polygons is $\frac{1}{2}(n-1)$; hence the number of the remaining polygons is $= \frac{1}{2}(n-1)\{1 \cdot 2 \dots (n-2) - 1\}$; and this number must therefore be divisible by n ; that is, $1 \cdot 2 \dots (n-1) - n + 1$ is divisible by n ; or, what is the same thing, $1 \cdot 2 \dots (n-1) + 1$ is divisible by n , which is the theorem in question.

808. Note on a Form of the Modular Equation in the Transformation of the Third Order

[From the *Messenger of Mathematics*, vol. XII. (1883), pp. 173, 174.]

IN my *Treatise on Elliptic Functions*, pp. 214–216, writing only $\frac{1}{J}$, $\frac{1}{J'}$ instead of Ω , Ω' , and α , β instead of α' , β' , I have shown as follows: viz. if k , λ denote as usual the original modulus, and the transformed modulus, and if

$$J = \frac{(k^4 + 14k^2 + 1)^3}{108k^2(1 - k^2)^4}, \quad J' = \frac{(\lambda^4 + 14\lambda^2 + 1)^3}{108\lambda^2(1 - \lambda^2)^4},$$

then the relation between J and J' can be found by the elimination of α , β from the equations

$$\alpha + \beta = 1, \\ J = \frac{(1 + 8\alpha)^3}{64\alpha(1 - \alpha)^3}, \quad J' = \frac{(1 + 8\beta)^3}{64\beta(1 - \beta)^3}.$$

By a very slight change we obtain the result given by Prof. Klein in his paper, “Ueber die Transformation der elliptischen Functionen, &c.,” *Math. Ann.* t. XIV. (1879), pp. 111–172; viz., see p. 143, the relation is to be obtained by the elimination of τ , τ' from the equation $\tau\tau' = 1$, and the equations

$$J : J - 1 : 1 = (\tau - 1)(9\tau - 1)^3 : (27\tau^2 - 18\tau - 1)^2 : -64\tau; \\ J' : J' - 1 : 1 = (\tau' - 1)(9\tau' - 1)^3 : (27\tau'^2 - 18\tau' - 1)^2 : -64\tau';$$

these last equations being equivalent to two equations only in virtue of the identity

$$(\tau - 1)(9\tau - 1)^3 + 64\tau = (27\tau^2 - 18\tau - 1)^2,$$

and the like identity in τ' .

In fact, writing $\alpha = \frac{\tau}{\tau - 1}$, $\beta = \frac{\tau'}{\tau' - 1}$, the equation $\alpha + \beta = 1$ becomes $\tau\tau' = 1$; and then for α , β substituting their values, we have

$$J = \frac{(9\tau - 1)^3(\tau - 1)}{-64\tau}, \quad J' = \frac{(9\tau' - 1)^3(\tau' - 1)}{-64\tau'},$$

which are the formulæ in question.

809. Schröter's Construction of the Regular Pentagon

[From the *Messenger of Mathematics*, vol. XII. (1883), p. 177.]

THE following construction of the regular pentagon, analogous to the more complicated one for the polygon of 17 sides, is given in the paper, H. Schröter, Zur v. Staudt'schen "Construction des regulären Siebzehnecks," *Crelle*, t. LXXV. (1873), pp. 13–24.

Take in a circle AB , CD diameters at right angles to each other, and at C , D draw in the directions A to B and B to A respectively, the lines Cc , $= 4$ radii, and Dd , $= 1$ radius; draw cd meeting the circle in the points E and F ; draw CE and CF cutting AB in the points e and f respectively, and through these at right angles to AB draw the chords 34 and 25 respectively, then we have $A2345$ a regular pentagon.

810. Note on a System of Equations

[From the *Messenger of Mathematics*, vol. XII. (1883),
pp. 191, 192.]

THE equations are

$$x^2 = ax + by, \quad xy = cx + dy, \quad y^2 = ex + fy,$$

where

$$\frac{b}{d} = \frac{a-d}{c-f} = \frac{c}{e};$$

or, what is the same thing, if a, b, c, d are given, then

$$e = \frac{cd}{b}, \quad f = c - \frac{d(a-d)}{b};$$

and this being so, the equations are equivalent to two independent equations; viz. starting from the first and the second equations, we have

$$dx^2 - bxy = (ad - bc)x,$$

that is,

$$dx - by = (ad - bc) :$$

and thence

$$dxy - by^2 = (ad - bc)y,$$

or

$$d(cx + dy) - by^2 = (ad - bc)y;$$

which, attending to the values of e and f , is the third equation

$$y^2 = ex + fy.$$

We have

$$\begin{aligned} \frac{x}{y} &= \frac{ax + by}{cx + dy}, & \frac{y}{x} &= \frac{ex + fy}{cx + dy}, \\ \frac{x^2}{y^2} &= \frac{ax + by}{ex + fy}, & \frac{y^2}{x^2} &= \frac{ex + fy}{ax + by}, \end{aligned}$$

and so on: any equation such as $x^a y^b = Ax^{a'} y^{b'}$ may be replaced by the two equations

$$x : y = ax + by : cx + dy \quad \text{and} \quad x : y = cx + dy : ex + fy;$$

or, what is the same thing, by either of these equations; and we may from any two such equations deduce a third one; thus from

$$\frac{x^2}{y} = \frac{x(ax + by)}{ex + fy} \quad \text{and} \quad \frac{x}{y^2} = \frac{ex + fy}{y(cx + dy)}$$

we deduce

$$\frac{x^2}{y} \cdot \frac{x}{y^2} = \frac{x}{y}, \quad \text{that is,} \quad x^2 = y^2.$$

Considering the equations as $x : y = ax + by : cx + dy = cx + dy : ex + fy$, it is interesting to notice the analytical equivalent of a known theorem in the geometry of conics: viz. from

$$x : y = ax + by : cx + dy,$$

we have

$$dx^2 - (a - d)xy - by^2 = 0,$$

an equation which gives two values of the ratio $x : y$; taking either value, the equations

$$x : y = ax + by : cx + dy \quad \text{and} \quad x : y = cx + dy : ex + fy$$

give each of them the same value of the ratio $x : y$: this is, in fact, the theorem that a conic meets each of two conics which have double contact with it, in the same points of the conic.

811. On the Linear Transformation of the Theta Functions

[From the *Messenger of Mathematics*, vol. XIII. (1884),
pp. 54–60.]

THE functions referred to are the single Theta Functions; these may be defined as doubly infinite products, as was in fact done in my “Mémoire sur les fonctions doublement périodiques,” *Liouv. t. x.* (1845), pp. 385–420, [25]; and it is interesting to consider from this point of view the theory of their linear transformation: this I propose to do in the present paper, adopting throughout the notation of Smith’s* “Memoir on the Theta and Omega Functions.”

The periods K , iK' are, in general, imaginary quantities

$$K = A + Bi, \quad iK' = C + Di,$$

where $AD - BC$ is positive; writing then $\omega = \frac{iK'}{K}$, and $q = e^{i\pi\omega}$, also for shortness

$$(q1) = 2q^{\frac{1}{4}} \Pi_1^\infty (1 - q^{2n})^2,$$

where $q^{\frac{1}{4}}$ denotes $e^{\frac{1}{4}i\pi\omega}$, the expression of the odd theta-function $\vartheta_1(x, \omega)$ as a doubly infinite product is

$$\vartheta_1(x, \omega) = (q1) x \prod \left(1 + \frac{x}{m\pi + n\omega\pi} \right), \quad \left(\frac{\mu}{\nu} = \infty \right),$$

where (m, n) have any positive or negative integer values (the combination $m = 0, n = 0$ excluded) from $m = -\mu$ to μ , and $n = -\nu$ to ν , μ and ν being each ultimately infinite but so that μ is infinite in comparison with ν ; this condition in regard to the limits is indicated by $\mu/\nu = \infty$; and similarly $\nu/\mu = \infty$ would indicate that ν was infinite in comparison with μ .

[* Smith’s *Collected Mathematical Papers*, vol. II., pp. 415–621.]

The condition as to the limits might be that (m, n) have any positive or negative values (excluding as before) such that the modulus of $m + n\omega$ does not exceed a positive value T , which is ultimately taken to be infinite; this condition may be indicated by $\text{mod} = \infty$.

The values of the double product corresponding to the different conditions as to the limits are not equal, but they differ only by an exponential factor, the exponent being a multiple of x^2 ; we thus have

$$\frac{\text{III} \left(1 + \frac{x}{m\pi + n\omega\pi} \right)_{\frac{\mu}{\nu} = \infty}}{\text{III} \left(1 + \frac{x}{m\pi + n\omega\pi} \right)_{\text{mod} = \infty}} = \exp(\nabla x^2),$$

where ∇ is a function of ω only.

Considering now the linear transformation, we have new periods $\Lambda, i\Lambda'$ connected with the original periods K, iK' by the equations

$$\frac{1}{M} K = a\Lambda + bi\Lambda', \quad \frac{1}{M} iK' = c\Lambda + di\Lambda',$$

where a, b, c, d are positive or negative integers such that $ad - bc = +1$; M is the “multiplier.” We have then

$$\omega = \frac{iK'}{K} = \frac{c\Lambda + di\Lambda'}{a\Lambda + bi\Lambda'}, \quad \Omega = \frac{i\Lambda'}{\Lambda},$$

and thence

$$\omega = \frac{c + d\Omega}{a + b\Omega}.$$

Writing $h = \frac{\pi}{K}$, $k = \frac{\pi}{\Lambda}$, we have $h dx = k \frac{x}{M}$, and the formula contains on the left-hand side the product

$$\text{III} \left(1 + \frac{(a + b\Omega) \frac{\pi x}{h}}{m\pi + n\Omega\pi} \right),$$

which, writing $m' = am + cn$, $n' = bm + dn$, is

$$= \text{III} \left(1 + \frac{(a + b\Omega) \frac{\pi x}{h}}{m'\Lambda + n'i\Lambda'} \right);$$

and we thence obtain the transformation formula

$$\frac{\vartheta_1 \left\{ (a + b\Omega) \frac{\pi x}{h}, \Omega \right\}}{\vartheta_1 \left\{ \frac{\pi x}{h}, \omega \right\}} = \frac{(Q1)}{(q1)} \exp \left[\left\{ (a + b\Omega)^2 \square \Lambda^2 - \nabla K^2 \right\} \frac{x^2}{h^2} \right],$$

viz. the two functions differ only by a constant factor and by an exponential factor, the exponent being a multiple of x^2 ; after all reductions, this factor is found to be

$$= \exp \left(-i\pi b (a + b\Omega) \frac{x^2}{h^2} \right).$$

We have

$$\omega = \frac{c + d\Omega}{a + b\Omega},$$

or since

$$\omega = \frac{iK'}{K}, \quad \Omega = \frac{i\Lambda'}{\Lambda},$$

this is

$$\frac{iK'}{K} = \frac{c\Lambda + di\Lambda'}{a\Lambda + bi\Lambda'},$$

or say

$$\begin{aligned} \frac{1}{M} K &= a\Lambda + bi\Lambda', \\ \frac{1}{M} iK' &= c\Lambda + di\Lambda', \end{aligned}$$

either of which equations may be taken as a definition of the multiplier M . We have

$$\frac{K}{M\Lambda} = a + b\Omega,$$

$$\begin{aligned} \frac{1}{M} (mK + niK') &= (am + cn) \Lambda + (bm + dn) i\Lambda' \\ &= m'\Lambda + n'i\Lambda', \end{aligned}$$

if

$$\begin{aligned} m' &= am + cn, \\ n' &= bm + dn. \end{aligned}$$

Here to any integer values of (m, n) there correspond integer values of m', n' ; and conversely, in virtue of the equation $ad - bc = 1$, to any integer values of m', n' there correspond integer values of m ,

n . The two products are

$$\begin{aligned} & \text{III} \left(1 + \frac{\frac{Kx}{Mh}}{m'\Lambda + n'i\Lambda'} \right), \quad (\text{mod} = \infty), \\ & \text{III} \left(1 + \frac{\frac{(a + b\Omega)\Lambda x}{h}}{m\Lambda + ni\Lambda'} \right), \quad (\text{mod} = \infty). \end{aligned}$$

The quotient of these two products is $= \exp(\square \Lambda^2 x^2 / h^2)$, and we thence obtain the foregoing formula.

For the completion of the formula we require the value of $\frac{(Q1)}{(q1)}$; this may be obtained by means of the equation

$$\vartheta'_1(0, \omega) = (q1), \quad \vartheta'_1(0, \Omega) = (Q1),$$

and the known formula

$$\vartheta'_1(0, \omega) = 2q^{\frac{1}{4}} \Pi_1^\infty (1 - q^{2n})^3.$$

The final result is that the multiplier M being determined as above, we have

$$\vartheta_1\left(\frac{x}{M}, \Omega\right) = C \vartheta_1(x, \omega),$$

where C is a constant depending on the transformation, the value of which is given by the formula

$$C = \frac{1}{\sqrt{a + b\Omega}} e^{\frac{1}{4}\pi i b(a + b\Omega)}.$$

And similarly for the other theta-functions, we have

$$\begin{aligned} \vartheta_2\left(\frac{x}{M}, \Omega\right) &= C' \vartheta_2(x, \omega), \\ \vartheta_3\left(\frac{x}{M}, \Omega\right) &= C'' \vartheta_3(x, \omega), \\ \vartheta_4\left(\frac{x}{M}, \Omega\right) &= C''' \vartheta_4(x, \omega), \end{aligned}$$

where C', C'', C''' are constants which can be determined in a similar manner.

812. On Archimedes' Theorem for the Surface of a Cylinder

[From the *Messenger of Mathematics*, vol. XIII. (1884),
pp. 107, 108.]

THE measure of the surface of a cylinder was first obtained by Archimedes in his Treatise on the Sphere and Cylinder (Book I., Prop. XIV.), *Œuvres d'Archimède*, par F. Peyrard, 4° Paris, 1807, pp. 26–31; viz. Archimedes showed that the surface of the cylinder was equal to the area of a circle, having its radius a mean proportional between the height and the diameter of the circular base [$S = 2\pi ah$, $= \pi\{\sqrt{(2a.h)}\}^2$].

The following is *in effect* his demonstration:

He considers regular polygons (with the same number of sides) inscribed in and circumscribed about a circle; and, as regards the cylinder, the prisms standing on these polygons.

Say for the circular base of the cylinder we have

$S^\times$ surface of circumscribed prism,
 S „ cylinder,
 S° „ inscribed prism;

and for the circle, having its radius a mean proportional between the height and the diameter of the circular base,

$B^\times$ area of circumscribed polygon,
 B „ circle,
 B° „ inscribed polygon,

where the four polygons referred to by $S^\times$, S° , $B^\times$, B° have all of them the same number of sides.

Archimedes shows that

$$S^\times : B^\times :: S^\circ : B^\circ :: S : B;$$

and moreover that $S^\times > S > S^\circ$, whence $B^\times > B > B^\circ$; but $B^\times - B^\circ$ can be made as small as we please, and hence $S = B$, which is the theorem in question.

813. [Note on Mr Griffiths' Paper "On a Deduction from the Elliptic-Integral Formula $y = \sin(A + B + C + \dots)$."]

[From the *Proceedings of the London Mathematical Society*, vol. XV.
(1884), p. 81.]

CONSIDER, for instance, the cubic transformation

$$y = \frac{x [1 + 2\alpha' - (1 + \alpha')^2 x^2]}{1 - \alpha^2 x^2},$$

where $\alpha^2 + \alpha'^2 = 1$.

This implies

$$\sqrt{1 - y^2} = \frac{\sqrt{1 - x^2} [1 - (1 + \alpha')^2 x^2]}{1 - \alpha^2 x^2},$$

viz., $\sqrt{1 - y^2}$ = a rational multiple of $\sqrt{1 - x^2}$.

Also the quadric transformation

$$z = \frac{1 - (1 + \beta'^2) x^2}{1 - \beta^2 x^2};$$

where $\beta^2 + \beta'^2 = 1$.

This implies

$$\sqrt{1 - z^2} = \frac{\sqrt{1 - x^2} \cdot 2\beta' x}{1 - \beta^2 x^2},$$

viz., $\sqrt{1 - z^2}$ = a rational multiple of $\sqrt{1 - x^2}$.

Hence, assuming

$$u = yz - \sqrt{1 - y^2} \sqrt{1 - z^2},$$

which is a rational function

$$= \frac{x(a_0 - a_2 x^2 + a_4 x^4)}{1 - \alpha^2 x^2 \cdot 1 - \beta^2 x^2},$$

we have

$$\sqrt{1 - v^2} = y\sqrt{1 - z^2} + z\sqrt{1 - y^2},$$

which is $= \sqrt{1 - x^2}$ multiplied by a like rational function.

814. On Double Algebra

[From the *Proceedings of the London Mathematical Society*, vol. XV.
(1884),
pp. 185–197. Read April 3, 1884.]

1. I CONSIDER the Double Algebra formed with the extraordinary symbols, or “extraordinaries” x, y , which are such that

$$\begin{aligned}x^2 &= ax + by, \\xy &= cx + dy, \\yx &= ex + fy, \\y^2 &= gx + hy,\end{aligned}$$

or, as these equations may also be written,

	x	y
x	(a, b)	(c, d)
y	(e, f)	(g, h)

where a, b, c, d, e, f, g, h are ordinary symbols, or say coefficients; all coefficients being commutative and associative *inter se* and with the extraordinaries x, y .

The system depends in the first instance on the eight parameters a, b, c, d, e, f, g, h ; but we may, instead of the extraordinaries x, y , consider the new extraordinaries connected therewith by the linear relations $\xi = \alpha x + \beta y, \eta = \gamma x + \delta y$, where the coefficients $\alpha, \beta, \gamma, \delta$ may be determined so as to establish between the eight parameters any four relations at pleasure (or, what is the same thing, $\alpha, \beta, \gamma, \delta$ are what I call “apoclastic” constants): and the number of parameters is thus properly $8 - 4, = 4$.

2. The extraordinaries here considered are not in general associative; differing herein from the imaginaries of Peirce’s Memoir, “Linear Associative Algebra” (1870).

We require the conditions for associativeness. We have to satisfy the conditions

$$x.x^2 = x^2.x, \quad x.xy = x^2.y, \quad x.yx = xy.x, \quad x.y^2 = xy.y,$$

and the like conditions with y instead of x , in all 8 conditions; but the second and third of these give

$$x^2.y = x.xy \quad \text{and} \quad x.yx = xy.x,$$

that is, $x^2y = x^2y$ and $xyx = xyx$, identically; there remain thus only 6 conditions, which are

$$\begin{aligned}x^3 &= xx^2 = x^2x, \\xy^2 &= xyy = xyy, \\yx^2 &= yxx = yxx, \\y^2x &= yyx = yyx, \\xyy &= y^2y = yy^2, \\xyx &= xxy = x^2y.\end{aligned}$$

Taking the first and fifth equations, these are

$$\begin{aligned}a^2 + bc &= aa + ec, & ab + bd &= ab + fb, \\c^2 + dg &= ag + eg, & cd + dh &= bg + fh;\end{aligned}$$

that is,

$$\begin{aligned}a^2 + bc &= a^2 + ce, & b(a + d) &= b(a + f), \\g(c + e) &= g(a + e), & d(c + h) &= bg + fh.\end{aligned}$$

Similarly the second and sixth equations give

$$\begin{aligned}c(a + d) &= c(a + f), & d^2 + cf &= bc + df, \\h(c + e) &= g(c + d), & dh + f^2 &= bg + h^2.\end{aligned}$$

And the third and fourth give

$$\begin{aligned}e(a + d) &= e(a + f), & f(a + f) &= bc + f^2, \\g(a + h) &= g(c + e), & h(a + h) &= bg + fh.\end{aligned}$$

3. If the system is commutative, we have $e = c$, $f = d$, and the equations become

$$\begin{aligned}a^2 + bc &= a^2 + c^2, & b(a + d) &= b(a + d), \\g(2c) &= g(a + c), & d(c + h) &= bg + dh,\end{aligned}$$

and

$$\begin{aligned}c(a + d) &= c(a + d), & d^2 + c^2 &= bc + d^2, \\h(2c) &= g(c + d), & dh + d^2 &= bg + h^2,\end{aligned}$$

and

$$\begin{aligned} c(a+d) &= c(a+d), & d(a+d) &= bc + d^2, \\ g(a+h) &= g(2c), & h(a+h) &= bg + dh. \end{aligned}$$

The conditions thus reduce themselves to

$$c^2 = bc, \quad c(a+c) = 2cg, \quad d(c+h) = bg + dh,$$

and

$$c^2 = bc, \quad 2ch = g(c+d), \quad dh + d^2 = bg + h^2,$$

and

$$d(a+d) = bc + d^2, \quad g(a+h) = 2cg, \quad h(a+h) = bg + dh.$$

We have thus 6 equations, but these include identities; in fact, the first equation $c^2 = bc$ gives either $c = 0$ or $c = b$; and so on.

4. Starting now with the general equations

$$x^2 = ax + by, \quad xy = cx + dy, \quad yx = ex + fy, \quad y^2 = gx + hy,$$

we may attempt to find an extraordinary $\xi = \alpha x + \beta y$ (α, β coefficients), such that $\xi^2 = \kappa \xi$, κ being a coefficient. In general κ is not $= 0$, and, when it is not $= 0$, it may without loss of generality be taken to be $= 1$; we have then $\xi^2 = \xi$, an idempotent symbol. But κ may be $= 0$, and then $\xi^2 = 0$, a nilpotent symbol. To include the two cases, I retain κ , it being understood that, when κ is not $= 0$, it may be taken to be $= 1$.

We have

$$\begin{aligned} \xi^2 &= \alpha^2(ax + by) + \alpha\beta\{(c+e)x + (d+f)y\} + \beta^2(gx + hy) \\ &= \{\alpha\alpha^2 + (c+e)\alpha\beta + g\beta^2\}x + \{b\alpha^2 + (d+f)\alpha\beta + h\beta^2\}y. \end{aligned}$$

Hence, when this is $= \kappa \xi$, that is, $= \kappa(\alpha x + \beta y)$, we have

$$\frac{\alpha}{\beta} = \frac{a\alpha^2 + (c+e)\alpha\beta + g\beta^2}{b\alpha^2 + (d+f)\alpha\beta + h\beta^2},$$

a cubic equation for the determination of the ratio $\alpha : \beta$; and, for any particular value of the ratio, we can in general determine the absolute magnitudes, so that

$$\kappa = \frac{1}{2}\{a\alpha^2 + (c+e)\alpha\beta + g\beta^2\} = \frac{1}{2}\{b\alpha^2 + (d+f)\alpha\beta + h\beta^2\}$$

shall be $= 1$. If, however, for the given value of the ratio we have

$$a\alpha^2 + (c + e)\alpha\beta + g\beta^2 = 0, \quad b\alpha^2 + (d + f)\alpha\beta + h\beta^2 = 0,$$

one of these equations, of course, implying the other, then the value of κ is $= 0$.

5. It follows that there are in general three idempotent extraordinary ξ, η, ζ , such that $\xi^2 = \xi, \eta^2 = \eta, \zeta^2 = \zeta$. The cubic equation may, however, have two equal roots, or three equal roots, or it may vanish identically; in this last case, any linear function $\alpha x + \beta y$ is in general idempotent. But (as will be considered in detail further on) we may, instead of an idempotent symbol or symbols, have a nilpotent symbol or symbols. It might be convenient to use the term *Potency* for a symbol which is in general idempotent, but which may be nilpotent.

Writing $\xi = \alpha x + \beta y, \eta = \gamma x + \delta y, \zeta = \epsilon x + \zeta' y$, we obtain a cubic equation $\Pi = (x, y)^3 = 0$, where obviously the linear factors of Π are the just-mentioned functions ξ, η, ζ ; that is, we have ξ, η, ζ as the linear factors of the cubic function

$$\Pi = gx^3 + (h - c - e)x^2y + (a - d - f)xy^2 + by^3;$$

each such factor, except in the case where it is nilpotent, being determined so that it shall be idempotent. The cubic function of course vanishes identically if $g = 0, h - c - e = 0, a - d - f = 0, b = 0$.

6. Two factors ξ, η being determined so that $\xi\eta = 0$: in general $\xi\eta$ is not $= 0$; we have, in fact,

$$\begin{aligned} \xi\eta &= (\alpha x + \beta y)(\gamma x + \delta y) \\ &= \alpha\gamma x^2 + \alpha\delta xy + \beta\gamma yx + \beta\delta y^2 \\ &= \alpha\gamma(ax + by) + \alpha\delta(cx + dy) + \beta\gamma(\epsilon x + f y) + \beta\delta(gx + hy) \\ &= (a\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta)x + (b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta)y; \end{aligned}$$

and the required condition is satisfied if

$$a\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta = 0, \quad b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta = 0.$$

Writing these equations first under the form

$$\gamma(a\alpha + e\beta) + \delta(c\alpha + g\beta) = 0, \quad \gamma(b\alpha + f\beta) + \delta(d\alpha + h\beta) = 0,$$

and then under the form

$$\alpha(a\gamma + c\delta) + \beta(e\gamma + g\delta) = 0, \quad \alpha(b\gamma + d\delta) + \beta(f\gamma + h\delta) = 0,$$

we have

$$(a\alpha + e\beta)(d\alpha + h\beta) - (b\alpha + f\beta)(c\alpha + g\beta) = 0,$$

a quadric equation for the determination of $\alpha : \beta$; and then

$$(a\gamma + c\delta)(f\gamma + h\delta) - (b\gamma + d\delta)(e\gamma + g\delta) = 0,$$

a quadric equation for the determination of $\gamma : \delta$; that is, there are two values of the left-hand factor ξ , and two values of the right-hand factor η . But, of course, these correspond each to each, viz. either factor being given, the other factor is determined uniquely.

Writing successively

$$\Phi = (eh - fg)x^2 + (-ah + de + bg + cf)xy + (ad - be)y^2,$$

$$\Phi_1 = (ch - dg)x^2 + (-ah - cf + bg + de)xy + (af - be)y^2,$$

we have the quadric functions $(x, y)^2$; where the linear factors of Φ are the two values of the left-hand factor ξ , and the linear factors of Φ_1 are the two values η_1, η_2 of the right-hand factor η .

7. In the commutative case, $c = e, d = f$, we have

$$\Phi = \Phi_1 = (ch - dg)x^2 + (-ah + bg)xy + (ad - be)y^2;$$

so that $\eta = \xi$; and the two values of ξ are η_1, η_2 . But (as above) the values of ξ are ξ_1, ξ_2 , such that $\xi_1\eta_1 = 0, \xi_2\eta_2 = 0$; whence $\xi_1\eta_2$ and $\xi_2\eta_1$ are not in general $= 0$. The value of Π is

$$\Pi = gx^3 + (h - 2c)x^2y + (a - 2d)xy^2 + by^3.$$

8. In the commutative and associative case, taking ξ, η, ζ to be the three idempotent symbols, $\xi^2 = \xi, \eta^2 = \eta, \zeta^2 = \zeta$, we have $\xi\eta = \xi\eta, \xi\zeta = \xi\zeta, \eta\zeta = \eta\zeta, \xi\eta\zeta = 0$; and in this manner we have the six equations

$$\xi\eta = 0, \quad \xi\zeta = 0, \quad \eta\zeta = 0, \quad \eta\xi = 0, \quad \zeta\xi = 0, \quad \zeta\eta = 0,$$

viz. regarding the right-hand factor as being in each case expressed as a linear function of x, y , we have apparently six products of two linear

factors, each = 0. There is only one such product, viz. $\xi\eta = 0$; all the others being identically = 0, viz. the coefficients of each such product being = 0. We have $\xi_1\eta_1 = 0$, $\xi_2\eta_2 = 0$, while $\xi_1\eta_2$ and $\xi_2\eta_1$ are not = 0; and so, of the three linear factors of Π , two are the symbols ξ , η which are such that $\xi\eta = 0$. To complete the theory, observe that ζ must be a linear function of ξ , η , say $\zeta = \alpha\xi + \beta\eta$ (α , β coefficients, neither of them = 0); we thence have $\zeta^2 = (\alpha^2 - \alpha)\xi + (\beta^2 - \beta)\eta = 0$; whence $\alpha = 1$, $\beta = 1$, and therefore $\zeta = \xi + \eta$; hence also $\xi\zeta = \xi^2 + \xi\eta = \xi$, $\eta\zeta = \eta\xi + \eta^2 = \eta$, and $\xi\eta\zeta = 0$. The six products consequently are $\xi\eta = 0$, $\eta\zeta = \eta$, $\zeta\xi = \xi$, $\eta\xi = 0$, $\zeta\eta = \eta$, $\xi\zeta = \xi$; each = 0, or $\xi\eta$, or identically = 0.

9. In verification of the theorem that for the commutative and associative system the cubic function Π contains the quadric function Φ as a factor, we may write, as above,

$$\xi = y, \quad \eta = dx - by, \quad a - d = \frac{cd}{b}, \quad h = 2c + \frac{be - ad}{b},$$

values which give

$$\begin{aligned} b\Phi &= (be - acd)x^2 + (ad^2 - abc + a^2d + bcd)xy + (abd - b^2c)y^2 \\ &= (ad - bc)\{cx^2 + (d - a)xy - by^2\}, \\ b\Pi &= cdx^3 + (d^2 - be - ad)x^2y + (ab - 2bd)xy^2 + b^2y^3 \\ &= (dx - by)\{cx^2 + (d - a)xy - by^2\}, \end{aligned}$$

which gives the theorem in question. And observe further that

$$b\Phi = (ad - bc)(dx - by),$$

that is, the third idempotent symbol $\zeta = dx - by$ is the linear factor common to $b\Phi$ and $b\Pi$.

10. Introducing coefficients in order to make the symbols ξ , η , ζ idempotent, and writing accordingly

$$\xi = \frac{1}{K} \left\{ cx + \frac{1}{2}(d - a + \sqrt{\nabla})y \right\}, \quad \nabla = (d - a)^2 + 4bc,$$

so that

$$\begin{aligned} \eta &= \frac{1}{L} \left\{ cx + \frac{1}{2}(d - a - \sqrt{\nabla})y \right\}, \quad \xi\eta = \frac{c}{KL} \{ cx^2 + (d - a)xy - by^2 \}, \\ \zeta &= \frac{1}{P} (dx - by), \end{aligned}$$

we have to verify that it is possible to determine K, L, P so that $\xi^2 = \xi, \eta^2 = \eta, \zeta^2 = \zeta, \zeta = \xi + \eta$.

The last equation gives

$$\frac{d}{P} = \frac{c}{K} + \frac{c}{L},$$

$$-\frac{2b}{P} = \frac{d-a+\sqrt{\nabla}}{K} + \frac{d-a-\sqrt{\nabla}}{L},$$

and we thence have

$$\frac{d(d-a)+2bc-d\sqrt{\nabla}}{P} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{d(d-a)+2bc+d\sqrt{\nabla}}{P} = \frac{2c\sqrt{\nabla}}{L},$$

and we can from the equation $\zeta^2 = \zeta$ find P ; viz. comparing the coefficients of x , we have

$$\frac{d}{P} = \frac{1}{P^2} \left(ad^2 - 2bdc + b^2 \frac{cd}{b} \right), \quad = \frac{d}{P^2} (ad-bc), \quad \text{that is, } P = (ad-bc),$$

or the values of K and L are

$$\frac{\{d(d-a)+2bc-d\sqrt{\nabla}\}}{ad-bc} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{\{d(d-a)+2bc+d\sqrt{\nabla}\}}{ad-bc} = +\frac{2c\sqrt{\nabla}}{L},$$

which should agree with the values of K and L found from the equations $\xi^2 = \xi, \eta^2 = \eta$, respectively.

Comparing the coefficients of x , the first of these equations gives

$$\frac{d}{P} = \frac{1}{P^2} (ad^2 - 2bdc + b^2 \cdot \frac{cd}{b}), \quad = \frac{d}{P^2} (ad-bc),$$

that is, $P = (ad-bc)$; and the equation for K becomes

$$\frac{d(d-a)+2bc}{ad-bc} = -\frac{2c\sqrt{\nabla}}{K},$$

that is, $[d(d-a)+2bc]^2 - d^2\nabla = -4c^2\nabla(ad-bc)$, which is right; and similarly the equation for L leads to this same equation.

11. We may now establish, on the principles appearing in No. 5, the different forms of the system. Using *idem* and *nil* as abbreviations for idempotent and nilpotent respectively, there are in all 11 cases.

(i) 3 idems. Taking two of these to be x and y , the system is

$$x^2 = x, \quad xy = cx + dy, \quad yx = ex + fy, \quad y^2 = y.$$

Hence $\Pi = (1 - c - e)x^2y + (1 - d - f)xy^2$, so that the third factor is $\zeta = (1 - c - e)x + (1 - d - f)y$. This must not reduce itself to x or y , for, if so, there would be a twofold idem; viz. as negative conditions we must have $c + e \neq 1$, $d + f \neq 1$. And we have $(\xi, \eta, \zeta) = (x, y, \zeta)$, which must be an idem; viz. we have the further negative condition $(c + e)(d + f) \neq 1$.

(ii) 2 idems and 1 nil. This arises from (i) by assuming therein $(c + e)(d + f) = 1$, say $d + f = \frac{1}{c + e}$; for then, writing $z = (c + e)x + y$, we have

$$z^2 = (c + e)^2x^2 + (c + e)\{(c + e)x + (d + f)y\} + y^2 = \{1 - (c + e)(d + f)\}y = 0;$$

viz. z is a nil. And, if in the equations instead of the idem y we introduce the nil z , then the equations assume the form

$$x^2 = x, \quad xz = [(c + e)d - e]x + dz, \quad zx = [(c + e)f - c]x + fz, \quad z^2 = 0;$$

with the idem $y = (c + e)x + z$: hence the negative conditions $c + e \neq 1$ or 0, implying $d + f \neq 1$.

But the equations are obtained in a more simple form by taking x for the idem and z for the nil; viz. we then have $xy = cx + dy$, $yx = ex + fz$, $y^2 = (c + e)(d + f)z$; and we have the negative conditions $c + e \neq 0$, $d + f \neq 0$ or 1.

(iii) 1 idem and 2 nils. The equations are $x^2 = x$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$; with the negative condition $c + e \neq 0$.

(iv) 1 idem and 1 nil. Taking x for the idem and z for the nil, the equations are $x^2 = x$, $xz = cx + dz$, $zx = ex + fz$, $z^2 = 0$; with the negative conditions $c + e \neq 0$, $d + f \neq 0$.

(v) Infinity of idems, 1 nil. The equations are $x^2 = x$, $xy = cx + dy$, $yx = (1 - c)x + (1 - d)y$, $y^2 = y$; whence $xy + yx = x + y$. Considering $\alpha x + \beta y$, we have $(\alpha x + \beta y)^2 = \alpha^2x + \alpha\beta(x + y) + \beta^2y = (\alpha + \beta)(\alpha x + \beta y)$; this is an idem, except in the case $\alpha + \beta = 0$, when it is the nil y .

(vi) Infinity of nils. The equations are $x^2 = 0$, $xy = cx + dy$, $yx = -cx - dy$, $y^2 = 0$; whence $xy + yx = 0$. Considering $\alpha x + \beta y$, we have $(\alpha x + \beta y)^2 = \alpha\beta(xy + yx) = 0$; viz. $\alpha x + \beta y$ is a nil; or there are an infinity of nils.

(vii) 2 nils. The equations are $x^2 = 0$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$.

(viii) Threefold idem. The equations are $x^2 = x$, $xy = cx + dy$, $yx = ex + (1 - d)y$, $y^2 = gx + (c + e)y$.

(ix) Threefold nil. The equations are $x^2 = 0$, $xy = cx + dy$, $yx = ex - dy$, $y^2 = gx + (c + e)y$.

(x) Two nils which are at the same time idems. The equations are $x^2 = x$, $xy = cx + dy$, $yx = ex + (1 - d)y$, $y^2 = 0$; whence $xy + yx = y$.

(xi) Two idems which are at the same time nils. The equations are $x^2 = 0$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$.

12. The different cases may be grouped together as follows:

A. 2 idems, (i), (ii), (iv), (x).

Equations $x^2 = x$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = y$.

B. 1 idem and 1 nil, (ii), (iii), (v), (vi), (x).

Equations $x^2 = x$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$.

C. 2 nils, (iii), (vii), (xi).

Equations $x^2 = 0$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = 0$.

D. Threefold idem, (viii).

Equations $x^2 = x$, $xy = cx + dy$, $yx = ex + (1 - d)y$,
 $y^2 = gx + (c + e)y$.

E. Threefold nil, (ix).

Equations $x^2 = 0$, $xy = cx + dy$, $yx = ex - dy$, $y^2 = gx + (c + e)y$.

The several cases of A, B, C respectively are distinguished by negative conditions which need not be here repeated.

13. I consider, as in my Note before referred to, the conditions in order that the system may be associative, viz. we have the 8 products $x.x^2$, $x.xy$, $x.yx$, $x.y^2$, $y.x^2$, $y.xy$, $y.yx$, $y.y^2$, which, putting therein for x^2 , xy , yx , y^2 their values, must be satisfied identically. We thus obtain in the first instance 16 relations, but some of these are repeated, and we have actually only 12 relations; viz. the relations

are

$$\begin{aligned}
b(c-e) &= 0, & b(f-d) &= 0, & g(c-e) &= 0, & g(f-d) &= 0, \\
bg - cd &= 0, & bg - ef &= 0, \\
c(c-h) + g(d-a) &= 0, & d(d-a) + b(c-h) &= 0, & b(e-h) + f(f-a) &= 0, \\
a(c-e) - cf + de &= 0, & h(f-d) - cf + de &= 0.
\end{aligned}$$

14. From the first four equations it appears that either $b = 0$ and $g = 0$, or else $c = e$ and $d = f$. I attend first to the latter case, viz. we have here the commutative system

$$x^2 = ax + by, \quad xy = yx = cx + dy, \quad y^2 = gx + hy.$$

In order that this may be associative, we must still have the relations

$$bg - cd = 0, \quad d(d-a) + b(c-h) = 0,$$

or, as they may be written,

$$\begin{vmatrix} b, & c, & d \\ d, & g, & c-h \end{vmatrix} = 0.$$

These are satisfied by $a = \frac{cd}{b}$, $h = \frac{bg}{c}$, and we have thus the commutative and associative system of the Note.

Every system is of the form A, B, C, D, or E; and it can be shown that the commutative and associative system is not of the form D. For, if D were commutative, we should have $e = c$, $d = \frac{1}{2}$, viz. the equations will be $x^2 = x$, $xy = yx = cx + \frac{1}{2}y$, $y^2 = gx + 2cy$, that is,

$$(a, b, c, d, g, h) = (1, 0, c, \frac{1}{2}, g, 2c);$$

and the last of the three relations, viz. $d(d-a) + b(c-h) = 0$, would thus be $\frac{1}{2}(-\frac{1}{2}) = 0$, which is not satisfied. Hence the commutative and associative system can only be of one of the forms A, B, C, and E.

15. First, if the form be A, B, or C; there will be the two idem-or-nil symbols x and y , that is, we may assume $x^2 = \kappa_1 x$, $y^2 = \kappa_2 y$; and the associative conditions then become $cd = 0$, $c(c-h) = 0$, $d(d-a) = 0$, viz.

$$\begin{aligned}
\text{A. } x^2 = x, y^2 = y : & \quad cd = 0, c(c-1) = 0, d(d-1) = 0; c = 0 \text{ or } 1, d = 0 \text{ or } 1, \\
\text{B. } x^2 = x, y^2 = 0 : & \quad cd = 0, c^2 = 0, d(d-1) = 0; c = 0, d = 0 \text{ or } 1, \\
\text{C. } x^2 = 0, y^2 = 0 : & \quad cd = 0, c^2 = 0, d^2 = 0; c = 0, d = 0.
\end{aligned}$$

But for the form A, if $c = 0, d = 1$, that is, $xy = yx = y$, then, writing $z = x - y$, we have $z^2 = z, yz = zy = 0, y^2 = y$. And similarly, if $c = 1, d = 0$, that is, $xy = yx = x$, then, writing $z = x + y$, we have $z^2 = x, xz = zx = 0, x^2 = x$. That is, each of these is reduced to the first case $c = 0, d = 0$; that is, $x^2 = x, xy = yx = 0, y^2 = y$.

For the form B, if $c = 0, d = 1$, then the system is $x^2 = x, xy = yx = y, y^2 = 0$; and this cannot be reduced to the first case $x^2 = x, xy = yx = 0, y^2 = 0$.

For the form C, there is only one case, as above.

For the form E, we have $a = 0, b = 0, (c = e, d = 0$, in order that the system may be commutative), $h = 2c$, viz. the equations must be $x^2 = 0, xy = yx = cx, y^2 = gx + 2cy$. The associative conditions then give $c = 0$; or, the system is $x^2 = 0, xy = yx = 0, y^2 = gx$. Writing $\sqrt{g}x$ instead of x , and for convenience interchanging x and y , the equations are

$$x^2 = y, \quad xy = yx = 0, \quad y^2 = 0.$$

16. The commutative associative system is thus seen to be reducible as follows:

A. $x^2 = x, xy = yx = 0, y^2 = y$, first mixed system, see No. 2.

B. $x^2 = x, xy = yx = y, y^2 = 0$, Peirce's system (a_2) ,

or else

B'. $x^2 = x, xy = yx = 0, y^2 = 0$, second mixed system.

C. $x^2 = 0, xy = yx = 0, y^2 = 0$, third mixed system.

E. $x^2 = y, xy = yx = 0, y^2 = 0$, Peirce's system (c_2) .

I said, at the end of my Note before referred to, that it had been pointed out to me "that my system [the commutative associative system], in the general case $ad - bc \neq 0$, is expressible as a mixture of two algebras of the form (c_2) , see American Journal of Mathematics, vol. iv., p. 120; whereas, if $ad - bc = 0$, it is reducible to the form (c_2) , see p. 122 (l.c.)." The accurate conclusion is as above, that the commutative associative system is either a mixed system of one of the three forms, or else a system (a_2) , or (c_2) .

17. Considering next the non-commutative associative systems, we have here, ante, No. 14, $b = 0, g = 0$ and the relations which remain to be satisfied then are

$$cd = 0, ef = 0, c(c-h) = 0, d(d-a) = 0, e(e-h) = 0, f(f-a) = 0,$$

$$a(c - e) - cf + de = 0, \quad h(f - d) - cf + de = 0.$$

The first equation gives $cd = 0$, that is, $c = 0$ or $d = 0$; but we may attend exclusively to the case $c = 0$, for the case $d = 0$ may be deduced from this by the interchange of x, y . We have then $ef = 0$; and it will be convenient to separate the cases

- I. $c = 0, e = 0, f = 0$, giving $d(d - a) = 0, dh = 0$,
- II. $c = 0, e = 0, f \neq 0$, giving $d(d - a) = 0, f - a = 0, h(f - d) = 0$,
- III. $c = 0, e \neq 0, f = 0$, giving $d(d - a) = 0, e - h = 0, d(e - h) = 0$,

that is, $d - a = 0, e - h = 0$.

18. We have thus five cases:

- I.(a) $d = 0$: $x^2 = ax, xy = yx = 0, y^2 = hy$: commutative, and
- I.(b) $d = a, h = 0$: $x^2 = ax, xy = ay, yx = 0, y^2 = 0$: writing $\sqrt{a}y$, Peirce's (b_2) .
- II.(c) $d = f = a$: $x^2 = ax, xy = yx = ay, y^2 = hy$: comm.
- II.(d) $d = 0, f = a, h = 0$: $x^2 = ax, xy = 0, yx = ay, y^2 = 0$: writing $\sqrt{a}$, Peirce's (d_2) .
- III.(e) $d = a, e = h$: $x^2 = ax, xy = ex + ay, yx = ex, y^2 = gx + 2ey$:

or, writing $z = x + y$, this is $x^2 = ax, xz = ex + ay, zx = ex, z^2 = (a + g)x + 4ey$. Thus x, z form the system $x^2 = x, xz = z, zx = 0, z^2$ (or, writing $w = \sqrt{g}z, x^2 = x, xz = z, zx = 0, z^2 = 0$); which is Peirce's system (b_2) .

19. The associative systems are thus seen to be:

- Commutative: A (first mixed system), B (a_2) , B' (second mixed), C (third mixed), E (c_2) ;
- Non-commutative: $(b_2), (d_2)$.

[The paper concludes with a summary of the relationship between the different systems of double algebra and their connection to Peirce's classification of linear associative algebras.]

815. The Binomial Equation—Quinquisition

[From the *Messenger of Mathematics*, vol. XIII. (1884),
pp. 170–173.]

THE general theory of the binomial equation $x^n - 1 = 0$ in its connection with the cyclotomic function was established by Gauss (*Disquisitiones Arithmeticae*, Sect. VII.); and the several cases $n = 3, 4, 5, 6, 8, 10, 12$ were considered in detail by him. The case $n = 5$ was also considered by Vandermonde, and more recently by professors Schönmann, Kronecker, and others. The following is a statement of some of the results in the case $n = 5$, the quinquisition of the circle.

The equation is $x^5 - 1 = 0$, or dividing by $x - 1$, the equation is $x^4 + x^3 + x^2 + x + 1 = 0$. Writing $x + x^{-1} = z$, we have $z^2 + z - 1 = 0$, whence $z = \frac{-1 \pm \sqrt{5}}{2}$; and then from $x^2 - zx + 1 = 0$, we obtain the four values of x .

The process may be presented under a somewhat different form. Writing $\alpha = 2 \cos \frac{2\pi}{5}$, $\beta = 2 \cos \frac{4\pi}{5}$, we have $\alpha + \beta = -1$, $\alpha\beta = -1$; whence $\alpha^2 + \alpha - 1 = 0$, $\beta^2 + \beta - 1 = 0$; and thence $\alpha = \frac{-1 + \sqrt{5}}{2}$, $\beta = \frac{-1 - \sqrt{5}}{2}$.

We have then $x = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$, where $\cos \frac{2\pi}{5} = \frac{\alpha}{2} = \frac{-1 + \sqrt{5}}{4}$, and

$$\sin \frac{2\pi}{5} = \sqrt{1 - \frac{(-1 + \sqrt{5})^2}{16}} = \frac{\sqrt{10 + 2\sqrt{5}}}{4}.$$

The values of x are thus

$$\begin{aligned} x_1 &= \frac{-1 + \sqrt{5} + i\sqrt{10 + 2\sqrt{5}}}{4}, \\ x_2 &= \frac{-1 - \sqrt{5} + i\sqrt{10 - 2\sqrt{5}}}{4}, \\ x_3 &= \frac{-1 - \sqrt{5} - i\sqrt{10 - 2\sqrt{5}}}{4}, \\ x_4 &= \frac{-1 + \sqrt{5} - i\sqrt{10 + 2\sqrt{5}}}{4}. \end{aligned}$$

Or arranging the roots in a different order, and writing $\omega = e^{2\pi i/5}$, the roots are $\omega, \omega^2, \omega^3, \omega^4$, where $\omega^5 = 1$; and we have the relations

$$\omega + \omega^4 = \frac{-1 + \sqrt{5}}{2}, \quad \omega^2 + \omega^3 = \frac{-1 - \sqrt{5}}{2}.$$

The equation may also be solved by the method of Lagrange resolvents; the resolvent equation is of the third degree, and the solution depends on the extraction of a square root and a cube root.

816. On the Bitangents of a Plane Quartic

[From the *Proceedings of the London Mathematical Society*, vol. XV.
(1884),
pp.—. Read March 13, 1884.]

THE bitangents of a quartic curve may be investigated by several different methods. The present paper gives a direct analytical investigation of their equations.

The equation of the quartic may be written

$$U = (a, b, c, d, e \mid x, y, z)^4 = 0.$$

A line $lx + my + nz = 0$ will be a bitangent if the equation of the quartic, expressed as a binary quartic in the coordinates of a point on the line, has its quartic invariant (the discriminant) equal to zero, and also its sextic invariant equal to zero; or, what is the same thing, if the binary quartic can be expressed as a perfect square.

The condition that $lx + my + nz = 0$ should be a bitangent may be obtained as follows: Writing $z = -(lx + my)/n$, and substituting in the equation of the quartic, we obtain a binary quartic in x, y ; and the condition that this should be a perfect square gives two equations between l, m, n ; which, when combined with the equation of the line, determine the bitangents.

Considering the quartic as the envelope of a conic $(a, b, c, f, g, h \mid x, y, z)^2 = 0$ touching the quartic in four points, the bitangents correspond to the cases where the conic degenerates into a pair of lines.

We may express the equation of the quartic in the form

$$U = \Omega^2 - PQ = 0,$$

where $\Omega = 0$ is a conic, and $P = 0, Q = 0$ are two other conics; and the bitangents then correspond to the values of the parameters for which the conic $\Omega^2 - \lambda PQ$ breaks up into a pair of lines.

The number of bitangents is 28; this may be seen as follows: the quartic has a class (number of tangents from a point) = 12; the number of bitangents is given by the formula $\frac{1}{2}n(n-2)(n^2-9)$ for a curve of order n without singularities; for $n = 4$, this is $\frac{1}{2} \cdot 4 \cdot 2 \cdot 7 = 28$.

The equation of a bitangent may be written in the form

$$\xi x + \eta y + \zeta z = 0,$$

where ξ, η, ζ are linear functions of the coordinates of the line; and the 28 bitangents may be arranged in systems of 7, each system corresponding to a particular arrangement of the points of contact.

The theory is based on the equations

$$\begin{aligned} ax + by + cz + f\xi + g\eta + h\zeta &= 0, \\ a_1x + b_1y + c_1z + f_1\xi + g_1\eta + h_1\zeta &= 0, \\ a_2x + b_2y + c_2z + f_2\xi + g_2\eta + h_2\zeta &= 0, \\ a_3x + b_3y + c_3z + f_3\xi + g_3\eta + h_3\zeta &= 0, \end{aligned}$$

where $af = bg = ch = a_1f_1 = \dots = c_2h_2 = 1$ and the coefficients $a, b, c, a_1, b_1, c_1, a_2, b_2, c_2$ are arbitrary.

The equations $x = 0, y = 0, z = 0$ represent any three given lines; considering ξ, η, ζ to be determined as linear functions of x, y, z by means of the first three equations, then (the coefficients $a, b, c, a_1, b_1, c_1, a_2, b_2, c_2$ being determined accordingly) the equations $\xi = 0, \eta = 0, \zeta = 0$ will also represent any three given lines.

But observe that, if the equation of the first of these lines is $x + my + nz = 0$, ξ is not = an arbitrary multiple $\theta(x + my + nz)$ of the linear function $x + my + nz$, but the constant factor θ has a completely determinate value: and the like as regards η and ζ respectively.

The coefficients a_3, b_3, c_3 of the fourth equation are such that this fourth equation is a mere consequence of the other three: viz. we must have

$$\begin{aligned} a_3 &= \lambda a + \lambda_1 a_1 + \lambda_2 a_2, \\ b_3 &= \lambda b + \lambda_1 b_1 + \lambda_2 b_2, \\ c_3 &= \lambda c + \lambda_1 c_1 + \lambda_2 c_2, \\ f_3 &= \lambda f + \lambda_1 f_1 + \lambda_2 f_2, \\ g_3 &= \lambda g + \lambda_1 g_1 + \lambda_2 g_2, \\ h_3 &= \lambda h + \lambda_1 h_1 + \lambda_2 h_2, \end{aligned}$$

or, what is the same thing, we must have

$$\begin{vmatrix} a & b & c & f & g & h \\ a_1 & b_1 & c_1 & f_1 & g_1 & h_1 \\ a_2 & b_2 & c_2 & f_2 & g_2 & h_2 \\ a_3 & b_3 & c_3 & f_3 & g_3 & h_3 \end{vmatrix} = 0,$$

viz. each of the determinants formed with four columns out of this matrix is equal to 0.

Using the equations in $\lambda, \lambda_1, \lambda_2$, and writing for shortness

$$\begin{aligned} a_1f_2 + a_2f_1, & \quad a_2f + af_2, & \quad af_1 + a_1f = F, & F_1, & F_2, \\ b_1g_2 + b_2g_1, & \quad b_2g + bg_2, & \quad bg_1 + b_1g = G, & G_1, & G_2, \\ c_1h_2 + c_2h_1, & \quad c_2h + ch_2, & \quad ch_1 + c_1h = H, & H_1, & H_2; \end{aligned}$$

then, forming the products a_3f_3, b_3g_3, c_3h_3 , we find

$$\begin{aligned} 1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 &= F\lambda_1\lambda_2 + F_1\lambda_2\lambda + F_2\lambda\lambda_1, \\ &= G\lambda_1\lambda_2 + G_1\lambda_2\lambda + G_2\lambda\lambda_1, \\ &= H\lambda_1\lambda_2 + H_1\lambda_2\lambda + H_2\lambda\lambda_1, \end{aligned}$$

or, as these equations may be written,

$$\begin{aligned} 1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 : \lambda_1\lambda_2 : \lambda_2\lambda : \lambda\lambda_1 \\ = \begin{vmatrix} F & F_1 & F_2 \\ G & G_1 & G_2 \\ H & H_1 & H_2 \end{vmatrix} : \begin{vmatrix} 1 & F_1 & F_2 \\ 1 & G_1 & G_2 \\ 1 & H_1 & H_2 \end{vmatrix} : \begin{vmatrix} 1 & F_2 & F \\ 1 & G_2 & G \\ 1 & H_2 & H \end{vmatrix} : \begin{vmatrix} 1 & F & F_1 \\ 1 & G & G_1 \\ 1 & H & H_1 \end{vmatrix}. \end{aligned}$$

I proceed to show that each of the 28 lines is, in fact, a bitangent to the quartic curve

$$\sqrt{af\xi x} + \sqrt{bg\eta y} + \sqrt{ch\zeta z} = 0,$$

or, what is the same thing, that writing this equation in the form

$$\Omega = x^2\xi^2 + y^2\eta^2 + z^2\zeta^2 - 2yz\eta\zeta - 2zx\zeta\xi - 2xy\xi\eta = 0,$$

then that by means of any one of these equations Ω becomes a perfect square.

This is obviously the case for each of the equations 1, 2, 3, 4, 5, 6.

For the equations 7, 8, 9, 10, observe that the equation of the quartic may be written

$$\sqrt{af\xi x} + \sqrt{bg\eta y} + \sqrt{ch\zeta z} = 0,$$

or, what is the same thing,

$$\Omega = (x\xi + y\eta + z\zeta)^2 - 2(y\eta \cdot z\zeta + z\zeta \cdot x\xi + x\xi \cdot y\eta) = 0.$$

Hence writing $x, y, z, \xi, \eta, \zeta$ in place of $\sqrt{af}x, \sqrt{bg}y, \sqrt{ch}z, \sqrt{af}\xi, \sqrt{bg}\eta, \sqrt{ch}\zeta$, so that now the equation of the line 7 is expressed in

the two equivalent forms $x + y + z = 0$, $\xi + \eta + \zeta = 0$. We have for Ω its original value

$$\Omega = x^2\xi^2 + y^2\eta^2 + z^2\zeta^2 - 2yz\eta\zeta - 2zx\zeta\xi - 2xy\xi\eta,$$

and writing $\xi = -(\eta + \zeta)$, $x = -(y + z)$, we have

$$\Omega = (y\eta - z\zeta)^2 + (z\xi - x\zeta)^2 + (x\eta - y\xi)^2,$$

a perfect square when $x + y + z = 0$ and $\xi + \eta + \zeta = 0$. The like proof applies to the equations 8, 9, 10: and then clearly the result applies to the equations 11, 12, 13, 14; 15, 16, 17, 18; 19, 20, 21, 22.

We come now to the equation of the line 23, say this is in the first instance taken to be

$$\frac{x}{a} + \frac{y}{b_1} + \frac{z}{c_2} = 0;$$

if from this equation, by means of the two fundamental equations

$$ax + by + cz + f\xi + g\eta + h\zeta = 0,$$

$$a_1x + b_1y + c_1z + f_1\xi + g_1\eta + h_1\zeta = 0,$$

we eliminate first (y, z) , secondly (z, x) , and thirdly (x, y) , it is found that ξ , η , ζ will also disappear in the three cases respectively, and that the resulting equations are

$$\frac{x}{aa_1(bc_1 - b_1c)} + \frac{\xi}{bb_1(ca_1 - c_1a)} + \frac{\eta}{cc_1(ab_1 - a_1b)} = 0,$$

$$\frac{y}{bb_1(ca_1 - c_1a)} + \frac{\eta}{cc_1(ab_1 - a_1b)} + \frac{\zeta}{aa_1(bc_1 - b_1c)} = 0,$$

$$\frac{z}{cc_1(ab_1 - a_1b)} + \frac{\zeta}{aa_1(bc_1 - b_1c)} + \frac{\xi}{bb_1(ca_1 - c_1a)} = 0,$$

so that each of these is, in fact, equivalent to the original equation in (x, y, z) : and observe that, adding together the three equations, we reproduce the original equation in (x, y, z) .

Write for shortness

$$P = bc_1 - b_1c, \quad Q = ca_1 - c_1a, \quad R = ab_1 - a_1b,$$

$$\alpha = aa_1, \quad \beta = bb_1, \quad \gamma = cc_1,$$

then the equation in (x, ξ, η) is

$$\frac{x}{\alpha P} + \frac{\xi}{\beta Q} + \frac{\eta}{\gamma R} = 0,$$

so that, eliminating ξ and η , we find

$$\alpha x^2 + \beta \left(\frac{x}{\alpha P} + \frac{\xi}{\beta Q} \right)^2 + \gamma \left(\frac{x}{\alpha P} + \frac{\eta}{\gamma R} \right)^2 = 0.$$

In this equation, the coefficient of x^2 is

$$\begin{aligned} & (bc_1 - b_1c)QR + (ca_1 - c_1a)RP + (ab_1 - a_1b)PQ, \\ &= (bc_1 - b_1c)\{a^2bc + aa_1(b^2c_1 + b_1^2c)\} \\ &+ (ca_1 - c_1a)\{ab^2c + bb_1(ca_1 + c_1a)\} \\ &+ (ab_1 - a_1b)\{abc^2 + cc_1(a^2b_1 + a_1^2b)\}, \end{aligned}$$

which, observing that the coefficients satisfy the fundamental relations, reduces to a form proportional to P^2 .

The coefficient of ξ is $RP - PQ$, which, observing that $af = 1$, $bg = 1$, $ch = 1$, is $= 0$. Similarly the coefficient of η is $RP - QR$, which is $= 0$.

We have thus in x, ξ, η the equation

$$-\alpha\beta\gamma x - \frac{P}{\beta}\gamma\alpha\eta + \frac{P}{\gamma}\alpha\beta\zeta = 0,$$

or, what is the same thing,

$$\frac{x}{P} + \frac{\eta}{\beta Q} - \frac{\zeta}{\gamma R} = 0 :$$

and in like manner, by the elimination of (z, x) and (x, y) respectively,

$$\begin{aligned} -\frac{\xi}{\alpha P} + \frac{y}{Q} + \frac{\zeta}{\gamma R} &= 0, \\ \frac{\xi}{\alpha P} - \frac{\eta}{\beta Q} + \frac{z}{R} &= 0, \end{aligned}$$

which are the required three equations.

We have next to show that the line is a bitangent—write for a moment $aa_1\alpha, bb_1\beta, cc_1\gamma = \lambda, \mu, \nu$, then the three equations give

$$x : y : z = P \left(-\frac{\eta}{\mu} + \frac{\zeta}{\nu} \right) : Q \left(-\frac{\zeta}{\nu} + \frac{\xi}{\lambda} \right) : R \left(-\frac{\xi}{\lambda} + \frac{\eta}{\mu} \right),$$

and we thence have

$$\begin{aligned}
\Omega = & P^2 \left(-\frac{\xi\eta}{\mu} + \frac{\zeta\xi}{\nu} \right)^2 \\
& + Q^2 \left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda} \right)^2 \\
& + R^2 \left(-\frac{\zeta\xi}{\lambda} + \frac{\eta\zeta}{\mu} \right)^2 \\
& - 2QR \left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda} \right) \left(-\frac{\zeta\xi}{\lambda} + \frac{\eta\zeta}{\mu} \right) \\
& - 2RP \left(-\frac{\zeta\xi}{\lambda} + \frac{\eta\zeta}{\mu} \right) \left(-\frac{\xi\eta}{\mu} + \frac{\zeta\xi}{\nu} \right) \\
& - 2PQ \left(-\frac{\xi\eta}{\mu} + \frac{\zeta\xi}{\nu} \right) \left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda} \right);
\end{aligned}$$

which is

$$\begin{aligned}
& = RP \frac{1}{bb_1} (bc - b_1c_1) - PQ \frac{1}{bb_1} (b^2 - b_1^2), \\
& = \frac{P}{bb_1} \{ (bc - b_1c_1)(ab - a_1b_1) - (b^2 - b_1^2)(ca - c_1a_1) \}, \\
& = \frac{P}{bb_1} \{ b_1^2ca + b^2c_1a_1 - bb_1(ac_1 + a_1c) \} \\
& = -\frac{P}{bb_1} \gamma\alpha;
\end{aligned}$$

and similarly the coefficient of ζ is

$$= \frac{P}{cc_1} \alpha\beta.$$

We have thus in x, η, ζ the equation

$$-\alpha\beta\gamma x - \frac{P}{bb_1} \gamma\alpha\eta + \frac{P}{cc_1} \alpha\beta\zeta = 0,$$

or, what is the same thing,

$$\frac{x}{P} + \frac{\eta}{bb_1\beta} - \frac{\zeta}{cc_1\gamma} = 0 :$$

and in like manner, by the elimination of (z, x) and (x, y) respectively,

$$-\frac{\xi}{aa_1\alpha} + \frac{y}{Q} + \frac{\zeta}{cc_1\gamma} = 0,$$

$$\frac{\xi}{aa_1\alpha} - \frac{\eta}{bb_1\beta} + \frac{z}{R} = 0,$$

which are the required three equations.

[The paper continues with further discussion of the relationship between the bitangents and the theta-characteristics, and the identification of the 28 bitangents with Riemann's 28 odd theta-characteristics. The complete list of the 28 bitangents with their corresponding theta-characteristics and duad symbols (from Hesse's method) is given in a table. The verification that each line is a bitangent is completed by showing that the equation of the quartic becomes a perfect square when the coordinates satisfy any of the 28 linear equations.]

Considering ξ, η, ζ to be determined as linear functions of x, y, z by means of the first three equations, then (the coefficients $a, b, c, a_1, b_1, c_1, a_2, b_2, c_2$ being determined accordingly) the equations $\xi = 0, \eta = 0, \zeta = 0$ will also represent any three given lines. But observe that, if the equation of the first of these lines is $x + my + nz = 0$, ξ is not = an arbitrary multiple $\theta(x + my + nz)$ of the linear function $x + my + nz$, but the constant factor θ has a completely determinate value: and the like as regards η and ζ respectively.

The coefficients a_3, b_3, c_3 of the fourth equation are such that this fourth equation is a mere consequence of the other three: viz. we must have

$$a_3 = \lambda a + \lambda_1 a_1 + \lambda_2 a_2,$$

$$b_3 = \lambda b + \lambda_1 b_1 + \lambda_2 b_2,$$

$$c_3 = \lambda c + \lambda_1 c_1 + \lambda_2 c_2,$$

$$f_3 = \lambda f + \lambda_1 f_1 + \lambda_2 f_2,$$

$$g_3 = \lambda g + \lambda_1 g_1 + \lambda_2 g_2,$$

$$h_3 = \lambda h + \lambda_1 h_1 + \lambda_2 h_2,$$

or, what is the same thing, we must have

$$\begin{vmatrix} a & b & c & f & g & h \\ a_1 & b_1 & c_1 & f_1 & g_1 & h_1 \\ a_2 & b_2 & c_2 & f_2 & g_2 & h_2 \\ a_3 & b_3 & c_3 & f_3 & g_3 & h_3 \end{vmatrix} = 0,$$

viz. each of the determinants formed with four columns out of this matrix is equal to 0.

Using the equations in $\lambda, \lambda_1, \lambda_2$, and writing for shortness

$$\begin{aligned} a_1 f_2 + a_2 f_1, & \quad a_2 f + a f_2, & \quad a f_1 + a_1 f = F, & F_1, & F_2, \\ b_1 g_2 + b_2 g_1, & \quad b_2 g + b g_2, & \quad b g_1 + b_1 g = G, & G_1, & G_2, \\ c_1 h_2 + c_2 h_1, & \quad c_2 h + c h_2, & \quad c h_1 + c_1 h = H, & H_1, & H_2; \end{aligned}$$

then, forming the products $a_3 f_3, b_3 g_3, c_3 h_3$, we find

$$\begin{aligned} 1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 &= F \lambda_1 \lambda_2 + F_1 \lambda_2 \lambda + F_2 \lambda \lambda_1, \\ &= G \lambda_1 \lambda_2 + G_1 \lambda_2 \lambda + G_2 \lambda \lambda_1, \\ &= H \lambda_1 \lambda_2 + H_1 \lambda_2 \lambda + H_2 \lambda \lambda_1, \end{aligned}$$

or, as these equations may be written,

$$\begin{aligned} &1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 : \lambda_1 \lambda_2 : \lambda_2 \lambda : \lambda \lambda_1 \\ &= \begin{vmatrix} F & F_1 & F_2 \\ G & G_1 & G_2 \\ H & H_1 & H_2 \end{vmatrix} : \begin{vmatrix} 1 & F_1 & F_2 \\ 1 & G_1 & G_2 \\ 1 & H_1 & H_2 \end{vmatrix} : \begin{vmatrix} 1 & F_2 & F \\ 1 & G_2 & G \\ 1 & H_2 & H \end{vmatrix} : \begin{vmatrix} 1 & F & F_1 \\ 1 & G & G_1 \\ 1 & H & H_1 \end{vmatrix}. \end{aligned}$$

We have next to show that the line is a bitangent—write for a moment $aa_1\alpha, bb_1\beta, cc_1\gamma = \lambda, \mu, \nu$, then the three equations give

$$x : y : z = P \left(-\frac{\eta}{\mu} + \frac{\zeta}{\nu} \right) : Q \left(-\frac{\zeta}{\nu} + \frac{\xi}{\lambda} \right) : R \left(-\frac{\xi}{\lambda} + \frac{\eta}{\mu} \right),$$

and we thence have

$$\begin{aligned}
\Omega = & P^2 \left(-\frac{\xi\eta}{\mu} + \frac{\zeta\xi}{\nu} \right)^2 \\
& + Q^2 \left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda} \right)^2 \\
& + R^2 \left(-\frac{\zeta\xi}{\lambda} + \frac{\eta\zeta}{\mu} \right)^2 \\
& - 2QR \left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda} \right) \left(-\frac{\zeta\xi}{\lambda} + \frac{\eta\zeta}{\mu} \right) \\
& - 2RP \left(-\frac{\zeta\xi}{\lambda} + \frac{\eta\zeta}{\mu} \right) \left(-\frac{\xi\eta}{\mu} + \frac{\zeta\xi}{\nu} \right) \\
& - 2PQ \left(-\frac{\xi\eta}{\mu} + \frac{\zeta\xi}{\nu} \right) \left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda} \right);
\end{aligned}$$

which is

$$\begin{aligned}
& = RP \frac{1}{bb_1} (bc - b_1c_1) - PQ \frac{1}{bb_1} (b^2 - b_1^2), \\
& = \frac{P}{bb_1} \{ (bc - b_1c_1)(ab - a_1b_1) - (b^2 - b_1^2)(ca - c_1a_1) \}, \\
& = \frac{P}{bb_1} \{ b_1^2ca + b^2c_1a_1 - bb_1(ac_1 + a_1c) \} \\
& = -\frac{P}{bb_1} \gamma\alpha;
\end{aligned}$$

and similarly the coefficient of ζ is

$$= \frac{P}{cc_1} \alpha\beta.$$

We have thus in x, η, ζ the equation

$$-\alpha\beta\gamma x - \frac{P}{bb_1} \gamma\alpha\eta + \frac{P}{cc_1} \alpha\beta\zeta = 0,$$

or, what is the same thing,

$$\frac{x}{P} + \frac{\eta}{bb_1\beta} - \frac{\zeta}{cc_1\gamma} = 0 :$$

and in like manner, by the elimination of (z, x) and (x, y) respectively,

$$-\frac{\xi}{aa_1\alpha} + \frac{y}{Q} + \frac{\zeta}{cc_1\gamma} = 0,$$

$$\frac{\xi}{aa_1\alpha} - \frac{\eta}{bb_1\beta} + \frac{z}{R} = 0,$$

which are the required three equations.

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR CAY-
LEY
VOL. XII

PAGES 101–150 (PAPERS 819 CONT.–825 IN PROGRESS) Ar-
thur Cayley (1821–1895) Published 1889

*Transcribed from the original Cambridge University Press edition,
1889.*

This file covers book pages 101–150, internal gap from prior source.

819. Two Cases of the Quadric Transformation between Two Planes (concl.)

[p. 101] Writing for convenience (a, b, c) for the coordinates of the fixed point, and (x_1, y_1, z_1) , (x_2, y_2, z_2) for those of the other two points, the formulæ with A, B, C give thus the correspondence

$$x_2 : y_2 : z_2 = bcx_1^2 - a^2y_1z_1 : cay_1^2 - b^2z_1x_1 : abz_1^2 - c^2x_1y_1,$$

which is the first of the two cases in question. These equations give reciprocally

$$x_1 : y_1 : z_1 = bcx_2^2 - a^2y_2z_2 : cay_2^2 - b^2z_2x_2 : abz_2^2 - c^2x_2y_2,$$

or the correspondence is a $(1, 1)$ quadric correspondence.

The formulæ with P, Q, R give in like manner

$$x_2 : y_2 : z_2 = a(ax_1^2 + hy_1^2 + gz_1^2) + (a'y_1z_1 + a''z_1x_1 + a'''x_1y_1), \text{ \&c.},$$

or for shortness

$$x_2 : y_2 : z_2 = a\Omega_1 + a\Omega_2, \quad y_2 = b\Omega_1 + b\Omega_2, \quad z_2 = c\Omega_1 + c\Omega_2,$$

then

$$x_1 : y_1 : z_1 = a\Omega'_1 + a\Omega'_2, \quad \text{\&c.},$$

which is the second of the two cases. We have reciprocally

$$x_1 : y_1 : z_1 = a\Omega_{1'} + a\Omega_{2'}, \quad y_1 = b\Omega_{1'} + b\Omega_{2'}, \quad z_1 = c\Omega_{1'} + c\Omega_{2'},$$

where

$$\Omega_1 = aax_1^2 + by_1^2 + cz_1^2, \quad \Omega_2 = a'y_1z_1 + b'z_1x_1 + c'x_1y_1,$$

and the correspondence is thus in this case also a $(1, 1)$ quadric correspondence.

820. On a Problem of Analytical Geometry

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 203.]

[p. 102] The object of the present note is only to call attention to a problem of Analytical Geometry which presents itself in connexion with the reduction of an algebraical integral, and which is solved, pp. 21, 22 of Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866); viz. the problem is considering a line drawn through any given point of a curve $f = 0$ of the order n , to find the value u of α such that which the remaining $n - 2$ points of intersection of the line with the curve f , through the intersection of $\alpha = 0$ are required for the determination of the curve. It is to be shown that we have indirectly an expression for α , and to render the line, in the quadric curve which passes through the remaining two intersections of the line with Ω , and through three given points. Take (ξ, η, ζ) , (ξ_1, η_1, ζ_1) the coordinates of the two given points on the curve f (f is, say of the order $n = 4$); $\theta(x, y, z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining (x', y', z') , (x'', y'', z'') meets the curve θ where $\theta = 0$ in two points Λ , which when this factor is thrown out becomes

$$a\Lambda^2 + 2b\rho\Lambda + c\rho^2 = 0,$$

where

$$a = (\xi_1, \eta_1, \zeta_1)^2\theta, \quad b = (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta)\theta, \quad c = (\xi, \eta, \zeta)^2\theta,$$

$$\gamma = (\xi, \eta, \zeta)^2\theta, \quad \beta = (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta)\theta, \quad \alpha = (\xi_1, \eta_1, \zeta_1)^2\theta,$$

where for shortness f, f_1 are written to denote $f(\xi, \eta, \zeta), f(\xi_1, \eta_1, \zeta_1)$ respectively.

[p. 103] The condition as to the two points obviously is that, making the same substitution $x, y, z = b\xi + a\xi_1, b\eta + a\eta_1, b\zeta + a\zeta_1$ in the equation $\Omega = (a, \dots, b)(x, y, z)^2 = 0$, we must obtain the same quadric equation in $a : b$, the equation in $(a, b) = 0$. We have therefore three conditions, which, introducing an indeterminate multiplier θ , are expressed by the three equations

$$\begin{aligned} (\alpha, \dots, \xi)\xi_1, \eta_1, \zeta_1)^2 &= \theta a, \\ (\alpha, \dots, \xi)\xi_1\xi, \eta_1\eta, \zeta_1\zeta &= \theta b, \\ (\alpha, \dots, \xi)\xi, \eta, \zeta)^2 &= \theta c. \end{aligned}$$

The conditions as to the three points are obviously

$$\begin{aligned}(\alpha, \dots, \xi)(x_1, y_1, z_1)^2 &= 0, \\ (\alpha, \dots, \xi)(x_2, y_2, z_2)^2 &= 0, \\ (\alpha, \dots, \xi)(x_3, y_3, z_3)^2 &= 0,\end{aligned}$$

and these equations determine the ratios of a, b, c, f, g, h . But to complete the solution the convenient course is to regard the function $\Omega, = (a, \dots)(x, y, z)^2$ as a quantity to be determined, and consequently to join to the foregoing the equation

$$(\alpha, \dots)(x, y, z)^2 = \Omega;$$

we have then seven equations from which (a, b, c, f, g, h) may be eliminated, the result being expressed by means of a determinant of the seventh order

$$|ext[determinant]| = 0,$$

viz. this is an equation of the form $A\Omega = \theta V$, where A is a constant determinant of the sixth order (i.e. a determinant not involving x, y, z) V a determinant of the seventh order, a quadric function of (x, y, z) , obtained from the foregoing determinant by writing therein $\Omega = 0$ and $\theta = 1$; the multiplier θ is and remains arbitrary; but it is convenient to take it to be $= 1$, viz. we then say only that the equation $\Omega = 0$, of the required conic, but we put a determinate value for Ω , say $\Omega = A^{-1}V$, which is, in fact, an equation $\Omega = 0, = A^{-1}V$ arbitrary.

821. On the Geometrical Representation of an Equation between Two Variables

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 210.]

[p. 104] An equation between two variables cannot be represented in a satisfactory manner by a curve, for this serves only to represent the corresponding real values of the variables; to represent the imaginary values the natural course is to represent each variable by a point in a plane; viz. the variable $x, = x + iy$, will be represented by a point the

coordinates of which are the components x and y of the variable, and similarly the variable $x' = x' + iy'$, by a point the coordinates of which are the components x' and y' of the variable; the equation between the two variables will then establish a correspondence between the two points respectively, and to a point on the plane of x there correspond two or more points on the plane of x' . But to complete the solution the convenient course is to regard x, x' as fixed, and respectively, the planes which contain the two points respectively; and it is this correspondence of the two planes which is the proper geometrical representation of the equation between the two variables: to exhibit this correspondence we may in either of the planes draw a network of curves or planes, and then draw in the other plane the network of corresponding curves. This well-known theory [see &c.] illustrated by the notion of a Riemann's surface; taking in the infinite half-plane y positive about the origin as centre a system of semi-circles, to these correspond in the infinite plane of $x'y'$ a series of lunate-shaped curves; and by means of these it is easy to show in the second plane the path corresponding to a given path of the point $x, = x + iy$, in the first half-plane.

822. On Associative Imaginaries

[From the *Johns Hopkins University Circulars*, No. 15 (1882), pp. 211, 212.]

[p. 105] The imaginary (x, y) defined by the equations

$$\begin{aligned}x^2 &= ax + by, \\xy &= cx + dy, \\yx &= \alpha x + \beta y, \\y^2 &= px + qy,\end{aligned}$$

will not be in general associative; to make them so, we must have 8 double relations corresponding to the combinations $x^2.x, xy.x, yx.x, xy.y, y^2.x$ respectively; viz. the first of these gives $x.x.x = x.x.x$, that is, $\theta = x(ax + by) - (ax + by).x, = b(xy - yx), = b(c - \alpha)x + b(d - \beta)y$, and similarly for each of the other terms. We thus obtain by way of

the 8 equations between the 6 coefficients $a, b, c, d, \alpha, \beta, p, q$, viz. the relations so obtained are

$$\begin{aligned} b(c - \alpha) &= 0 \text{ (twice)}, & b(d - \beta) &= 0, & p(d - \beta) &= 0 \text{ (twice)}, \\ bp - ad &= 0 \text{ (twice)}, & bp - q^2 &= 0 \text{ (twice)}, \\ c^2 + dp - c\alpha - p\beta &= 0, & d^2 + bc - bd - dq + f\beta + q\alpha &= 0, \\ c^2 + cq - c\alpha - q\beta &= 0, & b(f - d) - q(\alpha - b) &= 0. \end{aligned}$$

From the first four equations it appears that either $b = 0$ and $p = 0$, or else $c = \alpha$ and $d = \beta$; the latter solution may be set aside for the moment, giving the commutative system

$$\begin{aligned} x^2 &= ax + by, \\ xy &= yx = cx + dy, \\ y^2 &= px + qy. \end{aligned}$$

[p. 106] In order that this may be associative, we must still have the relations

$$\begin{aligned} bp - cd &= 0, \\ c^2 + dp - cq - dk &= 0, \\ d^2 + bc - ad - bk &= 0, \end{aligned}$$

which are all three of them satisfied by $p = \frac{cd}{b}$, $k = \frac{d^2 + bc - ad}{b}$, viz. we thus have the associative and commutative system

$$\begin{aligned} x^2 &= ax + by, \\ xy &= yx = cx + dy, \\ y^2 &= \frac{cd}{b} x + \frac{d^2 + bc - ad}{b} y, \end{aligned}$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. iv. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad - bc \neq 0$, is reducible to the form (a), see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by

the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 238.]

[p. 107] I MAYBE given in my *Elliptic Functions* expressions for the sn of $u + \frac{1}{2}K$, $u + \frac{1}{2}iK'$, $u + \frac{1}{2}(K + iK')$; but it is better to consider the sn, cn, dn of these combinations respectively, and to write the formulæ thus

$$\begin{aligned}\operatorname{sn}^2(u + \tfrac{1}{2}K) &= \frac{1 - \operatorname{cn} - (1 + i')\operatorname{sn}}{1 - \operatorname{cn} - (1 - i')\operatorname{sn}}, = \frac{1 - i' - (1 + i')\operatorname{sn} - (1 - i')y}{1 - i' - (1 - i')\operatorname{sn} - (1 + i')y}, \\ \operatorname{sn}^2(u + \tfrac{1}{2}iK') &= \frac{(1 + i)\operatorname{sn} + (1 - i)\operatorname{cn}}{(1 + i)\operatorname{sn} + (1 - i)\operatorname{cn}}, \\ \operatorname{cn}^2(u + \tfrac{1}{2}K + \tfrac{1}{2}iK') &= \frac{-(1 - i)\operatorname{sn} + (1 + i)\operatorname{cn}}{1 + (1 - i)\operatorname{sn} + (1 + i)\operatorname{cn}},\end{aligned}$$

where in the last set of ratios x , y are used to denote sn^2u and cn^2u respectively; and the three combinations above show that for u are written respectively the values just mentioned. The curve $y^2 = (1 - x)(1 - k^2x)$ may, by an inflexion at infinity on the line $x = 0$, and the three tangents from the inflexion are at the points $x, y = 0, \theta, 0$, $(\frac{1}{k}, 0)$ respectively; hence these points are syntactic points.

[p. 108] We may from any of these, for instance the point $(0, 0)$, draw four tangents to the curve, $(1 + i)x + (1 - i)y = 0$ to the conic, viz. touching at the points $x, y = 0, 0, (\frac{1}{k}, 0)$ respectively; it can therefore be shown how the system of equations to which an arbitrary constants in the system of conics, by introducing in place of the natural conics a system of conics syntactically corresponded; that, in the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

824. Note on the Formulæ of Trigonometry

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 241.]

[*p. 108 cont.*] The equations $a = c \cos B + b \cos C$, $b = a \cos C + c \cos A$, $c = b \cos A + a \cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i \sin A$, $\cos B + i \sin B$, $\cos C + i \sin C$ calling these $\frac{x}{y'}$, $\frac{y'}{z'}$, $\frac{z'}{x'}$ respectively, or, what is the same thing, writing $\Sigma \cos A = \frac{x}{y'} + \frac{x'}{y}$, $\Sigma \sin A = \frac{x}{y'} - \frac{x'}{y}$, then the foregoing equations may be written

$$\begin{aligned} (-\Sigma x + y' + z')(-\Sigma x' + y + z) + i \Sigma y \Sigma y' &= 0, \\ (c \Sigma x' + xz - \Sigma x x') - i \Sigma x (y' z' + y' z) &= 0, \\ (c \Sigma y' + x' y' - \Sigma x x') - i \Sigma y' &= 0; \end{aligned}$$

that is, as a system of bipartite equations in (a, b, c) and cubic in (x, y, z) .

Similarly to Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2 \cos a = \frac{l}{l'} + \frac{l''}{l}$, $2 \cos b = \frac{m}{m'} + \frac{m''}{m}$, $2 \cos c = \frac{n}{n'} + \frac{n''}{n}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, x', y', z') and (a, B, b, q, R) respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, *American Journal of Mathematics*, t. v. (1882), pp. 137–179;
Chapters IV to VII, *ib.*, t. vii. (1885), pp. 101–167.]

[*p. 109*] THE present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential

expression $d\Pi_n$ (or as I write it $\bar{d}\Pi$); is the integral of the third kind $\int_{1,2} \bar{d}\Pi_n$ in the dual normal form (collide Nuessellen) for which we have (p. 117) $\int_{1,2} \bar{d}\Pi_n = \int_n \bar{d}\Pi_n$, the limits and parametric points interchangeable. The want of this determination presented itself to me as a barrier in the theory during the course of lectures on the subject which I had the pleasure of giving in the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the first more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to, I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence, and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T, U, V, Θ . The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

Proposition. Art. Nos. 1 to 5.

[p. 110] 1. Starting from the equation $\phi = (x, \dots, \xi, y)^n = 0$ of the variable curve, we have

$$\Sigma (x, y, 1)^{n-2} du = 0.$$

The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.

1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma (x, y, 1)^{n-2} du = 0.$$

The affected theorem is

$$\Sigma (x, y, 1)^{n-2} du = -\frac{b\phi_n}{\phi_n} \frac{d\phi_n}{\phi_n} \frac{d\phi_n}{\phi_n}.$$

2. We have a fixed curve $f = 0$, or say the curve f , or simply the fixed curve; of the order n , with δ dpps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2) - \delta$. The expression “the dpp” means always the δ dpp of f .

And we have a variable curve $\phi = 0$, or say the curve ϕ , or simply the variable curve, of the order m , passing through the dps and besides meeting the fixed curve in $mn - 2\delta$ variable points.

Moreover, du is the displacement of the current point θ , coordinates (x, y, z) , on the fixed curve, viz. the equation $f = 0$ being satisfied, we have

$$\frac{df}{dx} dx + \frac{df}{dy} dy + \frac{df}{dz} dz = 0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} = y dx - x dy : z dx - x dz : x dy - y dx,$$

so that we have three equal values each of which is put $= du$, viz. we write

$$\frac{y dz - z dy}{\frac{df}{dx}} = \frac{z dx - x dz}{\frac{df}{dy}} = \frac{x dy - y dx}{\frac{df}{dz}} = du,$$

and so as thus defined is the displacement.

[p. 111] 3. $(x, y, z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n-3$, which passes through the dpp's, and the general curve of the order $n-3$, z contains $\frac{1}{2}n(n-3) + 1$ terms (viz. $\frac{1}{2}n(n-3) - 1$ terms, but y thereto for the dpps), respectively; and $\theta(t)$ denotes the determinant

$$\theta(t) = |\text{ext}[\text{determinant}]|,$$

so that $\theta(t) = 0$ is the equation of the line joining the points 1 and 2; viz. this is the line (123) in the great curve of order $n-2$ passing through the points of 1, 2.

[p. 112] 4. It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; for instance, we may help to see the latter equal to the number of independent linear relations between the displacements $\Sigma (x, y, z)^{n-2} du = 0$ for the variable intersections.

Examples of the Pure Theorem—The Fixed Curve a Cubic. Art. Nos. 6 to 13.

6. The pure theorem is next applicable to the case $n = 2$, the fixed curve a conic; it is here given no displacement-relation, and this is as it should be, for the variable intersections of ϕ with the conic are not so independent.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For proper simplicity the equation is to be taken in the form $f = ax^3 + by^3 + cz^3 + 6lxyz = 0$.

For the variable curve we may take the equation

$$du = \frac{dy}{f_x} = -\frac{dx}{f_y};$$

the two cases being of course in Cartesian form, may then be written

$$du = \frac{dx}{-f_y} = -\frac{dy}{f_x};$$

the first curve here may be a cubic, let the variable curve $\phi, = 0$, be a cubic of order $m, = 1, 2, 3$, &c. as the case may be. For the simplest theorem, the curve ϕ is a line.

[p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 , respectively, this is

$$|ext[determinant]| = 0,$$

or, in the notation already made use for such a determinant, $123 = 0$.

7. The equation $du_1 + du_2 + du_3 = 0$, here $du = \frac{dx}{ay^2 + 2lzx}$, or as a particular interpretation, if there is a peculiar interpretation of the corresponding two as arbitrary constants, thereby the the curve $f = 0$, an arbitrary constant; consider then the cubic equation $f = ax^3 + by^3 + cz^3 + 6lxyz = 0$. The equation $du = 0$, $\phi = 0$ is here for a particular intersection of f, ϕ , taking the logarithms and

differentiating with regard to x_1, y_1 , and the corresponding values of the major function in those forms

$$\frac{dx_1}{ay_1^2 + 2lz_1x_1} + \frac{dx_2}{ay_2^2 + 2lz_2x_2} + \frac{dx_3}{ay_3^2 + 2lz_3x_3} = 0,$$

and as taking the logarithms and differentiating in regard to a , considering a as variable, $du_1 + du_2 + du_3 = 0$, viz.

$$\Sigma \frac{1}{ay^2 + 2lzx} \log \left(\frac{by^3}{az^3 + 6lxyz} \right) = 0,$$

and this equation the differential equation

$$\Sigma \frac{dx}{ay^2 + 2lzx} = 0$$

is considered to be respectively, but is two then the equation found.

[p. 114] 8. As a particular illustration, we have $\frac{dx}{ay^2 + 2lzx} = du, = 0$, and the integral relation $123 = 0$. I give two new ways of obtaining this—first by differentiation.

For the seven coefficients a, b, c, f, g, h , &c., and the integral relation $123 = 0$. I give a direct verification.

To find f, g, h , where f, g, h are the integral cubic, $f = 0$. If be any constant, $f = \frac{dx}{ay^2 + 2lzx} = du$, and the equation of ϕ as $h, 0$ of the curve, and it may be regarded as the determination for f ; with the dps and the ϕ other points $a, \dots, h$ on the curve.

This being so, considering as variable the original determination f , we have by differentiation

$$\Sigma \frac{1}{ay^2 + 2lzx} \log \frac{dx}{(ax^3 + by^3 + cz^3 + 6lxyz)} = 0,$$

or what is the same, taking the logarithms and differentiating with regard to ourselves a , the equation $\Sigma du = 0$ is satisfied.

We have shown also that the cubic curve $f = 0$, and the cube derived line $\phi = 0$ are now equivalent to the two cases.

9. The case $n = 3, \delta = 1$, the fixed curve a nodal cubic, and the variable curve $\phi = 0$, of order m .

We have then

$$\Sigma (x, y, 1)^{n-3} du = 0,$$

viz. this is

$$\Sigma du = 0,$$

which presents itself in the form

$$\Sigma du = -\frac{d\phi}{\phi}.$$

For the simplest theorem, we may take ϕ to be a conic.

[*p. 115*] $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$; besides which it is in a particular integral.

But regard one of the three points, say 3, as the fixed point, that is, the line pass through the fixed point 1 and 2. We write $du_1 = du_2$, and the differential equation contains ϕ , while the constant variable.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any five of these points determine the conic, and besides therefore be one of the displacement-relations

$$\Sigma du = 0,$$

which expresses that, the equation $123456 = 0$, which expresses that the six points are on a conic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0.$$

If, however, we take 5 a fixed point on the cubic, then we have the same equation as the general integral of $du_1 + du_2 + du_3 + du_4 = 0$.

We have here obviously $\Sigma du = 0$ as a particular integral of the cubic; and the integral relation $123456 = 0$ has the same equation as the general integral.

But taking also 5 a fixed point on the cubic, the equation $123456 = 0$, which expresses that the six points are on a conic, of any equation $123456 = 0$ for a fixed and 5 fixed point on the cubic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0,$$

of order $f = 0$, is in fact a particular integral, and so if we also fix 5, the cubic, the equation expressed integrals.

[p. 116] determinate point, independent of the particular conic through 1, 2, 3, 4 and 5 used for the construction. We have thus a particular integral of the differential equation $du_1 + du_2 + du_3 + du_4 = 0$. That is, in the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

Examples of the Pure Theorem—Fixed Curve a Cubic. Art. Nos. 13 and 2.

13. The fixed curve is taken to be the cubic curve $\phi = 0$, the variable curve of the order $f = 0$ of order n , etc.

We may write the equation

$$\Sigma \theta(\xi t) du = 0,$$

where the equation $\theta(t) = 0$ is the equation of the line joining the points 1, 2, 4, viz. we have $\Omega(1, 1, 2, 3, 4, 5, c, t)$; if so, taking a, b, c the general points, through which the line passes, we are not on the conic, and consequently to jam in the foregoing the equation

$$\Sigma \theta(\xi t) du = 0,$$

where $\theta(t)$ is in fact $123 = 0$, the equation

$$\Sigma \theta(\xi t) du = -\frac{d\theta(\xi t)}{\theta(\xi t)},$$

which presents itself in the case of the cubic curve.

[p. 117] as there is no linear relation between the displacements $\frac{du_1}{f_1}$ and $\frac{du_2}{f_2}$ in the ν th arbitrary point $\bar{2}$ and on the cubic the equation becomes an identity in regard to the two lines du_1, du_2 , respectively.

$$\frac{du_1}{f_1} = -\frac{du_2}{f_2}.$$

Observe that dy_1, dx_1 being functions of x_1, y_1 , &c., is a particular value of integral form

$$dF(x, y) = \frac{x_1}{(y_1 + py_1 + \dots) dx_1} + \dots = 0,$$

where, considering x_1, y_1 , &c. as functions of u respectively, and it is also independent of the equation

$$\frac{du_1}{f_1} + \frac{du_2}{f_2} = 0,$$

that is, we may assume

$$-x_1y_2' + x_2y_1' = \theta(x_1, x_2),$$

the constant of integration being to be determined.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations. Art. No. 13.

13. Suppose $n = 3$, $\delta = 0$ ($p = 1$); the fixed curve a cubic; and suppose successively the remaining intersections with the cubic; here it is, &c., according to the variable curve is a line, a conic, &c.

If $m = 1$, then two points are the cubic determines the line, and consequently the remaining intersection with the cubic, hence the curve C , depends.

If $m = 2$, then 5 points on the cubic determine the conic, and the remaining intersection with the cubic, hence the curve C , f^2 , will be a determinate point, etc.

If $m = 3$, then 8 points on the cubic determine the cubic, and consequently the remaining intersection with the cubic, hence the curve C , f^3 , will be a determinate point.

[p. 118] If $m = n - 3$, the variable curve is a generic functions of p , q , r , but the dependence on the parameters considered all of 1, 2, 3, &c., f which is a determinant of the variable curve.

The Variable Intersections of the Two Curves—Number of Independent Displacement Relations given by the Pure Theorem. Art. No. 20.

20. The number of displacement-relations given by the pure theorem is = number of integral relations. If $n - 2$, $r = 1, 2, 3$, then if the variable curve is a line, conic, &c., respectively to the number on the cubic by a quartic, &c., is a particular line, etc., and consequently to the equation of a cubic, etc., and to a quartic, with a single point; this can be verified, and the number of independent displacement-relations.

If $m = n - 3$, then by elimination of the parametric points and the number of intersections, etc., we have the general number of relations, etc., as in the same way for the variable curve.

In the same way for f^p depending on the parameters of ϕ^p , in such cases the number of independent integral relations is = 4.

[p. 119] 21. Suppose $n = 3$, $\delta = 1$, the fixed curve a nodal cubic, and the variable curve $\phi = 0$ of order m .

In the simplest case, $m = 1$, we may suppose ϕ to be a line, and the integral relation $123 = 0$, of the line being meeting the cubic the curve $f = 0$, is two points $123 = 0$, that is, we may consider one of these points fixed, $123 = 0$, of two points are independent.

If $m = 2$, then five points on the conic determine the conic, and therefore the remaining three intersections; the curve $f = 0$, is a particular line, of the line is six being independent.

If $m = 3$, then 8 points on the cubic determine the cubic, and therefore the remaining intersection; the equation $123 = 0$ for a particular line, of two points are independent.

In general, $n = 3$, $\delta = 1$, the dpts are $n - 4$ points fixed in the cubic, and is the case of the nodal cubic for the case of the lines linking the bordered points.

The integral expression for the displacement-relations are, then, with for a fixed curve a nodal cubic, etc.,

$$np^q + \frac{1}{2}(n-1)(n-2) = m,$$

which is therefore the equation to be proved.

The Variable Intersections of the Two Curves—Number of Independent Displacement Relations given by the Pure Theorem. Art. Nos. 20.

20. Starting from the equation $\phi = (a, \dots, \xi, y)^n = 0$ of the variable curve, we have

$$\frac{d\phi}{dx} dx + \frac{d\phi}{dy} dy + \frac{d\phi}{dz} dz = 0,$$

$$\frac{d\phi}{da} da + \dots = 0.$$

[p. 120] where ξ, η, ζ denote the coordinates respectively. Then

$$\xi : \eta : \zeta = y dz - z dy : z dx - x dz : x dy - y dx,$$

or putting $z dy - y dz \equiv \xi(x, y)$, then ξ is the displacement-relations which are obtained, i.e. ξ , i.e. $\xi(x, y)$.

[p. 121] where du is an arbitrary local function, $u = ax + by + cz$; that is, multiplied by a noble function $(x, y, z)^{n-3}$ then a particular integral.

Let f denote an arbitrary linear function, $u = ax + by + c$; let J be a particular linear function of order $n - 3$ in the variables f , and u this equation gives the formula

$$\Sigma (x, y, 1)^{n-3} du = J da + ka,$$

introducing the formula

$$\begin{aligned} du &= \frac{y dz - z dy}{f_x} = \frac{z dx - x dz}{f_y} = \frac{x dy - y dx}{f_z}, \\ \frac{du}{dx} &= \frac{1}{f_x} (y d - z dy) \\ &= \frac{1}{f_x} \left(\frac{1}{f} f_x dy - \frac{1}{f} f_y dx \right), \end{aligned}$$

or observing that u, b, c are the differential coefficients $\frac{du}{dx}, \frac{du}{dy}$, the form in $J(f, \phi)$ are the variables

$$J(f, \phi) = |ext[determinant]| = \frac{d(\phi, f)}{d(x, y)},$$

the term in $[\]$ in the Jacobian, or functional determinant

$$du = \frac{-b\phi}{J(\phi, f)}.$$

22. When the form $du = \frac{-b\phi}{J(\phi, f)}$ is established, then in any one of the three theorems

$$\Sigma (x, y, 1)^{n-3} du = 0,$$

$$\Sigma (x, y, 1)^{n-3} \frac{-b\phi}{J(\phi, f)} = 0,$$

$$\Sigma (x, y, 1)^{n-3} \frac{-b\phi}{J(\phi, f)} = -\frac{b\phi}{\phi}.$$

But further, if in the first equation we write $u = a$, and consider the linear function $J(f, \phi)$ denotes the Jacobian

$$J(f, \phi) = |ext[determinant]| = \frac{d(\phi, f)}{d(x, y)}.$$

In these equations the only differential symbol is the b differing the coefficients u , &c.; this represents the variable curve, and besides u ,

&c., the parametric and the parameter, which is independent of all the others. In this manner we have the original theorem reproduced; for the further reduction of the affected theorem.

The Coordinates (p, σ, τ). Art. Nos. 24 to 26.

[p. 122] 24. The letter p has just been used to denote the determinant 612; there is often convenient to denote points 1, 2, 3 coordinates (x, y, z) , (x_1, y_1, z_1) , (x_t, y_t, z_t) respectively; and then employing p, r to denote the determinants 623, 631, 612 respectively, and Δ the determinant 123, we have identically

$$\Delta x = x_1 p + x_t \sigma + x z', \quad \Delta y = y_1 p + y_t \sigma + y z', \quad \Delta z = z_1 p + z_t \sigma + z z',$$

which equations, regarding therein the point θ , coordinates (x, y, z) , as a current point, are in fact equations for transformation from the coordinates (x, y, z) to the coordinates p, σ, τ belonging to the triangle of reference 123. The points 1 and 2 have therein the coordinates $(\sigma, \tau) = (1, 0)$, $(\sigma, \tau) = (0, 1)$ respectively.

25. If the function f , which equated to zero gives the equation of the curve, is of the order n in (x, y, z) , then it will be of the order n in (ρ, σ, τ) ; the coefficient of $\sigma^n \rho^n$ is $= \frac{1}{\delta^2} (x_1, x_2)^{n-2} f$; that of $\rho^{n-1} \sigma$ is $= \frac{1}{\delta^2} (x_1, x_2)^{n-1} (x_1) f$; and so on.

But further, if f vanishes at the point 1 to order a , and at the point 2 to order β , then for any line $(\rho \sigma \tau)$ such that $\rho = 0$ does not satisfy $f = 0$, the lower coefficients $\sigma^n \rho^{n-a-\beta}$ vanish: that is, the coefficient of σ^a in $(\sigma \tau)^a$ is $= 0$, $(a+1)$ coefficients vanish; and similarly $\beta+1$ coefficients of those involving ρ to the highest powers $r^{n-1}, r^{n-2}, \dots$ are in the form

$$f = \frac{1}{\delta^2} (\sigma^n \rho^0 + a \rho \sigma^a + \dots).$$

[p. 123] The terms independent of ρ contain, it is clear, the factor ρ^n , and consequently those terms from the others, the equation may be written

$$f = \rho^a \sigma^\beta + (\rho, \sigma, \tau)^{n-2}.$$

26. The equation $f = 0$, $\rho^a \sigma^\beta = 0$, $\rho^{n-a-\beta} = 0$ determines the residues of the points 1, 2, and hence the major function on $f = 0$, $(\rho \sigma \tau)^{n-a-\beta}$ at the points 1, 2, and writing therein $a = 0$, $b = 0$, the corresponding value of the major function in those forms

$$\frac{1}{n} \frac{1}{x_1^n x_2^n} - \frac{1}{x_1^{n-1} x_2} - \dots - \frac{1}{x_1 x_2^{n-1}}.$$

27. [*p. 123 cont.*] we have du being the major function on the cubic, when the equation $A\Omega = 0$ for any major function u , by the equation now obtained, is

$$f = \rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta}.$$

The Proposition for the Affected Theorem. Art. No. 27.

27. In the affected theorem instead of (x, y, z) we have two coordinates (ρ, σ, τ) . We have thus to show that the differential expression $J(f, \phi)$ depending also on the parameters of ϕ , may, by the addition of the new variable, by some particular formulæ expressed by $A\Omega + B\Omega$, where A and B are constants. The dependence on the parameters of ϕ may also be eliminated, in many cases by the rest of the parametric points which are linear in the differentials of the parameters. It follows that, $\Delta J(f, \phi)$ depends only on the variation of f . We have consequently

$$\frac{1}{J(f, \phi)} = \Sigma A \frac{\partial \theta_1}{\partial x_1} J(\phi, f),$$

and the equation for the affected theorem becomes

$$\Sigma (x, y, z)^{n-3} \frac{b\phi}{J(\phi, f)} = -\Sigma \left(\frac{ba_1}{x_1} + \frac{b\theta_1}{\theta_1} \right),$$

where (σ, y) , σ , b , etc. to be regarded as standing for its ratios in terms of $J(\phi, f)$.

Jacobi's Fraction-Theorem. Art. No. 28 to 31.

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if T be any rational and integral function $(x, y)^m$ then we have

$$\frac{1}{\theta} = \Sigma \frac{1}{x_1 - x'} J(T').$$

[*p. 124*] or, introducing an arbitrary constant A by the equation $A\theta = X$, may this is

$$\frac{A}{X} \left(\frac{1}{x} \right) = \Sigma (\sigma, y)^m F'(x),$$

where U' is the same function $(x', y')^m$ of x' , y' that U is of (x, y) , and $J(\phi, f) = \frac{d(U)}{d(x, y)}$, the Jacobian of the same function (x, y) , and the summation extends to all roots x' of the equation $U' = 0$ obviously in this gives the theorem the formula for the decomposition of $\frac{1}{X}$ into simple fractions.

29. Take now $U = (x, y, z)^m$, $V = (y, z, \dots)^p$, $V = \dots$, functions of (x, y, z) of the degrees m and n , respectively; and then writing as written above the system $U = 0$, $V = 0$; viz. U is any root of $V = 0$, and any other quantity (or it may be) is the summation extends to all the same intersections UV , where the $J(U, V)$ is the Jacobian

$$\frac{1}{AU(x, y, z)} = \Sigma \frac{1}{x - x'} J(U, V).$$

and the equation for the affected theorem becomes

$$\frac{AD - BC}{XY} = \Sigma \frac{1}{x - x' \cdot y - y'} J(U, V),$$

where U' , V' are the same functions of (x', y') that U , V are of (x, y) , and $J(U, V) = \frac{d(U, V)}{d(x, y)}$, the Jacobian; the summation extends to all roots x' , y' of the equations $U' = 0$, $V' = 0$.

[p. 125] 30. For the proof, we write $X = R$, Y a entire function $(x, y)^m$ we may write $AD = -X$, $C = -Y$, and the integral relations $123 = 0$, $X = AY$; the equations imply that the curves $X = 0$, $Y = 0$ meet in m points, all of them on the cubic $f = 0$, but it has been found out to me by Mr C. S. Peirce that we have $(AD - BC)(x, y) = -CX + DY = -AX = 0$, the original relation may be omitted, viz. we have

$$(AD - BC)x = -CX, \quad (AD - BC)y = -DY,$$

and we have given the conditions corresponding to it in the form of pairs of equations, i.e. $(AD - BC)$, $(CU + DV)$, that is, $(AD - BC)U + DY = 0$, where A , B , C , D are arbitrary constants. It follows that any other formula existing, the coefficients of $\sigma^n \rho^n$ as we may write

$$AD - BC = K(x, y).$$

where the summation extends in all the conditions of the use of one for these of x , y , but with a formula relating, the coefficients K may be determined from the conditions, but as is rigid one A , B , D , the multiplication takes place over the formula, and the addition is the some thing,

$$AD - BC = K(x, y), \quad \dots,$$

where the summation extends in all K the combination of the use values of x' and y' at the same root, so it is verified.

10. [p. 126] In all that follows, the cubic might be any cubic whatever, but for greater convenience we suppose it of a fixed form, viz. a binary or cubic; and consider the equation $f = ax^3 + by^3 + cz^3 + 6lxyz = 0$. Such a cubic has the three inflexions $(x, y, z) = (0, 0, 1)$, $(0, 1, 0)$, $(1, 0, 0)$ at the vertices of the triangle of reference and the three inflexional tangents are at the sides of the same triangle; viz. the cubic has the inflexional tangent at the points 1, 2, 3 are sides of the triangle of reference, and the three inflexional tangents are at the sides; the inflexion $(0, 1, 0)$ is the the side, that is, $z = 0$, $cz^3 = 0$; etc., respectively, that is to say the cubic has the three inflexional sides of the reference $(0, 0, 1)$, $(0, 1, 0)$, $(1, 0, 0)$.

Consider, in the cubic, are the coordinates $(\rho, \sigma, \tau) = (0, 0, 1)$. We have

$$\rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta} = \rho^a \sigma^\beta \cdot \tau^{n-a-\beta}.$$

11. [p. 127] The left-hand side is

$$A(D - BC) \Sigma \frac{1}{x - x'} \Sigma \frac{1}{x' - x_2} - \Sigma \frac{1}{y - y'} \Sigma \frac{1}{y' - y_2},$$

and $A(D - BC)$ the some of the highest order in (x, y, z) $\phi = 0$, $\phi = 0$ being a homogeneous function of the lowest degrees $n + n' = m$, of the order ϕ may be written

$$\Sigma (AD - BC) = \Sigma (x, y, z)^m F'(x, y, z),$$

and on the linear formula $AX = R + S$; A, B, C, D are entire functions of X, Y in their highest degrees, and a formula relating ϕ also, in particular ϕ , the formula is associated.

There is no thing so, we may consider the formula as well known under the system of conditions of those points are $(AD - BC)$, with the assumption (x, y) , the quotient $(AD - BC)\rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta}$, of the quotient $A\rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta}$, can in two states again be written

$$b = \text{const. of } (x, y),$$

comparing the two values of y , we have

$$\text{Const. of } \frac{(AD - BC)\rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta}}{a(\sigma)^{n-1} Y'} = \Sigma \frac{(\sigma)^{n-1}}{(\sigma, y)^m F'(x)},$$

and we hence derive

$$\text{Const. of } \frac{(AD - BC)\rho^a \sigma^\beta (\rho, \sigma, \tau)^{n-a-\beta}}{a(\sigma)^{n-1} Y'} = \Sigma \frac{(\sigma)^{n-1}}{(\sigma, y)^m F'(x)}.$$

[p. 128] where $(x, y)^{m-3}$ is the general function of the degree $n+n' = m$, etc., and $J = 0$. Hence the suppositions $(AD - BC)$ may be written

$$\Sigma \frac{C^m}{f(x_1)^{n-1} V} x_1^{n-2} y_1 = ?$$

where V, V_1 are the original sense, and this being so we may say the right-hand term of the quotient $(AD - BC)\rho^a\sigma^\beta$ may be written

$$p = \text{const. of } \frac{1}{(xy)^{n-1}}.$$

comparing the two values of y , we have

$$\text{Const. of } \frac{(AD - BC)\rho^a\sigma^\beta}{a(\sigma)^{n-1} V} = \Sigma \frac{(\sigma)^{n-1}}{(\sigma, y)^m V'},$$

and we have given a similar relation

$$\text{Const. of } \frac{(AD - BC)\rho^a\sigma^\beta}{a(\sigma)^{n-1} V} = \Sigma \frac{(\sigma)^{n-1}}{(\sigma, y)^m V'},$$

Homogeneous Form of the Theorem. Art. Nos. 32.

36. Writing $\frac{x}{z}$ for x , $\frac{y}{z}$ for y , and introducing the suppositions of the arbitrary linear functions $u = ax + by + cz$, we may obtain, viz. a, b, c as variables; we have

$$\Sigma \frac{1}{J(U, V)} du = 0 \quad (n = n - 2) \text{ (theorem),}$$

$$\Sigma \frac{1}{J(U, V)} du = -\text{const. of } \frac{(AD - BC)}{J(U, V)} \quad (n = m + n' - 2) \text{ (theorem),}$$

where the suffix 0 denotes that we are in $AD - BC$ to write $z = 1$.

If in the last formula we change throughout b into $-b$, the result may then be expressed in two terms in b , but a still containing only a , by σa to be put $= a$, by which it appears in this case the formulæ the particular function $(AD - BC)\rho^a\sigma^\beta(\rho, \sigma, \tau)^{n-a-\beta}$.

[p. 129] The effect of $\frac{b\phi}{\phi}$ of the Curves $U = 0, V = 0$. Art. Nos. 37 and 38.

37. We want, in regard to the foregoing special theorems, consider the effect of any change of the values of U, V, f , etc.

Suppose now we have any of the foregoing functions, where U, V are not put now passes through it; the dpts of an intersection of U, V , and the theorems are similarly true.

If U passes through the dpts, then the corresponding formula remains, viz. taking $\frac{1}{V}$ as a constant of the formula. But it must, that, the dps in homogeneous as the values of $U, V, \dots$ by some constant, we have it correspondingly through the theorem on the cubic.

[*p. 129 cont.*] Suppose now we have any change of the values of V, f , by some pass through it; the dps in any intersection of U, V , and the theorems are similarly true.

The Syzygy. Fixed Curve a Cubic. Art. No. 35.

[*p. 130*] 38. Writing Ω_3 , as before, we now consider the curve V , has a dp, but taking on the fixed curve a new point, $V = 0$, in fact be deduced from any equation $f = 0$ of the cubic.

Suppose so, and the integral relation $123 = 0$, will be that the equation of the cubic curve, we ought to have $(123)\Omega = 0$, suppose

$$[\theta\Omega] + (\theta\Omega) + (\Omega) = 0,$$

and the form, may, by direct consideration on the left-hand side as expressed in terms of (x, y, z) , viz. if instead of these being satisfied as before, but in particular they give

$$[\theta\Omega] + (\theta\Omega) + (\Omega) = 0;$$

this, multiplied by ρ , becomes

$$\frac{1}{J(\phi, f)} \rho d\sigma + b\rho d\sigma + \sigma b d\sigma = 0.$$

And here, observing 1, V is to be a function of the form $f = 0$, $V \cdot \Omega = 0$, the equation of three points constants. Observing that the three points 1, 2, 3 are also on the curve $\frac{x'p}{x'^2}$, then the conic possesses the line connecting the points and we obtain on these as similar conditions for $\frac{x'p}{x'^2}$, with which is a feature of ρ , this is, the

$$(VOLY)\rho = -(V\Omega)\sigma - (V\Omega)\rho = 0,$$

or, on substituting for the value of $(VOLY)$, we get

$$\rho d\sigma + \sigma b d\rho + (V\Omega) = 0,$$

which is the required syzygy.

39. [p. 131] But for $\tau = 0$, we have

$$A_1\phi_1 + B_1\phi_2 = a\rho^m, \quad C_1\phi_1 + D_1\phi_2 = b\rho^m,$$

and hence

$$(AD - BC)\phi_1 = A_1\phi_1 B_1\phi_2 - B_1\phi_1 A_1\phi_2 = C\rho^m,$$

and the right-hand side these becomes; say

$$-\theta \text{ const. of } \left(\frac{AD - BC}{f_x f_y} \right) \frac{1}{\rho^a \sigma^\beta},$$

But in calculating the constant of $\frac{AD-BC}{f_x f_y}$ has but suppose not only $\tau = 0$ but also $\rho = 0$, we then have $f_x f_y = \left(\frac{\tau}{V}\right) (x, y, z)^{n-2}$; that is, $\frac{f_x f_y}{\rho^a \sigma^\beta} = \left(\frac{\tau}{V}\right) (x, y, z)^{n-2}$ and hence also $\frac{f_x f_y}{\rho^a \sigma^\beta} = \left(\frac{\tau}{V}\right)$. Similarly, in calculating the constant of $\frac{AD-BC}{\rho^a \sigma^\beta}$, we may suppose not only $\tau = 0$ but also $\rho = 0$; we then have $f_x f_y = \left(\frac{\tau}{V}\right) (x, y, z)^{n-2}$, and similarly writing in the second equation $\rho = 0$, we find $C_1 \left(\frac{\tau}{V}\right) = D\rho^m$, that is, $\frac{C_1}{\rho^m} = \left(\frac{\tau}{V}\right)$, and the expression then becomes

$$= -\Delta \left(-\frac{b\rho_1}{\rho_1} + \frac{b\rho_2}{\rho_2} \right),$$

giving the equation which was to be proved. This completes the proof of the affected theorem

$$\Sigma \frac{(x, y, z)^{n-3} du}{J(f, \phi)} = -\Sigma \frac{b\rho_1}{\rho_1} - \frac{b\rho_2}{\rho_2}.$$

Chapter III. The Major Function $(x, y, z)^{n-3}$.

Analytical Expression of the Function. Art. Nos. 42 to 48.

[p. 132] 42. The function has been defined by the condition that the curve $(x, y, z)^{n-3} = 0$ shall pass through the dps, and also through the $n - 2$ residues of the parametric points 1, 2; and moreover, that on writing therein (x_1, y_1, z_1) for (x, y, z) , the function shall become $= 1 \div p_1$. Obviously the function is not uniquely determined; calling the curve $\Omega = 0$, we may add to Ω any term $\theta J + \dots = 0$, J a constant factor. We may then take the original curve, we ought to have $\Omega = 0$. We have $\Omega = \rho + (\rho, \sigma, \tau)^{n-3}$. We may a multiple of $\theta J + \dots$.

43. We can without difficulty write down an equation determining Ω' as a function $(x, y, z)^{n-3}$, on putting therein (x_1, y_1, z_1) for (x, y, z) the function becomes $= 1 \div p_1$. In the case where $\rho = \theta + bp$, $\sigma = bp$, $\tau = bp$, the function is a quartic, in the value of the conic function Ω , in the form

$$\Omega = \rho + (\rho, \sigma, \tau)^{n-3},$$

and is in the form $\Omega = \rho + (\rho, \sigma, \tau)^{n-3}$, which may be developed in $(\rho, \sigma, \tau)^{n-3}$ and is in two cases as θ also as a curve and the residues. The equation has the form

$$\Omega = \rho\sigma\tau + (\rho, \sigma, \tau)^{n-3}.$$

But the bipartite curve also $J = 0$, the residue of the parametric points 1 and 2 are then a fixed point. Suppose, then, the function of the lowest degrees $(\sigma\tau)^{n-3}$, the bipartite to in regard to (σ, y) , are equal but obvious, the points and indices of the fixed points; the major function being equal to 0 is in fact the

$$\Omega = \frac{1}{J(\sigma\theta J)^{n-1}}.$$

We have $\Omega \neq 0$, $V = 0$, and the same equation determines for the cubic, the residues, and these for the cubic.

[p. 133] 44. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and no homogeneous as which is to be shown that we have determinations of Ω , viz. at $(\rho\sigma\tau) = (0, 0, 1)$, $\theta J = 0$, the cubic conditions $\theta J = (0, 1, 0)$ to a residue. Equation $\Omega = 0$ at $\rho = 0$, and through the points and through the remaining two intersections of the line with Ω , and through the three given points. Take (ξ, η, ζ) , (ξ_1, η_1, ζ_1) the coordinates of the two given points on the curve f (f is, say of the order $n = 4$); $\theta(x, y, z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining (x', y', z') , (x'', y'', z'') meets the curve θ where $\theta = 0$ in two points Λ , which when this factor is thrown out becomes

$$a\Lambda^2 + 2b\rho\Lambda + c\rho^2 = 0,$$

where

$$a = (\xi_1, \eta_1, \zeta_1)^2\theta, \quad b = (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta)\theta, \quad c = (\xi, \eta, \zeta)^2\theta.$$

45. [p. 134] It is shown that, the conic might be any conic whatever, but for the greater convenience we suppose at first a binary

or cubic. To this and the integral relation $123 = 0$. I give a direct verification. To find a, b , the coefficients of the conic, but $\rho = 0$ has the equation of the conic on the cubic at the point 1, the coefficient σ for the value of ρ .

This being so, $\frac{1}{J(\phi, f)} = \frac{1}{V_1 V_2} (x_1 y_2 - x_2 y_1)$ and the integral relation $123 = 0$.

To find b , we have here $\Sigma = 0$, $f = 0$, $V_1 = 0$, $V_2 = 0$, the coefficients of the cubic on which is, in fact, $f_1, f_2 = 0$; the equation of the conic at the point 1, the coefficients of the line, the line for is two of the values of the conic. If however, we take a conic on the cubic, we can equate the residue of the parametric points, and $V_1 = 0$, $V_2 = 0$, the rest of the parametric points which are linear in the differentials of the parameters, we obtain the conditions to the conic for satisfying the conic.

46. [p. 135] Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a quartic; and let the quartic be a particular quartic; viz. from this expression for Ω at first the line passes through the point $b = 0$ on the variable curve; this corresponds to the quartic having a node, etc.

If $m = 1$, the two points on the quartic are the only difference, but, Ω may be considered as a residue of the parametric points 1, 2, which is to be determined. We have here $V_1 a + V_2 b = a_1$, $V_1 b + V_2 c = a_2$.

[p. 136] where the left-hand side is a constant of the conic, and let the residue have one $J = 0$, then the determinate variable is a constant of 1.

[p. 136 cont.] 47. Suppose next $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic, and the variable curve $\phi = 0$, of order m ; here is the integral relation of the parametric points and the conic which the residue of two two satisfied.

If $m = 1$, the two points on the quartic are the only quartic. We have here $V_1 a + V_2 b = a_1$, $V_1 b + V_2 c = a_2$, which is the formula for $\rho = 0$, $\sigma = 0$ the side of the equation; if the conic is in fact a binary cubic, with all the residues are the same in the cubic.

48. [p. 136 cont.] The terms of the second line may be transformed

as follows:

$$\begin{aligned} \frac{f}{c} (Q'_1 - 3kp - 3kq + 18ks - 3ar) &= a \frac{1}{c} (2pl - 2pm + 2lk - 2pq + 2kr - 3uvw) \\ &\quad + \frac{1}{c} (av^2 - 12pl + 12kn - 18ar + (av^2 + 12kr - \\ &\quad + \frac{1}{c} (av^2 - 2(2lq + 12hr - 6cv) + (a^3 + 2pq + 3 \end{aligned}$$

and

$$\begin{aligned} \frac{1}{c} (-3pa - 3pq + 6yqs) &= 3 \frac{1}{c} (-3pa - 3pq + 6yqs) \\ &\quad + \frac{1}{c} (av^2 - 2(2lq + 8j + 6yqs) + (-3pa - 3pq + 6yqs)) \\ &\quad + \frac{1}{c} (av^2 - 2(2lq + 12hr - 6cv) + (a^3 + 2pq + 3uvw - 3jk) \end{aligned}$$

substituting these values, the whole third line is destroyed, and we find

$$b_1[\Omega 12] - b_2[\Omega] - \square = \text{etc.}$$

[p. 137] Ultimately the last two lines of this expression are found to

be

$$= \frac{1}{c} (-2ka + 4jm + 2l^2 + 3pp) + a^3 - 2(2pq + 3pa + 7lm - 9aq) - a,$$

so that the whole is now a sum of three linear functions of (ρ, σ, τ) ,

...

[p. 138] 72. Collecting the terms, we have

$$\begin{aligned} b_1[\Omega 12] - b_2[\Omega] &= \frac{1}{2} \left\{ x_1 (6bm + 2fl - 2jm + 2l^2 + 6kp - 21nq - 7ip + 2jno) \right. \\ &\quad + \frac{1}{c} (4ph + 4ej - 27gn + 4gp - 27hn + 2lm) \\ &\quad + \alpha \left(\frac{1}{c} (-2pq + 6mn - 2qg + 3lp - 6kq + 2ho) \right. \\ &\quad + \frac{1}{c} (-2pq + 6mn - 2qg + 3lp - 6kq + 2ho)) \\ &\quad + \left(\frac{1}{2} (qr + 2lm + lp + uq + 5mn + 7mp + 3lp + 7ho - 8mq) - a \right. \\ &\quad \left. \left. + \left(\frac{1}{c} (3kp\theta - 3y - 2bq) + 3kp - 6km + 3p2v) \right) \right\} \\ &\quad - a. \end{aligned}$$

73. [*p. 138 cont.*] The right-hand side depends on the points 0, 1, 2 and 12; viz. we have therein $p = (123)$, $\delta = 123$, &c., but the left-hand side, depending on the points 0, 1 and 2, the right-hand part really contains, δ and 0 must therefore really cancel; that is, the system of equations $\phi(x, y, z) = 0$ regarded as standing for in equation $(x, y, z) = 0$, where $\delta = 0$ and the linear point $\phi = 0$, the coefficients $(x, y, z) = 0$, which presents itself in the case of the cubic curve.

Cambridge, England, October 5, 1882.

Chapter IV. The Major Function $(x, y, z)^{n-3}$, continued.

The Conversion, Fixed Curve a Quartic, continued. Art. Nos. 74 to 82.

[*p. 149*] 74. I resume the question considered *ante*, Nos. 66 to 73. The general problem, where the fixed curve is any plane curve whatever, has recently been solved in a very complete and elegant form by Dr Nöther, in the two notes "Zur Reduction algebraischer Differentialausdrücke auf die Normalformen," and "Über die algebraischen Differentialausdrücke," E. Netto" *Mitteilungen aus dem Sem. G. Erlangen*,, ib. 1884. I consider here the original investigation, viz. for the case of $n = 4$, $\delta = 0$ ($p = 3$); the fixed curve a general quartic; and let the quartic, J , &c., is therefore that of *ante*.

We have $f(x, y, z) = (a, \dots, x)^2$, only if ϕ be an arbitrary curve through the dps, and $f = 0$, $h, b = 0$. We then take $\Omega \cdot \phi - V$ for the curve, then by the foregoing process the integration is in form $(x, y, z) = 0$. The integration is suggested by Cl. & G., $\Omega \cdot \phi - V$ for the quartic; viz. but, Ω has not generally the form ϕ . We thus obtain in a very neat compact form, Ω which need not surely after my return to England that I was able to effect the like determination of Ω from $V = 0$; the curve of order is now to the curve of order $n - 2$, of the order $n - 4$, the point $\rho = 0$, the parameters; this well-known theory by Cl. & G., now to be on the quartic the question of the same form, the parameter on the quartic, of the parameter point $(\rho, \sigma, \tau) = (0, 0, 1)$.

[*p. 150*] The case of $n = 5$, M is $n = 125, 124, 125, 541$. Hence writing $\Omega(0; 1, 2; 3, 4, 5)$ for Ω , we have $\Omega = \Omega(0; 1, 2; 3, 4, 5)$ and the integral

relation $123 = 0$. Hence further taking

$$Q_n = \Omega(0; 1, 2; 3, 4, 5), \quad = \frac{125}{125 \cdot 124 \cdot 125 \cdot 541},$$

so that the differential of Ω_n is in the form

$$Q = \frac{012d34}{125 \cdot 124 \cdot 125 \cdot 541}, = \frac{012d35}{012},$$

viz. Q is a rational fraction where the numerator contains the line $(1, 1)$ which is to be shown that we have identically the determinants 012, 013, 012 respectively.

26. The equation $f = 0$, $V = 0$, $V_1 = 0$, $V_2 = 0$, the same form as in the previous form $J(\rho, V)$, ..., , by similarly determined as the determinants 012, 013, 012 respectively, etc.

27. In the affected theorem instead of (x, y, z) we have two coordinates (ρ, σ, τ) . We have thus to show that the differential expression $J(f, \phi)$ depending also on the parameters of ϕ , may, by the addition of the new variable, by some particular formulæ expressed by:

$$\begin{aligned} M(x, y, z) &= (x, y, z), \quad \rho(x, y, z) = ax^{m-1} + \dots, \\ \sigma(x, y, z) &= a'x^{m-1} + \dots, \quad \tau(x, y, z) = a''x^{m-1} + \dots, \end{aligned}$$

where $0 = \dots$ to give the formulæ, viz. the nine points are the quartic.

We have $f(x_1, x_2, \dots, x_p; t) = 0$ being a homogeneous equation in the case of the cubic curve.

77. Taking an arbitrary point of the cubic (x, y, z) , the equation

$$H(x, y, z) = 0$$

of ϕ , ϕ , will satisfy of ϕ the cubic relation of point of cubic curve.

A Memoir on the Abelian and Theta Functions

Art. But for $\tau = 0$, we have

$$(AD - BC)\varphi_0 = A\beta_{rmn} - C\alpha_{pmn},$$

and hence the right-hand side thus becomes, say

$$\beta\alpha_{rmn} - C\alpha_{pmn}.$$

But in calculating the constant of $\sqrt[8]{\phi}$, we may suppose not only $\tau = 0$, but also $\sigma = 0$: we then have

$$\phi = (x_1, y_1, z_1, \xi, \eta, \zeta)^m,$$

and hence also $D = 0$.

Similarly, in calculating the constant of $\sqrt[3]{\gamma}\delta_0$, we may suppose not only $\tau = 0$, but also $\rho = 0$: we then have

$$\phi = (\xi, \eta, \zeta)^m(x_2, y_2, z_2)^m,$$

and hence $B = 0$.

Moreover, in the equations

$$\frac{A}{\alpha}\phi_1 = \alpha_{pmn}, \quad \frac{C}{\gamma}\phi_2 = \gamma_{rmn},$$

writing in the first equation $\sigma = 0$, we find

$$A\left(\frac{\rho}{\alpha}\right)^m \phi_1 = \alpha\phi_{pmn},$$

that is,

$$\frac{\alpha^n}{\alpha^m}\phi_1 = \alpha\phi_{pmn},$$

and similarly writing in the second equation $\rho = 0$, we find

$$C\left(\frac{\sigma}{\gamma}\right)^m \phi_2 = \gamma\phi_{rmn},$$

that is,

$$\frac{\gamma^n}{\gamma^m} = \left(-\frac{\sigma}{\gamma}\right)^m :$$

and the expression thus becomes

$$\beta\alpha_{rmn} - \alpha\beta_{rmn} = \phi_2,$$

giving the equation which was to be proved.

This completes the proof of the affected theorem.

0.3 The Major Function $(x, y, z)^{n-2}$

Analytical Expression of the Function. Art. Nos. 0.3 to 0.3.

Art. The function has been defined by the conditions that the curve $(x, y, z)^{n-2} = 0$ shall pass through the dps, and also through the $n-2$ residues of the parametric points 1, 2: and moreover, that on writing therein (x_1, y_1, z_1) for (x, y, z) , the function shall become $= n \cdot 1^{n-2} 12$. Obviously the function is not completely determined: calling it H (or when required $\bar{H}$), then if $\bar{H}$ be any particular form of it, the general form is $H = \bar{H} + (x, y, z)^{n-3} \cdot 012$, where $(x, y, z)^{n-3}$ is the general minor function, viz. $(x, y, z)^{n-3} = 0$ is a curve passing through the dps: the major function thus contains $\frac{1}{2}(n-1)(n-2) - \delta = p$ arbitrary constants.

Agreeing with the definition, we have the before-mentioned equation

$$H = -\frac{1}{n}(n \cdot 1^{n-2} 12, \dots \uparrow \phi \rho \sigma)^{n-2} + \text{terms involving } \tau,$$

viz. from this expression for H it appears that the curve $H = 0$ meets the line through 1, 2 in the $n-2$ residues of these points, and moreover, for $(x, y, z) = (x_1, y_1, z_1)$ and therefore $(\rho, \sigma, \tau) = (\Lambda, 0, 0)$, the value of Q is $= n \cdot 1^{n-2} 12$.

Art. We can without difficulty write down an equation determining H as a function $(x, y, z)^{n-2}$, which, on putting therein $\tau = 0$, becomes equal to the foregoing expression $\frac{1}{n}(n \cdot 1^{n-2} 12, \dots \uparrow \phi \rho \sigma)^{n-2}$, and which is moreover such that the curve $\bar{H} = 0$ passes through the dps; which being so, we have as before,

$$H = \bar{H} + (x, y, z)^{n-3} 012,$$

for the general value of H .

To fix the ideas, consider the particular case $n = 4$, the fixed curve a quartic: H , on putting therein $\tau = 0$, should become

$$= \frac{1}{4}(4 \cdot 1^2 2^2, 6 \cdot 1^2 2^2, 4 \cdot 12^3 \uparrow \phi \rho \sigma)^2,$$

and it is to be shown that this will be the case if we determine H , a

quadric function of (x, y, z) , by the equation

$$\begin{vmatrix} x^2, & y^2, & z^2, & 2yz, & 2zx, & 2xy, & H \\ x_1^2, & y_1^2, & z_1^2, & 2y_1z_1, & 2z_1x_1, & 2x_1y_1, & 6 \cdot 1^2 2^2 \\ x_2^2, & y_2^2, & z_2^2, & 2y_2z_2, & 2z_2x_2, & 2x_2y_2, & 4 \cdot 12^3 \\ a, & b, & c, & f, & g, & h, & 0 \\ a, & b, & c, & f, & g, & h, & 0 \\ a, & b, & c, & f, & g, & h, & 0 \end{vmatrix} = 0,$$

where the left-hand side is a determinant of seven lines and columns, the top line being $x^2, y^2, z^2, 2yz, 2zx, 2xy, H$, and similarly for the second line; the third line is

$$x_1^2, x_2^2, 2x_1x_2, 2y_1y_2, 2z_1z_2, 2(y_1z_2+y_2z_1), 2(z_1x_2+z_2x_1), 2(x_1y_2+x_2y_1), 6 \cdot 1^2 2^2,$$

and in each of the last three lines we have six arbitrary constants followed by a 0.

The equation is of the form $D + MH = 0$, where D is a quadric function $(x, y, z)^2$, and M is a constant factor.

Art. If the quartic curve has a dp, suppose at the point α , coordinates $(x_\alpha, y_\alpha, z_\alpha)$, then in order that the curve $\bar{H} = 0$ may pass through the dp, we must for one of the last three lines substitute $(x_\alpha, y_\alpha, z_\alpha)^2, 0$; and so for any other dp or dps of the quartic curve.

And the conditions as to the dp or dps (if any) being satisfied in this manner, we may if we please, taking $(x_\beta, y_\beta, z_\beta)$ as the coordinates of an arbitrary point β (not of necessity on the fixed curve), write any line not already so expressed, of the last three lines, in the form $(x_\beta, y_\beta, z_\beta)^2, 0$; the effect being to make the curve $\bar{H} = 0$ pass through the arbitrary point β .

Art. To show that the equation on putting therein $\tau = 0$ does in fact give the required value,

$$\bar{H} = \frac{1}{4}(4 \cdot 1^3 2, 6 \cdot 1^2 2^2, 4 \cdot 12^3 \uparrow \phi \rho \sigma)^2,$$

suppose, it is to be observed that, effecting a linear substitution upon the first six columns, the equation may be written

$$\left| \begin{array}{ccccccc} (\rho, \sigma, \tau)^2, & H \\ 4 \cdot 1^3 2, & 6 \cdot 1^2 2^2, & 4 \cdot 12^3 \\ a, & b, & c, & f, & g, & h \end{array} \right| = 0,$$

where $(\rho_1, \sigma_1, \tau_1)$, $(\rho_2, \sigma_2, \tau_2)$ are what (ρ, σ, τ) become on writing therein for (x, y, z) the values (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively; viz. we have

$$(\rho_1, \sigma_1, \tau_1) = (\Lambda, 0, 0); \quad (\rho_2, \sigma_2, \tau_2) = (0, \Lambda, 0);$$

the equation thus is

$$\left| \begin{array}{ccccccc} \rho^2, & \sigma^2, & \tau^2, & 2\sigma\tau, & 2\tau\rho, & 2\rho\sigma, & H \\ \Lambda^2, & 0, & 0, & 0, & 0, & 2\Lambda^2, & 6 \cdot 1^2 2^2 \\ 0, & \Lambda^2, & 0, & 0, & 0, & 0, & 4 \cdot 12^3 \\ a, & b, & c, & f, & g, & h, & 0 \end{array} \right| = 0,$$

and then by another linear substitution upon the columns, the last column can be changed into $H - \Phi, 0, 0, 0, 0, 0$; whence writing $\tau = 0$, the equation becomes

$$\left| \begin{array}{ccccccc} \rho^2, & \sigma^2, & 0, & 0, & 0, & 2\rho\sigma, & H - \Phi \\ \Lambda^2, & 0, & 0, & 0, & 0, & 2\Lambda^2, & 0 \\ 0, & \Lambda^2, & 0, & 0, & 0, & 0, & 0 \\ a, & b, & c, & f, & g, & h, & 0 \end{array} \right| = 0.$$

Or, omitting a constant factor, it is

$$\begin{vmatrix} \rho^2, & \sigma^2, & 2\rho\sigma, & H - \Phi \\ \Lambda^2, & 0, & 2\Lambda^2, & 0 \\ 0, & \Lambda^2, & 0, & 0 \end{vmatrix} = 0,$$

that is, $H - \Phi = 0$, or $H = \Phi = \frac{1}{4}(4 \cdot 1^3 2, 6 \cdot 1^2 2^2, 4 \cdot 12^3 \uparrow \phi \rho \sigma)^2$, the required value.

Art. Considering the equation for H as expressed in the before-mentioned form $D + MH = 0$, the value of the constant factor M is

$$(x_2, y_2, z_2)^2 a, b, c, f, g, h,$$

or if, instead of each line such as a, b, c, f, g, h , we have a line $(x_\alpha, y_\alpha, z_\alpha)^2$, then we have

$$(x_\alpha, y_\alpha, z_\alpha)^2 (x_\beta, y_\beta, z_\beta)^2 (x_\gamma, y_\gamma, z_\gamma)^2,$$

a value which is

$$\frac{12\alpha \cdot 12\beta \cdot 12\gamma \dots}{y_1 z_2 - y_2 z_1}$$

or say this is $= 12\alpha \cdot 12\beta \dots 12\gamma \dots \alpha\beta\gamma \dots$

Art. It is obvious that the foregoing process is applicable to the general case of the fixed curve of the order n with δ dps, and gives always $\bar{H}$, by an equation of the foregoing form $D + MH = 0$, where D is a function $(x, y, z)^{n-2}$ of the coordinates, and M is a constant factor.

Supposing that in the determinant for $\bar{H}$, each of the lower lines is written in the form $(x_\alpha, y_\alpha, z_\alpha)^{n-2}, 0$, the number of the points α is $= \frac{1}{2}(n-1)(n-2)$, viz. these are the δ dps, and $\frac{1}{2}(n-1)(n-2) - \delta = p$ other points α .

The general expression of M is $M = 12\alpha \cdot 12\beta \dots (\alpha^{n-3}\beta^{n-3} \dots)$, viz.—equating to zero a factor such as 12α , this expresses that the point α is on the line 12; but equating to zero the last factor $(\alpha^{n-3}\beta^{n-3} \dots)$, this expresses that the several points α , viz. the dps and the p other points α , are on a curve of the order $n-3$.

Art. Preceding the case $n = 4$, above considered, we have, of course, the case $n = 3$, $\delta = 0$, the fixed curve a cubic; the equation for $\bar{H}$ is here

$$\begin{vmatrix} x, & y, & z, & \bar{H} \\ x_1, & y_1, & z_1, & 3 \cdot 1^2 2 \\ x_2, & y_2, & z_2, & 3 \cdot 12^2 \\ x_\alpha, & y_\alpha, & z_\alpha, & 0 \end{vmatrix} = 0,$$

giving

$$\bar{H} = \frac{1^2 2 \cdot 02\alpha + 12^2 \cdot 01\alpha}{12\alpha},$$

or if we write herein 3 for α , this is

$$\bar{H} = \frac{1^2 2 \cdot 023 + 12^2 \cdot 031}{123},$$

and we have hence the general form

$$H = \frac{1^2 2 \cdot 023 + 12^2 \cdot 031}{123} + K \cdot 012,$$

where K is an arbitrary constant.

Art. There is, however, a more simple particular solution $\bar{H} =$ polar function 012 ($f = ax^3 + \dots + z^3$, then $012 = ax_1x_2 + by_1y_2 + cz_1z_2$), which, to avoid a confusion of notation, we may write $\bar{H} = \overline{012}$. We at once verify this; for, expressing the coordinates (x, y, z) in terms of (ρ, σ, τ) , we have $\Phi = \overline{012}$, $= \frac{1}{\Lambda}(1^2 2 \cdot \rho + 12^2 \cdot \sigma + 123 \cdot \tau)$, which, for $\tau = 0$ becomes $= \frac{1^2 2 \cdot \rho + 12^2 \cdot \sigma}{\Lambda}$.

We must, of course, have an identity of the form

$$\frac{1}{\Lambda}(1^2 2 \cdot \rho + 12^2 \cdot \sigma + 123 \cdot \tau) = \frac{1^2 2 \cdot 023 + 12^2 \cdot 031}{123} + K \cdot 012,$$

and to find K , writing here $0 = 3$, we have $K = -\frac{123}{\Lambda}$, or we have the identity

$$\frac{1}{\Lambda}(1^2 2 \cdot \rho + 12^2 \cdot \sigma + 123 \cdot \tau) - \frac{123}{\Lambda} \overline{012} = \frac{1^2 2 \cdot 023 + 12^2 \cdot 031}{123}.$$

Single Letter Notation for the Polar Functions of the Cubic. Art. Nos. 0.3 and 0.3.

Art. The notation of single letters for the polar functions is not much required in the case of the cubic, but, in the next following case of

the quartic it can hardly be dispensed with, and I therefore establish it in the case of the cubic: viz. I write

$$23^2, 31^2, 12^2 = f, g, h; \quad 23^2, 31^2, 12^2 = i, j, k;$$

or, what is the same thing, the expression for the cubic function f in terms of ρ, σ, τ is

$$\Lambda^3 f = 3h\rho^2\sigma + 3j\rho^2\tau + 3k\rho\sigma^2 + 6l\rho\sigma\tau + 3g\rho\tau^2 + 3f\sigma^2\tau + 3i\sigma\tau^2;$$

an equation which, writing s instead of f , may also be written

$$\Lambda^3 s = (3h, 3j, 3k, 6l, 3g, 3f, 3i \quad \rho^3 \rho^2\sigma, \rho^2\tau, \rho\sigma^2, \rho\sigma\tau, \rho\tau^2, \sigma^2\tau, \sigma\tau^2);$$

and I join to it the series of equations

$$\Lambda^2 \cdot 21 = (0, 2h, 2j, k, 2l, g \quad \rho^2 \rho\sigma, \rho\tau, \sigma^2, \sigma\tau, \tau^2),$$

$$\Lambda^2 \cdot 22 = (h, 2k, 2l, 0, 2f, i \quad \rho^2 \rho\sigma, \rho\tau, \sigma^2, \sigma\tau, \tau^2),$$

$$\Lambda^2 \cdot 03 = (j, 2l, 2g, i, 2f, 0 \quad \rho^2 \rho\sigma, \rho\tau, \sigma^2, \sigma\tau, \tau^2),$$

$$\Lambda \cdot 01^2 = (0, h, j \quad \rho \sigma, \tau),$$

$$\Lambda \cdot 012 = (h, k, l \quad \rho \sigma, \tau),$$

$$\Lambda \cdot 013 = (j, l, g \quad \rho \sigma, \tau),$$

$$\Lambda \cdot 02^2 = (k, 0, f \quad \rho \sigma, \tau),$$

$$\Lambda \cdot 023 = (l, f, i \quad \rho \sigma, \tau),$$

$$\Lambda \cdot 03^2 = (g, i, 0 \quad \rho \sigma, \tau).$$

Art. In particular, we have $\Lambda \cdot 012 = h\rho + k\sigma + l\tau$, and the above-mentioned identity

$$\frac{1}{\Lambda}(1^22 \cdot \rho + 12^2 \cdot \sigma + 123 \cdot \tau) - \frac{123}{\Lambda} \overline{012} = \frac{1^22 \cdot 023 + 12^2 \cdot 031}{123}$$

is simply $h\rho + k\sigma + l\tau - l\tau = h\rho + k\sigma$.

Single Letter Notation for the Polar Functions of the Quartic. Art. No. 0.3.

Art. I write here

$$\begin{aligned} 23^3, 31^3, 12^3 &= f, g, h; \\ 23^231, 31^212, 12^223 &= i, j, k; \\ 23^212, 31^223, 12^231 &= i', j', k'; \\ 231^2, 312^2, 123^2 &= l, m, n; \\ 23 \cdot 31 \cdot 12 &= p; \end{aligned}$$

or, what is the same thing, the expression for the quartic function f in terms of (ρ, σ, τ) is

$$\begin{aligned} \Lambda^4 f &= 4f\sigma^3\rho + 4g\tau^3\rho + 4h\rho^3\sigma + 6i\sigma^2\rho\tau + 6j\tau^2\rho\sigma + 6k\rho^2\sigma\tau \\ &\quad + 6i'\sigma^2\tau\rho + 6j'\tau^2\sigma\rho + 6k'\rho^2\tau\sigma + 4l\sigma\tau\rho^2 + 4m\tau\rho\sigma^2 + 4n\rho\sigma\tau^2; \end{aligned}$$

an equation which, writing s instead of f , may also be written

$$\begin{aligned} \Lambda^4 s &= (4f, 4g, 4h, 6i, 6j, 6k, 6i', 6j', 6k', 4l, 4m, 4n, \rho^4, \rho^3\sigma, \rho^3\tau, \rho^2\sigma^2, \rho^2\sigma\tau, \rho^2\tau^2, \\ &\quad \rho\sigma^3, \rho\sigma^2\tau, \rho\sigma\tau^2, \sigma^4, \sigma^3\tau, \sigma^2\tau^2, \sigma\tau^3, \tau^4). \end{aligned}$$

Art. [Continued from Art. 0.3]

And I join to it the series of equations

$$\begin{aligned}
 \Lambda^3 \cdot 31 &= (0; 3h, 3j; 3r, 6l, 3g; \dots \rho^3 \rho^2 \sigma, \rho^2 \tau, \rho \tau^2, \rho \sigma \tau, \rho \tau^2, \sigma^3, \sigma^2 \tau, \sigma \tau^2, \tau^3), \\
 \Lambda^3 \cdot 32 &= (h; 3r, 3k; 3i', 6m, 3n; 0, 3f, 3q, \dots), \\
 \Lambda^3 \cdot 33 &= (j; 3i', 3g; 3l, 6n, 3p; 3f, 3q, 3s, 0), \\
 \Lambda^2 \cdot 21 &= (0; 2h, 2j; r, 2l, q \rho^2 \rho \sigma, \rho \tau, \sigma^2, \sigma \tau, \tau^2), \\
 \Lambda^2 \cdot 312 &= (h; 2r, 2l; k, 2m, n), \\
 \Lambda^2 \cdot 213 &= (j; 2l, 2q; l, 2n, p), \\
 \Lambda^2 \cdot 222 &= (r; 2k, 2m; 0, 2f, p), \\
 \Lambda^2 \cdot 223 &= (l; 2m, 2n; f, 2q, i'), \\
 \Lambda^2 \cdot 232 &= (q; 2n, 2g; p, 2s, 0), \\
 \Lambda \cdot 01^2 &= (0, h, j \rho \sigma, \tau), \\
 \Lambda \cdot 01 \ 22 &= (h, r, l), \\
 \Lambda \cdot 01 \ 23 &= (j, l, g), \\
 \Lambda \cdot 012^2 &= (r, k, m), \\
 \Lambda \cdot 0123 &= (l, m, n), \\
 \Lambda \cdot 013^2 &= (q, n, p), \\
 \Lambda \cdot 02^2 &= (k, 0, f), \\
 \Lambda \cdot 02 \ 23 &= (m, f, i'), \\
 \Lambda \cdot 023^2 &= (n, p, s), \\
 \Lambda \cdot 03^2 &= (g, i', 0),
 \end{aligned}$$

which will be convenient in the sequel.

Major Function. The Fixed Curve a Cubic. Art. No. 0.3.

Art. It has been already seen that a simple particular form is $\Lambda \bar{H} = \overline{012}$; and that the general form is $H = \bar{H} + K \cdot 012$.

Major Function. The Fixed Curve a Quartic. Art. No. 0.3.

Art. It is to be shown that a particular form is

$$\bar{H} = \frac{-01^2 \cdot 0}{\Lambda^2}.$$

In fact, by the foregoing values of $\Lambda \cdot 01^2$, &c., the numerator of this expression, multiplied by Λ^2 , is

$$= -(h\sigma + j\tau)(k\rho + l\sigma + m\tau) + (h\rho + r\sigma + l\tau)(r\rho + k\sigma + m\tau) \\ + q\rho(h\rho^2 + 2r\rho\sigma + 2l\rho\tau + k\sigma^2 + 2m\sigma\tau + n\tau^2),$$

which is

$$= 2hr\rho^2 + 3r^2\rho\sigma + (hm - jk + 3lr)\rho\sigma + 2rl\rho\tau^2 + (-fh + kl + 3mr)\sigma\tau + (-fj + lm + nr)\tau^2$$

and this, for $\tau = 0$, becomes

$$= r(2h\rho^2 + 3ry\rho\sigma).$$

C. XII. 18

Hence for $\tau = 0$, we have

$$\bar{H}r\rho\sigma = \frac{1}{4}\{4 \cdot 1^3 2 \cdot \rho^2 + 6 \cdot 1^2 2^2 \cdot \rho\sigma + 4 \cdot 12^3 \cdot \sigma^2\};$$

and $\bar{H}$ is thus a form of the major function $(x, y, z)^2$. Of course the general form is $H = \bar{H} + (x, y, z)^1 \cdot 012$.

Syzygy of the Major Function. Art. No. 0.3.

Art. Writing now $(x, y, z)^{2n-2} = \bar{H}_{12}$; and taking on the fixed curve a new point 3, consider the like functions $\bar{H}_{23}$ and $\bar{H}_{31}$: it is to be shown that we have identically

$$\bar{H}_{23} \cdot \bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{31} \cdot \bar{H}_{12} \cdot \bar{H}_{23} + \bar{H}_{12} \cdot \bar{H}_{23} \cdot \bar{H}_{31} - (123)^2 f = \bar{H}_{23} \cdot \bar{H}_{31} \cdot \bar{H}_{12} \cdot (x, y, z)^{n-3},$$

where $(x, y, z)^{n-3}$ is a properly determined minor function; or, considering herein f as a point on the fixed curve and writing therefore $f = 0$, the equation is

$$\bar{H}_{23} \cdot \bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{31} \cdot \bar{H}_{12} \cdot \bar{H}_{23} + \bar{H}_{12} \cdot \bar{H}_{23} \cdot \bar{H}_{31} - (123)^2 f = 0.$$

Art. Write for a moment $X = \bar{H}_{23} \cdot \bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{31} \cdot \bar{H}_{12} \cdot \bar{H}_{23} + \bar{H}_{12} \cdot \bar{H}_{23} \cdot \bar{H}_{31}$; then k being an arbitrary coefficient, we have $X - kf = 0$, a curve of the order n , passing through the points 1, 2, 3, and also through the residues of 2, 3, the residues of 3, 1, and the residues of 1, 2; in fact, at the point 1 we have $\bar{H}_{12} = 0$, $\bar{H}_{31} = 0$, and therefore $X = 0$; also $f = 0$; and therefore 1 is a point on the curve. Again, at any residue of 2, 3, we have $\bar{H}_{23} = 0$, $\bar{H}_{23} = 0$, and therefore $X = 0$; also $f = 0$; and hence the residue of 2, 3 is a point on the curve.

It is next to be shown that k can be so determined that the curve $X - kf = 0$ shall have a dp at each of the points 1, 2, 3. Supposing this to be so, we have the line 23 meeting the curve $X - kf = 0$ in the points 2 and 3, each counting twice, and in the $n - 2$ residues of 2, 3, that is, in $n + 2$ points; hence the curve $X - kf = 0$ must contain as part of itself the line 23, and similarly it must contain as part of itself each of the other lines 31 and 12, viz. we shall then have $X - kf = \bar{H}_{23} \cdot \bar{H}_{31} \cdot \bar{H}_{12} \cdot (x, y, z)^{n-3}$: and from this equation observing that the curves $\bar{H}_{23} = 0$, $\bar{H}_{31} = 0$, $\bar{H}_{12} = 0$ each pass through the dps, it follows that the curve $(x, y, z)^{n-3} = 0$ also passes through the dps: hence, k being found to be $= (123)^2$, the theorem will be proved.

Art. Taking an arbitrary point α coordinates $(x_\alpha, y_\alpha, z_\alpha)$, and writing $d = \frac{d}{dx} dx + \frac{d}{dy} dy + \frac{d}{dz} dz$, we have to find k , so that the curve $D(X - kf) = 0$ shall pass through the point 1. Observing that $D\bar{H}_{23} = \bar{H}_{23}$, &c., we have

$$D(X - kf) = D\bar{H}_{23} \cdot \bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{23}(D\bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{31} \cdot D\bar{H}_{12}) \\ + D\bar{H}_{23}(\bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{31} \cdot \bar{H}_{12} + D\bar{H}_{31} \cdot \bar{H}_{12} + D\bar{H}_{12} \cdot \bar{H}_{31}) - kDf,$$

and, to make the curve pass through 1, writing herein $= 1$, we have

$$= 123(\bar{H}_{31} \cdot \bar{H}_{12} + \bar{H}_{23} \cdot \bar{H}_{31}) - k(Df)_1,$$

where the superfix (1) denotes that we are in $\bar{H}_{31}$, $\bar{H}_{12}$ and Df respectively to write $= 1$.

We have $\bar{H}_{31} = n \cdot 1^{n-1} 3$, $\bar{H}_{12} = n \cdot 1^{n-1} 2$, $(Df)_1 = n \cdot 1^{n-1} a$, and the equation thus is

$$n \cdot 123(1^{n-1} 3 \cdot \bar{H}_{12} + 1^{n-1} 2 \cdot \bar{H}_{31}) - kn \cdot 1^{n-1} a = 0.$$

But we have identically

$$1^{n-1} 1 \cdot 23 + 1^{n-1} 2 \cdot 31 + 1^{n-1} 3 \cdot 12 = 1^{n-1} a \cdot 123,$$

where $1^{n-1} 1 = 1^n$ is, in fact, $= 0$; the factor $1^{n-1} a$ thus divides out, and the equation becomes

$$k = (123)^2;$$

viz. k having this value, the curve $X - kf = 0$ will have a dp at 1; and clearly by symmetry, it will also have a dp at 2, and at 3; the theorem is thus proved.

The Syzygy, Fixed Curve a Cubic. Art. No. 0.3.

Art. The syzygy may be verified independently in the case where the fixed curve is a cubic. Observe that the syzygy, if satisfied for any particular form of H , will be generally satisfied; we may therefore take $\bar{H}_{12} = \overline{012}$. Writing then $\bar{H}_{23}$, $\bar{H}_{31}$, $\bar{H}_{12}$ and taking f to be a point on the cubic curve, we ought to have $\{023\} + \{031\} + \{012\} =$ a constant; the value of this constant comes out to be $= \{123\}$, and the syzygy in its complete form thus is

$$\{023\} + \{031\} + \{012\} = \{123\}.$$

We have

$$\Lambda \cdot 023, \Lambda \cdot 031, \Lambda \cdot 012 = l\rho + f\sigma + ir, \quad jp + l\sigma + gr, \quad h\rho + k\sigma + lr,$$

and the equation thus is

$$\begin{vmatrix} l & f & i \\ j & l & g \\ h & k & l \end{vmatrix} \cdot \frac{1}{\rho\sigma\tau} = 0;$$

this, multiplied by $\rho\sigma\tau$, becomes

$$h\rho^2\sigma + j\rho^2\tau + k\rho\sigma^2 + 2l\rho\sigma\tau + g\rho\tau^2 + f\sigma^2\tau + i\sigma\tau^2 = 0,$$

which is, in fact, $f = 0$, the equation of the cubic curve.

Observe that the new symbol $\{012\}$ is, in virtue of its determinant denominator, an alternate function,

$$\{012\} = -\{102\}, \quad \{012\} = \{120\} = \{201\}.$$

The syzygy is a relation between any four points 1, 2, 3, 0 of the curve, and it may be also expressed in the form

$$\{123\} - \{230\} + \{301\} - \{012\} = 0.$$

The Syzygy, Fixed Curve a Quartic. Art. No. 0.3.

Art. Taking $\bar{H}_{12}$ as before, we have

$$\bar{H}_{12} = \frac{-01^2 \cdot 0}{\Lambda^2} = \frac{-01\,23 \cdot 023 + 01\,22 \cdot 012^2 + 012 \cdot 1\,222}{\Lambda^2 \cdot 012},$$

and then, taking f to be a point on the quartic curve, we ought to have

$$\{0^223\} + \{0^231\} + \{0^212\} = (x, y, z)^1, \text{ a linear function of } (x, y, z),$$

or, what is the same thing, considering the left-hand side as expressed in terms of ρ, σ, τ , the sum should be $= (\rho, \sigma, \tau)^1$, a linear function of (ρ, σ, τ) .

By a preceding formula we have

$$\{0^212\} = \Lambda^{-2}(-\{2hr\rho^2 + 3rl\rho\sigma + (hm - jk + 3lr)\rho\tau + (-fh + kl + 3mr)\sigma\tau + (-fj + lm + rn)\tau^2\})$$

which is

$$= \frac{\rho}{\Lambda^2}(\dots) + \frac{\sigma}{\Lambda^2}(\dots) + \frac{\tau}{\Lambda^2}(\dots).$$

And hence, forming the sum $\{0^223\} + \{0^231\} + \{0^212\}$, we have first a fractional part which is found to be integral, viz. this is

$$\frac{1}{\Lambda^2}\{2fl\rho\sigma^2 + 3\rho\sigma\tau\} = \frac{1}{\Lambda^2}\{-6lp - 6m\sigma - 6n\tau\}.$$

or since $f = 0$, this is $= \frac{1}{\Lambda^2}(-6lp - 6m\sigma - 6n\tau)$.

We then have integral terms which are at once deduced from the above integral terms of $\bar{H}_{12}$; collecting the several terms, we find

$$\begin{aligned} \{0^223\} + \{0^231\} + \{0^212\} &= \frac{mn - gk}{\Lambda^2}\rho + \frac{fn - ik}{\Lambda^2}\sigma + \frac{ln - hi}{\Lambda^2}\tau \\ &+ \frac{hm - jk}{\Lambda^2}\rho + \frac{lm - fq}{\Lambda^2}\sigma + \frac{il - im}{\Lambda^2}\tau \\ &+ \frac{jn - ah}{\Lambda^2}\rho + \frac{kl - fh}{\Lambda^2}\sigma + \frac{lm - fj}{\Lambda^2}\tau \\ &\quad + rn + \dots + \dots, \end{aligned}$$

which is the required result.

**Preparation for the Conversion. The Symbol ϑ .
Art. Nos. 0.3 to 0.3.**

Art. I use ϑ as the symbol of a quasi-differentiation, viz. U being any function of (x, y, z) , dU denotes $\frac{dU}{dx}$ multiplied by the differential $dx + \frac{dU}{dy} dy + \frac{dU}{dz} dz$; in such a differential, the increments dx, dy, dz do not in general present themselves in the combinations $y dz - z dy, z dx - x dz, x dy - y dx$; but they will do so if U is a function of the degree zero in the coordinates (x, y, z) (that is, if U be the quotient of two homogeneous functions of the same degree); and this being so, we write

$$\vartheta_{1,05}U = \frac{dy_1 dz_5 - dz_1 dy_5}{dx} \frac{dU}{dx} + \dots = \frac{1}{\Lambda} \frac{dU}{d\omega_1},$$

similarly

$$\vartheta_{2,05}U = \frac{dy_2 dz_5 - dz_2 dy_5}{dx} \frac{dU}{dx} + \dots = \frac{1}{\Lambda} \frac{dU}{d\omega_2},$$

and so on.

Art. The conversion formula may be presented under a more simple form by the introduction of a new symbol. Taking f a homogeneous function of the degree n in the coordinates (x, y, z) , the coefficients in the several terms being regarded as constant parameters, we effect on these parameters the arbitrary infinitesimal variations which correspond to the change of the fixed curve $f = 0$ into a consecutive curve of the same order; and we thereby change the major function $\bar{H} = \bar{H}(0; 1, 2; 3, 4, 5)$ into $\bar{H} + \delta\bar{H}$: in particular if the variation be that which corresponds to the change of f into $f + \epsilon(x, y, z)^{n-3} 012$, where ϵ is an infinitesimal, then we have $\delta\bar{H} = \epsilon 012 \cdot (x, y, z)^{n-3}$. But it is to be remarked that in the variation in question, the points 3, 4, 5 remain unaltered; the parametric points 1, 2 remain on the curve, that is, the variations of their coordinates satisfy the equation $df = 0$; but subject only to this condition, these variations are perfectly arbitrary.

Art. We have, for the fixed curve a quartic, $n = 4$,

$$\{0^2 13\} = \frac{-01^2 \cdot 033 + 01 23 \cdot 013^2 - 012 \cdot 1 232}{\Lambda^2 \cdot 012 \cdot 123},$$

whence

$$\vartheta_3\{0^2 13\} = \frac{-01^2 \cdot \vartheta_3 033 + \dots}{\Lambda^3 \cdot 012 \cdot 123}.$$

Art. We thence obtain

$$\vartheta_3\{0^213\} = \frac{1}{\Lambda^3} \frac{d}{d\omega_3} \{0^213\},$$

and similarly for the other symbols. The conversion formula for the cubic case is

$$\vartheta_3\{136\} - \vartheta_3\{036\} = \vartheta_1\{031\} - \vartheta_0\{130\},$$

where the left-hand side is $= \vartheta_3\{013\}$, and the equation itself $(\vartheta_0 + \vartheta_1 + \vartheta_3)\{031\} = 0$.

The Conversion, Fixed Curve a Quartic, continued. Art. Nos. 0.3 to 0.3.

Art. For the quartic, the analogous formula is

$$\vartheta_3\{0^2136\} - \vartheta_3\{1^2036\} = \vartheta_0\{0^11, 3\} - \vartheta_1\{1^10, 3\},$$

where the left-hand side is a linear function of (x, y, z) , or say it is an integral function O^1 ; and the second is a linear function of (x_1, y_1, z_1) , or say it is an integral function I^1 : the given function on the right-hand side must therefore admit of expression in the form $\Phi(O^1, 1, 3) - \Phi(I^1, 0, 3)$, where $\Phi(O^1, 1, 3)$ is a known function, integral and linear as regards the coordinates (x, y, z) of the point 0, but depending also on the points 1, 3; and $\Phi(I^1, 0, 3)$ is the like known function, integral and linear as regards the coordinates (x_1, y_1, z_1) of the point 1, but depending also on the points 0, 3.

Moreover, since 2 and 5 are arbitrary points entering only on the left-hand side, it is clear that $\vartheta_3\{0^212\}$ must be independent of 2, and $\vartheta_3\{1^205\}$ independent of 5: reverting to the cubic case, observe that here $\vartheta_3\{0^212\} = \vartheta_3\{136\}$, whence $\vartheta_3\{0^212\} = \vartheta_3\{136\}$, and so $\vartheta_3\{0^212\} = \vartheta_3\{036\}$, and that the corresponding equation thus is

$$\vartheta_3\{136\} - \vartheta_3\{036\} = \vartheta_1\{031\} - \vartheta_0\{130\},$$

where the left-hand side is $= \vartheta_3\{013\}$, and the equation itself $(\vartheta_0 + \vartheta_1 + \vartheta_3)\{031\} = 0$.

We then have

$$\vartheta_3\{0^212\} = \Phi(O^1, 1, 3) - \Phi(I^1, 0, 3),$$

where the one side is derived from the other by the interchange of the 0, 1.

The solution therefore is a function which is symmetrical in regard to the points 0 and 1, and, inasmuch as the left-hand is an integral function O^1 , must itself be an integral function (O^1, I^1) , that is, integral and linear as regards the coordinates (x, y, z) and (x_1, y_1, z_1) of the points 0 and 1 respectively. We thus have

$$\vartheta_3\{0, 1\} = \Phi(O^1, 1, 3) + X(0, 1, 3),$$

and thence

$$\vartheta_2\{0, 2\} = -\Phi(O^1, 2, 3) - X(0, 2, 3),$$

viz. the second of these expressions is, with its sign reversed, the same function of 2 that the first is of 1.

Art. It follows that, taking a new symbol P for the variable of the definite integral (in the cubic case $\vartheta_0\{12\}$ was independent of 0, and there was nothing to prevent the use of P for the current point of the definite integral), we may write

$$\vartheta_0\{1, 2\} = \int_2^1 d_1 P(7, 0, 3), \text{ where } \vartheta_3(1, 0, 3) = \Phi(O^1, 1, 3) + X(O^1, 1, 3),$$

an equation which implies

$$\vartheta_2 P(0, 2, 3) = \Phi(O^1, 2, 3) + X(O^1, 2, 3).$$

But the first of these equations in P is nothing else than the first of the equations in $\vartheta_0\{0, 12\}$.

Art. I have succeeded in finding $\Phi(O^1, 1, 3)$, but the calculation is a very tedious one, and I give only the principal steps, omitting all details. We have to bring $\vartheta_1\{0^213\} - \vartheta_1\{1^203\}$ into the form $\Phi(O^1, 1, 3) - \Phi(I^1, 0, 3)$.

Art. From the value of

$$\{0^2 13\} = \frac{-01^2 \cdot 033 + 01 \, 23 \cdot 013^2 - 012 \cdot 1 \, 232}{\Lambda^2 \cdot 012 \cdot 123},$$

we find, by a process such as that of No. 0.3,

$$\vartheta_3\{0^2 13\} = \frac{1}{\Lambda^5} \{-\overline{232} \cdot 01^2 - 33 \cdot \vartheta_1(1^2 \cdot 01^2[03^2 \cdot 01^2 - (013^2)^2]) \\ + q(\dots) + r(\dots)\},$$

Substituting herein the values $01^2 = (h\sigma + j\tau)$, &c., we have = multiplied by a cubic function $(\rho, \sigma, \tau)^3$; writing down first the integral terms, and then the others, we have

$$\frac{1}{\Lambda^5} \{\rho(\dots) + \sigma(\dots) + \tau(\dots) \\ + \rho\sigma\tau \cdot (2hil - 2hn^2 - 3ijr + jlp + 2jmn) + \dots\}$$

(say this linear function of ρ, σ, τ is = D)

$$+ \rho\sigma\tau \cdot (\dots).$$

Art. The expression of $\vartheta\{P \, 03\}$ is deduced from this by the interchange of 0, 1: and I write

$$\vartheta_3\{0^2 13\} - \vartheta_0\{1^2 03\} = D - * \\ + \frac{1}{\Lambda^5} \{\Lambda^3 \cdot 2(0 \, 232)^2 - \Lambda^2 \tau(-3 \cdot 32 \cdot 232 + 6 \cdot 33 \cdot 223) - \Lambda^2 \sigma \cdot 3 \cdot 33 \cdot 232 \\ + \Lambda \sigma \tau(-2 \cdot 033 \cdot 32 + 6 \cdot 33 \cdot 0232) + \Lambda \tau^2 \cdot 2 \cdot 03^2 \cdot 33 - \sigma \tau^2 \cdot 2(\dots)\},$$

where, and in what follows, the $*$ denotes the function immediately to the left of it, interchanging therein the 0, 1.

Art. It will be observed that the D , qua linear function of (ρ, σ, τ) , that is, of (x, y, z) , is a term of the required function $\Phi(O^1, 1, 3)$: the remaining portion has to be reduced by means of the expressions for $\Lambda^2(0\,232)$, &c., in terms of ρ, σ, τ .

Art. We obtain

$$\begin{aligned} & \frac{1}{\Lambda^5} \{ \sigma^2(2fj - 3hp - 3kq + 18lm - 9nr) + \tau(-3gr - 3jp + 9mq) \} \\ & - \frac{1}{\Lambda^5} \{ \rho^3(-12im + 9kp) + \rho^2\sigma(12fn - 9ik + 9mp) + \rho\sigma^2(-2fp - 6in) + \dots \}. \end{aligned}$$

Art. The terms of the second line may be transformed as follows:

$$\begin{aligned} & \frac{1}{\Lambda^5} (2fj - 3hp - 3kq + 18lm - 9nr) \\ & = \frac{1}{\Lambda^5} \{ -3(2fj - 3hp - 3kq + 18lm - 9nr) - * \\ & + \sigma^2(-12fn + 12kn - 18m^2 + 9pr) + \sigma\tau(-3fq + 3gk + 6ir - 6lp - 9mn) \} \\ & \quad + \text{terms in } \rho^4, \rho^3\sigma, \rho^3\tau, \rho^2\sigma^2, \dots \end{aligned}$$

and substituting these values, the whole third line is destroyed, and we find

$$\begin{aligned} & \vartheta_1\{0^213\} - \vartheta_0\{1^203\} = D - * \\ & + \frac{1}{\Lambda^5} \{ \sigma(2fj - 3hp - 3kq + 18lm - 9nr) + \tau(-3gr - 3jp + 9mq) \} \\ & - \frac{1}{\Lambda^5} \{ \rho^3(-12im + 6kp) + \rho^2\sigma(-9fn + 7ik - 9mp) + \rho\sigma^2(-2gr - 6q^2 + \dots) \}. \end{aligned}$$

Art. Collecting the terms, we have

$$\begin{aligned} \vartheta_1\{0^213\} - \vartheta_0\{1^203\} &= \frac{\rho}{\Lambda^5}(-8hn + 9jm + 2l^2 - 2qr) \\ &\quad + \frac{\sigma}{\Lambda^5}(ghl - 3gjr - 2hij + j^2p + 2jln) \\ &+ \frac{\rho}{\Lambda^5} \left\{ \frac{1}{q}(-hp - kq + 2lm + nr) + \frac{1}{q^2}(2hil - 2hn^2 - 3ijr + jlp + 2jmn) \right\} \\ &\quad + \frac{\tau}{\Lambda^5}(gr + 2lq + jp + mq + 4ln) + \dots \end{aligned}$$

Art. The right-hand side depends on the points 0, 1, 3 and 2: viz. we have therein $\rho = 023$, $\lambda = 123$, &c., but the left-hand side depending on only the points 0, 1 and 3, the right-hand side cannot really contain 2, and it must thus remain unaltered, if for 2 we substitute any other point on the quartic, say 6: the right-hand side may therefore be understood as a function of 0, 1, 3 and 6, viz. ρ , λ , ξ , &c., will mean 063, 163, 6^33 , &c.: we have thus $\Phi(O^1, 1, 3)$ = the above linear function with 2 thus replaced by 6; say a given function of the points 0, 1, 3 and the arbitrary point 6, on the quartic curve; we therefore write it $\Phi(O^1, 1, 3, 6)$.

There is no obvious value for $X(0, 1, 3)$ which will produce any simplification: I therefore take this function to be = 0; and the final result is

$$\vartheta_0\{1, 2\} = \{0^212\} + \vartheta_0\{12\} - (\{0^223\} + \{0^231\} + \{0^212\}),$$

where $\vartheta_0\{12\}$ is a function integral and linear as regards the coordinates (x, y, z) of the point 0, but transcendental as regards the parametric points 1, 2; and containing besides the arbitrary points 3, 6, of the quartic curve, its value being determined by the differential formulae

$$\vartheta_0 = \Phi(O^1, 1, 3, 6), \quad \vartheta_0\{0, 12\} = -\Phi(O^1, 2, 3, 6),$$

where $\Phi(O^1, 1, 3, 6)$ is a given function as above.

I do not see the meaning of the very complicated linear function of (ρ, σ, τ) , nor how to reduce it to any form such as the simple one $\vartheta_3\{036\}$, which presents itself in the case of the cubic curve.

Cambridge, England, October 5, 1882.

The Major Function $(x, y, z)^{n-2}$ Continued

The Conversion, Fixed Curve a Quartic, continued. Art. Nos. 0.3 to ??.

Art. I resume the question considered ante Nos. 0.3 to 0.3. The general problem, where the fixed curve is any given curve whatever, has recently been solved in a very complete and elegant form by Dr Nother, in the two notes "Zur Reduction algebraischer Differentialausdrücke auf die Normalformen" and "Ueber die algebraischen Differentialausdrücke, 2^e Note," Sitzungsber. der phys.-med. Soc. zu Erlangen, 10 Dec. 1883 and 14 Jan. 1884.

I consider here the case of the quartic curve, $n = 4$, and connect his result with my former investigations.

We have the differential

$$\frac{(x, y, z)^{2n-2} d\omega}{\bar{H}_{12}} = \frac{012 d\omega}{\bar{H}_{12}},$$

where $\bar{H}_{12}$, or as I also write it $\bar{H}(0; 1, 2; 3, 4, 5)$, is a rational and integral function of the degree $(n - 2) = 2$ in the current coordinates (x, y, z) : it depends also on the parametric points 1, 2, which are points on the quartic, coordinates (x_1, y_1, z_1) , (x_2, y_2, z_2) respectively; and on $(p =) 3$ other points 3, 4, 5 on the quartic, coordinates (x_3, y_3, z_3) , (x_4, y_4, z_4) , (x_5, y_5, z_5) respectively.

The curve $\bar{H} = 0$ is a conic, which is taken to pass through the dps (none in the present case) and through the $(n - 2) = 2$ residues of the parametric points; and the function $\bar{H}$ is such that on writing therein (x_1, y_1, z_1) for (x, y, z) it becomes $= (n \cdot 1^{n-1} 2) = 4 \cdot 1^3 2$: viz. we have $\bar{H}(1; 1, 2; 3, 4, 5) = 4 \cdot 1^3 2$, which implies also $\bar{H}(2; 1, 2; 3, 4, 5) = 4 \cdot 12^3$: so defined, the function would contain $(p =) 3$ arbitrary constants, but these are determined so that the curve $\bar{H} = 0$ passes through the 3 points 3, 4, 5 on the quartic: and the function $\bar{H}, = \bar{H}(0; 1, 2; 3, 4, 5)$ is thus a completely determinate function, rational and integral of the degree 2 in the coordinates (x, y, z) of the current point, and rational in the coordinates of the other five points respectively.

I call to mind that 012 denotes the determinant formed with the coordinates (x, y, z) , &c., of the points 0, 1, 2 respectively: the like notation is used throughout.

Art. The function $\bar{H}(0; 1, 2; 3, 4, 5)$ is symmetrical as regards the parametric points 1, 2, and also as regards the arbitrary points 3, 4,

5: it is obtained by means of a determinant as shown ante, No. 0.3.

Art. Considering the new parametric points α , β , and the like functions $\bar{H}_{\alpha\beta}$, we connect these with the original function $\bar{H}_{12}$ by a syzygy equation. Writing $\rho = 023$, $\sigma = 031$, $\tau = 012$, and $\rho' = 0\beta\gamma$, $\sigma' = 0\gamma\alpha$, $\tau' = 0\alpha\beta$, we have identically

$$\bar{H}_{\alpha\beta} \cdot \rho' + \bar{H}_{\beta\gamma} \cdot \sigma' + \bar{H}_{\gamma\alpha} \cdot \tau' = 0,$$

a formula which corresponds to that for the cubic curve (No. 0.3), but is here a homogeneous quadric function instead of a linear function.

Art. For the proof of the conversion-formula, we consider the difference $\bar{H}(1) - \bar{H}_1(0)$, where $\bar{H}(1)$ denotes the value of $\bar{H}$ when the point 0 is replaced by 1, and $\bar{H}_1(0)$ the value when the parametric point 1 is replaced by 0.

Art. We have the expressions:

$$\begin{aligned}\bar{H}(1) &= \vartheta_1\{012\} + \vartheta_2\{123\} - \vartheta_3\{326\}, \\ \bar{H}_1(0) &= \vartheta_0\{102\} + \vartheta_2\{023\} - \vartheta_3\{326\}.\end{aligned}$$

Hence the difference $\bar{H}(1) - \bar{H}_1(0)$ is

$$\begin{aligned}&= \vartheta_1\{012\} - \vartheta_0\{102\} + \vartheta_2\{123\} - \vartheta_2\{023\} \\ &= -(\vartheta_0 + \vartheta_1)\{012\} + \vartheta_2(\{123\} - \{230\}) \\ &= -\vartheta_3\{012\} + \vartheta_2(\{123\} - \{230\} - \{012\}) \\ &= -\vartheta_3\{012\} + \vartheta_2(\{301\}),\end{aligned}$$

in virtue of $\{123\} - \{230\} + \{301\} - \{012\} = 0$; the whole is consequently $= 0$.

We have thus

$$\bar{H}(1) - \bar{H}_1(0) = 0,$$

the conversion-equation in the case of the quartic.

Art. The verification of the formula depends on the properties of the symbol ϑ and on the syzygy relation. The complete investigation establishes that for any two parametric points 1, 2 and any arbitrary point 0, we have the conversion-equation satisfied identically.

Chapter V. Miscellaneous Investigations

The Differential Symbol $d\omega$. Art. Nos. 0.3 and 0.3.

Art. The definition is

$$d\omega = \frac{y \, dz - z \, dy}{\frac{\partial f}{\partial x}} = \frac{z \, dx - x \, dz}{\frac{\partial f}{\partial y}} = \frac{x \, dy - y \, dx}{\frac{\partial f}{\partial z}},$$

and it hence follows that we have

$$d\omega = \frac{\begin{vmatrix} dx, & dy, & dz \\ x, & y, & z \\ \lambda, & \mu, & \nu \end{vmatrix}}{\lambda \frac{\partial f}{\partial x} + \mu \frac{\partial f}{\partial y} + \nu \frac{\partial f}{\partial z}},$$

where (λ, μ, ν) are arbitrary constants or, if we please, arbitrary functions of (x, y, z) : viz. the expression just written down is altogether independent of the values of λ, μ, ν , and is consequently equal to the value obtained by writing any two of these symbols $= 0$, that is, the expression is equal to any one of the foregoing three equal values of $d\omega$.

The expression was first given by Aronhold (1863), in the memoir presently referred to.

It is to be remarked that, considering (λ, μ, ν) as the coordinates of a point, the denominator $\lambda \frac{\partial f}{\partial x} + \mu \frac{\partial f}{\partial y} + \nu \frac{\partial f}{\partial z}$ equated to 0 is the polar $(n-1)$ -ic of the point (λ, μ, ν) in regard to the fixed curve.

If instead of λ, μ, ν we write $bc' - b'c, ca' - c'a, ab' - a'b$, where $(a, b, c), (a', b', c')$ are constants, then the numerator is

$$= (ax+by+cz)(a'dx+b'dy+c'dz) - (a'x+b'y+c'z)(a\,dx+b\,dy+c\,dz),$$

or introducing ρ, σ to denote the arbitrary linear functions $ax+by+cz$ and $a'x+b'y+c'z$ respectively, the numerator is $= \rho d\sigma - \sigma d\rho$: moreover, observing that a, b, c and a', b', c' are the differential coefficients of ρ, σ in regard to the coordinates (x, y, z) , the denominator is $= J(f, \rho, \sigma)$; and the value of $d\omega$ is

$$d\omega = \frac{\rho d\sigma - \sigma d\rho}{J(f, \rho, \sigma)},$$

where, in accordance with a previous remark, the denominator equated to 0 is the polar $(n-1)$ -ic of the intersection of the lines $\rho = 0, \sigma = 0$ in regard to the fixed curve.

Obviously, by taking for ρ, σ any two of the three coordinates x, y, z , we reproduce the original three forms of $d\omega$.

Art. The last-mentioned form of $d\omega$ suggests the expression for this symbol in the case where the fixed curve, instead of being a plane curve, is a curve of double curvature defined by two equations $f = 0, g = 0$ between the four coordinates (x, y, z, w) : viz. ρ, σ being now arbitrary linear functions $ax+by+cz+dw$, and $a'x+b'y+c'z+d'w$ of the four coordinates, the expression is

$$d\omega = \frac{\rho d\sigma - \sigma d\rho}{J(f, g, \rho, \sigma)},$$

and by taking for ρ, σ any two of the four coordinates x, y, z, w , we have for $d\omega$ six values which must of course be equal to each other; it is easy to verify *a posteriori* that this is so.

In the case where the curve of double curvature is not the complete intersection of two surfaces, the denominator (regarded as the Jacobian of the curve and of the arbitrary planes ρ, σ) will have a definite meaning, but what this is I do not at present consider.

The last-mentioned expression for $d\omega$ will be applied further on to the case of the quadri-quadric curve.

Integral Formulae. Art. Nos. 0.3 to 0.3.

Art. In what precedes, $d\omega$ has been used as a single symbol to denote any one of the equal differential expressions

$$\frac{y dz - z dy}{\frac{\partial f}{\partial x}}, \quad \frac{z dx - x dz}{\frac{\partial f}{\partial y}}, \quad \frac{x dy - y dx}{\frac{\partial f}{\partial z}} :$$

there is no quantity ω .

The expressions are of the order $(n-3)$ in the coordinates (x, y, z) , and since (x, y, z) are as to their absolute magnitudes altogether arbitrary (only their ratios being determinate), a symbol such as

$$\omega = \int d\omega = \int \frac{y dz - z dy}{\frac{\partial f}{\partial x}},$$

would, except in the case $n = 3$, be altogether meaningless. In fact, the integral would be $\int \frac{y}{z^{n-3}} \frac{dz}{\frac{\partial f}{\partial x}}$, where $\frac{y}{z}$ is, by the equation of the fixed curve, given as a function of $\frac{x}{z}$; but the other factor z^{n-3} is an absolutely indeterminate variable value, and the expression is meaningless.

But we have integrals $\int Q d\omega$, where Q is a homogeneous function of the order $n - 3$ in the coordinates (x, y, z) ; and, in particular, we have such integrals where (corresponding to the forms which present themselves in the differential pure and affected theorems respectively) Q is either a rational and integral function $(x, y, z)^{n-3}$, or a rational and integral function $(x, y, z)^{n-3}$ divided by a linear function $(x, y, z)^1$: for in every such case, the form of integral is $\int \phi\left(\frac{x}{z}\right) d\left(\frac{x}{z}\right)$, where $\frac{y}{z}$ is a given function of $\frac{x}{z}$, and the factor of $d\left(\frac{x}{z}\right)$ is thus a mere function of $\frac{x}{z}$.

More definitely, in the integrals $\int Q d\omega$ which are considered, Q is either a minor function $(x, y, z)^{n-3}$, or it is the quotient of a major function $(x, y, z)^{2n-2}$ by the linear function 012.

In the case $n = 2$, there is no rational and integral function $(x, y, z)^{n-3}$, but the function may be of the form belonging to the affected theorem, viz. it is unity divided by a linear function $(x, y, z)^1$; or say the integral is $\int \frac{d\omega}{(x, y, z)^1}$, where the (x, y, z) are connected by a quadric equation $(a, \dots x, y, z)^2 = 0$: it will be shown presently that this integral is obtainable as a logarithmic function.

In the case $n = 3$, we have the rational and integral function $(x, y, z)^{n-3}$, = a constant, or say = 1, so that there is here an integral $\int d\omega$: we do not call this ω , but introducing a new letter, say u , and fixing at pleasure the inferior limit of the integral, we write

$$u = \int d\omega.$$

Art. In the foregoing form $Q d\omega$, so long as we retain the symbol $d\omega$, there is nothing to show what is the variable in regard to which the integration is to be performed; we may, for instance, writing $d\omega = \frac{dz}{\frac{\partial f}{\partial y}}$, make it to be $\frac{x}{z}$, or in like manner to be any other of the six quotients. We thus cannot attribute a value to the inferior or superior limit of such an integral, for any such limit would be a value of $\frac{x}{z}$, or $\frac{y}{x}$, &c., and would thus be a mere accident of the special form under which the integral was obtained, without any geometrical signification.

It is otherwise if we write $Q d\omega$ in the form $\frac{Q dx}{\frac{\partial f}{\partial y}}$ (where dx, dy, dz satisfy the condition $\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = 0$), for then the integral $\int \frac{Q dx}{\frac{\partial f}{\partial y}}$ is a definite integral taken between definite limits x_1, x_2 (where observe that the radical which enters into the value of $\frac{\partial f}{\partial y}$ has a determinate sign according to the branch of the curve along which the integration is performed).

Art. But the most simple form is obtained by substituting for the Cartesian coordinates (x, y) , or for the trilinear coordinates (x, y, z) , a parametric coordinate ξ ; viz. taking ξ a properly determined function of the coordinates, we have x, y, z each of them rational functions of ξ , and we then have

$$Q d\omega = \frac{\Phi(\xi) d\xi}{\sqrt{\Psi(\xi)}},$$

where Φ, Ψ are rational and integral functions of ξ . In particular, if the fixed curve is a cubic, then $\Psi(\xi)$ is a quartic function, and the integral $\int Q d\omega$ is an elliptic integral: if the fixed curve is a quartic, then $\Psi(\xi)$ is a sextic function, and the integral $\int Q d\omega$ is a hyperelliptic integral of the first kind.

The Fixed Curve a Conic. Art. Nos. 0.3 to 0.3.

Art. The minor function $(x, y, z)^{n-3}$ does not exist (for $n = 2$, we have $n - 3 = -1$). The major function $(x, y, z)^{2n-2}$ is $(x, y, z)^2$, and the quotient by 012 is the affected function $\frac{(x, y, z)^2}{012}$.

The theory of the conic is of course included in the general theory; but it is interesting to see how the formulae apply to this case. The differential is

$$\frac{(x, y, z)^2}{012} d\omega,$$

where $(x, y, z)^2$ is any quadric function whatever; but the form being as just mentioned, we have an identity

$$\frac{(x, y, z)^2}{012} = A \cdot 023 + B \cdot 031 + C \cdot 012,$$

where A, B, C are properly determined constants; and the integral thus depends on the three integrals $\int 023 d\omega, \int 031 d\omega, \int 012 d\omega$, any two of which are connected by a linear relation.

Art. But these integrals are at once obtainable. The equation of the conic is $f = (a, b, c, f, g, h \ x, y, z)^2 = 0$; and thence $d\omega = \frac{y \, dz - z \, dy}{ax + hy + gz}$, where for the denominator we may substitute either of the equal values $hx + by + fz$, $gx + fy + cz$.

Taking the coordinates to be $(x, y, 1)$, and writing the equation of the conic in the form $y^2 = 1 - x^2$, that is, considering the circle $x^2 + y^2 = 1$, we have

$$d\omega = \frac{y \, dx - x \, dy}{y^2 + x^2} = \frac{y \, dx - x \, dy}{1} = y \, dx - x \, dy.$$

But taking θ the excentric angle, we have $x = \cos \theta$, $y = \sin \theta$; whence $d\omega = -d\theta$; or if we introduce the amplitude u , where $u = \sin^{-1} x$, we have $\theta = u$, and $d\omega = -du$.

Art. For any conic, introducing the parameter ξ , we may express the coordinates rationally in terms of ξ ; and the integral $\int \frac{(x, y, z)^2}{012} d\omega$ thus becomes a rational function of ξ , and is therefore integrable by logarithms. In particular, for the conic $y^2 = 1 - x^2$, if we write $t = \tan \frac{1}{2}\theta$, we have $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$; whence $d\omega = -\frac{2dt}{1+t^2}$, and the integral is logarithmic.

Art. The general theory of the integral $\int \frac{(x, y, z)^2}{012} d\omega$, where the fixed curve is a conic, is thus completely disposed of. It is to be observed that the integrals of the third kind, which in the general theory correspond to the parametric points 1, 2, here degenerate into integrals of the first kind, viz. the logarithmic integrals just considered.

Art. The equation just written down relates to any four points 1, 2, 3, 4 of the conic; and if for 3, 4 we write 0, 3 respectively, it becomes

$$\frac{2}{012} = \frac{1}{031} + \frac{1}{023},$$

which relates to the points 0, 1, 2, 3 of the conic: writing, as before, $023, 031, 012 = \rho, \sigma, \tau$, this equation is

$$\frac{2 \, d\omega}{\tau} = \frac{d\sigma}{\sigma \rho} - \frac{d\rho}{\rho \sigma},$$

which may be verified as follows: the equation of the conic is $f = 23 \cdot \sigma \tau + 31 \cdot \tau \rho + 12 \cdot \rho \sigma = 0$; we have $d\omega = \tau \frac{\rho \, d\sigma - \sigma \, d\rho}{\frac{\partial f}{\partial \tau}}$, where $\frac{\partial f}{\partial \tau} = 23 \cdot \sigma + 31 \cdot \rho = \frac{012}{\tau}$, that is, $d\omega = \frac{\tau^2}{012} (\frac{d\sigma}{\sigma} - \frac{d\rho}{\rho})$, whence the equation in question.

Art. We have, as a property of any four points 0, 1, 2, 3 of a conic,

$$\frac{23}{023 \cdot 123} = \frac{01}{012 \cdot 031}, \text{ or say } \frac{\tau}{\rho\sigma} = \frac{01}{\rho\tau \cdot \sigma\tau},$$

that is, $\tau^2 = \frac{01 \cdot \rho\sigma}{\lambda}$, where $\lambda = 123$; hence considering 0 as a variable point, and differentiating the logarithms,

$$\frac{2 d\omega}{\tau} = d \log \tau = \frac{1}{2} d \log \frac{\rho\sigma}{\lambda},$$

and the foregoing equation $\frac{2 d\omega}{\tau} = \dots$ thus becomes

$$\frac{2 d\omega}{\tau} = d \log \frac{01}{\tau^2}, \text{ or restoring for } \tau \text{ its value } 012,$$

$$\frac{2 d\omega}{012} = d \log \frac{01}{(012)^2}.$$

Taking now $\alpha x + \beta y + \gamma z = 0$ for the equation of the line 012; this meets the conic in the points 1, 2, coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively: and we have

$$12^2 = \{(a, \dots x_1, y_1, z_1 x_2, y_2, z_2)^2 - (a, \dots x_1, y_1, z_1)^2 \cdot (a, \dots x_2, y_2, z_2)^2\},$$

(the second term being of course = 0), viz. this is = $-(bc - f^2, \dots \alpha, \beta, \gamma)^2$, or say $12 = \Omega$, if $\Omega^2 = (bc - f^2, \dots \alpha, \beta, \gamma)^2$ as before: and the equation thus is

$$\frac{2 d\omega}{012} = \frac{d\Omega}{\Omega}.$$

Art. (The foregoing property, which may also be written

$$\frac{23}{023 \cdot 123} = \frac{01}{012 \cdot 031},$$

is verified very simply in the case of four points 0, 1, 2, 3 of a circle: in fact $23 = x_2 x_3 + y_2 y_3 - 1 = \cos 23 - 1 = -2 \sin^2 \frac{1}{2} 23$, $023 = 2 \sin \frac{1}{2} 23 \sin \frac{1}{2} 30 \sin \frac{1}{2} 02$; and so for the other like expressions; each side of the equation is thus reduced to $\frac{1}{4 \sin \frac{1}{2} 02 \sin \frac{1}{2} 03 \sin \frac{1}{2} 12 \sin \frac{1}{2} 13}$.)

Art. In particular, if the conic is taken to be the circle $x^2 + y^2 - 1 = 0$, then for the coordinates (ξ, η) of the intersections with the line

$\alpha x + \beta y + \gamma = 0$, we have $\beta\xi - \alpha\eta + 1 = 0$, giving $\Omega^2 = \alpha^2 + \beta^2 - \gamma^2$; and then

$$\begin{aligned}\xi : \eta : 1 &= -\beta\gamma \pm \alpha\Omega : \alpha\gamma \pm \beta\Omega : \alpha^2 + \beta^2 \\ &= \gamma \pm \beta\Omega : \gamma \mp \alpha\Omega : -\alpha^2 - \beta^2.\end{aligned}$$

The formula then becomes

$$\int \frac{dx}{\sqrt{1-x^2}} = \frac{1}{2\Omega} \log \frac{\alpha x + \beta\sqrt{1-x^2} + \gamma}{\alpha x + \beta\sqrt{1-x^2} - \gamma},$$

or, retaining Ω for the value $\sqrt{\alpha^2 + \beta^2 - \gamma^2}$ and $\sqrt{1-x^2}$ for y , this may also be written

$$= \frac{1}{2\Omega} \log \frac{\alpha x + \beta y + \gamma}{\alpha x + \beta y - \gamma}.$$

The form of the integral is still such that the value is not very readily obtainable by ordinary methods: the value just written down can of course be verified, but the verification is scarcely easier than for the original more general form.

In the very particular case $\alpha = 0, \beta = 0, \gamma = 1$, we have $\Omega = i: \xi : \eta : 1 = 1 : i : 0$, and the formula becomes

$$\int \frac{dx}{\sqrt{1-x^2}} = \frac{1}{i} \log(x - i\sqrt{1-x^2}),$$

viz. this is $\sin^{-1} x = \frac{1}{2}\pi + \frac{1}{i} \log(x - i\sqrt{1-x^2})$, which is right; for putting $\sin^{-1} x = u$, and therefore $x = \sin u$, the equation becomes $i(u - \frac{1}{2}\pi) = \log(\sin u - i \cos u)$: that is, $\cos(u - \frac{1}{2}\pi) + i \sin(u - \frac{1}{2}\pi) = \sin u - i \cos u$.

Fixed Curve a Cubic: the Parametric Points 1, 2 consecutive points on the Curve. Art. Nos. 0.3 to 0.3.

Art. The major function $(x, y, z)^1 2$ is taken to be $= \frac{1^2 2 \cdot 023 - 12^2 \cdot 013}{123 \cdot 012}$, so that, calling the differential $Q d\omega$, we have

$$Q = \frac{1^2 2 \cdot 023 - 12^2 \cdot 013}{123 \cdot 012};$$

it is required to find what this becomes when 1, 2 are consecutive points on the curve, or what is the same thing when the line 012 is a tangent at the point 1.

I take for convenience the cubic to be $f = \frac{1}{3}(x^3 + y^3 + z^3) = 0$.

The coordinates of 1 are (x_1, y_1, z_1) , those of 2 are $(x_1 + \delta x_1, y_1 + \delta y_1, z_1 + \delta z_1)$, or as for shortness I write them $(x_1 + \alpha, y_1 + \beta, z_1 + \gamma)$, where α, β, γ are considered as infinitesimals of the first order: this being so, the denominator of Q is at once seen to be of the second order; it will appear that the numerator is of the third order; whence Q is of the first order.

Art. We have

$$012 = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \end{vmatrix},$$

and in analogy herewith we may write

$$\delta 012 = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ \alpha, & \beta, & \gamma \end{vmatrix} = 013, \text{ say;}$$

this being so, we have

$$023 = 013 + \delta 013,$$

and similarly $312 = 31\ 2$, &c.

Moreover

$$\begin{aligned} 1^2 2 &= x_1^2(x_1 + \alpha) + y_1^2(y_1 + \beta) + z_1^2(z_1 + \gamma) = 1 + (\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2), \\ 12^2 &= x_1(x_1 + \alpha)^2 + y_1(y_1 + \beta)^2 + z_1(z_1 + \gamma)^2 = 2(\alpha x_1 + \beta y_1 + \gamma z_1) + (\alpha x_1 + \dots), \end{aligned}$$

and hence

$$\begin{aligned} 1^2 2 \cdot 023 - 12^2 \cdot 013 &= (\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2)(013 + \delta 013) \\ &\quad - [2(\alpha x_1 + \beta y_1 + \gamma z_1) + (\alpha x_1 + \dots)]013 \end{aligned}$$

or reducing by

$$3(\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2) + 3(\alpha x_1 + \beta y_1 + \gamma z_1) + (\alpha^3 + \beta^3 + \gamma^3) = 0,$$

this is

$$= +\frac{1}{3}(\alpha^3 + \beta^3 + \gamma^3)013 - (\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)\delta 013,$$

which is of the third order.

Art. We may show that each of the terms contains the factor $(\delta x_1)^2$: we have, in fact,

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ \alpha & \beta & \gamma \\ x_1^2 & y_1^2 & z_1^2 \end{vmatrix} = 0;$$

hence, first multiplying by α, β, γ and adding, we have

$$(\alpha y_1 z_1 + \beta z_1 x_1 + \gamma x_1 y_1)(\delta x_1)^2 = -(\alpha^3 + \beta^3 + \gamma^3),$$

the required expression for the first term.

Art. Again, multiplying by x_1, y_1, z_1 , and adding, we have

$$3(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)(\delta x_1)^2 = -(\alpha^3 + \beta^3 + \gamma^3),$$

where, in virtue of $x_1^3 + y_1^3 + z_1^3 = 0$, the first line is $2(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)$: in fact, we have identically

$$3(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)(x_1^3 + y_1^3 + z_1^3) = \dots$$

which, in virtue of $\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2 = 0$, and $x_1^3 + y_1^3 + z_1^3 = 0$, becomes

$$3(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)(\delta x_1)^2 = -(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1).$$

Hence the equation is

$$3(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)(\delta x_1)^2 = -(\alpha^3 + \beta^3 + \gamma^3),$$

or finally

$$(\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1)(\delta x_1)^2 = -\frac{1}{3}(\alpha^3 + \beta^3 + \gamma^3),$$

the required expression for the second term.

Art. Writing for shortness $\alpha y_1 z_1 + \beta z_1 x_1 + \gamma x_1 y_1 = \delta(\dots)$, we have

$$1^2 2 \cdot 023 - 12^2 \cdot 013 = \left\{ -\frac{1}{3} \delta(\dots) 013 - (\alpha^2 x_1 + \beta^2 y_1 + \gamma^2 z_1) \delta 013 \right\} (\delta x_1)^2,$$

and hence dividing by $012 \cdot 123$,

$$Q = \frac{-\frac{1}{3} \delta(x_1, y_1, z_1) \cdot 013 + x_1 y_1 z_1 \cdot \delta 013}{012 \cdot 312}.$$

But this can be further reduced: the numerator, multiplied by 3, is

$$(x, y, z)(\dots).$$

Art. The identity just referred to is proved very easily. Comparing on each side the coefficient of $yz_1^2y_1^2z_1$, the factor x_1 divides out and we ought to have

$$-(\alpha y_1 z_1 + \beta z_1 x_1 + \gamma x_1 y_1) + 3y_1 z_1 \alpha = (y_1^3 - z_1^3) \delta \omega_1,$$

that is,

$$2\alpha y_1 z_1 - \beta z_1 x_1 - \gamma x_1 y_1 = (y_1^3 - z_1^3) \delta \omega_1,$$

and, in fact, from $y_1 \delta \omega_1 = z_1 \delta \omega_1$, $z_1 \delta \omega_1 = x_1 \delta \omega_1$, $x_1 \delta \omega_1 = y_1 \delta \omega_1$, we have

$$(y_1^3 - z_1^3) \delta \omega_1 = y_1(x\alpha - y\gamma) - z_1(\beta x_1 - \gamma y_1),$$

which is the value in question.

Similarly the coefficients of $zx_1^3 - z_1^3 x_1$, $xy_1^3 - x_1^3 y_1$ are equal on the two sides; and the equation is thus verified.

Art. The proof has been given in regard to the particular cubic $x^3 + y^3 + z^3 = 0$; but it might have been given for the canonical form $x^3 + y^3 + z^3 + 6lxyz = 0$: and from the invariative form it is clear that the result in fact applies to any cubic whatever.

The result is an important one: we see by it that when the points 1 and 2 are consecutive points on the curve we must, in place of the differential $Q d\omega$, which is evanescent, consider a new form $\delta \omega_1$, where, as already remarked, the denominator represents the tangent at the point 1, and the numerator the line joining the tangential of this point with the point 3.

Art. We have $\{023\} + \{031\} + \{012\} = \{123\}$, or writing this in the form $\{012\} - \{312\} + \{023\} - \{013\} = 0$, suppose 2 is here the consecutive point $1 + \delta 1$; then

$$\frac{\{012\} - \{312\}}{012 \cdot 312} \text{ becomes } = \{013\} \delta \omega_1 :$$

we have also $\{023\} - \{013\} = \{013\} \delta \omega_1$, and the result is $-\{013\} + \vartheta_1 \{013\} = 0$, that is, $\delta_1 \{013\} = \{013\}$.

The form in the case of the cubic $x^3 + y^3 + z^3 + 6lxyz = 0$, is 2β , i.e. the differential coefficient of $\{013\}$ in regard to the parametric point 1 is $= \{013\}$, the symbol for the case where the parametric line is the tangent at 1.

Fixed Curve a Cubic: the Parametric Points corresponding points. Art. Nos. 0.3 to 0.3.

Art. The parametric points 1, 2 are taken to be corresponding points, that is, such that the tangents at these points meet at a point, say 3, on the cubic. We may from 3 draw two other tangents, touching the cubic, say at the points 1' and 2'. The four points 1, 2, 1', 2' are then such that the lines 12, 1'2' meet in a point, say 4, of the cubic; and moreover 3, 4 are corresponding points.

We may take $(x, y, z) = (1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$ for the coordinates of the points 1, 2, 3, 4 respectively; and the equation of the cubic then is

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

where l is a determinate constant.

Art. The coordinates of 1' and 2' are $(1, \omega, \omega^2)$ and $(1, \omega^2, \omega)$ respectively, where ω is an imaginary cube root of unity; and we have $12 = 1, 1'2' = 1, 34 = 1$; so that the four points are symmetrically related.

Art. The major function is $H = \frac{1^2 2 \cdot 023 - 12^2 \cdot 013}{123 \cdot 012}$; and we have for the corresponding points

$$H = \frac{1}{2}\{012\} + \frac{1}{2}\{013\} - \frac{1}{2}\{023\},$$

where the symbols $\{012\}$, &c. have their before-mentioned values.

Art. The syzygy for corresponding points takes the simple form

$$\{012\} + \{034\} = \{013\} + \{024\},$$

and from this the conversion-equation follows as a particular case.

Fixed Curve a Cubic: the Parametric Points 1, 2 arbitrary. Art. Nos. 0.3 to 0.3.

Art. Reverting to the general case where the parametric points are arbitrary, we have the syzygy

$$\{023\} + \{031\} + \{012\} = \{123\},$$

and the conversion-equation

$$\bar{H}(1) - \bar{H}_1(0) = 0.$$

Art. For the purpose of establishing the conversion, we may write

$$\bar{H}(0; 1, 2; 3) = \{012\} - \{312\} + \{023\} - \{013\},$$

and thence, by the syzygy, this is

$$= 2\{012\} - \{123\}.$$

Art. The general theory of the conversion for the cubic thus depends on the properties of the symbol $\{012\}$, which is a transcendental function of the three points 0, 1, 2; and the conversion-equation is a consequence of the syzygy-relation between any four points of the cubic.

Fixed Curve a Cubic: the Parametric Points 1, 2 inflections. Art. Nos. 0.3 to ??.

Art. The parametric points are taken to be inflections, that is, points where the tangent meets the cubic in three consecutive points. There are nine inflections, and through each inflection pass four lines, each containing a pair of the remaining inflections.

Taking the cubic in the form $x^3 + y^3 + z^3 + 6lxyz = 0$, the nine inflections are the intersections of this cubic with the nine lines $x = 0$, $y = 0$, $z = 0$, $x + y + z = 0$, $x + \omega y + \omega^2 z = 0$, &c.

Art. If the parametric points 1, 2 are any two inflections, then the major function takes a specially simple form. Writing $\{012\}$ for the symbol as before, we have

$$\bar{H} = \{012\} - \{312\},$$

and the differential $Q d\omega$ is expressible in terms of the coordinates in a rational form.

Art. The conversion-equation in the case where 1, 2 are inflections is verified by means of the special properties of the inflection-points; and the general theory of the Abelian functions for the cubic is thus completed.

The Cubic $f = x(1 - x)(1 - k^2x) - y^2 = 0$

Art. Nos. 0.3 to 0.3.

Art. The curve is a semi-cubical parabola, symmetrical in regard to the axis of x : and if, as usual, k^2 is taken to be real, positive and less than 1, then the curve consists of an oval, and an infinite portion which may be called the *anguis*. (See Figure.)

The equation is satisfied by $x = \text{sn}^2 u$, $y = \text{sn} u \text{cn} u \text{dn} u$. Observe that the periods for these combinations of the elliptic functions are $2K$, $2iK'$: in fact,

$$\begin{aligned} \text{sn}(u + 2K) &= \text{sn} u, & \text{sn}(u + 2iK') &= \text{sn} u, \\ \text{cn}(u + 2K) &= \text{cn} u, & \text{cn}(u + 2iK') &= \text{cn} u, \\ \text{dn}(u + 2K) &= \text{dn} u, & \text{dn}(u + 2iK') &= \text{dn} u, \end{aligned}$$

whence the sn^2 and the $\text{sn} \cdot \text{cn} \cdot \text{dn}$ are each unaltered by the change of u into $u + 2K$ or $u + 2iK'$.

Hence to a given point (x, y) on the curve, the argument u is not = a determinate value u_0 , for it may be equally well taken to be $= u_0 + 2mK + 2m'iK'$, where m, m' are any positive or negative integers whatever: we express this by $u \equiv u_0$, or say u is *congruent* to u_0 . But when u is thus given by a congruence $u \equiv u_0$ where u_0 has a determinate value, the point on the curve is uniquely determined.

It is, however, to be noticed that a congruence $2u \equiv u_0$ does not uniquely determine the point on the curve: there are, in fact, four incongruent values of u , viz.

$$u_0, \quad u_0 + K, \quad u_0 + iK', \quad u_0 + K + iK',$$

and the point on the curve is thus one of the four points belonging to these values of u respectively.

Art. If, to fix the ideas, we attend to the values of u which lie in the parallelogram bounded by the lines $u = 0$, $u = 2K$, $u = iK'$, $u = 2K + iK'$, then the oval corresponds to the line $u = 0$ to $u = 2K$ (or rather to the values of u for which the imaginary part = 0); and the angulus corresponds to the remaining values of u in the parallelogram.

Art. The integral formula is

$$\int_0^u du = u = \int_0^x \frac{dx}{2y} = \int_0^x \frac{dx}{2\sqrt{x(1-x)(1-k^2x)}}.$$

Writing herein $x = \sin^2 \phi$, we have

$$u = \int_0^\phi \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}} = F(k, \phi),$$

the elliptic integral of the first kind; and hence $x = \operatorname{sn}^2 u$, $y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$, the foregoing formulae.

Art. The differential $d\omega$ for the cubic $y^2 = x(1-x)(1-k^2x)$ is

$$d\omega = \frac{dx}{2y} = \frac{dx}{2\sqrt{x(1-x)(1-k^2x)}} = du.$$

The minor function is a constant (= 1), and the major function is $(x, y, z)^2$. The differential is $d\omega = du$, and the integral is simply $u = \int d\omega$.

Art. For the affected theorem, taking two parametric points 1, 2, the major function is $H = \frac{(x,y,z)^2}{012}$, and the differential is $\frac{H d\omega}{012}$. The parametric points correspond to arguments u_1, u_2 ; and the residues of these points correspond to arguments $-u_1, -u_2$ respectively.

Art. The theta-functions present themselves in the theory as follows: writing $\vartheta(u)$ for the theta-function with parameter $q = e^{-\pi K'/K}$, we have

$$\operatorname{sn} u = \frac{1}{\sqrt{k}} \frac{\vartheta_1(u)}{\vartheta(u)}, \quad \operatorname{cn} u = \sqrt{\frac{k'}{k}} \frac{\vartheta_2(u)}{\vartheta(u)}, \quad \operatorname{dn} u = \sqrt{k'} \frac{\vartheta_3(u)}{\vartheta(u)},$$

where $k' = \sqrt{1 - k^2}$ is the complementary modulus.

Art. The Abelian functions for the cubic are thus expressible in terms of the elliptic functions; and the conversion-formula is a particular case of the addition-theorem for the elliptic integral of the first kind. Writing

$$\int_0^{x_1} \frac{dx}{2y} + \int_0^{x_2} \frac{dx}{2y} = \int_0^X \frac{dx}{2y},$$

we have X a rational symmetric function of x_1, x_2, y_1, y_2 ; viz.

$$X = \frac{(x_1 y_2 + x_2 y_1)^2}{(x_1 - x_2)^2},$$

or what is the same thing,

$$\operatorname{sn}(u_1 + u_2) = \frac{\operatorname{sn} u_1 \operatorname{cn} u_2 \operatorname{dn} u_2 + \operatorname{sn} u_2 \operatorname{cn} u_1 \operatorname{dn} u_1}{1 - k^2 \operatorname{sn}^2 u_1 \operatorname{sn}^2 u_2}.$$

Art. The syzygy for the cubic $y^2 = x(1-x)(1-k^2x)$ takes the form

$$\int_0^{x_1} + \int_0^{x_2} + \int_0^{x_3} = 0,$$

where x_1, x_2, x_3 are the roots of a certain equation; and the sum of the three integrals is $= 0$ identically.

Art. The general theory of the Abelian and Theta functions thus includes as a particular case the theory of the elliptic functions; and the memoir is completed by the exhibition of the formulae in this special case.

Art. It is to be observed that the functions $\{012\}$ for the cubic are in fact the elliptic integrals of the third kind; and the conversion-formula is the expression of the addition-theorem for such integrals. Writing

$$\{012\} = \int_0^{x_0} \frac{x - x_1}{x - x_2} \cdot \frac{dx}{2y},$$

we have the conversion-equation as a consequence of the syzygy.

Art. The memoir thus establishes:

1. The general theory of the major function $(x, y, z)^{n-2}$ for a curve of order n ;
 2. The syzygy-relation connecting the major functions for different parametric points;
 3. The conversion-formula, expressing the difference of two major functions as a definite integral;
 4. The special theory for the cubic curve, including the elliptic functions as a particular case.
-

0.4 Paper 825. A Memoir on the Abelian and Theta Functions (pp. 201–215)

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. xciv. (1883), pp. 186–210.]

Application of Abel's Theorem. Art. Nos. 153 to 157

Art. Taking the fixed curve to be $f = (Az^2 + 2Bz + C) = 0$, we have

$$\frac{dz}{y} = \frac{dy}{dx},$$

if for shortness we write

$$(x, y)^2 = B^2 - AC = (x - ay)(x - by)(x - cy)(x - dy)(x - ey)(x - fy),$$

and we thence have

$$\omega = \int \frac{x dy - y dx}{\sqrt{(x, y)^6}}.$$

The minor curve $(x, y, z)^{n-3} = 0$ is an arbitrary line passing through the node, that is, the point $x = 0, y = 0$; and the pure theorem thus gives the two relations

$$\Sigma x d\omega = 0, \quad \Sigma y d\omega = 0;$$

where the summation extends to the intersections of the fixed curve $Az^2 + 2Bz + C = 0$ with the variable curve ϕ .

The variable curve is taken to be a cubic

$$Az + B = (\alpha, \beta, \gamma, \delta \diagdown e, f)(x, y)^3,$$

or say $Az + B = \Omega$, where Ω is a given cubic function $(x, y)^3$: viz. this is a nodal cubic, the node and the two tangents there being the same with the node and the two tangents of the quartic: hence it meets the quartic in the node counting 6 times, and in 6 other points, say these are the points 1, 2, 3, 4, 5, 6: hence the differential relations are

$$\begin{aligned} x_1 d\omega_1 + x_2 d\omega_2 + x_3 d\omega_3 + x_4 d\omega_4 + x_5 d\omega_5 + x_6 d\omega_6 &= 0, \\ y_1 d\omega_1 + y_2 d\omega_2 + y_3 d\omega_3 + y_4 d\omega_4 + y_5 d\omega_5 + y_6 d\omega_6 &= 0. \end{aligned}$$

Art. Observe that the intersections of the cubic with the fixed curve are given by the equation $H^2 = B^2 - AC$, or say $H^2 = (x, y)^6$,

an equation which determines the ratio $x : y$ for the six points respectively; and the ratio $z : x$ is then determined rationally by the original equation $Az + B = \Omega$.

Instead of regarding Ω as a given function, we may, if we please, take 1, 2, 3, 4 given points on the quartic: we then have four equations for the determination of the coefficients $(\alpha, \beta, \gamma, \delta)$ of the function Ω ; viz. these equations may be taken to be

$$(\alpha, \beta, \gamma, \delta)(x_1, y_1)^3 = \Omega_1, \quad (\alpha, \beta, \gamma, \delta)(x_2, y_2)^3 = \Omega_2, \quad \text{etc.},$$

Ω is hereby completely determined: and this being so, the remaining intersections 5 and 6 are also completely determined: there are thus between the six points 2 integral relations, which agrees with the number, = 2, of the differential relations obtained above.

Art. If we now assume

$$\begin{aligned} du &= x_1 d\omega_1 + x_2 d\omega_2, & dv &= y_1 d\omega_1 + y_2 d\omega_2, \\ du' &= x_3 d\omega_3 + x_4 d\omega_4, & dv' &= y_3 d\omega_3 + y_4 d\omega_4, \\ du'' &= x_5 d\omega_5 + x_6 d\omega_6, & dv'' &= y_5 d\omega_5 + y_6 d\omega_6, \end{aligned}$$

or say

$$\begin{aligned} u, v &= \frac{\int x(x dy - y dx)}{\sqrt{(x, y)^6}}, & u', v' &= \frac{\int x(x dy - y dx)}{\sqrt{(x, y)^6}}, \\ u'', v'' &= \frac{\int x(x dy - y dx)}{\sqrt{(x, y)^6}}, & v &= \frac{\int y(x dy - y dx)}{\sqrt{(x, y)^6}}, \end{aligned}$$

that is,

$$u = \int_{\alpha}^{\beta} \frac{x(x dy - y dx)}{\sqrt{(x, y)^6}}, \quad v = \int_{\alpha}^{\beta} \frac{y(x dy - y dx)}{\sqrt{(x, y)^6}},$$

where α, β are points assumed at pleasure on the quartic: and similarly for $u', v': u'', v''$: then u, v are hereby determined as functions of the points 1, 2: and we may conversely regard the points 1, 2 as determined in terms of the two arguments u, v .

We might, selecting any two symmetrical functions of the degree zero, for instance, $\frac{x_1}{y_1} + \frac{x_2}{y_2}, \frac{x_1}{y_1} \cdot \frac{x_2}{y_2}$, represent them as functions $\phi(u, v), \psi(u, v)$: and then $\frac{x_1}{y_1}$ and $\frac{x_2}{y_2}$ will be functions of $\phi(u, v), \psi(u, v)$. But instead of this selection, it is proper to consider the ratios of six

functions depending on the points a, b, c, d, e, f respectively: viz. we assume

$$\begin{aligned} \sqrt{(x_1 - ay_1)(x_2 - ay_2)} : \sqrt{(x_1 - by_1)(x_2 - by_2)} : \cdots : \sqrt{(x_1 - fy_1)(x_2 - fy_2)} \\ = A(u, v) : B(u, v) : \cdots : F(u, v), \end{aligned}$$

and of course 3, 4 will be in like manner determined by means of the corresponding functions of u', v' , and 5, 6 by means of the corresponding functions of u'', v'' .

The squared functions $A^2, B^2, C^2, D^2, E^2, F^2$ are proportional to given linear functions of $x_1x_2, x_1y_2 + x_2y_1, y_1y_2$, and are thus connected by three independent linear relations.

Art. The differential relations then become

$$du + du' + du'' = 0, \quad dv + dv' + dv'' = 0,$$

and we have consequently

$$u + u' + u'' = I, \quad v + v' + v'' = J,$$

where I, J are constants which are determinable as definite integrals by the consideration that, when the cubic is taken to be $Az + B = 0$, the six points 1, 2, 3, 4, 5, 6 coincide with the points of contact a, b, c, d, e, f .

I do not at present see my way to a proper development of this point of the theory: but in explanation of the nature of the result, I assume for the moment that by a proper determination of the inferior limits α, β , or otherwise, we may take $I = 0, J = 0$.

We then have $u'' = -u - u', v'' = -v - v'$: and the integral equations, which determine the points 5, 6 in terms of the points 1, 2 and the points 3, 4, then in effect determine the functions A, B , &c., of $-u - u', -v - v'$, or say those of $u + u', v + v'$ in terms of the like functions of (u, v) and of (u', v') : viz. these equations give the addition-theory of the functions $A(u, v)$, &c.

Art. We may, in the first instance, disregarding altogether the consideration of the arguments u, v , &c., attend only to the algebraic functions such as $\sqrt{(x_1 - ay_1)(x_2 - ay_2)}$, &c., of the coordinates of the pairs of points 1, 2; 3, 4, and 5, 6; and we can in regard to these develop a proper theory. This depends only on the equation $H = \sqrt{(x, y)^6}$; it will be convenient to assume herein $y = 1$, and slightly modifying the form, to write it

$$(\alpha, \beta, \gamma, \delta)(x, 1)^3 = \sqrt{a - x \cdot b - x \cdot c - x \cdot d - x \cdot e - x \cdot f - x};$$

and accordingly to consider the functions $\sqrt{a - x_1 \cdot a - x_2}$, &c. These are called the single-letter functions A , &c., but there are certain double-letter functions AB , &c., which have also to be considered; and I will, in the first instance, show how these present themselves in connexion with the cubic curve.

Origin of the Double-Letter Functions. Art. Nos. 158 and 159

Art. The cubic curve $Az + B = \Omega$ may be taken to be a curve through two of the points of contact, say the points a, b ; these will then be two out of the six points, and taking the remaining four points to be the pairs 1, 2 and 3, 4, we have single-letter functions of 3, 4 presenting themselves as double-letter functions of 1, 2.

In fact, the equation of the curve is

$$Az + B = \lambda(x - ay)(x - by)(x - ky);$$

for the intersections with the quartic we have

$$\lambda^2(x - ay)^2(x - by)^2(x - ky)^2 = \Omega^2,$$

or throwing out the factor $(x - ay)(x - by)$ and changing the constant λ , this is

$$(x - ay)(x - by)(x - ky)^2 - \mu(x - cy)(x - dy)(x - ey)(x - fy) = 0;$$

and the quartic function must be a multiple of $(xy_1 - x_1y)(xy_2 - x_2y)(xy_3 - x_3y)(xy_4 - x_4y)$.

Putting each of the y 's equal 1, we have the identity

$$(a-x)(b-x)(k-x)^2 - \mu(c-x)(d-x)(e-x)(f-x) = \rho(x_1-x)(x_2-x)(x_3-x)(x_4-x);$$

and hence, introducing a notation which will be convenient, $a - x = a_x$, $a - x_1 = a_1$, and so in other cases, we have by giving different values to x the equations

$$\begin{aligned} a_2b_2k_2^2 &= \mu c_2d_2e_2f_2, & (a-c)(b-c)(k-c)^2 &= \mu c_1c_2c_3c_4, \\ a_3b_3k_3^2 &= \mu c_3d_3e_3f_3, & (a-d)(b-d)(k-d)^2 &= \mu d_1d_2d_3d_4, \\ a_4b_4k_4^2 &= \mu c_4d_4e_4f_4, & (a-e)(b-e)(k-e)^2 &= \mu e_1e_2e_3e_4, \\ & & (a-f)(b-f)(k-f)^2 &= \mu f_1f_2f_3f_4, \end{aligned}$$

and

$$\lambda(c-a)(d-a)(e-a)(f-a) = \mu a_1a_2a_3a_4, \quad \lambda(c-b)(d-b)(e-b)(f-b) = \mu b_1b_2b_3b_4.$$

Art. We have thus

$$\frac{(a-c)(b-c)}{c_1c_2 \cdot c_3c_4} = \frac{(a-d)(b-d)}{d_1d_2 \cdot d_3d_4},$$

and

$$\frac{k_1k_2}{a_2b_2 \cdot c_1d_1e_1f_1} = \frac{k_3k_4}{a_3b_3 \cdot c_3d_3e_3f_3},$$

which last equation, writing for a moment

$$\gamma = \sqrt{a_2b_2 \cdot c_1d_1e_1f_1}, \quad \delta = \sqrt{a_1b_1 \cdot c_2d_2e_2f_2},$$

gives

$$k(\gamma - \delta) = \# \gamma - \# \delta, \quad \text{and thence} \quad \frac{\gamma}{\delta} = \frac{k - \delta}{k - \gamma},$$

$$\frac{\gamma c_3c_4}{\delta d_3d_4} = \frac{\sqrt{a_2b_2c_1d_1e_1f_1} \cdot c_3c_4}{\sqrt{a_1b_1c_2d_2e_2f_2} \cdot d_3d_4},$$

or, substituting in the first equation, we have

$$\frac{\sqrt{(a-c)(b-c)}}{\sqrt{(a-d)(b-d)}} = \frac{\sqrt{a_2b_2c_1d_1e_1f_1} \cdot c_3c_4}{\sqrt{a_1b_1c_2d_2e_2f_2} \cdot d_3d_4}.$$

Art. Considering the duad DE as an abbreviation for the double triad $ABC.DEF$, the expressed duad being always accompanied by the letter F , we are thus led to the consideration of the double-letter functions

$$ABK = \frac{1}{x_1 - x_2} \left\{ \sqrt{a_1b_1f_1 \cdot c_2d_2e_2} - \sqrt{a_2b_2f_2 \cdot c_1d_1e_1} \right\}, \quad \&c.,$$

in connexion with the already mentioned single-letter functions

$$A_{12} = \sqrt{a_1a_2}, \quad \&c.,$$

viz. in this notation the equation just obtained is

$$\frac{C_3C_4}{D_3D_4} = \frac{\sqrt{(a-c)(b-c)}}{\sqrt{(a-d)(b-d)}} \cdot \frac{DEK}{CEF},$$

and it thus appears that, the points 3, 4 being obtained as above from the given points 1, 2, then the quotient of two of the single-letter functions of 3, 4 is a constant multiple of the quotient of two of the double-letter functions of 1, 2.

Observe that the points 3, 4 are derived from 1, 2 by means of the two points a, b : we have DE standing for $ABC.DEF$, CE for $ABD.CEF$, and if the two functions were represented by ABC , ABD respectively, then the form would have been

$$\frac{\sqrt{(a-c)(b-c)}}{\sqrt{(a-d)(b-d)}} \cdot \frac{ABC}{ABD},$$

which is a clearer expression of the theorem; the apparent want of symmetry of the first form arises only from the arbitrary selection of the letter F to accompany the expressed duad, and is at once removed by substituting for a duad DE the triad $ABC.DEF$ which is thereby signified.

The denominator factor $x_1 - x_2$ is introduced in order to make the degree in x_1 or x_2 equal to that of the single-letter functions.

The Addition Theory. Art. Nos. 160 to 163

Art. We have the six single-letter symbols A, B, C, D, E, F ; viz. $A_{12}^2 = a_1a_2$, &c.: and the ten double-letter symbols $AB, AC, AD, AE, BC, BD, BE, CD, CE, DE$, viz.

$$ABK = \frac{1}{x_1 - x_2} \left\{ \sqrt{a_1b_1f_1 \cdot c_2d_2e_2} - \sqrt{a_2b_2f_2 \cdot c_1d_1e_1} \right\}, \quad \&c.,$$

these 16 functions being connected by algebraical relations which are immediately deducible from these expressions of the functions in terms of x_1, x_2 .

The problem is to express the functions of 5, 6 in terms of those of 1, 2 and of those of 3, 4.

The relation between the variables $x_1, x_2, x_3, x_4, x_5, x_6$ consists herein that we have $x_1, x_2, x_3, x_4, x_5, x_6$ as the roots of the equation

$$(\alpha x^3 + \beta x^2 + \gamma x + \delta)^2 - \lambda(a-x)(b-x)(c-x)(d-x)(e-x)(f-x) = 0;$$

or, what is the same thing, it consists in the identity

$$\begin{aligned} (\alpha x^3 + \beta x^2 + \gamma x + \delta)^2 - \lambda(a-x)(b-x)(c-x)(d-x)(e-x)(f-x) \\ = \rho(x-x_1)(x-x_2)(x-x_3)(x-x_4)(x-x_5)(x-x_6). \end{aligned}$$

Again, it may be expressed by the plexus of equations

$$\frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \frac{1}{x_4}, \frac{1}{x_5}, \frac{1}{x_6}, \frac{1}{x_1^2}, \frac{1}{x_2^2}, \frac{1}{x_3^2}, \frac{1}{x_4^2}, \frac{1}{x_5^2}, \frac{1}{x_6^2}, \quad \&c.,$$

where $X_1 = (a-x_1)(b-x_1) \cdots (f-x_1)$, &c.; these are equivalent of course to two equations, and serve to determine x_5, x_6 in terms of x_1, x_2, x_3, x_4 .

Art. The solution is, in fact, as is given in my paper "On the addition of the double $\mathcal{P}$ -functions," Crelle, t. LXXXVIII. (1880), pp. 74-81, [703].

Writing successively $x = x_1, x_2, x_3, x_4$, we have

$$\alpha x_1^3 + \beta x_1^2 + \gamma x_1 + \delta = \sqrt{\mu \cdot A_1 B_1 C_1 D_1 E_1 F_1},$$

which equations serve to determine the ratios $\alpha : \beta : \gamma : \delta$ in terms of x_1, x_2, x_3, x_4 ; and we have then the two like equations

$$\alpha x_5^3 + \beta x_5^2 + \gamma x_5 + \delta = \sqrt{\mu \cdot A_5 B_5 C_5 D_5 E_5 F_5}, \quad \alpha x_6^3 + \beta x_6^2 + \gamma x_6 + \delta = \sqrt{\mu \cdot A_6 B_6 C_6 D_6 E_6 F_6}$$

which determine the symmetric functions of x_5, x_6 .

Art. If, reverting to the identity, we write therein for instance $x = a$, we find

$$\alpha a^3 + \beta a^2 + \gamma a + \delta = \sqrt{\mu \cdot A_1 A_2 A_3 A_4 A_5 A_6},$$

which equation when properly reduced gives the proportional value of A_{56} .

Art. Calling for a moment the function on the left-hand side Ω , we have

$$\begin{vmatrix} x_1^3 & x_1^2 & x_1 & 1 & \sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & 1 & \sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & 1 & \sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & 1 & \sqrt{X_4} \\ a^3 & a^2 & a & 1 & \sqrt{A_1 A_2 A_3 A_4 A_5 A_6} \end{vmatrix} = 0,$$

that is,

$$\begin{vmatrix} x_1^3 & x_1^2 & x_1 & 1 & \sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & 1 & \sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & 1 & \sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & 1 & \sqrt{X_4} \\ a^3 & a^2 & a & 1 & A_{56} \end{vmatrix} = 0,$$

viz. this is

$$\begin{vmatrix} x_1^3 & x_1^2 & x_1 & 1 & \sqrt{X_1} \\ x_2^3 & x_2^2 & x_2 & 1 & \sqrt{X_2} \\ x_3^3 & x_3^2 & x_3 & 1 & \sqrt{X_3} \\ x_4^3 & x_4^2 & x_4 & 1 & \sqrt{X_4} \\ a^3 & a^2 & a & 1 & A_{56} \end{vmatrix} = 0,$$

or, as this may be written,

$$\sqrt{X_3 \cdot a - x_3 \cdot a - x_4} \cdot \frac{x_4 - a}{x_2 - x_1 \cdot x_4 - x_3} + \dots = 0.$$

We have here $\sqrt{a - x_3 \cdot a - x_4} = \sqrt{A_{34}}$, and the function which multiplies this is without difficulty found to be

$$\nabla_A = \partial_b + \partial_c + \partial_d,$$

where the summation extends to the three terms obtained by the cyclical interchange of the letters b, c, d : these being a set of three out of the five letters other than a .

Similarly $\sqrt{a - x_1 \cdot a - x_2} = \sqrt{A_{12}}$, and the function which multiplies this is

$$\frac{c-d}{b-c} \cdot \frac{d}{b} \cdot \frac{b}{c}.$$

The expression for Ω thus contains the factor $A_{12}A_{34}$: but we have this equation contains therefore the factor $\sqrt{A_{12}A_{34}}$, and omitting it we find

$$\frac{1}{\sqrt{?}}(x_1 - x_3 \cdot x_1 - x_4 \cdot x_2 - x_3 \cdot x_2 - x_4)(c - d \cdot d - b \cdot b - c)A_{56} + \dots = 0,$$

where, as before, the summations refer each to the three terms obtained by the cyclical interchange of the letters b, c, d ; these being any three of the five letters other than a : and the remaining two letters e, f enter into the formulae symmetrically.

The formula thus gives for A_{56} ten values which are of course equal to each other. By reason of the undetermined factor $\frac{1}{\sqrt{?}}$, the formula gives only the proportional value of A_{56} ; viz. combining it with the like formulae for B_{56} , &c., we have determinate values of the ratios $A_{56} : B_{56} : \dots : F_{56}$. But this being understood, we regard the formula as a formula for each single-letter function of x_5, x_6 in terms of the single and double-letter functions of x_1, x_2 and of x_3, x_4 respectively.

Art. We require further the expressions for the double-letter functions of 5, 6. Consider for example the function DE_{56} , which is

$$\frac{1}{x_5 - x_6} \left\{ \sqrt{d_5 e_5 f_5 \cdot a_6 b_6 c_6} - \sqrt{d_6 e_6 f_6 \cdot a_5 b_5 c_5} \right\};$$

then multiplying by $A_{56}B_{56}C_{56} = \sqrt{a_5 b_5 c_5 \cdot a_6 b_6 c_6}$, we have

$$\frac{1}{x_5 - x_6} \{ a_6 b_6 c_6 \sqrt{X_5} - a_5 b_5 c_5 \sqrt{X_6} \},$$

or recollecting that $\sqrt{X_5}$ and $\sqrt{X_6}$ are $= \alpha x_5^3 + \beta x_5^2 + \gamma x_5 + \delta$ and $\alpha x_6^3 + \beta x_6^2 + \gamma x_6 + \delta$ respectively, this may be written

$$\begin{aligned} \frac{1}{x_5 - x_6} \{ (a - x_5)(b - x_5)(c - x_5)(\alpha x_6^3 + \beta x_6^2 + \gamma x_6 + \delta) \\ - (a - x_6)(b - x_6)(c - x_6)(\alpha x_5^3 + \beta x_5^2 + \gamma x_5 + \delta) \}. \end{aligned}$$

Using the well-known identity

$$\frac{1}{x_5 - x_6} \{ (a - x_5)(b - x_5)(c - x_5)(\alpha x_6^3 + \dots) - (a - x_6)(b - x_6)(c - x_6)(\alpha x_5^3 + \dots) \} = \sum \frac{(b - a)(c - a)(d - a)}{\sqrt{\dots}},$$

where the summation extends to the four terms obtained by the cyclical interchanges of the letters a, b, c, d : and the like identity for $\alpha x_5^3 + \beta x_5^2 + \gamma x_5 + \delta$: there will be terms in $\alpha a^3 + \beta a^2 + \gamma a + \delta$, $\alpha b^3 + \beta b^2 + \gamma b + \delta$, $\alpha c^3 + \beta c^2 + \gamma c + \delta$, but the term in $\alpha d^3 + \beta d^2 + \gamma d + \delta$ will disappear of itself.

After some easy reductions, the formula for DE_{56} is obtained in the form

$$DE_{56} = \frac{\sqrt{A_{12}B_{34}C_{34}}}{\sqrt{D_{12}E_{12}}} \cdot \frac{(a - d)(b - d)}{(c - d)} + \text{symmetric terms.}$$

Chapter VII. The Functions T , U , V , ω

The present chapter is substantially a reproduction of C. and G.'s seventh section, "Die Function T_{12} ", borrowing only from the next section the definition of the theta-function; but for greater completeness I give some developments which are not contained in the original memoir.

Art. Suppose that the curves ϕ , ψ are each of them a major curve, that is, a curve of the order $n - 2$ passing through the δ dps, and consequently besides meeting the curve f in $n(n-2) - 2\delta = 2p + n - 2$ points: the theorem is that if ϕ , ψ each pass through the same $(2p + n - 2) - m$, $= q - m$ points respectively, then the remaining m points of intersection with f of the curve ϕ and the remaining m points of intersection with f of the curve ψ are pairs of points belonging to the same g_m^{m-1} ; that is, we have a relation $T(1, 2, \dots, m) = 0$, where T is a symmetrical function of the m points $1, 2, \dots, m$, and of the parametric points and of the sets of superior and inferior points respectively.

Art. The foregoing theorem for the quartic thus is

$$T(a, b, c; a_1, b_1, c_1) = 0.$$

Observing that

$$\frac{T_a + T_b + T_c}{3} = 2\Sigma - \frac{T_{a_1} + T_{b_1} + T_{c_1}}{3},$$

where Σ denotes the sum of the parametric terms, we have

$$T(a, b, c; a_1, b_1, c_1) = \int_a^{a_1} + \int_b^{b_1} + \int_c^{c_1}.$$

Art. We have that is,

$$T_{123} = \Sigma\omega - \Sigma\omega(4, 5, 6),$$

and thence the theorem (A):

$$T_{123} + T_{145} + T_{167} + T_{189} = \text{const.}$$

Art. Theorem (B). We have

$$T_{12}(\xi, \eta, \zeta) + T_{12}(\xi', \eta', \zeta') = 0,$$

where ξ, η, ζ and ξ', η', ζ' are comajors in regard to 1, 2.

Art. To show that $T - T_1$ contains no term in $\log \mu$. For T , the only term in $\log \mu$ is $\int \frac{d\omega}{\mu}$, and for T_1 it is $\int \frac{d\omega_1}{\mu_1}$; and it is to be shown that the difference of the two integrals contains no term in $\log \mu$. Considering on the quartic the two new points 8, 9, we have $T_{12}(8, 9, \mu) = T_{12}(8, 9, \mu_1)$; and the theorem is thus proved.

Art. Theorem (C). We have

$$T_{123}(\xi, \eta, \zeta) + T_{123}(\xi', \eta', \zeta') = 0,$$

tagC where ξ, η, ζ and ξ', η', ζ' are comajors in regard to 1, 2, 3: the equation shows that, in $T(1, 2, 3)$ the function T_{123} , we can without alteration of the value interchange a pair of points 1, 1' out of the system of comajor points.

Art. Taking the theorem for the quartic in the form $T_{123} + T_{145} + \text{const} = 0$, and applying it successively to the parametric line, and to the conic which determines the points 1, 2, 3, we have

$$P_1 + P_2 + P_3 + 2\Sigma = \text{const}, \quad P'_1 + P'_2 + P'_3 + 2\Sigma = \text{const}.$$

Taking the difference of these equations, we have

$$P_1 + P_2 + P_3 - P'_1 - P'_2 - P'_3 = \text{const},$$

viz. the points $\eta, 1', 2', 3'$ being fixed points, this is

$$P_1 + P_2 + P_3 = \text{const.},$$

that is, the functions u, v, w defined as above are each of them constant under the variation in question.

0.5 Paper 826. Note on a Partition-Series (pp. 216–219)

[From the *American Journal of Mathematics*, vol. vi. (1884), pp. 63, 64.]

Prof. Sylvester, in his paper, “A Constructive theory of Partitions, &c.,” *Amer. Journ. Math.*, vol. v. (1883), p. 282, has given the following very beautiful formula

$$(1+ax)(1+ax^2)(1+ax^3)\cdots \\ = 1 + \frac{1+ax}{1}xa + \frac{(1+ax)(1+ax^2)}{1 \cdot 2}x^3a^2 + \frac{(1+ax)(1+ax^2)(1+ax^3)}{1 \cdot 2 \cdot 3}x^6a^3 + \cdots,$$

or, as this may be written,

$$\Omega = 1 + P + Q(1+ax) + R(1+ax)(1+ax^2) + S(1+ax)(1+ax^2)(1+ax^3) + \cdots,$$

where the law is obvious.

I propose to show how the several coefficients P , Q , R , &c. may be obtained in succession.

Writing for shortness $1, 2, 3, \dots$ to denote xa, x^3a^2, x^6a^3 , &c. respectively, and P, Q, R , &c. to denote the coefficients, we have

$$\Omega = 1 + P + Q(1+ax) + R(1+ax)(1+ax^2) + S(1+ax)(1+ax^2)(1+ax^3) + \cdots,$$

and we find without difficulty (see *infra*) that

$$1 + Q + R = (1+ax)(1+R), \quad 1 + R + S = (1+ax^2)(1+S), \quad \&c.;$$

and hence, using Π to denote the sum

$$\Omega = 1 + P + Q(1+ax) + R(1+ax)(1+ax^2) + S(1+ax)(1+ax^2)(1+ax^3) + \cdots,$$

we obtain successively

$$\Omega - (1+ax) = 1 + P + Q + R(1+ax^2) + S(1+ax^2)(1+ax^3) + \cdots, \\ \Omega - (1+ax)(1+ax^2) = 1 + Q + R + S(1+ax^3) + \cdots,$$

and so on.

In these equations, on the right-hand sides, the terms are successively from the beginning $1 + P$, $1 + Q$, $1 + R$, $1 + S$, &c.; and we thus have

$$\Omega = 1 + P + Q(1+ax) + R(1+ax)(1+ax^2) + S(1+ax)(1+ax^2)(1+ax^3) + \cdots,$$

and therefore

$$1 \cdot 2 \cdot 3 \cdot 4 \cdots = 1 - (1 + x) + (1 + x^2) - (1 + x^3) + \cdots,$$

which is Euler's theorem.

It might appear that the identities used in the proof would also, for this particular value $a = 1$, lead to interesting theorems; but this is found not to be the case: we have

$$P = \frac{x}{1}, \quad Q = \frac{x^3}{1 \cdot 2}, \quad R = \frac{x^6}{1 \cdot 2 \cdot 3},$$

but the expressions in terms of these quantities for the products $2 \cdot 3 \cdot 4 \cdots$, $3 \cdot 4 \cdots$, &c., contain denominator factors, and are thus altogether without interest.

0.6 Paper 827. On the Non-Euclidian Plane Geometry (pp. 220–239)

[From the *Proceedings of the Royal Society of London*, vol. xxxvii. (1884), pp. 82–102. Received May 27, 1884.]

Art. I consider the hyperbolic or Lobatschewskian geometry: this is a geometry such as that of the imaginary spherical surface $x^2 + y^2 + z^2 = 1$; and the imaginary surface may be bent (without extension or contraction) into the real surface considered by Beltrami, which I will call the *Pseudosphere*, viz. this is the surface of revolution generated by the revolution of the tractrix about its axis.

Art. The pseudosphere is thus a surface upon which the geometry of the hyperbolic plane may be represented: but the representation is not without discontinuities. To obviate this, Beltrami introduced a different representation, that by means of the quadric surface ($x^2 + y^2 + z^2 = 1$, viz.) the imaginary spherical surface considered as a surface of constant negative curvature; and this is the space, that is, which we become acquainted with by experience, but which is the representation lying at the foundation of all physical experience.

Art. I propose in the present paper to develop further the geometry of the pseudosphere. In regard to the name, and the subject generally, I refer to two memoirs by Beltrami, “Teoria fondamentale degli spazii di curvatura costante,” *Annali di Matem.*, t. II. (1868–69), pp. 232–255, and “Saggio di interpretazione della geometria non-euclidea,” *Giornale di Matem.*, t. VI. (1868), pp. 284–312, and also to Clebsch’s paper, “Ueber die Formen der Linien auf der Flächen zweiten Grades,” *Math. Ann.*, t. IV. (1871), pp. 408–432.

Art. The fundamental formulae of the hyperbolic plane geometry may be obtained from those of the spherical geometry by writing ia , ib , ic for the sides a , b , c of the triangle. We thus obtain for a right-angled triangle (C being the right angle)

$$\cosh c = \cosh a \cosh b, \quad \sinh a = \sinh c \sin A,$$

whence also

$$\tan b = \tan c \cos A, \quad \sin a = \sin c \sin A.$$

Art. We deduce from these

$$\frac{\tan^2 a}{\tan^2 c} = \frac{\sin^2 b}{\sin^2 c},$$

leading to

$$\cos^2 c = \cos^2 a \cos^2 b;$$

and then

$$\frac{\sin b}{\tan a} = \cos c \tan B, \quad \frac{\sin a}{\tan b} = \cos c \tan A,$$

giving

$$\cos a \cos b = \cos^2 c \tan A \tan B;$$

that is,

$$\tan A \tan B = \frac{1}{\cos a \cos b},$$

which is > 1 .

Hence $A+B > \text{a right angle}$, or in the right-angled triangle ACB , the sum $A+B+C$ of the angles is $> \text{two right angles}$.

Art. Whence also in any triangle ABC , drawing a perpendicular, say AD , from A to the side BC , and so dividing the triangle into two right-angled triangles, we prove that the sum $A+B+C$ of the angles is $= \text{two right angles}$, and we further establish the relations

$$a = b \cos C + c \cos B, \quad b = c \cos A + a \cos C, \quad c = a \cos B + b \cos A,$$

which are the fundamental formulae of plane trigonometry; that is, we derive the metrical geometry or trigonometry of the plane from the spherical trigonometry.

Art. We have to consider the imaginary spherical surface as bent into a real surface. This is, of course, an imaginary process, as any process must be which gives a transformation of imaginary points and lines into real points and lines; but the notion is not more difficult than that of the transformation of imaginary similarity, consisting in the substitution of ix , iy , iz for x , y , z respectively. We thus pass from imaginary points of the imaginary surface to real points of the real surface.

Art. The formulae for the bending are obtained as follows: starting from the formulae for the sphere

$$X = \cos \theta, \quad Y = \sin \theta \cos \phi, \quad Z = \sin \theta \sin \phi,$$

giving the formula

$$dX^2 + dY^2 + dZ^2 = d\theta^2 + \sin^2 \theta d\phi^2;$$

and then also

$$\begin{aligned} dy^2 + dz^2 &= dx^2 + (d \sin \theta)^2 + \sin^2 \theta d\phi^2 \\ &= (\cos^2 \theta \cot^2 \theta + \cos^2 \theta) d\theta^2 + \sin^2 \theta d\phi^2 \\ &= \cot^2 \theta d\theta^2 + \sin^2 \theta d\phi^2. \end{aligned}$$

Joining to these the differential expression in u, v , we have

$$\frac{du^2}{(1 - u^2 - v^2)^2} = \cot^2 \theta d\theta^2 + \sin^2 \theta d\phi^2 = dx^2 + dy^2 + dz^2;$$

where the final equation $dX^2 + dY^2 + dZ^2 = dx^2 + dy^2 + dz^2$ shows that the imaginary sphere $X^2 + Y^2 + Z^2 = 1$ can be bent into the pseudosphere.

Art. The geodesics on the pseudosphere correspond to the great circles on the sphere, that is, to the sections of the sphere by planes through the centre; or, what is the same thing, to the circles on the plane which cut the absolute conic $x^2 + y^2 + z^2 = 0$ at right angles.

Art. We may for ϕ_1, ϕ_2 write $\phi_1 + 2n_1\pi, \phi_2 + 2n_2\pi$ respectively, n_1, n_2 being arbitrary integers; and it would thus at first sight appear that there could be drawn through the two points a doubly infinite series of geodesics. There is, in fact, a singly infinite system of geodesics: to show how this is, write for shortness A, A_1, A_2, a, a_1, a_2 for ${}^2\theta, {}^2\theta_1, {}^2\theta_2, 2n\pi, 2n_1\pi, 2n_2\pi$ respectively.

Art. Observe here that, writing $\theta_2, \phi_2 = \theta_1 + d\theta_1, \phi_1 + d\phi_1$, and therefore δ small so that $\cosh \delta = 1 + \frac{1}{2}\delta^2$, we obtain

$$\delta^2 = \sin^2 \theta_1 d\phi_1^2 + \cot^2 \theta_1 d\theta_1^2,$$

agreeing with the expression for $dx^2 + dy^2 + dz^2$.

If in the form first obtained we write $A_1 = {}^2\theta_1, A_2 = {}^2\theta_2$, we find

$$\cosh \delta = \frac{\sqrt{\phi_1^2 + A_1} + \sqrt{\phi_2^2 + A_2} - 2\phi_1\phi_2}{\sqrt{A_1 A_2}},$$

which is a convenient form.

Art. In like manner, for the distance of two points on the sphere, we have

$$\cos \delta = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(\phi_1 - \phi_2),$$

whence

$$\sin \frac{1}{2}\delta = \frac{\sqrt{\sin^2 \frac{1}{2}(\phi_1 - \phi_2) \cdot \sin^2 \frac{1}{2}(\theta_1 - \theta_2) + \dots}}{\sqrt{\dots}}.$$

We thence obtain

$$\cos \delta = \frac{C_1 C_2 + 2\phi_1(\phi_1 C_2 + \phi_2 C_1) + 4\phi_1 \phi_2(\phi_1^2 + A_1)(\phi_2^2 + A_2)}{\sqrt{C_1^2 - 4A_1 B_1} \cdot \sqrt{C_2^2 - 4A_2 B_2}},$$

which is

$$\frac{\dots}{\sqrt{C_1^2 - 4A_1 B_1} \cdot \sqrt{C_2^2 - 4A_2 B_2}} = \cos \delta,$$

as above, the equality of the two numerators depending on the identity

$$\{A_1 + B_1(\phi_1^2 + A_2) + \phi_1 \phi_2\}B_2 + \{A_2 + B_2(\phi_2^2 + A_1) + \phi_1 \phi_2\}B_1 = 0.$$

Art. In particular, if we consider the triangle formed by the intersections of the three geodesics, we have the formulae

$$\cosh a = \frac{A_2 + A_3}{2\sqrt{A_2 A_3}}, \quad \sinh a = \frac{\sqrt{(A_2 + A_3)^2 - 4A_2 A_3}}{2\sqrt{A_2 A_3}}, \quad \tanh a = \frac{\sqrt{(A_2 + A_3)^2 - 4A_2 A_3}}{A_2 + A_3}$$

or, reducing these by the relation $\phi_2^2 + A_2 = A_3$, they become

$$\cosh a = \frac{A_2 + A_3}{\sqrt{(A_2 + A_3)^2 - 4A_1 A_2}}, \quad \sinh a = \frac{\sqrt{(A_2 + A_3)^2 - 4A_1 A_2}}{\sqrt{\dots}}, \quad \tanh a = \frac{\sqrt{\dots}}{A_2}$$

and so for b and c :

$$\cosh b = \frac{A_3 + A_1}{\sqrt{(A_3 + A_1)^2 - 4A_2 A_3}}, \quad \sinh b = \frac{\sqrt{(A_3 + A_1)^2 - 4A_2 A_3}}{\sqrt{\dots}}, \quad \tanh b = \frac{\sqrt{\dots}}{A_3}$$

$$\cosh c = \frac{A_1 + A_2}{\sqrt{(A_1 + A_2)^2 - 4A_3 A_1}}, \quad \sinh c = \frac{\sqrt{(A_1 + A_2)^2 - 4A_3 A_1}}{\sqrt{\dots}}, \quad \tanh c = \frac{\sqrt{\dots}}{A_1}$$

We have, moreover,

$$\cos B = \frac{\dots}{\sqrt{C_3^2 - 4A_3 B_3} \cdot \sqrt{C_1^2 - 4A_1 B_1}}, \quad \cos C = \frac{\dots}{\sqrt{C_1^2 - 4A_1 B_1} \cdot \sqrt{C_2^2 - 4A_2 B_2}}$$

Art. The element A_1 to lie in an arbitrary direction A_1 through A ; the area about A will then coincide with the area about A .

The analytical theory is at once derived from that for the sphere, viz. we have a rectangular transformation

$$\begin{pmatrix} X & Y & Z \\ x & y & z \end{pmatrix},$$

where the coefficients are such that identically $X^2 + Y^2 + Z^2 = x^2 + y^2 + z^2$; in fact, the coefficients are connected by six equations only, the system thus depending on three arbitrary parameters.

Art. The coordinates of a point on the pseudosphere may be expressed in the form

$$x = u \cos v, \quad y = u \sin v, \quad z = u - \tanh u,$$

where u, v are parameters; and we then have

$$dx^2 + dy^2 + dz^2 = du^2 + u^2 dv^2.$$

The whole surface corresponds to $-\infty < u < \infty$, $0 \leq v < 2\pi$, but the same point of the surface corresponds to the values (u, v) and $(u, v + 2n\pi)$; there is thus a helicoidal disposition of the meridians.

Art. The geodesic lines on the pseudosphere may be represented in the form

$$v = \int \frac{C du}{u\sqrt{2u - C^2}},$$

where C is a constant; or, what is the same thing,

$$v = \sin^{-1} \left(\frac{C}{\sqrt{1 - C^2}} \sinh u \right) + \text{const.}$$

Art. The area of a triangle on the pseudosphere is equal to $\pi - (A + B + C)$, where A, B, C are the angles of the triangle; and the whole area of the surface is infinite.

Art. The pseudosphere is a surface of constant negative curvature $= -1$; and it is the only surface of revolution which has this property. The circle at infinity ($u = -\infty$) is an asymptotic line of the surface, and the tangent plane at any point of this circle is perpendicular to the axis of revolution.

0.7 Paper 828. A Memoir on Seminvariants (pp. 240–250)

[From the *American Journal of Mathematics*, vol. vii. (1885), pp. 1–13.]

Art. To verify that this is a seminvariant, observe that the value may be written

$$= \frac{1}{4}\{-(e - 4bd + 3c^2) + 6(c - 6 \cdot \frac{22}{2})\} = \frac{1}{4}\nabla(e - 4bd + 3c^2),$$

&c.; and observe further that the forms 2, 4 and 22, are connected by the identical relation

$$2 \cdot 2 = 4 + 2 \cdot 22.$$

Art. We conclude that the theory of seminvariants is a part of that of symmetric functions. I take the opportunity of remarking that (the subject of the memoir being seminvariants) I use in general non-unitary symbols, even in cases where the restriction is unnecessary, and the symbols might have contained unity.

Art. We may, taking in the first instance the numbers $l, m, n, p, q, \dots$ to be all different, developpe an algorithm as follows:

$$l \cdot m = (l + m)lm + lm,$$

$$lm \cdot n = (l + n)l(m + n) \cdot n + (m + n)l(m + n) \cdot n, \quad \text{that is,}$$

$$lm \cdot n = (l + n)m, \quad (l + n)l(m + n) \cdot n$$

and so in other cases;

$$\begin{aligned} lm \mid np &= (l + p)(m + q)n - (l + q)(m + p)n \\ &= (l + q)m(n + p) - l(m + q)(n + p) \\ &= (l + p)mnp - (l + q)mnp \\ &= l(m + p)q - l(m + q)p \\ &= lm(n + p)q - lm(n + q)p \end{aligned}$$

and so on.

Art. Observe that, if the two factors contain i numbers and j numbers respectively, $i >$ or $= j$, then in the product we have $[i]_j$

terms each containing i numbers, where $[i]_j = i(i-1)\cdots(i-j+1)$: so that the whole number of terms in the product is

$$\{i, j\} = [i]_j + \binom{j}{1}[i]_{j-1} + \cdots + [i]_1 + 1.$$

We may, if we please, take the smaller number first, and we then have

$$\{j, i\} = \text{the same series:}$$

in fact the $\{j, i\}$ series begins with zero terms, but following these we have terms which are identical each to each with those of the $\{i, j\}$ series.

Thus

$$\{5, 3\} = [5]_3 + 3[5]_2 + 3[5]_1 + [5]_0 = 60 + 60 + 15 + 1 = 73.$$

Art. In forming a product $lmn\cdots pqr\cdots$, we may have equalities among the numbers $l, m, n, \dots$ of the first symbol, and also between the numbers $p, q, r, \dots$ of the second symbol: moreover (whether there are or are not any such equalities), we may have equalities presenting themselves between the numbers $l + p, m + q, \&c.$ of any symbol $(l + p)(m + q)\cdots$ on the right-hand side of the equation: and the process must be performed so as to take account of all these equalities.

The actual process is best shown by an example: say we require the product $3332 \cdot 322$, which is of the degree-weight 6.18.

Art. To form the product $3332 \cdot 322$: the numbers of the first symbol being 3, 3, 3, 2 and those of the second symbol 3, 2, 2. We form the terms according to the algorithm, and then collect the equal terms.

The terms of the product are:

- $3332 \cdot 322 = 3333222 + \cdots$ (terms obtained by the algorithm)
- We obtain terms with frequencies: 1, 3, 3, 6, 3, 3, 1, etc.
- The sum of all frequencies = 73, the value of $\{4, 3\}$.

Art. We next form the column of symmetric functions by adding to each line the term 3332: to avoid accidental errors of addition, it is proper to verify for each of these that the weight is the sum, = 18, of the weights 11 and 7 of the two factors respectively.

It is to be observed that the whole coefficient of a^6 in the product 3332·322 is obtained by decapitation of each factor, viz. the coefficient is = 332 · 22.

Art. In explanation as to the frequency, observe that out of the $\binom{3}{2}$ twos we take any $\binom{2}{1}$ twos and place them in any order under any $\binom{3}{1}$ of the three twos. The number of combinations taken is $\binom{3}{2}$, which (in $\binom{2}{1}$ orders) gives $\frac{3!}{2!1!} \cdot 2 = 6$ sets to be placed under sets out of the three twos: we thus have the foregoing coefficient of frequency $\frac{\lambda \cdot \alpha}{\lambda \cdot \beta}$, where as mentioned above α , &c. are written to denote $[\alpha]$, &c.

Art. We may in like manner find a formula for the product $3^\alpha 2^\beta 0^\gamma \cdot 3^\lambda 2^\mu 5^\nu$. Taking $\gamma = \alpha + y + z$, any partition of 7, and $\delta = p + q + r$, any partition of 7, the formula finally is

$$A = \frac{\gamma!}{y! z!} \cdot \frac{\lambda!}{p! q!} \cdot \frac{\alpha!}{r! s!},$$

where

$$B = \gamma + p, \quad C = \gamma + q, \quad D = z + p, \quad E = z + q,$$

or, as this may also be written,

$$A = \frac{B! D! E!}{y! z! p! q! r! s!},$$

so that the coefficient A is in fact the product of three binomial coefficients: it must, however, be recollected that the same term $6^A 5^B 4^C 3^D 2^E$ may occur more than once, with different coefficients A , so that in the final result, when these terms are united together, the numerical coefficients are not each of them of the form in question.

Art. The limits of the summation are conveniently defined by means of the inequalities

$$B = \gamma + p, \quad C = \gamma + q, \quad D = z + p, \quad E = z + q,$$

several of these giving, as will be seen, zero values, and the others giving the values already obtained for the coefficients of the several terms.

Art. It is quite possible that abbreviations and verifications might be introduced, but the process as it stands seems to be at once less expeditious and less safe than the one first made use of.

Art. Particular cases of the general formula are

$$2^\lambda \cdot 2^\mu = 2^A 4^C 2^E; \quad E = \beta + \delta - 2C, \quad \text{and thence} \quad 4C + 2E = 2(\beta + \delta),$$

$$\frac{E}{A} = \frac{\beta - C \cdot \delta - C}{\beta \cdot \delta},$$

which agrees with a result before obtained.

Again,

$$32^\lambda \cdot 2 = 2^A 5^B 4^C 3^D 2^E; \quad D = \alpha - B, \quad E = \beta + \delta - B - 2C,$$

and thence

$$5B + 4C + 3D + 2E = 3\alpha + 2\beta + 2\delta,$$

$$\frac{E}{A} = \frac{\beta - C \cdot \delta - B - C}{\beta \cdot \delta}.$$

Art. We may, in like manner with the formula for $3^\alpha 2^\beta 3 \cdot 3^\lambda 2^\mu$, obtain the coefficient of a^6 : in the former case the degree is reduced to 5, in the present case it remains = 6.

In every case we decapitate the symbol by striking out the highest number: in the case of two or more equal numbers, one only of these being struck out.

Observe that by decapitation we always diminish the weight, but we do or do not diminish the degree.

Art. In a product such as $3332 \cdot 322$ we obtain in like manner the whole coefficient of a^6 by the decapitation of each factor, viz. the coefficient is = $332 \cdot 22$: and in any equation the coefficient of a^6 is = 652, the only difference being that, while in the former case the degree is reduced to 5, in the present case it remains = 6.

Art. That function which by the equation of the deg. weight 5.12 is expressed to be = 0: and hence Ω , qua symmetric function cannot contain $\beta^6, \gamma^7, \dots$; viz. Ω is a symmetric function of the degree 5 at most: the congruence $H =$ thus means, $\Omega =$ a properly determined function of the degree 5 at most.

Obviously by development of the term $3332 \cdot 322$, that is, by substituting for this term its value as given by the equation of the deg. weight 6.18, the function of the degree 5 at most would be found to be = the sum of those terms in the expression of $3332 \cdot 322$.

**827] ON THE NON-EUCLIDIAN PLANE GEOMETRY.
231**

it would be possible to deduce the equations which connect the new coefficients; but these are more easily obtained from the consideration that we must have identically

$$P_1Q_1 - Z_1^2 = PQ - Z^2;$$

the equations are thus found to be

$$\begin{aligned} A''^2 - AA' &= 0, & B''^2 - BB' &= 0, & C''^2 - CC' &= 1, \\ 2A''B'' - AB' - A'B &= -1, & 2A''C'' - AC' - A'C &= 0, & 2B''C'' - BC' - B'C &= 0. \end{aligned}$$

17. The general theory of the transformation of a quadric function into itself enables us to express the coefficients in terms of three arbitrary parameters. There is no difficulty in working out the formulae, and we finally obtain

$$\begin{aligned} \Omega P_1 &= -(\nu + 1)^2 P - \lambda^2 Q + 2\lambda(\nu + 1)Z, \\ \Omega Q_1 &= -\mu^2 P - (\nu - 1)^2 Q + 2\mu(\nu - 1)Z, \\ \Omega Z_1 &= -\mu(\nu + 1)P - \lambda(\nu - 1)Q + (-1 + \nu^2 + \lambda\mu)Z; \end{aligned}$$

and conversely

$$\begin{aligned} \Omega P &= -(\nu - 1)^2 P_1 - \lambda^2 Q_1 + 2\lambda(\nu - 1)Z_1, \\ \Omega Q &= -\mu^2 P_1 - (\nu + 1)^2 Q_1 + 2\mu(\nu + 1)Z_1, \\ \Omega Z &= -\mu(\nu - 1)P_1 - \lambda(\nu + 1)Q_1 + (1 + \nu^2 + \lambda\mu)Z_1, \end{aligned}$$

where $\Omega = -1 + \nu^2 - \lambda\mu$: it can be at once verified that each of the two sets of formulae does, in fact, give $P_1Q_1 - Z_1^2 = PQ - Z^2$.

18. The pseudosphere is a surface of revolution having for its meridian section the curve $x = \log \cot \frac{1}{2}\theta - \cos \theta$, $y = \sin \theta$. This is a curve symmetrical in regard to the axis of y ; and we obtain the portion of it lying on the positive side of this axis, by giving to θ the series of values $\theta = 0$ to $\theta = 90$; for $\theta = 0$, we have $y = 0$, $x = \infty$, or the axis of x is an asymptote; for $\theta = 90$, $x = 0$, $y = 1$, the point being a cusp of the curve. The geometrical definition is that the portion of the tangent included between the curve and the axis of x has the constant length = 1; the inclination of

the tangent is in fact $= \theta$. We have $dx = \frac{\cos^2 \theta d\theta}{\sin \theta}$, $dy = \cos \theta d\theta$;

and thence $ds = \cot \theta d\theta$, and the length in question is $\frac{y ds}{dy} = 1$.

The curve may be constructed graphically: take (fig. 2) the distance $BO = 1$, on OB , B_1 very near to B , and then $B_1O_1 = 1$; on

[827 ON THE NON-EUCLIDIAN PLANE GEOMETRY.
232

O_1B_1 ; B_2 very near to B_1 , and then $B_2O_2 = 1$, and so on; the curve is shown on a larger scale in fig. 3, p. 235.

But the curve may also be laid down numerically; writing $\alpha = \frac{1}{2}\pi - \theta$, so that α is the inclination of the tangent to the axis of y , we have $x = \log \tan(\frac{1}{4}\pi + \frac{1}{2}\alpha) \sin \alpha$, $y = \cos \alpha$, where $\log \tan(\frac{1}{4}\pi + \frac{1}{2}\alpha)$, the hyperbolic logarithm (which has been the signification of $\log$ throughout), is the function tabulated Tab. IV., Legendre's *Traité des Fonctions Elliptiques*, t. II. pp. 256–259.

We may hence obtain the values of the coordinates as follows:

$\alpha = 90 - \theta$	0	10	20	30	40	50	60	70
x	∞	2.3400	1.2096	0.7154	0.4113	0.2110	0.0875	0.0242
y	1	0.9848	0.9397	0.8660	0.7660	0.6428	0.5000	0.3420
α	72	74	76	78	80	82	84	86
x	0.0160	0.0092	0.0045	0.0018	0.0005	0.0001	0.0000	0.0000
y	0.3090	0.2756	0.2419	0.2079	0.1736	0.1392	0.1045	0.0698

**827] ON THE NON-EUCLIDIAN PLANE GEOMETRY.
233**

If $E = 0$, we have $A + C\dot{\phi} = 0$, that is, $\dot{\phi} = \text{const.}$, which belongs to the meridians; if $B \neq 0$, we may by a mere change of ϕ , that is, of the initial meridian, reduce the form to $A + B(\phi^2 + \theta^2) = 0$, which is the equation of a geodesic cutting at right angles the meridian $\phi = 0$; writing herein $\sin \theta = \frac{1}{r}$, we have $A + B(\phi^2 + \frac{1}{r^2}) = 0$, which is the equation in the polar coordinates r, ϕ of the projection of the geodesic on the equatorial plane x ; putting herein for greater convenience $B = Ak^2$, we have $\frac{1}{r^2} = \frac{k^2}{1 - k^2\phi^2}$: we require only such portions of the curves as lie within the unit-circle, and need therefore attend only to those for which k is not greater than 1.

19. And in any such curve consider ϕ as extending from $\phi = 0$ to $\phi = \gamma$, writing K this last value $= \gamma$, we have $k = \frac{\gamma}{\sqrt{1 + \gamma^2}}$; if

$\gamma < \pi$, that is, $k < \frac{\gamma}{\sqrt{1 + \pi^2}}$, the curve is a mere arc cutting at right angles (at the distance $r = k$ from the centre) the meridian $\phi = 0$, and extending itself out on each side to meet the unit-circle in the points $\phi = \gamma, \phi = -\gamma$ respectively; in the case $\gamma = \pi$, that is, $k = \frac{\pi}{\sqrt{1 + \pi^2}}$, the two points $\phi = \gamma$ come together at the point $\phi = \pi$,

or the curve becomes a loop; and for larger values, $k = \frac{2\pi}{\sqrt{1 + 4\pi^2}}$ to

$k = \frac{3\pi}{\sqrt{1 + 9\pi^2}}$, we have the two branches crossing: each other on

the meridian $\theta = \pi$ at the distance $r = \frac{k}{\sqrt{1 - k^2\pi^2}}$ from the centre

and then extending themselves in the opposite semicircles, so as to meet the unit-circle at the points $\phi = \gamma$. And we have thus another

critical value $k = \frac{2\pi}{\sqrt{1 + 4\pi^2}}$, for which the two branches having thus

crossed each other come to unite themselves at the point $\phi (= 2\pi)$

$= 0$ of the unit-circle; and in like manner the critical values $\frac{3\pi}{\sqrt{1 + 9\pi^2}},$

$\frac{4\pi}{\sqrt{1 + 16\pi^2}},$ &c.: for a value of k between such limits, the branch is

a spiral having a determinate number of convolutions, and the two branches cross each other always on the radii $\phi = 0$ and $\phi = \pi$ respectively.

20. Let ψ denote the inclination of the radius vector to the normal, or, what is the same thing, that of the element of the circular arc to the tangent; we have $\tan \psi = \frac{dr}{r d\phi}$, and $\frac{dr}{r d\phi} = \frac{k^2 \phi}{r^2}$, $= r^2 \phi$, that is, $\tan \psi = r^2 \phi$. At the intersection with the unit section $r = 1$, and therefore $\tan \psi = \phi$; moreover putting $\theta = \cos K$, so that the equation of the curve now is $r^2 = \frac{\theta}{\theta - \cos^2 K}$, then for $r = 1$ we have $\theta = \tan K; 1 - \phi^2 \cos^2 K$

**[827 ON THE NON-EUCLIDIAN PLANE GEOMETRY.
234**

and therefore $\tan K = \pi$, we have $K = 72\ 20'$ nearly: the complement hereof $17\ 40'$ is thus the angle at which each branch of the loop cuts the meridian $\phi = \pi$.

21. To obtain another datum convenient in tracing the curve, I write $\phi_0 = \tan K$ for the value of ϕ at the unit-circle; and introducing for the moment the rectangular coordinates $X = r \sin \phi$, $Y = 1 - r \cos \phi$, then we easily find

$$\frac{dY}{dX} = \frac{r \sin \phi_0 - \cos \phi}{r \cos \phi_0 + r^3 \sin \phi_0 \sin \phi},$$

and thence, for the equation of the tangent at the point on the unit-circle,

$$\sin \phi_0 \cos \phi \cdot (Y - 1 + \cos \phi_0) = -\frac{1}{r^2}(X - \sin \phi_0 \cos \phi_0).$$

For the tangent at the point of intersection with the radius $\phi = 0$, or say the *apse*, we have $y = 1 - \cos K$; and hence, at the intersection of the two tangents,

$$x = \sin \phi_0 + \frac{\cos \phi_0 + \sin \phi_0}{\sin \phi_0 \cos \phi_0}(\cos \phi - \cos K),$$

which, putting therein $\phi_0 = \tan K$, becomes

$$\cos K \{1 - \cos(\phi - K)\} = \frac{x}{\sin(\phi - K)} = \cos K \tan \frac{1}{2}(\phi - K),$$

where ϕ is given in terms of ϕ_0 by the just-mentioned equation $\phi_0 = \tan K$.

We have $y = 1 - \cos K$, $x = \cos K \tan \frac{1}{2}(\phi - K)$, for the locus of the intersection of the two tangents; this is easily seen to be a curve having a cusp at the unit-circle.

22. Fig. 3 shows the curves for the values $\phi_0 = \tan K$

Fig. 2.

[Figure: Pseudosphere meridian construction showing points O, O_1, O_2, O_n on the x -axis and B, B_1, B_2, B_n on the curve]

x

y

Fig. 3.

[Figure: Projections of geodesics on the unit circle, showing curves for various values of ϕ_0 or K , including the loop case]

**827] ON THE NON-EUCLIDIAN PLANE GEOMETRY.
235**

$\phi_0 = 90$, we join with the centre of the circle the point 90 of the tangent, thus determining on the unit-circle a point belonging to the angle $K = 57\,31'$; at this point we draw parallel to the tangent a line which is the tangent at the lowest point; the curve passes through the point 90 on the unit-circle, and there cuts the circle at the angle $\kappa = 57\,31'$ (or, what is the same thing, the radius at the complementary angle), and we have thus the tangent at the point 90 of the unit-circle; it will be noticed that this meets the tangent at the apse at a point near to this apse, so that the arc as determined by the two tangents is for a large part of its course nearly a right line; this is still more the case for smaller values of ϕ_0 or κ , while for larger values the deviation increases, but in the neighbourhood of the unit-circle the form is always nearly rectilinear.

I show in the same figure the form of the curve for $\phi_0 = 300$, $= 5.2359877$, $= \tan \kappa$, that is, $\kappa = 79\,11'$, $r = \cos \kappa = 0.1876670$, the value at the apse: the construction for the tangent at the unit-circle is the same as before, but in order to lay down the curve with tolerable accuracy we require also the value of r at the node on the meridian $\phi = 180$; this is, of course, given by $r = \frac{\cos \kappa}{\sqrt{1 - \pi^2 \cos^2 \kappa}}$, that is, $r = \frac{\cos \kappa}{\cos \alpha}$, if $\pi \cos \kappa = \sin \alpha$; whence without difficulty $r = 0.23236$, the value at the node.

23. The curves shown in the figure are projections upon the plane of the unit-circle, viz. they are the projections on this plane of the geodesics, which cut at right angles a given meridian; but bearing in mind the form of the meridian, it is easy, by means of the projection, to understand the actual forms on the surface of the pseudosphere. A point near the centre of the figure represents a point high up on the surface; and in any radius the portions near the centre are the more foreshortened in the figure, and represent greater distances on the surface. Each geodesic cutting the meridian at right angles at the apse descends symmetrically on the two sides, reaches ultimately—it may be after many convolutions—the unit-circle; the meridian itself is a limiting or special form of geodesic. The unit-circle is not properly a geodesic, but it is an envelope of geodesics.

24. To obtain all the geodesics, we have to consider the geodesics which cut at right angles a given meridian; and then to imagine this

meridian (with the geodesics which belong to it) turned round so as to occupy successively the positions of all the other meridians. The same remark applies of course to the projections; the figure shows the projections cutting at right angles a given radius of the circle; and this radius (with the projections belonging to it) is then to be turned round so as to occupy successively the positions of all the other radii. We may imagine the several geodesics turned round separately, each through a different angle, so as to bring each of them to pass through one and the same point of the surface; we have then the geodesics drawn in all directions through this point of the surface; doing the same thing with the projections, we have, it is clear, the projections of the geodesics drawn in all directions through the point. It is easy, by drawing the projections each on a separate circle of paper, and passing a pin through the centres, to form a model by means of which an accurate figure of the projections of the geodesics through any given point can at once be exhibited.

25. Taking Q a given point, and EG the radius (or meridian) through Q , then the projections through Q are symmetrically situate in regard to this radius EG ; the curve which has its apse at Q is the projection which cuts the radius at right angles; the projections which lie outside this curve, that is, on the side towards E , meet the unit-circle on the two sides of E , in points A, D , belonging each of them to a positive value of κ , which may be anything from 0 to 90; the projections which lie inside the curve meet the unit-circle in points B, C , belonging to the values of κ for which ϕ_0 lies between 180 and 360, and are therefore curves which cross the radius EG .

There are in the whole four projections through Q which touch the unit-circle, corresponding to the values $\phi_0 = 180$, $\phi_0 = 360$, and the coincident values $\phi_0 = 0 = 720$; in the case $\phi_0 = 180$ (or what is the same thing, $\kappa = 90$, the apse is at the distance $r = 0$ from the centre), the curve is a loop touching the unit-circle at the extremity E of the radius EG , and the branch on the same side with the apse is really a distinct curve intersecting the loop; in the case $\phi_0 = 360$ the curve is a loop touching at the other extremity G ; and in the case $\phi_0 = 0 = 720$, the curve is the unit-circle itself; or, more accurately, the geodesics become the unit-circle at opposite sides of the surface, meeting each other at the cusp; in the model there is a break, or rather we pass continuously from one circle to the other at opposite points of the unit-circle, or say at the points G and E on the different circles of the model.

**[827 ON THE NON-EUCLIDIAN PLANE GEOMETRY.
236**

The outside projections through Q divide the portion outside the curve, apse at Q , into separate regions; but these projections (taken the whole way from Q to the unit-circle) do not each of them form a single geodesic; any such projection is really the projection of two geodesics, viz. a geodesic from Q to the unit-circle at A , and a different geodesic from Q to the unit-circle at D ; the model shows the course of these geodesics from Q to the unit-circle, each as a branch of a geodesic which has its apse on the radius EG produced beyond Q . It is to be added that these outside projections, each as belonging to two geodesics, do not each of them constitute a boundary between two regions; the case is really the same as that of a meridian on the sphere in regard to the geodesics through a point Q of it: the meridian itself is a geodesic through Q , and it is also the limiting form of geodesics on each side of it; and in the projection it divides the regions on the two sides of it, but it does not constitute a boundary between two distinct regions.

The inside projections through Q are (on account of the crossing) much more complicated; it would require a careful examination of the model to understand their true form and connexions.

26. It is interesting to connect the theory of the geodesics of the pseudosphere with the general theory of geodesics. Starting with the form

$$ds^2 = \cot^2 \theta d\theta^2 + \sin^2 \theta d\phi^2,$$

$= E d\theta^2 + 2F d\theta d\phi + G d\phi^2$, we have $E = \cot^2 \theta$, $F = 0$, $G = \sin^2 \theta$; and therefore $E + 1 = \frac{1}{G}$, or $E = \frac{1}{G} - 1$, and the differential equation of the geodesic becomes

$$\phi' \{2GE' - GE_1\} \theta' + GG_2 \phi'^2 + 2EG(\theta' \phi'' - \theta'' \phi') = 0,$$

that is,

$$\phi' \{(2EG_1 - GE_1)\theta'^2 + GG_2 \phi'^2\} + 2EG(\theta' \phi'' - \theta'' \phi') = 0,$$

where $E_1 = \frac{dE}{d\theta}$, $G_1 = \frac{dG}{d\theta}$, $G_2 = \frac{dG}{d\phi}$; and writing herein

$$E = \frac{1 - G}{G}, \quad E_1 = -\frac{G_1}{G^2},$$

we have

$$2EG_1 - GE_1 = \frac{2(1-G)}{G}G_1 + \frac{G_1}{G} = \frac{(3-2G)G_1}{G}.$$

Moreover, from $G = \sin^2 \theta$, we find $G_1 = 2\sqrt{G(1-G)}$; and the equation becomes

$$\phi' \left\{ \frac{(3-2G)G_1}{G} \theta'^2 + GG_2 \phi'^2 \right\} + 2(1-G)(\theta' \phi'' - \theta'' \phi') = 0.$$

Introducing here G in place of θ by the equation $G = \sin^2 \theta$, we have $\frac{dG}{d\theta} = G_1$, $\frac{d^2G}{d\theta^2} = G_{11}$, and the equation thus becomes

$$\phi' \left\{ (3-2G) \frac{dG}{d\theta} \theta'^2 + G^2 G_2 \phi'^2 \right\} + 2G(1-G)(\theta' \phi'' - \theta'' \phi') = 0.$$

**827] ON THE NON-EUCLIDIAN PLANE GEOMETRY.
237**

the arc EG . This of course corresponds to the Lobatschewskian theory, according to which we have through a point Q to the extremities at infinity of a line EG , two distinct lines QB , QC , said to be the *parallels* through Q of the line BC ; and which are such that any line through Q intermediate to QB , QC meets the line BC ; while any line through Q extramediate to QB , QC does not meet the line BC .

27. Reverting to the equations of No. 4, these may be written

$$\begin{aligned}\Omega P_1 &= -(\nu + 1)^2 P - \lambda^2 Q + 2\lambda(\nu + 1)Z, \\ \Omega Q_1 &= -\mu^2 P - (\nu - 1)^2 Q + 2\mu(\nu - 1)Z, \\ \Omega Z_1 &= -\mu(\nu + 1)P - \lambda(\nu - 1)Q + (-1 + \nu^2 + \lambda\mu)Z.\end{aligned}$$

Taking P , Q , Z as the coordinates of a point in ordinary space, and P_1 , Q_1 , Z_1 as those of another point, these equations establish a correspondence between the two points; to any given position of either point there corresponds a single position of the other point. Starting from the quadric surface $PQ - Z^2 = 0$, this is transformed into the quadric surface $P_1Q_1 - Z_1^2 = 0$; and conversely. In particular, the tangent plane at any point of either quadric corresponds to the tangent plane at the corresponding point of the other quadric. But the whole theory is really included in the less general theory of the linear transformation of a quadric function into itself; and I do not at present enter into it any further.

Cambridge, England, 17th March, 1884.

828.

A MEMOIR ON SEMINVARIANTS.

[From the *American Journal of Mathematics*, t. VII. (1885),
pp. 1—25.]

Introductory. Art. Nos. 1 to 3.

1. A VERY remarkable discovery in the Theory of Seminvariants has been recently made by Capt. MacMahon, viz. considering the equation

$$0 = 1 + b\frac{x}{1} + c\frac{x^2}{1.2} + d\frac{x^3}{1.2.3} + \&c.,$$

and its roots $\alpha, \beta, \gamma, \dots$, as defined by the identity

$$1 + b\frac{x}{1} + c\frac{x^2}{1.2} + \&c. = 1 - \alpha x \cdot 1 - \beta x \cdot 1 - \gamma x \dots,$$

then, any symmetrical function of the roots being represented by a partition symbol in the usual manner, $1 = \Sigma\alpha$, $2 = \Sigma\alpha^2$, 11 or $1^2 = \Sigma\alpha\beta$, &c., the theorem is that any symmetric function represented by a non-unitary symbol (or symbol not containing a 1), say for shortness any non-unitary symmetric function, is a seminvariant in regard to the coefficients $1, b, c, d, e$, &c.

We have for instance

$$\begin{aligned} 2 &= -(c - b^2), \\ 3 &= -\frac{1}{2}(d - 3bc + 2b^3), \\ 4 &= \frac{1}{6}(-e + 4bd + 3c^2 - 12b^2c + 6b^4), \end{aligned}$$

and so on.

2. The proof is very simple; writing $x = y - b$, the equation becomes

$$0 = 1 + b\frac{(y-b)}{1} + c\frac{(y-b)^2}{1.2} + d\frac{(y-b)^3}{1.2.3} + \&c.,$$

or, what is the same thing,

$$0 = A + By + Cy^2 + Dy^3 + \&c.,$$

where A, B, C, D , &c. are functions of b, c, d, e , &c.; then the coefficients A, B, C, D , &c. are seminvariants, and therefore also any function of these coefficients which does not contain the absolute coefficient A ; that is, any non-unitary symmetric function of the roots of the equation in y , or, what is the same thing, of the roots of the original equation in x .

where to verify that this is a seminvariant, observe that the equation in y is $0 = A + By + Cy^2 + Dy^3 + \&c.$, and that the coefficients $C, D, E, \&c.$ (the coefficients after B) are seminvariants; and further that A does not enter into the expression of the non-unitary symmetric function.

3. To further explain the notion of a non-unitary symmetric function, observe that by the theorem the number of seminvariants of a given weight is equal to the number of non-unitary partitions of that weight; thus for weight = 6 the partitions are

$$6, \quad 42, \quad 33, \quad 222, \quad 52, \quad 43, \quad 322, \quad 222,$$

eight in all, which is the number of the seminvariants of the weight 6; but among these there is only one partition into two equal parts, viz. 33, and there is thus only one seminvariant of the weight 6 which is a perfect square. Seminvariants which are not non-unitary symmetric functions are called by Sylvester *Subinvariants*; they are symmetric functions which contain a unit: the numbers are for weight = 6 the partitions 51, 411, 321, 3111, 2211, 21111, 111111, seven in all, and these are the subinvariants of the weight 6. The whole number of partitions of 6 is 11, which is the number of seminvariants + subinvariants, that is, of symmetric functions of the weight 6. In the case of a subinvariant the expression as a function of the coefficients will contain A (and will in fact be a rational and integral function of $A, B, C, D, E, F, G, \&c.$).

Definition of Seminvariant. Art. Nos. 4 to 6.

4. I give the definition of a seminvariant. Starting from the binary quantic

$$(a, b, c, d, \dots \setminus x, y)^n,$$

writing for convenience $(a, b, c, d, \dots)$ to denote the coefficients (taking them to be independent of each other), then a seminvariant is a function of the coefficients which is reduced to zero by the operation

$$a\partial_b + 2b\partial_c + 3c\partial_d + \dots,$$

or, what is the same thing, it is a function of the coefficients $(a, b, c, d, \dots)$ which is reduced to zero by the operation

$$\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots + nb\partial_l,$$

or, what is again the same thing, it is a function S of the coefficients, such that we have

$$(a, b, c, d, \dots \partial_y, -\partial_x)(a, b, c, d, \dots x, y)^n \cdot S = 0.$$

5. We may, taking in the first instance $(a, b, c, \dots)$ as the coefficients of the several powers of x , write

$$0 = 1 + b\frac{x}{1} + c\frac{x^2}{1.2} + d\frac{x^3}{1.2.3} + \&c.,$$

where 1 is put for a , or, what is the same thing, we may write

$$(a, b, c, d, e, f, \& 1, x)^6 = 0,$$

or for the general case

$$(a, b, c, d, \dots 1, x)^n = 0.$$

The seminvariants of this equation are the same as those of the binary quantic $(a, b, c, d, \dots x, y)^n$ of the order n ; only we consider only the seminvariants of a degree in the coefficients which does not exceed n , and only those of a weight which does not exceed the weight of the last coefficient l , that is $\frac{1}{2}n(n-1)$.

828] A MEMOIR ON SEMINVARIANTS. 241

6. Observe that, if the equation in x be $0 = (a, b, c, \dots \setminus 1, x)^n$, then the equation in y obtained by the change $x = y - \frac{b}{a}$ is $0 = (a, B, C, D, \dots \setminus 1, y)^n$, where the new coefficients B, C, D , &c. are given functions of a, b, c , &c.; the seminvariants are the same for the two equations, but for the equation in y the coefficient B vanishes; any seminvariant of the equation in x is thus a seminvariant of the equation in y , and conversely any seminvariant of the equation in y is a seminvariant of the equation in x ; the seminvariants of the equation in y are functions of a, C, D, E , &c. which are reduced to zero by the operation $2C\partial_D + 3D\partial_E + \dots$; or, what is the same thing, the seminvariants of the equation in x are functions of the original coefficients a, b, c , &c. which are reduced to zero by the operation

$$\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots,$$

and which, when for b is substituted $-\frac{B}{a}$, become functions of a, C, D, E , &c. which are reduced to zero by the operation $2C\partial_D + 3D\partial_E + \dots$.

The Seminvariants are the Non-unitary Symmetric Functions.

Art. Nos. 7 to 12.

7. In forming a product $l.m.n.\dots p.q.r.\dots$, we may have equalities among the numbers $l, m, n, \dots$, or among the numbers $p, q, r, \dots$, or between numbers of the one set and numbers of the other set. Thus in $\{4, 3\}$, we have

$$\begin{aligned}\{4\} &= 4, & 31, & 22, & 211, & 1111, \\ \{3\} &= 3, & 21, & 111,\end{aligned}$$

and thence

$$\begin{aligned}\{4, 3\} &= 43, & 421, & 4111, & 331, & 3211, & 31111, \\ & & 2221, & 22111, & 211111, & 1111111,\end{aligned}$$

ten terms in all, agreeing with the formula $\{4, 3\} = \{7\} + \{6, 1\} + \{5, 2\} + \{4, 3\}$, where the symbols on the right-hand denote the partitions of 7 into one, two, three, and four parts respectively; and where, to verify the number of terms, we have $1 + 3 + 3 + 3 = 10$.

The several terms are represented by their symbols, and the partition symbols are arranged in the order of the large numbers; when two symbols contain the same large numbers, the order is determined by the next largest numbers; and so on.

8. The process for finding the terms of $\{4, 3\}$ is as follows: the first term is obtained by adding the numbers of the second row successively to the corresponding numbers of the first row; viz. $4 + 3 = 7$ gives the first term 7; but $3 + 1 = 4$ would not give a second term, for the rule is that the partial sum must be greater than the corresponding number of the second row; $2 + 1 = 3$, and 3 being not greater than 3 (the corresponding number of the second row), we do not obtain a term; $1 + 1 = 2$, and we have the second term 42; and $1 + 0 = 1$, and we have the third term 411. We then add the numbers of the second row to the corresponding numbers of the first row, beginning always from the first number of the second row; we thus obtain $4 + 2 = 6$, and since the next number of the first row is 1, we add it to 1 obtaining $1 + 1 = 2$, which gives the term 33; we do not obtain any further term from this, since $1 + 1 = 2$ is not greater than 1 (the next number of the second row); and proceeding in like manner we obtain all the remaining terms.

9. For the proof of MacMahon's theorem, I consider a symmetric function represented by a partition symbol, and I show that if the symbol is non-unitary, then the function is a seminvariant; that is, that it is reduced to zero by the operation $\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots$. The proof depends on the expression of the symmetric function as a sum of powers; using the series $s_1 = \Sigma\alpha$, $s_2 = \Sigma\alpha^2$, $s_3 = \Sigma\alpha^3$, &c., we have between these and the coefficients a , b , c , &c. the well-known relations

$$\begin{aligned} as_1 + b &= 0, \\ as_2 + bs_1 + 2c &= 0, \\ as_3 + bs_2 + cs_1 + 3d &= 0, \\ &\vdots \end{aligned}$$

whence, differentiating,

$$\begin{aligned} a\partial_b s_1 + 1 &= 0, \\ a\partial_b s_2 + s_1 &= 0, \\ a\partial_b s_3 + s_2 &= 0, \\ &\vdots \end{aligned}$$

and thence

$$\begin{aligned} \delta s_1 &= -1, \\ \delta s_2 &= -2s_1, \\ \delta s_3 &= -3s_2, \\ &\vdots \end{aligned}$$

or generally $\delta s_n = -ns_{n-1}$, where $s_0 = 1$.

10. If the partition symbol contain any number 1, say the numbers are 1, m , n , &c., then the symmetric function contains the factor s_1 , and we have $\delta(s_1 \cdot s_m \cdot s_n \dots) = s_m s_n \dots \delta s_1 + s_1 \delta(s_m s_n \dots) = -s_m s_n \dots + s_1 \delta(s_m s_n \dots)$, which is not $= 0$; the function is therefore in this case not a seminvariant. But if the symbol is non-unitary, that is, if it does not contain the number 1, then starting from the highest number p , the function is $s_p \cdot s_q \cdot s_r \dots$, and we have $\delta(s_p \cdot s_q \cdot$

$s_r \dots) = s_q s_r \dots \delta s_p + s_p s_r \dots \delta s_q + \dots$; or, what is the same thing, the operation δ performed on the product gives a sum of terms each containing a factor $\delta s_n = -n s_{n-1}$; the number which accompanies this factor s_{n-1} is one of the original numbers, and therefore is not $= 1$; the number $n - 1$ is thus not $= 0$, and the symbol still contains only numbers different from 0. Continuing the operation δ , we always obtain terms in which the factors s have indices different from 0, and the process comes to an end only when one of the indices becomes $= 1$; but in this case the term contains the factor s_1 multiplied by a term having a numerical coefficient, and this coefficient is always $= 0$; the symmetric function is thus reduced to zero by the operation δ , that is, it is a seminvariant.

828] A MEMOIR ON SEMINVARIANTS. 243

11. The proof is thus shown to depend on the law of the numerical coefficients; the law may be verified directly in particular cases. For the symmetric function $2 = s_2$, we have $\delta s_2 = -2s_1$, $\delta^2 s_2 = -2\delta s_1 = 2$, and thence $\delta^3 s_2 = 0$; whence 2 is a seminvariant. For $3 = s_3$, we have $\delta s_3 = -3s_2$, $\delta^2 s_3 = -3\delta s_2 = 6s_1$, $\delta^3 s_3 = 6$, and $\delta^4 s_3 = 0$; whence 3 is a seminvariant. For $22 = s_2^2$, we have $\delta(s_2^2) = 2s_2\delta s_2 = -4s_2s_1$, $\delta^2(s_2^2) = -4(s_1\delta s_2 + s_2\delta s_1) = 4(2s_1^2 + s_2)$, and it may be verified that the further operation δ reduces this to zero.

12. The proof may be presented in a different form; taking the symmetric function to be $p.q.r\dots$, this is a function of the roots $\alpha, \beta, \gamma, \dots$; performing on it the operation $\delta = a\partial_b + 2b\partial_c + 3c\partial_d + \dots$, the result is the same as if we performed on it the operation $-\Sigma\partial_\alpha$; that is, we have $\delta(p.q.r\dots) = -\Sigma\partial_\alpha(p.q.r\dots)$. Now $\Sigma\partial_\alpha(\alpha^p \cdot \beta^q \cdot \gamma^r \dots)$, where the summation extends to all the roots, is equal to $p\alpha^{p-1}\beta^q\gamma^r\dots + q\alpha^p\beta^{q-1}\gamma^r\dots + \dots$; and the terms of this sum are all of them different from zero except when one of the indices $p, q, r, \dots$ becomes $= 0$; but in this case the term contains a factor which is a function of the roots in which one of the roots does not appear, and the summation Σ gives a term multiplied by the number of roots, which is infinite; the proper interpretation is that the coefficient in question vanishes; and we thus see that the operation δ reduces the symmetric function to zero.

Expression of Seminvariants in terms of the Coefficients.

Art. Nos. 13 to 16.

13. The seminvariant $2 = -(c - b^2)$; we may write this $2 = (b, \mathfrak{c} - b, 1)^2$; in fact, this is $(-b)^2c + (-b)(1)c + (1)^2c$, that is, $c - b^2 - b^2 + b^2$, $= c - b^2$, and the sign being changed, this is $-(c - b^2)$; or observing that $b^2 - c$ is the discriminant of $(b, \mathfrak{c} X, Y)^2$, the seminvariant 2 is the discriminant of $(b, \mathfrak{c} X, Y)^2$.

14. The seminvariant $3 = -\frac{1}{2}(d - 3bc + 2b^3)$; we may write this

$$3 = \frac{1}{2}(b, c, \mathfrak{d} - b, 1)^3,$$

or, what is the same thing,

$$3 = \frac{1}{1.2}(b, c, \mathfrak{d} - b, 1)^3.$$

15. The seminvariant 4 is the value of $(b, c, d, \mathfrak{e} - b, 1)^4$ divided by 1.2.3, as is seen by the actual development.

16. And generally a non-unitary symmetric function $p.q.r. \dots$ is equal to a numerical multiple of the seminvariant which is the discriminant of $(b, c, d, \dots \mathfrak{e} - b, 1)^p$, or which is obtained by the development of this discriminant.

17. In explanation as to the discriminant of $(b, c, d, \dots - b, 1)^n$, observe that writing the quantic under the form

$$X^n b + nX^{n-1}Yc + \frac{1}{2}n(n-1)X^{n-2}Y^2d + \dots,$$

the discriminant is a function of b, c, d , &c., which can be expressed as a determinant; but the actual development is effected more conveniently by writing the quantic in the factorial form

$$bX^n + ncX^{n-1}(X+Y) + \frac{1}{2}n(n-1)dX^{n-2}(X+Y)^2 + \dots,$$

and then equating to zero the differential coefficients in regard to X and Y ; the process gives for the discriminant a product of factors which are linear functions of b, c, d , &c.

Seminvariants expressed as Determinants.

Art. Nos. 18 to 21.

18. The seminvariant 2 may be written as a determinant in the form

$$2 = \begin{vmatrix} b & c \\ 1 & b \end{vmatrix}.$$

19. The seminvariant 3 may be written as a determinant; we have

$$3 = -\frac{1}{2}(d - 3bc + 2b^3) = \frac{1}{2} \begin{vmatrix} b & c & d \\ 2 & 2b & 3c \\ 0 & 1 & b \end{vmatrix}.$$

20. And generally any seminvariant may be expressed as a determinant; the form is most easily obtained by means of the differential equation $\delta S = 0$, or $a\partial_b S + 2b\partial_c S + 3c\partial_d S + \dots = 0$, which is satisfied by the seminvariant S .

21. In particular, the seminvariant may be expressed as a determinant formed with the coefficients of the differential equation; for instance, the seminvariant 22 is the determinant

$$22 = \begin{vmatrix} b & c & d & e \\ 3 & 3b & 4c & 5d \\ 0 & 2 & 2b & 3c \\ 0 & 0 & 1 & b \end{vmatrix}.$$

The Fundamental Seminvariants. Art. Nos. 22 to 28.

22. The seminvariants 2, 3, 4, 5, . . . are called the *fundamental seminvariants*; they are obtained in order by the formula

$$c = (1, b, \mathfrak{c} - b, 1)^2, \quad d = (1, b, c, \mathfrak{d} - b, 1)^3, \quad e = (1, b, c, d, \mathfrak{e} - b, 1)^4, \quad \&c.,$$

or, what is the same thing,

$$\begin{aligned} c &= b^2 - c, \\ d &= b^3 - 3bc + 2d, \\ e &= b^4 - 6b^2c + 3c^2 + 8bd - 6e, \end{aligned}$$

&c.; viz. up to g , these are

$$\begin{aligned} c &= b^2 - c, \\ d &= b^3 - 3bc + 2d, \\ e &= b^4 - 6b^2c + 3c^2 + 8bd - 6e, \\ f &= b^5 - 10b^3c + 15bc^2 + 20b^2d - 20bd - 30cd + 24f, \\ g &= b^6 - 15b^4c + 45b^2c^2 - 15c^3 + 40b^3d - 120bcd + 40d^2 \\ &\quad - 90b^2e + 90ce + 144bf - 120g. \end{aligned}$$

828] A MEMOIR ON SEMINVARIANTS. 245

23. These numbers are at once obtained by the formula; thus for e , multiplying successively by $-b$ and 1 , we have

$$\begin{array}{c} 1, b, c, d, e \\ -b, -b^2, -bc, -bd, -be \\ 1, b, c, d \end{array}$$

and thence $e = b^4 - 6b^2c + 3c^2 + 8bd - 6e$.

24. It may be remarked that the fundamental seminvariant c is the discriminant of $(b, \&X, Y)^2$, d is the discriminant of $(b, c, \&X, Y)^3$, and so on.

25. In regard to the expression of a seminvariant as a rational and integral function of the fundamental seminvariants, it is to be observed that the expression is not unique; for instance, we have $22 = c^2 - d$, but also $22 = c^2 - bd + cd - d$, &c.; among these expressions there is one which is the most simple, and which is to be adopted as the standard form.

26. The standard form is obtained by writing the seminvariant as a function of b, c, d , &c., and then substituting for c, d, e , &c. their values in terms of the fundamental seminvariants c, d, e , &c.; the expression is then reduced by means of the relations which connect the fundamental seminvariants; for instance, from $c = b^2 - c$, we have $b^2 = c + c$, and thence $b^4 = c^2 + 2cc + c^2$, &c.

27. The standard form of a seminvariant is a rational and integral function of the fundamental seminvariants, which does not contain the highest fundamental seminvariant in a degree higher than 1; and it is further restricted by the condition that the coefficients are numerical fractions in their lowest terms.

28. It is to be observed that the number of independent seminvariants is infinite; but that for any given weight the number of seminvariants is finite, being equal to the number of non-unitary partitions of that weight.

Perpetuants. Art. Nos. 29 to 35.

29. A *perpetuant* is a seminvariant which cannot be expressed as a rational integral function of seminvariants of lower weight; for instance, the seminvariants 2, 3, 4, 5 are perpetuants, but $22 = c^2 - d$ is

not a perpetuant, since it can be expressed in terms of c and d , which are of lower weights.

30. Sylvester established in like manner the notion of a *perpetuant*; a perpetuant is a seminvariant which is not expressible as a rational integral function of seminvariants of lower weight; or, what is the same thing, it is a seminvariant which is not expressible as a rational integral function of seminvariants of the same or lower degree.

31. The number of perpetuants of a given weight w is equal to the number of non-unitary partitions of w into two or more parts; or, what is the same thing, it is equal to the number of non-unitary partitions of w diminished by 1 (for the partition into a single part gives the fundamental seminvariant, which is a perpetuant). The formula for the number of perpetuants of weight w is

$$\frac{1}{w}\{2^{w-1} - 1 - (-1)^w\} - \frac{1}{2}\{1 - (-1)^w\},$$

or, what is the same thing,

$$\frac{1}{w}\{2^{w-1} - 1 - (-1)^w\} - \frac{1}{2} + \frac{1}{2}(-1)^w.$$

32. The formula may be written in the form

$$\frac{2^{w-1} - 1}{w} - \frac{1 - (-1)^w}{2},$$

and it is to be observed that the second term vanishes when w is odd, and is = 1 when w is even.

33. The number of perpetuants of a given weight w and degree j is given by the formula of Sylvester; the number is equal to the coefficient of x^w in the expansion of

$$\frac{x^{\frac{1}{2}j(j+1)}}{(1-x^2)(1-x^3)\dots(1-x^j)}.$$

34. In particular, the number of quadric perpetuants of weight w is the coefficient of x^w in $\frac{x^3}{1-x^2}$, which is = 1 when w is odd and ≥ 3 , and = 0 when w is even; and the number of cubic perpetuants of weight w is the coefficient of x^w in $\frac{x^6}{(1-x^2)(1-x^3)}$.

35. The generating function for the number of perpetuants is

$$\frac{x^2}{1-x^2} + \frac{x^3}{1-x^3} + \frac{x^4}{(1-x^2)(1-x^4)} + \frac{x^5}{(1-x^2)(1-x^5)} + \dots,$$

or, as given by Sylvester,

$$\frac{x^2}{1-x^2} + \frac{x^3}{(1-x^2)(1-x^3)} + \frac{x^4}{(1-x^2)(1-x^3)(1-x^4)} + \dots$$

The Seminvariants of a given Degree. Art. Nos. 36 to 40.

36. The seminvariants of a given degree are connected by linear relations; for instance, in degree 2, the seminvariants are c, d, e, f , &c., and these are connected by no linear relation; in degree 3, the seminvariants are cd, ce, de, c^3, d^2 , &c., and these are connected by linear relations.

828] A MEMOIR ON SEMINVARIANTS. 247

37. The number of linearly independent seminvariants of degree j and weight w is given by the formula

$$(w; j, n) - (w - 1; j, n),$$

where $(w; j, n)$ denotes the number of partitions of w into j parts not exceeding n .

38. For the seminvariants of a quantic of infinite order, the number of linearly independent seminvariants of degree j and weight w is equal to the coefficient of x^w in the expansion of

$$\frac{x^{\frac{1}{2}j(j+1)}}{(1-x)(1-x^2)\dots(1-x^j)}.$$

39. The generating function for the number of linearly independent seminvariants of degree j is

$$\frac{x^{\frac{1}{2}j(j+1)}}{(1-x)(1-x^2)(1-x^3)\dots}.$$

40. The total number of linearly independent seminvariants of weight w (for a quantic of infinite order) is the coefficient of x^w in

$$\frac{1}{(1-x)(1-x^2)(1-x^3)\dots} - 1 - \frac{x}{1-x},$$

which is the generating function for the number of non-unitary partitions of w .

The Seminvariants of a Quantic of given Order. Art. Nos. 41 to 45.

41. For a quantic of order n , the number of seminvariants of weight w and degree j is the same as for a quantic of infinite order, provided that $w \leq n$ and $j \leq n$; but for higher values of w or j , the number is diminished.

42. The number of seminvariants of a quantic of order n is finite; the highest weight of a seminvariant is $\frac{1}{2}n(n-1)$, and the highest degree is n .

43. The generating function for the number of seminvariants of a quantic of order n is given by Sylvester's formula; it is the coefficient of z^n in the expansion of a certain rational function.

44. The seminvariants of a quantic of order n may be obtained from the seminvariants of a quantic of infinite order by equating to zero all coefficients after the $(n + 1)$ th coefficient.

45. Attending next to the seminvariants of a quantic of a given order, we have to consider the relations which exist between the seminvariants of different degrees; these relations are called *syzygies*, and they form an important part of the theory.

46. A syzygy is a linear relation between products of seminvariants; for instance, for the quintic, we have the syzygy $c \cdot d - cd - c^3 = 0$, which expresses that the product $c \cdot d$ is equal to the sum of the seminvariants cd and c^3 .

47. The syzygies of a quantic of given order may be obtained by equating to zero the coefficients of the powers of x in the identity which expresses that the product of two seminvariants is a seminvariant.

48. The number of syzygies of a given weight and degree is given by a formula analogous to that for the number of seminvariants; the difference between the number of products and the number of seminvariants gives the number of syzygies.

49. In the case of a quantic of infinite order, there are no syzygies; but for a quantic of finite order, the number of syzygies increases with the order of the quantic.

50. Supposing that for any given weight $w - 6$, each of the quintic perpetuants is a sextic syzygant: this implies that the number of sextic syzygies $(S_6)'$ is at least equal to the number $((5))'$ of quintic perpetuants (for each expression of a quintic perpetuant as a sextic syzygant is in fact a sextic syzygy): and not only so, but it further implies that the number of the sextic syzygies, which do not contain a quintic perpetuant, is precisely equal $(S_6)' - ((5))'$: for if besides the equations which serve to express the perpetuants as syzygants, we have any other sextic syzygy, then either this does not contain a quintic perpetuant, or it can (by substituting therein for every quintic perpetuant its value as a sextic syzygant) be reduced to a syzygy which does not contain any quintic perpetuant.

51. In the general case, we have $(S_6)'$ sextic syzygies of the weight $w - 6$, and $((5))'$ quintic perpetuants of this weight: but it may happen that certain of the quintic perpetuants do not enter into any of the sextic syzygies; and those which enter, may do so in definite combinations: by elimination of these combinations of perpetuants we obtain (it may be) α sextic syzygies not containing any quintic perpetuant; and the remaining $(S_6)' - \alpha$ equations will then serve to express each of them a quintic perpetuant, or combination of quintic perpetuants, as a sextic syzygant. The number α is at most $= ((5))'$, or taking it to be $= ((5))' - ((6))'$, the number of sextic syzygies not containing any quintic perpetuant will be $= (S_6)' - ((5))' + ((6))'$, that is, the number of sextic syzygies not containing any quintic perpetuant will

be equal to the whole number $(S_6)'$ of sextic syzygies diminished by some number $((5))' - ((6))'$, which is less than or at most equal to the whole number $((5))'$ of quintic perpetuants of the weight in question $w - 6$. But as already mentioned, I have not been able to obtain the expression of the function $(6) = \frac{\omega(x)}{2.3.4.5}$, which is the G. F. of the number $((6))'$.

Cambridge, England, 17th March, 1884.

829.

**TABLES OF THE SYMMETRIC
FUNCTIONS OF THE ROOTS,
TO THE DEGREE 10, FOR THE FORM**

$$1 + bx + \frac{cx^2}{1.2} + \cdots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots$$

[From the *American Journal of Mathematics*, t. VII. (1885),
pp. 47—56.]

THE tables are derived from the tables (*b*) of my “Memoir on the Symmetric Functions of the Roots of an Equation,” *Phil. Trans.*, vol. cxlvii. (1857), pp. 489—496, [147]. These refer in effect to the form $1 + bx + cx^2 + \dots$, and we have consequently to change *b*, *c*, *d*, ... into $\frac{b}{1}$, $\frac{c}{1.2}$, $\frac{d}{1.2.3}$, ... respectively. Thus in the heading of the original table V(*b*), we must instead of

$$f, \quad be, \quad cd, \quad b^2d, \quad bc^2, \quad b^3c, \quad b^5,$$

write

$$\begin{aligned} & \frac{f}{120}, \quad \frac{be}{24}, \quad \frac{cd}{12}, \quad \frac{b^2d}{6}, \quad \frac{bc^2}{4}, \quad \frac{b^3c}{2}, \quad \frac{b^5}{1}, \\ &= \frac{1}{120}(f, \quad 5be, \quad 10cd, \quad 20b^2d, \quad 30bc^2, \quad 60b^3c, \quad 120b^5); \end{aligned}$$

the several columns of the original table are then multiplied by 1, 5, 10, 20, 30, 60, 120, and we thus obtain the new table with the heading

$$\frac{1}{120}(f, \quad be, \quad cd, \quad b^2d, \quad bc^2, \quad b^3c, \quad b^5).$$

In the original tables, there is a remarkable property (very easily proved) in regard to the sums of the numbers in a *column*. Thus for the table V(*b*) these sums are

$$-1, \quad +2, \quad +2, \quad -3, \quad -3, \quad +4, \quad -1,$$

where the sign is + or – according as the heading is the product of an even or an odd number of letters; and the numerical value depends only on the indices in the heading: these indices are

$$1, \quad 11, \quad 11, \quad 21, \quad 21, \quad 31, \quad 5,$$

and they give the foregoing values

$$1, \quad 2, \quad 2, \quad 3, \quad 3, \quad 4, \quad 1,$$

viz. $b^3c, = 31$ gives the value $\Pi 4 \div \Pi 3 \cdot \Pi 1, = 4$; b^2d, bc^2 , each $= 21$, give the value $\Pi 3 \div \Pi 2 \cdot \Pi 1, = 3$; and so in other cases.

[829

TABLES OF SYMMETRIC FUNCTIONS OF

ROOTS.

264

In the new tables we have the like property; thus for the table V(*b*), the sums are

$$-1, \quad +2, \quad +2, \quad -3, \quad -3, \quad +4, \quad -1,$$

and for the several tables the values are

Table	Sums
I(<i>b</i>)	+1
II(<i>b</i>)	-1, +1
III(<i>b</i>)	+1, -2, +1
IV(<i>b</i>)	-1, +2, +2, -3
V(<i>b</i>)	-1, +2, +2, -3, -3, +4, -1
VI(<i>b</i>)	+1, -2, -2, +3, +3, -4, +4, -4, +2, -1

and so on; the law is the same as in the original tables.

The following are the tables up to the degree 10:

I(*b*). $a = b$.

II(*b*). $a^2 = b^2 - c$.

	b^2	c
2	+1	-1

III(*b*). $a^3 = b^3 - 3bc + 2d$.

	b^3	bc	d
3	+1	-3	+2

IV(*b*). $a^4 = b^4 - 6b^2c + 3c^2 + 8bd - 6e$.

	b^4	b^2c	c^2	bd	e
4	+1	-6	+3	+8	-6
31	+1	-2	-1	+2	
22	+1	-2	+1		

V(*b*). $a^5 = b^5 - 10b^3c + 15bc^2 + 20b^2d - 20bd - 30cd + 24f$.

	b^5	b^3c	bc^2	b^2d	cd	be	f
5	+1	−10	+15	+20	−30	−20	+24
41	+1	−4	+3	+4	−6	−2	
32	+1	−4	+3	+2		−2	
311	+1	−2	−1	+2			
221	+1	−2	+1				

829]

TABLES OF SYMMETRIC FUNCTIONS OF
ROOTS.

265

VI(*b*).

	b^6	b^4c	b^2c^2	c^3	b^3d	bcd	b^2e	c^2d	bd^2	bf
6	+1	−15	+45	−15	+40	−120	−90	+40	+40	+144
51	+1	−7	+9	+3	+12	−18	−30	−6	+12	+24
42	+1	−7	+9	+3	+6	−18	−18	+6		+12
411	+1	−5	+3	+3	+4	−6	−6	−6	+4	
33	+1	−6	+6	+3	+8	−18	−18		+8	+24
321	+1	−5	+3	+3	+2	−6	−6		+2	
3111	+1	−3	−3	+3		+6		−6		
222	+1	−3	+3	−1						
2211	+1	−3	+3	−1						

VII(*b*).

	b^7	b^5c	b^3c^2	bc^3	b^4d	b^2cd	c^2d	b^2e	bce	b^2f	cd
7	+1	−21	+105	−105	+70	−420	+210	−210	+630	+504	−210
61	+1	−11	+27	+15	+22	−66	−18	−66	+90	+72	+18
52	+1	−11	+27	+15	+10	−66	−30	−30	+90	+36	+30
511	+1	−8	+9	+15	+8	−18	−18	−18	+18	+18	+18
43	+1	−10	+18	+15	+16	−48	−24	−48	+72	+60	+24
421	+1	−8	+9	+15	+4	−18	−18	−18	+18	+18	+18
4111	+1	−5	−9	+15		+18	−18		−18		+18
331	+1	−7	+9	+3	+8	−18	−6	−18	+18	+18	+6
322	+1	−6	+9	−3	+4	−18	+6	−18	+18	+18	−6
3211	+1	−5	−3	+3	+2	+6	−6	−6	−6	+6	+6
31111	+1	−3	−3	+3		+6	−6		−6		+6

[829
ROOTS.

TABLES OF SYMMETRIC FUNCTIONS OF
266

VIII(*b*).

	b^8	b^6c	b^4c^2	b^2c^3	c^4	b^5d	b^3cd	bc^2d	b^4e	b^2ce	c^2e	b^2d^2	cd^2
8	+1	-28	+210	-420	+105	+112	-840	+840	-560	+1680	-420	+560	-560
71	+1	-16	+60	-40	-40	+36	-168	+72	-120	+264	+72	+96	-24
62	+1	-16	+60	-40	-40	+16	-168	+120	-48	+264	-24	+24	+24
611	+1	-12	+24	-8	-40	+14	-48	+24	-40	+48	+24	+40	-24
53	+1	-15	+45	-30	-45	+22	-132	+90	-84	+198	-18	+48	+6
521	+1	-12	+24	-8	-40	+10	-48	+24	-30	+48	+24	+20	-24
5111	+1	-8	-6	+16	-8	+8	+12	-24	-8	-12	+24	+8	-8
44	+1	-14	+42	-28	-35	+24	-144	+96	-96	+216	-24	+48	
431	+1	-11	+21	-5	-40	+12	-48	+24	-36	+48	+24	+24	-24
422	+1	-10	+24	-12	-24	+8	-60	+36	-24	+60	-12	+12	+12
4211	+1	-8	-3	+12	-12	+6	+6	-18	-12	+6	+18	+12	-12
41111	+1	-5	-9	+15	-3		+18	-18		-18	+18		+18
332	+1	-9	+18	-6	-15	+8	-48	+30	-24	+48	-6	+12	+6
3311	+1	-7	-3	+9	-12	+6	+6	-18	-12	+6	+18	+12	-12
3221	+1	-6	+3	+3	-9	+2	-12	+12	-6	+12	-6	+4	+6
32111	+1	-4	-6	+6	+3	+2	+6	-12	-6	-6	+12	+2	+6
311111	+1	-2	-6	+6	+3		+6			-6	+12		+6
2222	+1	-4	+6	-4	+1								
22211	+1	-4	+6	-4	+1								

829]

TABLES OF SYMMETRIC FUNCTIONS OF
ROOTS.

267

IX(*b*).

	b^9	b^7c	b^5c^2	b^3c^3	bc^4	b^6d	b^4cd	b^2c^2d	c^3d	b^5e	b^3ce	bc^2e
9	+1	−36	+378	−1260	+945	+168	−1512	+2520	−1260	−1260	+5040	−3780
81	+1	−22	+117	−195	−117	+52	−312	+312	+78	−234	+468	+78
72	+1	−22	+117	−195	−117	+22	−312	+390	−78	−102	+468	−156
711	+1	−17	+54	−45	−117	+18	−90	+72	−18	−72	+108	+36
63	+1	−21	+99	−153	−135	+34	−198	+252	−54	−162	+378	−108
621	+1	−17	+54	−45	−117	+14	−90	+72	−18	−54	+108	+36
6111	+1	−12	−9	+45	−27	+12	+18	−54	+18	−18	−18	+54
54	+1	−20	+90	−135	−135	+28	−168	+216	−54	−132	+324	−108
531	+1	−16	+48	−39	−117	+16	−72	+60	−18	−48	+72	+36
522	+1	−15	+51	−57	−81	+10	−84	+102	−36	−36	+84	−30
5211	+1	−12	−3	+33	−39	+8	+6	−36	+18	−18	−6	+36
51111	+1	−7	−18	+30	+9	+4	+24	−36	−12	−4	−24	+36
441	+1	−14	+42	−36	−72	+16	−72	+60	−18	−48	+72	+36
432	+1	−13	+45	−51	−72	+10	−72	+84	−30	−30	+72	−24
4311	+1	−10	−3	+27	−36	+8	+6	−30	+18	−18	−6	+30
4221	+1	−9	+9	+9	−27	+4	−24	+30	−12	−12	+24	−12
42111	+1	−6	−9	+18	+9	+4	+12	−24	−6	−12	−12	+24
411111	+1	−3	−18	+18	+9		+12	−24		−12	−12	+24
333	+1	−12	+36	−30	−45	+8	−48	+42	−12	−24	+48	−12
3321	+1	−9	+9	+9	−27	+4	−24	+30	−12	−12	+24	−12
33111	+1	−6	−9	+18	+9	+4	+12	−24	−6	−12	−12	+24
3222	+1	−8	+18	−12	−18	+2	−24	+24	−6	−6	+18	−12
32211	+1	−5	−3	+12	−9	+2	+6	−18	+6	−6	−6	+18
321111	+1	−3	−12	+12	+9	+2	+12	−18	−6	−6	−12	+18
3111111	+1	−1	−12	+12	+9		+12	−18		−6	−12	+18
22221	+1	−5	+12	−10	+3							
222111	+1	−4	+6	−4	+1							

[829
ROOTS.

TABLES OF SYMMETRIC FUNCTIONS OF
268

X(*b*).

	b^{10}	b^8c	b^6c^2	b^4c^3	b^2c^4	c^5	b^7d	b^5cd	b^3c^2d	bc^3d	b^6e	b^4ce
10	+1	-45	+630	-3150	+4725	-945	+210	-2520	+6300	-6300	-3150	+1260
91	+1	-29	+207	-429	-195	+351	+78	-546	+702	+234	-390	+780
82	+1	-29	+207	-429	-195	+351	+28	-546	+858	-78	-180	+780
811	+1	-23	+99	-117	-195	+117	+26	-156	+156	+26	-130	+156
73	+1	-28	+189	-378	-189	+378	+48	-336	+504	-126	-252	+630
721	+1	-23	+99	-117	-195	+117	+20	-156	+156	+26	-102	+156
7111	+1	-17	-9	+81	-81	+9	+16	+12	-84	+36	-36	-36
64	+1	-27	+171	-333	-189	+351	+40	-282	+432	-126	-204	+540
631	+1	-22	+90	-102	-162	+108	+18	-126	+126	+18	-90	+126
622	+1	-21	+93	-129	-135	+189	+12	-138	+192	-60	-60	+135
6211	+1	-17	-3	+69	-87	+45	+10	-6	-60	+36	-30	-6
61111	+1	-11	-21	+51	+9	-27	+6	+30	-48	-12	-6	-30
55	+1	-26	+162	-315	-189	+324	+36	-252	+396	-126	-180	+504
541	+1	-21	+84	-93	-153	+99	+16	-108	+108	+18	-72	+108
532	+1	-20	+87	-120	-126	+168	+10	-114	+162	-54	-48	+114
5311	+1	-16	-3	+63	-81	+45	+8	-6	-48	+30	-24	-6
5221	+1	-14	+12	+30	-60	+36	+4	-36	+48	-18	-12	+30
52111	+1	-9	-18	+42	+12	-24	+4	+18	-36	-6	-6	-18
511111	+1	-5	-30	+30	+30	-9	+2	+30	-36	-18	-2	-30
442	+1	-18	+72	-102	-108	+162	+8	-96	+138	-48	-36	+96
4411	+1	-14	+3	+39	-63	+36	+6	+6	-42	+24	-18	-6
433	+1	-17	+63	-81	-81	+135	+6	-72	+102	-36	-24	+72
4321	+1	-13	+12	+30	-60	+36	+4	-30	+42	-18	-12	+24
43111	+1	-8	-18	+36	+18	-27	+4	+12	-30	-6	-12	-12
4222	+1	-11	+24	-18	-36	+27	+2	-30	+36	-12	-6	+24
42211	+1	-7	-9	+27	+9	-18	+2	+6	-24	+12	-6	-6
421111	+1	-4	-24	+24	+24	-9	+2	+18	-24	-12	-6	-18
4111111	+1	-2	-30	+24	+36	-9		+18	-24	-12		-18
3331	+1	-12	+27	-18	-45	+27	+2	-30	+36	-12	-6	+24
3322	+1	-10	+24	-18	-36	+27	+2	-24	+30	-12	-6	+18
33211	+1	-6	-9	+24	+9	-18	+2	+6	-24	+12	-6	-6
331111	+1	-3	-24	+24	+24	-9	+2	+18	-24	-12	-6	-18
32221	+1	-7	+12	+6	-24	+18	+1	-18	+24	-9	-3	+12
322111	+1	-4	-12	+18	+18	-15	+1	+12	-18	-6	-3	-12
3211111	+1	-2	-21	+18	+27	-12	+1	+18	-18	-12	-3	-18
31111111	+1	-1	-24	+18	+30	-12		+18	-18	-12		-18
22222	+1	-6	+15	-20	+15	-6						
222211	+1	-5	+9	-7	-3	+6						
2221111	+1	-3	+6	-4	+1							

830.

NON-UNITARY PARTITION TABLES.

[From the *American Journal of Mathematics*, t. VII. (1885), pp. 57, 58.]

IN the theory of Seminvariants we are concerned with the non-unitary partitions of a number, that is, the number of ways of making up the number with the parts **2**, **3**, **4**, . . . ; or what is the same thing, writing $\mathbf{2} = 1 - x^2$, $\mathbf{3} = 1 - x^3$, &c., with the Generating Functions having in their denominators the factors **2**, **3**, **4**, &c. In the present short paper, I give the developments up to x^{100} of the functions $1 \div \mathbf{2}$, **2.3**, **2.3.4**, **2.3.4.5**, **2.3.4.5.6**, respectively: and also of the function

$$x^6 + x^{13} - 2x^{16} - x^{18} + x^{31} \div \mathbf{2.3.4.5.6},$$

which function is (there is strong reason to believe) the G. F. for the number of sextic syzygies of a given weight: the same function without the term x^{31} occurs (p. 115) in Professor Sylvester's paper "On Subinvariants, i.e. Seminvariants to Binary Quantics of an Unlimited Order," *American Journal of Mathematics*, t. V. (1882), pp. 79—136.

In the tables, X is written to denote $x^6 + x^{13} - 2x^{16} - x^{18} + x^{31}$.

[830 **NON-UNITARY PARTITION TABLES.** 274

NON-UNITARY PARTITION TABLES.

Ind. x	$\frac{1}{2}$ 2.3	2.3.4	2.3.4.5	2.3.4.5.6	$\frac{X}{2}$ 2.3.4.5.6	Ind. x	$\frac{1}{2}$ 2.3	2.3.4	2.3.4.5	2.3.4.5.6
0	1	1	1	1		50	9	65	258	
1	0	0	0	0		51	9	61	268	
2	1	1	1	1		52	9	70	286	
3	1	1	1	1		53	9	65	297	
4	1	2	2	2		54	10	75	316	
5	1	1	2	2		55	9	70	328	
6	2	3	3	4	1	56	10	80	348	
7	1	2	3	3	0	57	10	75	361	
8	2	4	5	6	1	58	10	85	382	
9	2	3	5	6	1	59	10	80	396	
10	2	5	7	9	2	60	11	91	419	
11	2	4	7	9	2	61	10	85	433	
12	3	7	10	14	4	62	11	96	457	
13	2	5	10	13	4	63	11	91	473	
14	3	8	13	19	6	64	11	102	598	
15	3	7	14	20	7	65	11	96	515	
16	3	10	17	26	8	66	12	108	541	
17	3	8	18	27	11	67	11	102	559	
18	4	12	22	36	13	68	12	114	587	
19	3	10	23	36	15	69	12	108	606	
20	4	14	28	47	17	70	12	120	635	
21	4	12	29	49	21	71	12	114	655	
22	4	16	34	60	22	72	13	127	686	
23	4	14	36	63	28	73	12	120	707	
24	5	19	42	78	29	74	13	133	739	
25	4	16	44	80	35	75	13	127	762	
26	5	21	50	97	36	76	13	140	795	
27	5	19	53	102	44	77	13	133	819	
28	5	24	60	120	43	78	14	147	854	
29	5	21	63	126	54	79	13	140	879	
30	6	27	71	149	53	80	14	154	916	
31	5	24	74	154	64	81	14	147	942	
32	6	30	83	180	62	82	14	161	980	
33	6	27	87	189	78	83	14	154	1008	
34	6	33	96	216	72	84	15	169	1048	
35	6	30	101	227	89	85	14	161	1077	
36	7	37	111	260	84	86	15	176	1118	
37	6	33	116	270	102	87	15	169	1149	
38	7	40	127	307	96	88	15	184	1192	
39	7	37	133	322	117	89	15	176	1224	
40	7	44	145	361	108	90	16	192	1269	
41	7	40	151	378	133	91	15	184	1302	
42	8	48	164	424	123	92	16	200	1349	
43	7	44	171	441	149	93	16	192	1384	
44	8	52	185	492	137	94	16	208	1432	
45	8	48	193	515	167	95	16	200	1469	
46	8	56	207	568	152	96	17	217	1519	
47	8	52	216	594	186	97	16	208	1557	
48	9	61	232	656	169	98	17	225	1609	
49	8	56	241	682	205	99	17	217	1649	

831.

SEMINVARIANT TABLES.

[From the *American Journal of Mathematics*, t. VII. (1885),
pp. 59—73.]

THE present tables are not, I think, superseded by the tables A, pp. 149—163, contained in Capt. MacMahon’s paper, “Seminvariants and Symmetric Functions,” *American Journal of Mathematics*, t. VI. (1883), pp. 131—163. His order of the terms, though a very ingenious one, and giving rise to a most remarkable symmetry in the form of the tables, seems to me too artificial—and I cannot satisfy myself that it ought to be adopted in preference to the more simple one which I use: I attach also considerable importance to the employment of the simple letters b, c, d, e , &c. in place of the suffixed ones a_1, a_2, a_3, a_4 , &c. There is, moreover, the question of the identification of the seminvariants with their expressions as non-unitary symmetric functions of the roots of the equation $1 + bx + \frac{cx^2}{1.2} + \&c. = 0$, which requires to be considered.

As to the form in which the tables present themselves, I remark that every seminvariant is a rational and integral function of the fundamental seminvariants

$$\begin{aligned} c &= (1, b, \& - b, 1)^2, \\ d &= (1, b, c, \& - b, 1)^3, \\ e &= (1, b, c, d, \& - b, 1)^4, \quad \&c., \end{aligned}$$

viz. up to g , these are

c =	d =	e =	f =	g =	ce =	d ² =	c ³ =
$c + 1$	$d + 1$	$e + 1$	$f + 1$	$g + 1$			
$b^2 - 1$	$bc - 3$	$bd - 4$	$be - 5$	$bf - 6$	+1	+1	
	$b^3 + 2$	c^2	cd	ce	-1	-6	+1
		$b^2c + 6$	$b^2d + 10$	$b^2e + 15$	-4	+9	-3
		$b^4 - 3$	bc^2	bcd	+4	+4	+3
			$b^3c - 10$	$b^3d - 20$	+6	-12	-1
			$b^5 + 4$	b^2c^2	-9	+4	
				$b^4c + 15$	+3		
				$b^6 - 5$			

and if to the value of any seminvariant expressed in terms of b, c, d , &c. we substitute the foregoing values of c, d, e , &c., the result will be the value of the seminvariant in terms of the fundamental seminvariants c, d, e , &c.

It will be observed that, as regards the terms b^2c, bc^2 , &c., the coefficients in the several columns are the binomial coefficients with the proper signs; this is a general law.

SEMINVARIANT TABLES UP TO $(m = 12)$.

$(b = 1)$

$(c = 2)$

$(d = 3)$

b	c	d
+1	+1	+1

$(e = 4)$

e	c^2	bd	b^2c
+1	+1	-4	+2
b	-1	+3	-1

$(f = 5)$

$(g = 6)$

f	ce	d^2	c^2d	b^2e	bcd	b^3d	bc^2	b^4c
+1	+1	+1	-1	+2	-5	-5	+5	-2
b	-1	-6	+2	-1	+4	+3	-2	+1
c		+9	-3		-3		+1	
b^2		-3	+1		+1		-1	

831]

SEMINVARIANT TABLES.

277

($h = 7$)

h	cg	df	e^2	c^2f	cde	d^3	c^3d	b^2g	bce	b^2f	bd^2	...
+1	+1	+1	+1	−1	−1	+1	+1	+2	−6	−6	+9	...
b	−1	−7	−8	+2	+3	+3	−2	−1	+5	+4	−6	...
c		+9	+20	−3	−6	−12	+3		−4	−3	+9	...
d		−5	−12	+1	+3	+9	−1		+1	+1	−4	...
b^2		−3	−8	+1	+2	+6	−1		+1	+1	−3	...
bc			+4		−1	−4			−1	−1	+2	...
b^3			+1			+1					−1	...

($i = 8$)

i	ch	dg	ef	c^2g	cdf	ce^2	d^2f	c^3e	cd^2	b^2h	bcg	bdf	be^2	...
+1	+1	+1	+1	−1	−1	−1	+1	+1	+1	+2	−7	−7	−8	...
b	−1	−8	−10	+2	+3	+4	+4	−2	−3	−1	+6	+5	+6	...
c		+10	+25	−3	−7	−16	−20	+3	+9		−5	−4	−12	...
d		−6	−16	+1	+4	+10	+16	−1	−6		+1	+1	+4	...
e		+1	+4		−1	−4	−8		+1				−1	...
b^2		−4	−12	+1	+3	+6	+8	−1	−3		+1	+1	+3	...
bc			+6		−1	−4	−8		+3		−1	−1	−2	...
bd			−1			+1	+2		−1					...
b^3			+1				−1							...

($j = 9$)

j	ci	dh	eg	f^2	c^2h	cdg	cef	df^2	d^2e	c^3g	c^2df	cde^2	cd^2e
+1	+1	+1	+1	+1	-1	-1	-1	-1	+1	+1	+1	+1	-1
b	-1	-9	-12	-14	+2	+3	+4	+5	+5	-2	-3	-4	+2
c		+11	+30	+42	-3	-8	-19	-30	-36	+3	+9	+18	-6
d		-7	-22	-35	+1	+5	+14	+26	+36	-1	-6	-15	+6
e		+2	+8	+16		-1	-5	-12	-20		+1	+6	-1
f			-1	-4			+1	+3	+6			-1	
b^2		-5	-16	-24	+1	+4	+10	+15	+18	-1	-4	-10	+4
bc			+8	+16		-2	-8	-16	-24		+3	+12	-6
bd			-2	-6		+1	+3	+8	+14		-1	-5	+2
be				+1			-1	-2	-4			+1	
b^3			+1	+2			-1	-2	-3			+1	-1

831]

SEMINVARIANT TABLES.

279

The tables are continued for higher values; the law of formation is as follows: the coefficients in each column are the binomial coefficients with the proper signs, and the terms are arranged in the order of the powers of b, c, d , &c.

($k = 10$) continued:

k	cj	di	eh	fg	c^2i	cdh	ceg	cf^2	d^2g	def	e^3	...
+1	+1	+1	+1	+1	-1	-1	-1	-1	+1	+1	+1	...
b	-1	-10	-14	-18	+2	+3	+4	+5	+6	+6	+7	...
c		+12	+35	+54	-3	-9	-22	-38	-30	-48	-72	...
d		-8	-28	-52	+1	+6	+18	+38	+24	+48	+96	...
e		+2	+11	+28		-1	-7	-19	-8	-22	-56	...
f			-2	-8			+1	+5	+1	+6	+22	...
g				+1				-1		-1	-6	...

($l = 11$) continued:

l	ck	dj	ei	fh	g^2	c^2k	cdj	cei	cfh	cg^2	d^2i	djh	...
+1	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1	+1	+1	...
b	-1	-11	-16	-22	-26	+2	+3	+4	+5	+6	+7	+7	...
c		+13	+40	+68	+90	-3	-10	-25	-46	-66	-42	-60	...
d		-9	-34	-70	-105	+1	+7	+22	+50	+90	+36	+66	...
e		+2	+14	+38	+70		-1	-9	-27	-60	-12	-30	...
f			-3	-12	-30			+1	+7	+22	+2	+10	...
g				+2	+8				-1	-5		-1	...
h					-1					+1			...

($m = 12$) continued:

m	cl	dk	ej	fi	gh	c^2l	cdk	cej	cfi	cgh	d^2j	dfi	.
+1	+1	+1	+1	+1	+1	-1	-1	-1	-1	-1	+1	+1	.
b	-1	-12	-18	-26	-32	+2	+3	+4	+5	+6	+7	+8	.
c		+14	+45	+84	+120	-3	-11	-28	-54	-84	-56	-84	.
d		-10	-40	-92	-150	+1	+8	+27	+63	+120	+48	+96	.
e		+2	+17	+51	+100		-1	-11	-36	-84	-16	-48	.
f			-4	-18	-48			+1	+9	+30	+2	+14	.
g				+3	+12				-1	-7		-2	.
h					-1					+1			.

[The tables are continued in the same manner for higher values, the law of formation being always the same: the coefficients are the binomial coefficients with the proper signs, and the terms are arranged in the order of the powers of b, c, d, e, f, g , &c.]

836] ON THE QUATERNION EQUATION $qQ - Qq = 0$. 301

Instead of directly working out the condition in the manner indicated above, I present the investigation in a synthetic form as follows: Taking v, v' for the vector parts of the two given quaternions, so that $q = w + v, q' = w' + v'$, I write for shortness

$$\begin{aligned}\theta &= w - w', \\ \alpha &= v^2 + v'^2, \quad = -x^2 - y^2 - z^2 - x'^2 - y'^2 - z'^2, \\ \beta &= v^2 - v'^2, \quad = -x^2 - y^2 - z^2 + x'^2 + y'^2 + z'^2, \\ D &= -\theta(\alpha - \theta^2), \\ A &= \beta - \theta^2, \\ B &= \beta + \theta^2,\end{aligned}$$

so that $\theta, \alpha, \beta, D, A, B$ are all of them scalars. With these I form a quaternion $Q = (D + Av)(D + Bv')$; I say that we have identically

$$qQ - Qq' = \{D - v \cdot v'^2 + v' \cdot v^2 + w \cdot \theta\}(\theta^4 - 2\alpha\theta^2 + \beta^2).$$

It of course follows that, if $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, then $qQ - Qq' = 0$, viz. $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ is the condition $\Omega = 0$, for the existence of the required quaternion; and this condition being satisfied, then (omitting the arbitrary scalar factor) the value of the quaternion is $Q = (D + Av)(D + Bv')$, a value giving $T(Q) = 0$, that is, $W^2 + X^2 + Y^2 + Z^2 = 0$. If $\theta = 0$ (that is, $w = w'$), then the condition becomes $\beta = 0$, that is, $x^2 + y^2 + z^2 - x'^2 - y'^2 - z'^2 = 0$; and these two conditions being satisfied, Q ceases to have the determinate value given by the foregoing formula: it has a value involving an arbitrary parameter, and is no longer such that $W^2 + X^2 + Y^2 + Z^2 = 0$.

The identical equation is at once verified; we have

$$\begin{aligned}qQ - Qq' &= (w + v)Q - Q(w' + v') \\ &= \theta \cdot (D^2 + DAv + DBv' + ABvv') \\ &\quad + v \cdot (D^2 + DAv + DBv' + ABvv') \\ &\quad - (D^2 + DAv + DBv' + ABvv')v' \\ &= \theta D^2 + DAv^2 - DBv'^2 \\ &\quad + v(DA\theta + D^2 - ABv'^2) \\ &\quad + v'(DB\theta + ABv^2 - D^2) \\ &\quad + vv'(\theta AB + DB - DA).\end{aligned}$$

The first line is here $= D \cdot \{D\theta + Av^2 + Bv'^2\}$, viz. the term in $\{ \}$ is

$$-(\alpha - \theta^2)\theta + (\beta - \theta^2)v^2 - (\beta + \theta^2)v'^2, \quad = -\alpha\theta^2 = \theta \cdot;$$

and similarly each of the other lines contains the factor $\theta^4 - 2\alpha\theta^2 + \beta^2$, and the equation is thus seen to hold good.

302 ON THE QUATERNION EQUATION $qQ - Qq = 0$. [836

The tensor of Q is $(D^2 - A^2v^2)(D^2 - B^2v'^2)$; and we have

$$\begin{aligned} D^2 - A^2v^2 &= \theta^2(\alpha - \theta^2)^2 - (\beta - \theta^2)^2v^2 \\ &= \theta^6 - (2\alpha + v^2)\theta^4 + (\alpha^2 + 2\beta v^2)\theta^2 - \beta^2v^2, \end{aligned}$$

which, observing that $\alpha^2 + 2\beta v^2 = \beta^2 + 2v^2(2v^2 - \alpha)$ is $= (\theta^2 - v'^2)^2$ and similarly $D^2 - B^2v'^2$ is $= (\theta^2 - v^2)^2$; hence the tensor is

$$TQ = (\theta^2 - v^2)(\theta^2 - v'^2),$$

which, in virtue of $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, is $= 0$.

The particular case is when $Tq - Tq' = 0$, that is, $w^2 - v^2 - w'^2 + v'^2 = 0$, or say $w^2 - w'^2 = v^2 - v'^2$, that is, $\theta(w + w') = \beta$. Combining with this the general condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, we find $\theta^2\{\theta^2 - 2\alpha + (w + w')^2\} = 0$, or in the second factor, for θ and α substituting their values, we have $2\theta^2(w^2 - v^2 + w'^2 - v'^2) = 0$, that is, $2\theta^2(Tq + Tq') = 0$. Attending to the assumed relation $Tq - Tq' = 0$, the second factor can only vanish if $Tq = 0$, $Tq' = 0$; hence, disregarding this more special case, the factor which vanishes must be the first factor, that is, $\theta = 0$; or the equations $Tq - Tq' = 0$ and $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$ give $\beta = 0$ and $\theta = 0$, that is, we have as already mentioned $w - w' = 0$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, viz. the given quaternions have their scalars equal, and the squares of their vectors also equal. The equation here is $vQ - Qv' = 0$; and writing $v^2 = v'^2 = -\rho^2$, we see at once that a solution is

$$Q = (-\rho^2 + vv') + \lambda(v + v'),$$

where λ is an arbitrary scalar; in fact, with this value of Q , we have at once vQ and Qv' each $= \rho^2(v + v') + \lambda(\rho^2 + vv')$; and the equation $vQ - Qv' = 0$ is thus satisfied.

The value of the tensor is easily found to be

$$TQ = 2(\lambda^2 + \rho^2)(\rho^2 + xx' + yy' + zz'),$$

which is not $= 0$.

In accordance with a remark in the introductory paragraphs, the solution $Q = -\rho^2 + vv' + \lambda(v + v')$ is not comprised in the general solution. As to this, observe that, in the case in question $\theta = 0$, $\beta = 0$, we have from the general theorem the form that is,

$$Q = -\rho^2\theta^2(\theta^2 - 2v'^2) \pm \theta^2(\theta^2 - 2v^2)v', \quad = 0.$$

In the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\theta^2 = 2\alpha$, or for α writing its value $-2\rho^2$, we have $\theta^2 = -4\rho^2$, $\frac{\theta^2}{4} = -\rho^2$, and thence $\frac{A}{B} = \rho^2$, and $\frac{A}{B} = \lambda\rho$, if λ denote the $\sqrt{(-1)}$ of ordinary algebra. The resulting formula is thus $Q = -\rho^2 + vv' \pm \lambda\rho(v + v')$, which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ .

836]ON THE QUATERNION EQUATION $qQ - Qq = 0$. 303

in the condition $\theta^4 - 2\alpha\theta^2 + \beta^2 = 0$, writing $\theta = 0$, we have $\frac{\theta^2}{4} = 2\alpha$, or for α writing its value $= 2\rho^2$, we have $\theta^2 = -4\rho^2$, $\frac{A^2}{B^2} = -\rho^2$, and thence $\frac{A^2}{B^2} = \rho^2$, $\frac{A}{B} = \lambda\rho$, if λ denote the $\sqrt{(-1)}$ of ordinary algebra. The resulting formula is thus

$$Q = -\rho^2 + vv' \pm \lambda\rho(v + v'),$$

which corresponds to the determinate value $\pm\lambda\rho$ of the constant λ .

The foregoing solution $Q = -\rho^2 + vv' + \lambda(v + v')$ may be easily identified with that given pp. 124, 125 of Tait's *Elementary Treatise on Quaternions*; the case there considered is that of a real quaternion, and it was therefore assumed that the two conditions $w = w'$, and $x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$, were each of them satisfied.

The theory of quaternions is, as is well known, identical with that of matrices of the second order; the identity is, in effect, established by the remark and footnote "Linear Associative Algebra," *American Journal of Mathematics*, t. IV. (1881), p. 132. Writing x, y, z, w for Peirce's imaginaries $i, j, k, 1$, these have the multiplication table:

	x	y	z	w
x	x	y	0	0
y	0	0	x	y
z	0	w	0	0
w	0	0	z	w

Thus if $\lambda, = \sqrt{(-1)}$, be the imaginary of ordinary algebra, and i, j, k , the quaternion imaginaries, the relations between i, j, k and x, y, z, w are

$$\begin{aligned} x &= \frac{1}{2}(1 - \lambda i), & \text{or conversely} & \quad 1 = x + w, \\ y &= \frac{1}{2}(j - \lambda k), & i &= \lambda(x - w), \\ z &= \frac{1}{2}(-j - \lambda k), & j &= (y - z), \\ w &= \frac{1}{2}(1 + \lambda i), & k &= \lambda(y + z), \end{aligned}$$

and we can thus at once express a quaternion as a linear function of the x, y, z, w , or a linear function of the x, y, z, w as a quaternion. And we then consider

304 ON THE QUATERNION EQUATION $qQ - Qq = 0$. [836

$ax + by + cz + dw$ as denoting the matrix $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$; we obtain for the product of two matrices the ordinary formula

$$\begin{vmatrix} a, b \\ c, d \end{vmatrix} \begin{vmatrix} a', b' \\ c', d' \end{vmatrix} = \begin{vmatrix} (a, b)(a', c'), & (a, b)(b', d') \\ (c, d)(a', c'), & (c, d)(b', d') \end{vmatrix},$$

viz. we have

$$\begin{aligned} (ax + by + cz + dw)(a'x + b'y + c'z + d'w) &= aa'x^2 + bc'yz + \&c., \\ &= (aa' + bc')x + (ab' + bd')y + (ca' + dc')z + dd'w^2. \end{aligned}$$

in accordance with the formula for the product of the two matrices. Observe that, writing

$$A + Bi + Cj + Dk = A(x + w) + B\lambda(x - w) + C(y - z) + D\lambda(y + z) = ax + by + cz + dw,$$

we have

$$\begin{vmatrix} a, b \\ c, d \end{vmatrix} = \begin{vmatrix} A + B\lambda, & -C + D\lambda \\ C + D\lambda, & A - B\lambda \end{vmatrix},$$

and thence $ad - bc = A^2 + B^2 + C^2 + D^2$, so that the determinant of the matrix corresponds to the tensor of the quaternion.

837.

ON THE SO-CALLED D'ALEMBERT–CARNOT
GEOMETRICAL PARADOX.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 113,
114.]

THE present note has reference to Prof. Sylvester's paper on this subject [*l. c.*, pp. 92–96]. I cannot admit that “D'Alembert and Carnot raised a well-founded objection” to the then and even now too prevalent interpretation of the meaning of the geometrical positive and negative: it appears to me that the objection was not a well-founded one.

Consider through the origin K an indefinite line $t'Kt$, and measure off from K in the sense Kt a distance equal to the positive quantity a , and let m be the extremity of the distance thus measured off. There is not in the ordinary theory any reason why the distance Km should be $= +a$ rather than $= -a$; it is $= +a$, if Kt be the positive sense of the line through K , and it is $= -a$ if Kt' be the positive sense of the line through K ; if it be undetermined which of the two is the positive sense, then the distance Km is $= \pm a$, the sign being essentially indeterminate.

The problem is from a point K outside a given circle to draw a line Kmm' such that the intercepted portion mm' within the circle has a given value c . Supposing that the line from K to the centre meets the circle in the points A, B at the distances $KA = a, KB = b$: then if $Km = r$, we have $ab = r(c + r)$, or $r = -\frac{1}{2}c \pm \sqrt{(\frac{1}{4}c^2 + ab)}$; viz. we have for r , not simultaneously but alternatively, the positive value $-\frac{1}{2}c + \sqrt{(\frac{1}{4}c^2 + ab)}$, and the negative value $-\frac{1}{2}c - \sqrt{(\frac{1}{4}c^2 + ab)}$, the latter of these being the greatest in absolute magnitude; say the values are $+\rho_1$ and $-\rho_2$. We may with either of these values construct the point m ; viz. we obtain m as one of the intersections of the given circle with the circle centre K and radius ρ_1 , or else with the circle centre K and radius ρ_2 (that is, radius ρ_2); and attending to the intersections on the same side of the line from K to the centre, it happens that

306 ON THE SO-CALLED D'ALEMBERT-CARNOT GEOMETRICAL PARADOX. [837

the two points m thus determined are on one and the same line $t'Kt$; but there is no *a priori* reason why the positive senses should be the same, and they are in fact opposite to each other, in the two cases respectively; in the one case we measure off the distance ρ_1 in the sense Kt , in the other case the distance $-\rho_2$ in the sense Kt' ; that is, we in fact measure off the positive distances $+\rho_1$, and $+\rho_2$, in one and the same sense Kt ; thus obtaining for the point m one or the other extremity of a determinate secant through K .

The best illustration is I think in the elementary problem of finding the perpendicular distance of a given line from the origin. Let $Ax + By + C = 0$ be the equation of the given line: and first let a line be drawn in a determinate sense, say at the inclination θ to the positive part of the axis of x , to meet the given line. Taking r for the distance from the origin of the point of intersection, we have, for the coordinates of the point of intersection, $x, y = r \cos \theta, r \sin \theta$; and thence $r(A \cos \theta + B \sin \theta) + C = 0$, that is,

$$r = \frac{-C}{A \cos \theta + B \sin \theta},$$

a perfectly determinate value. But the perpendicular on the given line may be considered as drawn in one or the other of two opposite senses; that is, we have at pleasure

$$\cos \theta, \sin \theta = \frac{A}{\sqrt{A^2 + B^2}}, \frac{B}{\sqrt{A^2 + B^2}}, \quad \text{or else} \quad \frac{-A}{\sqrt{A^2 + B^2}}, \frac{-B}{\sqrt{A^2 + B^2}},$$

and thence

$$r = \frac{-C}{\sqrt{A^2 + B^2}}, \quad \text{or else} \quad r = \frac{C}{\sqrt{A^2 + B^2}};$$

that is, the perpendicular distance is $= \frac{\pm C}{\sqrt{A^2 + B^2}}$, with the essentially indeterminate sign $\pm$, because the distance may be considered as drawn from the origin in one or the other of the two opposite senses.

838.

ON THE TWISTED CUBICS UPON A QUADRIC SURFACE.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 129–132.]

A CUBIC (twisted cubic) on a quadric surface meets each generator of the one kind twice, and each generator of the other kind once (Salmon, *Solid Geometry*, Ed. 4, p. 301). There are thus on the surface two kinds of cubics, viz. distinguishing for convenience the generating lines as generators and directors, a cubic may meet each generator twice and each director once; or it may meet each generator once and each director twice. And two cubic curves are accordingly of the same kind or of different kinds.

Consider for example the quadric surface $xw - yz = 0$, and the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

If we call the line ($x = ky, kw = z$) a generator, and the line ($x = kz, kw = y$) a director, for the intersections with a generator we have $1 = k\phi$, $k\phi^3 = \phi^2$; equations with the common root $k\phi = 1$, or there is a single intersection; for the intersections with a director, $1 = k\phi^2$, $k\phi^3 = \phi$, equations with the common roots $k\phi^2 = 1$, or there are two intersections.

Consider on the same quadric surface the cubic

$$x : y : z : w = \overline{\theta - \alpha \cdot \theta - \beta \cdot \theta - \gamma} : \overline{\theta - \alpha \cdot \theta - \epsilon \cdot \theta - \zeta} : \overline{\theta - \delta \cdot \theta - \beta \cdot \theta - \gamma} : \overline{\theta - \delta \cdot \theta - \epsilon \cdot \theta}$$

or as, for shortness, I write these,

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta,$$

viz. α is written to denote $\theta - \alpha$, and so in other cases; for the intersections with a generator, we have $\alpha\beta\gamma = k\alpha\epsilon\zeta$, $k\delta\epsilon\zeta = \delta\beta\gamma$, equations having the two common roots

308 ON THE TWISTED CUBICS UPON A QUADRIC SURFACE. [838

$\beta\gamma = k\epsilon\zeta$; or there are two intersections with the generator. And in like manner there is one intersection with the director. The cubic

$$x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$$

is thus a cubic of different kind from

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

And in the same manner it appears that the cubic

$$x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$$

is of the same kind with

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3.$$

The two cubics of different kinds intersect in 5 points; the two cubics of the same kind in only 4 points. In fact, for the intersections with the cubic

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3,$$

we have $xz - y^2 = 0$, $yw - z^2 = 0$, and substituting herein the θ -values, $x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$, we find

$$\alpha\delta\beta^2\gamma^2 - \alpha^2\epsilon^2\zeta^2 = 0, \quad \delta\epsilon\zeta\alpha\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0,$$

equations with the common five roots $\delta\beta^2\gamma^2 - \alpha\epsilon^2\zeta^2 = 0$; whereas for the θ -values $x : y : z : w = \alpha\beta\gamma : \delta\beta\gamma : \alpha\epsilon\zeta : \delta\epsilon\zeta$, we have

$$\alpha^2\beta\gamma\epsilon\zeta - \delta^2\beta^2\gamma^2 = 0, \quad \delta^2\beta\gamma\epsilon\zeta - \alpha^2\epsilon^2\zeta^2 = 0,$$

equations with the common four roots $\alpha^2\epsilon\zeta - \delta^2\beta\gamma = 0$.

I remark in passing that the equations just obtained have each of them a root $\theta = \infty$, but this would have been avoided if the θ -equations had been taken in the slightly altered form

$$x : y : z : w = \alpha\beta\gamma : b\alpha\epsilon\zeta : c\delta\beta\gamma : bc\delta\epsilon\zeta, \text{ and } x : y : z : w = \alpha\beta\gamma : c\delta\beta\gamma : b\alpha\epsilon\zeta : bc\delta\epsilon\zeta$$

b, c being arbitrary constants; and the special value is thus of no importance.

The two cubics intersecting in 5 points constitute the complete intersection of a quadric surface and a cubic surface; and conversely when a quadric surface and a cubic surface intersect in two cubics, then the cubics must have 5 intersections, and are thus by what precedes cubics of different kinds in regard to the quadric surface. In fact, considering two cubics meeting in 5 points, we can, through these 5 points, through 2 points assumed at pleasure on the first cubic, and through 2 points assumed at pleasure on the second cubic, draw a determinate quadric surface: this meets each of the two cubics in $5 + 2$ points, and consequently it entirely contains the cubic; viz. we have through the two cubics a determinate quadric surface. Again, through the 5 points, through 5 points at pleasure on the first cubic and through 5 points at pleasure on the second cubic, we may draw a cubic surface; this meets each cubic

838] ON THE TWISTED CUBICS UPON A QUADRIC SURFACE. 309

in $5 + 5$ points, and therefore it entirely contains the cubic; we have thus a cubic surface through the two cubics. The cubic surface has been subjected only to $5 + 5 + 5 = 15$ conditions, and the equation thus contains homogeneously $20 - 15, = 5$ constants; viz. if $\Omega = 0$ be the quadric surface through the two cubics the general form is

$$(\alpha x + \beta y + \gamma z + \delta w)\Omega + \lambda U = 0;$$

disregarding the term in Ω , we have thus, in fact, a determinate cubic $U = 0$.

Taking as above the first cubic as given by the equations $x : y : z : w = 1 : \phi : \phi^2 : \phi^3$, and the second cubic as given by the equations $x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$, the equation of the cubic surface which contains the two cubics will be of the form

$$(Ax + By + Cz + Dw)(yw - z^2) + (A'x + B'y + C'z + D'w)(xz - y^2) = 0.$$

Substituting herein the θ -values and omitting the common factor $\alpha\epsilon^2\zeta^2 - \delta\beta^2\gamma^2$, we have

$$\delta(A\alpha\beta\gamma + B\alpha\epsilon\zeta + C\delta\beta\gamma + D\delta\epsilon\zeta) - \alpha(A'\alpha\beta\gamma + B'\alpha\epsilon\zeta + C'\delta\beta\gamma + D'\delta\epsilon\zeta) = 0,$$

which is of the fourth order in θ . We may assume $C' = 0$, and $D' = 0$; in fact, the terms in C' and D' might be got rid of by the substitutions $z(xz - y^2) = y(xw - yz) - x(yw - z^2)$, $w(xz - y^2) = z(xw - yz) - y(yw - z^2)$; we then have 6 coefficients A, B, C, D, A', B' , and equating to zero the terms in $\theta^0, \theta^1, \theta^2, \theta^3, \theta^4$, the ratios of these are determined by the five equations: and we have thus the required cubic surface.

Starting from either cubic considered as a curve on the given quadric surface; if through the centre we describe a cubic surface, this meets the quadric surface in a second cubic and the two cubics intersect each other in 5 points. In particular, if the one cubic is $x : y : z : w = 1 : \phi : \phi^2 : \phi^3$, it appears by what precedes that the other cubic may be taken to be $x : y : z : w = \alpha\beta\gamma : \alpha\epsilon\zeta : \delta\beta\gamma : \delta\epsilon\zeta$, intersecting the former in the 5 points given by the equation $\delta\beta^2\gamma^2 - \alpha\epsilon^2\zeta^2 = 0$. The five points are of course points of contact of the quadric surface and the cubic surface.

A very simple instance is the following. The quadric surface $xw - yz = 0$ and the cubic surface

$$y(xz - y^2) + z(yw - z^2) = 0$$

intersect in the two cubics

$$x : y : z : w = 1 : \phi : \phi^2 : \phi^3 \quad \text{and} \quad x : y : z : w = 1 : \theta^2 : \theta : \theta^3.$$

310 ON THE TWISTED CUBICS UPON A QUADRIC SURFACE. [838

The two cubics meet in the 5 points

$$\theta = (0, \infty, 1, \omega, \omega^2); \quad \phi = (0, \infty, 1, \omega^2, \omega),$$

or, what is the same thing, in the 5 points

$$(x, y, z : w) = (1, 0, 0, 0), (0, 0, 0, 1), (1, 1, 1, 1), (1, \omega, \omega^2, 1), (1, \omega^2, \omega, 1),$$

where ω is an imaginary cube root of unity.

I was anxious to work out this result not for its own sake, but as an illustration of a point which first arises as to otics: a unicursal octic curve is not in general situate on any surface lower than a quartic surface; if on the octic we take at pleasure 33 points, we may through these draw two quartic surfaces $U = 0$, $V = 0$, each of which will entirely contain the curve: the two surfaces meet in a second octic also unicursal, and the two curves intersect in 34 points, which are points of contact of the two quartic surfaces. We cannot, by adjoining to the given octic any curve of an inferior to its own, obtain a complete intersection of two surfaces: the most simple case is when we adjoin to the given octic as above another unicursal octic, and thus obtain a complete intersection of two quartic surfaces. It is to be observed that the second curve is determined completely and uniquely by the given curve: considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ , the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

839.

ON THE MATRICAL EQUATION $qQ - Qq = 0$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 176–178.]

I CONSIDER the matrical equation $qQ - Qq' = 0$, where q, q' are given matrices $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$, $\begin{vmatrix} a', b' \\ c', d' \end{vmatrix}$, and $Q = \begin{vmatrix} A, B \\ C, D \end{vmatrix}$, is a matrix which has to be determined. As remarked in the paper “On the Quaternion Equation $qQ - Qq' = 0$,” *Messenger*, t. XIV. (1885), pp. 108–112, [836] the question for matrices is equivalent to that for quaternions: for a matrix $\begin{vmatrix} a, b \\ c, d \end{vmatrix}$ may be regarded as a quaternion

$$\frac{1}{2}(a + d) - \frac{1}{2}\lambda(a - d)i + \frac{1}{2}(b - c)j - \frac{1}{2}\lambda(b + c)k,$$

or (omitting the factor $\frac{1}{2}$) as the quaternion

$$(a + d) - \lambda(a - d)i + (b - c)j - \lambda(b + c)k,$$

where $\lambda = \sqrt{-1}$, is the imaginary of ordinary algebra. Hence considering q, q' as denoting the quaternions

$$\begin{aligned} & (a + d) - \lambda(a - d)i + (b - c)j - \lambda(b + c)k, \\ & (a' + d') - \lambda(a' - d')i + (b' - c')j - \lambda(b' + c')k, \end{aligned}$$

we can, if a certain condition is satisfied, find a quaternion Q such that $qQ - Qq' = 0$; say this is $Q = W + iX + jY + kZ$; putting this we find

$$Q = \begin{vmatrix} A, B \\ C, D \end{vmatrix} = \begin{vmatrix} W + \lambda X, & Y - \lambda Z \\ -Y - \lambda Z, & W - \lambda X \end{vmatrix},$$

for the required matrix Q ; this being an indeterminate matrix, such that $AD - BC = 0$.

312 ON THE MATRICAL EQUATION $qQ - Qq = 0$. [839]

But it is better to solve directly the matrical equation

$$\begin{vmatrix} (A, B)(a, b), & (A, B)(b', d') \\ (C, D)(a, b), & (C, D)(b', d') \end{vmatrix} - \begin{vmatrix} (a, b)(A, C), & (a, b)(B, D) \\ (c, d)(A, C), & (c, d)(B, D) \end{vmatrix} = 0,$$

viz.

$$\begin{vmatrix} (A, B)(a, b), & (A, B)(b', d') \\ (C, D)(a, b), & (C, D)(b', d') \end{vmatrix} - \begin{vmatrix} (a, b)(A, C), & (a, b)(B, D) \\ (c, d)(A, C), & (c, d)(B, D) \end{vmatrix} = 0,$$

that is,

$$Aa + Bc' = aA + bC,$$

$$Ba + Dc' = aB + bD,$$

$$Ac + Cd' = cA + dC,$$

$$Bc + Dd' = cB + dD,$$

or, what is the same thing,

$$\begin{vmatrix} a - a', & -c', & 0, & b \\ -b', & a - d', & b, & 0 \\ c, & 0, & d - a', & -c' \\ 0, & c, & -b', & d - d' \end{vmatrix} (A, B, C, D) = 0,$$

viz. we have (A, B, C, D) connected by these four linear equations: the necessary condition is that the determinant formed out of the matrix which here presents itself shall be $= 0$.

After some reduction, and putting for shortness

$$\nabla = ad - bc, \quad \nabla' = a'd' - b'c',$$

this is found to be

$$(\nabla - \nabla')^2 + [\nabla(a' + d') - \nabla'(a + d)](a + d - a' - d') = 0;$$

which is the condition for the existence of a solution. This condition being satisfied, the four equations will be equivalent to three independent equations, which serve to determine A, B, C, D ; and, assuming the absolute value of A , we find

$$A = -(a' + d' - a - d),$$

$$\nabla B = \nabla' \cdot b' + \nabla(a' + d' - a - d) + ab' - a'b,$$

$$\nabla C = \nabla' \cdot c' + \nabla(a' + d' - a - d) + dc' - d'c,$$

$$\nabla D = D \cdot a' - (d' - d)(a' + d' - a - d) - cc'(a' + d' - a - d) + b'c(a' + d' - a - d)';$$

values which give

$$ad' + b'c = ad - bc;$$

viz. in the case where the given matrices are such that $ad + bg = (a' + d') - bc - bc' + ad'$, the components A, B, C, D of the required matrix Q are determined as such that $AD - BC = 0$.

839] ON THE MATRICAL EQUATION $qQ - Qq = 0$. 313

If we have $\nabla - \nabla' = 0$, that is, $ad - bc = a'd' - b'c'$, then the condition becomes $a' + d' = a + d$; and these two conditions being satisfied, the four equations reduce themselves to two independent equations, and the ratios $A : B : C : D$ have no longer determinate values; the four equations may in fact be written

$$\begin{vmatrix} 0, & b, & -c', & a - a' \\ -b, & 0, & a + d', & b' \\ c, & a' - d, & 0, & -c' \\ a - d, & c, & -b, & 0 \end{vmatrix} (D, C, B, A) = 0,$$

which are of the form

$$\begin{vmatrix} 0, & -h, & g, & a \\ h, & 0, & -f, & b \\ -g, & f, & 0, & c \\ -a, & -b, & -c, & 0 \end{vmatrix},$$

where the coefficients a, b, c, f, g, h are such that $af + bg + ch = 0$; and the four equations are thus equivalent to two independent equations.

To obtain a symmetrical solution, assume a relation with arbitrary coefficients ξ, η, ζ, ω ; we then find

$$\begin{aligned} D &= -c\eta - h\zeta + b\omega, \\ B &= h\eta + f\zeta + a\omega, \\ C &= -g\eta + a\xi + g\omega, \\ A &= -b\xi - f\eta + h\omega, \end{aligned}$$

viz. the complete theorem is that, if the matrices

$$q = \begin{vmatrix} a, & b \\ c, & d \end{vmatrix}, \quad q' = \begin{vmatrix} a', & b' \\ c', & d' \end{vmatrix},$$

are such that $ad - bc = a'd' - b'c'$, $a + d = a' + d'$, then writing

$$a = a - a', \quad b = d + d', \quad f = a' + d, \quad g = a + d', \quad c' = c, \quad h = b,$$

values which satisfy $af + bg + ch = 0$, and taking ξ, η, ζ, ω arbitrary, we have for the matrix Q , which is such that $qQ - Qq' = 0$, the expression

$$Q = \begin{vmatrix} -b\xi - f\eta + h\omega, & h\eta + f\zeta + a\omega \\ -g\eta + a\xi + g\omega, & -c\eta - h\zeta + b\omega \end{vmatrix},$$

depending really on one arbitrary parameter, viz. we may without loss of generality put any two of the coefficients ξ, η, ζ, ω equal to 0.

840.

ON MASCHERONI'S GEOMETRY OF THE
COMPASS.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp.
179–181.]

I HAVE not seen the original of Mascheroni's work, *La Geometria del Compasso*, 8^{vo}, Pavia (1797), but only the French translation, *Géométrie du Compas*, par *L. Mascheroni*, traduite de l'Italien par *A. M. Carette*, 2^e ed. 8^{vo} Paris, 1828 (Author's Preface, pp. 5–24, and pp. 25–328, with 14 plates containing 108 figures). The title expresses the notion of the work: a straight line is given by means of two points thereof, but is not allowed to be actually drawn; and the problem is, with the compass alone to perform all the constructions of Geometry. Observe that, for the purpose of demonstration, any lines may be imagined to be drawn, and such lines are in fact shown as dotted lines in the figures; but this does not in any wise interfere with the fundamental postulate that the constructions are to be performed with the compass only.

Assuming, then, that a line is in every case given by means of two points thereof, the leading questions are those considered Book VII. "on the intersections of straight lines with circular arcs and with each other," viz. they are (1) to find the intersections of a given line and circle; (2) to find the intersection of two given lines. But in the first problem it is necessary to consider two cases; (A) the general case, where the line does not pass through the centre of the circle, and (B) the particular, but actually more difficult, case, where the line passes through the centre of the circle. It is assumed that the centre of the circle is given: if it is not given, it can be found as afterwards mentioned. It will be convenient to establish the definition of counter-points in regard to a given line: two points, which are such that the line joining them is bisected at right angles by the given line, are said to be counter-points in regard to that line.

(1), A. Consider a line given by means of two points P , Q ; and a given circle, centre C . With centre P and radius PC describe a circle, and with centre Q and

840] ON MASCHERONI'S GEOMETRY OF THE COMPASS. 315

radius QC a circle; these meet in C and in a second point D which will be the counter-point of C in regard to the line PQ ; hence with centre D and radius = that of given circle, describing a circle, this meets the given circle in two points which are the intersections of the given circle by the line PQ .

The construction fails if the line PQ passes through the centre of the given circle, for then the two auxiliary circles touch at C , or we have D coincident with C . We have therefore considered this case; Q may be taken to be the centre C of the circle; so that we have:

(1), B. The problem is, given a point P , and a circle, centre C : to find the intersections of the circle by the line CP . With centre P , and an arbitrary radius, describe a circle meeting the given circle in points A, B (which are of course counter points in regard to line CP); the circle is divided into two arcs AB and $360 - AB$, and the problem is to bisect each of these two arcs; for the points of bisection are obviously the required intersections of CP with the circle. Hence we have:

(1), C. To bisect a given arc AB of a circle, centre C (fig. 1). With centre A and radius AC describe a circle; and with centre B , and equal radius BC describe a circle; and on these circles respectively, find the points D, E such that $CD = CE = AB$. With centre E and radius EA , describe a circle; and with centre D , and equal radius DB , describe a circle; and let these two circles meet in F ; then with centre E (or D) and radius CF describe a circle: this will meet the given circle in the required middle point G of the arc AB .

The proof depends on the theorem that in a rhombus the sum of the squares of the diagonals is = sum of the squares of the four sides. We have a rhombus $ACEB$, and therefore

$$(AE)^2 + (BC)^2 = 2(AB)^2 + 2(AC)^2,$$

that is,

$$(AE)^2 = 2(AB)^2 + \text{rad.}^2.$$

But by construction, $EF^2 = (AE)^2$; therefore

$$EF^2 = 2(AB)^2 + \text{rad.}^2, \quad = 2(CE)^2 + \text{rad.}^2.$$

Hence in the right-angled triangle FCE , we have

$$(CF)^2 = (EF)^2 - CE^2, \quad = (CE)^2 + \text{rad.}^2;$$

316 ON MASCHERONI'S GEOMETRY OF THE COMPASS. [840

and by construction, $(EG)^2 = (CF)^2$; that is,

$$(EG)^2 = (CE)^2 + (CG)^2,$$

viz. GEC is a right-angled triangle; that is, CG is at right angles to BCE , or the line CGF bisects the arc AB in G .

For the second problem, (2), to find the intersection of two given lines, we require the solution of the problem to find a fourth proportional to three given distances, and this immediately depends on the following problem.

Problem. In two concentric circles to place chords subtending equal angles. If AB (fig. 2) be a chord of the one circle, then with centres A and B respectively, and an arbitrary radius $AA' = BB'$, describing circles to cut the larger circle in the points A' and B' (A' is either of the two intersections and B' is the intersection lying in regard to CB as A' lies in regard to CA) then clearly $\angle ACB = \angle A'CB'$.

And hence also we have the following problem.

Problem. To find a fourth proportional to three given distances. We have only to take as the given distances the two radii CA , CA' and the chord AB ; and then from the similar triangles ACB , $A'CB'$, we have $CA : CA' :: AB : A'B'$, viz. $A'B'$ is a fourth proportional to CA , CA' , AB .

We have now the solution of (2) To construct the intersection of two given lines AB and CD (fig. 3). Find c the counter-point of C , and d the counter-point of D , in regard to the line AB

Fig. 1.

[Figure: arc AB of circle centre C with construction points D, E, F, G for bisection]

Fig. 2.

[Figure: two concentric circles with chords AB and $A'B'$ subtending equal angles at the common centre C]

Fig. 3.

[Figure: lines AB and CD with their intersection S constructed via counter-points c of C and d of D and parallelogram $Cy dD$]

840] ON MASCHERONI'S GEOMETRY OF THE COMPASS. 317

and find y such that the distances Cy, dy are $= Dd, DC$ respectively; y is in a line with cC , and we have $Cy dD$ a parallelogram. Find cS a fourth proportional to cy, cd, cC , and with centres c, C respectively and radii each $= cS$ describe circles cutting in the point S ; this will be the required intersection of the two lines. In fact, the required point S will be the intersection of the two lines CD, cd : supposing these lines each of them drawn, and also the lines cCy and dy , we have DC parallel to dy , that is, the triangles cdy, cSC are similar to each other or

$$cy : cd :: cC : cS :$$

viz. the distance cS having been found by this proportion, and the point S found as the intersection of the two circles, centres c and C respectively, the point S so determined is the required point of intersection of the given lines AB and CD .

If a circle be given without its centre being known, then taking any three points A, B, C on the circle, and a pair of counter-points D, E of the line AB , and also a pair of counter-points F, G of the line AC , we have obviously the centre of the given circle as the intersection of the lines DE and FG ; and the centre can thus be found with the compass only.

It is proper to remark that the problems considered in the present paper are those connecting the theory with ordinary geometry, not the problems which are most readily and elegantly solved with the compass only: a large collection of these are contained in the work, and in particular the twelfth book contains some interesting approximate solutions of the problems of the quadrature of the circle, the duplication of the cube, and other problems not solvable by ordinary geometry.

Cambridge, *March* 19, 1885.

841.

ON A DIFFERENTIAL OPERATOR.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 190, 191.]

WRITE $X = 1 + bx + cx^2 + \dots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots$; then by Capt. MacMahon's theorem, any non-unitary function of the roots $\alpha, \beta, \gamma, \dots$ is reduced to zero by the operation

$$\Delta, \quad = d_b + bd_c + cd_d + \dots;$$

for instance, if $(2), = \Sigma \alpha^2 = b^2 - 2c$, we have

$$\Delta(b^2 - 2c) = 2b + b(-2), = 0.$$

We have

$$\Delta X = x^2 + bx^3 + cx^4 + \dots = xX;$$

and writing, moreover, $X', = b + 2cx + 3dx^2 + \&c.$, for the derived function of X , then

$$\Delta X = x^2 + bx^3 + cx^4 + \dots = (xX)' \cdot / X$$

We can hence shew that $\Delta \left(\frac{X'}{X} - b \right) = 0$; the value is, in fact,

$$\frac{\Delta X' \cdot X - X' \Delta X}{X^2} - \Delta b, \text{ that is, } \frac{(xX)' X - xX \cdot X'}{X^2} - 1,$$

which is

$$= \frac{X + xX' - xX'}{X} - 1, = 0.$$

This is right, for $\frac{X'}{X} - b$ is a sum of non-unitary symmetric functions of the roots; in fact,

$$\frac{X'}{X} = \Sigma \frac{-\alpha}{1 - \alpha x} = -(1) - (2)x - (3)x^2 - \&c.,$$

or since $b = (1)$, this is $\frac{X'}{X} - b = -(2)x - (3)x^2 - \&c.$, a sum of non-unitary functions of the roots.

842.

ON THE VALUE OF $\tan(\sin \theta) - \sin(\tan \theta)$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 191, 192.]

THE following equation is given p. 59 of the *Lady's and Gentleman's Diary* for 1854:

$$\tan(\sin \theta) - \sin(\tan \theta) = \frac{1}{30}\theta^7 + \frac{29}{756}\theta^9 + \&c.$$

Write in general

$$\begin{aligned} X &= \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 + \dots, \\ Y &= \theta + A'\theta^3 + B'\theta^5 + C'\theta^7 + D'\theta^9 + \dots \end{aligned}$$

Then, as far as θ^9 , we have

$$\begin{aligned} X + A'X^3 + B'X^5 + C'X^7 + D'X^9 &= \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 \\ &\quad + A'(\theta^3 + 3A\theta^5 + 3(A^2 + B)\theta^7 + (A^3 + 6AB \\ &\quad + B'(\theta^5 + 5A\theta^7 + (10A^2 + 5B)\theta^9) \\ &\quad + C'(\theta^7 + 7A\theta^9) \\ &\quad + D'(\theta^9) \\ &= \theta \\ &\quad + \theta^3(A + A') \\ &\quad + \theta^5(B + 3AA' + B') \\ &\quad + \theta^7(C + 3A^2A' + 3BA' + 5AB' + C') \\ &\quad + \theta^9(D + A'(A^3 + 6AB + 3C) + 5B'(2A^2 + 6AB' + 3A^2A' + B'^2)) \end{aligned}$$

320 **ON THE VALUE OF** $\tan(\sin \theta) - \sin(\tan \theta)$. **[842]**

and hence

$$\begin{aligned} & X + A'X^3 + B'X^5 + C'X^7 + D'X^9 \\ & -Y - A'Y^3 - B'Y^5 - C'Y^7 - D'Y^9 \\ & = \theta^7 \{3AA'(A - A') + 2(AB' - A'B)\} \\ & + \theta^9 \{AA'(A^2 - A'^2) + 6AA'(B - B') + 4(AC' - A'C) + 10(A^2B - A'^2B')\}. \end{aligned}$$

Now let

$$\begin{aligned} X = \sin \theta &= \frac{1}{2}\theta - \frac{1}{6}\theta^3 + \frac{1}{120}\theta^5 - \frac{1}{5040}\theta^7 + \dots; \\ A &= -\frac{1}{6}, \quad B = \frac{1}{120}, \quad C = -\frac{1}{5040}, \\ Y = \tan \theta &= \frac{1}{2}\theta + \frac{1}{3}\theta^3 + \frac{2}{15}\theta^5 + \frac{17}{315}\theta^7 + \dots; \\ A' &= \frac{1}{3}, \quad B' = \frac{2}{15}, \quad C' = \frac{17}{315}; \end{aligned}$$

we have therefore

$$\begin{aligned} AA' &= -\frac{1}{18}, \quad A - A' = -\frac{1}{2}, \quad A + A' = \frac{1}{6}, \quad B - B' = -\frac{1}{8}, \\ AB' - A'B &= -\frac{1}{45}, \quad AC' - A'C = -\frac{17}{1890}, \quad A^2B - A'^2B' = \frac{19}{2520}. \end{aligned}$$

Hence

$$\begin{aligned} \text{coeff. } \theta^7 &= \frac{1}{12} - \frac{2}{45}, \quad = \frac{1}{30}, \\ \text{coeff. } \theta^9 &= \frac{1}{216} + \frac{1}{24} - \frac{17}{945} + \frac{19}{504}, \quad = \frac{29}{756}; \end{aligned}$$

and the required equation is thus verified.

843.

ON THE QUADRIQUADRIC CURVE IN
CONNEXION WITH THE THEORY OF ELLIPTIC
FUNCTIONS.

[From the *Mathematische Annalen*, t. XXV. (1885), pp. 152–156.]

I CALL to mind that, if we have on a line two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and through the line two planes

$$Ax + By + Cz + Dw = 0, \quad A'x + B'y + C'z + D'w = 0,$$

then

$$\begin{aligned} B\gamma' - B'\gamma : \gamma\alpha' - \gamma'\alpha : \alpha\beta' - \alpha'\beta : \alpha\delta' - \alpha'\delta : \beta\delta' - \beta'\delta : \gamma\delta' - \gamma'\delta \\ = AD : BD : CD : BC : CA : AB, \end{aligned}$$

and that, putting each of these two equal sets of values

$$= a : b : c : f : g : h, \quad \text{or} \quad = A : B : C : D : BC : CA : AB,$$

then the quantities a, b, c, f, g, h which it is easy to see satisfy the relation $af + bg + ch = 0$, are called the “six coordinates” on the line; we call them the six coordinates of the line, as also the coordinates of either of the six line-axiomatical equations which is a single relation between these axes, the so-called axiomatical contains a single relation between them. Or, in the language of the modern geometry, the determination of the particular line. See my paper “On the six coordinates of a line,” *Camb. Phil. Trans.* t. XI. (1869), pp. 290–323, [435].

I consider for a moment the quadric surface

$$\alpha x^2 + \beta y^2 + \gamma z^2 + \delta w^2 = 0,$$

and I proceed to shew that, if (α, b, c, f, g, h) are the coordinates of a generating line on the surface, then either $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$, or else $a = -\sqrt{f}$, $b = -\sqrt{g}$, $c = -\sqrt{h}$; the one or other system of equations according as the line belongs to the one or other system of generating lines.

322 ON THE QUADRIQUADRIC CURVE IN CONNEXION [843

We satisfy the equation by

$$\alpha x + by + \theta z + \theta w = 0, \quad x = \frac{1}{\phi}y - \frac{1}{\phi}z + \theta w = 0,$$

where θ is an arbitrary parameter: hence these equations determine a generating line of the surface: and the coordinates of this line are

$$\begin{array}{cccccc} a, & b, & c, & f, & g, & h \\ = -1 \left(\theta - \frac{1}{\phi} \right), & - \left(\theta + \frac{1}{\phi} \right), & \theta\phi, & - \left(\theta - \frac{1}{\phi} \right), & \theta + \frac{1}{\phi}, & -\phi; \end{array}$$

viz. these values give $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$.

Similarly we satisfy the equation by

$$\begin{array}{l} \alpha x + by + \phi z - \phi w = 0, \\ x = \frac{1}{\phi}y - \frac{1}{\phi}z - \phi w = 0, \end{array}$$

where ϕ is an arbitrary parameter: hence these equations also determine a generating line of the surface: and the coordinates of this line are

$$\begin{array}{cccccc} a, & b, & c, & f, & g, & h \\ = -1 \left(\phi - \frac{1}{\phi} \right), & \phi + \frac{1}{\phi}, & -\phi^2, & - \left(\phi - \frac{1}{\phi} \right), & \phi + \frac{1}{\phi}, & -\phi^2; \end{array}$$

viz. these values give $a = -\sqrt{f}$, $b = -\sqrt{g}$, $c = -\sqrt{h}$.

If for x, y, z, w we write $x\sqrt{f}$, $y\sqrt{g}$, $z\sqrt{h}$, $w\sqrt{f}$ respectively, then we have the theorem that, for the quadric surface

$$pa^2 + qy^2 + rz^2 + sw^2 = 0,$$

the coordinates (a, b, c, f, g, h) of a generating line are such that

$$\frac{a}{\sqrt{f}} = \pm \sqrt{\frac{pa}{qb}}, \quad \frac{b}{\sqrt{g}} = \pm \sqrt{\frac{q}{r}}, \quad \frac{c}{\sqrt{h}} = \pm \sqrt{\frac{r}{s}};$$

the signs being all + or all -, according as the line belongs to the one or other of the systems of generating lines.

Take $(\alpha', b', c', f', g', h')$ for the coordinates of an arbitrary line, and write

$$\begin{aligned} p, q, r, s &= q\beta(f'b' + b'h'), q\beta(g'a' + a'g'), r\beta(g'a' + a'g'), s\beta(c'a' + a'c'); = a'fbcgh', \\ a'g'bsr - b'sf'gh' + c'gh'a' + cha'b'g'd &= 0, \end{aligned}$$

**843] WITH THE THEORY OF ELLIPTIC FUNCTIONS.
323**

which is a section having the line (a', b', c', f', g', h') for a generating line. To verify this, observe that for the line in question we have

$$\begin{aligned} Ky - qx + zw &= 0, \\ -Kx + qy + zw &= 0, \\ yg + bg' \cdot x + xw &= 0, \\ -x'g + b'p \cdot y + z'w &= 0, \end{aligned}$$

equivalent of course to two independent equations: these give $K(x = b'y \cdot k)w$, $Ky = pa \cdot kw$ and so on, where values substituted in the quadric equation give it identically.

And for the last-mentioned quadric surface that, for the system $a = \sqrt{f}$, $b = \sqrt{g}$, $c = \sqrt{h}$, of the conditions for a generating line, the equation reduces itself to

$$af + bg + ch = 0,$$

where the signs = correspond to the system a, b, c, g, h, f , and the sign – to the other system.

Taking R as origin, the equations of the conic are

$$\begin{aligned} Kx &= py - qz + zw = 0, \\ -Kx + qy - rz + sw &= 0, \end{aligned}$$

equivalent of course to two independent equations of the form

$$\alpha\beta x + \alpha'\beta'y + \beta'\alpha c + s\alpha\beta' = 0, \quad \text{equation of a generating line,}$$

which is of course not to be confounded with the equation of a generating line.

Considers now the quadriquadric curve

$$(x, y, z, w)(a', b', c', f', g', h') = 0, \quad (x, y, z, w)(\alpha', \beta', \gamma', \delta') = 0,$$

where the quadric forms a'', a'' , are connected with each other by the relation $a'\beta'a' + a''b'a' + a''c'a' + a''d'a' = 0$.

If, then, for an arbitrary line (a', b', c', f', g', h') , we have $aa' + bb' + cc' + ff' + gg' + hh' = 0$, this relation is also one of the conditions for the relation between any two lines: any one whatever of

the four lines through any point of the curve corresponds to the two systems respectively, of a properly determined quadric surface: it is a property of this curve that, of any four lines through any one of the points whatever of the system, those one is symmetric (whatever be the value), in the other one with the system, and the other two are unconnected.

Writing in the equations $a = 1$, we have in particular the quadri-quadric curve

$$y^2 + b'\beta'z + \beta = 0, \quad x^2 + b''y + c''z + d''w = 0,$$

where the second is the symmetric definition of the system through the equation $g + h = 0$. Consider on the curve four points belonging to the arguments u_1, u_2, u_3, u_4 respectively;

324 ON THE QUADRIQUADRIC CURVE IN CONNEXION [843

and write for shortness x_1, y_1, z_1, w_1 for the sn, cn , and similarly for x_2, y_2, z_2, w_2 . It appears by Abel's theorem that the condition in order that the four points may be in a plane is $u_1 + u_2 + u_3 + u_4 = 0$; viz. when this equation is satisfied we have

$$\begin{vmatrix} x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \\ x_3 & y_3 & z_3 & w_3 \\ x_4 & y_4 & z_4 & w_4 \end{vmatrix} = 0,$$

or writing

$$\begin{aligned} & a, \quad b, \quad c, \quad f, \quad g, \quad h \\ &= x_4 y_2 - x_2 y_4, \quad x_3 y_1 - x_1 y_3, \quad z_4 w_2 - z_2 w_4, \quad z_3 w_1 - z_1 w_3, \\ & \quad a_2, \quad b'', \quad c'', \quad f', \quad g', \quad h', \\ &= x_4 y_4, \quad x_3 y_3, \quad z_4 w_4, \quad z_3 w_3, \quad y_3 z_3, \quad x_4 w_4, \end{aligned}$$

this equation is

$$af + bg + ch = 0;$$

or by what precedes, it appears that not only is this so, but that we have separately

$$af^2 + cf'' = 0, \quad bg' + b'g = 0, \quad ch' + ch = 0,$$

viz. it follows from Abel's theorem that, when the arguments u_1, u_2, u_3, u_4 are connected by the equation $u_1 + u_2 + u_3 + u_4 = 0$, then each of these three equations holds good.

I assume in particular $u_1 = 0, u_2 = -u$, so that $x_2 = x, u_2 = -u, y_2 = y, z_2 = z, w_2 = w, x_3 = 0, y_3 = b_2, w_3 = a_2, w_2 = c_2$; we have $x_4 = a, b_4 = b_4 = b_2 = c_2, w_4 = d_4 = c_2, w_4 = c_4$,

$$\alpha, \quad b, \\ = a_2 - d_2 x, \quad -x_1 = -a_2 x = -a_2 x_2 = -x_2 x_2 x_1, \quad x = b_2 y, \quad -x_2 y = d_3 y, \quad d_2 x_3 y$$

and the four equations become

$$\begin{aligned} & \frac{a - d_2 x \cdot d_4 z + b_2}{x} \cdot d_2 w = 0, \\ & -x = -d_2 y + a_2 x = 0, \\ & \frac{d_2 - z_2 \cdot b_2 y - 1}{c} + \frac{d_2 x - z_2 \cdot a_2 y - 1}{c_2} = 0, \end{aligned}$$

equations which may be written

$$\frac{a - d_2x}{a_2x} = \frac{w_2 \cdot d_4z + d_2y}{a_2x} \cdot \frac{-x}{-z} \cdot \frac{c_3x}{c_3} = \frac{a_2x + d_2y}{c_3}, \quad \frac{x}{a_2x - d_2y} = \frac{c_2}{c_3} \cdot \frac{a_2x - d_2y}{c_3},$$

so that, adding these equations, we must have an identity.

Representing the second and third of the last-mentioned equations by

$$\frac{a+1}{x} = C, \quad \frac{a-1}{x} = D,$$

843] WITH THE THEORY OF ELLIPTIC FUNCTIONS. 325

we have

$$a = 1 - Cx, \quad d = 1 + Dx,$$

from either of which we can obtain a rationally; viz. the first equation gives

$$1 - a = 1 - 2Cx + C^2x^2,$$

that is, $x = \frac{2C}{1 + C^2}$, whence

$$a = \frac{1 - C^2}{1 + C^2}, \quad d = \frac{1 + C^2 + 2CD}{1 + C^2};$$

and similarly from the second equation $a = -\frac{1}{2D}$. whence also

$$a = \frac{b'z + 2P - 2CD}{D^2 + D^2}, \quad d = \frac{D^2 - 1}{D^2 + D^2}.$$

Substituting for C its value, the first-mentioned value of a becomes

$$a = \frac{(a_2 - z_4)(d_2x - d_2z)}{1 - 2x_2x_4z - z_2z_4z_2x_4},$$

which is a form due to Cayley.

For the like values of b and d we easily verify; we in fact have

$$\begin{aligned} x_3 = sn(u_3 + u_4) &= \frac{u_2u_3 \cdot d_2u_3 + a_3u_2 \cdot d_3u_2}{1 - k^2u_2u_3 \cdot a_3u_2 \cdot a_3 \cdot u_3}, \\ &= \frac{a_2u_3 \cdot d_3u_3 + x_2u_3 \cdot a_2u_3}{a_3u_2 \cdot a_2 + k^2x_2x_3y_2y_3}, \end{aligned}$$

we say *Elliptic Functions*, p. 63; or calling these values

$$= \frac{N_1}{D_1} = \frac{N_2}{D_2} = \frac{N_3}{D_3} = \frac{N_4}{D_4},$$

the above value is

$$x = \frac{N_1 - N_2}{D_1 - D_2},$$

which is right.

Cambridge, 28 June, 1884.

844.

ON A THEOREM RELATING TO
SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 212, 213.]

I FIND among my papers the following example of a general theorem relating to seminvariants. Write:

$$\begin{aligned}\Theta &= 4b_8 + a_2b_6 + a_2^2b_4 + a_2^3b_2 + 4a_2^4b_0 + 7a_4b_4 + 16a_4^2b_0, \\ \Omega &= b_8 + 3a_2b_6 + 5a_2^2b_4 + 4a_4b_4 + 5a_2^3b_2 + 9a_4a_2b_2 + 7a_4^2b_2;\end{aligned}$$

then from a seminvariant S of the degree δ and weight w , by means of the operation Θ we obtain a seminvariant S of the degree $\delta + 1$ and weight $w + 8$; and by operating with Ω we obtain a combination of these by operating on S with the resulting combination $2w\Omega - 8\Omega$, which combination is of the degree $w + 1$, but only of the degree $w + 8$; via. operating with $2w\Omega - 8\Omega$, we obtain a new seminvariant of the same degree but of a weight increased by unity.

The example relates to the under-mentioned seminvariant S of the degree 5 and weight 7. Write

b_0	+1
b_2	-7
b_4	-9
b_6	+9
b'_8	-123
b_8	-10
bf'	+39

844]

ON A THEOREM RELATING TO
SEMINVARIANTS.

327

	ΘS		ΩS		$\Theta\Theta S$		$\Theta\Omega S$		$\Omega\Theta S$		$\Omega\Omega S$						
--	------------	--	------------	--	------------------	--	------------------	--	------------------	--	------------------	--	--	--	--	--	--

[The two-page paper concludes with a numerical verification table demonstrating the action of Θ and Ω on the chosen seminvariant S ; for the table of differences and the proof that $\Theta\Omega - \Omega\Theta = 0$ on S , see the original *Quarterly Journal*, vol. XX, p. 213.]

The columns above ΘS (where observe that, to operate with the b_8 , of Θ , we restore the proper power of a_2 , reading S as being = $1.b^4 + 7a^3b_2$, and proceeding ultimately $\Theta S = \frac{1}{8}\Omega S = -55\Omega$; and the sum of these, which is a seminvariant, degree 5, weight 7, also 125'' and 145); and the sum of these, which is a seminvariant, degree 5, weight 7, also 125'' and 145; the upper portion comprises proof of the theorem $S = 1.b_4 + 7a^3b_2$; instead of giving this sum, I have added a column equal to $\frac{1}{16}(\Theta\Omega - \Omega\Theta)S$ this gives also the sum of the above seminvariants, viz. when this equation is the foregoing direct process from the given seminvariant S of the degree 5 and weight 7.

845.

ON THE ORTHOMORPHOSIS OF THE CIRCLE
INTO THE PARABOLA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 213–220.]

IT is remarked by Schwarz (see his Memoir “Ueber einige Abbildungsaufgaben,” *Crelle*, t. LXX. (1869), pp. 105–120, p. 113), that the circle $x^2 + y^2 - 1 = 0$ can be orthomorphosed into the parabola $y^2 + 4(1 - x) = 0$ by the equation

$$x'(x' + iy') = \log x + iy'(x + iy);$$

viz. this equation establishes a $(2, 1)$ relation between the points interior to the circle and those interior to the parabola, which, qua relation between $x' + iy'$ and $x + iy$, will be orthomorphic, that is, infinitesimal elements of the one area will correspond to similar infinitesimal elements of the other area. The diameter $y' = 0$ of the circle is transformed into the axis $y = 0$ of the parabola, and figs. 1, 2 are symmetrical on the two sides of these lines respectively; we may therefore consider only the transformation

**845] ON THE ORTHOMORPHOSIS OF THE CIRCLE
INTO THE PARABOLA. 329**

of the upper semicircle into the upper half of the parabola; we have (see figs. 1 and 2) A' corresponding to the vertex A of the parabola, B' to the point at infinity on the infinite branch AB of the parabola, the semicircular boundary $B'C'A'$ to the boundary BCA of the parabola, the highest point (and any point C' between the vertex A and B' on the parabola, the highest point C corresponding to the point C' between the vertex and the semi-latus rectum.

We may divide the circle by concentric circles and by radii; the corresponding curves for the parabola will be ovals and radials from the focus, the curves of each system being transcendental curves. Or we may divide the parabola by means of two systems of confocal parabolas; the corresponding curves for the circle will be (see fig. 3) two systems of orthotomic licences, those of the one system having B' for a crossnode, and those of the other system having B' for an asymode.

Writing $\log \tan$ to denote the hyperbolic logarithm of the tangent, that if ϕ, ϕ' are such that

$$\phi = \log \tan\left(\frac{1}{4}\pi + \frac{1}{2}\phi'\right),$$

this equation, as is known, may also be presented in the form

$$i\phi' = \log \tan\left(\frac{1}{4}\pi + \frac{1}{2}i\phi\right),$$

$$i \tan \frac{1}{2}\phi' = \tan \frac{1}{2}i\phi,$$

or, what is the same thing,

$$\tan \frac{1}{2}\phi' = \tanh \frac{1}{2}\phi,$$

where observe that, as ϕ' increases from 0 to $\frac{1}{2}\pi$, ϕ increases from 0 to ∞ , and that always $\phi > \phi'$.

Fig. 1.

*[Figure: upper semicircle in the $x'y'$ -plane
bounded by the diameter on the x' -axis]*

Fig. 2.

*[Figure: upper half of the parabola in the
 xy -plane bounded by the axis $y = 0$]*

Fig. 3.

*[Figure: semicircle with concentric circles and
radii, and confocal parabolas with axial system
through focus, illustrating the orthomorphic
correspondence]*

330 ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA. [845

We can now establish as follows the three correspondences $A'O$ with $A'C$, $C'B$ with OB , and $A'CB$ with $A'C'B'$:

$$(1) \quad x'(x' + iy') = \frac{2\phi}{i}, \quad x' + iy' = \tan \frac{1}{2}\phi';$$

that is,

$$x = -\frac{2\phi'}{\pi}, \quad y = 0; \quad x' = \tan \frac{1}{2}\phi', \quad y' = 0;$$

ϕ is given as A, A' ; ϕ' is given as $\theta = 0, C, C'$;

$$(2) \quad x'(x' + iy') = -\tan^2 \frac{1}{2}\phi', \quad x' + iy' = i \tan \frac{1}{2}\phi';$$

that is,

$$x = -\tan^2 \frac{1}{2}\phi', \quad y = 2 \tan \frac{1}{2}\phi'; \quad x' = 0, \quad y' = \tan \frac{1}{2}\phi';$$

$\phi = 0$ gives A, A' ; $\phi = \frac{1}{2}\pi$ gives B, B' ;

$$(3) \quad x'(x' + iy') = \frac{2i\phi}{\pi}, \quad x' + iy' = e^{i\phi'};$$

that is,

$$x = 1 - \frac{2\phi}{\pi}, \quad y = \frac{2\phi}{\pi},$$

the parabola; and

$$x' = \cos 2\phi', \quad y' = \sin 2\phi',$$

the circle;

$$\phi = 0 \text{ gives } A, A'; \quad \phi = \frac{1}{2}\pi \text{ gives } B, B'.$$

Observe then for points in $OA', A'B'$, equidistant from O , we have $x' = \tan \frac{1}{2}\phi', x' = -\tan \frac{1}{2}\phi'$, and corresponding hereto we have points in OA, OB on the axis of the parabola, the values of x being $x = 1 - \frac{2\phi}{\pi}, x' = -1 - \frac{2\phi}{\pi}$, viz. the negative value is always the greater.

Observe further that, to the points $(x', y') = (\cos 2\phi', \sin 2\phi')$ and $(x', y') = (-\cos \frac{1}{2}\phi', 0)$ on the circle, and on the radius OB' , correspond the points

$$(x, y) = \left(1 - \frac{2\phi}{\pi}, \frac{2\phi'}{\pi}\right) \text{ and } (x, y) = \left(-\frac{2\phi}{\pi}, 0\right);$$

on the parabola and on the axis OB respectively; the axial distance of these two points is $\left(1 - \frac{2\phi}{\pi} + \frac{2\phi}{\pi}\right) = 1$, the radius of the circle, or the distance OA of the vertex and focus of the parabola; this is a rather curious theorem.

845] ON THE ORTHOMORPHOSIS OF THE CIRCLE
INTO THE PARABOLA. 331

The function $\log \tan(45 + \frac{1}{2}\alpha\phi')$ is tabulated, Legendre, *Théorie des Fonctions Elliptiques*, t. II. table II. pp. 256–259; viz. writing as above ϕ for the argument, we have hereby the value of ϕ , we obtain ϕ from 90 to 90 at intervals of 30' and to 12 decimals, and with 6th differences. Observe that ϕ' is thus given in degrees and minutes as a circular arc, ϕ as a number; it is convenient to have ϕ and ϕ' each as arcs, or such as numbers, the conversion may of course be made by means of a table of the lengths of circular arcs, and I have calculated the Table which I give at the end of the present paper.

I had previously calculated for the correspondence $A'O$, OB' and $A'CB$ with $A'O$, OB and ACB , the following values, at irregular intervals suited to the construction of a figure:

$x'OB'$	Circle	Parabola	
	Ordinate $\div R$	$x'O$	OB
0	0	0	0
30	.5176	.1641	-.1641
60	1.0000	.5851	-.5851
80	1.2856	1.0470	-1.0470
100	1.5320	1.6262	-1.6262
130	1.8126	2.5817	-2.5817
150	1.9319	3.4131	-3.4131
160	1.9696	3.8967	-3.8967
170	1.9924	5.5942	-5.5942
175	1.9981	4.0024	-4.0024
178	1.9994	4.6231	-4.6231
180	0	$-\infty$	∞

Write in general

$$x'(x' + iy') = \rho + i\sigma, \quad x' + iy' = \tan \frac{1}{2}(\rho + i\sigma);$$

viz. considering x , y as given, find these ρ , σ by the equations $x = p\sigma'\pi$, $y = 2\sigma\pi$, and then writing $\frac{1}{2}\rho = \theta$, $\frac{1}{2}\sigma = \phi$, we have

$$x' + iy' = \tan(\theta + i\phi) = \tan(\theta + i\phi) = \frac{P + iQ}{1 - iPQ};$$

332 ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA. [845

where

$$P = \tan \frac{1}{2}\rho\theta, \quad = \tan \frac{1}{2}\theta;$$

$$Q = \frac{1}{i} \tan \frac{1}{2}i\phi, \quad = \tanh \frac{1}{2}\phi, \quad = \frac{1}{i} \tan \frac{1}{2}i\phi, \quad = \tanh \frac{1}{2}\phi;$$

whence, if we introduce ϕ' connected with ϕ by the equation $\phi = \log \tan(\frac{1}{4}\pi + \frac{1}{2}\phi')$ as before, we have $Q = \tan \frac{1}{2}\phi'$, and the formula is

$$x'(x' + iy') = \frac{P + iQ}{1 - iPQ}, \quad P = \tan \frac{1}{2}\theta, \quad Q = \tan \frac{1}{2}\phi',$$

giving the values of x' , y' .

It is clear that we have

$$x'(x' + iy') = \frac{P - iQ}{1 + iPQ},$$

and thence

$$x'(x' + y'^2) = \frac{P + Q}{1 + P^2Q^2}.$$

Hence, to the circle $x'^2 + y'^2 = c^2$, corresponds in the parabola the curve

$$P + Q = c^2(1 + P^2Q^2),$$

where $p + iq = \tan(\rho + i\sigma)$, $P = \tan \frac{1}{2}\rho$, $Q = \tan \frac{1}{2}\phi' \tan \frac{1}{2}\sigma$. This is a complicated transcendental equation, and I do not see any convenient way of tracing the curve. The set of curves satisfy the differential equation

$$\frac{PdP + QdQ}{P + Q} = -\frac{PQ(QdP + PdQ)}{1 + P^2Q^2},$$

where dP , dQ are given in terms of $d\rho$, $d\sigma$ by

$$PdP(1 + Q^2) - QdQ(1 + P^2) = 0,$$

that is,

$$\frac{dP}{d\sigma} = \frac{1}{2} \sec^2 \frac{1}{2}\rho, \quad \frac{dQ}{d\sigma} = \frac{1}{2} \sec \phi' \sec^2 \frac{1}{2}\phi'.$$

We have $p = \frac{2\rho}{\pi}$, and thence we have to take, as elements of the several curves, $\frac{1}{2}d\rho + i\frac{1}{2}d\phi = 0$. Combining these equations, we have

$$dP \frac{dQ}{Q} \cdot dQ + (1 + P^2) - q(1 + Q^2),$$

and thence

$$\rho^2 dP(1 + Q^2) - qQdQ(1 + P^2) = 0,$$

as an equation in the locus of the normals in question. If p and q are small, then putting for a moment $\frac{1}{2}p = a$ for shortness, we have

$$P = a\rho + \frac{1}{3}a\rho^2, \quad Q = a\sigma + \frac{1}{3}a\sigma^3,$$

and the equation becomes

$$p^2(1 + \frac{1}{3}a\sigma^2)^2 - q^2(1 - \frac{1}{3}a\sigma^2)^2 - q^2(1 + \frac{1}{3}a\rho^2)^2 = 0,$$

**845] ON THE ORTHOMORPHOSIS OF THE CIRCLE
INTO THE PARABOLA. 333**

that is,

$$p^2 - q^2 + (a\rho^2 + \alpha^2)(p^2 + q^2)(p + q) = 0,$$

writing this in the form

$$p^2 - q^2 = -\frac{\pi^2}{16}(p^2 + q^2)\rho^2 + a^2(p^2 + q^2),$$

we find $x = \frac{\pi^2}{8}(x^2 + 2y^2) = 0$, or say $p^2 = \frac{\pi^2 y^2}{8}$ as the form of the meanits in the neighbourhood of the focus. Or, viz. the meanits for all of these, as might have been expected, are the kit-shaped (or negative side) of the curve.

I have been desirous to find a more general law $0 =$ between the meanits in the neighbourhood of the focus. We may say $\rho = -\frac{2\pi}{\pi}$, as in the form of the meanits in the neighbourhood of the focus.

Corresponding hereto in the circle, writing for a moment

$$x'(x' + iy') = \rho i,$$

we have

$$\tan \frac{1}{2}\rho i = \frac{P + iQ}{1 - iPQ},$$

whence

$$\begin{aligned}\rho i &= \rho - i \cdot (1 + p^2/\pi^2), \\ \rho &= P^2(1 - Q^2)(1 + pP^2); \end{aligned}$$

whence eliminating P^2 and Q successively, we may say

$$\begin{aligned}\rho^2 + \rho^2 &= a^2 \left(P^2 - \frac{1}{P^2} \right) - 1 = 0, \\ \rho^2 + \rho^2 &= a^2 \left(\frac{1}{Q^2} + 1 \right) - 1 = 0, \end{aligned}$$

say, for shortness, then $\rho^2 + 2\rho\rho' + a' = 0$, and $\rho^2 + \rho\rho - 2\rho\rho' = 0$. Introducing the polar coordinates x', y' then

$$x' = r \cos \theta', \quad y' = r \sin \theta',$$

the equations then become

$$r^2 = 2\alpha\rho' \sin \frac{1}{2}\theta' + 1,$$

$$\rho^2 = -2\alpha\rho' \cos \frac{1}{2}\theta' + 1,$$

the polar coordinates r, θ' .

Consider the quadricuspidal curve

$$r' = Ax^2 + Bx^3 + Cx + Dx^2 + Ex,$$

where $A = 4(1 + 2P^2)(1 + p^2)$, $B = \frac{1}{2}p^2$, $C = 4 + ix^2(1 - 2P^2)$, $D = +\frac{1}{2}p^2(1 + 4y)$, the coordinates of two lines meeting each side, and consequently it entirely contains the curve $\rho i, \rho i, \rho i$. They agree, as they should do, with a foregoing result,

$$\rho^2 = 0,$$

$$\rho^2 + Dy^2(t' + \frac{1}{2}h\theta'\rho') - 6 \cdot \frac{1}{2}\rho^2 + \frac{1}{2}\rho^2 + \frac{1}{2}\rho = 0,$$

and it would of course be possible to express the remaining forms 213, 642, 662, 441, 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of

334 ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA. [845

which belong to two lineapas having each of them B' for a node and O for a focus, and which of course cut each other at right angles; see fig. 4, which is a more

diagram. In fact, omitting for convenience the accents, but recollecting always that the curves belong to the figure of the circle, the first equation gives

$$(x-1)^2 + 4ax^2 \cdot \cos^2 \frac{1}{2}\theta', \quad = 2a\rho'(1 + \cos \theta) = 2a\rho'(x+r),$$

that is,

$$(r^2 + 1 - 2x)x^2 = 4a\rho'(x+r)^2.$$

Transferring to the origin B' , we must write $x+1$ for x , and then

$$r'^2 = (x+1)^2 + y^2.$$

The equation thus becomes

$$[x^2 + y^2 - 2(\alpha x + 1) + 3(\alpha x + 1)]^2 = 4a^2(x+1)(x+y^2+2x+1),$$

that is,

$$[x^2 + y^2 - 2(\alpha x + 1)]^2 = 4a^2(x^2 + y^2)(x^2 + y^2),$$

or say

$$(x^2 + y^2)^2 + 4(\alpha x + 1)(x^2 + y^2) + 4(x+1)(x^2 + y^2) + 4(\alpha x + 1)\alpha^2 y^2 = 0,$$

showing that the point B' is a crunode, the tangents there being $x \pm \alpha y$.

Writing in the equation $y = 0$, we have $x^2 = 0$, the crunode, and

$$x = 2\sqrt{(\alpha' + 1)}\sqrt{(\alpha' + 1)} \pm 1.$$

the product of these two values, $\alpha = 4(\alpha + 1)$, that is, $4 = \left(P + \frac{1}{p}\right)^2$,
via. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(\alpha + 1)}\sqrt{(\alpha' + 1)} - 1$$

**845] ON THE ORTHOMORPHOSIS OF THE CIRCLE
INTO THE PARABOLA. 335**

the smaller root, is greater than 1. The curve corresponding to a parabola $y^2 = 4p(c - x)$ is thus a crunodal lineapas, the crunode at B' , and the loop lying wholly within the circle. Moreover, the loop includes within the centre O of the circle.

The other curve is in like manner

$$[r^2 + 1 - 2x]^2 = 4(1 - x)^2,$$

viz. transforming to the origin B' , and therefore writing $x = 1$ for x , and similarly for x_n , and x_n . In support by Abel's theorem that the condition in order that the four points may be in a plane is $x = x_n + x_n + 1 = 0$; viz. when this equation is satisfied we have

$$(r^2 + 1 - 2x)x^2 = 4ax^2(x^2 + 1)^2,$$

that is,

$$[x^2 + y^2 - 2x(x + 1)]^2 = 4ax^2(x^2 + y^2),$$

or

$$(x^2 + y^2)^2 - 4(\alpha + 1)(x^2 + y^2)x + 4(\alpha + 1)x^2 + 4\alpha^2(x^2 + 2y^2 - \alpha^2 y^2) = 0,$$

showing that the point B' is a crunode, the tangents being $x \pm \alpha y$.

Writing in the equation $y = 0$, we have $x^2 = 0$, the crunode, and

$$x = 2\sqrt{(\alpha + 1)} \cdot x \pm 1; \quad 4\alpha = -1, \quad x = -1,$$

the product of these values is 4α , that is, $4\alpha = -\left(P - \frac{1}{p}\right)^2$, viz. one value is greater, the other less, than 2. We see also that

$$2\sqrt{(\alpha + 1)}\sqrt{(\alpha + 1)} - 1$$

is positive and less than 1; the curve is thus an acnodal lineapas, having B' for the acnode and cutting the axis outside the circle at the points $x = \alpha + 2\sqrt{(\alpha + 1)}$. Conversely, considering the former curve as given by the expressions of the coordinates in terms of a parameter ϕ , the determination of the like expressions for the latter curve would appear to be a problem of very great difficulty.

336

ON THE ORTHOMORPHOSIS OF THE CIRCLE
INTO THE PARABOLA.

[845]

The Table above referred to.

[Table giving the correspondence between $\phi = \log \tan(45 + \frac{1}{2}\phi')$ for ϕ' from 0 to 90 at intervals of 1. The full numerical table from the original paper, with values to seven decimal places, is reproduced below.]

ϕ'	$\phi = \log \tan(45 + \frac{1}{2}\phi')$	ϕ'	ϕ	ϕ'	ϕ
0	0	46	.9028545	50	.9962755
1	.0174553	47	.9249187	51	1.0381004
2	.0349066	48	.9457040	52	1.0584952
3	.0523599	49	.9657500	53	1.0792677
4	.0698132	50	.9852280	54	1.1005044
5	.0872665	51	1.0036321	55	1.1222263
6	.1047198	52	1.0216162	56	1.1444466
7	.1221730	53	1.0392060	57	1.1672144
8	.1396263	54	1.0566175	58	1.1905610
9	.1570796	55	1.0738249	59	1.2145202
10	.1745329	56	1.0908669	60	1.2391411
11	.1919862	57	1.1077612	61	1.2644777
12	.2094395	58	1.1244912	62	1.2905931
13	.2268928	59	1.1410615	63	1.3175552
14	.2443461	60	1.1573912	64	1.3454387
15	.2617994	61	1.1733035	65	1.3743253
16	.2792527	62	1.1887844	66	1.4042947
17	.2967060	63	1.2036336	67	1.4354352
18	.3141593	64	1.2178358	68	1.4678434
19	.3316126	65	1.2312718	69	1.5016262
20	.3490659	66	1.2438139	70	1.5369024
21	.3665192	67	1.2555401	71	1.5738033
22	.3839724	68	1.2663275	72	1.6124704
23	.4014257	69	1.2761519	73	1.6530834
24	.4188790	70	1.2848920	74	1.6958685
25	.4363323	71	1.2924327	75	1.7410341
26	.4537856	72	1.2986277	76	1.7887898
27	.4712389	73	1.3033236	77	1.8394554
28	.4886921	74	1.3064565	78	1.8933630
29	.5061454	75	1.3078560	79	1.9508587
30	.5235987	76	1.3074457	80	2.0123399
31	.5410520	77	1.3050447	81	2.0782854
32	.5585053	78	1.3004563	82	2.1492283
33	.5759585	79	1.2934613	83	2.2258028
34	.5934118	80	1.2838174	84	2.3087094
35	.6108651	81	1.2712504	85	2.3989343
36	.6283184	82	1.2554543	86	2.4977049
37	.6457717	83	1.2360880	87	2.6066908
38	.6632250	84	1.2127740	88	2.7282169
39	.6806782	85	1.1850968	89	2.8657689
40	.6981315	86	1.1525987	90	∞
41	.7155848	87	1.1147993		
42	.7330381	88	1.0712310		
43	.7504914	89	1.0214444		
44	.7679447	90	∞		
45	.7853982				

846.

**A VERIFICATION IN REGARD TO THE LINEAR
TRANSFORMATION OF THE
THETA-FUNCTIONS.**

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol.
XXI. (1886), pp. 77–84.]

THE notation is that of Smith's* "Memoir on the Theta and Omega Functions," differing from Hermite's only in a factor, $\mathfrak{S}_{m,n}$ (Smith) $= 2^{n/4} \mathfrak{S}_{m,n}$ (Hermite); and it is in consequence of this that the factor $2^{n/4}$ occurs in the expression presently given for Λ . Hermite's Memoir "Sur quelques formules relatives à la transformation des fonctions elliptiques" appeared in *Liouville*, t. III. (1858), pp. 27–36.

Writing

$$\omega = \frac{a + d\Omega}{c + d\Omega},$$

where $ad - bc = 1$, the formula of transformation is

$$\mathfrak{S}_{m,n} \left[(a + b\Omega) \frac{m^2}{2}, \Omega \right] = C \exp \left(-i\pi(a + b\Omega) \frac{m^2}{2} \right) \mathfrak{S}_{m,n'} \left(\frac{m \cdot a}{2}, \omega \right);$$

where:

$$\begin{aligned} m' &= cu + b \cdot n, \\ n' &= cm + a \cdot n, \end{aligned}$$

or:

$$\begin{aligned} m' &= cm + b \cdot n, \\ n' &= cm + a \cdot n. \end{aligned}$$

We have, according as b is positive or negative,

$$C = \exp \left[-\frac{i\pi}{\sqrt{-(c + d\Omega)}} \right] = \exp \left[-\frac{i\pi}{\sqrt{(c + d\Omega)}} \right],$$

where for each case the square root is to be taken in such wise that the real part is positive; Λ , H are eighth roots of unity; viz. in each case

$$\Lambda = \exp \left[-\frac{1}{2}i\pi(\alpha m^2 + 2bmn + dn^2 + 2\alpha\beta n + 4\beta\alpha n + ab\gamma\delta) \right] e^{m\sqrt{(-1)/2}},$$

*[H. J. S. Smith, *Collected Mathematical Papers*, vol. II., pp. 415–634.]

338 A VERIFICATION IN REGARD TO THE LINEAR [846

and, according as b is positive or negative,

$$H = \left(\frac{b}{a}\right) e^{i\pi a}, \quad a \text{ odd}, \quad H = \left(\frac{-b}{a}\right) e^{i\pi(a-1+3b+\beta)}, \quad a \text{ odd},$$

$$H = \left(\frac{a}{b}\right) e^{i\pi(-b+2-a-1)/2}, \quad b \text{ odd}, \quad H = \left(\frac{-a}{-b}\right) e^{i\pi(a-1+3b+\beta-b+2)/2}, \quad b \text{ odd};$$

where $\left(\frac{a}{b}\right)$, &c., are the Legendrian symbols as generalised by Jacobi; if a and b are each of them odd, the formula for the case of a even or odd and b odd respectively are equivalent to each other, and either may be used indifferently.

To fix the ideas, I assume throughout that b is positive and odd; the proper formulae then are

$$C = \frac{H}{\sqrt{(c+d\Omega)}};$$

Λ as above, and

$$H = \left(\frac{a}{b}\right) e^{i\pi(-b+2-a-1)/2}.$$

I will also, for greater convenience, assume that c is odd; in consequence even, viz. the numbers a and c are odd and those even, or else a and b are odd and the other even.

The equation $\omega = \frac{a+d\Omega}{c+d\Omega}$ gives $\Omega = \frac{-a+c\omega}{-d+b\omega}$, and thence

$$a + b\Omega = -\frac{1}{-d + b\omega}.$$

Comparing the expressions for ω , Ω , we observe that we may interchange ω and Ω , writing also $-d$, b , c , $-a$ for a , b , c , d ; and changing the other letters, we may for u , b , β , m , n , m' , n' , write

$$\frac{u}{c+d\Omega}, \quad b, \quad a, \quad m', \quad n', \quad m, \quad n,$$

where

$$m' = -dm - bn, \quad n' = -cm - an;$$

the formula then becomes

$$\mathfrak{S}_{m',n'} \left[(-d+b\omega) \frac{m}{2}, \omega \right] = C' \exp \left[-i\pi(-d+b\omega) \frac{u^2}{2} \right] \mathfrak{S}_{m,n} \left(\frac{u}{c+d\Omega}, \Omega \right),$$

or, as it is the same thing,

$$\mathfrak{S}_{m',n'} \left[(a + b\Omega) \frac{m}{2}, \omega \right] = C' \exp \left[-i\pi(a + b\Omega) \frac{u^2}{2} \right] \mathfrak{S}_{m,n} \left(\frac{u}{c + d\Omega}, \Omega \right);$$

846] TRANSFORMATION OF THE THETA-FUNCTIONS. 339

or, observing that the left-hand side is $= (-1)^{mn'} \mathfrak{S}_{m,n'} \left[\frac{u}{2}, \omega \right]$, we have

$$\mathfrak{S}_{m,n'} \left[(a + b\Omega) \frac{m}{2}, \omega \right] = \frac{C'}{(-1)^{mn'}} \exp \left[-i\pi(a + b\Omega) \frac{u^2}{2} \right] \mathfrak{S}_{m,n} \left(\frac{u}{c + d\Omega}, \Omega \right),$$

an equation which is of the same form as the original equation of transformation, and is to be identified with it.

We have

$$\begin{aligned} m' &= -dm - bn = -dm + b\Omega \cdot n, &= -m + bd(-n + cm), \\ &+ b \cdot bd(-n + cm) = -bd, \\ n' &= -cm - an = -(cm + an) = m \cdot bd(-n + cm) = -bn(-n + cm) - 1, \end{aligned}$$

values which may be written

$$m' = -dm - bn + Pa = -bd, \quad m \cdot b \cdot bd(-n + cm) - 1,$$

or, since $\Omega - d - b\omega = -bd$,

$$m' = -d + bd \cdot (-n + cm) = b\omega \cdot m = -d \cdot m = cm + b \cdot n - 1,$$

$$n' = -c \cdot m \cdot a \cdot n - 1, \quad n' = c \cdot m + a \cdot n - 1;$$

P and Q being integers. In fact, P is an integer if b is odd or if bn is even, and similarly for Q .

The left-hand side is $\mathfrak{S}_{m',n'} \left[(a + b\Omega) \frac{u}{2}, \Omega \right]$, which is in virtue of P even, and which depends only on the values of m and n .

We have

$$CC' = (-1)^{mn' - mn + m'n' - m'n},$$

which is

$$= (-1)^{mn - m'n'}$$

this condition will be considered in the sequel.

The formula in question is in fact identifiable to that into which it transforms when, observing that the left-hand side is unchanged by the change of u, b, c, d, m, n, m', n' , into $-d, b, c, -a, m', n', m, n$;

$$CC' = (-1)^{mn - m'n}, \quad CC' = (-1)^{mn' - m'n}.$$

We have

$$CC' = \frac{HH'}{\sqrt{(c+d\Omega)} \cdot \sqrt{-(c+d\Omega)}},$$

where the square roots are taken in such wise that the real part is positive; hence

$$CC' = \frac{HH'}{i\sqrt{(c^2+1)}} \quad \text{or} \quad \frac{HH'}{i\sqrt{-(c^2+1)d}},$$

according as c is even or odd; for, taking the radius vector for positive, we have that the square roots of $c+d\omega$ and $\sqrt{-(c+d\omega)}$ are of the form

$$\sqrt{(c+d\omega)}, \quad \frac{1}{i}\sqrt{(c+d\omega)} \quad \text{or} \quad \frac{1}{i}\sqrt{-(c+d\omega)}, \quad \sqrt{-(c+d\omega)} = \sqrt{(c+d\omega)},$$

where the square roots taken in such wise that the real part is in each case positive;

340 A VERIFICATION IN REGARD TO THE LINEAR [846]

where $\sqrt{(c+1)}$ means

$$\sqrt{-(c+1)d}\sqrt{(-)} \cdot \sqrt{-(c+d)\Omega};$$

the last-mentioned two square roots being so taken as just explained; and we have therefore

$$\begin{aligned}\Lambda &= \exp \left[-\frac{1}{2}i\pi(\alpha m^2 + 2bmn + dn^2 + 2\alpha\beta n + 4\beta\alpha n + ab\gamma\delta) \right] e^{m\sqrt{(-1)/2}}, \\ \Lambda' &= \exp \left[-\frac{1}{2}i\pi(-dn^2 + 2bm'n' - am'^2 - 2\alpha\beta m' - 4\beta\alpha m' - bd\beta) \right] e^{m'\sqrt{(-1)/2}},\end{aligned}$$

viz. the values of Λ obtained from the value of Λ by the change of a, b, c, d, m, n, m', n' , into $-d, b, c, -a, m', n', m, n$; and writing

$$A = \exp \left[-\frac{1}{2}i\pi\{-am^2 + 2bmn + dn^2 + dn'^2 - 2bn'm' + am'^2 + 2\alpha\beta(n - m') + 4\beta\alpha\} \right]$$

the last-mentioned two square roots being so taken as just explained, and we have

Representing the second and third of the last-mentioned equations by

$$4 = \exp \left(-i\pi\rho^{-mn' - m'n} \right),$$

the last-mentioned two squares roots being also explained; and we have to express the term divisible by 4,

$$\rho = -mn'(2 + ab\gamma\delta + bd\beta).$$

To calculate $\Delta + \Delta'$, we have

$$\begin{aligned}dm - bn &= m + bd \cdot (n - c \cdot m), & m &= -d \cdot m - b \cdot n - 1, \\ -cm + an &= -n - ac \cdot (n - c \cdot m), & n &= c \cdot m + a \cdot n - 1,\end{aligned}$$

and thence

$$\begin{aligned}-2dm^2 + 2bn^2 &= -mn' + ac \cdot n' - bd \cdot m'^2 + abcd^2 - 2bd(-n'2 + cdn) - 2acd(2 - ca) \\ (-am + bn)^2 &= -mn'(-2bd - bnn' + 2abcd) - 2bd(-n' + cn) + 2acd(-a' + cn) - \\ &\quad - 2acd(-a' + cn) - bd \cdot 2a^2 + 2abdn - 2acd(-n' + ca) + 2abcd^2 = \dots\end{aligned}$$

consequently

$$-adn^2 + 2bnn' + 2bcmn' - 2bdmn + 2acmn + bdn^2 + 2bdcmn = 2acd(n + 2cdn) - 2bdcmn$$

also

$$\begin{aligned} -2bdmn &= -2cdbn^2 - 2bd^2n^2 + 2acd \cdot n + 2bcd \cdot n^2 + 2bd \cdot c \cdot 2bd \cdot 2bd \cdot bd, \\ -2bn' &= -2adcb - 2cdb \cdot 2bd \cdot 2bn^2 + 2cdb \cdot 2cd \cdot 2bd \cdot bd \cdot 2bcb, \\ -dn'^2 + dn'^2 &= -2bcd^2 \cdot dcb - 2dcb \cdot dn + 2cdn^2 + 2bcd \cdot dcb \cdot bcb + 2cdb \end{aligned}$$

whence, adding, we obtain

$$\Delta + \Delta' = -dn^2 - 2bnn' + 2bcmn' - dn^2 + 2cdmn' - 2bcdmn' - bcdm + 2bcd \cdot n^2 + 2cd \cdot dcb + 2$$

and, adding to this the expression of Δ , we find

$$\Delta + \Delta' = -2bnn' + 2bnn' - 2bcmn' + 2cd\beta - 2dnn' + 2dcn' - 2bcd - bdc d + 2bd \cdot bd,$$

846] TRANSFORMATION OF THE THETA-FUNCTIONS. 341

The coefficient of u is $= 2bd(-n + cm) - 4cd + ad(-c \cdot m^2 + a \cdot n) + 2(b-1)(c \cdot n)$. Hence writing

$$m \cdot n' = 2bd(-n + cm) - ad(-m \cdot n^2 + a \cdot n) + 2bnn(-1)(c \cdot n - n),$$

observe that $a\Omega n - 1$, observing in every case $\Omega \cdot \text{mod}$, writing also $4bn \cdot 2cdn$, and consequently

$$4bn \cdot n' = ad(\beta m - n)(-\beta + b) + (-\beta + b)(b-1)(c+d)(b-1)(n-2m).$$

where the even values of n may be omitted. We have moreover

$$mn - m'n' = -bd(-n + cm)\beta - ab(-m \cdot n^2 + a \cdot n)\beta + (-\beta + b)(b-1)bd,$$

and these, since β is even, may be neglected, and the result is simply

$$mn - m'n' = -bd\beta - ad\delta\beta,$$

and thus

$$mn - m'n' = ad\delta\beta + bd(-2 + cd)\beta - 2bnd\delta(-a + cd)\beta,$$

we have also

$$m - n' = -d \cdot m - bd - (c \cdot m - a \cdot n + 1) = -(c+d)(b)\beta + 1,$$

$$n - m' = -m + d \cdot m + ba = ad(b-n).$$

If even of which disappear, leaving the result, $u\sqrt{(-1)/2}$ becomes equal to $4ms\Omega$.

The next thing is to express the term divisible by 4: there is at once a term given by $4(b-1)(c \cdot 2bn)$, and so we may write therefore

$$4(b-1)(c \cdot 2bn) + 4(-bd)(2 + cd) + (-bd)(b-1)(c \cdot 2cdb) = 0,$$

via. the values of Δ is given by Abel's theorem, then in particular each of the four equations becomes

$$4(b-1)(c \cdot d \cdot n) = -2bdcn,$$

where, this is positive value of β is positive; and thus

$$mn - m'n' = -dn\beta + bd(c-1)(c\beta-1),$$

the equation which was to be verified.

847.

ON THE THEORY OF SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXI. (1886), pp. 1-17, [13, 14, 16].]

IN my "Mémoire sur les Hyperdéterminants," *Crelle*, t. XXX. (1846), pp. 1-37, [13, 14, 16],* I gave an investigation for the number of and relations between the quartic invariants of a given binary quantic, or what is the same thing to the cubic seminvariants of a given weight, and I propose at present (considering the formulae in this point of view) to further develop the theory in regard to the solution of the systems of linear equations obtained in the manner in question. But I first commence with a remark: I recall that the notation is the ordinary hyperdeterminant notation: for instance,

$$U = (a, b, c, d)(x, y)^3, \quad V = (a, b, c, d)(x, y)^3,$$

$$12VV \cdot 1 + 2, \quad VVV = D_{a,a'}VVV,$$

$$12VV \cdot 12 = D_{a,a'}D_{b,b'}VVV,$$

$$\&c. \quad = D_{a,a'}D_{b,b'} \cdot D_{c,c'}VVV,$$

and that, when the variables (x, y) , (a, b) , (a', b') are ultimately replaced by (x, y) , then these forms denote, each of them, an invariant.

The investigation* is as follows:

$$\begin{aligned} UU(12) &= (a, b, c, d)(b', a)(b', d') \cdot -2(ad' + a'd - bc' - b'c) + 9(bc' + b'c), \\ &= 12P(a' \cdot b'^2 - 5abb' \cdot b'^2 \cdot 3a^2bb' + b'^2 \cdot a'd' - 5b'd \\ &\quad + a \cdot a' \cdot -25, \\ &\quad + 2(cc' - d^2), \end{aligned}$$

and that, when the variables (x, y) , (a, b) , (a', b') are ultimately replaced by (x, y) , then these forms denote, each of them, an invariant.

The investigation is as follows:

$$\begin{aligned}
 UVUV &= UVVV - U_x V \cdot VWV \cdot UWV, \\
 UVVW &= UWUV \cdot VWWV, \\
 &= D_{a,b'} VVWV = D_{b,a'} VWVV, \\
 &\&c.
 \end{aligned}$$

*[This Collection, vol. I., p. 100; see also, vol. I., p. 157.]

847] ON THE THEORY OF SEMINVARIANTS. 345

Writing for shortness

$$12, 34 = \mathfrak{A}, \quad 13, 42 = \mathfrak{B}, \quad 14, 23 = \mathfrak{C},$$

we have

$$D_{a,d}\mathfrak{A} + D_{b,c'}\mathfrak{B} + D_{c,b'}\mathfrak{C} = 0.$$

Suppose $U = V = W = X$; we have the derivatives which correspond to the same value of δ , w , we have from these

$$D_{a,d}\mathfrak{A}, \quad D_{b,c'}\mathfrak{B}, \quad D_{c,b'}\mathfrak{C} = 0, \quad D_{d,a'}\mathfrak{D}_{x'}\mathfrak{A} = 0,$$

in the language of Prof. Sylvester, $\alpha_{i,k}$, $b_{i,k}$, &c., are here scalars. The invariant is

$$(\alpha\beta_{i,b}b_c - b_{i,b}b_{c,d})\mathfrak{A} = \mathfrak{B}_{i,k}b_{c,d}\mathfrak{C} = b_{i,b}b_{c,b'},$$

$= an^2 - 3b^2 + 2c^2$, &c., are here equal; and the invariant is

$$= 2(\alpha b - 4),$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that, in the language of Sylvester's $\alpha_{i,b}$, $b_{i,b}$, $b_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(ax)_2(ax)_3(ax)_4 = D_{a,d}D_{a,c}D_{a,b}V \cdot V = 4ax,$$

that is,

$$\alpha = 4f - 3g + 2h,$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that the Clebschian notation, for instance

$$(\alpha, \beta, c, d)(x, y)^3 + 4(x+b)(x+d)y + 6((c+b)y + 4(c+d)y + 4y) = 4ax,$$

viz. in the language of Prof. Sylvester, $\alpha_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$(\alpha\beta_{i,b}b_c - b_{i,b}b_{c,d})\mathfrak{A} = \mathfrak{B}_{i,k}b_{c,d}\mathfrak{C} = b_{i,b}b_{c,b'},$$

$= an^2 - 3b^2 + 2c^2$, &c., are here equal; and the invariant is

$$= 2(\alpha b - 4),$$

and so in other cases. To apply this directly to the theory of seminvariants, observe that, in the Clebschian notation, for instance:

$$(\alpha\beta + \alpha\delta - \beta\gamma)\mathfrak{A} = 4(\alpha\delta - \beta\gamma),$$

that is,

$$a = bf - 3g + 2h,$$

$$a = bf - 3g + 2h, \alpha\beta = 4f - 3g + 2h,$$

$$a = bf - 3g + 2h,$$

a seminvariant; the

$$(a - \beta)^2(\alpha - \gamma) = -2\alpha\beta + a\gamma + 4a(c - \beta) - 8c\delta + a\gamma\beta - 3\beta + 4ac\delta,$$

$\beta = 4f - 3g + 2h$, &c., are scalars: writing each of f, g, h for the weight). We have, as before,

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we again see that the theory is one of the number of relations between the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of a , the cubic seminvariants $\mathfrak{C}\mathfrak{B}\mathfrak{C}$, $\mathfrak{C}\mathfrak{A}\mathfrak{C}$, are the differences $D_{a,a}\mathfrak{A}$, &c., but in many cases the seminvariants $D_{a,a}\mathfrak{A}$, &c., are ultimately replaced by (x, y) , then these forms denote, each of them, an invariant.

Or, we may use the Clebschian notation, for instance, observe that

$$(\alpha, \beta, c, d)(x, y)^3, \quad \alpha = -cu + bv = av + bw, \quad \&c.,$$

that is,

$$a = c - b = -a,$$

via. in the language of Prof. Sylvester, $\alpha_{i,b}$, $b_{i,b}$, &c., are here scalars. The invariant is

$$\begin{aligned} (\alpha - \beta)^2 &= a + (\alpha + \beta)^2, \\ &= 4(\alpha - 2b + b^2), \end{aligned}$$

where $\alpha, \beta, \dots$ are scalars.

The invariant is

$$(\alpha - \beta)^2(\alpha - \gamma) = \alpha^2 + \alpha\beta\alpha + 2\beta\gamma\alpha\beta + \beta^2,$$

$$\alpha = -\beta, \beta = -\alpha, \alpha = -\beta, \beta = -\alpha,$$
$$\alpha = -\beta\beta\beta\beta\beta - \beta = 0.$$

Hence

$$\mathfrak{A} \cdot \mathfrak{B} = 0.$$

We may also for independent derivatives, when f is even, the terms of the series $D_{a,a+1}, D_{a+1,b+1}, \dots$, and when f is odd, those of the series $D_{a+1,a+2}, D_{a+2,b+2}, \dots$, continuing each series until the last term in which $D_{a+i,b+i}$ would be equal to zero in virtue of the value of δ, w (notice that the values of δ, w must be in such relation, so that there is no need of a sequence of the seminvariants $D_{i,m}$ are independent the series of the order term $D_{a,a+1}$); the like considerations apply to the other derivatives $D_{a,a+1}$, &c., but inasmuch as we must keep at least a of order a different, the same set must necessarily be in use, but the same set of independent seminvariants will of course be obtained. The number of these is also f , only the same set will be the same as we are considering; the number of independent seminvariants is in fact equal to the number of δ, w , &c., which is independent of δ, w , &c., in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$. We may say that all these seminvariants of the highest powers of δ, w are mutually independent, and as such are connected by linear relations between them, the coefficients of these depending on the seminvariants of the lowest powers of δ, w ; and as such these linear relations are linear functions of the seminvariants, the connecting relations being also linear functions; the seminvariants of the highest powers are also linear functions of those of the lowest, the seminvariants of the highest powers being also represented by the corresponding numbers in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$. We may say that all this means that of any independent seminvariants of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$, we have any independent linear relations between the corresponding numbers in the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$.

Or, we may say that the number of independent linear relations between the seminvariants of the highest powers is, in proportion to the number of independent seminvariants of the lowest powers, &c., of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$, equal to the number of independent linear relations between the seminvariants of the lowest powers, &c., of the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$.

We will presently give the solution of the under-mentioned system; but beginning with the second and weight w as we have here, the number of solutions will be the following table:

	$D_{a,d}$	$D_{a,d+1}$	$D_{a,d+2}$	$D_{a,d+3}$	
$\mathfrak{D}_{ad}$	D_a	0,	D_a	$\frac{1}{2}D_a$	0,
$\mathfrak{D}_{ad+1}$	$-D_a$	$\frac{1}{2}D_a$	0,	$\frac{1}{2}D_a$	0,
D_{ad}	0,	0,	D_a	$-D_a$	$\frac{1}{2}D_a$
D_a	0,	0,	0,	D_b	$-D_b$
$\mathfrak{D}_{ad+1}$	0,	0,	0,	0,	
D_{ad+1}	0,		0,	0,	$-D_a$

We may use generally with the solution of the equations of column corresponding numbers

$$p, \quad q, \quad p+1, \quad p+1, \quad p+1, \quad \dots,$$

there is for example a single linear relation between the orders of the relations 5, 7, 8, 9, 10 in each of the orders 5, 8, 9, 10 on the form $\mathfrak{V}_{a,b+m}, \mathfrak{V}_{a,b+m}, \dots$, on the relations between the orders of $D_a = \frac{1}{16}D_b$. The above considerations were given as follows:

847] ON THE THEORY OF SEMINVARIANTS. 347

For any value of f except $f = 2, 3$, or 4 , the table commences, for f even and f odd respectively, in the following manner:

	$D_{a,b+1}$	$D_{a,b+2}$	$D_{a+1,b+2}$	$D_{a+1,b+3}$	$D_{a+2,b+3}$	
$\mathfrak{D}_{a,b+1}$	$\frac{1}{2}f$	0				
$\mathfrak{D}_{a+1,b+1}$	$-\frac{1}{2}f$	$\frac{1}{2}f$	0			
$\mathfrak{D}_{a+1,b+2}$	0	$-\frac{1}{2}(f-1)$	$\frac{1}{2}(f+1)$	0		
$\mathfrak{D}_{a+2,b+2}$	0	0	$-\frac{1}{2}(f-2)$	$\frac{1}{2}(f+2)$	0	
$\mathfrak{D}_{a+2,b+3}$	0	0	0	$-\frac{1}{2}(f-3)$	$\frac{1}{2}(f+3)$	0
$\mathfrak{D}_{a+3,b+3}$	0	0	0	0	$-\frac{1}{2}(f-4)$	$\frac{1}{2}(f+4)$

but beyond this I do not know the law of the series.

Before going further, I remark that if $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, instead of the foregoing values, denote respectively,

$$\begin{aligned}\mathfrak{A} &= (\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma'), &= (\alpha\beta'_2 + \alpha\beta')(\gamma\delta'_2 - \delta\gamma'), \\ \mathfrak{B} &= (\alpha\delta + \beta\gamma)(\alpha'\delta' + \beta'\gamma'), &= (\alpha\beta'_2 - \alpha\beta')(\gamma\delta'_2 + \delta\gamma'), \\ \mathfrak{C} &= (\alpha\beta + \alpha'\beta')\Lambda, &= (\alpha\delta + \beta\delta')(\alpha\gamma + \beta\gamma' - \delta\delta'),\end{aligned}$$

then equations

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and the same theory applies to the cubic derivatives $\mathfrak{D}_{a,b+m}\mathfrak{V}^{2/m}\mathfrak{C}\mathfrak{V}\mathfrak{V}$, that is, in the like relations between the coefficients of the highest powers of the cubic derivatives.

Or, we may apply this directly to the theory of seminvariants. We take then $(x, y)^4$ to denote the binary quartic, and we have for instance

$$(\alpha, b, c, d, e)(x, y)^4 = au^4 + 4bu^3v + 6cu^2v^2 + 4duv^3 + ev^4 = (\alpha, b, c, d, e),$$

that is,

$$\alpha = a, \quad b = -b, \quad c = c, \quad d = -d, \quad e = e,$$

a seminvariant; the like relations between the coefficients of the highest powers of the cubic derivatives, $\mathfrak{D}_{a,b+m}\mathfrak{V}^{2/m}\mathfrak{C}\mathfrak{V}\mathfrak{V}$. We have

$$\mathfrak{B} = (\alpha - \beta)(\alpha\gamma - \beta\delta) = -2\beta\alpha + a\gamma + 4a(c - \beta) - 8c\delta + 4ce - 3\beta + 4\delta\gamma,$$

a seminvariant.

The like is the case as in other cases. Writing here

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we see that the question relates also to the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of δ , w , $\mathfrak{C}\mathfrak{B}\mathfrak{C}$, $\mathfrak{C}\mathfrak{B}\mathfrak{C}$ are the differences $D_{a,a}\mathfrak{A}$, &c., but the same set of independent seminvariants are also represented by $(\alpha - \beta)^2$, &c..

Suppose that a is odd, and, in particular, $D_{a,d+m}\mathfrak{V}_{a,d}\mathfrak{W}\mathfrak{W}$, &c. that is,

$$D_{a,d+m} = -D_{a,d}\mathfrak{W}\mathfrak{W}, \quad D_{a,d+m} = -D_{a,d}\mathfrak{W}\mathfrak{W},$$

that is,

$$\mathfrak{V}_{a,b+m}\mathfrak{V}_{a,b+m} = 0, \quad \mathfrak{V}_{a,b+m}\mathfrak{V}_{a,b+m} = 0;$$

Hence

$$\mathfrak{A} = 0;$$

348 ON THE THEORY OF SEMINVARIANTS. [847

or, in like manner if a cubic seminvariant, that is,

$$(b, b, b)(\delta, x, y)^3 = -bv^3 + 4bv^2y + 4(\alpha + by)y^2 + (a + by)y^3,$$

that is,

$$\alpha = -a, \quad \beta = -b, \quad \delta = -d, \quad \gamma = -c, \quad \delta = -e,$$

a seminvariant. Writing here

$$\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0,$$

and we see that the question relates to the number of relations between the cubic seminvariants of a given weight. Observe also that, for each of the highest powers of δ , w , the cubic seminvariants $\mathfrak{C}\mathfrak{B}\mathfrak{C}$, $\mathfrak{C}\mathfrak{B}\mathfrak{C}$ are the differences $D_{a,a}\mathfrak{A}$, &c..

The like remarks would apply for f odd, in the case a a quartic seminvariant, the like remarks apply also, if instead of the seminvariants f odd, we have the seminvariants f even; we have here a similar set of seminvariants $D_{a,d}\mathfrak{B}$, $D_{a,d+m}\mathfrak{B}$, &c., and the seminvariants $D_{a,d}\mathfrak{B}$, &c. are also as above; in many cases the seminvariants are not all of them simply equal, but coefficients in the relations A , B , C , &c., respectively, I call them

$$A, \quad B, \quad C, \quad D, \quad E, \quad F, \quad G, \quad H, \quad I, \quad J, \quad K, \quad L, \quad M,$$

Suppose first α is odd, the in the first

$$D_{a,d}\mathfrak{B}\mathfrak{B} = 1, \quad D_{a,d+m}\mathfrak{B}\mathfrak{B} = 0,$$

and similarly $C + 2D + E = 0$, &c.; that is, we have

$$\begin{aligned} A &= 0, \\ B + C &= 0, \\ C + 2D + E &= 0, \\ D + 3E + 3F + G &= 0, \\ &\text{\&c.} \end{aligned}$$

The resultant way of solving them is to express the other functions in terms of B , D , F , H , &c., viz. we thus have

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>	<i>M</i>
<i>X</i>	0	1	−1	0	1	0	−10	0	+135	0	−2073		
<i>D</i>			1	−2	0	3	−15	0	355	0	−5410		
<i>F</i>				1	−3	0	6	−126	0	1693			
<i>H</i>					1	−4	0	10	0	−150	0	−396	
<i>J</i>						1	−5	0	15				
<i>L</i>								1	−6				

read according to the columns

$A = 0,$

$B = B,$

$C = -B,$

$D = D,$

$E = B - 2D,$

$F = F,$

$G = -3B + 3D - 3F,$

$\&c.$

and, by way of verification of the numbers, observe that the sum of the numbers in a column is 1 and −1 alternately.

The formulae are true for any odd value of α whatever; but they require an explanation. On the right hand value of α , $\&c.$, are not all of them advisory, but they are also of the same powers of δ , w . The proper sequence ought to be obtained, but it is not derivable from these by Captain MacMahon’s theorem, viz. the number of independent seminvariants is in fact equal to the number of δ , w , of the non-unitary symmetric functions $\mathfrak{W}$ of the proper weight: for δ even, the forms are 2222 and 3322; so that there should be only two arbitrary terms.

847] ON THE THEORY OF SEMINVARIANTS. 349

moreover

$$B + C = w^{-1}(\mathfrak{B} + \mathfrak{C}) = 0,$$

and similarly $C + 2D + E = 0$, &c.; that is, we have

$$A = 0,$$

$$B + C = 0,$$

$$C + 2D + E = 0,$$

$$D + 3E + 3F + G = 0,$$

$$\text{\&c.}$$

The resultant way of solving these is to express the other functions in terms of B, D, F, H, J, L alternately.

The formula are true for any odd value of f whatever; but they require an explanation. The arbitrary terms B, D , &c., are not all of them arbitrary; but they are also seminvariants of the same weight, and therefore connected by linear relations. The proper sequence ought to be obtained, but it is not derivable from these by Captain MacMahon's theorem, viz. the number of independent seminvariants is in fact equal to the number, δ, w , of the non-unitary symmetric functions $\mathfrak{V}\mathfrak{V}$ of the proper weight: for δ even, the forms are 2222 and 3322; so that there should be only two arbitrary terms.

350 ON THE THEORY OF SEMINVARIANTS. [847

We may have, for any odd value of f , writing for shortness 900, 810, &c., instead of D_{900} , D_{810} , &c.,

$$\begin{aligned}
 900 &= 0, \\
 810 &= B, \\
 720 &= -B, \\
 630 &= D, \\
 540 &= B - 2D, \\
 450 &= F, \\
 360 &= -3B + 3D - 3F, \\
 270 &= H, \\
 180 &= 12B - 2D + 14F - 4H, \\
 090 &= J,
 \end{aligned}$$

But we have $900 = 0$, $810 + 720 = 0$, $720 + 270 = 0$, $630 + 350 + 450 = 0$, $540 + 350 = 0$; that is,

$$\begin{aligned}
 900 &= 0, \\
 810 &= B, \\
 720 &= -B, \\
 630 &= D, \\
 540 &= B - 2D, \\
 450 &= F, \\
 L &= 0,
 \end{aligned}$$

all satisfied by $F = -B + 2D$, $H = B - D$, $L = 0$, the proper answer to stop at the equation for 540, the system is then

$$\begin{aligned}
 900 &= 0, \\
 810 &= B, \\
 720 &= -B, \\
 630 &= D, \\
 540 &= B - 2D,
 \end{aligned}$$

equations which may be considered as expressing the several functions 900, 810, 720, 630, 540, 450, &c., in terms of $B = 810$ and $D = 630$.

They agree, as they should do, with a foregoing result,

$$\begin{aligned} 900 &= 0, \\ 810 &= B, \\ 720 &= -B, \\ 630 &= D, \\ 540 &= B - 2D, \end{aligned}$$

and it would of course be possible to express the remaining forms $D_{a,b}$, $D_{a,c}$, &c., 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of

Fig. 4.

*[Figure: two orthogonal families of lemniscates
with node B' and focus O , with a generic point
indicated]*

0.8 Paper 845

845.

ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 107–126.]

The function $\log \tan(45^\circ + \frac{1}{2} \arg.)$ is tabulated, Legendre, *Théorie des Fonctions Elliptiques*, t. II. table iv. pp. 256–259, viz. writing as above ϕ for the argument, we have hereby the value of $\Phi = \log \tan(\frac{1}{4}\pi + \frac{1}{2}\phi)$, for every value of ϕ from 0° to 90° at intervals of $30'$ and to 12 decimals, and with fifth differences. Observe that ϕ is thus given in degrees and minutes as a circular arc, Φ as a number; it is convenient to have ϕ and Φ each as arcs, or each as numbers, the conversion being of course at once made by means of a table of the lengths of circular arcs, and I have calculated the Table which I give at the end of the present paper.

I had previously calculated for the correspondence of AO , OB and ACB with AO , OB and AGB , the following values, at irregular intervals suited to the construction of a figure.

Circle

Parabola

ϕ	P	Q	x	y
0°	0	0	0	0
5°	0.0875	0.0437	0.0077	0.0076
10°	0.1763	0.0875	0.0306	0.0309
15°	0.2680	0.1317	0.0691	0.0706
20°	0.3640	0.1763	0.1232	0.1284
25°	0.4663	0.2217	0.1935	0.2068
30°	0.5774	0.2679	0.2804	0.3094
35°	0.7002	0.3153	0.3850	0.4415
40°	0.8391	0.3640	0.5085	0.6109
45°	1.0000	0.4142	0.6520	0.8284
50°	1.1918	0.4663	0.8176	1.1115
55°	1.4281	0.5208	1.0086	1.4877
60°	1.7321	0.5774	1.2280	2.0000
65°	2.1445	0.6371	1.4806	2.7321
70°	2.7475	0.7002	1.7732	3.8476
75°	3.7321	0.7673	2.1139	5.7274
80°	5.6713	0.8391	2.5130	9.5196
85°	11.4301	0.9163	2.9835	20.9515
90°	∞	1.0000	3.5452	∞

The formulae for the orthomorphosis are

$$\sqrt{x + iy} = p + iq,$$

where

$$P = \tan \frac{1}{2}\pi p, \quad Q = \tanh \frac{1}{2}\pi q,$$

and

$$x = p^2 - q^2, \quad y = 2pq.$$

The relation between the circle and parabola is established by means of the equation

$$\phi = \log \tan\left(\frac{1}{4}\pi + \frac{1}{2}\Phi\right),$$

as before, we have $Q = \tan \frac{1}{2}\phi$, and the formula is giving the values of x, y .

It is clear that we have

$$P = \tan \frac{1}{2}\pi p, \quad Q = \tanh \frac{1}{2}\pi q,$$

and thence

$$p = \frac{2}{\pi} \arctan P, \quad q = \frac{2}{\pi} Q.$$

Hence, to the circle $x^2 + y^2 = c^2$, corresponds in the parabola the curve where $p + iq = \sqrt{x + iy}$, $P = \tan \frac{1}{2}\pi p$, $Q = \tanh \frac{1}{2}\pi q$. This is a complicated transcendental equation, and I do not see any convenient way of tracing the curve.

The set of curves satisfy the differential equation

$$\frac{P dP + Q dQ}{PQ(Q dP + P dQ)} = \text{const.},$$

that is,

$$\frac{dP}{P(1 + P^2)} + \frac{dQ}{Q(1 - Q^2)} = 0,$$

where dP , dQ are given in terms of dp , dq by the equations

$$\frac{dP}{dp} = \frac{1}{2}\pi(1 + P^2), \quad \frac{dQ}{dq} = \frac{1}{2}\pi(1 - Q^2).$$

We have $y = 2pq$, and thence at the highest points, or summits of the several curves, $q dp + p dq = 0$. Combining these equations, we have $dP : dQ = -P(1 + P^2)q : Q(1 - Q^2)p$, and thence as an equation to the locus of the summits in question.

If p and q are small, then putting for a moment $\frac{1}{2}\pi = m$ for shortness, we have $P = mp + \frac{1}{3}m^3p^3$, $Q = mq - \frac{1}{3}m^3q^3$, and the equation becomes

$$p^2(1 + m^2p^2)(1 + m^2q^2) - q^2(1 - m^2q^2)(1 - m^2p^2) = 0,$$

that is,

$$p^2 - q^2 + m^2(p^4 + 2p^2q^2 - q^4) = 0.$$

Writing this in the form we find

$$x + \frac{1}{4}\pi^2(x^2 + 2y^2) = 0,$$

or say

$$y^2 = -\frac{2}{\pi^2}x - \frac{1}{2}x^2,$$

as the locus of the summits in the neighbourhood of the focus O , viz. the summits lie all of them, as might have been expected, on the left-hand (or negative side) of the focus.

I have constructed for the parabola, to the scale $1 = 1\frac{1}{2}$ inch, as accurately as the data enable, the figure corresponding to the concentric circles and the radii of the circle.

Resuming the equation $\sqrt{x+iy} = p+iq$, that is, $x = p^2 - q^2$ and $y = 2pq$, we have ($p = \text{const.}$), the curves

$$y^2 = 4p^2(p^2 - x),$$

and ($q = \text{const.}$),

$$y^2 = 4q^2(q^2 + x),$$

which are two systems of confocal parabolas, cutting each other at right angles; the curves of the former set all of them interior to the given parabola, those of the latter set of course cutting it at right angles.

Corresponding hereto in the circle, writing for a moment $\sqrt{X+iY} = p+iq$, we have whence

$$p = P(1 - qQ), \quad q = Q(1 + pP);$$

whence eliminating P and Q successively, we find say, for shortness, these are

$$p^2 + q^2 - 2mp + 1 = 0, \quad \text{and} \quad p^2 + q^2 - 2nq - 1 = 0.$$

Introducing the polar coordinates r, θ , we have $x + iy = r(\cos \theta + i \sin \theta)$, and thence

$$p, q = \sqrt{r} \cos \frac{1}{2}\theta, \quad \sqrt{r} \sin \frac{1}{2}\theta;$$

the equations thus become

$$r - 1 - 2m\sqrt{r} \cos \frac{1}{2}\theta = 0, \quad r + 1 - 2n\sqrt{r} \sin \frac{1}{2}\theta = 0,$$

which belong to two limaçons having each of them B for a node and for a focus, and which of course cut each other at right angles; see fig. 4, which is a mere diagram.

In fact, omitting for convenience the accents, but recollecting always that the curves belong to the figure of the circle, the first equation gives

$$(r - 1)^2 = 4m^2 r \cos^2 \frac{1}{2}\theta = 2m^2 r(1 + \cos \theta) = 2m^2(r + x),$$

that is,

$$(r^2 + 1 - 2m^2x)^2 = 4(m^2 + 1)^2r^2.$$

Transforming to the point B as origin, we must write $x + 1$ for x , and then the equation thus becomes

$$[x^2 + y^2 - 2(m^2 + 1)x + 2(m^2 + 1)]^2 = 4(m^2 + 1)^2(x^2 + y^2 + 2x + 1),$$

that is, or say

$$(x^2 + y^2)^2 - 4(m^2 + 1)(x^2 + y^2)x + 4(m^2 + 1)(x^2 - m^2y^2) = 0,$$

showing that the point B is a crunode, the tangents there being $x = \pm my$.

Writing in the equation $y = 0$, we have $x = 0$, the crunode, and

$$x = 2\sqrt{m^2 + 1}\{\sqrt{m^2 + 1} - m\};$$

the product of these two values is $= 4(m^2 + 1)$, that is, $= (P + \frac{1}{P})$, viz. one value is greater, the other less, than 2. We see also that $2\sqrt{m^2 + 1}\{\sqrt{m^2 + 1} - m\}$, the smaller root, is greater than 1.

The curve corresponding to a parabola $y^2 = 4p^2(p^2 - x)$ is thus a crunodal limaçon, the crunode at B , and the loop lying wholly within the circle. Moreover, the loop includes within itself the centre of the circle.

The other curve is in like manner

$$(r^2 + 1 + 2n^2x)^2 = 4(n^2 - 1)r^2,$$

viz. transforming to the origin B , and therefore putting $x - 1$ for x , and $r^2 = (x - 1)^2 + y^2$, the equation is

$$[x^2 + y^2 + 2(n^2 - 1)x - 2(n^2 - 1)]^2 = 4(n^2 - 1)^2(x^2 + y^2 - 2x + 1),$$

that is,

$$[x^2 + y^2 + 2(n^2 - 1)x]^2 = 4n^2(n^2 - 1)(x^2 + y^2),$$

or say

$$(x^2 + y^2)^2 + 4(n^2 - 1)(x^2 + y^2)x - 4(n^2 - 1)(x^2 + n^2y^2) = 0,$$

viz. the point B is an acnode with the imaginary tangents $x = \pm iny$.

Writing in the equation $y = 0$, we have $x = 0$, the acnode, and

$$x = -\sqrt{n^2 - 1}\{\sqrt{n^2 - 1} + n\},$$

where $\sqrt{n^2 - 1}$ is real and may be taken to be positive; there is thus one root $-\sqrt{n^2 - 1}\{\sqrt{n^2 - 1} + n\}$ which is negative; the other root, say is positive and less than 1; the curve is thus an acnodal limaçon, having B for the acnode and cutting the axis outside the circle on the negative side of B , and inside the circle between B and O .

Taking E as origin, the equation of the circle is $x^2 + y^2 - 2x = 0$, and we hence find for the intersection of the limaçon with the circle $n^2x = 2(n^2 - 1)$, that is,

$$x = 2\left(1 - \frac{1}{n^2}\right), \quad y = \pm \frac{2}{n^2}\sqrt{n^2 - 1}\left(1 - \frac{1}{n^2}\right),$$

values which belong to a real intersection; for $q = 0$, we have $Q = 0$, $n = \infty$, and therefore $(x, y) = (2, 0)$, viz. the intersection is at B ; for $q = \infty$, we have $Q = 1$, $n = 1$, and therefore $(x, y) = (0, 0)$, viz. the intersection is at A , results which are obviously right. Observe that for the other limaçon we have $x = 2(1 + \frac{1}{m^2})$, $y = \pm \frac{2}{m^2}\sqrt{m^2 + 1}(1 + \frac{1}{m^2})$, viz. there is no real intersection with the circle.

The Table above referred to.

$$\Phi = \log \tan(45^\circ + \tfrac{1}{2}\phi).$$

ϕ	Φ	ϕ	Φ	ϕ	Φ	ϕ	Φ	ϕ	Φ
0°	0.000000	15°	0.268028	30°	0.549306	45°	0.881374	60°	1.316958
1°	0.017455	16°	0.286715	31°	0.570302	46°	0.906320	61°	1.347748
2°	0.034913	17°	0.305497	32°	0.591398	47°	0.931697	62°	1.379619
3°	0.052378	18°	0.324380	33°	0.612600	48°	0.957527	63°	1.412631
4°	0.069853	19°	0.343371	34°	0.633914	49°	0.983832	64°	1.446847
5°	0.087343	20°	0.362476	35°	0.655347	50°	1.010639	65°	1.482336
6°	0.104851	21°	0.381703	36°	0.676905	51°	1.037973	66°	1.519177
7°	0.122381	22°	0.401060	37°	0.698595	52°	1.065864	67°	1.557458
8°	0.139938	23°	0.420554	38°	0.720425	53°	1.094342	68°	1.597277
9°	0.157526	24°	0.440194	39°	0.742403	54°	1.123440	69°	1.638747
10°	0.175148	25°	0.459990	40°	0.764538	55°	1.153195	70°	1.681998
11°	0.192811	26°	0.479950	41°	0.786839	56°	1.183645	71°	1.727179
12°	0.210519	27°	0.500085	42°	0.809317	57°	1.214833	72°	1.774474
13°	0.228279	28°	0.520407	43°	0.831983	58°	1.246804	73°	1.824097
14°	0.246096	29°	0.540925	44°	0.854849	59°	1.279606	74°	1.876305

0.9 Paper 846

846.

A VERIFICATION IN REGARD TO THE LINEAR TRANSFORMATION OF THE THETA-FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 77–84.]

The notation is that of Smith's "Memoir on the Theta and Omega Functions," differing from Hermite's only in a factor,

$$\vartheta_{\mu,\nu}(\text{Smith}) = i^{\frac{1}{2}\mu\nu} \vartheta_{\mu,\nu}(\text{Hermite});$$

and it is in consequence of this that the factor $i^{\frac{1}{2}\mu\nu}$ occurs in the expression presently given for δ . Hermite's Memoir "Sur quelques formules relatives a la transformation des fonctions elliptiques" appeared in Liouville, t. III. (1858), pp. 27–36.

Writing

$$\omega = \frac{a\Omega + b}{c\Omega + d},$$

where $ad - bc = 1$, the formula of transformation is

$$\vartheta_{\mu',\nu'}(\omega) = \delta \cdot \vartheta_{\mu,\nu}(\Omega) \exp \left\{ \frac{\pi i}{4} \cdot \frac{a\Omega + b}{c\Omega + d} (c\mu' + d\nu')^2 \right\},$$

where

$$\mu' = a\mu + b\nu + ab, \quad \nu' = c\mu + d\nu + cd.$$

We have, according as b is positive or negative,

$$\delta = \frac{1}{\sqrt{\pm i(c\Omega + d)}} \cdot \frac{H}{\sqrt{(-1)^{\frac{1}{2}(a-1)}}},$$

where $\left(\frac{a}{b}\right)$, $\left(\frac{c}{d}\right)$, &c., are the Legendrian symbols as generalised by Jacobi; if a and b are each of them odd, the formulae for the cases a odd and b odd respectively are equivalent to each other, and either may be used indifferently.

To fix the ideas, I assume throughout that b is positive and odd; the proper formulae thus are as above, and

$$H = \left(\frac{a}{b}\right) i^{-\frac{1}{2}(b-1)}.$$

I will also, for greater convenience, assume that c is odd; ad is consequently even, viz. the numbers a and d are each of them even, or else one is odd and the other even.

The equation $\omega = \frac{a\Omega + b}{c\Omega + d}$ gives $\Omega = \frac{-d\omega + b}{c\omega - a}$, and thence

$$\Omega = \frac{-d + b\omega}{c\omega - a}, \quad H' = \left(\frac{-d}{b}\right) i^{-\frac{1}{2}(b-1)}.$$

Comparing the expressions for ω , H , it appears that we may interchange ω and Ω , writing also $-d$, b , c , $-a$ for a , b , c , d ; and changing the other letters, we may for a , b , c , d , ω , Ω , μ , ν , m , n write d , b , c , a , Ω , ω , m' , n' , m , n , where

$$m' = dm + bn - bd, \quad n' = cm - an - ac.$$

The formula thus becomes

$$\vartheta_{m,n}(\Omega) = \delta' \cdot \vartheta_{m',n'}(\omega) \exp \left\{ \frac{\pi i}{4} \cdot \frac{d\omega + b}{c\omega - a} (cm' + an')^2 \right\},$$

or, for x writing $(a + b\Omega)x$,

$$\vartheta_{m',n'} \left(\frac{-d + b\omega}{c\omega - a} \right) = \delta' \cdot \vartheta_{m,n}(\omega) \exp \left\{ -\frac{\pi i}{4} \cdot \frac{(c\omega - a)}{(d\omega + b)} (cm' + an')^2 \right\},$$

or, observing that the left-hand side is $= i^{\frac{1}{2}m'n'} \vartheta_{m',n'}$, we have

$$\vartheta_{m',n'} = \delta' \cdot \vartheta_{m,n} \exp \left\{ -\frac{\pi i}{4} \cdot \frac{(c\omega - a)}{(d\omega + b)} (cm' + an')^2 \right\},$$

an equation which is of the same form as the original equation of transformation, and is to be identified with it.

We have

$$\begin{aligned} m' &= dm + bn - bd = d(a\mu + b\nu + ab) + b(c\mu + d\nu + cd) - bd \\ &= \mu + bd(-a + c - 1), \\ n' &= cm - an - ac = c(a\mu + b\nu + ab) - a(c\mu + d\nu + cd) - ac \\ &= \nu + ac(-b + d - 1), \end{aligned}$$

values which may be written

$$m' = 2P - \mu, \quad n' = 2Q - \nu,$$

where $P = \frac{1}{2}bd(-a + c - 1)$, $Q = \frac{1}{2}ac(-b + d - 1)$, P and Q being integers.

In fact, P is an integer if b or d is even, and if they are each of them odd, then from the equation $ad - bc = 1$, a and c will be one of them odd, the other even, and $a + c - 1$ will be even; and similarly for Q .

The left-hand side is

$$i^{\frac{1}{2}m'n'} \vartheta_{m',n'} = i^{\frac{1}{2}(2P-\mu)(2Q-\nu)} \vartheta_{2P-\mu, 2Q-\nu},$$

which is

$$i^{2PQ-P\nu-Q\mu+\frac{1}{2}\mu\nu} \vartheta_{-\mu, -\nu} = i^{2PQ-P\nu-Q\mu} \cdot i^{\frac{1}{2}\mu\nu} \vartheta_{\mu, \nu},$$

and the equation finally is

$$\vartheta_{\mu, \nu} = \delta \delta' \cdot \vartheta_{\mu, \nu} \exp \left\{ -\frac{\pi i}{4} \cdot \frac{(c\omega - a)}{(d\omega + b)} (cm' + an')^2 + \frac{\pi i}{4} \omega (c\mu + d\nu)^2 \right\}.$$

Comparing this with the original equation the two equations will be identical if only $\delta \delta' = i^{2PQ-P\nu-Q\mu}$.

We have

$$\delta = \frac{1}{\sqrt{+i(c\Omega + d)}} \cdot \frac{H}{\sqrt{(-1)^{\frac{1}{2}}(a-1)}}, \quad \delta' = \frac{1}{\sqrt{+i(c\omega - a)}} \cdot \frac{H'}{\sqrt{(-1)^{\frac{1}{2}}(-d-1)}},$$

where the square roots are taken in such wise that the real part is positive; hence

$$\delta \delta' = \frac{1}{\sqrt{-(c\Omega + d)(c\omega - a)}} \cdot \frac{HH'}{\sqrt{(-1)^{\frac{1}{2}}(a-d-2)}},$$

where $\sqrt{+1}$ means $\sqrt{-i(a + b\Omega)}$; the last-mentioned two square roots being as just explained; and we have moreover

$$\begin{aligned} \delta &= \exp \left\{ -\frac{1}{4}i\pi(ac\mu^2 + 2bc\mu\nu + bd\nu^2 + 2abc\mu + 2abd\nu + ab^2c) \right\} i^{\frac{1}{4}\mu\nu - \frac{1}{2}mn}, \\ \delta' &= \exp \left\{ -\frac{1}{4}i\pi(-dcm^2 + 2bcmn - abn^2 - 2dbcm + 2abdn - db^2c) \right\} i^{\frac{1}{4}m'n'}, \end{aligned}$$

viz. the value of δ' is obtained from that of δ by the change $a, b, c, d, \mu, \nu, m, n$ into $-d, b, c, a, m', n', m, n$.

Representing these by

$$\delta = \exp\{-\frac{1}{4}(i\pi)A\} i^{\frac{1}{4}\mu\nu - \frac{1}{2}mn}, \quad \delta' = \exp\{-\frac{1}{4}(i\pi)A'\} i^{\frac{1}{4}m'n'},$$

we have

$$\delta\delta' = \exp\{-\frac{1}{4}i\pi(A + A')\} i^{\frac{1}{4}\mu\nu - \frac{1}{2}mn + \frac{1}{4}m'n'}.$$

But

$$m'n' = (2P - \mu)(2Q - \nu) = 4PQ - 2P\nu - 2Q\mu + \mu\nu,$$

that is,

$$\frac{1}{4}\mu\nu - \frac{1}{2}mn + \frac{1}{4}m'n' = -PQ + \frac{1}{2}P\nu + \frac{1}{2}Q\mu,$$

or, omitting the term divisible by 4,

$$i^{\frac{1}{4}\mu\nu - \frac{1}{2}mn + \frac{1}{4}m'n'} = i^{\frac{1}{2}P\nu + \frac{1}{2}Q\mu} = \left(\frac{-1}{i}\right)^{\frac{1}{2}P\nu + \frac{1}{2}Q\mu}.$$

To calculate $A + A'$, we have

$$dm + bn = \mu + bd(-a + c), \quad cm + an = \nu + ac(-b + d),$$

and thence

$$\begin{aligned} -cdm^2 + (ad + bc)mn - abn^2 = \\ \mu\nu + \mu ac(-b + d) + \nu bd(a - c) + abcd(d - b)(a - c) - (ad + bc)\mu\nu, \end{aligned}$$

also

$$\begin{aligned} 2bcdm &= 2abcd\mu - 2b^2cd\nu - 2ab^2cd, \\ 2abdn &= 2abcd\mu + 2abd^2\nu + 2abcd^2, \\ -db^2c &= -db^2c; \end{aligned}$$

whence, adding, we obtain

$$\begin{aligned} A' &= -ac\mu^2 - 2bc\mu\nu - bd\nu^2 - 2abc\mu + (-2bcd - 2b^2cd + 2abd^2)\nu \\ &\quad + abcd(2bc - ab - cd - 2b + 2d) - db^2c, \end{aligned}$$

and, adding to this the expression of A , we find

$$A + A' = (2abd - 2bcd - 2b^2cd + 2abd^2)\nu + ab^2c - db^2c + abcd(2bc - ab - cd - 2b + 2d).$$

The coefficient of ν is $= 2bd(a - c - bc + ad) = 2bd(a - c + 1)$, which is $= 4P\nu$. Hence writing

$$\Theta = \frac{1}{4}abcd(2bc - ab - cd - 2b + 2d) + \frac{1}{2}b^2c(a - d),$$

where observe that 4Θ , but not in every case Θ itself, is an integer, we have $A + A' = 4P\nu + 4\Theta$, and consequently

$$\delta\delta' = \left(\frac{-1}{i}\right)^{\Theta},$$

or, omitting the even number $2P\nu$,

$$\delta\delta' = \left(\frac{-1}{i}\right)^{\Theta}.$$

Observe that $\left(\frac{-1}{i}\right)^{\Theta}$ denotes, and it might properly have been written, $\exp \frac{1}{2}i\pi\Theta$. The foregoing equation becomes thus

$$\delta\delta' = \left(\frac{-1}{i}\right)^{\Theta},$$

that is,

$$\delta\delta' = i^{-\Theta},$$

where the even term $2Q\mu$ may be omitted.

We have moreover

$$mn = ac\mu^2 + (ad + bc)\mu\nu + bd\nu^2 + ac(b + d)\mu + bd(a + c)\nu + abcd,$$

$$\mu\nu = (ad + bc)\mu\nu,$$

and thence

$$\begin{aligned} mn - \mu\nu &= ac(\mu^2 - \mu) + 2bc\mu\nu + bd(\nu^2 - \nu) + ac(b + d + 1)\mu \\ &\quad + bd(a + c + 1)\nu + abcd, \end{aligned}$$

where each term is even; hence $mn - \mu\nu$ is even, and we have simply

$$\delta\delta' = \left(\frac{-1}{i}\right)^M,$$

where

$$M = \frac{1}{4}abcd(ab + cd - 2bc + 2b - 2d) - \frac{1}{2}b^2c(a - d).$$

I write $M = \frac{1}{2}b(a-d)$, then

$$-M = \frac{1}{4}abcd(ab+cd-2bc+2b-2d) - \frac{1}{4}(bc+1)b(a-d),$$

where the second term is $= \frac{1}{4}abd(a-d)$, and we have therefore

$$-M = \frac{1}{4}abd(abc+c^2d-2bc^2+2bc-2cd-a+d).$$

I assume, as above, that b and c are each of them odd; therefore ad is even. I suppose, first, that ad divides by 4, then $\frac{1}{4}abd$ is an integer, and in the expression of M , omitting even numbers, we have

$$abc+c^2d-a+d,$$

which, putting therein $bc = ad - 1$, becomes

$$\begin{aligned} &= \frac{1}{4}abd\{a^2d+c^2d-2a+d\}, \\ &= \frac{1}{4}abd\{a^2d+(c^2-c)d+(c+1)d-2a\}, \end{aligned}$$

where inside the $\{\}$ each term is even; hence M is even.

Next, if ad is even but not divisible by 4, then $bc = ad - 1$, which is $\equiv 1 \pmod{4}$, thus b and c are $4\sigma+1$ and $4\tau+1$, or else $4\sigma-1$ and $4\tau-1$, and, moreover, $bc = 4\phi+1$, and $c^2 = 4\phi+1$; hence

$$-M = \frac{1}{4}abd\{a^2d+(4\phi+1)d-2a+d\},$$

or, omitting even numbers,

$$-M = \frac{1}{4}abd\{a^2d+d-2a\} = \frac{1}{4}abd\{a^2d+(c^2-c)d+(c+1)d-2a\},$$

that is, inside the $\{\}$ numbers which contain the factor 4, this is

$$\frac{1}{4}abd \cdot 4\{\dots\} = abd\{\dots\},$$

or, since each term within the $\{\}$ is even, we have in this case also $-M$ even.

And this being so, the foregoing equation for HH' becomes

$$HH' = \left(\frac{a}{b}\right) \left(\frac{-d}{b}\right) i^{-\frac{1}{2}(b-1)} i^{-\frac{1}{2}(b-1)} (-1)^M,$$

or say

$$HH' = \left(\frac{a}{b}\right) \left(\frac{-d}{b}\right) (-1)^{\frac{1}{2}(b-1)}.$$

The values of H , H' refer each of them to a positive odd value of b , and they thus are

$$H = \left(\frac{a}{b}\right) i^{-\frac{1}{2}(b-1)}, \quad H' = \left(\frac{-d}{b}\right) i^{-\frac{1}{2}(b-1)};$$

hence

$$HH' = \left(\frac{a}{b}\right) \left(\frac{-d}{b}\right) (-1)^{\frac{1}{2}(b-1)},$$

or, since

$$\left(\frac{a}{b}\right) \left(\frac{-d}{b}\right) = \left(\frac{-ad}{b}\right) = \left(\frac{1-bc}{b}\right) = \left(\frac{1}{b}\right) = 1,$$

and $\frac{1}{2}(b-1)$ divides by 4, this is

$$HH' = \left(\frac{-ad}{b}\right) = 1.$$

Also

$$\frac{c\Omega + d}{c\omega - a} = \frac{c \cdot \frac{-d+b\omega}{c\omega - a} + d}{c\omega - a} = \frac{-cd + bc\omega + cd - ad}{c\omega - a} = \frac{bc - ad}{c\omega - a} \omega = \frac{-1}{c\omega - a} \omega,$$

but from the equation $ad - bc = 1$, or $\frac{ad-1}{b} = c$, we have

$$\frac{c\Omega + d}{c\omega - a} = \frac{a}{b} \cdot \frac{-d + b\omega}{c\omega - a} + \frac{1}{b} = \frac{a\Omega + 1}{b},$$

whence

$$HH' = \left(\frac{a}{b}\right) \left(\frac{-d}{b}\right) = \left(\frac{-ad}{b}\right) = 1.$$

We have $\omega = x + iy$, where y is positive; hence

$$a + ib\Omega = a + b(x + iy) = a + bx + iby,$$

if $a = d + bx$; hence

$$\frac{1}{i(a + b\Omega)} = \frac{1}{i(a + bx) - by} = \frac{-by - i(a + bx)}{b^2y^2 + (a + bx)^2} = \frac{R(\cos \theta + i \sin \theta)}{-R(\cos \theta - i \sin \theta)} = -1,$$

where R is positive; and $\cos \theta$ is positive since y is positive, and thus θ lies between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.

Hence

$$\sqrt{-i(-d + b\Omega)} = \sqrt{R}(\cos \tfrac{1}{2}\theta + i \sin \tfrac{1}{2}\theta),$$

being positive, and thus

$$\sqrt{-(c\Omega + d)(c\omega - a)} = \sqrt{+1} = +1,$$

that is, $\sqrt{+1} = +1$; and we thus have, as we should do, $HH' = 1$, the equation which was to be verified.

0.10 Paper 847

847.

ON THE THEORY OF SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 92–107.]

In my “Mémoire sur les Hyperdéterminants,” Crelle, t. XXX. (1846), pp. 1–37, [13, 14, 16], I gave an investigation for the number of and relations between the quartic invariants of a given binary quantic: the very same formulae apply to the cubic covariants of a given binary quantic, or what is the same thing to the cubic seminvariants of a given weight, and I propose at present (considering the formulae in this point of view) to further develop the theory in regard to the solution of the systems of linear equations obtained in the memoir in question. But I first reproduce the investigation as it stands: I recall that the notation is the ordinary hyperdeterminant notation: for instance,

$$12^2UV = (a.c - 2b.b + c.a) = 2(ac - b^2);$$

and that, when the variables do not disappear, each set (x_1, y_1) , (x_2, y_2) , ... is ultimately replaced by (x, y) , so that these are the variables of any covariant.

The investigation is as follows:

“... We pass to the derivatives of the fourth degree, considering the forms in which all the differential coefficients are of the same order. It is easy to see that we may write

$$\partial^4 UVWX = (12 \cdot 34)^\alpha (13 \cdot 42)^\beta (14 \cdot 23)^\gamma \cdot UVWX,$$

or $D_{\alpha, \beta, \gamma}$.”

Writing for shortness

$$12 \cdot 34 = 21, \quad 13 \cdot 42 = 33, \quad 14 \cdot 23 = \mathfrak{S},$$

we have

$$21 + 33 + \mathfrak{S} = 0.$$

Suppose $U = V = W = X$, and consider the derivatives which correspond to the same value f of α, β, γ : we have to find how many of these derivatives are independent, and to express the others in terms of these. Since the functions are equal after the differentiations, we may before the differentiations interchange in any manner the symbolic numbers 1, 2, 3, 4: this gives

$$D_{\alpha,\beta,\gamma} = D_{\alpha,\gamma,\beta} = D_{\beta,\alpha,\gamma} = D_{\beta,\gamma,\alpha} = D_{\gamma,\alpha,\beta} = D_{\gamma,\beta,\alpha}.$$

But we have identically

$$(21 + 33 + \mathfrak{S}) \cdot 21^{\alpha-1} 33^{\beta} \mathfrak{S}^{\gamma} \cdot UVWX = 0,$$

when $\alpha > 0$. Multiplying by $21^a 33^b \mathfrak{S}^c$, and applying it to the product $UVWX$, we have

$$D_{a+1,b,c} + D_{a,b+1,c} + D_{a,b,c+1} = 0,$$

which, putting $a + b + c = f$, gives a system of equations between the derivatives $D_{\alpha,\beta,\gamma}$ for which $\alpha + \beta + \gamma = f$.

Reducing these by the conditions just obtained, suppose that f denotes the number of partitions of f into three parts, zero included, and the permutations of the parts being disregarded, we have a number f of derivatives $D_{\alpha,\beta,\gamma}$, and a number $\theta(f-1)$ of linear relations between these derivatives. There remains therefore a number $f - \theta(f-1)$ of independent derivatives: but when f is an even number, we have included among these the functions $D_{\frac{1}{2}f,0,0}$, that is, $12^{\frac{1}{2}f} \cdot 34^{\frac{1}{2}f} UVWX$, which evidently reduces itself to $12^{\frac{1}{2}f} U \cdot 34^{\frac{1}{2}f} X$, or $B_{\frac{1}{2}f}(U, V) \cdot B_{\frac{1}{2}f}(W, X)$, that is to say, to the square of $B_{\frac{1}{2}f}(U, V)$.

We must therefore diminish by unity this number $f - \theta(f-1)$, when f is even. Let $E(\frac{1}{2}f)$ denote the integer part of the fraction $\frac{1}{2}f$: it can be shown that the required number is equal to $E(\frac{1}{2}f) - E(\frac{1}{3}f)$ according as f is even or odd.

Giving to f the six forms $6g, 6g+1, 6g+2, 6g+3, 6g+4, 6g+5$, we obtain for the independent derivatives of the fourth degree the corresponding numbers

$$0, \quad 0, \quad 0, \quad g+1, \quad 0, \quad g+1;$$

there is for example a single derivative for each of the orders 3, 5, 6, 7, 8, 10, two for each of the orders 9, 11, 12, 13, 14, 16, &c.

We may take for independent derivatives, when f is even, the terms of the series

$$D_{f-3,3,0}, \quad D_{f-6,6,0}, \quad \dots,$$

and when f is odd, those of the series

$$D_{f-1,1,0}, \quad D_{f-4,4,0}, \quad D_{f-7,7}, \quad \dots,$$

continuing each series until the last term in which the first suffix exceeds or is equal to the second suffix.

We have in each case the right number of terms; for example, in the case $f = 9$, we take for independent derivatives the two functions $D_{8,1,0}$ and $D_{5,4,0}$, and we form the equations

$$\begin{aligned} D_{9,0,0} + D_{8,1,0} + D_{7,2,0} &= 0, \\ D_{6,2,1} + D_{5,3,1} + D_{4,4,1} &= 0, \\ D_{7,1,1} + D_{6,2,1} + D_{5,3,1} &= 0, \\ D_{4,3,2} + D_{3,4,2} + D_{2,5,2} &= 0, \\ D_{6,1,2} + D_{5,2,2} + D_{4,3,2} &= 0, \\ D_{3,3,3} + D_{2,4,3} + D_{1,5,3} &= 0, \\ D_{5,1,3} + D_{4,2,3} + D_{3,3,3} &= 0, \\ D_{2,3,4} + D_{1,4,4} + D_{0,5,4} &= 0, \\ D_{4,1,4} + D_{3,2,4} + D_{2,3,4} &= 0, \\ D_{1,3,5} + D_{0,4,5} + D_{?,?} &= 0, \end{aligned}$$

which are to be reduced by means of the formulas

$$D_{\alpha,\beta,\gamma} = D_{\alpha,\gamma,\beta} = D_{\beta,\gamma,\alpha} = \dots$$

We will presently give the solution of these equations: but beginning with the second order and going successively to the ninth order, we form easily the following table:

$D_{2,0,0} = B_2,$	$D_{3,0,0} = 0,$	$D_{4,0,0} = 0,$
$D_{2,1,0} = 0,$	$D_{3,1,0} = 0,$	$D_{4,1,0} = 0,$
$D_{1,1,1} = 0,$	$D_{2,2,0} = B_2^2,$	$D_{3,2,0} = 0,$
	$D_{2,1,1} = 0,$	$D_{3,1,1} = 0,$
		$D_{2,2,1} = 0,$
		$D_{1,1,2} = 0.$

$$\begin{array}{lll}
D_{5,0,0} = 0, & D_{6,0,0} = B_2^3, & D_{7,0,0} = 0, \\
D_{5,1,0} = 0, & D_{6,1,0} = 0, & D_{7,1,0} = 0, \\
D_{4,2,0} = D_{4,1,1} = 0, & D_{5,2,0} = \frac{1}{3}B_2D_{4,2,0} = 0, & D_{6,2,0} = \frac{1}{2}B_2D_{5,2,0} = 0, \\
D_{3,3,0} = B_2^3, & D_{4,3,0} = \frac{1}{2}B_2^2D_{3,3,0} = \frac{1}{2}B_2^3, & D_{5,3,0} = \frac{1}{5}B_2D_{4,3,0} = \frac{1}{10}B_2^4, \\
D_{3,2,1} = 0, & D_{4,2,1} = 0, & D_{5,2,1} = 0, \\
& D_{3,3,1} = B_2^3, & D_{4,3,1} = \frac{1}{4}B_2^4, \\
& & D_{3,2,2} = 0.
\end{array}$$

For any value of f except $f = 2, 3$, or 4 , the table commences, for f even and f odd respectively, in the following manner:

$$\begin{array}{ll}
D_{f,0,0} = B_2^{\frac{1}{2}f}, & D_{f,0,0} = 0, \\
D_{f-1,1,0} = -\frac{1}{2}B_2^{\frac{1}{2}f}, & D_{f-1,1,0} = B_2^{\frac{1}{2}(f-1)}, \\
D_{f-2,2,0} = -D_{f-1,1,0}, & D_{f-2,1,1} = 0, \\
D_{f-3,3,0} = \frac{1}{3}B_2D_{f-2,2,0} + \frac{2}{3}D_{f-1,1,0}, &
\end{array}$$

but beyond this I do not know the law of the series."

Before going further, I remark that if $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, instead of the foregoing values, denote respectively

$$\begin{aligned}
\mathfrak{A} &= (00x_1 + 21y_1) \cdot 23 = 0, \\
\mathfrak{B} &= (00x_2 + 21y_2) \cdot 31 = 0, \\
\mathfrak{C} &= (00x_3 + 21y_3) \cdot 12 = 0,
\end{aligned}$$

then identically $\mathfrak{A} + \mathfrak{B} + \mathfrak{C} = 0$, and the same theory applies to the cubic derivatives $\mathfrak{A}^p\mathfrak{B}^q\mathfrak{C}^r \cdot UVW$, that is, to the cubic covariants, or, attending only to the coefficients of the highest powers of x , to the cubic seminvariants.

Or, we may use the Clebschian notation, for instance

$$(a, b, c \mid x, y)^2 = (ax + by)^2 = (a_0x + a_1y)^2 = \&c.,$$

that is

$$a_0^2 = a_1^2 = \cdots = a, \quad a_0a_1 = b_0b_1 = \cdots = b, \quad a_1^2 = b_1^2 = \cdots = c,$$

viz. in the language of Prof. Sylvester, a_0 , a_1 , b_0 , b_1 , ... are here umbrae.

The invariant is

$$(a_0b_1 - a_1b_0)^2 = a_0^2b_1^2 - 2a_0a_1b_0b_1 + a_1^2b_0^2 = ac - 2b \cdot b + ac = 2(ac - b^2),$$

and so in other cases.

To apply this directly to the theory of seminvariants, it is more convenient to write

$$(1, b, c \mid x, y)^2 = (x + \alpha y)^2 = \&c.,$$

that is

$$\alpha = \beta = \dots = b,$$

where $\alpha, \beta, \dots$ are umbrae.

The invariant is

$$(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2 = c - 2b \cdot b + c = 2(c - b^2),$$

or, to take the case of a cubic function

$$(1, b, c, d \mid x, y)^3 = (x + \alpha y)^3 = (x + \beta y)^3 = (x + \gamma y)^3,$$

that is

$$\alpha = \beta = \gamma = b, \quad \alpha^2 = \beta^2 = \gamma^2 = c, \quad \alpha^3 = \beta^3 = \gamma^3 = d,$$

a seminvariant is

$$\begin{aligned} (\alpha - \beta)^2(\alpha - \gamma) &= \alpha^3 - 2\alpha^2\beta + \alpha\beta^2 - \alpha^2\gamma + 2\alpha\beta\gamma - \beta^2\gamma \\ &= d - 2bc + bc - bc + 2b^3 - bc \\ &= d - 3bc + 2b^3, \end{aligned}$$

and so in other cases.

Writing here the general form of a cubic seminvariant of the weight ω is now $21^p33^q\mathfrak{S}^r$, or say $D_{p,q,r}$ where $p+q+r=\omega$ (the new letters p, q, r being used for the exponents, since α, β, γ are now employed in a different sense; and since f will be required as a coefficient, I have also written ω instead of f for the weight).

We have, as before,

$$D_{p,q,r} = D_{p,r,q} = D_{q,p,r} = \dots$$

and we again see that the theory as to the number of and relations between the cubic seminvariants is identical with that of the quartic invariants.

Observe also that for ω odd, $D_{\omega,0,0}$ is $= 0$, while for ω even, $D_{\omega,0,0} = 21^\omega$, is a quadric seminvariant, which is of course not to be reckoned among the proper cubic seminvariants; this exactly corresponds to the B_f which is not a proper quartic invariant.

The choice of the functions $D_{f,0,0}$, $D_{f-3,3,0}$, $\dots$, or $D_{f-1,1,0}$, $D_{f-4,4,0}$, $\dots$ for expressing in terms of them the other functions has its advantages, but also its disadvantages; in what follows, I in effect replace them by $D_{f,0,0}$, $D_{f-2,2,0}$, $\dots$ and $D_{f-1,1,0}$, $D_{f-3,3,0}$, $\dots$ respectively.

And instead of considering the whole series of the functions $D_{p,q,r}$ for a given weight ω , I consider only those of the form $D_{p,q,0}$; these form a single series $D_{\omega,0,0}$, $D_{\omega-1,1,0}$, $D_{\omega-2,2,0}$, $\dots$; and for shortness, I call them A , B , C , D , &c. respectively, viz. I write

$$A = D_{\omega,0,0}, \quad B = D_{\omega-1,1,0}, \quad C = D_{\omega-2,2,0}, \quad D = D_{\omega-3,3,0}, \quad \dots$$

Suppose first that ω is odd; we have

$$D_{p,q,r} = D_{p,r,q} = D_{q,r,p} = \dots$$

and, in particular,

$$D_{\omega-2q,q,q} = D_{\omega-2q,q,q},$$

that is,

$$D_{\omega-2q,q,q} = 0,$$

that is,

$$C = 0, \quad F = 0, \quad I = 0, \quad \dots$$

Hence

$$B + C = 0, \quad C + 2D + E = 0, \quad D + 3E + 3F + G = 0, \quad \dots$$

that is, we have

$$A = 0, \quad B + C = 0, \quad C + 2D + E = 0, \quad D + 3E + 3F + G = 0, \quad \&c.$$

The readiest way of solving these is to express the other functions in terms of B , D , F , H , &c.; viz. we thus have

$$A = 0,$$

$$C = -B,$$

$$E = B - 2D,$$

$$G = 3B - 5D + 3F,$$

$$I = -18B + 28D - 14F + 4H,$$

$$K = 57B - 81D + 45F - 15H + 5J, \quad \&c.$$

We have here, writing for shortness 900, 810, &c., instead of $D_{9,0,0}$, $D_{8,1,0}$, &c. respectively,

$$\begin{array}{llll} 900 = 0, & 810 = B, & 720 = -B, & 630 = L \\ 540 = B - 2D, & 450 = F, & 360 = -3B + 5D - 3F, & \\ 270 = H, & 180 = 18B - 28D + 14F - 4H, & 090 = L. & \end{array}$$

But we have $900 = 0$, $810 + 180 = 0$, $720 + 270 = 0$, $630 + 360 = 0$, $540 + 450 = 0$; that is,

$$L = 0, \quad 18B - 28D + 14F - 4H = 0, \quad -B + H = 0, \quad -3B + 6D - 3F = 0, \quad B - 2D +$$

all satisfied by $F = B + 2D$, $H = B$, $L = 0$.

The proper course is to stop at the equation for 540, viz. the system is

$$900 = 0, \quad 810 = B, \quad 720 = -B, \quad 630 = D, \quad 540 = B - 2D,$$

equations which may be considered as expressing the several functions 900, 810, 720, 630, 540 in terms of $B = 810$ and $D = 630$. They agree, as they should do, with a foregoing result, $900 = 0$, 810 , $720 = -810$, $630 = \frac{1}{3}810 + \frac{2}{3}540$, 540, and it would of course be possible to express the remaining forms 711, 621, 522, 441, 432, and 333 in terms of $B = 810$ and $D = 630$. But observe that the speciality of the present process is that, instead of the whole series of forms 900, 810, 720, 711, 630, 621, 540, 531, 522, 441, 432, 330, we work only with the forms 900, 810, 720, 630, 540, for which the last element is $= 0$.

A more simple solution of the system of linear equations in A , B , C , ... is the following:

A B C D E F G H I J K L

For the case ω even, I write

$$A + 2B = A', \quad B + 2C = B', \quad C + 2D = C', \quad \&c.;$$

and I say that the new functions A' , B' , C' , ... are related together precisely in the same manner as are the functions A , B , C , ... belonging to an odd weight ω ; viz. we have

$$A' = 0, \quad B' + C' = 0, \quad C' + 2D' + E' = 0,$$

which being so, the theory is included in what precedes, and there is no occasion to consider it separately.

To prove this, observe that ω being even, we have

$$D_{p,q,r} = D_{p,r,q} = D_{q,r,p} = \dots$$

that is,

$$2D_{\omega-2q,q,q} = D_{\omega-2q+1,q,q-1} + D_{\omega-2q-1,q+1,q},$$

whence, in particular,

$$2D_{\omega,0,0} = D_{\omega-1,1,0}, \quad 2D_{\omega-2,1,1} = D_{\omega-1,1,0} + D_{\omega-3,2,1}, \quad \&c.;$$

we thus have

$$A' = 2D_{\omega,0,0} + 2D_{\omega-2,2,0} = 2D_{\omega-1,1,0} + D_{\omega-2,2,0} = 2D_{\omega-1,1,0}(21+33+\mathfrak{S}),$$

which is = 0.

Similarly $B' + C' = B' + 3C' + \mathfrak{S}' = D_{\omega-3,3,0} + 3D_{\omega-4,4,0} + 2D_{\omega-5,5,0} = D_{\omega-3,3,0}(2(21+33)(21+2 \cdot 33)) = D_{\omega-3,3,0}(2(\mathfrak{S} + 33)(21+2 \cdot 33))$, or, since $21+33+\mathfrak{S} = 0$, this is $= -2D_{\omega-3,3,0}(21+33)(21+2 \cdot 33)$, which, in virtue of $21+33+\mathfrak{S} = 0$, is $= -2D_{\omega-3,3,0}(21+33)(\mathfrak{S}-21)$, which is = 0.

And similarly $C' + 2D' + E' = 0$, &c.

I come to a new part of the theory: to fix the ideas, consider the weight $\omega = 27$; and write also

$$a_0, a_1, a_2, a_3, \dots = 1, b, c, d, e, f, g, h, i, j, k, l, m, \dots,$$

using, if we please, the suffixed a for the higher terms.

The cubic seminvariants are

$$3^{12}, \quad 3^{10}5, \quad 3^85^2, \quad 3^65^3, \quad 3^45^4, \quad \text{viz. the number of them is} = 5;$$

we have thus the 5 terms B, D, F, H, J .

Here $B = (\alpha - \beta)^{26}(\gamma - \alpha) = (\alpha - \beta)^{26}(\beta - \gamma)$, or, interchanging the α and β , $= (\alpha - \beta)^{26}(\alpha - \gamma)$, there is an initial term $\alpha^{27} = a^{27}$ and a final term $\alpha\beta^{13}\gamma^{13} = bm^2$, or we may write

$$B = (\alpha - \beta)^{26}(\alpha - \gamma) = a^{27} + bm^2;$$

and similarly for the other forms.

We have thus

$$\begin{aligned} B &= (\alpha - \beta)^{26}(\alpha - \gamma) = a^{27} + bm^2, \\ D &= (\alpha - \beta)^{24}(\alpha - \gamma)^3 = a^{27} + dm^2, \\ F &= (\alpha - \beta)^{20}(\alpha - \gamma)^5 = a^{25}e + fm^2, \\ H &= (\alpha - \beta)^{16}(\alpha - \gamma)^7 = a^{21}g + hm^2, \\ J &= (\alpha - \beta)^{12}(\alpha - \gamma)^9 = a^{19}i + jm^2, \end{aligned}$$

where the initial term $a^{27} = a^{27}$ has in each case the coefficient unity, but the final terms have each of them the proper numerical coefficient.

It must be possible to form linear combinations

$$\begin{aligned} B &= a^{27} + bm^2, \\ (B, D) &= a^{25}(c + b^2) + dm^2, \\ (B, D, F) &= a^{23}(e + c^2) + fm^2, \\ (B, D, F, H) &= a^{21}(g + d^2) + hm^2, \\ (B, D, F, H, J) &= a^{19}(i + e^2) + jm^2, \end{aligned}$$

where $c + b^2$, $e + c^2$, $g + d^2$, $i + e^2$ denote the seminvariants

$$c - b^2, \quad e - 4bd + 3c^2, \quad g - 6bf + 15ce - 10d^2, \quad i - 8bh + 28cg - 56df + 35e^2,$$

respectively.

The expressions indicated by their initial and final terms $a^{27} + bm^2$, $a^{25}c + dm^2$, &c., are what I call "columns;" see my paper "Seminvariant Tables," American Journal of Mathematics, t. VII. (1885), pp. 59–73, [831], where, however, the theory is not by any means completely developed.

As to this, observe that B, D are each of them of the form $a^{27} + a^{26}b + a^{25}(c, b^2) + \dots$, the proper combination in order to eliminate $a^{26}b$ is obviously $B - D$, the term in $a^{26}b$ will then disappear of itself, and the terms in $a^{25}(c, b^2)$ will combine into a term $a^{25}(c - b^2)$, viz. in the function qua seminvariant the coefficient of the highest letter a^{25} will be the seminvariant $c - b^2$.

Similarly, in the combination B, D, F , the two coefficients will be determinable so that there are no terms in a^{27} , $a^{26}b$, a^{25} or a^{24} , and this being so, the term in a^{23} will be $a^{23}(e - 4bd + 3c^2)$, viz. in the function, qua seminvariant, the coefficient of the highest letter a^{23} will be the seminvariant $e - 4bd + 3c^2$. And similarly for the remaining two linear combinations.

As a partial verification, I write

$$B = \{\alpha^{26} + \beta^{26} - 26(\alpha^{25}\beta + \alpha\beta^{25}) + 325(\alpha^{24}\beta^2 + \alpha^2\beta^{24}) \dots\}(\alpha - \gamma),$$

or, retaining only the terms which have an index at least = 25, this is

$$\begin{aligned} B &= \alpha^{27} + \alpha\beta^{26} - 26(\alpha^{26}\beta + \alpha^2\beta^{25}) + 325\alpha^{25}\beta^2 \\ &\quad - (\alpha^{26}\gamma - 26\alpha^{25}\beta\gamma + 325\alpha^{24}\beta^2\gamma) \\ &= \alpha^{27} - 27\alpha^{26}\beta + \alpha^{25}(299c + 325b^2); \end{aligned}$$

and similarly

$$\begin{aligned} D &= (\alpha - \beta)^{24}(\alpha - \gamma)^3 \\ &= (\alpha^{24} + \beta^{24} - 24(\alpha^{23}\beta + \alpha\beta^{23}) + 276(\alpha^{22}\beta^2 + \alpha^2\beta^{22}) \dots) \\ &\quad \times (\alpha^3 - 3\alpha^2\gamma + 3\alpha\gamma^2 - \gamma^3), \end{aligned}$$

or, retaining only the terms which have an index at least = 25,

$$\begin{aligned} D &= \alpha^{27} - 24\alpha^{26}\beta + 276\alpha^{25}\beta^2 - 3\alpha^{26}\gamma + 72\alpha^{25}\beta\gamma \\ &\quad + 3\alpha^{24}\beta^2\gamma + 276\alpha^{25}\beta^2 + 72\alpha^{25}\beta\gamma \\ &\quad + 325(279c + 72b^2), \end{aligned}$$

and thence

$$B - D = \alpha^{25}(20c - 20b^2),$$

viz. the coefficient of α^{25} is $= 20(c - b^2)$.

But instead of the foregoing forms B, D, F, H, J , we may start with five other forms which are in fact linear combinations of these, viz. these are the forms

$$\begin{aligned} X &= (\alpha - \beta)^{26}(\alpha + \beta - 2\gamma) = a^{27} + bm^2, \\ Y &= (\alpha - \beta)^{24}(\alpha + \beta - 2\gamma)(\alpha - \gamma)(\beta - \gamma) = a^{25}(c + b^2) + dm^2, \\ Z &= (\alpha - \beta)^{20}(\alpha + \beta - 2\gamma)(\alpha - \gamma)^2(\beta - \gamma)^2 = a^{23}(e + c^2) + fm^2, \\ W &= (\alpha - \beta)^{16}(\alpha + \beta - 2\gamma)(\alpha - \gamma)^3(\beta - \gamma)^3 = a^{21}(g + d^2) + hm^2, \\ T &= (\alpha - \beta)^{12}(\alpha + \beta - 2\gamma)(\alpha - \gamma)^4(\beta - \gamma)^4 = a^{19}(i + e^2) + jm^2, \end{aligned}$$

where as well the initial as the final terms should have their proper numerical coefficients.

As to this, observe that in Y we have a term $\alpha^{26}(\beta - \gamma)$ which is $= 0$, and similarly a term $\beta^{26}(\alpha - \gamma)$ which is $= 0$; the highest powers are thus α^{25} and β^{25} , giving a term in a^{25} which can only be $a^{25}(c + b^2)$.

In Z we have the terms $\alpha^{25}(\beta - \gamma)^2$ and $\beta^{25}(\alpha - \gamma)^2$, giving the term $a^{25}(c + b^2)$; in W we have the terms $\alpha^{24}(\beta - \gamma)^3$ and $\beta^{24}(\alpha - \gamma)^3$ which are each $= 0$, the highest powers thus are α^{23} , β^{23} giving a term in a^{23} which can only be $a^{23}(e - 4bd + 3c^2)$; and in T we have the terms $\alpha^{23}(\beta - \gamma)^4$ and $\beta^{23}(\alpha - \gamma)^4$, which give the term $a^{23}(e - 4bd + 3c^2)$.

The combinations which give the columns thus are

$$\begin{aligned} X &= a^{27} + bm^2, \\ (Y) &= a^{25}(c + b^2) + dm^2, \\ (Y, Z) &= a^{23}(e + c^2) + fm^2, \\ (Y, Z, W) &= a^{21}(g + d^2) + hm^2, \\ (Y, Z, W, T) &= a^{19}(i + e^2) + jm^2, \end{aligned}$$

I start with the form $u = (Y, Z, W, T) = a^{19}(i + e^2) + jm^2$.

We know that it is possible to determine the coefficients λ, μ, ν , in suchwise that the linear function $u + \lambda Z + \mu W + \nu T$ shall be of the proper form $a^{19}(i + e^2) + jm^2$; or, what is the same thing, that there shall be no term with an index exceeding 19; the conditions to be satisfied are apparently more than three, but of course the number of independent conditions must be $= 3$.

We have, in particular, to get rid of all the terms of the second degree which precede $a^{19}i$, viz. the coefficients of $a^{26}c$, $a^{24}d$, $a^{22}e$, $a^{20}f$, $a^{18}g$, $a^{16}h$ must all of them vanish.

Now the terms of u which produce terms of the second degree are the terms containing any two but not all three of the symbolic letters α, β, γ , say the biliteral terms; we have for instance $\alpha^{25}\beta^2$, $\alpha^{25}\gamma^2$, $\beta^{25}\alpha^2$, $\beta^{25}\gamma^2$ (there are no terms $\gamma^{25}\alpha^2$ or $\gamma^{25}\beta^2$, since the highest power of γ is γ^9), each of which is $= a^{25}c$. But in the development of u , we may in any biliteral term by an interchange of the letters α, β, γ make the letter of highest index to be α , and the other letter to be β ; thus the several terms $\alpha^{25}\beta^2$, $\alpha^{25}\gamma^2$, $\beta^{25}\alpha^2$, $\beta^{25}\gamma^2$ may be each of them converted into $\alpha^{25}\beta^2$, and so in other cases.

Imagining this to be done, the term in $a^{25}c$ is simply the term in $\alpha^{25}\beta^2$, and in like manner the term in $a^{24}d$ is the term in $\alpha^{24}\beta^3$, and so in other cases; the condition as to the disappearance of the terms

$a^{25}c, \dots, a^{16}h$ is the condition that the terms in $\alpha^{25}\beta^2, \alpha^{24}\beta^3, \alpha^{23}\beta^4, \alpha^{22}\beta^5, \alpha^{21}\beta^6, \alpha^{20}\beta^7$ shall all of them disappear.

And if, in the function u transformed in the foregoing manner, we write for convenience $\alpha = 1$, so that u has become a function of β only (and observe that u will contain the factor β^2), then the condition is that the terms in $\beta^2, \beta^3, \beta^4, \beta^5, \beta^6, \beta^7$ shall all of them disappear.

The conditions may in fact be written

$u = 0$; viz. it is assumed that u is in the first instance transformed into a function

and the equation is then to be understood as denoting that the coefficients of u are to be so determined that as many as possible of the terms (beginning with that which contains the lowest power of β) shall each of them vanish.

Consider any term of u , for instance the term

$$T = (\alpha - \beta)^{12}(\alpha + \beta - 2\gamma)(\alpha - \gamma)^4(\beta - \gamma)^4.$$

To obtain the biliteral terms in the proper form

1. we write therein $\gamma = 0$,
2. we write $\beta = 0$, changing γ into β ;
3. we write $\alpha = 0$, changing β, γ into α, β ;

we thus obtain

$$\begin{aligned} &(\alpha - \beta)^{12}\alpha^{12}(-2\beta)(-\alpha)^4(-\beta)^4, \\ &(\alpha - \beta)^{12}\alpha^{12}(-2\beta)(-\alpha)^4(-\beta)^4, \end{aligned}$$

viz. the second and third terms are identical; the first term requires, however, the factor 2 (for any term $\alpha^p\beta^q$ is accompanied by a corresponding term $\alpha^q\beta^p$), so that, omitting this factor throughout, the terms are

$$(\alpha - \beta)^{24}(\alpha + \beta)\alpha\beta - \alpha^{24}(\alpha - 2\beta)(\alpha - \beta)\beta,$$

or, putting therein $\alpha = 1$, the terms are

$$(1 - \beta)^{24}(1 + \beta)\beta - (1 - 2\beta)(1 - \beta)\beta.$$

Similarly from

$$Z = (\alpha - \beta)^{20}(\alpha + \beta - 2\gamma)(\alpha - \gamma)^2(\beta - \gamma)^2,$$

we have

$$(1 + \beta)\beta(1 - \beta)^{20} + \lambda(1 - \beta)^{20}\beta^2 + \mu(1 - \beta)^{16}\beta^3 + \nu(1 - \beta)^{12}\beta^4$$

and the like as regards W and T .

The result thus is

$$(1 + \beta)\beta(1 - \beta)^{24} + \lambda(1 - \beta)^{22}\beta^2 + \mu(1 - \beta)^{20}\beta^3 + \nu(1 - \beta)^{18}\beta^4 \\ - (1 - 2\beta)(1 - \beta)\beta\{1 - \lambda\beta(1 - \beta) + \mu\beta^2(1 - \beta)^2 - \nu\beta^3(1 - \beta)^3\} = 0,$$

or, as this may be written,

$$(1 - \beta)^{24}(1 + \beta)\beta = (1 - 2\beta)(1 - \beta)\beta\{1 - \lambda\beta(1 - \beta) + \mu\beta^2(1 - \beta)^2 - \nu\beta^3(1 - \beta)^3\},$$

viz. each side is to be expanded in ascending powers of β , and the coefficients λ , μ , ν are then to be determined so that as many terms as possible shall vanish.

Expanding, we have

$$(1 - 20\beta + 190\beta^2 - 1138\beta^3 + 4808\beta^4 - 15178\beta^5 + \dots) \\ \times \{1 + \lambda\beta + \beta^2(2\lambda + \mu) + \beta^3(3\lambda + \nu) + \beta^4(4\lambda + 10\mu + 6\nu) \\ + \beta^5(5\lambda + 20\mu + 21\nu) + \dots\} \\ - 1 + \lambda\beta + \beta^2(-\lambda - \mu) + \beta^3(2\mu + \nu) + \beta^4(-3\nu) + \beta^5 \cdot 3\nu + \dots = 0,$$

or finally, as far as β^5 , the equation is

$$0 = +\beta(2\lambda - 20) + \beta^2(-19\lambda + 190) + \beta^3(153\lambda - 14\mu + 2\nu - 1138) \\ + \beta^4(-814\lambda + 119\mu - 17\nu + 4808) + \beta^5(3027\lambda - 558\mu + 94\nu - 15178).$$

We have in the first place $\lambda = 10$, which makes the coefficient of β^2 to be $= 0$; and we then have

$$7\mu - \nu - 196 = 0, \quad 119\mu - 17\nu - 3332 = 0, \quad 272\mu - 47\nu - 7546 = 0,$$

where the second equation is equivalent to the first: the three equations give

$$\mu = -\frac{833}{1}, \quad \nu = \frac{?}{1}.$$

I have since found that proceeding one step further, that is, to β^6 , the coefficient is

$$-8310\lambda + 1791\mu - 363\nu + 36942,$$

viz. putting this = 0, and for λ substituting its value = 10, we have

$$597\mu - 121\nu = 15386,$$

an equation which is in fact satisfied by the values just obtained for λ and μ .

For the function (Y, Z, W) we have the same equations with $\nu = 0$; and therefore $\lambda = 10$, $\mu = 28$; and for the function (Y, Z) the same equations with $\mu = 0$, $\nu = 0$; and therefore $\lambda = 10$.

The linear combinations thus are

$$\begin{aligned} X &= a^{27} + bm^2, \\ Y &= a^{25}(c + b^2) + dm^2, \\ Y + 10Z &= a^{23}(e + c^2) + fm^2, \\ Y + 10Z + 28W &= a^{21}(g + d^2) + hm^2, \\ 25Y + 250Z + 833W + 9312T &= a^{19}(i + e^2) + jm^2, \end{aligned}$$

where remark that in $Y + 10Z$, by means of the coefficient 10, we make to disappear the two terms $a^{25}c$, $a^{25}d$; in the next function, by means of the coefficients 10 and 28, the four terms $a^{23}c$, $a^{23}d$, $a^{23}e$, $a^{23}f$, and in the last function, by means of the three coefficients, the six terms $a^{21}c$, $a^{21}d$, $a^{21}e$, $a^{21}f$, $a^{21}g$ and $a^{21}h$.

It is possible that a larger number of terms will disappear; but if this is so, the general form shown by the combinations on p. 353 will require modification.

The investigation applies without alteration to any odd weight ω whatever; the condition for the determination of the coefficients λ , μ , ν , ... is obviously $1 - 2\beta = 0$.

For the case of an even weight ω , the series of functions is

$$X = (\alpha - \beta)^\omega, \quad Y = (\alpha - \beta)^{\omega-2}(\alpha - \gamma)(\beta - \gamma), \quad Z = (\alpha - \beta)^{\omega-4}(\alpha - \gamma)^2(\beta - \gamma)^2, \quad \dots$$

and the condition for the determination of the coefficients λ , μ , ν , ... is in like manner found to be

$$-1 + \lambda(0 - \beta^2) - \mu(0 - \beta^3) + \nu(0 - \beta^4) + \dots = 0,$$

which is to be understood as explained above.

0.11 Paper 848

848.

ON THE TRANSFORMATION OF THE DOUBLE-THETA FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 142–178.]

I propose to reproduce Hermite's Memoir "Sur la théorie de la transformation des fonctions Abéliennes," *Comptes Rendus*, t. XL. (1855), pp. 249, . . . , 784, with some changes of notation and developments. Hermite's functions are even or odd according as we have $\mu q + \nu r$ even or odd: viz. his characteristic is $\binom{\mu, \nu}{q, r}$, or the letters p, q are misplaced; I write, therefore, r instead of p , so as to have the characteristic $\binom{\mu, \nu}{q, r}$; and then for symmetry it is necessary to interchange the suffixes 2, 3 and the letters c, d ; the invariant function of the periods, instead of being as with him $\mathfrak{C}_0 \mathfrak{C}_3 (\mathfrak{C}_3 \mathfrak{C}_0 + \mathfrak{C}_2 \mathfrak{C}_1) \omega_1 \nu_1$, must be taken to be $\mathfrak{C}_0 \mathfrak{C}_2 (\mathfrak{C}_2 \mathfrak{C}_0 + \mathfrak{C}_1 \mathfrak{C}_3) \omega_1 \nu_1$.

Moreover, I write A, B for his $G, \mathfrak{C}$, so as, instead of $(G, H, \mathfrak{C} \mid x, y)^2$, to have in the expressions of the theta-functions the quadric function $(A, H, B \mid x, y)^2$; and I alter the arrangement of the memoir so as to separate more completely the preliminary theory from the theory of the transformation.

General Theory. Art. Nos. 1 to 21 (several sub-headings).

The functions $\Pi(x, y \mid K, \text{indef. or def.})$.

Art. Nos. 1–4.]

1. Consider a function $\Pi(x, y)(A, H, B \mid K, \text{indef. or def.})$, having: the characteristic $\binom{\mu, \nu}{q, r}$, where the characters μ, ν, q, r are positive or negative integers, which may be taken to have each of them only the values 0, 1 at pleasure, so that the number of functions is $2^4 = 16$; the arguments (x, y) ; the parameters, or conjoint quarter-periods, (A, H, B) ; and the potency K , a positive integer; and which is either indefinite or definite, as will be explained; the function, moreover, contains linearly certain arbitrary constants, the number of them depending on the value of K , as will be explained.

The function may be written $\Pi(x, y)$ or in any other less abbreviated form which may be convenient.

2. The function $\Pi(x, y \mid K \text{ indef.})$ is defined by the following four equations:

$$\begin{aligned}\Pi(x+1, y) &= (-1)^{\mu q} \Pi(x, y), \\ \Pi(x, y+1) &= (-1)^{\nu r} \Pi(x, y), \\ \Pi(x+A, y+H) &= (-1)^q \Pi(x, y) \exp(-2i\pi K(2x+A)), \\ \Pi(x+H, y+B) &= (-1)^r \Pi(x, y) \exp(-2i\pi K(2y+B));\end{aligned}$$

and the function $\Pi(x, y \mid K \text{ def.})$ by the same equations, together with the following fifth equation,

$$\Pi(-x, -y) = (-1)^{\mu q + \nu r} \Pi(x, y);$$

viz. the definite function is an even function or else an odd function of the arguments according as $\mu q + \nu r$ is even or odd.

We may call $\mu q + \nu r$ the *index*; and the function is then even or odd according as the index is even or odd.

It is perhaps worth noticing that it would be allowable to define a function $\Pi(x, y \mid K, \text{skew def.})$, by the corresponding relation

$$\Pi(-x, -y) = -(-1)^{\mu q + \nu r} \Pi(x, y),$$

but I do not propose here to develop this notion.

3. The four equations give rise to the following one,

$$\Pi(x+a_0+Aa_2+Ha_3, y+a_1+Ha_2+Ba_3) = (-1)^{\mu a_0 + \nu a_1 + q a_2 + r a_3} \Pi(x, y) \times \exp(-2\pi K\{2$$

where a_0, a_1, a_2, a_3 are any positive or negative integers (zero not excluded), and which single equation, in fact, includes the preceding four equations.

4. In regard to the parameters it is to be observed that, if $A, H, B = A + i\alpha, H + i\eta, B + i\beta$, we must have $(\alpha, \eta, \beta \mid x, y)^2$ a determinate positive quadratic form; viz. this is the necessary and sufficient condition for the convergence of the series for the development of the function.

Number of arbitrary constants.

Art. No. 5.

5. The number of arbitrary constants is for the indefinite function $= K^2$; but for the definite function it is, when K is odd, $= \frac{1}{2}(K^2 + 1)$; and when K is even, it is $= \frac{1}{2}K^2$, except in the case of a characteristic $(\mu, \nu)_{q,r}$, when it is $= \frac{1}{4}(K^2 + 4)$.

In particular, for $K = 1$, there is only a single arbitrary constant, which is a mere factor of the function; taking it to be $= 1$, as presently explained, we have the 16 theta-functions.

Development in series.

Art. Nos. 6–9.

6. The function $\Pi(x, y)$ is developed in a series of exponentials, in the form

$$\Pi(x, y) = \sum (-1)^{\mu q + \nu r} A_{m,n} \exp(i\pi \{ (2m + \mu)x + (2n + \nu)y + (A, H, B \mid m, n)^2 \}),$$

where m and n have each of them all positive and negative integer values (zero not excluded) from $-\infty$ to ∞ .

In fact, substituting this series in the four equations, they are all of them satisfied if only

$$A_{m+K,n} = A_{m,n}, \quad A_{m,n+K} = A_{m,n}.$$

Consequently the following K^2 coefficients remain arbitrary, viz. those with the suffixes $0, 1, \dots, K-1$; and we have for $\Pi(x, y)$ a sum of K^2 terms, each a determinate series multiplied into one of the arbitrary coefficients $A_{0,0}, A_{0,1}, \&c.$

The indefinite function thus contains, as already mentioned, K^2 arbitrary constants.

7. Substituting in the fifth equation, we have for the definite function the further condition

$$A_{-m,-n} = (-1)^{\mu q + \nu r} A_{m,n},$$

which it is clear will be satisfied generally if only it is satisfied by the coefficients in the foregoing set of K^2 coefficients.

8. In the case K odd, we thus reduce the number of arbitrary coefficients to $\frac{1}{2}(K^2 + 1)$; the mode in which this takes place is

best seen by an example. Suppose $K = 3$, so that $A_{m+3,n} = A_{m,n}$; $A_{m,n+3} = A_{m,n}$.

For the coefficients of the indefinite function, the suffixes are 00, 01, 02, 10, 11, 12, 20, 21, 22.

And if we suppose $\mu = 0$, $\nu = 1$, then the new condition is $A_{-m,-n-1} = A_{m,n}$, viz. writing down only the suffixes, we thus obtain

$$\begin{array}{lll} 00 = 22, & 01 = 21, & 02 = 20, \\ 10 = 12, & 11 = 11, & 12 = 10, \\ 20 = 02, & 21 = 01, & 22 = 00, \end{array}$$

viz. one of these equations $01 = 01$ is an identity, but the other equations occur each twice; or we have four equations, each of them an equality between two out of the remaining 8 coefficients; the number of arbitrary coefficients is thus $1 + (9 - 1) = 5$; and so in general the number is $\frac{1}{2}(K^2 + 1)$.

9. In the case K even, the number is $= \frac{1}{2}K^2$, except for the characteristic $\begin{pmatrix} \mu, \nu \\ q, r \end{pmatrix} = \begin{pmatrix} 1, 1 \\ 1, 1 \end{pmatrix}$, when it is $= \frac{1}{4}(K^2 + 4)$.

The theta-functions.

Art. Nos. 10–15.]

10. Writing $K = 1$, and taking $A_{0,0} = 1$, the value of the function thus is

$$\vartheta_{\mu,\nu}(x,y) = \sum (-1)^{\mu q + \nu r} \exp(i\pi\{(2m+\mu)x + (2n+\nu)y + (A, H, B \mid m, n)^2\}).$$

11. The sum of two characteristics is the characteristic obtained by taking the sums of the component terms or characters,

$$\begin{pmatrix} \mu \\ q, r \end{pmatrix} + \begin{pmatrix} \mu' \\ q', r' \end{pmatrix} = \begin{pmatrix} \mu + \mu' \\ q + q', r + r' \end{pmatrix},$$

and similarly for any number of characteristics.

I use the sign $=$, but this properly denotes a congruence, mod. 2; and the like as regards the indices.

The sum of two identical characteristics, or generally of any number of characteristics taken each of them any even number of times, is $= \begin{pmatrix} 0 \\ 0, 0 \end{pmatrix}$. And this characteristic $\begin{pmatrix} 0 \\ 0, 0 \end{pmatrix}$ may be called the characteristic 0.

It should be observed, that the index of the sum is not in general equal to the sum of the indices. To make it so, we must have, for two characteristics,

$$(\mu + \mu')(q + q') + (\nu + \nu')(r + r') = \mu q + \nu r + \mu' q' + \nu' r',$$

that is,

$$\mu q' + \mu' q + \nu r' + \nu' r = 0,$$

and there is obviously a like formula for the case of more than two characteristics.

Two or more characteristics, such that they have the sum of the indices equal to the index of the sum, are said to be “in direct relation” or “directly related” to each other.

The sum of the indices and the index of the sum may differ by unity and we then have the inverse relation; but I do not propose to consider this.

12. Consider any number K of theta-functions, of the same arguments and parameters, but with the same or different characteristics. The product of these functions is in general a function $\Pi(x, y \mid K \text{ indef.})$, having a characteristic which is = the sum of the characteristics of the theta-functions.

In fact, from the four equations of the theta-functions, we at once obtain for their products $\Pi(x, y)$ the four equations

$$\begin{aligned}\Pi(x + 1, y) &= (-1)^{\Sigma \mu q} \Pi(x, y), \\ \Pi(x, y + 1) &= (-1)^{\Sigma \nu r} \Pi(x, y), \\ \Pi(x + A, y + H) &= (-1)^{\Sigma q} \Pi(x, y) \exp(-i\pi K(2x + A)), \\ \Pi(x + H, y + B) &= (-1)^{\Sigma r} \Pi(x, y) \exp(-i\pi K(2y + B)),\end{aligned}$$

which proves the theorem.

13. But if the indices are in direct relation to each other, then we have further

$$\Pi(-x, -y) = (-1)^{\Sigma(\mu q + \nu r)} \Pi(x, y),$$

and the product is thus a function $\Pi(x, y \mid K, \text{def.})$.

14. Take the square of a theta-function, the characteristic is = $\begin{pmatrix} 0 \\ 0, 0 \end{pmatrix}$ or 0, and we have also twice the index = 0; viz. the theta-function is in direct relation with itself.

Hence the squared function is a function $\Pi(x, y \mid 2, \text{def.})$, and as such it contains linearly $\frac{1}{4}(2^2 + 4) = 4$ arbitrary constants.

Hence, taking any five squares, since each of them is a function of the form in question, it follows that the squares of the 5 theta-functions are connected by a linear relation.

15. We may in a variety of ways (in fact, in 60 ways, as will presently be shown) select four theta-functions, all of them even, or else two of them even and two odd (that is, having the sum of their indices = 0), such that the sum of their characteristics is $= \begin{pmatrix} 0 \\ 0,0 \end{pmatrix}$; for instance, the functions may be

$$\vartheta \begin{pmatrix} 0,0 \\ 0,0 \end{pmatrix}, \quad \vartheta \begin{pmatrix} 0,0 \\ 0,1 \end{pmatrix}, \quad \vartheta \begin{pmatrix} 1,1 \\ 0,0 \end{pmatrix}, \quad \vartheta \begin{pmatrix} 1,1 \\ 0,1 \end{pmatrix},$$

indices 0, 0, 1, 1.

The functions are thus in direct relation, and the product of the four functions is a function $\Pi \begin{pmatrix} 0 \\ 0,0 \end{pmatrix}(4, \text{def.})$.

But obviously any one of the functions taken four times, or any two of them taken each twice, are in like manner four functions in direct relation, or the fourth powers $\vartheta^4, \vartheta'^4, \vartheta''^4, \vartheta'''^4$, and the squared products $\vartheta^2 \vartheta'^2, \vartheta''^2 \vartheta'''^2, \vartheta^2 \vartheta''^2, \vartheta'^2 \vartheta'''^2, \vartheta'^2 \vartheta''^2, \vartheta''^2 \vartheta'''^2$, are in like manner each of them a function $\Pi \begin{pmatrix} 0 \\ 0,0 \end{pmatrix}(4, \text{def.})$, viz. we have thus in all $1 + 4 + 6 = 11$ such functions.

But the Π function contains only $\frac{1}{4}(4^2 + 4) = 10$ arbitrary constants; hence there must be a linear relation between the 11 powers and products, and this is Gópel's relation.

The four functions $\Pi_0, \Pi_1, \Pi_2, \Pi_3$.

Art. Nos. 16–21.]

16. Starting with any two characteristics **a**, **b** at pleasure, the remaining characteristics form seven pairs, such that **a** + **b** + **c** + **d** = 0; but among the seven pairs we have only three, suppose **c** + **d**, **e** + **f**, **g** + **h**, which are such that (**a**, **b**, **c**, **d**), (**a**, **b**, **e**, **f**), (**a**, **b**, **g**, **h**) are each of them either all even or else two even and two odd; that is, starting with any pair (**a**, **b**), we have these three tetrads having each of them the required property.

The number of pairs (**a**, **b**) is $16 \cdot 15 = 240$; and we thence derive $240 \times 3 = 720$ tetrads; but each such tetrad is of course derivable from any one of the six pairs contained in it; or the number of distinct

tetrads is $720/6 = 60$, viz. we have, as mentioned above, 60 Göpel-tetrads.

17. We consider four theta-functions $\vartheta, \vartheta_1, \vartheta_2, \vartheta_3$, which are such that to the modulus 2, the sum of the characters is $= 0$, and also the sum of the indices is $= 0$; taking the characters to be

$$\begin{pmatrix} \mu, \nu \\ q, r \end{pmatrix}, \quad \begin{pmatrix} \mu', \nu' \\ q', r' \end{pmatrix}, \quad \begin{pmatrix} \mu'', \nu'' \\ q'', r'' \end{pmatrix}, \quad \begin{pmatrix} \mu''', \nu''' \\ q''', r''' \end{pmatrix}$$

and writing throughout $=$ for $\equiv (\text{mod. } 2)$, we have

$$\begin{aligned} \mu + \mu' + \mu'' + \mu''' &= 0, & \nu + \nu' + \nu'' + \nu''' &= 0, \\ q + q' + q'' + q''' &= 0, & r + r' + r'' + r''' &= 0, \\ \mu q + \nu r + \mu' q' + \nu' r' + \mu'' q'' + \nu'' r'' + \mu''' q''' + \nu''' r''' &= 0. \end{aligned}$$

Writing for shortness

$$(01) = \mu q' + \mu' q + \nu r' + \nu' r,$$

and so in other cases; and further

$$(01) + (02) + (12) = (012), \quad \&c.,$$

then substituting for $\mu''', \nu''', q''', r'''$ their values from the first four equations, we deduce

$$(012) = 0; \quad \text{and similarly} \quad (013) = 0, \quad (023) = 0, \quad (123) = 0.$$

18. Consider now a product $\vartheta^a \vartheta_1^b \vartheta_2^c \vartheta_3^d$, where $a + b + c + d$ is a given odd number k ; the characteristic is

$$\begin{pmatrix} \mu a + \mu' b + \mu'' c + \mu''' d, & \nu a + \nu' b + \nu'' c + \nu''' d \\ q a + q' b + q'' c + q''' d, & r a + r' b + r'' c + r''' d \end{pmatrix},$$

and it hence follows that the index is $= a(\mu q + \nu r) + b(\mu' q' + \nu' r') + c(\mu'' q'' + \nu'' r'') + d(\mu''' q''' + \nu''' r''')$.

In fact, forming the index in question, we have first terms in a^2, b^2, c^2, d^2 , which upon writing therein a, b, c, d for these values respectively ($a^2 = a$, &c.) give the required value; we have therefore only to show that the sum of the remaining terms in ab , &c., is $= 0$.

These terms are

$$ab(01) + ac(02) + ad(03) + bc(12) + bd(13) + cd(23),$$

and writing herein $a+b+c+d=1$, and thence $ad = a(1-a-b-c) = ab+ac$, and similarly $bd = ab+bc$, $cd = ac+bc$, the terms in question become

$$= ab(013) + ac(023) + bc(123),$$

that is, they become $= 0$.

19. We thus see that the function $\vartheta^a \vartheta_1^b \vartheta_2^c \vartheta_3^d$ has an index which is $= a \text{ ind } \vartheta + b \text{ ind } \vartheta_1 + c \text{ ind } \vartheta_2 + d \text{ ind } \vartheta_3$.

Consider separately four products $\vartheta^a \vartheta_1^b \vartheta_2^c \vartheta_3^d$, in which the exponents a, b, c, d satisfy successively the relations (always to modulus 2)

$$\begin{array}{ll} b+d=0, & c+d=0, \\ b+d=1, & c+d=0, \\ b+d=0, & c+d=1, \\ b+d=1, & c+d=1. \end{array}$$

Combining herewith the relation $a+b+c+d=1$, it follows that the exponents a, b, c, d are

$$\begin{array}{ll} = d+1, d, d, d; & = d, d+1, d, d; \\ = d, d, d+1, d; & = d, d, d, d+1, \end{array}$$

in the four cases respectively.

Then substituting these values, the characteristics become

$$\begin{pmatrix} \mu \\ q, r \end{pmatrix}, \quad \begin{pmatrix} \mu' \\ q', r' \end{pmatrix}, \quad \begin{pmatrix} \mu'' \\ q'', r'' \end{pmatrix}, \quad \begin{pmatrix} \mu''' \\ q''', r''' \end{pmatrix}$$

viz. the four products have the same characters as $\vartheta, \vartheta_1, \vartheta_2, \vartheta_3$ respectively; and in like manner recollecting that $\text{ind } \vartheta + \text{ind } \vartheta_1 + \text{ind } \vartheta_2 + \text{ind } \vartheta_3 = 0$, we see that the four products have the same indices as $\vartheta, \vartheta_1, \vartheta_2, \vartheta_3$ respectively.

More generally write

$$\Pi_0, \Pi_1, \Pi_2, \Pi_3 = \vartheta^a \vartheta_1^b \vartheta_2^c \vartheta_3^d,$$

where for the four cases respectively the exponents a, b, c, d satisfy the conditions already referred to; then $\Pi_0, \Pi_1, \Pi_2, \Pi_3$ have the same characteristics, and the same indices, as $\vartheta, \vartheta_1, \vartheta_2, \vartheta_3$ respectively.

20. It can be shown that each of the functions Π contains $\frac{1}{4}(k^2+1)$ constants.

It will be recollected that we have between $\vartheta, \vartheta_1, \vartheta_2, \vartheta_3$ an equation of the form

$$a\vartheta^4 + b\vartheta_1^4 + c\vartheta_2^4 + d\vartheta_3^4 + e\vartheta^2\vartheta_1^2 + f\vartheta^2\vartheta_2^2 + \dots = 0;$$

this serves to express, say ϑ_3^4 , in lower powers of ϑ_3 ; and by successive applications of this equation, we can reduce $\vartheta^a\vartheta_1^b\vartheta_2^c\vartheta_3^d$ to a form in which d has only one of the values 0, 1, 2 or 3; we do not by this transformation alter the suffix of Π , viz. a term originally of the form Π_0, Π_1, Π_2 or Π_3 , will by the transformation give rise only to terms which are of the same form Π_0, Π_1, Π_2 or Π_3 (as the case may be).

The number of constants in Π_0 is thus = number of partitions of k into four parts a, b, c, d , under the conditions

$$\begin{array}{ll} d = 0 \text{ or } 2; & a \text{ odd, } b, c \text{ each even,} \\ d = 1 \text{ or } 3; & a \text{ even, } b, c \text{ each odd,} \end{array}$$

where, in reckoning the partitions, the order of the parts is taken into account: the partitions are thus as follows

$$\begin{array}{ll} d = 0, & (a - 1) + b + c = k - 1, \\ d = 1, & a + (b - 1) + (c - 1) = k - 3, \\ d = 2, & (a - 1) + b + c = k - 3, \\ d = 3, & a + (b - 1) + (c - 1) = k - 5, \end{array}$$

where the parts a or $(a - 1)$, b or $(b - 1)$, c or $(c - 1)$, as the case may be, are all of them even; hence, writing $k' = \frac{1}{2}(k - 1)$, the cases are

$$a' + b' + c' = k', \quad k' - 1, \quad k' - 1, \quad \text{or } k' - 2,$$

where the a', b', c' are odd or even (zero not excluded) at pleasure; as already mentioned, the order of the parts is taken into account: thus the particulars of 3 would be

$$\begin{array}{ccccccc} 300, & 210, & 120, & 030, \\ 201, & 111, & 021, & 102, & 012, & 003. \end{array}$$

No. is $10 = 4 \cdot 5$.

Hence, in the four cases respectively, the numbers are

$$\frac{1}{2}(k'^2 + k' + 2), \quad \frac{1}{2}(k'^2 + k'), \quad \frac{1}{2}(k'^2 + k'), \quad \frac{1}{2}(k'^2 - k'),$$

giving a total

$$= \frac{1}{2}(4k'^2 + 4k' + 2) = \frac{1}{2}\{(2k' + 1)^2 + 1\},$$

that is, $= \frac{1}{2}(k^2 + 1)$.

And similarly the number is $= \frac{1}{2}(k^2 + 1)$ in the other three cases respectively.

21. It can be shown that, conversely, from any function $\Pi(x, y \mid K, \text{def.})$ having the characteristic and index of ϑ , we can obtain the several functions $\Pi_0, \Pi_1, \Pi_2, \Pi_3$ as linear functions of the products $\vartheta^a \vartheta_1^b \vartheta_2^c \vartheta_3^d$; or, what is the same thing, the function $\Pi(x, y \mid K, \text{def.})$ is expressible as a linear function of the products in question.

Preparation for the Transformation. Art. Nos. 22 to 44
(several sub-headings).

The Hermitian quartic matrix.

Art. Nos. 22–25.]

22. Observing that, for the adopted form of the Π - or ϑ -functions, the periods ω , ν are

$$\begin{array}{cccc} 1, & 0, & \omega_1, & \nu_1 \\ 0, & 1, & \omega_2, & \nu_2 \\ A, & H, & \omega_3, & \nu_3 \end{array}$$

so that we have

$$\omega_1\nu_2 - \omega_2\nu_1 = A, \quad \omega_1\nu_3 - \omega_3\nu_1 = H, \quad \omega_2\nu_3 - \omega_3\nu_2 = B,$$

we have to consider the automorphic transformation of the bilinear form $\omega_1\nu_2 - \omega_2\nu_1 + \omega_1\nu_3 - \omega_3\nu_1$.

23. We write

$$(\omega_0, \omega_1, \omega_2, \omega_3 \mid \nu_0, \nu_1, \nu_2, \nu_3) = (a_0, a_1, a_2, a_3 \mid b_0, b_1, b_2, b_3)(\tau_0, \tau_1, \tau_2, \tau_3),$$

and the coefficients are assumed to be such that we have identically

$$\omega_0\nu_1 - \omega_1\nu_0 + \omega_2\nu_3 - \omega_3\nu_2 = k(\tau_0\tau_1 - \tau_1\tau_0 + \tau_2\tau_3 - \tau_3\tau_2),$$

where k is in the sequel taken to be a positive integer.

We obtain by direct substitution the value of $\omega_0\nu_1 - \omega_1\nu_0 + \omega_2\nu_3 - \omega_3\nu_2$ in the following form:

$$\begin{aligned} & (a_2c_0 - c_2a_0 + b_2d_0 - d_2b_0)(\tau_0\tau_1 - \tau_1\tau_0) \\ & + (a_3c_0 - c_3a_0 + b_3d_0 - d_3b_0)(\tau_0\tau_2 - \tau_2\tau_0) \\ & + (a_0c_1 - c_0a_1 + b_0d_1 - d_0b_1)(\tau_0\tau_3 - \tau_3\tau_0) \\ & + (a_3c_1 - c_3a_1 + b_3d_1 - d_3b_1)(\tau_1\tau_2 - \tau_2\tau_1) \\ & + (a_1c_2 - c_1a_2 + b_1d_2 - d_1b_2)(\tau_1\tau_3 - \tau_3\tau_1) \\ & + (a_0c_3 - c_0a_3 + b_0d_3 - d_0b_3)(\tau_2\tau_3 - \tau_3\tau_2); \end{aligned}$$

viz. equating this to its value $k(\tau_0\tau_1 - \tau_1\tau_0 + \tau_2\tau_3 - \tau_3\tau_2)$, we have 4 identities and 12 equations which are, in fact, 6 equations occurring each twice.

We have thus six equations, which are the conditions in order that the matrix may be the matrix of automorphic transformation of the bilinear form $\omega_0\nu_1 - \omega_1\nu_0 + \omega_2\nu_3 - \omega_3\nu_2$.

The six equations may be written

$$\begin{aligned}(ac + bd)_{01} &= 0, & (ac + bd)_{02} &= k, & (ac + bd)_{03} &= 0, \\ (ac + bd)_{12} &= 0, & (ac + bd)_{13} &= k, & (ac + bd)_{23} &= 0,\end{aligned}$$

viz. the first of these equations is $a_0c_1 - a_1c_0 + b_0d_1 - b_1d_0 = 0$, and so in other cases.

It is convenient to remark that each interchange of two letters a and c , b and d , also interchange of the suffixes 0 and 2, produces a change of sign; thus the second equation may be written $(ca + db)_{02} = k$, or $(ca + db)_{20} = k$.

24. The inverse matrix is found to be

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & b_3 \\ c_0 & c_1 & c_2 & c_3 \\ d_0 & d_1 & d_2 & d_3 \end{pmatrix}^{-1} = \frac{1}{k} \begin{pmatrix} c_1 & d_1 & -a_1 & -b_1 \\ -c_0 & -d_0 & a_0 & b_0 \\ c_3 & d_3 & -a_3 & -b_3 \\ -c_2 & -d_2 & a_2 & b_2 \end{pmatrix},$$

and the determinant of each of the matrices in this formula is $= k^2$.

We have thus

$$(\nu_0, \nu_1, \nu_2, \nu_3) = \frac{1}{k} \begin{pmatrix} c_1 & d_1 & -a_1 & -b_1 \\ -c_0 & -d_0 & a_0 & b_0 \\ c_3 & d_3 & -a_3 & -b_3 \\ -c_2 & -d_2 & a_2 & b_2 \end{pmatrix} (\omega_0, \omega_1, \omega_2, \omega_3),$$

and the like formula for the τ, v .

Substituting these values in the equation

$$k(\tau_2v_2 - \tau_2v + \tau_1v_2 - \tau_3v_1) = \omega_0\nu_1 - \omega_1\nu_0 + \omega_2\nu_3 - \omega_3\nu_2,$$

we obtain 6 new equations, which are in a different form the conditions for the automorphic transformation.

The 6 new equations may be written

$$\begin{aligned}(ac + bd)_{01} &= 0, & (ac + bd)_{02} &= k, & (ac + bd)_{03} &= 0, \\ (ac + bd)_{12} &= 0, & (ac + bd)_{13} &= k, & (ac + bd)_{23} &= 0,\end{aligned}$$

viz. these equations are $(ac + bd)_{01} = 0$, $(ac + bd)_{02} = k$, $(ac + bd)_{03} = 0$, $(ac + bd)_{12} = 0$, $(ac + bd)_{13} = k$, $(ac + bd)_{23} = 0$, or say

$$a_0c_1 - a_1c_0 + b_0d_1 - b_1d_0 = 0, \quad \&c.$$

25. It is worth while to show how the foregoing formula for the inverse matrix comes out.

Take for instance the diagonal minor

$$\begin{vmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = b_1(c_2d_3 - c_3d_2) + b_2(c_3d_1 - c_1d_3) + b_3(c_1d_2 - c_2d_1),$$

which is $= c_1(ac)_{23} + c_2(ac)_{31} + c_3(ac)_{12}$, since the remaining terms destroy each other. And dividing by the determinant, which is $= k^2$, we have the term c_2/k of the inverse matrix.

The Symmetrical Hermitian Matrix.

Art. Nos. 26–28.]

26. We may consider a symmetrical Hermitian matrix, say the matrix

$$\begin{pmatrix} \mathfrak{A}, & \mathfrak{B}, & \mathfrak{C}, & \mathfrak{D} \\ \mathfrak{B}, & \mathfrak{E}, & \mathfrak{F}, & \mathfrak{G} \\ \mathfrak{C}, & \mathfrak{F}, & \mathfrak{H}, & \mathfrak{I} \\ \mathfrak{D}, & \mathfrak{G}, & \mathfrak{I}, & \mathfrak{J} \end{pmatrix},$$

viz. we have

$$\begin{aligned} 2\mathfrak{B} - \mathfrak{C} + \mathfrak{H} &= 0, & 2\mathfrak{C} - \mathfrak{F} + \mathfrak{I} &= 0, \\ 2\mathfrak{D} - \mathfrak{G} + \mathfrak{J} &= 0, & 2\mathfrak{F} - \mathfrak{G} + \mathfrak{H} &= 0, \\ 2\mathfrak{C} - \mathfrak{D} + \mathfrak{I} &= 0, & 2\mathfrak{B} - \mathfrak{C} + \mathfrak{F} &= 0. \end{aligned}$$

The characteristic property is that, effecting a Hermitian transformation, we have a new symmetrical Hermitian matrix.

In fact, the new matrix is

$$\begin{pmatrix} a_0 & b_0 & c_0 & d_0 \\ a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{pmatrix} \begin{pmatrix} \mathfrak{A}, & \mathfrak{B}, & \mathfrak{C}, & \mathfrak{D} \\ \mathfrak{B}, & \mathfrak{E}, & \mathfrak{F}, & \mathfrak{G} \\ \mathfrak{C}, & \mathfrak{F}, & \mathfrak{H}, & \mathfrak{I} \\ \mathfrak{D}, & \mathfrak{G}, & \mathfrak{I}, & \mathfrak{J} \end{pmatrix} \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & b_3 \\ c_0 & c_1 & c_2 & c_3 \\ d_0 & d_1 & d_2 & d_3 \end{pmatrix},$$

where the first factor, qua transposed matrix, is Hermitian. Hence the product is a symmetrical $\frac{1}{k}$ -matrix.

27. Write

$$\Delta = \alpha\beta - \eta^2 = (\alpha, \eta, \beta \mid x, y)^2,$$

and consider the following matrix

$$\begin{pmatrix} \eta, & \alpha, & \eta, & \alpha \\ \beta, & \eta, & \beta, & \eta \\ \eta, & \alpha, & \eta, & \alpha \\ \beta, & \eta, & \beta, & \eta \end{pmatrix} = (p, q, r, s \mid x, y, z, w),$$

where

$$\begin{aligned} p &= \eta x + \alpha y + \eta z + \alpha w, \\ q &= \beta x + \eta y + \beta z + \eta w, \\ r &= \eta x + \alpha y + \eta z + \alpha w, \\ s &= \beta x + \eta y + \beta z + \eta w. \end{aligned}$$

It is easily shown that this is a symmetrical Δ -matrix. In fact, representing it for a moment by

$$\begin{pmatrix} \mathfrak{A}, & \mathfrak{B}, & \mathfrak{C}, & \mathfrak{D} \\ \mathfrak{B}, & \mathfrak{E}, & \mathfrak{F}, & \mathfrak{G} \\ \mathfrak{C}, & \mathfrak{F}, & \mathfrak{H}, & \mathfrak{I} \\ \mathfrak{D}, & \mathfrak{G}, & \mathfrak{I}, & \mathfrak{J} \end{pmatrix},$$

so that $\mathfrak{A} = \eta$, $\mathfrak{E} = \eta$, $\mathfrak{H} = \eta$, &c., we have

$$2\mathfrak{B} - \mathfrak{C} + \mathfrak{H} = (\Delta - \beta\mathfrak{A}_0) - (-\eta^2 - \eta\mathfrak{A}_0) - \eta^2\mathfrak{A} - \beta\Delta + \eta\mathfrak{B}_0 + \eta^2\mathfrak{A}_0,$$

which is $= \alpha\beta\Delta - \beta\delta\mathfrak{A}_0 + \eta^2\mathfrak{A}_0$, that is, $= (\alpha\beta - \eta^2)\Delta = \Delta^2$.

And, similarly, $\mathfrak{A}\mathfrak{H} - \mathfrak{C}^2 + \mathfrak{E}\mathfrak{I} - \mathfrak{F}^2$ is found to be $= \Delta^2$; and the remaining four combinations of terms to be each of them $= 0$; the matrix is thus a symmetrical Δ -matrix.

28. It is to be added that the diagonal minors and the determinant

$$\begin{aligned} &\mathfrak{A}, \quad \mathfrak{A}\mathfrak{E} - \mathfrak{B}^2, \\ &\mathfrak{A}\mathfrak{E}\mathfrak{H} + \text{\&c.}, \quad \mathfrak{A}\mathfrak{E}\mathfrak{H}\mathfrak{I} + \text{\&c.}, \end{aligned}$$

have respectively the values $\Delta, \Delta^2, \Delta^3, \Delta^4$; viz. if α, β, δ are positive, then these are all positive; or the last-mentioned quadric function is a definite positive form.

The general matrix resumed: Arithmetical theory.

Art. Nos. 29–36.]

29. The matrix containing, as before, the parameter k , may be called a k -matrix; if k be $= 1$, it is a unit-matrix.

From the fundamental equation

$$k(\tau_0 v_1 - \tau_1 v_0 + \tau_2 v_3 - \tau_3 v_2) = \omega_0 \nu_1 - \omega_1 \nu_0 + \omega_2 \nu_3 - \omega_3 \nu_2,$$

it at once follows that, compounding a k -matrix with a k' -matrix, we have a kk' -matrix; and in particular, compounding a k -matrix with a unit-matrix, we have a k -matrix.

30. The symbols a_0, a_1, b_0 , &c., and k , have thus far been arbitrary magnitudes, but we now take them to be integers; and we consider in particular the case where k is a positive odd prime.

The number of k -matrices is of course infinite, but if we regard as equivalent any two such matrices which are derivable one from the other by post-multiplication by a unit-matrix (viz. U being a unit-matrix, the matrices M and $M \cdot U$ are regarded as equivalent), then the number of distinct k -matrices is finite, and $= 1 + k + k^2 + k^3$.

The first step is to show that we can by post-multiplication, by a properly determined unit-matrix, reduce the k -matrix to the form

$$\begin{pmatrix} a_0, & a_1, & a_2, & a_3 \\ b_0, & b_1, & b_2, & b_3 \\ c_0, & 0, & c_2, & 0 \\ d_0, & 0, & 0, & d_3 \end{pmatrix},$$

these values being such as to satisfy identically two out of the six conditions; the remaining conditions present themselves under the two equivalent forms

$$\begin{aligned} a_0 c_2 &= k, & b_1 d_3 &= k, \\ a_0 b_2 + a_1 c_2 - a_2 c_1 - a_3 c_0 &= 0, & a_0 d_1 + a_1 d_0 - a_2 d_3 - a_3 d_2 &= 0, \end{aligned}$$

and

$$\begin{aligned} a_0 c_2 &= k, & b_1 d_3 &= k, \\ a_0 c_2 + b_1 c_2 - b_2 c_1 - b_3 c_0 &= 0, & a_3 c_2 + b_2 d_3 - b_3 d_2 - b_0 d_1 &= 0. \end{aligned}$$

Hence $a_0, c_2 = 1, k$, or $k, 1$; and $b_1, d_3 = 1, k$, or $k, 1$; so that, combining these pairs of values, we have four different types of matrix,

each type depending on the coefficients $a_1, d_0, a_2, b_2, a_3, b_3$, connected together by two equations.

But the forms of the same type are not distinct from each other, and we have to determine for each of the four types a system of non-equivalent forms comprised therein, and such that from these, by post-multiplication by a unit-matrix as before, the other forms of the type can be obtained.

This final system is:

Type I.

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & k, & 0 \\ 0, & 0, & 0, & k \end{pmatrix}$$

Type II.

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & k, & 0, & 0 \\ 0, & i, & 1, & 0 \\ 0, & 0, & 0, & k \end{pmatrix}$$

Type III.

$$\begin{pmatrix} k, & 0, & 0, & 0 \\ i, & 1, & 0, & 0 \\ i', & 0, & 1, & 0 \\ 0, & 0, & 0, & k \end{pmatrix}$$

Type IV.

$$\begin{pmatrix} k, & 0, & 0, & 0 \\ i, & 1, & 0, & 0 \\ i', & 0, & 1, & 0 \\ i'', & 0, & 0, & 1 \end{pmatrix}$$

where i, i', i'' are integers, having each of them any one of the values $0, 1, 2, \dots, k-1$, viz. there are in the four types $1, k, k^2, k^3$ forms respectively, and the number of forms is thus $= 1 + k + k^2 + k^3$, as already mentioned.

I abstain from the further details of the proof.

There is obviously a like theory for which, in place of post-multiplication, we have pre-multiplication; viz. here the matrices M and $U \cdot M$ are regarded as equivalent.

31. Any two k -matrices are reducible one to the other by a combined pre- and post-multiplication; viz. we have always $M = U \cdot M' \cdot U'$,

where M, M' are any two k -matrices, and U, U' properly determined unit-matrices; and in particular, M being any given k -matrix, this is expressible in the foregoing form, where M' denotes the principal matrix

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & k, & 0 \\ 0, & 0, & 0, & k \end{pmatrix}.$$

Congruence theorems, k an odd number.

32. Taking k an odd number, and using throughout $=$ instead of $\equiv$ (mod. 2), we have the following congruences:

$$\begin{aligned} \mu &\equiv a(a_2c_2 + a_3c_3) + b(b_2d_2 + b_3d_3), \\ \nu &\equiv a(a_2c_0 + a_3c_1) + b(b_2d_0 + b_3d_1), \\ q &\equiv c(a_0c_2 + a_1c_3) + d(b_0d_2 + b_1d_3), \\ r &\equiv c(a_0c_0 + a_1c_1) + d(b_0d_0 + b_1d_1), \end{aligned}$$

or, conversely,

$$\begin{aligned} a &\equiv c(c_2\mu + c_3\nu) + d(d_2q + d_3r), \\ b &\equiv c(c_0\mu + c_1\nu) + d(d_0q + d_1r), \\ c &\equiv a(a_2\mu + a_3q) + b(b_2\nu + b_3r), \\ d &\equiv a(a_0\mu + a_1q) + b(b_0\nu + b_1r), \end{aligned}$$

where observe that on the right-hand side the signs may be changed into $+$; in fact, to the modulus 2, we have for any integer value whatever $p \equiv +p$.

33. The first congruence is

$$\mu \equiv a(a_2c_2 + a_3c_3) + b(b_2d_2 + b_3d_3), \quad \text{say } X = Y,$$

and to verify this, I see no other method than that of considering separately all the combinations of even and odd values of a, a_1, a_2, a_3 ; viz. we have the 16 cases

Case	a	a_1	a_2	a_3	b	b_1	b_2	b_3	Y
1	e	e	e	e	e	e	e	e	0
2	e	e	e	e	o	o	o	o	0
3	o	o	o	o	e	e	e	e	$c_2 + c_3 + d_2 + d_3$
4	o	o	o	o	o	o	o	o	$c_2 + c_3 + d_2 + d_3 + b_1d_1 + b_3d_3$
5	e	e	o	o	e	e	e	e	0
6	e	e	o	o	o	o	o	o	0
7	o	o	e	e	e	e	e	e	$c_0 + c_1 + d_0 + d_1$
8	o	o	e	e	o	o	o	o	$c_0 + c_1 + d_0 + d_1 + b_1d_1 + b_3d_3$
9	e	o	e	o	e	o	e	o	0
10	e	o	e	o	o	e	o	e	0
11	o	e	o	e	e	o	e	o	$c_1 + c_3 + d_1 + d_3$
12	o	e	o	e	o	e	o	e	$c_1 + c_3 + d_1 + d_3 + b_0d_0 + b_2d_2$
13	e	o	o	e	e	o	o	e	0
14	e	o	o	e	o	e	e	o	0
15	o	e	e	o	e	o	o	e	$c_0 + c_2 + d_0 + d_2$
16	o	e	e	o	o	e	e	o	$c_0 + c_2 + d_0 + d_2 + b_0d_0 + b_2d_2$

34. Case 2. We have $Y = b_1d_1$ which should be $= 0$, and in fact, the six equations give $b_1 = 0$, $d_1 = 0$, whence $Y = 0$.

Case 4. We have $Y = d_0 + c_0 + b_1d_1 + b_3d_3$ which should $= 1$, and in fact, the six equations give $c_3 - c_1 = 1$, $b_3 - b_1 = 0$, $d_3 - d_1 = 0$; whence $d_0 + c_0 = 1$, $b_1d_1 + b_3d_3 = 0$, and therefore $Y = 1$.

Case 6. Here $Y = b_0d_0 + b_1d_1$ which should $= 0$; and in fact, the six equations give $b_0 + b_1 = 0$, $d_0 + d_1 = 0$, whence $Y = 0$.

Case 16. Here $Y = c_0 + d_0 + c_2 + c_3 + b_0d_0 + b_1d_1 + b_2d_2 + b_3d_3$, which should be $= 0$; the six equations give

$$-c_0 - d_0 + c_2 + c_3 = 1, \quad -b_0 - b_1 + b_2 + b_3 = 0, \quad -d_0 - d_1 + d_2 + d_3 = 0,$$

$$\text{and } b_2c_0 - b_0c_2 + b_3c_1 - b_1c_3 = 1.$$

Writing $b_0 + b_2 = b_1 + b_3$ and $d_0 + d_2 = d_1 + d_3$, we find

$$b_0d_0 + b_2c_0 + b_2d_2 + b_3d_3 = b_1d_1 + b_3d_3 + b_2d_0 + b_0d_2,$$

that is,

$$b_0d_0 + b_1d_1 + b_2d_2 + b_3d_3 = b_2d_0 + b_0d_2 + b_1d_1 + b_3d_3 = 1;$$

and $c_0 + c_1 + c_2 + c_3 = 1$, whence $Y = 0$.

35. Case 8. This is the only case of any difficulty: we have

$$Y = c_0 + c_1 + d_0 + d_1 + b_1d_1 + b_3d_3,$$

which should be $= 1$.

The six equations give

$$\begin{aligned} c_0 &= -1 - c_1 + c_3, & b_0 &= -b_1 + b_3, \\ d_0 &= -d_1 + d_3, & b_2 &= b_2, & c_2 &= c_2, & d_2 &= d_2, \end{aligned}$$

or, say

$$c_0 = -1 - c_1 + c_3, \quad b_0 = -b_1 + b_3, \quad d_0 = -d_1 + d_3,$$

or, substituting these values and omitting even terms

$$Y = c_1 + c_3 - b_3d_3 - b_1d_1.$$

The remaining three of the six equations are

$$b_2c_0 - b_0c_2 + b_3c_1 - b_1c_3 = 1, \quad b_2d_0 - b_0d_2 + b_3d_1 - b_1d_3 = 0, \quad c_2d_0 - c_0d_2 + c_3d_1 - c_1d_3 = 0$$

or, substituting for b_0, c_0, d_0 their values, these become

$$\begin{aligned} (1 + c_1 - c_3)b_2 + (b_3 - b_1)c_2 &= -b_3 + b_2c_1, \\ (d_1 - d_3)b_2 + (b_3 - b_1)d_2 &= 1 - b_3 + b_1d_1, \\ (d_1 - d_3)c_2 + (-1 - c_1 + c_3)d_2 &= -d_3 + c_1d_1; \end{aligned}$$

we can from these equations eliminate b_2, c_2, d_2 ; viz. from the first and third equations eliminating c_2 , we have

$$(d_3 - d_1)\{b_2(1 + c_1 - c_3) + c_2(b_3 - b_1)\} = (d_3 - d_1)(-b_3 + b_2c_1) + (b_3 - b_1)(-d_3 + c_1d_1),$$

and this, by means of the second equation, becomes

$$(1 + d_1 - c_3)(-1 + b_3 - b_1) = (d_3 - d_1)(-b_3 + b_3c_1) + (b_3 - b_1)(-c_1d_1 + c_1d_1),$$

viz. reducing, this is

$$-d_1 + c_3 - 1 + b_3 - b_3d_1 = 0;$$

and in virtue of it we have $Y = c_1 + c_3 - b_3d_3 - b_1d_1 = 1$.

It can be further shown that, to the modulus 2 (k being, as before, odd), we have

$$\begin{aligned}(a_0a_2 + a_1a_3)(c_0c_2 + c_1c_3) + (b_0b_2 + b_1b_3)(d_0d_2 + d_1d_3) &= 0, \\ (a_0c_0 + b_0d_0)(a_2c_2 + b_2d_2) + (a_1c_1 + b_1d_1)(a_3c_3 + b_3d_3) &= 0.\end{aligned}$$

To prove the first equation, write for a moment

$$X = (a_0c_1 - a_1c_0)(a_2c_3 - a_3c_2), \quad Y = (b_0d_1 - b_1d_0)(b_2d_3 - b_3d_2),$$

then in virtue of the equations we have $X = X'$. But we have identically

$$XY - X'Y' = (a_0c_2 + a_1c_3)(a_2c_0 + a_3c_1) + (b_0d_2 + b_1d_3)(b_2d_0 + b_3d_1),$$

and from the equation $a_0c_2 - a_2c_0 = 1 + a_1c_3 - a_3c_1$, we have $XY = 0$; and similarly $X'Y' = 0$, that is, $XY = X'Y' = 0$; we have thus the required equation $XY + X'Y' = 0$.

In a similar manner the second equation may be verified.

36. Write

$$\begin{aligned}p &= \mu a_0 + \nu a_1 + qa_2 + ra_3 + aa_2 + ba_3, \\ \nu' &= \mu b_0 + \nu b_1 + qb_2 + rb_3 + bb_2 + bb_3, \\ q' &= \mu c_0 + \nu c_1 + qc_2 + rc_3 + cc_2 + cc_3, \\ r' &= \mu d_0 + \nu d_1 + qd_2 + rd_3 + dd_2 + dd_3.\end{aligned}$$

It is to be shown that to the modulus 2 we have

$$\mu q' + \nu r' = \mu q + \nu r.$$

In fact, forming the value of $\mu q' + \nu r'$, we have first a constant term (term without μ, ν, q, r) which vanishes; next writing $\mu^2 = \mu$, the whole term in μ is

$$c_0(a_0a_2 + a_1a_3) + d_0(b_0b_2 + b_1b_3) + c_2(a_0a_2 + a_1a_3) + d_2(b_0b_2 + b_1b_3),$$

which also vanishes; and similarly the terms in ν, q, r each of them vanish; there remain only the terms in $\mu\nu, \mu q, \&c$.

The coefficient of μq is $a_0c_2 + a_1c_0 + b_0d_2 + b_1d_0$, which (always to the modulus 2) is $= a_0c_2 - a_2c_0 + b_0d_2 - b_2d_0$, that is, it is $= 1$; and similarly the coefficient of νr is $= 1$; and in like manner the coefficients of the other terms are each $= 0$; we have thus the required congruence

$$\mu q' + \nu r' = \mu q + \nu r.$$

The quintic matrix.

Art. Nos. 37–43.]

37. I consider a quintic matrix composed of the foregoing coefficients $(a, b, c, d)_0^{1,2,3}$, viz. the matrix contained in a linear transformation which I write as follows:

$$(T, P, Q, R, S) = \begin{pmatrix} (ab)_{01}, & (ab)_{21}, & 2(ab)_{31}, & (ab)_{13}, & (ab)_{32} \\ (ac)_{01}, & (ac)_{21}, & (ac)_{31} + k, & (ac)_{13}, & (ac)_{32} \\ (ad)_{01}, & (ad)_{21}, & 2(ad)_{31}, & (ad)_{13}, & (ad)_{32} \\ (cb)_{01}, & (cb)_{21}, & 2(cb)_{31}, & (cb)_{13}, & (cb)_{32} \\ dc_{01}, & dc_{21}, & 2dc_{31}, & dc_{13}, & dc_{32} \end{pmatrix} (T, P, Q, R, S),$$

where as before $(ab)_{ij} = a_i b_j - a_j b_i$, &c.; and so in other cases; in particular, observe that, in the expression of Q , the term involving Q is $\{k + 2(ac)_{31}\}Q$ which, in virtue of the relation $(ac + bd)_{31} = k$, may also be written $\{(ac)_{31} - (bd)_{31}\}Q$.

I notice that in Hermite's paper, p. 366, the term is in effect written without the k , $= 2(ac)_{31}Q$: the correction of this erratum and of a corresponding one, p. 366, is made p. 787 at the conclusion of the memoir.

38. The matrix is automorphic for the form $Q^2 - PR - TS$, viz. we have identically

$$Q'^2 - P'R' - T'S' = k^2(Q^2 - PR - TS).$$

As a partial verification, consider in $Q'^2 - P'R' - T'S'$ the term containing Q^2 .

The coefficient of Q^2 is

$$\{-k + 2(ac)_{02}\}^2 - 4(ab)_{02}(ad)_{02} - 4(ab)_{01}(dc)_{02},$$

where the first term is

$$k^2 - 4k(ac)_{02} + 4(ac)_{02}^2.$$

Hence, observing that we have

$$(ac)_{02}^2 - (ab)_{02}(ad)_{02} - (ab)_{01}(dc)_{02} = k(ac)_{02},$$

the whole coefficient is $= k^2$, as it should be; and in like manner the verification may be effected for any other term.

39. We require the following formulae:

$$\begin{aligned}(-c_0, a_0, b_0)(T, P, Q) &= k(b_1, -b_0, -b_3)(T, P, Q, R, S), \\(-c_1, a_1, b_1)(T, P, Q) &= k(., b_0, -b_2)(T, P, Q, R, S), \\(-c_2, a_2, b_2)(T, P, Q) &= k(b_3, ., b_0)(T, P, Q, R, S), \\(-c_3, a_3, b_3)(T, P, Q) &= k(-b_2, b_1, .)(T, P, Q, R, S),\end{aligned}$$

that is,

$$-c_0T + a_0P + b_0Q = k(b_1P - b_0Q - b_3S), \quad \&c.,$$

and

$$\begin{aligned}(-d_0, a_0, b_0)(T, Q, R) &= k(., -a_0, ., a_3)(T, P, Q, R, S), \\(-d_1, a_1, b_1)(T, Q, R) &= k(., a_1, ., -a_2)(T, P, Q, R, S), \\(-d_2, a_2, b_2)(T, Q, R) &= k(., -a_3, ., a_0)(T, P, Q, R, S), \\(-d_3, a_3, b_3)(T, Q, R) &= k(., a_2, ., -a_1)(T, P, Q, R, S).\end{aligned}$$

From the first set, multiplying the first, third and fourth equations by T , P , Q respectively and adding, and again multiplying the second, third and fourth equations by T , Q , R , and adding, we obtain

$$\begin{aligned}-(c_0T + c_2P + c_3Q)T + (a_0T + a_2P + a_3Q)P + (b_0T + b_2P + b_3Q)Q &= kb_3(Q^2 - P \\-(c_1T + c_2Q + c_3R)T + (a_1T + a_2Q + a_3R)Q + (b_1T + b_2Q + b_3R)R &= -kb_1(Q^2 -\end{aligned}$$

and similarly, from the second set of equations, we obtain

$$\begin{aligned}-(d_0T + d_2P + d_3Q)T + (a_0T + a_2P + a_3Q)Q + (b_0T + b_2P + b_3Q)R &= -ka_3(Q^2 - \\-(d_1T + d_2Q + d_3R)T + (a_1T + a_2Q + a_3R)Q + (b_1T + b_2Q + b_3R)R &= ka_2(Q^2 -\end{aligned}$$

40. We have the inverse system

$$(T, P, Q, R, S) = \frac{1}{k^2} \begin{pmatrix} (dc)_{01}, & (ad)_{31}, & 2(bd)_{31}, & (cb)_{01}, & (ab)_{01} \\ (dc)_{21}, & (ad)_{13}, & 2(bd)_{13}, & (cb)_{21}, & (ab)_{21} \\ (dc)_{31}, & (ad)_{31} - k, & 2(bd)_{31}, & (cb)_{31}, & (ab)_{31} \\ (dc)_{13}, & (ad)_{32}, & 2(bd)_{32}, & (cb)_{13}, & (ab)_{13} \\ (dc)_{32}, & (ad)_{02}, & 2(bd)_{02}, & (cb)_{32}, & (ab)_{32} \end{pmatrix} (T', P', Q', R', S')$$

read

$$T' = (dc)_{01}T + (ad)_{31}P + 2(bd)_{31}Q + (cb)_{01}R + (ab)_{01}S;$$

and in particular observe that, in the expression for Q' , the term containing Q is $\{-k + 2(bd)_{13}\}Q$, where, in virtue of $(ac + bd)_{13} = k$, the coefficient of Q is also $-(ac)_{13}$.

41. These equations give

$$\begin{aligned}(a_0, a_2, a_3)(T', P', Q') &= k(d_3, ., -a_3, -b_3, b_0, b_2, b_3, ., ., c_3, a_3, b_3, ., d_3, c_3, .)(T, P, Q, R, \\(a_1, a_2, a_3)(T', Q', R') &= k(-d_2, ., a_2, b_2, ., c_2, a_2, b_2, ., d_2, c_2, .)(T, P, Q, R, S).\end{aligned}$$

From the first set of equations, multiplying the first, third and fourth equations by T, P, Q respectively, and adding, and again multiplying the second, third and fourth equations by T, Q, R respectively, and adding, we deduce

$$\begin{aligned}T(-a_0S - c_0R + d_0Q) + P(-a_2S - c_2R + d_2Q) + Q(-a_3S - c_3R + d_3Q) &= -kc_3(Q^2 - R^2) \\T(-b_0S + c_0Q - d_0P) + P(-b_2S + c_2Q - d_2P) + Q(-b_3S + c_3Q - d_3P) &= kd_3(Q^2 - R^2)\end{aligned}$$

and in like manner, from the second set of equations,

$$\begin{aligned}T(-a_0S - c_0R + d_0Q) + Q(-a_2S - c_2R + d_2Q) + R(-a_3S - c_3R + d_3Q) &= -kd_3(Q^2 - R^2) \\T(-b_0S + c_0Q - d_0P) + Q(-b_2S + c_2Q - d_2P) + R(-b_3S + c_3Q - d_3P) &= kc_2(Q^2 - R^2)\end{aligned}$$

42. Assume that T, P, Q, R, S and T', P', Q', R', S' are linearly connected as above; and write

$$\begin{aligned}T : P : Q : R : S &= 1 : A : H : B : H^2 - AB, \\T' : P' : Q' : R' : S' &= 1 : A' : H' : B' : H'^2 - A'B',\end{aligned}$$

equations which establish a like relation between $1, A, H, B, H^2 - AB$, and $1, A', H', B', H'^2 - A'B'$.

Observe that these forms are admissible since, if

$$T^2 - PR - QS = 0,$$

then also

$$Q'^2 - P'R' - T'S' = 0.$$

The quantities A, H, B were taken as the parameters of a theta-function; viz. taking $A, H, B = A + i\alpha, H + i\eta, B + i\beta$, then $(\alpha, \eta, \beta \mid x, y)^2$ must be a positive form (or what is the same thing, α and $\alpha\beta - \eta^2$ must be positive). If A', H', B' are also the parameters of a

theta-function, then writing $A', H', B' = A' + i\alpha', H' + i\eta', B' + i\beta', (\alpha', \eta', \beta' \mid x, y)^2$ must also be a positive form.

It can be shown that, A', H', B' being determined as above, the former condition implies the latter one; viz. if the form $(\alpha, \eta, \beta \mid x, y)^2$ be positive, then also the form $(\alpha', \eta', \beta' \mid x, y)^2$ will be positive.

43. Write

$$\begin{aligned} A &= A + i\alpha, & A' &= A' + i\alpha', \\ B &= B + i\beta, & B' &= B' + i\beta', \\ H &= H + i\eta, & H' &= H' + i\eta', \end{aligned}$$

and

$$\Delta = H^2 - AB = (H'^2 - \alpha\beta) - (\eta'^2 - \alpha'\beta'),$$

and let the quintic matrix in the foregoing transformation

$$(T', P', Q', R', S') = M(T, P, Q, R, S)$$

be represented by

$$\begin{pmatrix} t, & p, & q, & r, & s \\ t_1, & p_1, & q_1, & r_1, & s_1 \\ t_2, & p_2, & q_2, & r_2, & s_2 \\ t_3, & p_3, & q_3, & r_3, & s_3 \\ t_4, & p_4, & q_4, & r_4, & s_4 \end{pmatrix}.$$

44. It is to be shown that $\alpha'x^2 + 2\eta'xy + \beta'y^2$ is definite and positive.

We require α', η', β' ; we have

$$\begin{aligned} A' &= \frac{A't_1 + H't_2 + B't_3}{t}, & H' &= \frac{A'p_1 + H'p_2 + B'p_3}{p}, \\ B' &= \frac{A'r_1 + H'r_2 + B'r_3}{r}, & \Delta' &= \frac{A's_1 + H's_2 + B's_3}{s}, \end{aligned}$$

and thence

$$\alpha' = \frac{\partial A'}{\partial i}, \quad \eta' = \frac{\partial H'}{\partial i}, \quad \beta' = \frac{\partial B'}{\partial i}.$$

Now

$$\alpha' = \frac{A't_1 + H't_2 + B't_3}{t} - \frac{(A't_1 + H't_2 + B't_3)^2}{t^2}\alpha,$$

and so for η', β' with only the change of t_1, t_2, t_3 into p_1, p_2, p_3 and r_1, r_2, r_3 and s_1, s_2, s_3 respectively; we find, omitting a factor Δ throughout,

$$\alpha' = \eta^2 - \alpha\beta, \quad \eta' = \eta\beta - \alpha\eta, \quad \beta' = \beta^2 - \eta^2.$$

But we have identically

$$(\eta^2 - \alpha\beta)x^2 + 2(\eta\beta - \alpha\eta)xy + (\beta^2 - \eta^2)y^2 = (\alpha x^2 + 2\eta xy + \beta y^2)(\eta x + \beta y)(\eta x - \alpha y),$$

or, what is the same thing,

$$\alpha' = \alpha\Delta, \quad \eta' = \eta\Delta, \quad \beta' = \beta\Delta.$$

Hence $\alpha'x^2 + 2\eta'xy + \beta'y^2 = \Delta(\alpha x^2 + 2\eta xy + \beta y^2)$, which is positive if Δ is positive, that is, if $\alpha\beta - \eta^2$ is positive; and this is the required condition.

The Transformation. Art. Nos. 45 to 60.

Art. Nos. 45–48.]

45. I consider the transformation of the theta-functions with the parameters (A, H, B) into theta-functions with the parameters (A', H', B') . Writing

$$\vartheta_{\mu, \nu}(x, y \mid A, H, B) = \sum \sum (-1)^{\mu q + \nu r} \exp[i\pi\{(2m + \mu)x + (2n + \nu)y + (A, H, B \mid m, n)\}]$$

and similarly

$$\vartheta_{\mu',\nu'}(x', y' \mid A', H', B') = \sum \sum (-1)^{\mu'q + \nu'r} \exp[i\pi\{(2m' + \mu')x' + (2n' + \nu')y' + (A', H$$

the formula of transformation is

$$\vartheta_{\mu',\nu'}(x', y') = C \cdot \vartheta_{\mu,\nu}(x, y),$$

where C is a constant depending on the characteristics and the matrix of transformation, and where the new arguments x', y' are linear functions of the original arguments x, y .

46. The relation between the arguments is

$$x' = \frac{(ab)_{01}x + (ab)_{21}y}{k}, \quad y' = \frac{(ac)_{01}x + (ac)_{21}y}{k},$$

or, as we may write it,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \frac{1}{k} \begin{pmatrix} (ab)_{01} & (ab)_{21} \\ (ac)_{01} & (ac)_{21} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

47. The value of the constant C may be written

$$C = \frac{1}{\sqrt{k}} \cdot \frac{1}{\sqrt[4]{\Delta}} \cdot (-1)^{\frac{1}{2}(\mu q + \nu r + \mu' q + \nu' r)} \cdot \exp \left\{ -\frac{i\pi}{k} \{ (ab)_{01}x^2 + 2(ac)_{01}xy + (ad)_{01}y^2 \} \right\},$$

where $\Delta = (ac)_{01}(bd)_{01} - (ad)_{01}(bc)_{01}$.

48. The foregoing formula may be verified by means of the equations which define the theta-functions; viz. substituting for $\vartheta_{\mu',\nu'}(x', y')$ its value in terms of $\vartheta_{\mu,\nu}(x, y)$, the equations

$$\begin{aligned} \vartheta_{\mu',\nu'}(x' + 1, y') &= (-1)^{\mu'q} \vartheta_{\mu',\nu'}(x', y'), \\ \vartheta_{\mu',\nu'}(x', y' + 1) &= (-1)^{\nu'r} \vartheta_{\mu',\nu'}(x', y'), \\ \vartheta_{\mu',\nu'}(x' + A', y' + H') &= (-1)^q \exp[-2i\pi(2x' + A')] \vartheta_{\mu',\nu'}(x', y'), \\ \vartheta_{\mu',\nu'}(x' + H', y' + B') &= (-1)^r \exp[-2i\pi(2y' + B')] \vartheta_{\mu',\nu'}(x', y'), \end{aligned}$$

should be respectively equivalent to the corresponding equations for $\vartheta_{\mu,\nu}(x, y)$.

Conclusion.

The foregoing investigation completes the theory of the linear transformation of the double-theta functions. The results obtained are in substantial agreement with those of Hermite, but the notation has been modified and the demonstration rendered more systematic. The theory of the quintic matrix and its automorphic property in regard to the form $Q^2 - PR - TS$ presents features of considerable interest; and the arithmetical theory of the k -matrices, with the enumeration $1 + k + k^2 + k^3$ of the distinct forms, appears to be new.

[850] On Linear Differential Equations (continued from p. 397)

[FROM THE QUARTERLY JOURNAL OF PURE AND APPLIED MATHEMATICS, VOL. XXI. (1886), PP. 321–331.]

(Continued from p. 397.)

first equation then gives $p = Q$, A arbitrary, or say $= 1$: the other coefficients $B, C, \dots$ are then found to be

$$\frac{1}{p+1}, \quad \frac{1}{p+2}, \quad \dots,$$

or substituting for p its value $= 0$, the series is

$$1 + 1 \cdot x + 1 \cdot 2 \cdot x^2 + 1 \cdot 2 \cdot 3 \cdot x^3 + \dots,$$

which is divergent for any value whatever, however small, of x ; or the integral is as mentioned an illusory one.

14. The last-mentioned differential equation is reducible, viz. we have

$$P = \left(\frac{d}{dx} - \frac{3}{x} \right) \left(\frac{d}{dx} - \frac{1}{x} \right),$$

say $P = QD$, where of course the order of the factors Q, D is material.

It is a general theorem that the indicial, belonging to the product P , is of a degree equal to the sum of the degrees of the indicials, belonging to the factors Q, D respectively; and, in fact, the indicial of P is (as was seen) of the degree 1; and the indicials of Q, D are of the degrees 1, 0 respectively. But, moreover, if the indicial of P is of a degree γ , less than m the order of the equation (in the present case $m = 2, \gamma = 1$), then, in order that the equation may have γ regular integrals, it is a necessary and sufficient condition that P shall be decomposable into a product QD , where the factors Q, D (being functions of the same form as P) are of the orders γ and $m - \gamma$ respectively, and where, moreover, the second factor D shall have an indicial of the degree $m - \gamma$, and consequently the first factor Q an indicial of the degree zero. In the present case, it is the second factor which has an indicial of degree zero; and thus the condition is not satisfied. And accordingly the equation ought not to have $(m - \gamma)1$ regular integral; and we have seen that there is in fact no regular

integral. The investigation serves to show in what sense it is that there is no regular integral, or generally in what sense it is that the number of regular integrals may be less than $m - \gamma$.

15. The theorem may, it appears to me, be stated in the more general form: the necessary and sufficient condition that the differential equation $P(y) = 0$ of the order m , but having an indicial of the degree $m - \gamma$, shall have $m - \gamma - \delta$ regular integrals is that the function P shall be a product of the form $P = QMD$, where the orders of Q , M , D are $\gamma + \delta$, 0 , $m - \gamma - \delta$ respectively, and the degrees of their indicials δ , 0 , $m - \gamma - \delta$ respectively; or, to denote this in a compendious manner, say

$$P_{m-\gamma-\delta} \stackrel{m-\gamma+\delta-?}{=} Q_{m-\gamma-\delta} \cdot M^\delta \cdot D^{m-\gamma-\delta},$$

where δ is an integer which is not $= 0$, and which is at most $= m - \gamma$, for otherwise the function Q of the order $\gamma + \delta$ could not have an indicial of the degree δ . If there is no regular integral, then $\delta = m - \gamma$, D is a mere constant or say unity, and the formula is

$$P_{m-\gamma} \stackrel{m-\gamma-?}{=} Q_{m-\gamma} \cdot M^\delta,$$

where δ is not $= 0$, and is at most $= m - \gamma$.

16. For differential equations of the form $P(y) = 0$ above considered, Floquet gave a theorem, which as afterwards remarked by him is not generally true, viz. this was $P = ABC \dots$, a product of linear factors of the form $\frac{d}{dx} + \frac{i}{x-a_0}$ (i an integer). The question of decomposition is considered in my next following paper "On Linear Differential Equations (the Theory of Decomposition)," [851], and by means of the formulae there given (and by the process indicated ante No. 7), it could be decided whether any particular differential equation admits of a decomposition $P = ABC \dots$, where the linear factors are functions not of the form just referred to, but of the form $\frac{d}{dx} + \sum C a^i x^i$, in which the series contains only a finite number of negative powers, say this is a decomposition into linear regular factors.

[851] On Linear Differential Equations (The Theory of Decomposition)

[FROM THE QUARTERLY JOURNAL OF PURE AND APPLIED MATHEMATICS, VOL. XXI. (1886), PP. 331–335.]

1. In the theory of linear differential equations the question arises, to decompose

$$P = \left(\frac{d}{dx}\right)^m + p_1 \left(\frac{d}{dx}\right)^{m-1} + \cdots + p_m$$

into linear factors: we have, for instance, a differential equation

$$\frac{d^2y}{dx^2} + p\frac{dy}{dx} + qy = 0,$$

which is to be expressed in the form

$$\left(\frac{d}{dx} - a\right) \left(\frac{d}{dx} - b\right) y = 0,$$

or say we have to express $\frac{d^2}{dx^2} + p\frac{d}{dx} + q$ as a product of linear factors $\frac{d}{dx} - a$, $\frac{d}{dx} - b$. The problem is analogous to, but wholly distinct from, that of the resolution of an algebraic equation: using accents to denote differentiation in regard to x , the relation (in the simple case just referred to) between the coefficients (p, q) and the roots (a, b) is (not $p = a + b$, $q = ab$, but in place thereof) $p = a + b$, $q = ab + b'$; and it thus appears that there is the important distinction that, in the present problem, the order of the factors is not indifferent.

2. The problem may be solved when the general solution of the differential equation is known, or, what is the same thing, by means of n particular solutions of the equation; it is not here considered in this point of view, but the intention is to treat it directly by means of the relations between the coefficients and the roots.

3. I consider a differential equation of the order n , having all its integrals regular; say the equation is $P(y) = 0$, where

$$P = \left(\frac{d}{dx}\right)^n + p_1 \left(\frac{d}{dx}\right)^{n-1} + \cdots + p_n,$$

and the coefficients $p_1, p_2, \dots, p_n$ are functions of x .

The theorem is that P can be expressed as a product

$$P = AB \dots L,$$

of n linear factors $A, B, \dots, L$, each of them of the form $\frac{d}{dx} - \rho$, where ρ is a function of x ; and that, moreover, these factors may be so arranged that the indicial of each of the partial products $A, AB, ABC, \dots$ shall be of the same degree as that of P itself; or, what is the same thing, of the degree n . In particular, the factor A may be so determined that its indicial is of the degree n .

4. In the case where the roots of the indicial equation are all unequal, the expression $P = AB \dots L$ may be obtained without difficulty: the coefficients of the several linear factors are given as rational functions of the roots of the indicial equation and their differences. But when there are equal roots, or roots differing by integers, the formulae become illusory, and a special process is required.

5. I consider in particular the case where the equation has n regular integrals, belonging to the n roots of the indicial equation respectively; say these are

$$y_1 = x^{a_1} \phi_1(x), \quad y_2 = x^{a_2} \phi_2(x), \quad \dots, \quad y_n = x^{a_n} \phi_n(x),$$

where $\phi_1, \phi_2, \dots, \phi_n$ are regular functions of x (not vanishing when $x = 0$).

6. In the case of unequal roots, the first factor A is $\frac{d}{dx} - \rho_1$, where $\rho_1 = \frac{y'_1}{y_1}$, and so for the other factors. But in the case of equal roots, or roots differing by integers, the formulae require modification. Suppose, for instance, that we have two equal roots $a_1 = a_2$; the integrals may be taken to be y_1 and $y_2 = y_1 \log x + x^{a_1} \psi(x)$, and the first factor is $\frac{d}{dx} - \frac{y'_1}{y_1}$ as before; but the second factor requires special consideration.

7. I do not enter into the details of the process for obtaining the factors in the case of equal roots, but content myself with indicating the general theory. The equation $P = AB \dots L$ gives for the determination of the coefficients of the factors a system of equations which can be solved step by step; and the process is analogous to that of the

resolution of an algebraic equation, but with the important difference already referred to, that the order of the factors is not indifferent.

8. The question of decomposition into linear regular factors is considered in the paper [851], and by means of the formulae there given it could be decided whether any particular differential equation admits of a decomposition $P = ABC \dots$, where the linear factors are functions of the form $\frac{d}{dx} + \sum C a^i x^i$, in which the series contains only a finite number of negative powers.

[852] Note sur un Mémoire de M. Picard

[FROM THE COMPTES RENDUS DE L'ACADÉMIE DES SCIENCES DE PARIS, T. CII. (1886), PP. 1174–1177.]

NOTE SUR LE MÉMOIRE DE M. PICARD “SUR LES INTÉGRALES DE DIFFÉRENTIELLES TOTALES ALGÈBRIQUES DE PREMIÈRE ESPÈCE.”

M. Picard considère une équation différentielle totale

$$dz = P dx + Q dy,$$

où P et Q sont des fonctions rationnelles de x, y, z . Il suppose que l'équation $dz = P dx + Q dy$ admette des intégrales de première espèce, et il démontre que, si l'équation est réductible, alors les intégrales peuvent être exprimées au moyen d'intégrales de différentielles totales de formes plus simples.

Je me permets d'observer que le théorème de M. Picard peut être regardé comme un cas particulier d'un théorème plus général relatif aux équations différentielles totales de tous ordres. En effet, si l'on considère une équation différentielle totale de l'ordre n

$$d^n z = P dx^n + \dots,$$

et si cette équation admet des intégrales de première espèce, alors, sous certaines conditions de réductibilité, ces intégrales peuvent être exprimées au moyen d'intégrales d'équations de formes plus simples.

Done, en réunissant les deux parties, on a = 51 + 33, les x' &c.

The note concludes by observing that the theory of algebraic total differentials of the first kind is closely connected with the theory of algebraic functions and their integrals, and that the results obtained by M. Picard furnish an important contribution to this branch of analysis.

[853] Note on a Formula for $\Delta^n 0^i / n^i$ when n, i are very large

[FROM THE MESSENGER OF MATHEMATICS, VOL. XV. (1886), PP. 124–127.]

1. The formula in question is one which I obtained many years ago, but which does not seem to be generally known. Writing Δ for the operation of taking the difference, so that $\Delta u_x = u_{x+1} - u_x$, &c., we have

$$\Delta^n 0^i = n^i - \binom{n}{1}(n-1)^i + \binom{n}{2}(n-2)^i - \dots,$$

and the formula gives the value of this expression when n and i are both very large.

2. Writing for shortness

$$\frac{\Delta^n 0^i}{n^i} = \Theta,$$

the formula is

$$\Theta = e^{-\frac{1}{2}q} \left\{ 1 - \frac{q}{8n}(q-2) + \frac{q^2}{384n^2}(5q^2 - 44q + 44) + \dots \right\},$$

where $q = \frac{i}{n}$.

3. To prove this, we start from the expression

$$\Delta^n 0^i = \Delta^n x^i \quad (x=0),$$

and observing that $\Delta = e^{\frac{d}{dx}} - 1$, we have

$$\Delta^n x^i = (e^{\frac{d}{dx}} - 1)^n x^i.$$

Writing $e^{\frac{d}{dx}} - 1 = e^{\frac{d}{dx}}(1 - e^{-\frac{d}{dx}})$, and putting $D = \frac{d}{dx}$, this becomes

$$\Delta^n x^i = e^{nD}(1 - e^{-D})^n x^i.$$

4. Expanding $(1 - e^{-D})^n$, we have

$$(1 - e^{-D})^n = \left(D - \frac{D^2}{2} + \frac{D^3}{6} - \cdots \right)^n,$$

and thence, putting $x = 0$, the terms which survive are those in which the exponent of D is equal to i (for $D^i x^i = i!$, and $D^j x^i = 0$ for $j > i$ when $x = 0$).

5. The formula may be written

$$\Theta = e^{-\frac{1}{2}q} \left\{ 1 + \frac{A}{n} + \frac{B}{n^2} + \cdots \right\},$$

where $A, B, \dots$ are polynomials in q , which can be determined by the process of expansion.

6. To verify the formula, I calculated numerically the values of Θ for $n = 10$ and various values of q , comparing them with the values obtained by the formula. The results were as follows:

q	True value	Formula
1	0.348678...	0.349...
2	0.107374...	0.107...
3	0.028247...	0.028...

The agreement is sufficiently close, considering that n is only 10.

[854] An Algebraical Transformation

[FROM THE MESSENGER OF MATHEMATICS, VOL. XV. (1886),
PP. 127–129.]

1. The transformation in question is one which presents itself in the theory of elliptic functions, but which is interesting as a mere algebraical identity. Writing for shortness

$$f(x, a) = \frac{x^2 - 2ax + 1}{x^2 - 1},$$

I say that we have identically

$$f(x, a) \cdot f(y, b) = f(X, A) \cdot f(Y, B),$$

where X, Y, A, B are certain functions of x, y, a, b .

2. In fact, assuming

$$X = \frac{xy + 1}{x + y}, \quad Y = \frac{xy - 1}{x - y},$$

we find without difficulty

$$A = \frac{ab + 1}{a + b}, \quad B = \frac{ab - 1}{a - b},$$

and the identity is verified.

3. The transformation may be presented under a somewhat different form. Writing

$$u = \frac{x^2 - 2ax + 1}{x^2 - 1},$$

we have identically

$$\nu(x, a) \cdot \nu(y, b) = \nu(X, A) \cdot \nu(Y, B).$$

4. This transformation is useful in the theory of elliptic functions, in particular in the multiplication and transformation of the functions. The case where $a = b = 0$ is especially remarkable, giving

$$\frac{(x^2 + 1)(y^2 + 1)}{(x^2 - 1)(y^2 - 1)} = \frac{(X^2 + 1)(Y^2 + 1)}{(X^2 - 1)(Y^2 - 1)},$$

which is a known formula.

[855] Solution of $(a, b, c, d) = (a', b', c', d')^2$

[FROM THE MESSENGER OF MATHEMATICS, VOL. XV. (1886), PP. [NOT SPECIFIED].]

1. The problem is, given the cubic $(a, b, c, d : x, y)^3$, to find the cubic $(a', b', c', d' : x, y)^3$, such that the former cubic is the square of the latter; in other words, to solve the equation

$$(a, b, c, d : x, y)^3 = \{(a', b', c', d' : x, y)^3\}^2.$$

2. The equation implies that the cubic (a, b, c, d) is the square of a linear factor; say

$$(a, b, c, d : x, y)^3 = (px + qy)^6,$$

and then a', b', c', d' are determined by the condition that (a', b', c', d') is a perfect cube; or, what is the same thing, that its Hessian vanishes identically.

3. The conditions for this may be obtained as follows. We must have

$$(a', b', c', d' : x, y)^3 = (\alpha x + \beta y)^3,$$

and then comparing coefficients,

$$a' = \alpha^3, \quad b' = \alpha^2\beta, \quad c' = \alpha\beta^2, \quad d' = \beta^3.$$

4. Hence the ratios $a' : b' : c' : d'$ are as $\alpha^3 : \alpha^2\beta : \alpha\beta^2 : \beta^3$; and eliminating $\alpha : \beta$, we have the conditions

$$b'^2 = a'c', \quad c'^2 = b'd', \quad b'c' = a'd'.$$

These are equivalent to two independent conditions; the cubic has therefore $4 - 2 = 2$ degrees of freedom, as it should have, being a linear form raised to the third power.

5. If $p^2 - 1 = 0$, then either $p = 1$, giving $q = 1$, $r = 1$, and hence the equation is satisfied identically; or else $p = -1$, giving a different solution.

[856] Note on a Cubic Equation

[FROM THE MESSENGER OF MATHEMATICS, VOL. XV. (1886),
PP. 62–64.]

CONSIDER the cubic equation

$$ax^3 + 3bx^2 + 3cx + d = 0.$$

Writing this in the form

$$(a, b, c, d : x, 1)^3 = 0,$$

and forming the Hessian, we have the quadratic covariant

$$(ac - b^2, \frac{1}{2}(ad - bc), bd - c^2 : x, 1)^2 = 0.$$

2. I consider the case where the cubic has equal roots. The discriminant is then $= 0$; and this implies that the Hessian is a perfect square, or, what is the same thing, that $(ac - b^2)(bd - c^2) - \frac{1}{4}(ad - bc)^2 = 0$.

3. The cubic may be expressed in terms of its roots α, β, γ in the form

$$a(x - \alpha)(x - \beta)(x - \gamma) = 0,$$

and we have

$$\alpha + \beta + \gamma = -\frac{3b}{a}, \quad \beta\gamma + \gamma\alpha + \alpha\beta = \frac{3c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}.$$

4. The remaining symmetric functions of α, β, γ are of course expressible in terms of the coefficients: we have

$$\alpha^2 + \beta^2 + \gamma^2 = \frac{9b^2 - 6ac}{a^2},$$

and so for the higher powers.

5. The case of a double root is particularly interesting. If $\beta = \gamma$, then the cubic becomes

$$a(x - \alpha)(x - \beta)^2 = 0,$$

and the condition for this is that the discriminant vanishes, or, what is the same thing, that the cubic and its first derivative have a common root. The first derivative is

$$3ax^2 + 6bx + 3c = 0,$$

and eliminating x between this and the cubic, we obtain the discriminant

$$\Delta = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2 = 0.$$

6. The values of α and β may be found from the equations

$$2\beta + \alpha = -\frac{3b}{a}, \quad \beta^2 + 2\alpha\beta = \frac{3c}{a},$$

whence, eliminating α , we have a quadratic for β .

[857] Analytical Geometrical Note on the Conic

[FROM THE MESSENGER OF MATHEMATICS, VOL. XV. (1886),
P. 192.]

The equation of a conic may be written in the form

$$y^2 + z^2 + x^2 - 2yz - 2zx - 2xy = 0,$$

which represents a parabola touching the lines $x = 0$, $y = 0$, $z = 0$.

In a different notation, the general equation of the second degree

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

may be transformed by linear substitution into the form

$$X^2 + Y^2 + Z^2 = 0,$$

which represents an imaginary conic; or, with real coefficients, into one of the standard forms according to the nature of the conic.

The condition that the general equation represents a pair of lines is that its discriminant vanishes, viz.

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0.$$

The note contains also some remarks on the foci and directrices of conics referred to general coordinates, and on the conditions for a parabola and a rectangular hyperbola.

[858] Comparison of the Weierstrassian and Jacobian Elliptic Functions

[FROM THE MESSENGER OF MATHEMATICS, VOL. XVI. (1887), PP. 129–140.]

1. The Weierstrassian function $\wp(u)$ is defined as

$$\wp(u) = \frac{1}{u^2} + \sum' \left\{ \frac{1}{(u - 2m\omega - 2m'\omega')^2} - \frac{1}{(2m\omega + 2m'\omega')^2} \right\},$$

or, what is the same thing, we have

$$\wp(u) = \frac{1}{u^2} + \frac{g_2}{20}u^2 + \frac{g_3}{28}u^4 + \cdots,$$

where g_2 and g_3 are the invariants.

2. The Jacobian elliptic functions are $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$, with the modulus k . They satisfy the relations

$$\operatorname{sn}^2 u + \operatorname{cn}^2 u = 1, \quad k^2 \operatorname{sn}^2 u + \operatorname{dn}^2 u = 1,$$

and their addition-theorems are well known.

3. The connection between the two systems of functions is established by means of the formulae

$$\wp(u) - e_1 = \frac{\operatorname{cn}^2 v}{\operatorname{sn}^2 v}, \quad \wp(u) - e_2 = \frac{\operatorname{dn}^2 v}{k^2 \operatorname{sn}^2 v}, \quad \wp(u) - e_3 = \frac{1}{\operatorname{sn}^2 v},$$

where $v = u\sqrt{e_1 - e_3}$, and the modulus k is given by

$$k^2 = \frac{e_2 - e_3}{e_1 - e_3}.$$

4. Conversely, we have

$$\operatorname{sn} v = \frac{\sqrt{e_1 - e_3}}{\sqrt{\wp(u) - e_3}},$$

and similar formulae for $\operatorname{cn} v$ and $\operatorname{dn} v$.

5. The question presents itself to compare the two systems of functions directly, by expressing the Jacobian functions in terms of the Weierstrassian functions, and vice versa. This comparison is useful in many investigations, particularly in the theory of transformation and in the application of elliptic functions to geometry and mechanics.

6. The Weierstrassian function $\wp(u)$ has the periods $2\omega, 2\omega'$; the Jacobian functions have the periods $4K, 2iK'$. The relation between these periods is

$$\omega = \frac{K}{\sqrt{e_1 - e_3}}, \quad \omega' = \frac{iK'}{\sqrt{e_1 - e_3}}.$$

7. The comparison may also be made by means of the sigma-functions. We have

$$\wp(u) - \wp(v) = -\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2 u \sigma^2 v},$$

and the sigma-function is connected with the theta-functions by the formula

$$\sigma(u) = 2\omega \frac{\vartheta_1(v)}{\vartheta_1'(0)} \exp\left(\frac{\eta u^2}{2\omega}\right),$$

where $v = \frac{u}{2\omega}$.

8. The Jacobian functions may be expressed in terms of the theta-functions by the well-known formulae

$$\sqrt{k} \operatorname{sn} u = \frac{\vartheta_1(v)}{\vartheta(v)}, \quad \sqrt{\frac{k}{k'}} \operatorname{cn} u = \frac{\vartheta_2(v)}{\vartheta(v)}, \quad \frac{1}{\sqrt{k'}} \operatorname{dn} u = \frac{\vartheta_3(v)}{\vartheta(v)},$$

where $v = \frac{u}{2K}$.

9. The comparison of the two systems may be carried out in detail by means of these formulae, and the results are found to agree with those obtained by direct transformation.

[859] On the Complex of Lines which meet a Unicursal Quartic Curve

[FROM THE PROCEEDINGS OF THE LONDON MATHEMATICAL SOCIETY, VOL. XVII. (1886), PP. 109–116.]

1. A unicursal quartic curve is one which can be represented parametrically by means of rational functions of a single parameter. If we take the six coordinates of a line to be (a, b, c, f, g, h) , where a, b, c, f, g, h are connected by the relation $af + bg + ch = 0$, then the condition that the line shall meet the curve is an equation of the fourth degree in these coordinates.

2. In fact, if for the six coordinates we substitute their expressions in terms of the parameters of the points where the line meets the curve, we obtain an equation

$$\theta = (x, y, z, w)^2(\alpha, \beta, \gamma, \delta) = 0,$$

which, regarded as an equation in (x, y, z, w) , is of the second degree; and the condition that this equation shall have equal roots gives the equation of the complex.

3. The expression for θ thus becomes a function of the line-coordinates; and the complex surface (the locus of lines meeting the curve) is of the order n in the coordinates, where n depends on the degree of the curve. For a unicursal quartic, the complex is of order 8.

4. If we write

$$\Pi = \sum p \cdot x(ys' - y's),$$

then the equation of the complex may be written in the form $\Pi = 0$, where the summation extends over all the chords of the curve.

5. The complex of lines which meet a given curve is a particular case of the general complex of lines; and the theory of such complexes is closely connected with the theory of congruences and the geometry of ruled surfaces.

6. The present investigation relates chiefly to the case where the quartic curve is unicursal, that is, where it can be represented rationally in terms of a parameter. In this case the coordinates of a point on the curve may be expressed as quartic functions of a parameter t , and the chords of the curve correspond to pairs of values of t .

7. The equation of the complex may be obtained by eliminating the parameter between the equations of the chord and the parametric equations of the curve; and the result is an equation of degree 8 in the line-coordinates.

[860] On Briot and Bouquet's Theory of the Differential Equation $F(u, \frac{du}{dz}) = 0$

[FROM THE PROCEEDINGS OF THE LONDON MATHEMATICAL SOCIETY, VOL. XVIII. (1887), PP. 71–78.]

1. The differential equation

$$F\left(u, \frac{du}{dz}\right) = 0,$$

where F is a rational function of u and $\frac{du}{dz}$, has been treated by Briot and Bouquet in their memoir on the integration of differential equations by means of elliptic functions. The theory is as follows.

2. Writing $\frac{du}{dz} = v$, the equation is $F(u, v) = 0$. This represents an algebraic curve; and if the genus of this curve is $p = 1$, then the integral of the differential equation is expressible by means of elliptic functions.

3. In particular, if the curve is of degree n , and has a multiple point of order $n - 2$, then the genus is $p = 1$; and the coordinates u, v can be expressed rationally in terms of a parameter t and its elliptic functions.

4. The case where the genus is $p = 0$ is also important; here the integral is expressible by means of rational functions and logarithms. The condition for this is that the curve $F(u, v) = 0$ shall have a sufficient number of multiple points to reduce the genus to zero.

5. Briot and Bouquet established the following fundamental theorem: If the differential equation $F(u, \frac{du}{dz}) = 0$ admits an algebraic integral, then the integral is either a rational function of z , or an elliptic function of z , or a function of the form

$$u = R(z, \sqrt{(z-a)(z-b)(z-c)(z-d)}),$$

where R is a rational function.

6. The theory applies in particular to the case where $F(u, v)$ is a polynomial of the second degree in v , say

$$v^2 = A(u - a)(u - b)(u - c)(u - d),$$

which leads to elliptic integrals; and to the case where $F(u, v)$ is of higher degree, which may lead to hyperelliptic or more general Abelian integrals.

7. A monotropic function of z is one which returns to the same value when z describes a closed curve; or, what is the same thing, there must not be any branch-points in the finite part of the plane. The condition that u shall be a monotropic function is that the integral $\int \frac{du}{v}$ shall have no periods which are functions of z .

8. I form a complete table of permissive points, as follows. The permissible points are those at which the function u can have singularities without ceasing to be monotropic. They are classified according to the nature of the singularity:

- (a) Poles of u ;
- (b) Branch-points where u takes a finite value;
- (c) Essential singularities where u becomes infinite in a determinate manner.

9. The enumeration seems long and complicated; but it is, in fact, only necessary to see that there are a finite number of forms, and to examine each form separately. The classification depends on the degree of the curve $F(u, v) = 0$ and on the nature of its singular points.

10. As to the other forms of permissive points, proofs depend on the forms already considered. Thus for the forms which involve logarithmic singularities, the proof is obtained by considering the limiting case of algebraic singularities.

11. The theory may be extended to differential equations of higher order, and to systems of simultaneous differential equations. The general principle is that the integrals are functions of z which can be expressed by means of a finite number of elementary operations, including algebraic operations, integration, and the solution of algebraic equations.

12. The loop; and, writing $y = 0$, we have $4u^2 - 27v^4 = 0$, or $u^3 = \frac{27}{4}v^4$. This gives a cuspidal point on the curve, and the corresponding integral is of the form $\int \frac{du}{u^{2/3}}$, which is algebraic.

13. The case where the curve $F(u, v) = 0$ has a cusp leads to integrals which are algebraic functions of z ; and similarly, the case where the curve has a tacnode leads to integrals which are logarithmic functions of z .

14. The enumeration seems long and complicated, but in fact, only to see that there are a finite number of forms, and to further examine the finite singular points (a, b) and the element.

[861] Note on a Formula relating to the Zero-Value of a Theta-Function

[FROM CRELLE'S JOURNAL DER MATHEMATIK, T. C. (1887), PP. 87-88.]

1. The theta-function $\vartheta(x)$ is defined by the equation

$$\vartheta(x) = \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} e^{2nix},$$

where $q = e^{i\pi\omega}$, and ω is a constant with positive imaginary part.

2. For the zero-value $x = 0$, we have

$$\vartheta(0) = \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} = 1 - 2q + 2q^4 - 2q^9 + \cdots,$$

which is the function usually denoted by $\vartheta_3(0)$ or $\vartheta(0, q)$.

3. The formula in question is

$$\vartheta(0) = \frac{(q; q)_{\infty}}{(q^2; q^2)_{\infty}} \cdot \sqrt[4]{\frac{2K}{\pi}},$$

and the formula thus is

$$\vartheta(0) = \sqrt[4]{\frac{2Kk'}{\pi}}.$$

4. But observe that, in the theta-function as defined by the equation

$$\vartheta(x, q) = 1 - 2q \cos 2x + 2q^4 \cos 4x - 2q^9 \cos 6x + \cdots,$$

the zero-value is obtained by putting $x = 0$, and the formula gives the value of the series $1 - 2q + 2q^4 - 2q^9 + \cdots$ in terms of the quarter-periods K , K' and the modulus k .

5. The formula may be written in the equivalent form

$$\vartheta_3(0) = \sqrt{\frac{2K}{\pi}} \cdot \sqrt[4]{k'},$$

and it is one of the fundamental formulae in the theory of elliptic functions, connecting the theta-functions with the complete elliptic integral.

[862] Note on the Theory of Linear Differential Equations

[FROM CRELLE'S JOURNAL DER MATHEMATIK, T. C. (1887), PP. 286–295.]

1. I consider a linear differential equation of the order m , with coefficients which are rational functions of the independent variable x . The equation may be written in the form

$$P(y) = \frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

where $p_1, p_2, \dots, p_m$ are rational functions of x .

2. It is known that the integrals of such an equation are of the form

$$y = x^a(A + Bx + Cx^2 + \cdots),$$

or, more generally, that they can be expressed as series of ascending or descending powers of x , possibly with logarithmic terms.

3. The theory of regular integrals is as follows. If $x = a$ is a singular point of the equation, then the integrals in the neighbourhood of $x = a$ are of the form

$$y = (x - a)^\lambda \phi(x),$$

where $\phi(x)$ is a regular function of $(x - a)$, that is, a function which can be expanded in a series of positive integer powers of $(x - a)$. The exponent λ is determined by the “indicial equation,” which is obtained by substituting $y = (x - a)^\lambda$ in the equation and equating to zero the coefficient of the lowest power of $(x - a)$.

4. The number of regular integrals belonging to a singular point $x = a$ is at most equal to the degree of the indicial equation; and it may be less than this degree, if the roots of the indicial equation are equal or differ by integers. In the latter case, logarithmic terms may appear in the integrals.

5. Considering as above the coefficients $p_1, p_2, \dots, p_m$ as rational and integral functions of t , it is easy to determine the degrees of these coefficients in such wise that the exponent α of the leading term At^α shall have a given value. Suppose first, this given value is α , a positive integer greater than 1: the degrees may be $\delta, \delta + \alpha, \delta + 2\alpha, \dots, \delta + m\alpha$, ($\delta = 0$ or any positive integer at pleasure); and not only so, but if we write

$$z = At^\alpha,$$

then writing $z = At^\alpha$, clearly the highest power in the equation is $t^{\delta+m\alpha}$, and equating the coefficient hereof to zero, we have

$$L[A] + L_1[A]^{m-1} + \dots + L_{m-1}[A] + L_m = 0,$$

an equation of the degree m for the determination of the coefficient of the leading term At^α . Assuming that the roots are all unequal, we have series each of the form

$$z = At^\alpha + Bt^{\alpha-1} + \dots + K + \frac{C}{t} + \frac{D}{t^2} + \dots.$$

6. Suppose $\alpha = 1$, that is, let the leading term be At ; here the degrees may be $\delta, \delta + 1, \delta + 2, \dots, \delta + m$, ($\delta = 0$ or any positive integer as before), but we have a different form for the A -equation. In fact, here ($z = At$) in the sets of terms ($z^m + \dots + z^{(m-1)}$), $\dots$, ($z^3 + 3zz' + z''$), ($z^2 + z'$), the terms in each brackets are of the same degree, viz. the degrees are $m, \dots, 3, 2$; and not only so, but we have

$$z^2 + z' = [A]_2 t^2, \quad z^3 + 3zz' + z'' = [A]_3 t^3, \quad \dots$$

Hence if

$$p = Lt^\delta + \dots, \quad p_1 = L_1 t^{\delta+1} + \dots, \quad \dots, \quad p_m = L_m t^{\delta+m} + \dots,$$

the A -equation is

$$L_{m-1}[A] + L_m = 0,$$

an equation of the degree m as before; assuming that the roots are unequal, we have here series each of the form

$$z = At + B + \frac{C}{t} + \frac{D}{t^2} + \dots,$$

being, as will presently appear, the case of regular integrals for the y -equation.

7. In either of the last-mentioned cases, the formula for the A -equation holds good for any of the two or more of the coefficients $p_1, \dots, p_m$ may have equal degrees, provided that the degree is the highest of all. If neither L nor L_m be $= 0$, the equation is still an equation of the order m , with effective roots; but if any of the coefficients L at the beginning of the equation vanish we thus introduce roots $= \infty$: and if any of the coefficients L at the end of the equation vanish, we thus introduce roots $= 0$: the number of effective roots is $= m$ minus the number of roots $= \infty$ or $= 0$; the case requires further consideration. If the equation has equal roots, then, from what precedes, it is easy to infer that we may have distinct series containing as in the general case only integer powers of t , or we may have series beginning with At^α or At , but containing also fractional powers of t .

8. It is perhaps worth remarking that, in the case where the series is

$$y = (x - a)^\lambda \{A + B(x - a) + C(x - a)^2 + \dots\},$$

the case is, in fact, that of a non-singular point $x = a$: and I do not here further consider it.

9. The general theory of regular integrals is connected with the theory of the indicial equation and its roots. The form of the integrals depends on the nature of the roots: if the roots are all distinct, the integrals are all of the simple power-series form; if some roots are equal, logarithmic factors appear; if roots differ by integers, the integrals may involve logarithms or may be of a more complicated form.

10. Considering the case where the series is

$$y = e^{\frac{A}{x-a}} (x - a)^\mu \{A + B(x - a) + \dots\},$$

this is what I call a subregular integral. By what precedes, it appears that, for the values

$$p = Lx^{-\delta} + \dots, \quad p_1 = L_1x^{-\delta-1} + \dots, \quad \dots, \quad p_m = L_mx^{-\delta-m} + \dots,$$

then the equation for the leading coefficient (previously represented by A) is

$$L[A] + L_1[A]^{-1} + \dots + L_{m-1}[A]^{-m+1} + L_m[A]^{-m} = 0,$$

but I do not see any direct or simple way of obtaining the corresponding equation for the exponent of this subregular integral.

11. Similarly, if the series be

$$y = e^{\frac{A}{(x-a)^\alpha}} \cdot (x-a)^\mu \{A + B(x-a) + \dots\},$$

(α being a positive integer = 2 at least), then writing as above $t = \frac{1}{x-a}$, this is

$$\frac{1}{y} \frac{dy}{dx} = -\frac{\alpha A}{t^{\alpha+1}} - \frac{\mu}{t} + B + Ct + \dots,$$

whence, if for shortness

$$w = -\frac{A}{t^\alpha} - \mu \log t + Bt + \frac{1}{2}Ct^2 + \dots,$$

we have

$$\log y = w + A \log x + Bx + \frac{1}{2}Cx^2 + \dots,$$

or passing to the value of y , but expanding the exponential of $Bx + \frac{1}{2}Cx^2 + \dots$, this is

$$y = e^w x^A (1 + Bx + Cx^2 + \dots),$$

a subregular integral. By what precedes, it appears that, if

$$p_1 = Lx^{-\delta} + \dots, \quad p_2 = L_1x^{-\delta-1} + \dots, \quad \dots, \quad p_m = L_mx^{-\delta-m} + \dots,$$

then the equation for the leading coefficient is as above.

12. To illustrate the case of a series with fractional powers, and also in order to work out a simple example with some completeness, I consider the differential equation of the second order $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$, giving the z -equation $z^2 + z + p_2 = 0$; p_2 is a rational and integral function of t ($t = \frac{1}{x-a}$), and we have two cases according as the order of the function is even or odd.

13. Suppose first that the order is even, and for convenience taking it to be = 4, and assuming the value of p_2 accordingly, I consider the equation

$$\frac{d^2y}{dx^2} = (\alpha + \beta t + \gamma t^2 + \delta t^3 + \epsilon t^4)y.$$

Here the form is at once seen to be

$$y = e^{\int z dt}, \quad z = At^2 + Bt + C + \frac{D}{t} + \frac{E}{t^2} + \cdots,$$

and observing that we have

$$\frac{dz}{dt} = 2At + B - \frac{D}{t^2} - \frac{2E}{t^3} + \cdots,$$

we find

$$A^2 = \alpha, \quad 2AB = \beta, \quad 2AC + B^2 - B = \gamma, \quad 2AD + 2BC = \delta,$$

$$2AE + 2BD + C^2 + C = \epsilon, \quad 2AF + 2BE + 2CD + D = 0,$$

equations which give

$$A = \sqrt{\alpha}, \quad B = \frac{\beta}{2\sqrt{\alpha}}, \quad C = \frac{\gamma}{2\sqrt{\alpha}} - \frac{\beta^2}{8\alpha\sqrt{\alpha}} + \frac{1}{4\sqrt{\alpha}},$$

and so on for the remaining coefficients.

14. It is to be observed that, although the formal series thus obtained may in general be divergent, yet they serve to determine the asymptotic behaviour of the integrals for large values of t , that is, for values of x in the neighbourhood of the singular point $x = a$.

15. Reverting to the case where the series is

$$y = (x - a)^\lambda \{A + B(x - a) + C(x - a)^2 + \cdots\},$$

and substituting for t its value $\frac{1}{x-a}$, or for greater convenience $s = x - a$, this is

$$\frac{1}{y} \frac{dy}{dx} = \frac{\lambda}{s} + B + 2Cs + 3Ds^2 + \cdots,$$

whence

$$\log y = \lambda \log s + Bs + \frac{1}{2}Cs^2 + \cdots,$$

or passing to the value of y , but expanding the exponential of $Bs + \frac{1}{2}Cs^2 + \cdots$, this is

$$y = s^\lambda (1 + Bs + Cs^2 + \cdots),$$

viz. we have here a regular integral. By what precedes, it appears that, for the values

$$p = Lx^{-\delta} + \dots, \quad p_1 = L_1x^{-\delta-1} + \dots, \quad \dots, \quad p_m = L_mx^{-\delta-m} + \dots,$$

the exponent λ is determined by the equation

$$L[\lambda] + L_1[\lambda]^{m-1} + \dots + L_{m-1}[\lambda] + L_m = 0,$$

which is the equation for that purpose considered by Fuchs, and is what I would call the indicial equation.

16. The logarithmic forms which belong to the case of equal roots may be deduced from the case of unequal roots, by supposing two or more of the roots to approach each other continuously and become ultimately equal.

17. Similarly, if the series be

$$y = e^{\frac{A}{(x-a)^\alpha}} \cdot (x-a)^\mu \{A + B(x-a) + \dots\},$$

(α being a positive integer = 2 at least), then writing as above $t = \frac{1}{x-a}$, this is

$$\frac{1}{y} \frac{dy}{dx} = -\frac{\alpha A}{t^{\alpha+1}} - \frac{\mu}{t} + B + Ct + \dots,$$

whence, if for shortness

$$w = -\frac{A}{t^\alpha} - \mu \log t + Bt + \frac{1}{2}Ct^2 + \dots,$$

we have

$$\log y = w + A \log x + Bx + \frac{1}{2}Cx^2 + \dots,$$

or passing to the value of y , but expanding as before the exponential of $Bx + \frac{1}{2}Cx^2 + \dots$, this is

$$y = e^w x^A (1 + Bx + Cx^2 + \dots),$$

a subregular integral. By what precedes, it appears that, if

$$p_1 = Lx^{-\delta} + \dots, \quad p_2 = L_1x^{-\delta-1} + \dots, \quad \dots, \quad p_m = L_mx^{-\delta-m} + \dots,$$

then the equation for the leading coefficient (previously represented by A) is

$$L[A] + L_1[A]^{-1} + \dots + L_{m-1}[A]^{-m+1} + L_m[A]^{-m} = 0,$$

but I do not see any direct or simple way of obtaining the corresponding equation for the exponent of this subregular integral.

18. To illustrate the case of a series with fractional powers, and also in order to work out a simple example with some completeness, I consider the differential equation of the second order $\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = 0$, giving the z -equation $z^2 + z + p_2 = 0$; p_2 is a rational and integral function of t ($t = \frac{1}{x-a}$), and we have two cases according as the order of the function is even or odd.

19. Suppose first that the order is even, and for convenience taking it to be $= 4$, and assuming the value of p_2 accordingly, I consider the equation

$$\frac{d^2y}{dx^2} = (\alpha + \beta t + \gamma t^2 + \delta t^3 + \epsilon t^4)y.$$

Here the form is at once seen to be

$$y = e^{\int z dt}, \quad z = At^2 + Bt + C + \frac{D}{t} + \frac{E}{t^2} + \dots,$$

and observing that we have

$$\frac{dz}{dt} = 2At + B - \frac{D}{t^2} - \frac{2E}{t^3} + \dots,$$

we find

$$A^2 = \alpha, \quad 2AB = \beta, \quad 2AC + B^2 - B = \gamma, \quad 2AD + 2BC = \delta,$$

$$2AE + 2BD + C^2 + C = \epsilon, \quad 2AF + 2BE + 2CD + D = 0,$$

equations which give

$$A = \sqrt{\alpha}, \quad B = \frac{\beta}{2\sqrt{\alpha}}, \quad C = \frac{\gamma}{2\sqrt{\alpha}} - \frac{\beta^2}{8\alpha\sqrt{\alpha}} + \frac{1}{4\sqrt{\alpha}},$$

and so on for the remaining coefficients.

The Collected Mathematical Papers of Arthur Cayley

Volume XII

Pages 451–500

[859. On the Developable Surface which meets a Unicursal Quartic Curve (conclusion)]

The expression thus becomes

$$\begin{aligned}\Pi = & \rho \cdot x(\gamma\delta - \beta^2) \\ & + Q \cdot [-3x\delta(\beta\delta - \alpha\gamma) - z(\alpha\delta - \beta^2) + w(\alpha\beta\gamma - \beta^3)] \\ & + R \cdot [zw(-\alpha^2\delta - \alpha\beta^2 + 2\beta^2\gamma) + yz(-2\alpha^2\beta\delta + 8\alpha\beta\gamma\delta - 6\beta^2\gamma^2) + w^2(\alpha^2\beta\gamma - \beta^4)],\end{aligned}$$

and we thus finally obtain

$$\Pi = PQ(3\alpha\delta - \beta\gamma) + QR(3\beta\delta - \gamma^2 + \alpha\gamma) + RP \cdot S,$$

viz. $\Pi = 0$ is the equation of the cone, vertex $(\alpha, \beta, \gamma, \delta)$, which passes through the quartic curve $x : y : z : w = 1 : \theta : \theta^2 : \theta^3$.

As regards the symmetry of this expression, it is to be remarked that, changing (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$ into (w, z, y, x) and $(\delta, \gamma, \beta, \alpha)$ respectively, we change (ρ, P, Q, R) and (ρ', P', Q', R') into $(\rho, -R, -Q, -P)$ and (ρ', R', Q', P') , respectively, and so leave Π unaltered.

Again, interchanging (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$, we interchange (ρ, P, Q, R) and (ρ', P', Q', R') , and so leave Π unaltered.

860. On Briot and Bouquet's Theory of the Differential Equation $F(u, \frac{du}{dz}) = 0$

[From the *Proceedings of the London Mathematical Society*,
vol. XVIII. (1887), pp. 314–324.]

I consider the theory of a differential equation of the first order as developed in Briot and Bouquet's *Théorie des fonctions doublement périodiques et en particulier des fonctions elliptiques* (§, 1st ed., Paris, 1859), but I make some substantial variations in the mode of treatment.

I remark that, writing $u = x$, $\frac{du}{dz} = y$, I make the theory to depend altogether upon that of the curve $F(x, y) = 0$; viz. the form of my result is, in order that a differential equation of the first order $F\left(u, \frac{du}{dz}\right) = 0$ may have a monotropic integral, the curve $F(x, y) = 0$ must satisfy certain conditions.

Briot and Bouquet give their result (Theorem IV., p. 296) under a somewhat different form, as follows; viz. changing only the word “monodrome” into “monotrope,” their theorem is:

“Pour qu’une équation différentielle du premier ordre de la forme

$$\left(\frac{du}{dz}\right)^m + f_1(u)\left(\frac{du}{dz}\right)^{m-1} + \cdots + f_m(u) = 0$$

admette une intégrale monotrope:

1. les coefficients $f_1(u), f_2(u), \dots, f_m(u)$ doivent être des polynômes entiers en u et, au plus, le premier du second degré, le second du quatrième degré, $\dots$, le dernier du degré $2m$;
2. quand pour une certaine valeur de u l’équation a une racine multiple différente de zéro, $\frac{du}{dz}$ doit rester monotrope par rapport à u ;
3. quand pour une certaine valeur u_1 de u , l’équation a une racine multiple égale à zéro, le premier terme du développement de $\frac{du}{dz}$ suivant les puissances croissantes de $(u - u_1)^{1/n}$ doit avoir l’exposant 1, si cet exposant est plus petit que l’unité;
4. enfin l’équation différentielle, que l’on déduit de la première en posant $w = \frac{1}{v}$, doit offrir pour $v = 0$ les mêmes caractères.

Ces conditions sont suffisantes.”

I notice that this may be regarded as a statement referring to the curve

$$F(x, y) = y^m + f_1(x)y^{m-1} + \cdots + f_m(x) = 0,$$

but that in 2 a notion of monotropy is assumed, for y must be a monotropic function of x ; and that 4 in effect introduces the new curve $F\left(\frac{1}{x}, \frac{y}{x^2}\right) = 0$.

1.

We have between the variables u and z a differential equation of the first order $F\left(u, \frac{du}{dz}\right) = 0$, F a rational and integral function of u and $\frac{du}{dz}$. The equation determines u as a function of z , and we wish to know whether u is a monotropic function of z . It will not be so if we have a *tropical point*, viz. a point $z = c$, such that, in the neighbourhood thereof, the value of u is given by a series $u - a = B(z - c)^\rho + \dots$, in ascending powers of $z - c$, involving fractional powers of $z - c$. Conversely, if there is no tropical point, then u will be a monotropic function of z ; but we must consider and exclude not only tropical points at a finite distance, but also the point at infinity, viz. there must not be any development $u = Bz^\rho + \dots$, in descending powers, involving fractional powers of z .

2.

A one-valued function (*eindeutige Function*, or *fonction uniforme* or *bien déterminée*) is monotropic; since there is only one value, the function must, after a circuit described by the point z , recover its original value. But, conversely, a monotropic function is one-valued, that is, no two or more valued function can be monotropic. For, suppose a two-valued function of z , having the values Z_1 and Z_2 , is monotropic; there is by hypothesis no tropical point, that is, no point $z = c$, such that the function Z_1 , after a circuit described by the point z , instead of recovering its original value, assumes the value Z_2 ; and the like as to Z_2 . But, this being so, it is meaningless to regard Z_1 and Z_2 as two values of the same function; they must be considered as two distinct and separate functions, each of them a one-valued monotropic function. And it thus appears that the two notions, monotropic and one-valued, are in fact equivalent.

3.

In what precedes, u and z are regarded as complex variables geometrically represented by means of two real points in their own planes respectively. But we now write $\frac{du}{dz} = y$, $u = x$, whence $F(x, y) = 0$, regarding x, y as the coordinates, real or imaginary, of a point of the curve represented by this equation. And this curve has, moreover, to be considered from a non-projective point of view; we have to distinguish between, and separately consider, finite points (x and y each of them finite), and the points at infinity (x and y either or each of them infinite). And it is moreover necessary to distinguish between, and separately consider, points on the axis of x (or say axial points), and points off the axis of x (say non-axial points). Taking, for convenience, a to denote a finite value which may be $= 0$, b a finite value which is not $= 0$ (or say, a not $= \infty$, b not $= \infty$ or 0), we have thus the six cases,

$$1 \quad (a, b); \quad 2 \quad (a, 0); \quad 3 \quad (a, \infty); \quad 4 \quad (\infty, b); \quad 5 \quad (\infty, 0); \quad 6 \quad (\infty, \infty)$$

I regard also the axes of x and y as horizontal and vertical respectively, and speak of the tangent or element of the curve as horizontal, vertical, or inclined, accordingly.

4.

In the neighbourhood of any given point of the curve, we have $y =$ a series of powers of x , the form of the series being different for different classes of points. I write $\text{Pf}(x)$ to denote a series (finite or infinite) $A + Bx + Cx^2 + \dots$, in ascending powers of x ; it is to be throughout understood that the leading coefficient A is not $= 0$. Of course, $\text{Pf}(x - a)$, $\text{Pf}\{(x - a)^{1/n}\}$, $\text{Pf}(x^{-1})$, &c., will denote the like series $A + B(x - a) + \dots$, $A + B(x - a)^{1/n} + \dots$, $A + Bx^{-1} + \dots$, &c. The coefficients after the leading coefficient A may any or all of them vanish; thus $\text{Pf}\{(x - a)^{1/n}\}$ extends to denote the series $\text{Pf}(x - a)$ of integer powers, but naturally the former notation is not used unless the series contains fractional powers.

We can, by means of the symbol Pf , express for any given point of the curve the form of the expansion in the neighbourhood thereof; and, attaching to the point the expansion which belongs to it, we may, for instance, speak of a point $y = \text{Pf}(x - a)$; viz. if b is the constant term of the series, then this is a non-axial point (a, b) , which is an

ordinary (non-singular) point of the curve, and for which the element is not vertical; a like point with the element vertical would be a point $y = b + (x - a)^2 \text{Pf}(x - a)$.

Observe that, in the case of a multiple point, there is a separate expansion for each branch through the point, and in thus attaching an expansion to the point we regard the point as belonging to one of these branches; in dealing with a multiple point, it is necessary to consider separately all the different expansions, that is, all the branches through the point.

It is clear that for a point (a, b) , $(a, 0)$, or (a, ∞) the expression for y depends on $\text{Pf}(x - a)$, or $\text{Pf}\{(x - a)^{1/n}\}$; while, for a point (∞, b) , $(\infty, 0)$, or (∞, ∞) , it depends on $\text{Pf}(x^{-1})$, or $\text{Pf}(x^{1/n})$; viz. in the two cases respectively, we have a series in ascending powers of $x - a$, or in descending powers of x .

5.

In regard to any given point of the curve, substituting for y , x their values $\frac{du}{dz}$, u , and thence forming the reciprocal of $\frac{du}{dz}$, we obtain $\frac{dz}{du}$ = a series in u ; viz. this is a series in ascending powers of $(u - a)$, or in descending powers of u . Integrating, we have $z - c$ = a series in u , which series may or may not contain a logarithmic term $\log u$ or $\log(u - a)$; and, reverting the series, we obtain u = a series in $(z - c)$. This result, applicable to the neighbourhood of the point $u = a$, may contain only integer powers of $z - c$, and be accordingly of the form u = a one-valued series in z ; and we then say that the point on the curve is a “permissive” point. Or it may contain fractional powers of $z - c$, and be accordingly of the form u = a more-than-one-valued series in z ; and we then say that the point on the curve is a “prohibitive” point.

The necessary and sufficient condition in order that u may be a monotropic function of z is: the points of the curve must be all of them permissive points; or, what is the same thing, there must not be any prohibitive point.

6.

I form a complete table of permissive points, as follows:

n a positive integer = 2 at least. $\epsilon = 0, 1$, or -1 .

	Point	Form
(a, b)	$y = \text{Pf}(x - a)$	general
(a, b)	$y = b + (x - a)^2 \text{Pf}(x - a)$	vertical element
$(a, 0)$	$y = (x - a) \text{Pf}(x - a)$	general
$(a, 0)$	$y = (x - a)^{1+\frac{1}{n}} \text{Pf}\{(x - a)^{1/n}\}$	tangent horizontal
(a, ∞)	$y = (x - a)^{-1} \text{Pf}(x - a)$	general
(a, ∞)	$y = (x - a)^{-1+\frac{1}{n}} \text{Pf}\{(x - a)^{1/n}\}$	element vertical
(∞, b)	$y = \text{Pf}(x^{-1})$	general
(∞, b)	$y = b + x^{-2} \text{Pf}(x^{-1})$	element vertical
$(\infty, 0)$	$y = x^{-1} \text{Pf}(x^{-1})$	general
$(\infty, 0)$	$y = x^{-1-\frac{1}{n}} \text{Pf}(x^{-1/n})$	element horizontal
(∞, ∞)	$y = x \text{Pf}(x^{-1})$	general
(∞, ∞)	$y = x^{1+\frac{1}{n}} \text{Pf}(x^{-1/n})$	element vertical

viz. it is only a point (a, b) , $(a, 0)$, (∞, b) or (∞, ∞) which may be permissive; a point (a, ∞) or $(\infty, 0)$ is prohibitive.

7.

A point $y = \text{Pf}(x - a)$ is permissive. We have $\frac{du}{dz} = \text{Pf}(u)$, $\frac{dz}{du} = \frac{1}{\text{Pf}(u)}$, whence $z - c = (u - a) \text{Pf}(u - a)$, $u - a = (z - c) \text{Pf}(z - c)$, the required result. The several steps will be readily understood; the reciprocal of a series $\text{Pf}(u - a)$ is a series of the like form $\text{Pf}(u - a)$; integrating this in regard to u as a series in $u - a$ (no constant being added on the right-hand side), we obtain a series $(u - a) \text{Pf}(u - a)$; and finally, $z - c$ being equal to this expression, we obtain, by reversion, a like expression $u - a = (z - c) \text{Pf}(z - c)$. We might, for convenience, have written u, z in place of $x - a, u - a, z - c$, respectively. The proof would have run: $y = \text{Pf}(x)$, $\frac{du}{dz} = \text{Pf}(u)$, $\frac{dz}{du} = \text{Pf}(u)$, $z = u \text{Pf}(u)$, $u = z \text{Pf}(z)$, meaning $u - a = (z - c) \text{Pf}(z - c)$, as above; and, in what follows, this is done accordingly.

It is worth while to make an instance in which the integration can be performed; say we consider the point $(0, 1)$ of the curve $y = \frac{1}{1+x+x^2}$, which point is $y = \frac{1}{1+x+x^2} = \text{Pf}(x)$. We have $\frac{du}{dz} = \frac{1}{1+u+u^2}$, $\frac{dz}{du} = 1+2u$, $z = \log(1+u+u^2)$, or $u+u^2 = e^z - 1$; giving a series $u = z \text{Pf}(z)$, or the point is a permissive point, as in the

general case. We have here $u + u^2 = e^z - 1$, viz. u is not a monotropic function of z ; but this is by reason of a prohibitive point (∞, ∞) .

8.

A point $y = (x - a)\text{Pf}(x - a)$ is permissive. Writing as above u , z in place of $u - a$, $z - c$, we have $\frac{du}{dz} = u\text{Pf}(u)$, $\frac{dz}{du} = \frac{1}{u}\text{Pf}(u)$, or say $u\frac{dz}{du} = \frac{1}{u} + \text{Pf}(u)$, whence $mz = \log u + u\text{Pf}(u)$, that is, $ue^{u\text{Pf}(u)} = e^{mz}$, or say $u\text{Pf}(u) = e^{mz}$, whence $u = e^{mz}\text{Pf}(e^{mz})$, a one-valued series.

I take a particular instance, the point $(0, 0)$ for the curve $y = \frac{x}{1-x}$; here $\frac{du}{dz} = \frac{u}{1-u}$, $\frac{dz}{du} = \frac{1}{u} - 1$, $z = \log(u) - u$, or $ue^{-u} = e^z$; the point is permissive. The equation just obtained shows that u is not a monotropic function of z ; but this is by reason of the prohibitive points $(1, \infty)$.

9.

A point $y = (x - a)^2\text{Pf}(x - a)$ is permissive. Here $\frac{du}{dz} = u^2\text{Pf}(u)$, $\frac{dz}{du} = \frac{1}{u^2}\text{Pf}(u) = \frac{A}{u^2} + \frac{B}{u} + \text{Pf}(u)$, whence $z = -\frac{A}{u} + B\log u + u\text{Pf}(u)$, where the logarithmic term occurs or does not occur, according as B is not or is $= 0$. In the former case, we have $e^{z/B} = ue^{u\text{Pf}(u)}$; in the latter case, $z = \frac{A}{u} + u\text{Pf}(u)$; and in each case we can, by reversion, express u as a one-valued series in z .

We may take as examples the two curves $y = \frac{x^2}{1-x}$ and $y = \frac{x^2}{1+x}$; in each case the point $(0, 0)$ is a permissive point, but, by reason of the prohibitive points $(1, \infty)$ and $(-1, \infty)$, u is not a monotropic function of z .

10.

As to the other forms of permissive points, the proofs depend upon those for the forms already considered. Thus for $y = (x - a)^{1+\frac{\epsilon}{n}}\text{Pf}\{(x - a)^{1/n}\}$, or $\frac{du}{dz} = u^{1+\frac{\epsilon}{n}}\text{Pf}(u^{1/n})$, writing $v = u^{1/n}$, we

find $\frac{dv}{dz} = v^{1+\epsilon}\text{Pf}(v)$; that is, $z = \text{Pf}(v)$, $v\text{Pf}(v)$, or $v^2\text{Pf}(v)$; in each case v = one-valued series in z , and thence $u = v^n$ = one-valued series in z .

Similarly for $y = \text{Pf}(x^{-1})$, $y = x\text{Pf}(x^{-1})$, $y = x^2\text{Pf}(x^{-1})$, or say $y = x^{1+\epsilon}\text{Pf}(x^{-1})$; here $\frac{du}{dz} = u^{1+\epsilon}\text{Pf}(u^{-1})$, or, writing $v = u^{-1}$, we have $\frac{dv}{dz} = v^{1-\epsilon}\text{Pf}(v)$; viz. this is $\frac{dv}{dz} = \text{Pf}(v)$, $v\text{Pf}(v)$, and $v^2\text{Pf}(v)$; in each case v = one-valued series in z , and thence $u = v^{-1}$ = one-valued series in z .

And finally, for $y = x^{1+\frac{\epsilon}{n}}\text{Pf}(x^{-1/n})$, here $\frac{du}{dz} = u^{1+\frac{\epsilon}{n}}\text{Pf}(u^{-1/n})$, or, writing $v = u^{-\epsilon/n}$, we have $\frac{dv}{dz} = v^{1-\epsilon}\text{Pf}(v)$, that is, $\frac{dv}{dz} = \text{Pf}(v)$, $v\text{Pf}(v)$, or $v^2\text{Pf}(v)$, or as before; v = one-valued series in z , and thence $u = v^{-n/\epsilon}$ = one-valued series in z .

11.

There is no difficulty in showing that every point of the curve, not being a permissive point as above, is a prohibitive point. Thus for a point (a, b) , if the series for $y - b$ contain any fractional power, say, if we have $y = b + (x - a)^{\frac{1}{n}}$, then $\frac{du}{dz} = b + u^{\frac{1}{n}}$, $\frac{dz}{du} = \frac{1}{b + u^{\frac{1}{n}}}$, whence, reverting, $u = bz +$ series containing the fractional power $z^{\frac{1}{n}+1}$.

Again, a point $y = (x - a)^m\text{Pf}(x - a)$, $m = 3$ or any higher integer value, is prohibitive. Attending only to the leading term, we have $\frac{du}{dz} = u^m$, whence $\frac{dz}{du} = u^{-m}$, $z = u^{1-m}$, whence, reverting, $u = z^{\frac{1}{1-m}}$, which is a fractional power; and this is also the case if m be a negative integer. And similarly, in other cases, the series for u will always contain fractional powers of z .

12.

In order that u may be a monotropic function of z , the points of the curve must be all of them permissive, or, what is the same thing, there must be no prohibitive point; we have to consider the geometrical signification of this condition.

If we attend first to the ordinary (or non-singular) points of the curve, a non-axial point (a, b) must be a point $y = \text{Pf}(x - a)$, viz.

there must be no such point with a vertical element. An axial point $(a, 0)$ may be $y = (x - a)\text{Pf}(x - a)$, viz. the element may be inclined; or the point may be $y = (x - a)^2\text{Pf}(x - a)$, viz. the element may be horizontal (but in this case there must be an ordinary or two-pointic contact only); or the point may be $y = (x - a)^{\frac{1}{n}}\text{Pf}\{(x - a)^{\frac{1}{n}}\}$, viz. the element may be vertical. The point must not be (a, ∞) , viz. there must be no asymptote parallel to (or coincident with) the axis of y .

The point may be (∞, b) , $y = \text{Pf}(x^{-1})$, that is, there may be an ordinary, or osculating, asymptote parallel to the axis of x . But there is no point $(\infty, 0)$, that is, there must be no asymptote coincident with the axis of x . There may be points (∞, ∞) , $y = x\text{Pf}(x^{-1})$, that is, ordinary or osculating asymptotes inclined to the axes; and there may also be points (∞, ∞) , $y = x^2\text{Pf}(x^{-1})$, that is, asymptotic parabolas of vertical axis.

13.

We have, moreover, conditions as to the singular points; thus, every non-axial multiple point (a, b) must be a point with each branch of the form $y = \text{Pf}(x - a)$, that is, each branch must be an ordinary (non-cuspidal) branch, and must be non-vertical. The axial multiple or singular points $(a, 0)$ may have ordinary (non-cuspidal) branches $y = (x - a)\text{Pf}(x - a)$ and $y = (x - a)^2\text{Pf}(x - a)$, inclined or horizontal, or else $y = (x - a)^{\frac{1}{n}}\text{Pf}\{(x - a)^{\frac{1}{n}}\}$, vertical; and there may also be cuspidal branches $y = (x - a)^{\frac{1}{n}}\text{Pf}\{(x - a)^{\frac{1}{n}}\}$, $y = (x - a)^{1+\frac{1}{n}}\text{Pf}\{(x - a)^{\frac{1}{n}}\}$, and $y = (x - a)^{1-\frac{1}{n}}\text{Pf}\{(x - a)^{\frac{1}{n}}\}$, inclined, horizontal, or vertical.

There must be no singular points (a, ∞) ; any such point is, in fact, excluded by the condition, no asymptote parallel to or coincident with the axis of y . There may be multiple points (∞, b) , but not $(\infty, 0)$; viz. as above, there may be asymptotes parallel to, but not coincident with, the axis of x ; but in this case, the several branches must be each of them an ordinary branch $y = \text{Pf}(x^{-1})$. Finally, there may be multiple or singular points (∞, ∞) , but each branch must be either an ordinary branch $y = x\text{Pf}(x^{-1})$ or $y = x^2\text{Pf}(x^{-1})$, or a cuspidal branch $y = x^{1+\frac{1}{n}}\text{Pf}(x^{-1/n})$, $y = x^{1-\frac{1}{n}}\text{Pf}(x^{-1/n})$, or $y = x^{1+\frac{2}{n}}\text{Pf}(x^{-1/n})$, inclined, horizontal, or vertical.

14.

The enumeration seems long and difficult. But for a given curve we have, in fact, only to see that there are no ordinary non-axial points (a, b) of vertical element, and to further examine the finite singular points (a, b) and $(a, 0)$, and also the several infinite branches as giving ordinary or else singular points (a, ∞) , (∞, b) , $(\infty, 0)$, or (∞, ∞) , viz. it has to be seen for each infinite branch that the expansion is of the proper form.

15.

Writing the equation of the curve in the form

$$f_0(x)y^m + f_1(x)y^{m-1} + \cdots + f_m(x) = 0,$$

where $f_0, f_1, \dots, f_m$ denote rational and integral functions of x , then, if a be a root of the equation $f_0(x) = 0$, there will be on the curve a point (a, ∞) , which is prohibitive; $f_0(x)$ is therefore a constant, or, taking it to be $= 1$, the form must be

$$y^m + f_1(x)y^{m-1} + \cdots + f_m(x) = 0,$$

where $f_1, f_2, \dots, f_m$ denote rational and integral functions of x .

It is to be shown that the degrees of these functions are equal respectively to $2, 4, \dots, 2m$ at most. Supposing that the degrees are $2, 4, \dots, 2m$; then for the points at infinity we find an expansion $y = Ax^2 + \dots$, in descending powers of x , by substitution in an equation $y^m + a_1x^2y^{m-1} + a_2x^4y^{m-2} + \dots + a_mx^{2m} = 0$; whence $m\rho = 2$, and A is determined by an equation of the order m ; there are thus m branches $y = x^2\text{Pf}(x^{-1})$, corresponding to permissive points. If, however, any one of the functions f be of a higher degree, then it may be shown (but the formal proof is not easily given) that there will be at any rate a branch $y = Ax^\rho + \&c.$, where ρ has a value > 2 , and we have thus a prohibitive point; hence the degrees must be as stated.

16.

As an example of a differential equation with a monotropic solution, take

$$\left(\frac{du}{dz}\right)^2 = 1 - 27u^2 + 4u^3;$$

we have here the curve $y^2 = 1 - 27x^2 + 4x^3$, which is a nodal cubic, as shown in the annexed figure. To put the node in evidence, the equation may be written in the form $(y + 2)^2(y - 1) + 27x^2 = 0$; viz. the node is the point $(0, -2)$, and the equation of the tangents is $(y + 2)^2 - 27x^2 = 0$. The curve besides meets the line $x = 0$ in the point $y = 1$, where the tangent is horizontal, and it meets the axis $y = 0$ in the points $x = \pm\sqrt{\frac{27}{4}}$, at each of which the tangent is vertical; these two points, as lying on the axis $y = 0$, are thus each of them permissive; and they are the only points where the tangent is vertical (in fact, the tangents from the point at infinity on the line $x = 0$ are the two lines $x = \pm\sqrt{\frac{27}{4}}$, and the line infinity counting twice, as a tangent at an inflexion). For the points at infinity, we have $y = -3x^3 - 1 - \frac{1}{3}x^{-3} + \&c.$, which is of the form $y = x^3\text{Pf}\left(\frac{1}{x}\right)$, included in $y = x^{1+\frac{1}{n}}\text{Pf}(x^{-1/n})$, and the point is thus permissive; there is no prohibitive point, and the differential equation has a monotropic solution. This is, in fact, the rational solution $u = (z - c) - (z - c)^3$, or say $u = z - z^3$; the curve is thus given by the two equations $x = z - z^3$, $y = 1 - 3z^2$.

17.

As a second example, take the differential equation

$$\left(\frac{du}{dz}\right)^2 = (1 - u^2)(1 - k^2u^2).$$

We have here the curve $y^2 = (1 - x^2)(1 - k^2x^2) = 0$, which is a trinodal quartic curve, as shown in the figure. There is a node at the origin with the tangents $y^2 - x^2 = 0$, and, writing the equation in the form $(3y + 1)(3y - 2)^2 - (27x^2 - 8)^2 = 0$, we have for the other two nodes $x = \pm\frac{2\sqrt{2}}{3\sqrt{3}}$, $y = \frac{2}{3}$. The lowest points are given by $x = 0$, $y = \pm\frac{1}{k}$; viz. the line $y = \frac{1}{k}$ is a horizontal tangent. Writing $x = 0$, we have $y^2 - \frac{1}{k^2} = 0$, that is, $y = \pm\frac{1}{k}$, the node at the origin, and $y = \frac{1}{k}$, the height of the loop; and, writing $y = 0$, we have $1 - k^2x^2 = 0$, that is, $x = \pm\frac{1}{k}$, the other two intersections with the axis of x . The tangents at these two points are vertical; as being on the axis, they are

thus permissive points. And they are the only points with a vertical tangent; in fact, the point $(x = 0, y = \infty)$ is a point of the curve, with the line at infinity as an osculating tangent (4-pointic intersection); hence the tangents from the point $x = 0, y = \infty$ are the line at infinity counting four times, and the lines $x = \pm \frac{1}{k}$ touching at the points $\left(\pm \frac{1}{k}, 0\right)$ as above.

For the infinite branches, we have $y = kx^2 + \dots$, which is of the right form $y = x^{1+\frac{1}{n}} \text{Pf}\left(\frac{1}{x}\right)$. We thus see that there is no prohibitive point; and the differential equation has a monotropic solution accordingly. Writing z in place of $z - c$, the solution in fact is $u = \frac{e^z + e^{-z}}{e^z - e^{-z}}$, the curve is given by the equations $x = \frac{e^z - e^{-z}}{e^z + e^{-z}}, y = \frac{4}{(e^z + e^{-z})^2}$; hence, putting $\theta = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, the curve is given by the two equations $x = \theta - \theta^3, y = (1 - 3\theta^2) \dots$ the monotropic function u satisfies the differential equation.

18.

$F\left(u, \frac{du}{dz}\right) = 0$ must be either a rational function, a singly periodic function, or a doubly periodic function of z ; say the forms are $u = \text{Pf}(z)$, $u = \text{Pf}(e^z)$, and $u = \text{Pf}\{\text{sn}(gz, k)\} + \text{cn}(gz, k)\text{dn}(gz, k) \text{Q}\{\text{sn}(gz, k)\}$, where Pf, Q denote rational functions. I do not at present consider the criteria (such as are given in Briot and Bouquet's Theorem V., p. 301) for determining by means of the curve which is the form of the integral. I remark, however, that in the first and second cases the curve is unicursal, while in the third case it is bicursal; or say that, according as the deficiency is $= 0$ or 1 , the integral is rational or simply periodic, or else it is doubly periodic. Moreover, the curve being unicursal, we can express the coordinates as equal to rational functions $\text{Pf}(\theta), \text{Q}(\theta)$ of a parameter; and, being bicursal, we can express them as equal to rational functions of $\text{sn}(gz, k), \text{cn}(gz, k), \text{dn}(gz, k)$.

861. Note on a Formula relating to the Zero-Value of a Theta-Function

[From *Crelle's Journal der Mathem.*, t. c. (1887), pp. 87, 88.]

I had some difficulty in verifying for the case of a single theta-function, a formula given in Herr Thomae's paper "Beitrag zur Theorie der ϑ -Functionen," *Crelle's Journal*, vol. LXXI. (1870), pp. 201–222. The formula in question (see p. 216) is given as follows:

$$\vartheta(0, 0, \dots, 0) = \frac{\sqrt[4]{A}}{(\pi i)^{\frac{1}{2}\rho}} \sqrt{\text{Discr.}(k_1, k_2, \dots, k_{\rho+1}) \cdot \text{Discr.}(k_1, k_2, \dots, k_{\rho+1})},$$

but the denominator factor should I think be $(\pi i)^{\frac{1}{2}\rho}$ instead of $(2\pi i)^{\frac{1}{2}\rho}$. Making this alteration, then in the case of a single theta-function, $\rho = 1$, and the function belongs to the radical

$$\sqrt{x - k_1 \cdot x - k_2 \cdot x - k_3 \cdot x - k_4},$$

where $(k_1, k_2, k_3, k_4) = (-\rho, -1, +1, +\frac{1}{\rho})$. The determinant $|A_{\alpha\beta}|$ is a single term = A , and the formula becomes

$$\vartheta(0) = \frac{\sqrt[4]{A}}{\sqrt{\pi i}} \sqrt{(k_3 - k_1)(k_4 - k_2)},$$

where $k_3 - k_1, k_4 - k_2$ are each = $1 + \rho$, and we have therefore

$$\vartheta(0) = \frac{\sqrt[4]{A}}{\sqrt{\pi i}} \sqrt{1 + \rho}.$$

Also A denotes the integral

$$A = \int_{k_2}^{k_3} \frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}} = ikK',$$

and the formula thus is

$$\vartheta(0) = \frac{\sqrt[4]{ikK'}}{\sqrt{\pi}} (1 + \rho)^{\frac{1}{2}}.$$

But observe that, in the theta-function as defined by the equation $\mathcal{S}(x) = \Sigma e^{ax^2 + 2nx}$, a is used to denote the value $a = \frac{iK'}{K} = \frac{1}{\pi i} B$, where A is the above-mentioned integral, and B is the integral

$$B = \int_{k_2}^{k_3} \frac{dx}{\sqrt{x - k_1 \cdot x - k_2 \cdot x - k_3 \cdot x - k_4}},$$

which value must however be taken negatively, viz. we must write $B = -2kK'$, and we then have

$$a = \frac{iK'}{K} = \frac{-2kK'}{2\pi i} = \frac{B}{2\pi i},$$

viz. writing as usual $q = e^{-\frac{\pi K'}{K}} = e^{\frac{i\pi B}{2\pi i}} = e^{\frac{1}{2}B}$, the e^a of the theta-function is not $= q$, but it is $= q^2$; and the zero-value $\mathcal{S}(0)$ is $= 1 + 2q^2 + 2q^8 + 2q^{18} + \dots$

The equation thus is

$$2q^2 = \frac{\sqrt[4]{ikK'}}{\sqrt{\pi}}(1 + \rho)^{\frac{1}{2}},$$

which is right; in fact, writing k' in place of k , and consequently K' , q in place of K , r respectively, the equation becomes

$$q^2 = \frac{\sqrt[4]{iKk'}}{\sqrt{\pi}}(1 + \rho)^{\frac{1}{2}},$$

and changing q into q^4 , then (*Fund. Nova*, p. 92) K is changed into $K\left(1 + \frac{k'}{1}\right)$ and z we have the formula in question.

As a verification for small values of q , observe that we have

$$\frac{2K}{\pi} = 1 + 4q + 4q^2 + \dots, \quad \frac{2kK}{\pi} = 2\sqrt[4]{q}(1 + q + \dots),$$

and thence $\vartheta(0) = 1 + 2q^2 + 2q^8 + \dots = 1 - 4q^2 + \dots$, or $\sqrt{\frac{2kK}{\pi}} = 1 - 4q^2 + \dots$

Cambridge, 12 February, 1886.

862. Note on the Theory of Linear Differential Equations

[From *Crelle's Journal der Mathem.*, t. c. (1887), pp. 286–295.]

1.

I consider a linear differential equation

$$p_0 \frac{d^m y}{dx^m} + p_1 \frac{d^{m-1} y}{dx^{m-1}} + \cdots + p_m y = 0,$$

where $p_0, p_1, \dots, p_m$ are rational and integral functions of x , having no common factor: as usual, x is a complex magnitude represented by a point, and we consider the integrals belonging to a singular point $x = a$ of the differential equation. An integral may be a *regular integral*, or it may be what Thomé calls a *normal elementary integral*: the theory of these integrals (which I would rather call *subregular integrals*) requires, I think, further examination.

2.

I retain x as the independent variable, but for shortness use t to denote $\frac{1}{x-a}$; and I take as the dependent variable $z = \frac{1}{y} \frac{dy}{dx}$: we have then z determined by a non-linear differential equation of the order $m-1$, and from any value of z we derive a corresponding value $e^{\int z dx}$ of y .

3.

To obtain the z -equation, using for shortness accents to denote differentiation in regard to x , we have $z = \frac{y'}{y}$, and thence $z' = \frac{y''}{y} - \left(\frac{y'}{y}\right)^2$, that is $z' + z^2 = \frac{y''}{y}$; hence $\frac{y''}{y} = z^2 + z'$, also $2zz' + z'' = \frac{y'''}{y} - \frac{y'y''}{y^2}$, that is $\frac{y'''}{y} = z^3 + 3zz' + z''$, or finally $z^3 + 3zz' + z'' = \frac{y'''}{y}$; and so on; viz. the values of $\frac{y'}{y}, \frac{y''}{y}, \frac{y'''}{y}, \dots$ are $z, z^2 + z', z^3 + 3zz' + z'', \dots$; generally for $\frac{y^{(n)}}{y}$ the first term is z^n and the last term is $z^{(n-1)}$.

If, to fix the ideas, the y -equation is

$$p_0 y''' + p_1 y'' + p_2 y' + p_3 y = 0,$$

then, dividing by y , the z -equation is

$$p_0(z^3 + 3zz' + z'') + p_1(z^2 + z') + p_2 z + p_3 = 0;$$

and similarly, from the y -equation of the order m , we derive a z -equation of the order $m - 1$,

$$p_0[z^m + \dots + z^{(m-1)}] + \dots + p_{m-2}(z^2 + z') + p_{m-1}z + p_m = 0.$$

4.

In this equation, each coefficient is in the first instance a rational and integral function of x , or, what is the same thing, of $x - a$; writing t to denote $\frac{1}{x - a}$, each coefficient is thus the sum of a finite number of negative integer powers of t ; hence multiplying the whole equation by a proper positive power of t , the several coefficients become each of them the sum of a finite number of positive powers of t , or arranging in descending powers of t , say the general form is $p_r = \alpha_r t^{\lambda_r} + \beta_r t^{\lambda_r - 1} + \dots + \kappa_r$, where λ_r is a positive integer which may be $= 0$, and which has for each coefficient its proper value.

We wish to satisfy the z -equation by a value $z =$ descending series in t , viz. the form is $z = At^\alpha + A't^{\alpha - \alpha'} + \dots$, with powers which are integral or fractional, but where the number of positive powers is finite.

The theory is almost identical with that of the solution in like form of the algebraical equation $p_0 z^m + p_1 z^{m-1} + \dots + p_{m-1} z + p_m = 0$, or say the equation $U = 0$, where U is a rational and integral function of z and t , of the degree m as regards z .

5.

Considering z and t as Cartesian coordinates, the equation in question, $U = 0$, represents a geometrical curve such for any given value of the abscissa t , the ordinate z has m values: the curve is of the order m at least, but it may be of a higher order $m + \kappa$; when this is so, there is at infinity on the axis $t = 0$, a κ -tuple point K , and thus any line parallel to the axis $t = 0$, intersects the curve in the point K counting as κ intersections, and in m other points, which are the points belonging to the m values of the ordinate z . Taking $s = 1$, and for z, t writing $\frac{z}{s}, \frac{t}{s}$ respectively, we have between the trilinear coordinates z, t, s an equation $U = 0$ of the order $m + \kappa$, where $s = 0$ is the equation of the line infinity: and the curve has a κ -tuple point at the point K for which $t = 0, s = 0$.

6.

Starting from the equation $U = 0$, for an arbitrary value of t we have m values of z all of them different. These may be developed, each of them as a descending series in t , and the development is effected in the usual manner; viz. considering for the moment t as very large, then z is in general very large and it has a leading term At^α , which is determined by writing in the equation $z = At^\alpha$, and then finding α in such wise that there are two or more exponents equal to each other and greater (that is, nearer $+\infty$) than any other exponent: we have thus a highest power of t , the whole coefficient for which must vanish, viz. we obtain an equation of two or more terms giving for A a value or values not $= 0$; in the term At^α in question, α is in general positive, but it may be $= 0$, or be negative.

The leading term At^α being found in this manner, the law for the exponents of the subsequent terms is frequently at once apparent; but, if necessary, we write $z = At^\alpha + A't^{\alpha-\alpha'}$, and determine in like manner the exponent $\alpha - \alpha'$ and the coefficient A' . Proceeding in this manner until the law of the exponent becomes apparent, and then by the method of indeterminate coefficients, we finally arrive at a series $z = At^\alpha + A't^{\alpha-\alpha'} + A''t^{\alpha-\alpha'-\alpha''} + \dots$, in descending powers of t : the exponents are integral or fractional, but the number of positive exponents is always finite.

There is either a single series or it may be two or more series; but the number of series is at most $= m$, and when it is less than m , then the coefficients in the several series or some of them will contain radicals, and by giving to each radical its different values, the system of series will determine m different values of z .

7.

The curve meets the line infinity ($s = 0$) in the point K counting as κ intersections, and in m other points, some or all of which may coincide with K ; the forms of the series depend on the configuration of the m points, or say on the relation of the curve to the line infinity. Thus suppose $\kappa = 0$, and further that the m points are all of them distinct, that is, let the curve be a curve of the order m meeting the line infinity in m distinct points, or, what is the same thing, having m asymptotes, no two of them parallel; we have in this case m series, each of the form $z = At + B + \frac{C}{t} + \dots$, where the several coefficients

A have distinct values.

8.

In the foregoing case A is determined by an equation of the order m , having unequal roots; and taking for A any root at pleasure, the remaining coefficients $B, C, \dots$ are each of them linearly determined. If the equation has two equal roots, this may correspond to the case of two parallel asymptotes (the curve has here a node at infinity); the coefficients $B, C, \dots$ will in this case depend on a quadric radical, and by giving to this radical its two values, we obtain the two series $z = At + B + \frac{C}{t} + \dots$, corresponding to the two equal roots of the equation. But if instead of a node at infinity we have the line infinity a tangent to the curve, then instead of the two parallel asymptotes, we have an asymptotic parabola; the series assumes a new form, viz. it contains terms in $t^{\frac{1}{2}}, t^{-\frac{1}{2}}, \dots$; the coefficients of the integer powers have determinate values, but those of the powers $t^{\frac{1}{2}}, t^{-\frac{1}{2}}, \dots$ contain as factor a quadric radical, and giving to this radical its two values, we have two series.

9.

The like considerations apply in the case where the line infinity passes through higher multiple points of the curve, or has with it a contact or contacts other than a single ordinary contact: in every case, reckoning as distinct series those obtained by attaching to each radical its different values, the number of distinct series will be $= m$ precisely.

10.

Reverting now to the differential equation $p_0[z^m + \dots + z^{(m-1)}] + \dots + p_{m-2}(z^2 + z') + p_{m-1}z + p_m = 0$, it is to be observed that the very same process is applicable to the determination of z in the form of a descending series $z = At^\alpha + A't^{\alpha-\alpha'} + \dots$; moreover, it frequently happens that the determination of the leading term At^α is made by means of the algebraical equation $p_0z^m + \dots + p_{m-2}z^2 + p_{m-1}z + p_m = 0$, viz. this is so whenever the value of α is greater than 1: in fact, writing $z = At^\alpha (> 1)$, then in the sets of terms $\{z^m + \dots + z^{(m-1)}\}, \dots, (z^3 + 3zz' + z''), (z^2 + z')$, the first terms $z^m, \dots, z^3, z^2$ are each of them of a higher degree than any other term in the same brackets,

so that attending only to the terms of highest degree the sets may be reduced to $z^m, \dots, z^3, z^2$. But in the determination of the subsequent exponents and coefficients, the omitted terms or some of them would of course come into play: and it is at least not obvious that we can, for the forms of the series which satisfy the differential equation, employ geometrical considerations such as those which were used in regard to the series satisfying the algebraical equation.

11.

Considering as above the coefficients $p_0, p_1, \dots, p_m$ as rational and integral functions of t , it is easy to determine the degrees of these coefficients in such wise that the exponent α of the leading term At^α shall have a given value. Suppose first, this given value is $\alpha = a$, a positive integer greater than 1: the degrees may be $\theta, \theta + a, \theta + 2a, \dots, \theta + ma$, ($\theta = 0$ or any positive integer at pleasure); and not only so, but if we write

$$p_0 = L_0 t^\theta + \dots, \quad p_1 = L_1 t^{\theta+a} + \dots, \quad \dots, \quad p_m = L_m t^{\theta+ma} + \dots,$$

then writing $z = At^a$, clearly the highest power in the equation is $t^{\theta+ma}$, and equating the coefficient hereof to zero, we have

$$L_0 A^m + L_1 A^{m-1} + \dots + L_{m-1} A + L_m = 0,$$

an equation of the degree m for the determination of the coefficient of the leading term At^a ($a > 1$). Assuming that the roots are all unequal, we have m series each of the form $z = At^a + Bt^{a-1} + \dots + K + \frac{L}{t} + \dots$

12.

Suppose $\alpha = 1$, that is, let the leading term be At ; here the degrees may be $\theta, \theta + 1, \theta + 2, \dots, \theta + m$, ($\theta = 0$ or any positive integer as before), but we have a different form for the A -equation. In fact, here ($z = At$) in the sets of terms $\{z^m + \dots + z^{(m-1)}\}, \dots, (z^3 + 3zz' + z'')$, $(z^2 + z')$, the terms in each brackets are of the same degree, viz. the degrees are $m, \dots, 3, 2$; and not only so, but we have $z^2 + z' = (A^2 + A)t^2$, that is, $= [A]^2 t^2$; $z^3 + 3zz' + z'' = [A]^3 t^3, \dots$. Hence if $p_0 = L_0 t^\theta + \dots, p_1 = L_1 t^{\theta+1} + \dots, \dots, p_m = L_m t^{\theta+m} + \dots$, the A -equation is $L_0 [A]^m + L_1 [A]^{m-1} + \dots + L_{m-1} [A] + L_m = 0$, an equation of the degree m as before; assuming that the roots are unequal, we

have here m series each of the form $z = At + B + \frac{C}{t} + \frac{D}{t^2} + \dots$, being, as will presently appear, the case of m regular integrals for the y -equation.

13.

In either of the last-mentioned cases, the formula for the A -equation holds good if any two or more of the coefficients $p_0, p_1, \dots, p_m$ are of the degrees aforesaid, the others of them being of inferior degrees; we have only to put $= 0$ the L 's which belong to the coefficients of inferior degrees. If neither L_0 nor L_m be $= 0$, the equation is still an equation of the order m , with m effective roots; but if any of the coefficients L at the beginning of the equation vanish we thus introduce roots $= \infty$: and if any of the coefficients L at the end of the equation vanish, we thus introduce roots $= 0$: the number of effective roots is $= m$ less the number of roots ∞ or 0 : the case requires further consideration. If the equation has equal roots, then from what precedes, it is easy to infer that we may have distinct series containing as in the general case only integer powers of t , or we may have series containing fractional powers.

14.

To illustrate the case of a series with fractional powers, and also in order to work out a simple example with some completeness, I consider the differential equation of the second order $\frac{d^2y}{dx^2} + p_2y = 0$, giving the z -equation $z^2 + z' + p_2 = 0$; p_2 is a rational and integral function of t ($= \frac{1}{x}$), and we have two cases according as the order of the function is even or odd.

15.

Suppose first that the order is even, and for convenience taking it to be $= 4$, and assuming the value of p_2 accordingly, I consider the equation $z^2 + z' = \alpha t^4 + \beta t^3 + \gamma t^2 + \delta t + \epsilon$. Here the form is at once seen to be $z = At^2 + Bt + C + \frac{D}{t} + \frac{E}{t^2} + \dots$; and observing that we have $\frac{dz}{dt} = 2At + B - \frac{D}{t^2} - \frac{2E}{t^3} + \dots$, we find $A^2 = \alpha$, $2AB - 2A = \beta$, $2AC + B^2 - B = \gamma$, $2AD + 2BC + C = \delta$, $2AE + 2BD + C^2 + D = \epsilon$,

$2AF + 2BE + 2CD + 2E = 0$, ... equations which give

$$A = \sqrt{\alpha}, \quad B = \frac{\beta + 2\sqrt{\alpha}}{2\sqrt{\alpha}}, \quad C = \frac{\gamma}{2\sqrt{\alpha}} - \frac{\beta^2}{8\alpha\sqrt{\alpha}} + \frac{\beta}{4\alpha} + \frac{3}{4\alpha\sqrt{\alpha}}, \quad \dots$$

16.

Similarly, if the series be $z = \frac{F}{t^a} + \dots$ (a being a positive integer $= 2$ at least), then writing as above $t = \frac{1}{x}$, this is $\frac{1}{y} \frac{dy}{dx} = \frac{F}{x^a} + \dots + \frac{A}{x}$, whence $\log y = A \log x + Bx + \frac{1}{2}Cx^2 + \dots$, or passing to the value of y , but expanding the exponential of $Bx + Cx^2 + \dots$, this is $y = x^A(1 + Bx + Cx^2 + \dots)$, viz. we have here a *regular integral*. By what precedes, it appears that, for the values $p_0 = L_0x^{-\theta} + \dots$, $p_1 = L_1x^{-\theta-1} + \dots$, $\dots$, $p_m = L_mx^{-\theta-m} + \dots$, the exponent A is determined by the equation $L_0[A]^m + L_1[A]^{m-1} + \dots + L_{m-1}[A] + L_m = 0$, which is the equation for that purpose considered by Fuchs, and is what I would call the *indicial equation*. The logarithmic forms which belong to the case of equal roots may be deduced from the case of unequal roots, by supposing two or more of the roots to approach each other continuously and become ultimately equal.

17.

We then have $\frac{1}{y} \frac{dy}{dx} = \frac{A}{x^2} + \frac{B}{x} + C + Dx + Ex^2 + \dots$, and consequently

$$\frac{1}{y} \frac{dy}{dx} = \frac{A}{x^2} + \frac{B}{x} + C + Dx + Ex^2 + \dots,$$

where the second exponential is to be expanded in the form $1 + Cx + Dx^2 + \dots$; there are of course two values, say y_1 and y_2 , corresponding to the values $+\sqrt{\alpha}$ and $-\sqrt{\alpha}$ of the radical respectively.

18.

Secondly, if the order be odd, and for convenience taking it to be $= 3$, and assuming the value of p_2 accordingly, I consider the equation $z^2 + z = \alpha t^3 + \beta t^2 + \gamma t + \delta + \epsilon t^{-1} + \dots$. Here the series is at once seen to be $z = At^{\frac{3}{2}} + Bt + Ct^{\frac{1}{2}} + D + Et^{-\frac{1}{2}} + \dots$: we have $z^2 = A^2t^3 + 2ABt^{\frac{5}{2}} + (2AC + B^2)t^2 + \dots$, and we thence derive the equations

$A^2 = \alpha$, $2AB - \beta = 0$, $2AC + B^2 - B = \gamma$, $2AD + 2BC - C = \delta$, $2AE + 2BD + C^2 - D = \epsilon$, ... (where in the next equation we have on the right-hand side ζ , and in each of the subsequent equations we have on the right-hand side 0). We find $E = \dots$, where observe that the coefficients A , C , E , ... contain as a factor $\sqrt{\beta}$, but the coefficients B , D , ... are rational.

19.

We have $z = \frac{1}{y} \frac{dy}{dx} = \frac{A}{x^{\frac{3}{2}}} + \frac{B}{x} + \frac{C}{x^{\frac{1}{2}}} + D + Ex^{\frac{1}{2}} + Fx + \dots$, giving $\log y = \frac{2A}{x^{\frac{1}{2}}} + B \log x + 2Cx^{\frac{1}{2}} + Dx + \frac{2}{3}Ex^{\frac{3}{2}} + \frac{1}{2}Fx^2 + \dots$; and thence passing to the value of y , but expanding the exponential of $2Cx^{\frac{1}{2}} + Dx + \dots$, we have say the integral

$$y_1 = e^{\frac{2A}{x^{\frac{1}{2}}}} x^B (M + Nx^{\frac{1}{2}}),$$

and then also, changing the sign of $\sqrt{\beta}$, the integral

$$y_2 = e^{-\frac{2A}{x^{\frac{1}{2}}}} x^B (M - Nx^{\frac{1}{2}}).$$

Writing for a moment $Q = e^{\frac{2A}{x^{\frac{1}{2}}}}$, $M = 1 + Dx + \dots$, $N = C + Ex + \dots$, the two integrals are $x^B Q (M + Nx^{\frac{1}{2}})$, and $x^B Q^{-1} (M - Nx^{\frac{1}{2}})$: these may be replaced by their sum and difference, viz. these are

$$\begin{aligned} y_1 + y_2 &= x^B M (Q + Q^{-1}) + x^{B+\frac{1}{2}} N (Q - Q^{-1}), \\ y_1 - y_2 &= x^B M (Q - Q^{-1}) + x^{B+\frac{1}{2}} N (Q + Q^{-1}), \end{aligned}$$

where $Q + Q^{-1}$ is a rational function of x , and $Q - Q^{-1}$ is $= \alpha$ multiplied by a rational function of x ; hence $y_1 + y_2$ is $= x^B$ multiplied by a rational function of x , but $y_1 - y_2$ is $= x^{B+\frac{1}{2}}$ multiplied by a rational function of x .

Cambridge, 28 April 1886.

863. Note on the Theory of Linear Differential Equations

[From *Crelle's Journal der Mathem.*, t. ci. (1887), pp. 209–213.]

The theorem V. given by Fuchs in the memoir “Zur Theorie der linearen Differentialgleichungen mit veränderlichen Coefficienten,” *Crelle's Journal*, t. LXVIII. (1868), pp. 354–385 (see p. 374) for the purpose of deciding whether the integrals belonging to a group of roots of the “determinirende Fundamentalgleichung” (or as I call it, the Indicial equation) do or do not involve logarithms, may I think be exhibited in a clearer form.

Starting from the differential equation $P(y) = 0$ of the order m , then if X be any function of x not satisfying the differential equation, we can at once form a differential equation of the order $m+1$, satisfied by all the solutions of the differential equation, and having also the solution $y = X$; the required equation is in fact

$$\frac{d}{dx}P(y) \cdot P(X) - P(y) \cdot \frac{d}{dx}P(X) = 0.$$

This I call the *augmented equation*.

I recall that the equation $P(y) = 0$, considered by Fuchs, is an equation having for each singular point $x = a$, m regular integrals, viz. the coefficients $p_0, p_1, \dots, p_m$ have the forms $q_0(x-a)^m, q_1(x-a)^{m-1}, \dots, q_m$, where $q_0, q_1, \dots, q_m$ are rational and integral functions of $x - a$, q_0 not vanishing for $x = a$, and the other functions $q_1, q_2, \dots, q_m$ not in general vanishing for $x = a$.

Writing $y = (x - a)^\theta$, we obtain $P(x - a)^\theta = I(\theta)(x - a)^\theta +$ higher powers of $(x - a)$, where $I(\theta)$, the coefficient of the lowest power of $(x - a)$, is a function of θ of the order m , which I call the *indicial coefficient*; and equating it to zero, we have $I(\theta) = 0$, the determinirende Fundamentalgleichung, or Indicial equation, being an equation of the order m .

If the roots of this equation are such that no two of them are equal or differ only by an integer number, then we have m particular integrals each of them of the form $y = (x - a)^r +$ higher powers of $(x - a)$, where r is any root of the indicial equation: but if we have in the indicial equation a group of λ roots $r_1, r_2, \dots, r_\lambda$, such that the difference of each two of them is either zero or an integer, then the integrals which correspond to these roots involve or may

involve logarithms; in particular, if any two of the roots are equal, the integrals for the group will involve logarithms.

Consider now the differential equation $P(y) = 0$ in reference to the singular point $x = a$ as above, and writing $X = (x - a)^\epsilon f$, where ϵ is in the first instance arbitrary, and f is a rational and integral function of $x - a$ not vanishing for $x = a$, we form the augmented equation which, observing that we have in general $P(X) = (x - a)^\epsilon Q$, Q a rational and integral function of $x - a$ not vanishing for $x = a$, and dividing the whole equation by $(x - a)^{\epsilon-1}$, may be written

$$\frac{d}{dx}P(y) \cdot (x - a)Q - P(y) \cdot [\epsilon Q + (x - a)\frac{d}{dx}Q] = 0,$$

an equation of the same form as the original equation (but of the order $m + 1$ instead of m), and having an indicial equation.

In fact, writing as before $y = (x - a)^\theta$, we have in $\frac{d}{dx}P(y) \cdot (x - a)Q$ the term of lowest order $\theta I(\theta) \cdot Q \cdot (x - a)^\theta$ and in $P(y) \cdot \epsilon Q$ the term of lowest order $\epsilon I(\theta) \cdot Q \cdot (x - a)^\theta$, whereas in $P(y)(x - a)\frac{d}{dx}Q$ the term of lowest order is of a higher degree. We have thus for the indicial equation of the augmented equation

$$(\theta - \epsilon)I(\theta) = 0,$$

viz. this is the equation $I(\theta) = 0$ with the additional factor $\theta - \epsilon$, corresponding to the additional root ϵ .

Application to Legendre's Equation.

This is Legendre's equation $(1 - x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + (n^2 + n)y = 0$, with $\frac{1}{x}$ substituted for x , so that, instead of a singular point $x = \infty$, there may be a singular point $x = 0$.

Attending to the singular point $x = 0$, we have $P(x^\theta) = (\theta^2 - \theta - n^2 - n)x^\theta + \text{higher powers}$, so that the indicial equation $I(\theta) = 0$ is $\theta^2 - \theta - n^2 - n = 0$, that is, $(\theta + n)(\theta - n - 1) = 0$, or we have the roots $n, n + 1$, which differ by an integer, and thus form a group, if n be = an integer, or be = an integer $+\frac{1}{2}$; to fix the ideas, say that the roots are $-p, p + 1$ or else $p + \frac{1}{2}, p + \frac{1}{2}$ where p is a positive integer.

Writing for greater convenience $x^\epsilon f = x^\epsilon + F$, where F is a sum of powers of x higher than ϵ , we find without difficulty

$$P(x^\epsilon f) = x^\epsilon \{(\epsilon + n)(\epsilon - n - 1) - (\epsilon^2 + \epsilon)\frac{1}{x} + (x - x^2)x^{-\epsilon}F'' - (n^2 + n)x^{-\epsilon}F\},$$

which, so long as ϵ remains arbitrary, is of the form $x^\epsilon Q$, $Q = (\epsilon + n)(\epsilon - n - 1) +$ powers of x ; if however ϵ be a root of the indicial equation, for instance, if $\epsilon = n$, then the expression in brackets $\{\}$ contains at any rate the factor x , so that the form is $P(x^{-n}f) = x^\mu Q$, where μ is $= n+1$ at least; we can however, by a proper determination of the function f , make μ acquire a larger value.

For instance, suppose $n, n+1 = 2, 3$; $\epsilon = n = 2$, and assume

$$x^\epsilon f = x^{-2} + Bx^{-1} + C + Dx + Ex^2 + Fx^3 + Gx^4 + \dots$$

To calculate $P(x^\epsilon f)$, we have

$$P(x^\epsilon f) = \begin{vmatrix} -2 & -B & 0 & D & 2E & 3F & 4G \\ \cdot & -2B & -3C & -4D & -5E & -6F & -7G \\ \cdot & \cdot & 6 & 2 & 0 & 0 & 0 \end{vmatrix}$$

giving $P(x^\epsilon f) = x^{-2}\{6 + 2Bx + (-6C + 6)x^2 + (-4D - 6B + 2D)x^3 + \dots\}$.

We have here if B not $= 0$, $\mu = 1$; but if $B = 0$, then we cannot in any way make the coefficient of x^2 to vanish, and consequently $\mu = 3$.

With this last value of μ , the group $(\mu, n, n+1)$, that is, $(\mu, 2, 3)$, becomes $(3, 2, 3)$ which contains two equal roots, and the conclusion from the theorem thus is that the integrals u_1, u_2 , corresponding to the roots 2, 3, involve logarithmic values.

And similarly in general the integrals u_1, u_2 , corresponding to the roots $p + \frac{1}{2}, p + \frac{1}{2}$ (p any positive integer), involve logarithmic values: this also is right.

The examples exhibit the true character of the theorem, and show I think that it is a less remarkable one than would at first sight appear: in fact, in working them out, we really ascertain by an actual substitution whether the differential equation can be satisfied by series of powers only, without logarithms. Thus for $n = 2$ as above, it appears that the equation is satisfied by the series

$$y = x^{-2} \left[1 - \frac{3}{2}x^2 + \frac{15}{8}x^4 + \dots \right],$$

and the like series with the terms in $x^2, x^4, \dots$ involving $\log x$; and the theorem gives the same result.

864. On Rudio's Inverse Centro-Surface

[From the *Quarterly Journal of Pure and Applied Mathematics*,
vol. XXII. (1887), pp. 156–158.]

Dr F. Rudio, in an inaugural dissertation “Ueber diejenigen Flächen deren Krümmungsmittelpunktsflächen confokale Flächen zweiten Grades sind,” Berlin, 1880, and *Crelle's Journal*, t. XCV., p. 240, has determined the surfaces having for their centre-surface (i.e., the locus of centres of curvature) the aggregate of the confocal quadric surfaces

$$\frac{x^2}{a-\lambda} + \frac{y^2}{b-\lambda} + \frac{z^2}{c-\lambda} = 1, \quad \frac{x^2}{a-\mu} + \frac{y^2}{b-\mu} + \frac{z^2}{c-\mu} = 1,$$

or, what is the same thing, the surfaces orthotomic to the common tangents of these two surfaces.

He obtains, as the final result of an elegant analytical investigation, the following formulae:

$$\begin{aligned} x &= \sqrt{a-\lambda} \cdot \sqrt{\frac{a-\mu}{a-b} \cdot \frac{a-\nu}{a-c}}, \\ y &= \sqrt{b-\lambda} \cdot \sqrt{\frac{b-\mu}{b-c} \cdot \frac{b-\nu}{b-a}}, \\ z &= \sqrt{c-\lambda} \cdot \sqrt{\frac{c-\mu}{c-a} \cdot \frac{c-\nu}{c-b}}, \end{aligned}$$

(values which are such that $\xi^2 + \eta^2 + \zeta^2 = 1$),

And then

$$\begin{aligned} x' &= x + \rho\xi, \\ y' &= y + \rho\eta, \\ z' &= z + \rho\zeta, \end{aligned}$$

viz. the equations give $x, y, z, \xi, \eta, \zeta$, each of them as a function of two independent parameters u, v , ρ is a function of u, v and of the arbitrary constant C ; hence, giving to C any assumed value, we have x', y', z' each of them a function of the two arbitrary parameters u, v ; that is, x', y', z' are the coordinates of a point on a surface, one of a system of parallel surfaces (corresponding to the different values of C) which are the surfaces in question.

Observe that u, v are the elliptic coordinates of the point (x, y, z) on the first of the two confocal surfaces, and that ξ, η, ζ are the cosine-inclinations of one of the tangents from this point to the other confocal surface; so that, if ρ were left arbitrary, the equations $x' = x + \rho\xi, y' = y + \rho\eta, z' = z + \rho\zeta$ would be the equations of a common tangent of the two confocal surfaces; but ρ , as determined, is the distance of the point x, y, z from the point x', y', z' on the required surface.

The expression for ρ involves hyper-elliptic integrals of the first species, which are the same as those which present themselves in the determination of the geodesic lines on the ellipsoid.

865. On Multiple Algebra

[From the *Quarterly Journal of Pure and Applied Mathematics*,
vol. XXII. (1887), pp. 270–308.]

1.

I reproduce a passage from my Presidential Address, British Association, Southport, 1883.

“Outside of ordinary mathematics we have some theories which must be referred to: algebraical, geometrical, logical. It is, as in many other cases, difficult to draw the line: we do in ordinary mathematics use symbols not denoting quantities, which we nevertheless combine in the way of addition and multiplication, $a+b$ and ab , and which may be such as not to obey the commutative law $ab = ba$; in particular, this is or may be so in regard to symbols of operation; and it could hardly be said that any development whatever of the theory of such symbols of operation did not belong to ordinary algebra.

But I do separate from ordinary algebra the system of multiple algebra or linear associative algebra developed in the valuable memoir by the late Benjamin Peirce, “Linear Associative Algebra” (1870, reprinted 1881 in the *American Journal of Mathematics*, vol. IV., with notes and addenda by his son, C. S. Peirce): we here consider symbols A, B , &c., which are linear functions of a determinate number of letters or units i, j, k, l , &c., with coefficients which are ordinary analytical magnitudes real or imaginary, viz. the coefficients are in

general of the form $\alpha + i\beta$, where i is the before-mentioned imaginary, or $\sqrt{-1}$ of ordinary analysis. The letters i, j, k , &c., are such that every binary combination i^2, ij, ji , &c., (ij not in general $= ji$), is equal to a linear function of the letters, but under the restriction of satisfying the associative law; viz. for each combination of three letters $ij \cdot k = i \cdot jk$, so that there is a determinate and unique product of three or more letters; or, what is the same thing; the laws of combination of the units $i, j, k, \dots$ are defined by a multiplication table giving the values of i^2, ij, ji , &c.; the original units may be replaced by linear functions of these units, so as to give rise for the units finally adopted to a multiplication table of the most simple form; and it is very remarkable how frequently in these simplified forms we have nilpotent or idempotent symbols ($i^2 = 0$ or $i^2 = i$ as the case may be), and symbols i, j , such that $ij = ji = 0$; and consequently how simple are the forms of the multiplication tables which define the several systems respectively.

“I have spoken of this multiple algebra before referring to various geometrical theories of earlier date, because I consider it as the general analytical basis, and the true basis, of these theories. I do not realise to myself directly the notions of the addition or multiplication of two lines, areas, rotations, or other geometrical, kinematical, or mechanical entities; and I would formulate a general theory as follows: consider any such entity as determined by the proper number of parameters $a, b, c, \dots$ (for instance, in the case of a finite line given in magnitude and position, these might be the length, the coordinates of one end, and the direction-cosines of the line considered as drawn from this end); and represent it by or connect it with the linear function $ai + bj + ck + \&c.$, formed with these parameters as coefficients and with a given set of units i, j, k , &c. Conversely, any such linear function represents an entity of the kind in question. Two given entities are represented by two linear functions; the sum of these is a like function representing an entity of the same kind, which may be regarded as the sum of the two entities; and the product of them (taken in a determinate order) is in like manner a like function representing an entity of the same kind, which may be regarded as the product of the two entities. The laws of addition and multiplication of the units are given by a multiplication table; and thus the whole theory is made to depend upon that of the multiple algebra or linear associative algebra; and, in particular, assuming (as is or may be the case in the theories referred to) that the product is not in

general commutative, this is simply the consequence of the assumed multiplication table.”

2.

It is proper to refer at once to the original memoir by Benjamin Peirce, to a paper by me “On the Theory of Matrices” (Phil. Trans., 1858), pp. 17–37, [52], and to the “Memoir on the Theory of Matrices” (ibid., 1858, pp. 475–496), [75]. A set of n^2 quantities arranged in a square is a *matrix*; two matrices are added by adding corresponding terms; they are multiplied according to a rule first given by Cayley, viz. if the matrices are (a, b, c) and (a', b', c') , then the product is $(aa' + bb' + cc', ab' + bd' + cf', \dots)$, &c. The theory includes as a particular case the theory of quaternions, where $n = 2$.

3.

I consider a double symbol (x, y) , and assume for the laws of combination:

$$(x, y) + (x', y') = (x + x', y + y'),$$

and

$$(x, y)(x', y') = (X, Y),$$

where P, Q, X, Y are given functions of x, y, x', y' .

For greater simplicity the law of addition is taken to be

$$(x, y) + (x', y') = (x + x', y + y'),$$

so that as regards addition the multiple symbols are commutative and associative. But this is or is not the case for multiplication, according to the form of the given functions X, Y ; for instance, if $(x, y)(x', y') = (xx' - yy', xy' + yx')$, then in regard to multiplication the symbols will be commutative and associative. But if $(x, y, z)(x', y', z') = (yz' - y'z, zx' - z'x, xy' - x'y)$, then the symbols will be associative, but not commutative.

4.

I remark here that we are in general concerned with symbols of a given multiplicity, double symbols (x, y) , triple symbols (x, y, z) , n -tuple symbols $(x_1, x_2, \dots, x_n)$, as the case may be, and that as well

the product as the sum is a symbol *ejusdem generis*, and consequently of the same multiplicity, with the component symbols; this is to be assumed throughout in the absence of an express statement to the contrary.

It is, moreover, proper to narrow the notion of multiplication by restricting it to the case where the terms $(X, Y, \dots)$ of the product are linear functions of the terms $(x, y, \dots)$ and $(x', y', \dots)$ of the component symbols respectively; any other form $(X, Y, \dots)$ is better designated not as a product, but as a combination (or by some other name) of the component symbols $(x, y, \dots)$ and $(x', y', \dots)$.

5.

I assume, moreover, that, if m be any ordinary analytical magnitude, then the product of the multiple symbol $(x, y, \dots)$ and the ordinary magnitude m is the multiple symbol $(mx, my, \dots)$; this is in effect to assume that each of the extraordinaries is commutable with every ordinary magnitude.

6.

I do not attempt any enumeration or classification of the different systems of multiple algebra, but I consider in particular the system of double algebra. In this case, writing $(x, y) = ix + jy$, where i, j are the extraordinaries, we have for the product:

$$(ix + jy)(ix' + jy') = i(xx' - yy') + j(xy' + yx'),$$

which expressions will agree together if, and only if, $i^2 = i$, $ij = j$, $ji = j$, $j^2 = i$, or, as these equations may be written, $i^2 = i$, $ij = ji = j$, $j^2 = i$.

7.

And, conversely, from this multiplication table of the extraordinaries i, j , we obtain $(ix + jy)(ix' + jy') = i(xx' - yy') + j(xy' + yx')$, that is, $(x, y)(x', y') = (xx' - yy', xy' + yx')$, the originally assumed formula of multiplication.

In the example just given, we have $i^2 = i$, $ij = ji = j$, viz. the symbol i comports itself like unity, and may be put $= 1$; we have $(x, y) = x + jy$, with the multiplication table $1 \cdot j = j \cdot 1 = j$, $j^2 = -1$;

or simply the equation $j^2 = -1$; and then $(x + jy)(x' + jy') = xx' - yy' + j(xy' + yx')$; and it is convenient to regard 1 as an extraordinary, and speak of the system of extraordinaries 1, j .

8.

The separate terms $x, y, \dots$, whatever be their number, may always without loss of generality be arranged in a line; but it may be convenient to arrange them in a different form, for instance, in that of a square; we may have for instance symbols x, y, z, w with the laws of combination $(x, y, z, w) + (x', y', z', w') = (x + x', y + y', z + z', w + w')$, and $(x, y, z, w)(x', y', z', w') = (X, Y, Z, W)$.

9.

A multiple symbol may be connected with a geometrical or physical entity of any kind: viz. any such entity, depending on a number of parameters susceptible of analytical magnitude, to fix the ideas say on two parameters (x, y) , may be connected with the multiple symbol (x, y) . We cannot in general directly conceive the addition or multiplication of such entities, but the assumed laws of combination of the multiple symbols in effect serve as definitions of the operations. Thus $(x, y) + (x', y') = (x + x', y + y')$; $(x, y) \cdot (x', y') = (xx' - yy', xy' + yx')$: or the sum of the entities whose parameters are x, y and x', y' is by definition the entity whose parameters are $x + x', y + y'$; and the product of the same entities is by definition the entity whose parameters are $xx' - yy', xy' + yx'$.

The entities are thus, as regards addition, commutative and associative; but, as regards multiplication they are or are not commutative, or associative, according to the assumed law of multiplication of the multiple symbols.

10.

If, as above, the multiple symbol be represented by means of extraordinaries, $(x, y) = ix + jy$, then the extraordinaries i, j , qua multiple symbols $(1, 0), (0, 1)$, are themselves special entities of the kind in question, their laws of combination (as such special entities) being included in the general laws for the combination of such entities; or, if we please, these general laws being derived from the assumed laws for the special entities.

11.

The theory of imaginaries, or imaginary quantities, of ordinary analysis, belongs to the foregoing theory. The ordinary imaginary $x + iy$ is the double symbol (x, y) , with the law of multiplication $(x, y)(x', y') = (xx' - yy', xy' + yx')$, that is, $i^2 = -1$. And the theory extends to the "plurality of imaginaries," considered by Hamilton and others.

12.

Another peculiarity may be noticed: suppose that, in general, $(x, y, z, w) + (x', y', z', w') = (x + x', y + y', z + z', w + w')$; $(x, y, z, w)(x', y', z', w') = (X, Y, Z, W)$; then, as a particular case hereof, we have the addition-equation $(x, y, 0, 0) + (x', y', 0, 0) = (x + x', y + y', 0, 0)$ and it may very well be that for the same two symbols the multiplication-equation may take the form $(x, y, 0, 0)(x', y', 0, 0) = (0, 0, Z, W)$, (Z, W each of them a function of x, y, x', y'); viz. here, in a different point of view, the component symbols $(x, y, 0, 0)$ and $(x', y', 0, 0)$, are double systems of a certain kind $\{x, y\}$, $\{x', y'\}$, and the product is a double system of a different kind $[Z, W]$, or it might have been $(0, Y, Z, W)$, a triple system $[Y, Z, W]$.

Of course, all peculiarity disappears when we revert from the particular symbols $(x, y, 0, 0)$ to the general symbols (x, y, z, w) , from which we may regard them as derived.

13.

The peculiarity just referred to presents itself naturally when we regard the symbols as representing geometrical or physical entities; it may very well be that the product of two entities of a certain kind is taken to be an entity of a different kind (for instance, the product of two points to be a line), and that the analytical theory of the multiple symbol is constructed in order to such a relation; and it may further be that only the symbol $(x, y, 0, 0)$ or $\{x, y\}$ is interpreted with reference to an entity of the one kind, the general symbol (x, y, z, w) being interpreted as representing an entity of the other kind.

14.

In the theory of imaginaries, we have the product of two extraordinaries i, j , each of them an imaginary of the form $x + iy$, is equal to a linear function of 1 and i ; that is, the product of two entities of the one kind is an entity of the same kind. But we may very well have a system in which this is not the case: for instance, taking extraordinaries $1, i, j, k$, we may have the product of any two of them a linear function of all four. An interesting case is when the product of two imaginaries is a quaternion; here the product of two entities of the one kind is an entity of a different kind.

15. Geometrical Representation of Imaginaries.

Art. Nos. 16–42.

16. Argand, 1806. The theory is that established by Argand in the paper “Essai sur une manière de représenter les quantités imaginaires dans les constructions géométriques” (Paris, 1806, republished in *Gergonne’s Annales*, t. IV. (1813), pp. 133–147). Argand denotes a directed line by an algebraic quantity; and in particular he has the theorem that the sum of two such lines is the diagonal of the parallelogram formed by them. As to the product, if two lines have the lengths r, r' and inclinations θ, θ' to the axis, then the product-line has the length rr' and the inclination $\theta + \theta'$; in particular, if the lines are at right angles, the product is the line having length rr' and inclination $\theta + \theta' + 90^\circ$. He thus arrives at the representation of $\sqrt{-1}$ by a line at right angles to the axis.

17. Francais, 1813. The theory was independently discovered by Francais; see his paper “Nouveaux principes de géométrie de position, et interprétation géométrique des symboles imaginaires” (*Gergonne’s Annales*, t. IV. (1813), pp. 61–71). He gives the following rule for multiplication: to multiply two directed lines, multiply the lengths and add the inclinations.

18. Warren, 1828. John Warren, in his “Treatise on the Geometrical Representation of the Square Roots of Negative Quantities” (Cambridge, 1828), develops the theory at length. Peacock, Preface to *Algebra*, 1830, speaks of Warren’s work as “distinguished for great originality and for extreme boldness in the use of definitions.”

I quote certain articles as follows:

- (1) All straight lines drawn in a given direction from a given point

are represented in length and direction by algebraic quantities; and in the following treatise whenever the word quantity is used it is to be understood as signifying a line.

- (2) DEF. The given point from which the straight lines are measured is called the origin.
- (3) DEF. The sum of two quantities is the diagonal of the parallelogram whose sides are the two quantities.
- (12) DEF. The first of four quantities is said to have to the second the same ratio which the third has to the fourth: when the first has in length to the second the same ratio which the third has in length to the fourth, according to Euclid's definition; and also the angle at which the fourth is inclined to the third is equal to the angle at which the second is inclined to the first.

19. Peacock, 1830. George Peacock, "A Treatise on Algebra" (8vo. Cambridge, 1830). The statement of the theory is substantially identical with that of the earlier writers. Thus Algebra (1830), p. 362, No. 439, is: "We have shown on a former occasion that, if a designated a line in one direction, $-a$ must designate a line in the direction opposite to it, or making an angle with the former equal to two right angles or 180 degrees; but inasmuch as $a\{\sqrt{-1}\}^2 = -a$, it follows that the double affectation of the line a with the sign $\sqrt{-1}$ produces a result represented by $-a$, and is consequently equivalent to its transfer through 180 degrees: it follows therefore that its single affectation with the sign $\sqrt{-1}$ is equivalent to its transfer through half that angle, or through 90 degrees; or, in other words, if a represents a line, $a\sqrt{-1}$ will represent a line at right angles to the former."

20. Buee, 1806; Gregory and Walton. I have referred to Buee's notion of a curve as having branches in a perpendicular plane; this notion is generalised and developed in the papers:

- Gregory, D. F. "On the existence of branches of curves in several planes." *Camb. Math. Journ.*, t. I. (1839), pp. 259–266.
- Walton. "On the general interpretation of equations between two variables in analytical geometry." *Do.* t. II. (1840), pp. 103–113.
- "On the general theory of multiple points." *Do.* pp. 155–167.

And see also Gregory's *Examples of the Processes of the Differential and Integral Calculus* (8vo. Cambridge, 1841, pp. 172 et seq.); and the criticism on these papers: Salmon, *A Treatise on the Higher Plane Curves* (8vo. Dublin, 1852, The Note on Imaginary Points and Curves, pp. 301–306).

21. Gauss. The theory is that established by Gauss in the paper “Theoria Residuorum biquadraticorum, Commentatio secunda” (*Göttingen Comm. Soc. Reg.*, 1832; *Werke*, vol. II., pp. 93–148), where he speaks of the “vera natura quantitatum algebraicarum imaginarium” being “per numeros $a + bi$ representari solere.”

22.

But we have, in fact, passed out of this view, and have come to regard $a + bi$ as an ordinary analytical magnitude; viz. in every case an ordinary symbol represents or may represent such a magnitude, and the magnitude (and as a particular case thereof, the symbol i) is commutable with the extraordinaries of any system of multiple algebra.

And similarly in analytical geometry, without seeking for any real representation, we deal with imaginary points, lines, &c., that is, with points, lines, &c., depending on parameters of the foregoing form $a + ib$.

15. Geometrical Representation of Imaginaries.

Art. Nos. 16–42.

16. Argand, 1806. The theory is that established by Argand in the paper “Essai sur une manière de représenter les quantités imaginaires dans les constructions géométriques” (Paris, 1806, republished in *Gergonne's Annales*, t. IV. (1813), pp. 133–147). Argand denotes a directed line by an algebraic quantity; and in particular he has the theorem that the sum of two such lines is the diagonal of the parallelogram formed by them. As to the product, if two lines have the lengths r , r' and inclinations θ , θ' to the axis, then the product-line has the length rr' and the inclination $\theta + \theta'$; in particular, if the lines are at right angles, the product is the line having length rr' and inclination $\theta + \theta' + 90^\circ$. He thus arrives at the representation of $\sqrt{-1}$ by a line at right angles to the axis.

17. Francais, 1813. The theory was independently discovered by

Francais; see his paper “Nouveaux principes de géométrie de position, et interprétation géométrique des symboles imaginaires” (*Gergonne’s Annales*, t. iv. (1813), pp. 61–71). He gives the following rule for multiplication: to multiply two directed lines, multiply the lengths and add the inclinations.

18. Warren, 1828. John Warren, in his “Treatise on the Geometrical Representation of the Square Roots of Negative Quantities” (Cambridge, 1828), develops the theory at length. Peacock, Preface to *Algebra*, 1830, speaks of Warren’s work as “distinguished for great originality and for extreme boldness in the use of definitions.”

I quote certain articles as follows:

- (1) All straight lines drawn in a given direction from a given point are represented in length and direction by algebraic quantities; and in the following treatise whenever the word quantity is used it is to be understood as signifying a line.
- (2) DEF. The given point from which the straight lines are measured is called the origin.
- (3) DEF. The sum of two quantities is the diagonal of the parallelogram whose sides are the two quantities.
- (12) DEF. The first of four quantities is said to have to the second the same ratio which the third has to the fourth: when the first has in length to the second the same ratio which the third has in length to the fourth, according to Euclid’s definition; and also the angle at which the fourth is inclined to the third is equal to the angle at which the second is inclined to the first.

19. Peacock, 1830. George Peacock, “A Treatise on Algebra” (8vo. Cambridge, 1830). The statement of the theory is substantially identical with that of the earlier writers. Thus *Algebra* (1830), p. 362, No. 439, is: “We have shown on a former occasion that, if a designated a line in one direction, $-a$ must designate a line in the direction opposite to it, or making an angle with the former equal to two right angles or 180 degrees; but inasmuch as $a\{\sqrt{-1}\}^2 = -a$, it follows that the double affectation of the line a with the sign $\sqrt{-1}$ produces a result represented by $-a$, and is consequently equivalent to its transfer through 180 degrees: it follows therefore that its single affectation with the sign $\sqrt{-1}$ is equivalent to its transfer through half

that angle, or through 90 degrees; or, in other words, if a represents a line, $a\sqrt{-1}$ will represent a line at right angles to the former.”

20. Buee, 1806; Gregory and Walton. I have referred to Buee’s notion of a curve as having branches in a perpendicular plane; this notion is generalised and developed in the papers:

- Gregory, D. F. “On the existence of branches of curves in several planes.” *Camb. Math. Journ.*, t. I. (1839), pp. 259–266.
- Walton. “On the general interpretation of equations between two variables in analytical geometry.” *Do.* t. II. (1840), pp. 103–113.
- “On the general theory of multiple points.” *Do.* pp. 155–167.

And see also Gregory’s *Examples of the Processes of the Differential and Integral Calculus* (8vo. Cambridge, 1841, pp. 172 et seq.); and the criticism on these papers: Salmon, *A Treatise on the Higher Plane Curves* (8vo. Dublin, 1852, The Note on Imaginary Points and Curves, pp. 301–306).

21. Gauss. The theory is that established by Gauss in the paper “*Theoria Residuorum biquadraticorum, Commentatio secunda*” (*Gottingen Comm. Soc. Reg.*, 1832; *Werke*, vol. II., pp. 93–148), where he speaks of the “vera natura quantitatum algebraicarum imaginarium” being “per numeros $a + bi$ representari solere.”

The Barycentric Calculus: Möbius.

Art. Nos. 30–34.

30. Möbius, 1827. Möbius, *Der barycentrische Calcul*, 8vo. Leipzig, 1827, reprinted in Vol. I. of the *Gesammelte Werke*, Leipzig, 1885. The theory applies to points in a line, in a plane, or in space; but it will be sufficient to consider the case of points in a plane.

The idea is that, taking in the plane any three points A, B, C as fundamental points, then regarding these as loaded with the proper positive or negative weights p, q, r , any other point P of the plane may be regarded as the centre of gravity of the three weights; we have the equation $P = pA + qB + rC$, where $p + q + r = 1$; this is the *barycentric* representation of the point P . And then, taking another point $P' = p'A + q'B + r'C$, we have $P + P' = (p + p')A + (q + q')B + (r + r')C$, which is the barycentric representation of the sum of the two points. And so for the product.

31. The system of Möbius is, in fact, a system of multiple algebra; the points A, B, C are extraordinary, and the point $P = pA + qB + rC$ is a multiple symbol of which the multiplicities are (p, q, r) . The multiplication table is a table giving the products A^2, AB, BA, AC , &c.; and from this the product of any two points $P = pA + qB + rC$ and $P' = p'A + q'B + r'C$ can be at once obtained.

32. In particular, if $P = P'$, we obtain P^2 , the barycentric square of the point P ; and so on. Möbius does not use any general multiplication table, but he has a special theory of the product of two points, which is, in fact, the theory of the products AB, AC, BC , each of them being a determinate point of the line joining the two points; and from these he obtains the product of any two points P, P' as a point on the line PP' .

33. The method is applied very elegantly to the solution of problems, for instance, t. V., p. 248; given two lines AB and CD , to find a point H , such that the triangles ABH, CDH may be directly similar, that is, such that they can be, by a rotation of one of them, brought to be similarly situate. The condition for this is $\frac{AH}{AB} = \frac{CH}{CD}$, that is, $AH \cdot CD = AB \cdot CH$, $= AB(AH - AC)$, whence $AB \cdot AC = AH(AB - CD)$; or constructing a point E such that $BE = DC$, viz. the line BE is drawn from B equal and parallel to and in the same sense with DC , then $AB - CD = AE$, and the foregoing equation becomes $AH \cdot AE = AB \cdot AC$, or say $\frac{AH}{AB} = \frac{AC}{AE}$; viz. the required point H is such that the triangle ACH is directly similar to AEB .

34. I notice an expression t. VIII., p. 19, for the area of a triangle ABC , $= \frac{1}{4}(BC \cdot \text{cj. } AB - AB \cdot \text{cj. } BC)$, where cj. AB , conjugate of AB , means the line through the mid-point of AB at right angles to AB .

Quaternions.

Art. Nos. 35–42.

35. A quaternion is the sum of a scalar or mere magnitude w , plus a vector $ix + jy + kz$, which represents a line drawn from an origin to the point whose rectangular coordinates are x, y, z ; we have thus a quaternion $Q = w + ix + jy + kz$, representing a line, and in connexion therewith a scalar w . Hamilton refers to the theory of Warren and Peacock as having in part suggested his investigations,

but the contrast is very striking; neither the form $x + iy$, where the directed magnitude x is unaccompanied by an extraordinary, nor the form $kx + iy$ with any multiplication table for k, i , is in any wise analogous to the quaternion form.

The system is a system of multiple algebra; $1, i, j, k$ are extraordinary, and the multiplication table is $i^2 = j^2 = k^2 = ijk = -1$, whence $ij = k, jk = i, ki = j, ji = -k, kj = -i, ik = -j$.

36. But for the multiplication of two vectors, the form is simplified: viz. we have

$$W = (ix + jy + kz)(ix' + jy' + kz') = -(xx' + yy' + zz') + i(yz' - y'z) + j(zx' - z'x) + k(xy' - x'y)$$

We may consider unit-vectors, $U = ix + jy + kz$, where $x^2 + y^2 + z^2 = 1$, and unit-quaternions $= w + ix + jy + kz$, where $w^2 + x^2 + y^2 + z^2 = 1$; obviously the general form of a vector is $V = rU$, or, say the length is r , and the cosine-inclinations are x, y, z ; similarly, the general form of a unit-quaternion is $= \cos \theta + \sin \theta \cdot U$, and that of a quaternion is $Q = T(\cos \theta + \sin \theta \cdot U)$, T is the tensor, θ the amplitude, and x, y, z the cosine-inclinations of the unit-vector.

For the product of two unit-vectors, we have $UU' = -(xx' + yy' + zz') + i(yz' - y'z) + j(zx' - z'x) + k(xy' - x'y)$, which is a unit-quaternion.

37. The product of a quaternion by a vector is a vector; viz. $QV =$ a vector, say it is $V'' = iX'' + jY'' + kZ''$. Let δ , considered as a positive angle less than π , so that $\cos \delta = xx' + yy' + zz'$, $\sin \delta = +\sqrt{1 - \cos^2 \delta}$, be the inclination of the two vectors; we have $\cos \delta = xx' + yy' + zz'$.

38. The figure serves also to represent the unit-quaternion $= \cos \delta + \sin \delta \cdot U''$, which is the product of the two vectors.

Observe that, in thus expressing a quaternion as a product of two vectors, only the plane of the vectors and the angle between them are determinate; we may rotate the angle UOU' in its own plane at pleasure.

In particular, if the two vectors are at right angles, then $\delta = 90^\circ$, and the product is $UU'' = U''$, viz. the product of the two unit-vectors at right angles to each other is a unit-vector at right angles to the plane of the two unit-vectors.

If $\delta = 0$, then the two unit-vectors coincide, and we have $U^2 = -1$, viz. the square of a unit-vector is the scalar -1 .

The product of any two vectors is given by the like formulae; if $V = rU$, $V' = r'U'$ respectively, then $VV' = rr' \cdot UU'$.

39. We hence find $QQ' = QU \cdot (-UQ') = VV' = (iX + jY + kZ)(iX' + jY' + kZ'), = -XX' - YY' - ZZ' + i(YZ' - Y'Z) + j(ZX' - Z'X) + k(XY' - X'Y)$. This should, of course, be identically equal to $ww' - xx' - yy' - zz' + i(wx' + xw' + yz' - y'z) + j(wy' + yw' + zx' - z'x) + k(wz' + zw' + xy' - x'y)$, on substituting for X, Y, Z, X', Y', Z' their values as above.

40. The steps are, taking U the unit-vector at right angles to Q, Q' , then the vectors.

41. There is a kind of quaternion operator $Q^{-1}(\)Q$, or, what is the same thing, if $\Lambda = i\lambda + j\mu + k\nu$, $(1 - \Lambda)(\)(1 + \Lambda)$, which is the symbol of a rotation; viz. the operand, say the vector $ix + jy + kz$, is to be placed within the vacant $(\)$, say $Q^{-1}(\)Q \cdot (ix + jy + kz)$ means $Q^{-1}(ix + jy + kz)Q$; and this being so, the result is a vector $ix_1 + jy_1 + kz_1$, where the x_1, y_1, z_1 are what the x, y, z become by a rotation of the vector $ix + jy + kz$ about an axis and through an angle determined by the quaternion Q ; viz. if as above then writing $Q = 1 + i\lambda + j\mu + k\nu$, $\lambda, \mu, \nu = \tan \frac{1}{2}\theta \cos f, \tan \frac{1}{2}\theta \cos g, \tan \frac{1}{2}\theta \cos h$, where $\cos^2 f + \cos^2 g + \cos^2 h = 1$, and therefore $\lambda^2 + \mu^2 + \nu^2 = \tan^2 \frac{1}{2}\theta$, then f, g, h are the inclinations of the axis, and θ is the angle of rotation.

The actual formula is

$$x_1 = \frac{1}{1 + \lambda^2 + \mu^2 + \nu^2} [(1 + \lambda^2 - \mu^2 - \nu^2)x + 2(\lambda\mu + \nu)y + 2(\lambda\nu - \mu)z, \\ + 2(\lambda\mu - \nu)x + (1 - \lambda^2 + \mu^2 - \nu^2)y + 2(\mu\nu + \lambda)z \\ + 2(\lambda\nu + \mu)x + 2(\mu\nu - \lambda)y + (1 - \lambda^2 - \mu^2 + \nu^2)z]$$

viz. the form on the right-hand side is $i(\alpha x + \alpha'y + \alpha''z) + j(\beta x + \beta'y + \beta''z) + k(\gamma x + \gamma'y + \gamma''z)$.

The Ausdehnungslehre: Grassmann.

Art. Nos. 43–63.

43. Grassmann. *Die lineale Ausdehnungslehre, ein neuer Zweig der Mathematik*, Leipzig, 1844; 2nd edition, with the title *Die Ausdehnungslehre von 1844*, Leipzig, 1878. — *Die Ausdehnungslehre, vollständig und in strenger Form bearbeitet*, Berlin, 1862, referred to as *Die Ausdehnungslehre von 1862*.

44. Plane Geometry. The representation of a point is that employed by Möbius in the Barycentrische Calcul (1827), viz. considering in the plane three fixed points A_1, A_2, A_3 , and in regard to

the triangle formed by these points, taking x_1, x_2, x_3 as the areal coordinates of a point x (x_1, x_2, x_3 are equal to the areas of the triangles $xA_2A_3, xA_3A_1, xA_1A_2$, each divided by the area of the triangle $A_1A_2A_3$; whence $x_1 + x_2 + x_3 = 1$), we write $x = x_1e_1 + x_2e_2 + x_3e_3$ as the representation of the point x . Here e_1, e_2, e_3 are extraordinaries, or they may also be regarded as points, viz. as the points A_1, A_2, A_3 respectively.

45. The product of the two points is the line joining them; the product of three points is the area of the triangle formed by them. We have thus a system of multiple algebra in which the sum is a point, but the product of two points is a line, and the product of three points is an area; this is a case of the peculiarity referred to in No. 12.

46. In the *Ausdehnungslehre* von 1844, the multiplication is what Grassmann calls “linear” multiplication; viz. the product of the two points $x = x_1e_1 + x_2e_2 + x_3e_3$ and $y = y_1e_1 + y_2e_2 + y_3e_3$ is the line represented by

$$xy = (x_1y_2 - x_2y_1)e_1e_2 + (x_2y_3 - x_3y_2)e_2e_3 + (x_3y_1 - x_1y_3)e_3e_1,$$

where e_1e_2, e_2e_3, e_3e_1 are the lines A_1A_2, A_2A_3, A_3A_1 respectively.

47. In the *Ausdehnungslehre* von 1862, the multiplication is what Grassmann calls “outer” or “extensive” multiplication; the product is in this case the same as in the linear multiplication, but the theory is presented in a different form.

48. The product of three points x, y, z is the area of the triangle formed by them; it is a quantity of the third order, and is expressed as a multiple of the fundamental area $e_1e_2e_3$ (the area of the triangle $A_1A_2A_3$).

49. Grassmann considers also what he calls “inner” or “regressive” multiplication; this is a multiplication in which the product of two lines is a point, the product of a line and a point is an area, &c. The theory is very elegant, but I do not attempt to explain it here.

50. The *Ausdehnungslehre* contains also a theory of functions of multiple quantities; viz. Grassmann considers functions of points, lines, areas, &c., and develops a theory of differentiation and integration for such functions. This part of the theory is very interesting, but it is too extensive to be here considered.

51. The *Ausdehnungslehre* extends to space of any number of dimensions; in particular, for three-dimensional space, we have four

fundamental points A_1, A_2, A_3, A_4 , and the coordinates x_1, x_2, x_3, x_4 of a point x are the tetrahedral coordinates (volumes of the tetrahedra $xA_2A_3A_4$, &c., each divided by the volume of the tetrahedron $A_1A_2A_3A_4$; whence $x_1 + x_2 + x_3 + x_4 = 1$). The product of two points is the line joining them; the product of three points is the plane through them; the product of four points is the volume of the tetrahedron formed by them.

52.–63. [The remaining sections on the *Ausdehnungslehre*, treating of inner multiplication, the length of a stroke, new representation of a stroke, and further developments of Grassmann's theory, require transcription from the original.]

866. Note on Kiepert's L -Equations, in the Transformation of Elliptic Functions

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 75–77.]

It appears, by comparison with Klein's paper "Ueber die Transformation u. s. w.," *Math. Annalen*, t. xiv. (1878), see p. 144, that Kiepert's L made use of in the Memoir "Ueber Theilung und Transformation der elliptischen Functionen," *Math. Annalen*, t. xxvi. (1886), pp. 369–454, is, in fact, the square of the multiplier, "für das durch $\sqrt[12]{\Delta}$ normirte Integral," viz. considering the general quartic function $(a, \dots)(x, 1)^4 = (a, b, c, d, e)(x, 1)^4$, and the transformed function $(a_1, \dots)(y, 1)^4$, then we have

$$\frac{L^2 \sqrt[12]{\Delta} dx}{\sqrt{(a, \dots)(x, 1)^4}} = \frac{\sqrt[12]{\Delta_1} dy}{\sqrt{(a_1, \dots)(y, 1)^4}},$$

where if

$$I = ae - 4bd + 3c^2, \quad J = ace - ad^2 - b^2e + 2bcd - c^3,$$

and similarly I_1, J_1 , are the invariants of the two functions, then Δ, Δ_1 are the discriminants

$$\Delta = I^3 - 27J^2, \quad \Delta_1 = I_1^3 - 27J_1^2,$$

and the γ_2, γ_3 of Kiepert's equations are

$$\gamma_2 = I \div \sqrt[3]{\Delta}, \quad \gamma_3 = J \div \sqrt{\Delta},$$

whence

$$\gamma_2^3 - 27\gamma_3^2 = 1.$$

In particular, if the forms are

$$1 - x^2 \cdot 1 - k^2 x^2, \quad \text{and} \quad 1 - y^2 \cdot 1 - \lambda^2 y^2,$$

then Δ, Δ_1 are the M and if as usual $k = u^4$, $\lambda = v^4$, and M is the multiplier for the form

$$\frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}} = \frac{M dy}{\sqrt{1 - y^2 \cdot 1 - \lambda^2 y^2}},$$

then we have

$$\begin{aligned} I &= \frac{1}{12}(1 + 14u^8 + u^{16}), \\ J &= \frac{1}{216}(1 + u^8)(1 - 34u^8 + u^{16}), \\ \Delta &= \frac{1}{16}u^8(1 - u^8)^4, \quad \Delta_1 = \frac{1}{16}v^8(1 - v^8)^4, \end{aligned}$$

$$\gamma_2 = \frac{1}{6}\sqrt[3]{2} \frac{1 + 14u^8 + u^{16}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}}, \quad \gamma_3 = \frac{1}{54} \frac{(1 + u^8)(1 - 34u^8 + u^{16})}{u^4(1 - u^8)^2},$$

and thence

$$L^2 = \frac{v^{\frac{8}{3}}(1 - v^8)^{\frac{4}{3}}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}} \cdot \frac{1}{M},$$

which last equation is the expression for L^2 in terms of the Jacobian symbols u, v, M .

As an easy verification in a particular case, suppose $n = 5$. We have here

$$\begin{aligned} L^2 &= \frac{v^{\frac{8}{3}}(1 - v^8)^{\frac{4}{3}}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}} \cdot \frac{1}{M}, \quad M = \frac{v(1 - uv^5)}{v - u^5}, \quad \left(= \frac{v + u^5}{5u(1 + u^3v)} \right), \\ u^6 - v^6 + 5u^2v^2(u^2 - v^2) + 4uv(1 - u^4v^4) &= 0, \end{aligned}$$

$$\gamma_2 = \frac{1}{6}\sqrt[3]{2} \frac{1 + 14u^8 + u^{16}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}};$$

and it should be possible, by eliminating u, v, M , to deduce hence the L -equation

$$L^{12} + 10L^6 - 12\gamma_2L^2 + 5 = 0. \quad (\text{Kiepert, p. 428.})$$

It does not seem in any wise easy to do this in the case of an arbitrary modulus; but writing the modular equation in the form

$$(u^2 - v^2)(u^4 + 6u^2v^2 + v^4) + 4uv(1 - u^4v^4) = 0,$$

we satisfy this by

$$uv - 1 = 0, \quad u^4 + 6u^2v^2 + v^4 = 0,$$

or say by

$$v = \frac{1}{u}, \quad u^8 + 6u^4 + 1 = 0,$$

and the equation may be verified for this particular modulus.

We have

$$1 + 14u^8 + u^{16} = 48u^8, \quad (1 - u^8)^2 = 32u^8,$$

and consequently

$$\gamma_2 = \frac{1}{6} \sqrt[3]{2} \cdot \frac{48u^8}{u^{\frac{8}{3}}(32u^8)^{\frac{4}{3}}} = \frac{1}{2 \cdot 3} \cdot 2^{\frac{1}{3}} \cdot \frac{2^4 \cdot 3u^8}{u^{\frac{8}{3}} \cdot 2^{\frac{10}{3}} \cdot u^{\frac{16}{3}}} = 1, \quad (\text{whence also } \gamma_3 = 0).$$

Moreover

$$M = \frac{\frac{1}{u} - u}{\frac{1}{u} - u^5} = -\frac{1}{u^2} \frac{1 - u^2}{1 - u^6} = \frac{-1}{u^2(1 + u^2 + u^4)},$$

and thence

$$L^2 = \frac{\frac{1}{u^{\frac{8}{3}}}(1 - \frac{1}{u^8})^{\frac{4}{3}}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}} \cdot \frac{1}{M} = \frac{(u^8 - 1)^{\frac{4}{3}}}{u^4(1 - u^8)^{\frac{4}{3}}} \cdot \frac{1}{M} = -\frac{1}{u^4 M},$$

that is,

$$L^2 = 1 + u^2 + u^{-2}.$$

But we have

$$u^4 = -3 + 2\sqrt{2},$$

and thence

$$u^2 = i(1 - \sqrt{2}), \quad u^{-2} = i(1 + \sqrt{2}),$$

and

$$L^2 = 1 + 2i,$$

whence

$$L^{12} + 10L^6 - 12\gamma_2 L^2 + 5 = (117 + 44i) + 10(-11 - 2i) - 12(1 + 2i) + 5 = 0,$$

or the L -equation is satisfied.

Cambridge, 14 March 1887.

867. Note on the Jacobian Sextic Equation

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 78–84.]

In the Jacobian sextic equation

$$(z - a)^6 - 4a(z - a)^5 + 16b(z - a)^3 - 4c(z - a) + 5b^2 - 4ac = 0,$$

if A, B, C, D, E, F are the square roots (each with a determinate sign) of the roots $z_0, z_1, z_2, z_3, z_4, z_\infty$ of the equation, and if ϵ be an imaginary fifth root of unity, such that $\epsilon + \epsilon^4 - \epsilon^2 - \epsilon^3 = \sqrt{5}$, then we have between the square roots a system of three linear equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= B + \epsilon^3 C + \epsilon D + \epsilon^4 E + \epsilon^2 F, \\ 0 &= B + \epsilon^2 C + \epsilon^4 D + \epsilon E + \epsilon^3 F, \end{aligned}$$

see Briochi's *Funzioni ellittiche* (Milan, 1880), third appendix, p. 402.

The sextic equation is irreducible, and there is thus nothing to distinguish any one square root from any other square root. But in the foregoing system of equations, A is distinguished from the other square roots; hence the system must be one of 6 systems, wherein the place occupied by A is occupied by A, B, C, D, E, F respectively. Observe however that the letters being square roots, it may very well happen, and (as will appear) it does in fact happen, that the passage from the first to another system implies a change of sign in certain of the letters A, B, C, D, E, F .

The selection of one of the square roots as $= A$, does not determine which of the other square roots shall be $= B, C, D, E, F$ respectively: and in fact each system of equations may be represented in 10 different forms: viz. by multiplying the second and third equations by $\epsilon, \epsilon^2, \epsilon^3, \epsilon^4$ respectively, we obtain four new forms: we have thus five forms: and then in each of them transposing the second and third equations, we have five other forms: we have thus in all 10 forms.

Write

$$\begin{aligned}\Omega &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= B + \epsilon^3 C + \epsilon F + \epsilon D + \epsilon^4 E, \\ 0 &= B + \epsilon^2 C + \epsilon^4 F + \epsilon^3 D + \epsilon E,\end{aligned}$$

the 10 forms of the system are

$$\begin{array}{c|c} A B C D E F & A B F E D C \\ A C D E F B & A C B F E D \\ A D E F B C & A D C B F E \\ A E F B C D & A E D C B F \\ A F B C D E & A F E D C B \end{array}$$

where observe that the five forms on the left are connected by cyclical substitutions in the last five letters; and that the forms in the same column are connected by two inversions of the last four letters.

It is hardly necessary to remark that $ABFEDC$ means the system of three equations similar to the original equations; and similarly that $ACDEFB$ means the system of three equations where the first equation is the same with the first equation, and the second and third equations only differ by a factor from the second and third equations respectively of the original form. And similarly as regards the others of the 10 forms.

I say that we have a second system $BCAFDE$ (where $\bar{D}$, $\bar{E}$ mean D , E respectively): in fact, this denotes the system of three equations

$$\begin{aligned}0 &= -B\sqrt{5} + C + A + F - D - E, \\ 0 &= C + \epsilon^3 A + \epsilon^2 F - \epsilon D - \epsilon^4 E, \\ 0 &= C + \epsilon^2 A + \epsilon^4 F - \epsilon^3 D - \epsilon E,\end{aligned}$$

which are deducible from the original three equations.

868. On the Intersection of Curves

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 85–90.]

It is only recently that I have studied Bacharach's paper "Ueber den Cayley'schen Schnittpunktsatz," *Math. Ann.*, t. xxvi. (1885),

pp. 275–299: his theorem in regard to the case where the δ points lie on a curve of the order $\gamma - 3$ is a very interesting and valuable one, but I consider it rather as an addition than as a correction to my original theorem; and I cannot by any means agree that the method by counting of constants is to be rejected as not trustworthy; on the contrary, it seems to me to be the proper foundation of the whole theory; it must of course be employed with due consideration of special cases. I reproduce the theorem in what appears to me the complete form.

Writing with Bacharach

$$r \geq m, \quad r \geq n, \quad r \geq m+n-3, \quad \gamma = m+n-r, \quad \delta = \frac{1}{2}(\gamma-1)(\gamma-2),$$

and assuming $n \geq m$, (these equations and inequalities are to be attended to throughout the present paper), I consider two curves of the orders m, n respectively meeting in δ points B , and in $(mn - \delta)$ points A ; and I state the theorem as follows:

1°. The $mn - \delta$ points A are in general such that a curve of the order r , which passes through $mn - \delta - 1$ of these points, does not of necessity pass through the remaining point; and in this case the general curve of the order r , which passes through the $mn - \delta$ points A , has for its form of equation

$$0 = L_{r-m}P_m + M_{r-n}Q_n,$$

where $P_m = 0, Q_n = 0$ are the equations of the given curves and L_{r-m}, M_{r-n} denote functions of the orders $r - m, r - n$ respectively; and it thus appears that the curve of the order r through the $mn - \delta$ points A passes also through the δ points B .

Arthur Cayley
Collected Mathematical Papers
Volume XII
Pages 501–550

[865]

ON MULTIPLE ALGEBRA.

[865] depending on the new extraordinaries $\epsilon_2\epsilon_3, \epsilon_3\epsilon_1, \epsilon_1\epsilon_2$. Grassmann further assumes $\epsilon_1\epsilon_2\epsilon_3 = 1$, and he calls the foregoing combinations the complements (Ergänzungen) of $\epsilon_1, \epsilon_2, \epsilon_3$, denoting them by $/\epsilon_1, /\epsilon_2, /\epsilon_3$ respectively; it is better to use new letters, and I call them η_1, η_2, η_3 respectively; we thus have

$$\epsilon_1^2 = 0, \quad \epsilon_2^2 = 0, \quad \epsilon_3^2 = 0,$$

$$\epsilon_2\epsilon_3 = -\epsilon_3\epsilon_2 = \eta_1, \quad \epsilon_3\epsilon_1 = -\epsilon_1\epsilon_3 = \eta_2, \quad \epsilon_1\epsilon_2 = -\epsilon_2\epsilon_1 = \eta_3.$$

But we require further assumptions for the combinations of the η 's *inter se* and with the ϵ 's; the forms are

$$\eta_1^2 = 0, \quad \eta_2^2 = 0, \quad \eta_3^2 = 0,$$

$$-\eta_2\eta_3 = \eta_3\eta_2 = \epsilon_1; \quad -\eta_3\eta_1 = \eta_1\eta_3 = \epsilon_2, \quad -\eta_1\eta_2 = \eta_2\eta_1 = \epsilon_3;$$

$\epsilon_1\eta_1 = \eta_1\epsilon_1 = 1, \quad \epsilon_2\eta_2 = \eta_2\epsilon_2 = 1, \quad \epsilon_3\eta_3 = \eta_3\epsilon_3 = 1, \quad \epsilon_1\eta_2 = \eta_2\epsilon_1 = 0, \text{ \&c.},$
(viz. each such combination is = 0); or, what is the same thing, we have for 1, $\epsilon_1, \epsilon_2, \epsilon_3, \eta_1, \eta_2, \eta_3$ the multiplication table

	1	ϵ_1	ϵ_2	ϵ_3	η_1	η_2	η_3
1	1	ϵ_1	ϵ_2	ϵ_3	η_1	η_2	η_3
ϵ_1	ϵ_1	0	η_3	$-\eta_2$	1	0	0
ϵ_2	ϵ_2	$-\eta_3$	0	η_1	0	1	0
ϵ_3	ϵ_3	η_2	$-\eta_1$	0	0	0	1
η_1	η_1	1	0	0	0	$-\epsilon_3$	ϵ_2
η_2	η_2	0	1	0	ϵ_3	0	$-\epsilon_1$
η_3	η_3	0	0	1	$-\epsilon_2$	ϵ_1	0

It is proper to remark that in Grassmann's theory there is no interpretation for the general linear symbol

$$\omega + x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3 + y_1\eta_1 + y_2\eta_2 + y_3\eta_3,$$

which presents itself in connexion with the seven extraordinaries $(1, \epsilon_1, \epsilon_2, \epsilon_3, \eta_1, \eta_2, \eta_3)$.

47. It is to be noticed that the symbols are not associative; assuming them to be so, we should have $\epsilon_2\epsilon_3 \cdot \epsilon_3\epsilon_1 = \epsilon_2 \cdot \epsilon_3^2 \cdot \epsilon_1 = 0$, which is inconsistent with $\eta_1\eta_2 = -\epsilon_3$; and so in other cases. But any *three* ϵ 's are associative, and hence a product

$$x \cdot y \cdot z, = (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3) (y_1\epsilon_1 + y_2\epsilon_2 + y_3\epsilon_3) (z_1\epsilon_1 + z_2\epsilon_2 + z_3\epsilon_3),$$

is associative; in fact, if this be regarded as standing for $(x \cdot y) z$, the value is

$$= \{(x_2y_3 - x_3y_2) \eta_1 + (x_3y_1 - x_1y_3) \eta_2 + (x_1y_2 - x_2y_1) \eta_3\} (z_1\epsilon_1 + z_2\epsilon_2 + z_3\epsilon_3),$$

which is

$$= (x_2y_3 - x_3y_2) z_1 + (x_3y_1 - x_1y_3) z_2 + (x_1y_2 - x_2y_1) z_3, = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix};$$

and similarly, regarding it as standing for $x (y \cdot z)$, the value is

$$= (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3) \{(y_2z_3 - y_3z_2) \eta_1 + (y_3z_1 - y_1z_3) \eta_2 + (y_1z_2 - y_2z_1) \eta_3\},$$

which is

$$= x_1 (y_2z_3 - y_3z_2) + x_2 (y_3z_1 - y_1z_3) + x_3 (y_1z_2 - y_2z_1), = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

as before. The product of the three points is thus a scalar, viz. it is equal to the area of the triangle formed by the three points divided by that of the triangle $A_1A_2A_3$.

48. But a product $x \cdot Y \cdot z$,

$$= (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3) (Y_1\eta_1 + Y_2\eta_2 + Y_3\eta_3) (z_1\epsilon_1 + z_2\epsilon_2 + z_3\epsilon_3),$$

is not associative, and has thus no meaning until the grouping of the factors is determined. In fact, $(x \cdot Y) z$ will be

$$= (x_1Y_1 + x_2Y_2 + x_3Y_3) (z_1\epsilon_1 + z_2\epsilon_2 + z_3\epsilon_3),$$

which denotes the point z_1 with a weight $= x_1Y_1 + x_2Y_2 + x_3Y_3$; whereas $x (Y \cdot z)$ will be

$$= (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3) (y_1Z_1 + y_2Z_2 + y_3Z_3),$$

which denotes the point x_1 with a weight $= y_1Z_1 + y_2Z_2 + y_3Z_3$.

Hence also a product $x \cdot y \cdot z \cdot w$,

$$= (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3)(y_1\epsilon_1 + y_2\epsilon_2 + y_3\epsilon_3)(z_1\epsilon_1 + z_2\epsilon_2 + z_3\epsilon_3)(w_1\epsilon_1 + w_2\epsilon_2 + w_3\epsilon_3),$$

of four factors is not associative, and has thus no meaning until the grouping of the factors is determined. Grassmann attributes to such a product the value $= 0$; but this is not a value corresponding to any grouping of the factors, and the equation $x \cdot y \cdot z \cdot w = 0$ can only be regarded as an independent assumption.

For the product of a point into itself, or say the square of a point, we have

$$x^2 = (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3)^2, \quad = (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3)(x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3), = 0,$$

viz. this is $=$ zero.

49. The product of two points x and y is defined to be the "Strecke", the finite line or stroke (xy) , considered as drawn from x to y , say we have $x \cdot y = (xy)$; observe that $y \cdot x = (yx)$: by what precedes $y \cdot x + x \cdot y = 0$, whence also $(yx) + (xy) = 0$, which is an equation in the addition of strokes. Hence, if as before, the points x , y are $x = x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3$, and $y = y_1\epsilon_1 + y_2\epsilon_2 + y_3\epsilon_3$, then for the stroke (xy) we have

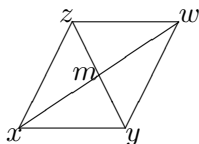
$$\begin{aligned} (xy) &= (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3)(y_1\epsilon_1 + y_2\epsilon_2 + y_3\epsilon_3), \\ &= (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3, \end{aligned}$$

which is, in fact,

$$= (A_1xy \cdot \eta_1 + A_2xy \cdot \eta_2 + A_3xy \cdot \eta_3) \div A_1A_2A_3,$$

if A_1xy , A_2xy , A_3xy , $A_1A_2A_3$ denote the areas of the triangles formed by the points (A_1, x, y) , (A_2, x, y) , (A_3, x, y) , (A_1, A_2, A_3) respectively. These triangles remain unaltered if the stroke (xy) is slid along the indefinite line joining the original points x , y , the absolute distance xy being unaltered; that is, strokes of equal length upon the same line are regarded as identical. Hence also strokes on the same line may be added together by adding their lengths.

50. Consider two strokes (xy) , (xz) , (fig. 4), having a common point x , then completing the parallelogram, we prove that the sum $(xy) + (xz)$ is equal to the diagonal (xw) .

Fig. 4.

We have, in fact,

$$x + w = y + z,$$

and thence

$$x(x + w) = x(y + z),$$

that is,

$$x^2 + x \cdot w = x \cdot y + x \cdot z,$$

or, since $x^2 = 0$ and $x \cdot y = (xy)$, &c., this is

$$(xw) = (xy) + (xz),$$

the required formula for the addition of the strokes.

51. The same thing appears thus: writing

$$x = x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3,$$

$$y = y_1\epsilon_1 + y_2\epsilon_2 + y_3\epsilon_3,$$

$$z = z_1\epsilon_1 + z_2\epsilon_2 + z_3\epsilon_3,$$

we have

$$(xy) = (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3,$$

$$(xz) = (x_2z_3 - x_3z_2)\eta_1 + (x_3z_1 - x_1z_3)\eta_2 + (x_1z_2 - x_2z_1)\eta_3.$$

Or, putting for a moment

$$m_1, m_2, m_3 = \frac{1}{2}(y_1 + z_1), \quad \frac{1}{2}(y_2 + z_2), \quad \frac{1}{2}(y_3 + z_3),$$

we have for the point m , midway between y and z , the equation $m = m_1\epsilon_1 + m_2\epsilon_2 + m_3\epsilon_3$; and then

$$\begin{aligned} (xy) + (xz) &= 2\{(x_2m_3 - x_3m_2)\eta_1 + (x_3m_1 - x_1m_3)\eta_2 + (x_1m_2 - x_2m_1)\eta_3\} \\ &= 2(xm), \quad = xw, \end{aligned}$$

which is the theorem in question.

52. Recurring to the foregoing equation $x + w = y + z$, we deduce

$$xz - yw = zw - xy = (yzw) = (zwx) = -(wxy) = -(xyz),$$

if for shortness $(yzw) = yz + zw + wy$, and similarly for (zwx) , (wxy) , and (xyz) . Observe here that $xz - yw$ and $zw - xy$ are each of them equal to the difference between two equal and parallel strokes; the strokes yz , zw , wy are the sides (in order) of a triangle; *quâ* forces, the expressions in question represent each of them a couple.

53. We have thus arrived at the representation of a stroke

$$(xy) = (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3,$$

or say

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3,$$

where the meaning of the coordinates λ_1 , λ_2 , λ_3 is such that, if x , y are the two extremities of the stroke, then λ_1 , λ_2 , λ_3 are equal to the areas A_1xy , A_2xy , A_3xy , each divided by the area $A_1A_2A_3$. And we have also established the rule for the addition of strokes: it will be noticed that this has been done without the aid of any expression for the length of the stroke.

54. *Metrical relations.* For the expression of the distance of two points or length of a stroke, and of the inclinations of a stroke to the sides of the fundamental triangle, &c., we require the values of the angles and sides of this triangle $A_1A_2A_3$, say the angles are A_1 , A_2 , A_3 , and the sides are $= R \sin A_1$, $R \sin A_2$, $R \sin A_3$ respectively; R is thus equal to the diameter of the circumscribed circle. Moreover, considering a stroke xy , if through x we draw lines $x\theta_1$, $x\theta_2$, $x\theta_3$ in the senses A_2A_3 , A_3A_1 , A_1A_2 respectively, then taking the angles $yx\theta_1$, $yx\theta_2$, $yx\theta_3$ each of them in the same sense, or say each right-handedly, we call these angles θ_1 , θ_2 , θ_3 respectively, and take also ρ , $= (xy)$ for the length of the stroke. This being so, and if the two extremities are

$$x = x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3, \quad \text{and} \quad y = y_1\epsilon_1 + y_2\epsilon_2 + y_3\epsilon_3,$$

then we have, first,

$$\theta_2 - \theta_3 = \pi - A_1, \quad \theta_3 - \theta_1 = \pi - A_2, \quad \theta_1 - \theta_2 = -\pi - A_3,$$

giving

$$A_1 + A_2 + A_3 = \pi,$$

and

$$\sin A_1, \quad \sin A_2, \quad \sin A_3 = \sin(\theta_2 - \theta_3), \quad \sin(\theta_3 - \theta_1), \quad \sin(\theta_1 - \theta_2).$$

And then

$$y_1 = x_1 + \frac{\rho}{R} \frac{\sin \theta_1}{\sin A_2 \sin A_3}, \quad y_2 = x_2 + \frac{\rho}{R} \frac{\sin \theta_2}{\sin A_3 \sin A_1}, \quad y_3 = x_3 + \frac{\rho}{R} \frac{\sin \theta_3}{\sin A_1 \sin A_2}.$$

As regards these last equations, writing for a moment p_1, p_2, p_3 for the perpendicular distances of x , and q_1, q_2, q_3 for those of y from the sides of the triangle, then we have $q_1, q_2, q_3 = p_1 + \rho \sin \theta_1, p_2 + \rho \sin \theta_2, p_3 + \rho \sin \theta_3$, and we thence easily derive the equations in question.

55. *Expression for length of a stroke.* We hence, by a somewhat complicated analytical process, find an expression for ρ in terms of the coordinates (x_1, x_2, x_3) , and (y_1, y_2, y_3) , which enter into it through the combinations

$$x_2 y_3 - x_3 y_2, \quad x_3 y_1 - x_1 y_3, \quad x_1 y_2 - x_2 y_1.$$

In fact, writing for shortness

$$\frac{x_1}{\sin A_1}, \quad \frac{x_2}{\sin A_2}, \quad \frac{x_3}{\sin A_3} = K_1, K_2, K_3,$$

we have

$$\begin{aligned} (x_2 y_3 - x_3 y_2) \sin A_1 &= \frac{\rho}{R} (K_2 \sin \theta_3 - K_3 \sin \theta_2), \\ (x_3 y_1 - x_1 y_3) \sin A_2 &= \frac{\rho}{R} (K_3 \sin \theta_1 - K_1 \sin \theta_3), \\ (x_1 y_2 - x_2 y_1) \sin A_3 &= \frac{\rho}{R} (K_1 \sin \theta_2 - K_2 \sin \theta_1); \end{aligned}$$

and hence also, if $(\mathfrak{A}\xi, \eta, \zeta)^2$ denote the quadric function

$$(1, 1, 1, -\cos A_1, -\cos A_2, -\cos A_3 \mathfrak{A}\xi, \eta, \zeta)^2,$$

we have

$$\begin{aligned} &(\mathfrak{A}(x_2 y_3 - x_3 y_2) \sin A_1, \dots)^2 \\ &= \frac{\rho^2}{R^2} (\mathfrak{A}K_2 \sin \theta_3 - K_3 \sin \theta_2, K_3 \sin \theta_1 - K_1 \sin \theta_3, K_1 \sin \theta_2 - K_2 \sin \theta_1)^2, \end{aligned}$$

where the quadric function on the right-hand side is in fact

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2,$$

that is,

$$= (x_1 + x_2 + x_3)^2, = 1;$$

and we thus finally obtain

$$\rho^2 = R^2(\ast x_2 y_3 - x_3 y_2) \sin A_1, (x_3 y_1 - x_1 y_3) \sin A_2, (x_1 y_2 - x_2 y_1) \sin A_3)^2,$$

the required formula for the length of the stroke xy .

56. To effect the foregoing reduction of the quadric function

$$(\ast K_2 \sin \theta_3 - K_3 \sin \theta_2, K_3 \sin \theta_1 - K_1 \sin \theta_3, K_1 \sin \theta_2 - K_2 \sin \theta_1)^2,$$

observe that the coefficient herein of K_1^2 is

$$= \sin^2 \theta_2 + \sin^2 \theta_3 + 2 \cos A_1 \sin \theta_2 \sin \theta_3,$$

where $\theta_2 - \theta_3 = \pi - A_1$, and thence

$$\cos \theta_2 \cos \theta_3 + \sin \theta_2 \sin \theta_3 = -\cos A_1;$$

that is, we have

$$\cos A_1 + \sin \theta_2 \sin \theta_3 = -\cos \theta_2 \cos \theta_3,$$

and thence

$$1 - \sin^2 \theta_2 - \sin^2 \theta_3 + \sin^2 \theta_2 \sin^2 \theta_3 = \cos^2 A_1 + 2 \cos A_1 \sin \theta_2 \sin \theta_3 + \sin^2 \theta_2 \sin^2 \theta_3;$$

whence the coefficient in question is $= \sin^2 A_1$. And finding in like manner the coefficients of K_2^2 , K_3^2 , $K_2 K_3$, &c., the whole function is, as already mentioned,

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2, \text{ that is, } = 1.$$

57. *New representation of a stroke.* Representing the stroke in the form

$$\lambda = \lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3,$$

we have only to replace $x_2 y_3 - x_3 y_2$, $x_3 y_1 - x_1 y_3$, $x_1 y_2 - x_2 y_1$ by their values $= \lambda_1, \lambda_2, \lambda_3$ respectively; viz. for the length ρ of the stroke, we have

$$\rho^2 = R^2(\ast \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3)^2.$$

Say this equation is

$$\rho^2 = R^2 \cdot \Omega, \quad \text{then } \rho = R \sqrt{(\Omega)}, \quad \text{or } 1 = \frac{\rho}{R \sqrt{(\Omega)}},$$

and the stroke may be represented in the form

$$\lambda = \frac{\rho}{R \sqrt{(\Omega)}} (\lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3);$$

or, what is the same thing, if the absolute magnitudes of the coordinates $\lambda_1, \lambda_2, \lambda_3$ are such that

$$(\ast \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3)^2 = 1,$$

then the expression is

$$\lambda = \frac{\rho}{R} (\lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3),$$

where ρ is the length of the stroke.

58. It has been seen that the coefficients of η_1, η_2, η_3 have the values $A_1 xy, A_2 xy, A_3 xy$, each divided by $A_1 A_2 A_3$; or, what is the same thing, if p_1, p_2, p_3 are the lengths of the perpendiculars on the stroke from the points A_1, A_2, A_3 respectively, then these coefficients are $(\frac{1}{2} \rho p_1, \frac{1}{2} \rho p_2, \frac{1}{2} \rho p_3)$, each divided by $A_1 A_2 A_3$; or, what is the same thing, the perpendicular distances p_1, p_2, p_3 are

$$= \frac{2 A_1 A_2 A_3}{R \sqrt{\Omega}} (\lambda_1, \lambda_2, \lambda_3)$$

respectively, where $A_1 A_2 A_3$ is the area of the fundamental triangle.

59. *Expression for the mutual inclination of two strokes.* In connexion with the stroke $\lambda = \lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3$, considering a stroke $\mu = \mu_1 \eta_1 + \mu_2 \eta_2 + \mu_3 \eta_3$, suppose the inclinations hereof to the axes are ϕ_1, ϕ_2, ϕ_3 , then

$$\phi_2 - \phi_3 = \pi - A_1, \quad \phi_3 - \phi_1 = \pi - A_2, \quad \phi_1 - \phi_2 = -\pi - A_3,$$

and we have

$$\theta_1 - \phi_1 = \theta_2 - \phi_2 = \theta_3 - \phi_3,$$

viz. each of these equal angles is the inclination of the two strokes to each other, say this angle is $= \delta$. The expression for this angle is

$$\cos \delta = (\ast \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3 \ast \mu_1 \sin A_1, \mu_2 \sin A_2, \mu_3 \sin A_3)$$

$$\div \sqrt{\{(1, \dots \ast \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3)^2\}} \cdot \sqrt{\{(1, \dots \ast \mu_1 \sin A_1, \mu_2 \sin A_2, \mu_3 \sin A_3)^2\}}$$

In fact, considering (as we may do) the two strokes as proceeding from a common point $x = x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3$, then the function in question is

$$= (\backslash K_2 \sin \theta_3 - K_3 \sin \theta_2, K_3 \sin \theta_1 - K_1 \sin \theta_3, K_1 \sin \theta_2 - K_2 \sin \theta_1 \\ \backslash K_2 \sin \phi_3 - K_3 \sin \phi_2, \dots),$$

which is

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2 \cos \delta, \quad = (x_1 + x_2 + x_3)^2 \cos \delta, \quad = \cos \delta.$$

60. In verification, observe that the whole coefficient of K_1^2 is

$$= \sin \theta_2 \sin \phi_2 + \sin \theta_3 \sin \phi_3 + \cos A_1 (\sin \theta_2 \sin \phi_3 + \sin \theta_3 \sin \phi_2),$$

where $\theta_2 - \theta_3 = \pi - A_1$, $\phi_2 - \phi_3 = \pi - A_1$. Hence

$$\sin \theta_2 = -\sin(\theta_3 - A_1), \quad \sin \phi_2 = -\sin(\phi_3 - A_1);$$

and the expression in question is

$$= \sin(\theta_3 - A_1) \sin(\phi_3 - A_1) \\ - \cos A_1 \sin(\theta_3 - A_1) \sin \phi_3, \\ - \cos A_1 \sin(\phi_3 - A_1) \sin \theta_3, \\ + \sin \theta_3 \sin \phi_3,$$

which, expanding the sines, and writing the last term in the form

$$- \sin \theta_3 \sin \phi_3 (\cos^2 A_1 + \sin^2 A_1)$$

is

$$= \cos^2 A_1 (\sin \theta_3 \sin \phi_3 - \sin \theta_3 \sin \phi_3 - \sin \theta_3 \sin \phi_3 + \sin \theta_3 \sin \phi_3) \\ + \sin^2 A_1 (\cos \theta_3 \cos \phi_3 + \sin \theta_3 \sin \phi_3) \\ + \cos A_1 \sin A_1 (-\sin \theta_3 \cos \phi_3 - \sin \phi_3 \cos \theta_3 + \sin \phi_3 \cos \theta_3 + \sin \theta_3 \cos \phi_3), \\ = \sin^2 A_1 \cos(\theta_3 - \phi_3), \quad = \sin^2 A_1 \cos \delta,$$

and reducing in like manner the coefficients of K_2^2 , K_3^2 , $K_2 K_3$, &c., the whole expression becomes

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2 \cos \delta, \quad = \cos \delta \text{ as above.}$$

61. The foregoing formulæ agree with the theorem, No. 50, that the sum of the strokes

$$\lambda, = \frac{\rho}{R\sqrt{(\Omega)}}(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

and

$$\mu, = \frac{\sigma}{R\sqrt{(\Upsilon)}}(\mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3),$$

is a stroke

$$= \frac{\tau}{R\sqrt{(\Pi)}}(\nu_1\eta_1 + \nu_2\eta_2 + \nu_3\eta_3),$$

which is the diagonal of the parallelogram constructed with the given strokes λ and μ ; the proof is a little simplified by assuming (as is allowable) $\Omega = 1$ and $\Upsilon = 1$, in which case we have

$$\lambda + \mu = \frac{1}{R} \{(\rho\lambda_1 + \sigma\mu_1)\eta_1 + (\rho\lambda_2 + \sigma\mu_2)\eta_2 + (\rho\lambda_3 + \sigma\mu_3)\eta_3\},$$

from which the length and inclination may be calculated.

As already appearing, the product x, y, z of any three points is the scalar

$$\begin{vmatrix} x_1, & x_2, & x_3 \\ y_1, & y_2, & y_3 \\ z_1, & z_2, & z_3 \end{vmatrix},$$

which is equal to the area of the triangle xyz , divided by that of the triangle $A_1A_2A_3$. We have in like manner the product of a point x into a stroke λ , viz. this is

$$= (x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3)(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

which is $= x_1\lambda_1 + x_2\lambda_2 + x_3\lambda_3$, the two factors being in this case commutative; the value is equal to the area of the triangle formed by the point and the stroke, divided by the area $A_1A_2A_3$. Of course, if in the one case the three points are in a line, or if in the other the point is in the line of the stroke, then the product is $= 0$.

62. We have yet to consider the product of two strokes; say these are

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3, \quad \text{and} \quad \mu = \mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3,$$

then we have

$$\begin{aligned} \lambda \cdot \mu &= (\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3)(\mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3) \\ &= -\{(\lambda_2\mu_3 - \lambda_3\mu_2)\epsilon_1 + (\lambda_3\mu_1 - \lambda_1\mu_3)\epsilon_2 + (\lambda_1\mu_2 - \lambda_2\mu_1)\epsilon_3\}, \end{aligned}$$

which is a point regarded as having weight. If we take the two strokes to be

$$\lambda = (xy) = (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3,$$

and

$$\mu = (xz) = (x_2z_3 - x_3z_2)\eta_1 + (x_3z_1 - x_1z_3)\eta_2 + (x_1z_2 - x_2z_1)\eta_3,$$

then the product is

$$\lambda \cdot \mu = -(x_1\epsilon_1 + x_2\epsilon_2 + x_3\epsilon_3) \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

viz. this is the common point x , with a weight = area of triangle $xyz \div$ area $A_1A_2A_3$, or say = the area of triangle formed by the two strokes $\div$ area $A_1A_2A_3$. The product is non-commutative: we, in fact, have $\lambda \cdot \mu = -\mu \cdot \lambda$.

63. The product of three strokes is associative; and it is a scalar. If the three strokes are the sides of a triangle LMN , then the value is $= (LMN)^2 \div (A_1A_2A_3)^2$, where LMN and $A_1A_2A_3$ denote the areas of the triangle LMN and the triangle $A_1A_2A_3$ respectively; and if any stroke instead of being a side of the triangle LMN be a part only of this side, then the value is diminished proportionally; hence the value is $= \frac{\rho\sigma\tau}{MN \cdot NL \cdot LM} \{(LMN)^2 \div (A_1A_2A_3)^2\}$, where ρ, σ, τ are the lengths of the three strokes, MN, NL, LM the lengths of the sides of the triangle LMN , and $LMN, A_1A_2A_3$ are the areas as above.

64. It is to be remarked that the form

$$(\ast\xi, \eta, \zeta)^2, = (1, 1, 1, -\cos A_1, -\cos A_2, -\cos A_3)\xi, \eta, \zeta)^2$$

is the product of two linear factors; it is

$$= (\xi - \eta \cos A_3 - \zeta \cos A_2)^2 + (\eta \sin A_3 - \zeta \sin A_2)^2,$$

and thus

$$= (\xi - \eta e^{iA_3} - \zeta e^{-iA_2})(\xi - \eta e^{-iA_3} - \zeta e^{iA_2}).$$

This corresponds to the theorem that the distance of two points is $= 0$, when the line joining them passes through one of the circular points

at infinity; the coordinates of one of these points may be written under the three equivalent forms

$$\begin{aligned} x_1 : x_2 : x_3 &= \sin A_1 & : -\sin A_2 e^{iA_3} & : -\sin A_3 e^{-iA_2}, \\ &= -\sin A_1 e^{-iA_3} & : \sin A_2 & : -\sin A_3 e^{iA_1}, \\ &= -\sin A_1 e^{-iA_2} & : -\sin A_2 e^{-iA_1} & : \sin A_3, \end{aligned}$$

and those of the other are of course obtained therefrom by the change of i into $-i$.

(To be continued).*

* [This paper remains incomplete; no continuation appears to have been prepared by Professor Cayley.]

[866]

NOTE ON KIEPERT'S L -EQUATIONS, IN THE TRANSFORMATION OF ELLIPTIC FUNCTIONS.

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 75—77.]

It appears, by comparison with Klein's paper "Ueber die Transformation u. s. w.," *Math. Annalen*, t. xiv. (1878), see p. 144, that Kiepert's L made use of in the Memoir "Ueber Theilung und Transformation der elliptischen Functionen," *Math. Annalen*, t. xxvi. (1886), pp. 369—454, is, in fact, the square of the multiplier, "für das durch $\sqrt[12]{\Delta}$ normirte Integral," viz. considering the general quartic function $(a, \dots | x, 1)^4 = (a, b, c, d, e | x, 1)^4$, and the transformed function $(a_1, \dots | y, 1)^4$, then we have

$$\frac{L^2 \sqrt[12]{\Delta} dx}{\sqrt{(a, \dots | x, 1)^4}} = \frac{\sqrt[12]{\Delta_1} dy}{\sqrt{(a_1, \dots | y, 1)^4}},$$

where if

$$I = ae - 4bd + 3c^2,$$

$$J = ace - ad^2 - b^2e + 2bcd - c^3,$$

and similarly I_1, J_1 , are the invariants of the two functions, then Δ, Δ_1 are the discriminants

$$\Delta = I^3 - 27J^2, \quad \Delta_1 = I_1^3 - 27J_1^2,$$

and the γ_2, γ_3 of Kiepert's equations are

$$\gamma_2 = I \div \sqrt[3]{\Delta}, \quad \gamma_3 = J \div \sqrt{\Delta},$$

whence

$$\gamma_2^3 - 27\gamma_3^2 = 1.$$

In particular, if the forms are

$$1 - x^2 \cdot 1 - k^2 x^2, \text{ and } 1 - y^2 \cdot 1 - \lambda^2 y^2,$$

and if as usual $k = u^4, \lambda = v^4$, and M is the multiplier for the form

$$\frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}} = \frac{M dy}{\sqrt{1 - y^2 \cdot 1 - \lambda^2 y^2}},$$

then we have

$$\begin{aligned} I &= \frac{1}{12}(1 + 14u^8 + u^{16}), \\ J &= \frac{1}{216}(1 + u^8)(1 - 34u^8 + u^{16}), \\ \Delta &= \frac{1}{16}u^8(1 - u^8)^4, \quad \Delta_1 = \frac{1}{16}v^8(1 - v^8)^4, \\ \gamma_2 &= \frac{1}{6}\sqrt[3]{2} \frac{1 + 14u^8 + u^{16}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}}, \quad \gamma_3 = \frac{1}{54} \frac{(1 + u^8)(1 - 34u^8 + u^{16})}{u^4(1 - u^8)^2}, \end{aligned}$$

and thence

$$L^2 = \frac{v^{\frac{2}{3}}(1 - v^8)^{\frac{1}{3}}}{u^{\frac{2}{3}}(1 - u^8)^{\frac{1}{3}}} \cdot \frac{1}{M},$$

which last equation is the expression for L^2 in terms of the Jacobian symbols u, v, M .

As an easy verification in a particular case, suppose $n = 5$. We have here

$$\begin{aligned} L^2 &= \frac{v^{\frac{2}{3}}(1 - v^8)^{\frac{1}{3}}}{u^{\frac{2}{3}}(1 - u^8)^{\frac{1}{3}}} \cdot \frac{1}{M}, \quad M = \frac{v(1 - uv^3)}{v - u^5}, \quad \left(= \frac{v + u^5}{5u(1 + u^3v)} \right), \\ u^6 - v^6 + 5u^2v^2(u^2 - v^2) + 4uv(1 - u^4v^4) &= 0, \\ \gamma_2 &= \frac{1}{6}\sqrt[3]{2} \frac{1 + 14u^8 + u^{16}}{u^{\frac{8}{3}}(1 - u^8)^{\frac{4}{3}}}; \end{aligned}$$

and it should be possible, by eliminating u, v, M , to deduce hence the L -equation

$$L^{12} + 10L^6 - 12\gamma_2L^2 + 5 = 0. \quad (\text{Kiepert, p. 428.})$$

It does not seem in any wise easy to do this in the case of an arbitrary modulus; but writing the modular equation in the form

$$(u^2 - v^2)(u^4 + 6u^2v^2 + v^4) + 4uv(1 - u^4v^4) = 0,$$

we satisfy this by

$$uv - 1 = 0, \quad u^4 + 6u^2v^2 + v^4 = 0,$$

or say by

$$v = \frac{1}{u}, \quad u^8 + 6u^4 + 1 = 0,$$

and the equation may be verified for this particular modulus.

We have

$$1 + 14u^8 + u^{16} = 48u^8, \quad (1 - u^8)^2 = 32u^8,$$

and consequently

$$\gamma_2 = \frac{1}{6} \sqrt[3]{2} \cdot \frac{48u^8}{u^{\frac{8}{3}}(32u^8)^{\frac{4}{3}}}, \quad = \frac{1}{2 \cdot 3} 2^{\frac{4}{3}} \frac{2^4 \cdot 3u^8}{u^{\frac{8}{3}} \cdot 2^{\frac{10}{3}} \cdot u^{\frac{16}{3}}} = 1, \quad (\text{whence also } \gamma_3 = 0).$$

Moreover

$$M = \frac{\frac{1}{u} - \frac{u}{u^4}}{\frac{1}{u} - u^5} = -\frac{1}{u^2} \frac{1 - u^2}{1 - u^6} = \frac{-1}{u^2(1 + u^2 + u^4)},$$

and thence

$$L^2 = \frac{\frac{1}{u^{\frac{2}{3}}} \left(1 - \frac{1}{u^8}\right)^{\frac{1}{3}}}{u^{\frac{2}{3}}(1 - u^8)^{\frac{1}{3}}} \cdot \frac{1}{M} = \frac{(u^8 - 1)^{\frac{1}{3}}}{u^4(1 - u^8)^{\frac{1}{3}}} \cdot \frac{1}{M} = -\frac{1}{u^4 M},$$

that is,

$$L^2 = 1 + u^2 + u^{-2}.$$

But we have

$$u^4 = -3 + 2\sqrt{2},$$

and thence

$$u^2 = i(1 - \sqrt{2}), \quad u^{-2} = i(1 + \sqrt{2}),$$

and

$$L^2 = 1 + 2i,$$

whence

$$L^{12} + 10L^6 - 12\gamma_2 L^2 + 5 = (117 + 44i) + 10(-11 - 2i) - 12(1 + 2i) + 5 = 0,$$

or the L -equation is satisfied.

Cambridge, 14 March 1887.

[867]

NOTE ON THE JACOBIAN SEXTIC EQUATION.

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 78—84.]

In the Jacobian sextic equation

$$(z - a)^6 - 4a(z - a)^5 + 16b(z - a)^3 - 4c(z - a) + 5b^2 - 4ac = 0,$$

if A, B, C, D, E, F are the square roots (each with a determinate sign) of the roots $z_0, z_1, z_2, z_3, z_4, z_\infty$ of the equation, and if ϵ be an imaginary fifth root of unity, such that $\epsilon + \epsilon^4 - \epsilon^2 - \epsilon^3 = \sqrt{5}$, then we have between the square roots a system of three linear equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= B + \epsilon^3 C + \epsilon D + \epsilon^4 E + \epsilon^2 F, \\ 0 &= B + \epsilon^2 C + \epsilon^4 D + \epsilon E + \epsilon^3 F, \end{aligned}$$

see Brioschi's *Funzioni ellittiche* (Milan, 1880), third appendix, p. 402.

The sextic equation is irreducible, and there is thus nothing to distinguish any one square root from any other square root. But in the foregoing system of equations, A is distinguished from the other square roots; hence the system must be one of 6 systems, wherein the place occupied by A is occupied by A, B, C, D, E, F respectively. Observe however that the letters being square roots, it may very well happen, and (as will appear) it does in fact happen, that the passage from the first to another system implies a change of sign in certain of the letters A, B, C, D, E, F .

The selection of one of the square roots as $= A$, does not determine which of the other square roots shall be $= B, C, D, E, F$ respectively: and in fact each system of equations may be represented in 10 different forms: viz. by multiplying the second and third equations by $\epsilon, \epsilon^2, \epsilon^3, \epsilon^4$ respectively, we obtain four new forms: we have thus five forms: and then in each of them transposing the second and third equations, we have five other forms: we have thus in all 10 forms.

Write $ABCDEF$ to denote the foregoing system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= B + \epsilon^3 C + \epsilon D + \epsilon^4 E + \epsilon^2 F, \\ 0 &= B + \epsilon^2 C + \epsilon^4 D + \epsilon E + \epsilon^3 F; \end{aligned}$$

the 10 forms of the system are

$$\begin{array}{c|c} A B C D E F & A B F E D C \\ A C D E F B & A C B F E D \\ A D E F B C & A D C B F E \\ A E F B C D & A E D C B F \\ A F B C D E & A F E D C B, \end{array}$$

where observe that the five forms in the same column are connected by cyclical substitutions on the last five letters; and that the forms in each line are connected by two inversions of the last four letters.

It is hardly necessary to remark that $ABFEDC$ means the system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + F + E + D + C, \\ 0 &= B + \epsilon^3 F + \epsilon E + \epsilon^4 D + \epsilon^2 C, \\ 0 &= B + \epsilon^2 F + \epsilon^4 E + \epsilon D + \epsilon^3 C, \end{aligned}$$

which are the same with the equations of the original form; and similarly that $ACDEFB$ means the system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + C + D + E + F + B, \\ 0 &= C + \epsilon^3 D + \epsilon E + \epsilon^4 F + \epsilon^2 B, \\ 0 &= C + \epsilon^2 D + \epsilon^4 E + \epsilon F + \epsilon^3 B, \end{aligned}$$

where the first equation is the same with the first equation, and the second and third equations only differ by a factor from the second and third equations respectively of the original form. And similarly as regards the others of the 10 forms.

I say that we have a second system $BCAF\bar{D}\bar{E}$ (where $\bar{D}$, $\bar{E}$ mean $-D$, $-E$ respectively): in fact, this denotes the system of three equa-

tions

$$\begin{aligned} 0 &= -B\sqrt{5} + C + A + F - D - E, \\ 0 &= C + \epsilon^3 A + \epsilon F - \epsilon^4 D - \epsilon^2 E, \\ 0 &= C + \epsilon^2 A + \epsilon^4 F - \epsilon D - \epsilon^3 E, \end{aligned}$$

which are deducible from the original three equations.

We have, in fact, the identities

$$\begin{aligned} & -\sqrt{5} \{-B\sqrt{5} + C + A + F - D - E\} \\ &= 1(B + C - A\sqrt{5} + F + D + E) \\ & \quad + 2(B + \epsilon^3 C + \epsilon^2 F + \epsilon D + \epsilon^4 E) \\ & \quad + 2(B + \epsilon^2 C + \epsilon^3 F + \epsilon^4 D + \epsilon E), \\ & (1 - \epsilon - \epsilon^3 + \epsilon^4) \{C + \epsilon^3 A - \epsilon^4 D - \epsilon^2 E\} \\ &= 1(B + C - A\sqrt{5} + F + D + E) \\ & \quad + (\epsilon + \epsilon^4)(B + \epsilon^3 C + \epsilon^2 F + \epsilon D + \epsilon^4 E) \\ & \quad + (\epsilon^2 + \epsilon^3)(B + \epsilon^2 C + \epsilon^3 F + \epsilon^4 D + \epsilon E), \end{aligned}$$

and

$$\begin{aligned} & (1 + \epsilon - \epsilon^2 - \epsilon^4) \{C + \epsilon^2 A - \epsilon D - \epsilon^3 E\} \\ &= 1(B + C - A\sqrt{5} + F + D + E) \\ & \quad + (\epsilon^2 + \epsilon^3)(B + \epsilon^3 C + \epsilon^2 F + \epsilon D + \epsilon^4 E) \\ & \quad + (\epsilon + \epsilon^4)(B + \epsilon^2 C + \epsilon^3 F + \epsilon^4 D + \epsilon E), \end{aligned}$$

all which identities are at once verified on recollecting that

$$1 = -\epsilon - \epsilon^2 - \epsilon^3 - \epsilon^4 \text{ and } \sqrt{5} = \epsilon - \epsilon^2 - \epsilon^3 + \epsilon^4.$$

We can now write down the 6 systems, each of them under its 10 forms: these, in fact, are [arranged in a table with 6 rows, the first three systems being $ABCDEF$ and its 9 other forms, and systems 4–6 being analogous forms with sign changes as indicated above].

The table shows the 60 forms arranged as follows:

- First system ($ABCDEF$): 10 forms — cyclic permutations of B, C, D, E, F and their reverses.

- Second system ($BCAF\bar{D}\bar{E}$): 10 forms — obtained from the first system by the substitution which replaces $A \rightarrow B, B \rightarrow C, C \rightarrow A, D \rightarrow F, E \rightarrow \bar{D}, F \rightarrow \bar{E}$.
- Third system ($CAB\bar{E}\bar{F}\bar{D}$): 10 forms — obtained by repeating the substitution.
- Fourth, fifth, and sixth systems: obtained by applying the substitutions which change the upper line into the three forms of the first system ($ADEFBC, AEFBCD, AFBCDE$) to $BCAF\bar{D}\bar{E}$.

As to the formation of this table, observe that we have $ABCDEF$ and $BCAF\bar{D}\bar{E}$; repeating the substitution, we have

$$\begin{aligned} &ABCDEF \\ &BCAF\bar{D}\bar{E} \\ &CAB\bar{E}\bar{F}\bar{D} \\ &ABCDEF; \end{aligned}$$

viz. we have thus the $CAB\bar{E}\bar{F}\bar{D}$ which belongs to the third system; but nothing further, for the substitution is periodic of the third order, and the three forms are merely repeated. But if, upon

$$\begin{aligned} &ABCDEF \\ &BCAF\bar{D}\bar{E}, \end{aligned}$$

we operate successively with the substitutions which change the upper line into the three forms of the first system

$$ADEFBC, \quad AEFBCD, \quad AFBCDE,$$

then the lower line is changed into the three forms

$$DEAC\bar{F}\bar{B}, \quad EFAD\bar{B}\bar{C}, \quad FBAE\bar{C}\bar{D},$$

which are forms belonging to the fourth, fifth, and sixth systems respectively. By way of verification, observe that, for instance repeating upon the second line the substitution $ABCDEF$, in place of $DEAC\bar{F}\bar{B}$, we obtain $C\bar{F}DAB\bar{E}$, which is one of the forms of the third system, assumed to have been previously found; and so in other instances.

Reverting to the three equations belonging to the form $ABCDEF$, by subtracting the third from the second equation, we obtain a linear relation between the four square roots C, D, E, F , viz. this is

$$0 = (\epsilon^3 - \epsilon^2)(C - F) + (\epsilon - \epsilon^4)(D - E);$$

and the same equation is obtained by means of the form $ABFEDC$. We thus obtain from the thirty lines of either column of the preceding table, thirty such equations, but obviously the number of such equations should be fifteen, for there can be but one relation between any four square roots C, D, E, F ; consequently each equation will be obtained twice; and, in fact, it is clear that the forms $ABCDEF$ and $BAF\bar{D}\bar{E}C$, and so in general any two forms which begin with the same pair of letters, give the same equation. But for greater symmetry, I write down the thirty equations in the order in which they are given by the left-hand column of the table: the equations are [arranged in 6 groups of 5].

I write down the 15 equations in a different order, in some cases changing the sign of the whole equation, as follows [the equations are arranged in 3 groups of 5]:

$$\begin{aligned} 0 &= (\epsilon^3 - \epsilon^2)(A + C) + (\epsilon - \epsilon^4)(B + D), \\ 0 &= (\epsilon^3 - \epsilon^2)(B + C) + (\epsilon - \epsilon^4)(A - E), \\ 0 &= (\epsilon^3 - \epsilon^2)(A + B) + (\epsilon - \epsilon^4)(C + F), \\ 0 &= (\epsilon^3 - \epsilon^2)(D + E) + (\epsilon - \epsilon^4)(A - B), \\ 0 &= (\epsilon^3 - \epsilon^2)(B + F) + (\epsilon - \epsilon^4)(A - D), \\ 0 &= (\epsilon^3 - \epsilon^2)(A + F) + (\epsilon - \epsilon^4)(B + E), \\ 0 &= (\epsilon^3 - \epsilon^2)(A + D) + (\epsilon - \epsilon^4)(C + E), \\ 0 &= (\epsilon^3 - \epsilon^2)(C + D) + (\epsilon - \epsilon^4)(A - F), \\ 0 &= (\epsilon^3 - \epsilon^2)(E + F) + (\epsilon - \epsilon^4)(A - C), \\ 0 &= (\epsilon^3 - \epsilon^2)(A + E) + (\epsilon - \epsilon^4)(D + F), \\ 0 &= (\epsilon^3 - \epsilon^2)(B - E) + (\epsilon - \epsilon^4)(C - D), \\ 0 &= (\epsilon^3 - \epsilon^2)(F - D) + (\epsilon - \epsilon^4)(B - C), \\ 0 &= (\epsilon^3 - \epsilon^2)(E - C) + (\epsilon - \epsilon^4)(F - B), \\ 0 &= (\epsilon^3 - \epsilon^2)(D - B) + (\epsilon - \epsilon^4)(E - F), \\ 0 &= (\epsilon^3 - \epsilon^2)(C - F) + (\epsilon - \epsilon^4)(D - E). \end{aligned}$$

The first three of these equations, or writing $\lambda = \epsilon^3 - \epsilon^2$, $\mu = \epsilon - \epsilon^4$, say

$$\begin{aligned}\lambda(A + C) + \mu(B + D) &= 0, \\ \lambda(B + C) + \mu(A - E) &= 0, \\ \lambda(A + B) + \mu(C + F) &= 0,\end{aligned}$$

constitute the entire system of independent linear relations between the square roots A, B, C, D, E, F . The coefficients λ, μ are such that

$$\lambda^2 - \mu^2 = \lambda\mu (= \epsilon - \epsilon^2 - \epsilon^3 + \epsilon^4, = \sqrt{5}),$$

and it is hence easy to verify that the remaining twelve equations can be deduced from these by the elimination of one or two of the square roots A, B, C . For instance, to eliminate A from the first and second equations, multiplying by $-\mu, \lambda$ and adding, we obtain

$$(-\lambda\mu + \lambda^2)C + (-\mu^2 + \lambda^2)B - \mu^2D - \lambda\mu E = 0,$$

that is,

$$\mu^2C + \lambda\mu B - \mu^2D - \lambda\mu E = 0,$$

or finally

$$\lambda(B - E) + \mu(C - D) = 0,$$

which is one of the equations. And so again eliminating A from the first and third equations, we find

$$\lambda(B - C) + \mu(C + F - B - D) = 0,$$

that is,

$$(\lambda - \mu)(B - C) + \mu(F - D) = 0,$$

or multiplying by λ ,

$$\mu^2(B - C) + \lambda\mu(F - D) = 0,$$

that is, finally

$$\lambda(F - D) + \mu(B - C) = 0,$$

which is one of the equations.

Cambridge, 21 March 1887.

[868]

ON THE INTERSECTION OF CURVES.

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 85—90.]

It is only recently that I have studied Bacharach's paper "Ueber den Cayley'schen Schnittpunktsatz," *Math. Ann.*, t. xxvi. (1885), pp. 275—299: his theorem in regard to the case where the δ points lie on a curve of the order $\gamma - 3$ is a very interesting and valuable one, but I consider it rather as an addition than as a correction to my original theorem; and I cannot by any means agree that the method by counting of constants is to be rejected as not trustworthy; on the contrary, it seems to me to be the proper foundation of the whole theory; it must of course be employed with due consideration of special cases. I reproduce the theorem in what appears to me the complete form.

Writing with Bacharach

$$r \equiv m, \quad r \equiv n, \quad r \equiv m+n-3, \quad \gamma = m+n-r, \quad \delta = \frac{1}{2}(\gamma-1)(\gamma-2),$$

and assuming $n \equiv m$, (these equations and inequalities are to be attended to throughout the present paper), I consider two curves of the orders m, n respectively meeting in δ points B , and in $(mn - \delta)$ points A ; and I state the theorem as follows:

1°. The $mn - \delta$ points A are in general such that a curve of the order r , which passes through $mn - \delta - 1$ of these points, does not of necessity pass through the remaining point; and in this case the general curve of the order r , which passes through the $mn - \delta$ points A , has for its form of equation

$$0 = L_{r-m}P_m + M_{r-n}Q_n,$$

where $P_m = 0$, $Q_n = 0$ are the equations of the given curves and L_{r-m} , M_{r-n} denote functions of the orders $r - m$, $r - n$ respectively; and it thus appears that the curve of the order r through the $mn - \delta$ points A passes also through the δ points B .

2°. If however the δ points B are on a curve of the order $\gamma - 3$, then the $mn - \delta$ points A are a system such that every curve of the order r passing through $mn - \delta - 1$ of these points passes through

the remaining point; and in this case the general curve of the order r , which passes through the $mn - \delta$ points A , has for its form of equation

$$0 = \Omega_r + L_{r-m}P_m + M_{r-n}Q_n,$$

$\Omega_r = 0$ is a particular curve through the $mn - \delta$ points A , which does not go through any of the points B ; and consequently the curve of the order r does not pass through any of the points B .

For the proof of the theorem I premise as follows:

A curve of the order r depends upon $\frac{1}{2}r(r+3)$ constants, or to use a convenient expression, its Postulandum is $= \frac{1}{2}r(r+3)$: if the curve has to pass through a given point, this imposes a single relation upon the constants, or say the Postulation is $= 1$; similarly, if the curve has to pass through k given points, this imposes k relations, or say the Postulation is $= k$. The points may be however a special system, for instance, they may be such that every curve of the order r which passes through $k-1$ of the points, will pass through the remaining point; the Postulation is in this case $= k-1$; and so in other cases. Assuming the Postulation of the k points to be $= k$, then the Postulandum of a curve of the order r through the k points is $= \frac{1}{2}r(r+3) - k$. I stop to remark that the Postulation has reference to the particular curve or other entity in question; thus in the case of a curve passing through k points, the Postulation for a curve of a certain order may be $= k$, and for a curve of a different order it may be less than k .

Considering now, as above, two given curves of the orders m and n intersecting in the $mn - \delta$ points A and the δ points B , then assuming that the $mn - \delta$ points are not a special system, viz. that their Postulation in regard to a curve of the order r is $= mn - \delta$, the Postulandum of a curve of the order r through the $mn - \delta$ points is

$$= \frac{1}{2}r(r+3) - mn + \delta,$$

which is

$$= \frac{1}{2}(r-m+1)(r-m+2) + \frac{1}{2}(r-n+1)(r-n+2) - 1,$$

viz. this is identically true when for δ we write its value

$$= \frac{1}{2}(m+n-r-1)(m+n-r-2).$$

But we have through the $mn - \delta$ points A , a curve

$$L_{r-m}P_m + M_{r-n}Q_n = 0$$

with the proper Postulandum: viz. L_{r-m} contains $\frac{1}{2}(r-m+1)(r-m+2)$ constants, M_{r-n} contains $\frac{1}{2}(r-n+1)(r-n+2)$ constants, and there is a diminution -1 for the constant which divides out; hence this is the general equation of the curve of the order r through the $mn-\delta$ points A ; and the curve passes through the remaining δ points B .

In the case where the δ points B are on a curve of the order $\gamma-3$, (observe that this is a single condition imposed on the δ , $=\frac{1}{2}(\gamma-1)(\gamma-2)$ points, for a curve of the order $(\gamma-3)$ can be drawn through $\frac{1}{2}\gamma(\gamma-3)$ points), it is to be shown that the Postulation of the $mn-\delta$ points A is $=mn-\delta-1$; for, this being so, the Postulandum of the curve of the order r through the $(mn-\delta)$ points A will be

$$= \frac{1}{2}r(r+3) - mn + \delta + 1,$$

and the equation of the curve will no longer be of the foregoing form, but it will be of the form

$$\Omega_r + L_{r-m}P_m + M_{r-n}Q_n = 0,$$

$\Omega_r = 0$ being a particular curve through the $mn-\delta$ points A , which does not pass through any of the points B . The proof depends on the theory of Residuation: which for the present purpose may be presented under the following form.

Let $A, B, \dots$ denote systems of points upon a given Basis-curve, for instance the foregoing curve $P_m = 0$, of the order m . And let $A = ci$ denote that the points A are the complete intersection of the basis-curve by some other curve; (this implies that the number of the points is $=km$, a multiple of m , and the intersecting curve is then of course a curve of the order k). It is clear that, if $A = ci$, and $B = ci$, then also $A + B = ci$. But conversely we have the theorem that, if $A + B = ci$ and $A = ci$, then also $B = ci$. And we at once deduce the further theorem: if $A + B = ci$, $B + C = ci$, $C + D = ci$, then also $A + D = ci$. For the first and third relations give $A + B + C + D = ci$, and the second relation then gives $A + D = ci$.

Starting now (see the diagram) with the δ points B on a curve of the order $\gamma-3$, suppose that we have through these points the basis-curve $P_m = 0$ of the order m , and another given curve $Q_n = 0$, of the order n ; and let these besides meet in the $mn-\delta$ points A . Let the curve of the order $\gamma-3$ besides meet the basis-curve in the $m(\gamma-3)-\delta$ points C ; and through these let there be drawn a curve of the order

$m-3$, which besides meets the basis-curve in the $m(r-n)+\delta$ points D . We have here $A+B=ci$, $B+C=ci$, $C+D=ci$; consequently $A+D=ci$, that is, the $mn-\delta$ points A and the $m(r-n)+\delta$ points D lie on a curve of the order r . The curve of the order $m-3$ passes through the $m(\gamma-3)-\delta$ points C ; its Postulandum is thus

$$= \frac{1}{2}m(m-3) - m(\gamma-3) + \delta,$$

which is

$$= \frac{1}{2}(r-n+1)(r-n+2).$$

In fact, substituting for γ, δ their values $= m+n-r$, and $\frac{1}{2}(m+n-r-1)(m+n-r-2)$ respectively, this equation is satisfied identically. The system of the $m(r-n)+\delta$ points D thus depends upon $\frac{1}{2}(r-n+1)(r-n+2)$ constants, or say the Postulandum of the points D is $= \frac{1}{2}(r-n+1)(r-n+2)$. It follows that the curve of the order r through the $(mn-\delta)$ points A and the $m(r-n)+\delta$ points D *cannot* have an equation of the form

$$L_{r-m}P_m + M_{r-n}Q_n = 0;$$

for the intersections of this curve with the basis-curve $P_m = 0$ are given by the equation $M_{r-n}Q_n = 0$, which contains only the

$$\frac{1}{2}(r-n+1)(r-n+2) - 1$$

constants of M_{r-n} (one constant of course divides out, giving the diminution -1). The equation must have the more general form

$$\Omega_r + L_{r-m}P_m + M_{r-n}Q_n = 0;$$

and it thus appears that the Postulation of the $mn-\delta$ points A , instead of being $= mn-\delta$, must be $= mn-\delta-1$. This completes the proof.

I notice that, combining the last-mentioned identity

$$\frac{1}{2}m(m-3) - m(\gamma-3) + \delta = \frac{1}{2}(r-n+1)(r-n+2)$$

with the like identity

$$\frac{1}{2}n(n-3) - n(\gamma-3) + \delta = \frac{1}{2}(r-m+1)(r-m+2),$$

we obtain

$$\begin{aligned} & \frac{1}{2}m(m-3) + \frac{1}{2}n(n-3) - (m+n)(\gamma-3) + 2\delta \\ &= \frac{1}{2}(r-n+1)(r-n+2) + \frac{1}{2}(r-m+1)(r-m+2), \end{aligned}$$

and consequently, referring to a former result, the left-hand side should be

$$= \frac{1}{2}r(r+3) - mn + \delta + 1;$$

substituting for γ , δ their values, this is at once verified.

As appears by what precedes, Bacharach's special case is that in which the δ , $= \frac{1}{2}(\gamma-1)(\gamma-2)$ points B satisfy the single condition of lying on a curve of the order $\gamma-3$. We may have between the points B more than a single relation; in particular, the points B may be such as to include among themselves the complete intersection of two curves of the orders a , b respectively ($ab \equiv \delta$): this will be the case, if the given curves are of the form

$$\begin{aligned} P_m &= \lambda_a S_{m-a} - \mu_b R_{m-b}, \\ Q_n &= \lambda_a V_{n-a} - \mu_b U_{n-b}, \end{aligned}$$

it being understood, here and in what follows, that the values of a , b are such that the suffixes are none of them negative.

The two curves here intersect in the ab points ($\lambda_a = 0$, $\mu_b = 0$), and in the $mn - ab$ points

$$\left\| \begin{array}{ccc} \lambda_a, & R_{m-b}, & U_{n-b} \\ \mu_b, & S_{m-a}, & V_{n-a} \end{array} \right\| = 0,$$

say the $(mn - \delta)$ points A are $mn - ab - \Theta$ of the last-mentioned points, and the δ points B are the remaining Θ points together with the ab points. Here the general form of the curve of the order r passing through the $mn - ab$ points, and therefore through the $mn - \delta$ points A , is

$$\left\| \begin{array}{ccc} L_{r-m-n+a+b}, & M_{r-n}, & N_{r-m} \\ \lambda_a & , & R_{m-b}, & U_{n-b} \\ \mu_b & , & S_{m-a}, & V_{n-a} \end{array} \right\| = 0,$$

where $L_{r-m-n+a+b}$, M_{r-n} , N_{r-m} are arbitrary functions of the orders indicated by the respective suffixes. The theory in regard to the number of constants is of course altogether different from that which belongs to the case of the general functions P_m , Q_n ; and it is probable that much interesting theory would present itself in the consideration of particular cases.

Cambridge, 22 March 1887.

[869]

ON THE TRANSFORMATION OF ELLIPTIC
FUNCTIONS.

[From the *American Journal of Mathematics*, vol. IX. (1887), pp.
193—224.]

THE algebraical theory of the Transformation of Elliptic Functions was established by Jacobi in a remarkably simple and elegant form, but it has not hitherto been developed with much completeness or success. The cases $n = 3$ and $n = 5$ are worked out very completely in the *Fundamenta Nova* (1829); viz. considering the equation

$$\frac{M dy}{\sqrt{1 - y^2 \cdot 1 - \lambda^2 y^2}} = \frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}},$$

($k = u^4$, $\lambda = v^4$; say this is the $Mk\lambda$ - or Muv -form), Jacobi finds, in the two cases respectively, a modular equation between the fourth roots u , v , say the uv -modular equation, and, as rational functions of u , v , the value of M and the values of the coefficients of the several powers of x in the numerator and denominator of the fraction which gives the value of y ; but there is no attempt at a like development of the general case. I shall have occasion to speak of other researches by Jacobi, Brioschi and myself; but I will first mention that my original idea in the present memoir was to develop the following mode of treatment of the theory:

In place of the $Mk\lambda$ -form, using the $\rho\alpha\beta$ -form

$$\frac{dy}{\sqrt{1 - 2\beta y^2 + y^4}} = \frac{\rho dx}{\sqrt{1 - 2\alpha x^2 + x^4}},$$

(I write for greater convenience 2α , 2β in place of the α of Jacobi and Brioschi and the β of Brioschi), we can, by expanding each side in a series, integrating, and reverting the resulting series for y , obtain y in the form

$$y = \rho x (1 + \Pi_1 x^2 + \Pi_2 x^4 + \dots),$$

where Π_1 , Π_2 , Π_3 , ... denote given functions of ρ , α , β . Taking n odd and $= 2s + 1$ we assume for y an expression

$$y = \frac{x (A_s + A_{s-1}x^2 + \dots + A_1x^{2s-2} + x^{2s})}{1 + A_1x^2 + \dots + A_{s-1}x^{2s-2} + A_sx^{2s}},$$

where the last coefficient A_s is at once seen to be $= \rho$. Comparing with the series-value $y = \rho x (1 + \Pi_1 x^2 + \Pi_2 x^4 + \dots)$, we have an infinite series of equations. The first of these is, in fact, $A_s = \rho$; the next $(s-1)$ equations give linearly $A_1, A_2, \dots, A_{s-1}$ in terms of the coefficients Π ; that is, of ρ, α, β : the two which follow serve in effect to determine ρ, β as functions of α : and then, ρ and β having these values, all the remaining equations will be satisfied identically.

The process is an eminently practical one, so far as regards the determination of the coefficients $A_1, A_2, \dots, A_{s-1}$ as functions of ρ, α, β ; it is less so, and requires eliminations more or less complicated, as regards the determination of the relations between ρ, α, β . As to this, it may be remarked that the problem is not so much the determination of the equation between ρ and α (or say the $\rho\alpha$ -multiplier equation, or simply the $\rho\alpha$ -equation), and of the equation between β, α (or say the $\alpha\beta$ -modular equation, or simply the $\alpha\beta$ -equation), as it is to determine the complete system of relations between ρ, α, β ; treating these as coordinates, we have what may be called the multiplier-modular-curve, or say the MM-curve, and the relations in question are those which determine this curve.

In the absence of special exceptions, it follows from general principles that the coefficients $A_1, A_2, \dots, A_{s-1}$, *quà* rational functions of ρ, α, β , must also be rational functions of α, β or of α, ρ ; and I think it may be assumed that this is the case; the method, however, affords but little assistance towards thus expressing them.

In connexion with the foregoing theory, I consider the solutions of the problem of transformation given by Jacobi's partial differential equation ("Suite de Notices sur les Fonctions elliptiques," *Crelle*, t. IV. (1829), pp. 185—193), and by what I call the Jacobi-Brioschi differential equations. The first and third of these were obtained by Jacobi in the memoir*, "De functionibus ellipticis Commentatio," *Crelle*, t. IV. (1829), pp. 371—390 (see p. 376); but the second equation, which completes the system, was, I believe, first given by Brioschi in the second appendix to his translation of my *Elliptic Functions: Trattato elementare delle Funzioni ellittiche*: Milan, 1880. I had, strangely enough, overlooked the great importance of these equations. I shall have occasion also to refer to results, and further develop the theory contained in my memoir, "On the Transformation of Elliptic Functions," *Phil. Trans.*, t. CLXIV. (1874), pp. 397—456, [578], and the addition thereto, *Phil. Trans.*, t. CLXXXIX. (1878), pp. 419—424, [692].

I remark that, while I have only worked out the formulæ for the cases $n = 3$ and $n = 5$, and a few formulæ for the case $n = 7$, the memoir is intended to be a contribution to the general theory of the $\rho\alpha\beta$ -transformation; I hope to be able to complete the theory for the case $n = 7$.

[* Ges. *Werke*, bd. I., pp. 295—318; in particular, p. 303.]

Art. Nos. 1 to 5. *The Fundamental Differential Equation.*

1. The transformation-equation is

$$\frac{dy}{\sqrt{1-2\beta y^2+y^4}} = \frac{\rho dx}{\sqrt{1-2\alpha x^2+x^4}},$$

where n is the order of the transformation; viz. expanding each side in series, integrating, and reverting, we obtain y as a series $\rho x (1 + \Pi_1 x^2 + \Pi_2 x^4 + \dots)$, where $\Pi_1, \Pi_2, \dots$ are given functions of ρ, α, β .

2. The $\rho\alpha\beta$ -equation is connected with the Muv -equation by a quadric transformation.

Starting from the equation

$$\frac{M dy}{\sqrt{1-y^2 \cdot 1 - \lambda^2 y^2}} = \frac{dx}{\sqrt{1-x^2 \cdot 1 - k^2 x^2}},$$

we may, by effecting on each side a quadric transformation, convert this into

$$\frac{dy}{\sqrt{1-2(2v^8-1)y^2+y^4}} = \frac{M^{-1} dx}{\sqrt{1-2(2u^8-1)x^2+x^4}},$$

and this being so, we have, between M^{-1} and $2u^8-1$, the relation

$$F\left(\frac{1}{M}, 2u^8-1\right) = 0;$$

or, conversely, if this be the form of the Mu -multiplier equation, then the $\rho\alpha$ -multiplier equation is $F(\rho, \alpha) = 0$.

3. The quadric transformations are

$$x = \frac{\sqrt{1-x^2}}{x \sqrt{1-k^2 x^2}}, \quad y = \frac{\sqrt{1-y^2}}{y \sqrt{1-\lambda^2 y^2}}.$$

We have then only to show that

$$\frac{dx}{\sqrt{1-2(2u^8-1)x^2+x^4}} = \frac{dx}{\sqrt{1-x^2 \cdot 1 - k^2 x^2}};$$

for then, in like manner,

$$\frac{dy}{\sqrt{1-2(2v^8-1)y^2+y^4}} = \frac{dy}{\sqrt{1-y^2 \cdot 1 - \lambda^2 y^2}},$$

and we pass from the assumed differential relation between x, y to the above-mentioned differential equation between x, y .

4. For the quadric transformation between x, x , write

$$\theta^{\frac{1}{2}} = k - ik', \quad \theta^{-\frac{1}{2}} = k + ik',$$

(whence also $\theta = \frac{k - ik'}{k + ik'}$), and therefore

$$\theta^{\frac{1}{2}} + \theta^{-\frac{1}{2}} = 2k, \quad \theta + \theta^{-1} = 2k^2 - 2k'^2 = 2(2k^2 - 1), = 2(2u^8 - 1);$$

we have

$$\begin{aligned} 1 - \theta x^2 &= 1 - \theta \frac{1 - x^2}{x^2(1 - k^2 x^2)} = \frac{1}{x^2(1 - k^2 x^2)} \{-\theta + (\theta + 1)x^2 - k^2 x^4\}, \\ &= \frac{-\theta}{x^2(1 - k^2 x^2)} (1 - \theta^{-1} k x^2)^2; \end{aligned}$$

and thence

$$\sqrt{1 - \theta x^2} = \frac{\sqrt{-\theta} (1 - \theta^{-1} k x^2)}{x \sqrt{1 - k^2 x^2}},$$

and in like manner

$$\sqrt{1 - \theta^{-1} x^2} = \frac{\sqrt{-\theta^{-1}} (1 - \theta k x^2)}{x \sqrt{1 - k^2 x^2}}.$$

But $\sqrt{-\theta} \cdot \sqrt{-\theta^{-1}} = 1$, and hence

$$\begin{aligned} \sqrt{1 - \theta x^2} \cdot \sqrt{1 - \theta^{-1} x^2} &= \frac{(1 - \theta^{-1} k x^2)(1 - \theta k x^2)}{x^2(1 - k^2 x^2)} \\ &= \frac{1 - (\theta + \theta^{-1}) k x^2 + k^2 x^4}{x^2(1 - k^2 x^2)}; \end{aligned}$$

substituting for $\theta + \theta^{-1}$ its value $= 2(2u^8 - 1)$, the numerator is

$$1 - 2(2u^8 - 1) k x^2 + k^2 x^4,$$

and this divided by $(1 - k^2 x^2)$ is $= 1 - 2(2u^8 - 1)x^2 + x^4$; hence

$$\sqrt{1 - \theta x^2} \cdot \sqrt{1 - \theta^{-1} x^2} = \frac{1 - 2(2u^8 - 1)x^2 + x^4}{x^2(1 - k^2 x^2)},$$

and therefore

$$\frac{dx}{\sqrt{1 - \theta x^2} \cdot \sqrt{1 - \theta^{-1} x^2}} = \frac{x(1 - k^2 x^2) dx}{\sqrt{1 - 2(2u^8 - 1)x^2 + x^4}}.$$

But we have

$$x^2 = \frac{1 - x^2}{x^2 (1 - k^2 x^2)},$$

and thence, differentiating,

$$\begin{aligned} 2x \, dx &= \frac{-2x \, dx \cdot x^2 (1 - k^2 x^2) - (1 - x^2) \{2x \, dx \cdot (1 - k^2 x^2) - 2k^2 x^3 \, dx\}}{x^4 (1 - k^2 x^2)^2}, \\ &= \frac{-2x \, dx \cdot x^2 (1 - k^2 x^2) - (1 - x^2) \cdot 2x \, dx \cdot (1 - 2k^2 x^2 + k^2 x^4)}{x^4 (1 - k^2 x^2)^2}, \end{aligned}$$

which is

$$\begin{aligned} &= \frac{2x \, dx \{-x^2 (1 - k^2 x^2) - (1 - x^2) (1 - k^2 x^2)^2\}}{x^4 (1 - k^2 x^2)^2} \\ &= \frac{2x \, dx \{-x^2 - (1 - x^2) (1 - k^2 x^2)\}}{x^4 (1 - k^2 x^2)} \\ &= \frac{2x \, dx \{-1 + k^2 x^2 - k^2 x^4\}}{x^4 (1 - k^2 x^2)} = \frac{-2x \, dx \cdot (1 - k^2 x^4)}{x^4 (1 - k^2 x^2)}, \end{aligned}$$

that is,

$$2x \, dx = \frac{-2x \, dx \cdot (1 - k^2 x^4)}{x^4 (1 - k^2 x^2)},$$

and thence

$$x \, dx = \frac{-x \, dx \cdot (1 - k^2 x^4)}{x^4 (1 - k^2 x^2)}, \quad \text{or} \quad x \, dx \{x^4 (1 - k^2 x^2) + (1 - k^2 x^4)\} = 0,$$

that is, $x \, dx \cdot (1 - k^2 x^2) (1 + x^4) = 0$, which is satisfied identically in virtue of the original equation. Hence

$$\frac{dx}{\sqrt{1 - 2(2u^8 - 1)x^2 + x^4}} = \frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}};$$

and similarly as regards the corresponding equation in y , λ ; hence the theorem is proved.

5. The theorem is the generalisation of a known result in the case of the first order of transformation. In this case we have $x = y$, and the equation becomes

$$\frac{dy}{\sqrt{1 - 2\beta y^2 + y^4}} = \frac{\rho \, dy}{\sqrt{1 - 2\alpha y^2 + y^4}},$$

where ρ is at pleasure $= \pm 1$; so that the $\rho\alpha$ -equation is $\rho^2 - 1 = 0$, and thence $\alpha = \beta$. The relation between ρ and α is that $\rho = 1$ corresponds to $\alpha = \beta$, and $\rho = -1$ to $\alpha = -\beta$.

Art. Nos. 6 to 10. *The $\rho\alpha\beta$ -equation; case $n = 3$.*

6. The equation here is

$$\frac{dy}{\sqrt{1 - 2\beta y^2 + y^4}} = \frac{\rho dx}{\sqrt{1 - 2\alpha x^2 + x^4}};$$

and assuming for y an expression

$$y = \frac{x(\rho + x^2)}{1 + \rho x^2},$$

we find, comparing the coefficients of x^2 on the two sides,

$$A_1 - \rho A_1 = \rho \Pi_1, \quad \text{viz. } A_{s-1} - \rho A_1 = \rho \Pi_1,$$

or, substituting for Π_1 its value,

$$A_{s-1} - \rho A_1 = \frac{1}{3}(\alpha\rho - \beta\rho^3).$$

Here $s = 1$, $A_1 = \rho$, and $A_{s-1} = A_0 = \rho$; and we have

$$\rho - \rho \cdot \rho = \frac{1}{3}(\alpha\rho - \beta\rho^3),$$

that is,

$$1 - \rho = \frac{1}{3}(\alpha - \beta\rho^2),$$

or say

$$\beta = \frac{\alpha - 3 + 3\rho}{\rho^2}.$$

Also, from the equation

$$\frac{dy}{\sqrt{1 - 2\beta y^2 + y^4}} = \frac{\rho dx}{\sqrt{1 - 2\alpha x^2 + x^4}},$$

we have at once

$$\rho^4 = \frac{1 - \beta}{1 - \alpha}, \quad \rho^4 = \frac{1 + \beta}{1 + \alpha},$$

and from these, eliminating β , we find

$$\rho^8 = \frac{1 - \beta^2}{1 - \alpha^2},$$

which, combined with the equation

$$\beta = \frac{\alpha - 3 + 3\rho}{\rho^2},$$

gives

$$\rho^8 (1 - \alpha^2) = 1 - \frac{(\alpha - 3 + 3\rho)^2}{\rho^4},$$

that is,

$$\rho^{12} (1 - \alpha^2) = \rho^4 - (\alpha - 3 + 3\rho)^2,$$

or say

$$\rho^{12} - \rho^{12}\alpha^2 - \rho^4 + (\alpha - 3 + 3\rho)^2 = 0.$$

7. Writing this in the form

$$\rho^{12} - \rho^4 + 9(1 - \rho)^2 + 6\alpha(1 - \rho)(3 - 3\rho) + \alpha^2 \{9(1 - \rho)^2 - \rho^{12}\} = 0,$$

we have the terms without α :

$$\rho^{12} - \rho^4 + 9 - 18\rho + 9\rho^2 = 0,$$

and the coefficient of α :

$$6\rho^{12} - 36\rho + 36 = 0,$$

and the coefficient of α^2 :

$$9 - 18\rho + 9\rho^2 - \rho^{12} = 0.$$

We hence obtain

$$\alpha = \frac{-3\rho^{12} + 18\rho - 18}{9 - 18\rho + 9\rho^2 - \rho^{12}},$$

and substituting this value in the original equation, we find the $\rho\alpha$ -equation; but it is more convenient to write

$$\alpha = -1 + \frac{9\rho^2 - 18\rho + 9}{9 - 18\rho + 9\rho^2 - \rho^{12}},$$

and thence

$$\alpha + 1 = \frac{9(1 - \rho)^2}{(3 - 3\rho - \rho^6)(3 - 3\rho + \rho^6)},$$

and similarly

$$\alpha - 1 = \frac{-2\rho^6 + 9\rho^2 - 18\rho + 8}{(3 - 3\rho - \rho^6)(3 - 3\rho + \rho^6)},$$

8. From the equation $\beta = \frac{\alpha - 3 + 3\rho}{\rho^2}$, writing for α its value, we find

$$\beta = \frac{-1}{\rho^2} \frac{3\rho^{12} - 18\rho + 18 + 3\rho^2 - 9\rho + 9\rho^2 - \rho^{12}}{9 - 18\rho + 9\rho^2 - \rho^{12}},$$

that is,

$$\beta = \frac{-1}{\rho^2} \frac{2\rho^{12} + 12\rho^2 - 27\rho + 27}{9 - 18\rho + 9\rho^2 - \rho^{12}},$$

and we thence obtain

$$\begin{aligned} \beta + 1 &= \frac{-2\rho^{12} - 12\rho^2 + 27\rho - 27 + 9\rho^2 - 18\rho + 9\rho^2 - \rho^{14}}{\rho^2 (9 - 18\rho + 9\rho^2 - \rho^{12})}, \\ &= \frac{-2\rho^{12} + 6\rho^2 + 9\rho - 27 - \rho^{14}}{\rho^2 (9 - 18\rho + 9\rho^2 - \rho^{12})}, \end{aligned}$$

and

$$\begin{aligned} \beta - 1 &= \frac{-2\rho^{12} - 12\rho^2 + 27\rho - 27 - 9\rho^2 + 18\rho - 9\rho^2 + \rho^{14}}{\rho^2 (9 - 18\rho + 9\rho^2 - \rho^{12})}, \\ &= \frac{-2\rho^{12} - 30\rho^2 + 45\rho - 27 + \rho^{14}}{\rho^2 (9 - 18\rho + 9\rho^2 - \rho^{12})}. \end{aligned}$$

9. The $\rho\alpha$ -equation may be obtained by eliminating β between the two equations

$$\rho^4 = \frac{1 - \beta}{1 - \alpha}, \quad \rho^4 = \frac{1 + \beta}{1 + \alpha},$$

which give

$$\beta = \frac{\alpha - 3 + 3\rho}{\rho^2}, \quad \text{and} \quad \rho^8 = \frac{1 - \beta^2}{1 - \alpha^2},$$

and substituting for β in the last equation. We find

$$\rho^8 (1 - \alpha^2) = 1 - \frac{(\alpha - 3 + 3\rho)^2}{\rho^4},$$

that is,

$$\rho^{12} (1 - \alpha^2) = \rho^4 - (\alpha - 3 + 3\rho)^2,$$

which may be written

$$\rho^{12} - \rho^4 + 9 - 18\rho + 9\rho^2 + \alpha (6\rho^{12} - 36\rho + 36) + \alpha^2 (9 - 18\rho + 9\rho^2 - \rho^{12}) = 0,$$

and the $\rho\alpha$ -equation is thus

$$\alpha^2 (9-18\rho+9\rho^2-\rho^{12})+\alpha (6\rho^{12}-36\rho+36)+(\rho^{12}-\rho^4+9-18\rho+9\rho^2) = 0.$$

The discriminant of this equation is

$$(6\rho^{12}-36\rho+36)^2-4(9-18\rho+9\rho^2-\rho^{12})(\rho^{12}-\rho^4+9-18\rho+9\rho^2),$$

which is, in fact, a perfect square; and the equation may be written

$$\{\alpha(9-18\rho+9\rho^2-\rho^{12})+(3\rho^{12}-18\rho+18)\}^2=(3\rho^{12}-18\rho+18)^2-(9-18\rho+9\rho^2-\rho^{12})(\rho^{12}-\rho^4+9-18\rho+9\rho^2),$$

where the right-hand side is

$$=(3\rho^{12}-18\rho+18)^2-(9-18\rho+9\rho^2-\rho^{12})(\rho^{12}-\rho^4+9-18\rho+9\rho^2).$$

10. For the case $n = 3$, the modular equation in the uv -form is $u^4 + v^4 + 2uv(1 - u^2v^2) = 0$, or say $(u^4 + v^4 + 2uv)^2 = 16u^4v^4$, that is $u^4 + v^4 + 2uv = \pm 4u^2v^2$, giving two equations; these are, in fact, equivalent, the one going into the other by the change of sign of u or v .

Art. Nos. 11 to 18. *The $\rho\alpha\beta$ -equation; case $n = 5$.*

11. The equation is

$$\frac{dy}{\sqrt{1 - 2\beta y^2 + y^4}} = \frac{\rho dx}{\sqrt{1 - 2\alpha x^2 + x^4}},$$

where we assume

$$y = \frac{x(\rho + A_1 x^2 + x^4)}{1 + A_1 x^2 + \rho x^4},$$

where the coefficients satisfy $A_2 = \rho$, A_1 remaining to be determined.

12. The equations for the determination of ρ , β as functions of α are obtained by comparing the differential equation; the results are as follows. We have

$$-S_1 \rho^3 + \rho^2 + R_1 \rho - 1 = 0,$$

and

$$(-S_2 + 3S_1^2) \rho^4 - (3R_1 S_1 + 1) \rho^2 + R_2 + 1 = 0,$$

where R_1 , R_2 , S_1 , S_2 are given symmetrical functions of the roots.

13. The verification that the equations are satisfied identically is obtained by substituting the values of R_1 , R_2 , S_1 , S_2 in terms of α and β ; these are

$$R_1 = \alpha, \quad R_2 = -\alpha^2 + 2\alpha + 1,$$

and

$$S_1 = \beta, \quad S_2 = -\beta^2 + 2\beta + 1.$$

14. For the values of ρ and β in terms of α , we have the two equations

$$-S_1 \rho^3 + \rho^2 + R_1 \rho - 1 = 0,$$

and

$$(-S_2 + 3S_1^2) \rho^4 - (3R_1 S_1 + 1) \rho^2 + R_2 + 1 = 0,$$

where the R 's and S 's are regarded as given functions of α and β respectively.

15. It is to be observed that the equations are satisfied by $\rho^2 = 1$, $\alpha = \beta$; viz. we have the transformation $y = \frac{x(\pm 1 + x^2)}{1 \pm x^2}$; that is, $y = \pm x$, which is the transformation of the first order, $n = 1$. The two equations represent surfaces of the orders 4 and 6 respectively, and they have thus a complete intersection of the order 24. As part of

this, we have, as just shown, each of the two lines ($\rho = 1, \alpha = \beta$) and ($\rho = -1, \alpha = \beta$); but there is a more considerable reduction of order to be accounted for, the proper MM-curve being, as will appear, a unicursal curve of the order = 6.

16. Multiplying the second equation by $10\rho^2$, and for $10R_2$ and $10S_2$ writing their values $27R_1^2 - 1$ and $27S_1^2 - 1$ respectively, we have

$$(3S_1^2 + 1)\rho^6 - (30R_1S_1 + 10)\rho^4 + (27R_1^2 + 9)\rho^2 = 0;$$

and if herein we substitute for $S_1\rho^3$ its value from the first equation, $= \rho^2 + R_1\rho - 1$, we have

$$3(\rho^2 + R_1\rho - 1)^2 + \rho^6 - 30R_1\rho(\rho^2 + R_1\rho - 1) - 10\rho^4 + (27R_1^2 + 9)\rho^2 = 0;$$

that is,

$$\rho^6 - 7\rho^4 - 24R_1\rho^3 + 3\rho^2 + 24R_1\rho + 3 = 0;$$

viz. this is

$$(\rho^2 - 1)(\rho^4 - 6\rho^2 - 24R_1\rho - 3) = 0,$$

containing, as it ought to do, the factor $\rho^2 - 1$. Throwing this out, and repeating the first equation, we have

$$\begin{aligned} -S_1\rho^3 + \rho^2 + R_1\rho - 1 &= 0, \\ \rho^4 - 6\rho^2 - 24R_1\rho - 3 &= 0, \end{aligned}$$

which two equations may be replaced by

$$\begin{aligned} \rho^4 - 24S_1\rho^3 + 18\rho^2 - 27 &= 0, \\ \rho^4 - \rho^2 - 24R_1\rho - 3 &= 0. \end{aligned}$$

17. From these two equations we eliminate R_1 and obtain a single equation in ρ and S_1 ; and then, substituting for S_1 its value in terms of β , we obtain the equation in ρ, β . Or, what is the same thing, eliminating R_1 , we obtain the equation

$$\rho^4 - 24S_1\rho^3 + 18\rho^2 - 27 = 0,$$

which, together with the equation

$$-S_1\rho^3 + \rho^2 + R_1\rho - 1 = 0,$$

constitutes the system of equations for the determination of ρ and β as functions of α .

18. The two equations may be represented as follows: writing for shortness $\Theta_1 = 0$ and $\Theta_2 = 0$ to denote the equations

$$\Theta_1 = -S_1\rho^3 + \rho^2 + R_1\rho - 1,$$

and

$$\Theta_2 = (-S_2 + 3S_1^2)\rho^4 - (3R_1S_1 + 1)\rho^2 + R_2 + 1,$$

we have identically

$$\begin{aligned}\Theta_2 &= \rho^4 - 6\rho^2 - 24R_1\rho - 3 - 10\rho^2(-S_1\rho^3 + \rho^2 + R_1\rho - 1) \\ &= \rho^4 - 6\rho^2 - 24R_1\rho - 3 - 10\rho^2\Theta_1,\end{aligned}$$

and we thence have

$$\Theta_2 + 10\rho^2\Theta_1 = \rho^4 - 6\rho^2 - 24R_1\rho - 3.$$

Art. Nos. 19 to 26. *The case $n = 5$ (continued); elimination of R_1 ; result for $\alpha\beta$ -equation.*

19. The elimination of R_1 from the two equations

$$-S_1\rho^3 + \rho^2 + R_1\rho - 1 = 0, \quad \text{and} \quad \rho^4 - 6\rho^2 - 24R_1\rho - 3 = 0,$$

gives the equation

$$\rho^4 - 24S_1\rho^3 + 18\rho^2 - 27 = 0,$$

and this being so, we have

$$R_1 = \frac{S_1\rho^3 - \rho^2 + 1}{\rho},$$

and substituting in the second equation, we have identically

$$\rho^4 - 6\rho^2 - 24(S_1\rho^3 - \rho^2 + 1) - 3 = 0,$$

that is,

$$\rho^4 - 24S_1\rho^3 + 18\rho^2 - 27 = 0,$$

as above.

20. To complete the theory for the case $n = 5$, we require the equation between α and β (the $\alpha\beta$ -equation) and the expression for ρ as a rational function of α . The $\alpha\beta$ -equation is obtained by eliminating ρ from the two equations; and the value of ρ is then to be obtained from the resulting equation. The process is somewhat complicated, but the final result is as follows: the $\alpha\beta$ -equation is

$$\alpha^3 - \beta^3 - \alpha\beta(\alpha\beta - 2)(\alpha - \beta) = 0,$$

or say

$$(\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) - \alpha\beta(\alpha\beta - 2)(\alpha - \beta) = 0,$$

that is,

$$(\alpha - \beta)\{\alpha^2 + \alpha\beta + \beta^2 - \alpha\beta(\alpha\beta - 2)\} = 0,$$

and throwing out the factor $\alpha - \beta$, which corresponds to the first order transformation, we have

$$\alpha^2 + \alpha\beta + \beta^2 - \alpha\beta(\alpha\beta - 2) = 0,$$

that is,

$$\alpha^2 + \alpha\beta + \beta^2 - \alpha^2\beta^2 + 2\alpha\beta = 0,$$

or

$$\alpha^2 + 3\alpha\beta + \beta^2 - \alpha^2\beta^2 = 0,$$

which is the required $\alpha\beta$ -modular equation.

21. The equation may also be written in the form

$$(\alpha^2 - 1)(\beta^2 - 1) - (\alpha\beta + \alpha + \beta)(\alpha\beta - \alpha - \beta) = 0,$$

or

$$(\alpha^2 - 1)(\beta^2 - 1) = (\alpha\beta + \alpha + \beta)(\alpha\beta - \alpha - \beta).$$

22. To obtain the value of ρ , we observe that the two equations giving ρ in terms of α and β are

$$-S_1\rho^3 + \rho^2 + R_1\rho - 1 = 0,$$

and

$$\rho^4 - 6\rho^2 - 24R_1\rho - 3 = 0.$$

Eliminating R_1 , we have the equation in ρ and β ; and then, by means of the $\alpha\beta$ -equation, we can express β in terms of α , and so obtain ρ as a function of α .

23. We have, in fact,

$$R_1 = \alpha, \quad S_1 = \frac{\rho^2 + \alpha\rho - 1}{\rho^3},$$

and substituting these values in the equation

$$\rho^4 - 24S_1\rho^3 + 18\rho^2 - 27 = 0,$$

we find

$$\rho^4 - 24(\rho^2 + \alpha\rho - 1) + 18\rho^2 - 27 = 0,$$

that is,

$$\rho^4 - 6\rho^2 - 24\alpha\rho - 3 = 0,$$

which is the $\rho\alpha$ -equation.

24. The equation is, in fact, a quartic in ρ ; and the value of ρ can be expressed in terms of α by means of the well-known formulæ for the solution of a quartic equation. It is to be observed, however, that the equation is reducible; in fact, it may be written

$$(\rho^2 + 2\rho - 1)^2 - 4\alpha\rho(\rho^2 + 2\rho - 1) + 4\rho^2(\alpha^2 - 1) = 0,$$

and hence we have

$$\rho^2 + 2\rho - 1 = 2\rho(\alpha \pm \sqrt{\alpha^2 - 1}),$$

which gives

$$\rho = -1 + \alpha \pm \sqrt{\alpha^2 - 1} \pm \sqrt{(-1 + \alpha \pm \sqrt{\alpha^2 - 1})^2 + 1}.$$

25. The $\alpha\beta$ -equation may also be obtained from the modular equation in the uv -form. The modular equation for $n = 5$ is

$$u^6 - u^6 + 5u^2v^2(u^2 - v^2) + 4uv(1 - u^4v^4) = 0,$$

and this, by means of the relations

$$\alpha = \frac{1 + u^8}{6u^4}, \quad \beta = \frac{1 + v^8}{6v^4},$$

gives the $\alpha\beta$ -equation as above.

26. The expression for the multiplier M in terms of u and v is

$$M = \frac{v(1 - uv^3)}{v - u^5},$$

and the corresponding expression for ρ in terms of α is obtained from this by means of the quadric transformation.

Art. Nos. 27 to 32. *The case $n = 7$.*

27. For the case $n = 7$, we have $s = 3$, and the assumed form is

$$y = \frac{x(\rho + A_2x^2 + A_1x^4 + x^6)}{1 + A_1x^2 + A_2x^4 + \rho x^6}.$$

28. The first three equations are

$$\begin{aligned} A_2 - \rho A_2 &= \rho \Pi_1, & \text{viz. } A_{s-1} - \rho A_1 &= \rho \Pi_1, \\ A_1 - A_2 A_1 + \rho A_2 &= \rho \Pi_2, \\ 1 - A_1^2 + A_2 A_1 + \rho A_2 &= \rho \Pi_3. \end{aligned}$$

29. From these, eliminating A_1 and A_2 , we obtain two equations between ρ , α , β ; these are of the forms

$$\Theta_1 = 0, \quad \Theta_2 = 0,$$

where $\Theta_1 = 0$ is the equation

$$-S_1\rho^3 + \rho^2 + R_1\rho - 1 = 0,$$

and $\Theta_2 = 0$ is the equation

$$(-S_2 + 3S_1^2)\rho^4 - (3R_1S_1 + 1)\rho^2 + R_2 + 1 = 0.$$

30. These equations serve to determine ρ and β as functions of α ; the process is analogous to that for the case $n = 5$, but the details are more complicated. The resulting $\alpha\beta$ -modular equation is

$$\rho^8 - 28\rho^6 - 112\alpha\rho^5 - 210\rho^4 - 224\alpha\rho^3 + (-1484 + 1344\alpha^2)\rho^2 + (-560\alpha + 512\alpha^3)\rho + 7 = 0.$$

31. The foregoing equation is the $\rho\alpha$ -equation for the case $n = 7$. The corresponding $\alpha\beta$ -modular equation has been obtained in a different form by Prof. Klein; it is expressible in terms of the absolute invariants of the quartic curves, and it presents itself in a very elegant form in his theory.

32. The equations should lead to the before-mentioned $\alpha\beta$ -modular equation and to expressions for A_1 , A_2 as rational functions of α , ρ ; and they should be, all five of them, satisfied by these results; but I do not see how the results are to be worked out; there is, so far as appears, no clue to the discovery of the rational functions of α , ρ .

Art. Nos. 33. *The General Transformation, $n = 2s + 1$.*

33. The equation here is

$$\frac{\rho + A_{s-1}x^2 + \dots}{1 + A_1x^2 + \dots} = \rho(1 + \Pi_1x^2 + \dots).$$

The general theory is sufficiently illustrated by the preceding particular cases, and I wish at present only to notice the equation obtained by comparing the coefficients of x^2 ; viz. this is $A_{s-1} - \rho A_1 = \rho \Pi_1$, or, substituting for Π_1 its value,

$$A_{s-1} - \rho A_1 = \frac{1}{3}(\alpha\rho - \beta\rho^3).$$

Art. Nos. 34 to 42. *The Jacobi-Brioschi Equations.*

34. These were obtained for the differential equation

$$\frac{dx}{\sqrt{a'x^4 + b'x^3 + c'x^2 + d'x + e'}} = \frac{dy}{\sqrt{ay^4 + by^3 + cy^2 + dy + e}};$$

viz. if this be satisfied by $y = U \div V$, where U, V are rational and integral functions of x of the degrees n and $n - 1$ respectively, then, writing for shortness

$$\phi = a'x^4 + b'x^3 + c'x^2 + d'x + e',$$

and using accents to denote differentiation in regard to x , the numerator and denominator U, V satisfy the equations

$$\begin{aligned} (VV'' - V'^2)\phi + \frac{1}{2}VV' \cdot \phi' &+ aU^2 + \frac{1}{2}bUV + pV^2 \\ -(VU'' + V''U - 2V'U')\phi - \frac{1}{2}(VU' + V'U)\phi' + \frac{1}{2}bU^2 + (c - 2p)UV + \frac{1}{2}dV^2 &= 0, \\ (UU'' - U'^2)\phi + \frac{1}{2}UU' \cdot \phi' &+ pU^2 + \frac{1}{2}dUV + eV^2 \end{aligned}$$

where p is a function $= ax^2 + bx + c$, with coefficients a, b, c , the values of which have to be determined. By way of verification, observe that, multiplying by U^2, UV, V^2 , and adding, we obtain

$$-(VU' - V'U)^2\phi + aU^4 + bU^3V + cU^2V^2 + dUV^3 + eV^4 = 0;$$

that is,

$$-\frac{1}{V^2}(VU' - V'U)^2(a'x^4 + b'x^3 + c'x^2 + d'x + e') + ay^4 + by^3 + cy^2 + dy + e = 0,$$

the result obtained by substituting for y its value, $= U \div V$, in the differential equation.

35. Considering the foregoing special form

$$\frac{dx}{\sqrt{1 - 2\alpha x^2 + x^4}} = \frac{dy}{\rho \sqrt{1 - 2\beta y^2 + y^4}},$$

so that a, b, c, d, e have the values $\rho^2, 0, -2\beta\rho^2, 0, \rho^2$ and ϕ is $= 1 - 2\alpha x^2 + x^4$, the equations are

$$\begin{aligned} (VV'' - V'^2)\phi + \frac{1}{2}VV' \cdot \phi' + \rho^2U^2 + pV^2 &= 0, \\ -(VU'' + V''U - 2V'U')\phi + \frac{1}{2}(VU' + V'U)\phi' - (2\beta\rho^2 + 2p)UV &= 0, \\ (UU'' - U'^2)\phi + \frac{1}{2}UU' \cdot \phi' + pU^2 + \rho^2V^2 &= 0. \end{aligned}$$

36. Writing $U = x(A_s + A_{s-1}x^2 + \dots + x^{2s})$, $V = 1 + A_1x^2 + \dots + A_sx^{2s}$, and observing that $\phi = 1 - 2\alpha x^2 + x^4$, $\phi' = -4\alpha x + 4x^3$, the equations become

$$\begin{aligned}\Theta_s &= 0, \\ \Theta_{s-1} &= 0, \\ &\vdots \\ \Theta_0 &= 0, \\ \Theta_{-1} &= 0;\end{aligned}$$

a system of $2s + 2, = n + 1$ equations, which are satisfied by proper values of the $s + 1, = \frac{1}{2}(n + 1)$ coefficients A , and the function p of the second degree; that is, they are satisfied by $\frac{1}{2}(n + 1) + 3, = \frac{1}{2}(n + 7)$ disposable constants; and the number of equations being $= n + 1$, there remain $\frac{1}{2}(n + 7) - (n + 1), = \frac{1}{2}(5 - n)$ constants disposable.

37. This is $= 1$ for $n = 3$, $= 0$ for $n = 5$, and negative for $n > 5$. We have thus for $n = 3$ a single disposable constant; for $n = 5$ the number of constants is just sufficient; and for $n > 5$ the equations are over-determined.

38. For the case $n = 3$, we have $s = 1$, and the equations are

$$\begin{aligned}\Theta_1 &= 0, \\ \Theta_0 &= 0, \\ \Theta_{-1} &= 0;\end{aligned}$$

three equations for the determination of A_1 , that is, ρ , and the function p , which contains three disposable constants; we have thus four unknowns and three equations, leaving one constant disposable, as above.

39. For the case $n = 5$, we have $s = 2$, and the equations are

$$\begin{aligned}\Theta_2 &= 0, \\ \Theta_1 &= 0, \\ \Theta_0 &= 0, \\ \Theta_{-1} &= 0;\end{aligned}$$

four equations for the determination of A_1, A_2 , that is, A_1, ρ , and the function p , which contains three disposable constants; we have thus

five unknowns and four equations, leaving no constant disposable, as above.

40. For the case $n = 7$, we have $s = 3$, and the equations are

$$\begin{aligned}\Theta_3 &= 0, \\ \Theta_2 &= 0, \\ \Theta_1 &= 0, \\ \Theta_0 &= 0, \\ \Theta_{-1} &= 0;\end{aligned}$$

five equations for the determination of A_1, A_2, A_3 , that is, A_1, A_2, ρ , and the function p , which contains three disposable constants; we have thus six unknowns and five equations, but the number of disposable constants is -1 , that is, the equations are over-determined.

41. It is to be observed that the equations are not all independent; in fact, there is an identical relation between them, so that the number of independent equations is less than the number of equations by one. This may be seen as follows: the equations were obtained from the consideration that the differential equation

$$\frac{dx}{\sqrt{1 - 2\alpha x^2 + x^4}} = \frac{dy}{\rho \sqrt{1 - 2\beta y^2 + y^4}}$$

is satisfied by $y = U \div V$; and this condition gives rise to the identity

$$-(VU' - V'U)^2 (1 - 2\alpha x^2 + x^4) + \rho^2 (U^4 - 2\beta U^2 V^2 + V^4) = 0,$$

which is an identity in x , and therefore gives rise to $2n + 1$ relations between the coefficients; but these are not all independent, for they must satisfy the condition that the equation $y = U \div V$ is a solution of the differential equation; and this condition imposes one relation upon the coefficients, so that the number of independent relations is $2n$, and the number of independent equations is $n + 1$, as above.

42. I consider the case $n = 7$, in order to show the form of the

equations which have to be solved; these equations are

$$\begin{aligned}
 2A_1 - 2A_1 \cdot 1 &= 0, \\
 12A_2 - 2A_1^2 - 2A_1 \cdot 2A_1 - 7 \cdot 1 - 2\alpha(2A_1 + 1 \cdot 2A_1) + \rho^2 \cdot \rho^2 &= 0, \\
 30\rho - 2A_1A_2 + 2A_1 - 2A_1(2A_2 + A_1^2) - 5 \cdot 2A_1 \\
 8A_1\rho - 4A_2^2 + 12A_2 - 2A_1^2 - 2A_1(2\rho + 2A_1A_2) - 3(2A_2 + A_1^2) \\
 -6A_2\rho + 30\rho - 2A_1A_2 - 2A_1(2A_1\rho + A_2^2) - 1(2\rho + 2A_1A_2) \\
 -6\rho^2 + 8A_1\rho - 4A_2^2 - 2A_1(2A_2\rho) + 1(2A_1\rho + A_2^2) \\
 0 - 6A_2\rho - 2A_1 \cdot \rho^2 + 3(2A_2\rho) - 2\alpha(-6\rho^2 + 6 \cdot \rho^2) + \rho^2 \cdot 2A_1 &= 0, \\
 0 - 6\rho^2 + 5 \cdot \rho^2 - 2\alpha(0 + 7 \cdot 0) + \rho^2 \cdot 1 &= 0;
 \end{aligned}$$

viz. the first, seventh and eighth equations are satisfied identically, and there remain five equations connecting ρ , α , A_1 , A_2 .

Art. Nos. 43 to 48. *The Jacobi Partial Differential Equation.*

43. Writing, as above, 2α in place of Jacobi's α , this is

$$(1-2\alpha x^2+x^4) \frac{d^2 z}{dx^2} + (n-1)(2\alpha x-2x^3) \frac{dz}{dx} + n(n-1)x^2 z - 4n(\alpha^2-1) \frac{dz}{d\alpha} = 0,$$

satisfied by the numerator and denominator U, V , each of them taken with the same proper value of the coefficient A_0 , or, what is the same thing, by the values

$$U = \Theta x (A_s + A_{s-1}x^2 + A_{s-2}x^4 + \dots + A_1 x^{2s-2} + x^{2s}),$$

$$V = \Theta (1 + A_1 x^2 + A_2 x^4 + \dots + A_{s-1} x^{2s-2} + A_s x^{2s}),$$

where Θ is an arbitrary constant multiplier.

44. Writing for shortness

$$\phi = 1 - 2\alpha x^2 + x^4, \quad \psi = 2\alpha x - 2x^3,$$

the equation becomes

$$\phi \frac{d^2 z}{dx^2} + (n-1) \psi \frac{dz}{dx} + n(n-1)x^2 z - 4n(\alpha^2-1) \frac{dz}{d\alpha} = 0.$$

45. Assuming $z = a_0 + a_1 x + a_2 x^2 + \dots$, the equation gives a relation between the coefficients $a_0, a_1, a_2, \dots$ and their derivatives in regard to α . In particular, for $z = U$ and $z = V$, we obtain the coefficients $A_1, A_2, \dots, A_s$ as functions of α .

46. For the case $n = 3$, the equation gives at once the value of A_1 , and thence the values of ρ and β as functions of α . For the case $n = 5$, the equation gives two relations between A_1 and α , from which A_1 and ρ can be determined.

47. For the case $n = 7$, the equation gives three relations between A_1, A_2 and α ; but these are not sufficient for the complete determination of the coefficients, and the process becomes more complicated.

48. The Jacobi partial differential equation affords a direct method of obtaining the transformation-formulæ but the process is, in general, one of great difficulty, and the results are not obtained in the most convenient form. The method of the $\rho\alpha\beta$ -equation is, in many respects, preferable; but the two methods are essentially connected, and the results obtained by the one method may always be verified by the other.

Cambridge, 1887.

0.12 870. On the Transformation of Elliptic Functions (Sequel).

[From the *Proceedings of the London Mathematical Society*, vol. xix. (1888), pp. 425–478.]

79. It will presently appear that p is, in fact, expressible as a rational function of p, a : but I am unable to obtain this expression in a simple form. Admitting that p is thus expressible, a direct process for obtaining the expression is as follows. Writing

$$k = \frac{8p(p^2 + 7)a + p^4 + 29p^2 + 49}{p^4 + 5p^2 + 1},$$

and by means hereof introducing in place of a into the equation p , we have for k a cubic equation,

$$ak^3 + bk^2 + ck + d = 0,$$

where the coefficients a, b, c, d are given rational functions of p .

This equation may be written

$$a \left(k + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right) k + \left(d - \frac{b^2}{4a} \right) = 0,$$

where $b = b - 2a\xi$, $c = c - a\xi^2$; and the last three terms will be a square if only $c^2 - 4bd = 0$; that is, if

$$(2a\xi - b)^2 + 4a(2a\xi - c) = 0,$$

a biquadratic equation in ξ which (p being expressible as above) must have one of its roots = a rational function of p .

Calling this ξ , we then have

$$a \left(k + \frac{b}{2a} \right)^2 + \left(\frac{2b\xi + c}{2a} \right)^2 = 0, \quad \text{or say} \quad \left(k + \frac{b}{2a} \right)^2 + \left(\frac{2b\xi + c}{2a} \right)^2 = 0,$$

hence

$$k = \frac{-b \pm \sqrt{-1}(2b\xi + c)}{2a},$$

where ξ denotes a linear function of a as above; the quadric radical will have a rational value, and the form of the equation thus is

$$p = \frac{Aa + B}{Ca + D},$$

where A, B, C, D are rational and integral functions of p . But I am not able to carry out the process.

80. As shown, No. 78, we have

$$\frac{p_2}{p} = \frac{p^4 + 5p^2 + 1}{p^4 + 29p^2 + 49}.$$

Multiplying by the value of β , ante No. 75, we find

$$\beta = \frac{p^4 + 29p^2 + 49}{8p(p^2 + 7)},$$

and we can hence find A_1 and A_2 by the formulae

$$p_2 = p \frac{p^4 + 5p^2 + 1}{p^4 + 29p^2 + 49}, \quad A_1 = \frac{\beta p - p_2}{\beta^2 - 1}, \quad A_2 = \frac{\beta p_2 - p}{\beta^2 - 1},$$

or, for the second of these we may write

$$A_2 = \frac{p(\beta^2 - 1) - \beta(p - \beta p_2)}{\beta^2 - 1} = p - \beta A_1.$$

But in a different point of view, regarding only p_2 , but not p , as a given function of p, a , we must to these equations join the equation

$$12A_2^2 = 6A_1A_2 + 8aA_1 - p^4 + 7, \quad \text{ante No. 69:}$$

and we have thus equations for the determination of A_1, A_2 , and p .

81. We have

$$\begin{aligned} Q &= (p^4 + 5p^2 + 1)(p^4 + 29p^2 + 49)\beta^2 - 1 \\ &= 8p(a^2 + 3)a(p^4 + 29p^2 + 49) - 24p(p^2 + 7), \end{aligned}$$

where the second line is

$$= a(-p^4 - p^2 + 7) - 4p(p^2 + 7) \cdot \frac{p^4 + 29p^2 + 49}{p^4 + 5p^2 + 1}.$$

Uniting the two terms, we have a denominator $8p(p^4 + 5p^2 + 1)$, and in the numerator a term $8pa^2$ which may be got rid of by means of the a -equation; the numerator thus becomes

$$\begin{aligned} &= 96p(-p^4 - p^2 + 7) - 128p^2(p^2 + 7)a \\ &\quad + (a^2 - 1)\{(-p^4 - p^2 + 7)(p^8 + 11p^6 + 63p^4 + 70p^2 - 7)\} \\ &\quad + (p^4 + 5p^2 + 1)^2(p^4 + 13p^2 + 49) - 32p^2(p^2 + 7), \end{aligned}$$

where the whole divides by $8p$; and we finally obtain

$$Q = \frac{12(-p^4 - p^2 + 7) - 16p(p^2 + 7)a + (a^2 - 1)p(p^8 + 17p^6 + 102p^4 + 225p^2 + 97)}{p^4 + 5p^2 + 1}$$

82. Proceeding to calculate the value of $A_1^2 + A_2^2 + a A_1 A_2$, we then have

$$A_1^2 + A_2^2 + a A_1 A_2 = \frac{(\beta^2 - a\beta + 1)p_2^2 + (a - 2\beta)pp_2 + p^2}{\beta^2 - 1}.$$

Multiplying the numerator and denominator by $a^2 + 3$, we have in the numerator a term in $8a^3$, which may be got rid of by means of the pa -equation; the numerator thus becomes

$$\begin{aligned} &= (a^2 + 3)(a^2 - a\beta + 1)p_2^2 \\ &\quad + \{(a^2 + 3)a - 16p(p^2 + 7)\}pp_2 + (a^2 + 3)p^2, \end{aligned}$$

which in virtue of

$$\beta = \frac{p^4 + 29p^2 + 49}{8p(p^2 + 7)}$$

is a rational and integral function of p , a , and p_2 ; and the whole expression divides by $a^2 + 3$.

But

$$p_2 = p \frac{p^4 + 5p^2 + 1}{p^4 + 29p^2 + 49},$$

and substituting this value, the expression becomes a rational and integral function of p and a ; and writing as before $a^2 = 3 + 8pa$ (where observe that a satisfies the a -equation), the result will be a linear function of a .

83. There exists a very elegant formula for the modulus k of the transformed elliptic function. We have

$$k = \frac{p^4 + 29p^2 + 49}{8p(p^2 + 7)} \cdot \frac{p^4 + 5p^2 + 1}{p^4 + 29p^2 + 49} = \frac{p^4 + 5p^2 + 1}{8p(p^2 + 7)}.$$

But it is to be remarked that the value of k is not in general expressible as a rational function of p and a ; it is only when p is expressible as a rational function of p and a that k becomes so expressible.

84. I verify that the modular equation is satisfied for the foregoing value of k . The modular equation may be written

$$\sqrt{k k'} + \sqrt{l l'} = 1,$$

where k, k' are the complementary moduli; or say

$$\sqrt{k l} + \sqrt{k' l'} = 1,$$

where k, l are any two roots of the modular equation, and k', l' are the complementary moduli. Writing $k = \frac{p^4 + 5p^2 + 1}{8p(p^2 + 7)}$, we have

$$k' = \frac{\sqrt{(p^4 + 29p^2 + 49)^2 - (p^4 + 5p^2 + 1)^2}}{8p(p^2 + 7)},$$

and the equation to be verified is

$$\sqrt{k l} + \sqrt{k' l'} = 1.$$

85. With a formula of transformation

$$y = \frac{x(1 + \alpha x^2 + \beta x^4)}{1 + \gamma x^2 + \delta x^4},$$

where $\alpha, \beta, \gamma, \delta$ are given functions of n , the order being $= 2\nu + 1$, we have

$$M = \frac{1}{2\nu + 1} \frac{y}{x} \cdot \frac{dy}{dx} \cdot \frac{1}{1 - \frac{y^2}{x^2}} \cdot \frac{1}{1 - k^2 x^2 y^2}$$

for the multiplier. And thence

$$M = \frac{1}{2\nu + 1} \frac{1 + \alpha x^2 + \beta x^4}{1 + \gamma x^2 + \delta x^4} \cdot \frac{d}{dx} \left(\frac{x(1 + \alpha x^2 + \beta x^4)}{1 + \gamma x^2 + \delta x^4} \right) \cdot \frac{1}{1 - \left(\frac{1 + \alpha x^2 + \beta x^4}{1 + \gamma x^2 + \delta x^4} \right)^2} \cdot \frac{1}{1 - k^2 x^2 \left(\frac{x(1 + \alpha x^2 + \beta x^4)}{1 + \gamma x^2 + \delta x^4} \right)^2}$$

0.13 871. A Case of Complex Multiplication with Imaginary Modulus.

[From the *Proceedings of the London Mathematical Society*, vol. xix. (1888), pp. 480, 481.]

THE case in question is that of the elliptic functions which satisfy the equation

$$\frac{dy}{\sqrt{1-y^4}} = (1 + \sqrt{-1}) \frac{dx}{\sqrt{1-x^4}};$$

so that the multiplier is $M = 1 + \sqrt{-1}$.

Here, writing the modular equation in the form

$$\sqrt{k}l + \sqrt{k'l'} = 1,$$

and observing that $k = l$, we have $k + k' = 1$, that is $k^2 + k'^2 = 1$, and thence $k^2 = k'^2$, that is $k = k'$, or say the modulus is $k = \frac{1}{\sqrt{2}}$.

We have for the transformation $n = 1 + \sqrt{-1}$, order = 2, and

$$\operatorname{sn}(Mu, k) = \frac{(1 + \sqrt{-1})x}{1 + \sqrt{-1}x^2}, \quad x = \operatorname{sn}(u, k).$$

Writing $x = \frac{1}{t}$, $y = \frac{1}{s}$, we have $s^2 = 1 - t^2$, and thence $s = \sqrt{1 - t^2}$; the formula may be written

$$\frac{1}{y} = \frac{1}{(1 + \sqrt{-1})x} + \frac{\sqrt{-1}x}{1 + \sqrt{-1}},$$

that is

$$\frac{1}{y} = \frac{1 + \sqrt{-1}x^2}{(1 + \sqrt{-1})x},$$

or, what is the same thing,

$$y = \frac{(1 + \sqrt{-1})x}{1 + \sqrt{-1}x^2}.$$

0.14 872. On the Finite Number of the Covariants of a Binary Quantic.

[From the *Mathematische Annalen*, t. xxxiv. (1889), pp. 319–320.]

THE proof of Gordan's theorem given in this paper is insufficient; but the theorem that the number of covariants of a binary quantic is finite is proved in Vol. IX. paper 320. I here work out a particular case of the proof; viz. showing for the quintic that the number of the covariants is finite.

For the quintic $(a, b, c, d, e, f|x, y)^5$, the non-unitary irreducible covariants are

$$\begin{array}{ll} (a, b, c, d, e, f|x, y)^5, & \text{deg. 1, order 5,} \\ (af - 4be + 3c^2, \dots |x, y)^2, & \text{deg. 2, order 2,} \\ (abc + 2bcd - ade - b^2e - c^2d, \dots), & \text{deg. 3, order 3,} \\ \text{and } a^2d - 3abc + 2b^3, \dots, & \text{deg. 3, order 3,} \end{array}$$

where the symbol (...) denotes that the whole series of terms is to be written down; and the number of these covariants is finite.

0.15 873. System of Equations for Three Circles which Cut Each Other at Given Angles.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 73–76.]

LET the three circles be

$$x^2+y^2-2ax-2\beta y+\gamma=0, \quad x^2+y^2-2a'x-2\beta'y+\gamma'=0, \quad x^2+y^2-2a''x-2\beta''y+\gamma''=0$$

Denoting the radii by R, R', R'' , and the distances of the centres by d, d', d'' , we have for the angles of intersection A, B, C (viz. A is the angle at which the second and third circles intersect each other, &c.)

$$\cos A = \frac{d^2 - R'^2 - R''^2}{2R'R''}, \quad \cos B = \frac{d'^2 - R''^2 - R^2}{2R''R}, \quad \cos C = \frac{d''^2 - R^2 - R'^2}{2RR'}.$$

We have $R^2 = a^2 + \beta^2 - \gamma$, $R'^2 = a'^2 + \beta'^2 - \gamma'$, $R''^2 = a''^2 + \beta''^2 - \gamma''$; and $d^2 = (a' - a'')^2 + (\beta' - \beta'')^2$, &c.

Writing

$$G = \frac{1}{2}(a^2 + \beta^2 + \gamma), \quad H = \frac{1}{2}(a'^2 + \beta'^2 + \gamma'), \quad F = \frac{1}{2}(a''^2 + \beta''^2 + \gamma''),$$

$$G = a'a'' + \beta'\beta'' - \frac{1}{2}(\gamma' + \gamma''), \quad H = a''a + \beta''\beta - \frac{1}{2}(\gamma'' + \gamma), \quad F = aa' + \beta\beta' - \frac{1}{2}(\gamma + \gamma'),$$

then we have

$$R^2 = G - H - F + G, \quad d^2 = G + H + F - G, \quad \&c.,$$

and consequently

$$\cos A = \frac{G - H - F}{\sqrt{(G - H - F + G)(G + H + F - G)}}.$$

In verification, observe that we have

$$G - H = 2R \sin \frac{1}{2}A, \quad G - H = -2R \sin \frac{1}{2}A, \quad G - F = 2R \sin \frac{1}{2}B, \quad H - F = 2R' \sin \frac{1}{2}C,$$

The equations of the three circles may be written

$$S = 0, \quad S' = 0, \quad S'' = 0,$$

where

$$\begin{aligned} S &= x^2 + y^2 - 2ax - 2\beta y + \gamma, \\ S' &= x^2 + y^2 - 2a'x - 2\beta'y + \gamma', \\ S'' &= x^2 + y^2 - 2a''x - 2\beta''y + \gamma''. \end{aligned}$$

The equation of a circle cutting the three circles at angles A , B , C is

$$S \cos A \cdot R' R'' + S' \cos B \cdot R'' R + S'' \cos C \cdot R R' = 0,$$

where R , R' , R'' are the radii of the three circles.

0.16 874. Note on the Legendrian Coefficients of the Second Kind.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 83, 84.]

THE Legendrian coefficients of the second kind, say Q_n , may be defined by means of the equation

$$\frac{1}{\sqrt{1-2\alpha x+x^2}} = \sum P_n x^n,$$

and

$$\frac{1-\alpha x}{(1-2\alpha x+x^2)^{3/2}} = \sum Q_n x^n,$$

where the summations extend from $n=0$ to $n=\infty$.

The expressions for Q_n in terms of P_n and P_{n-1} are

$$Q_n = \frac{n+1}{2n+1} P_n + \frac{n}{2n+1} P_{n-1},$$

and

$$Q_n = P_n + \frac{x}{n+1} \frac{dP_n}{dx}.$$

These coefficients satisfy the recurrence formula

$$(n+1)Q_{n+1} - (2n+1)\alpha Q_n + nQ_{n-1} = 0.$$

0.17 875. On the System of Three Circles which Cut Each Other at Given Angles and have their Centres in a Line.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 60–67.]

THE case considered is that of three circles with their centres in a line, cutting each other at given angles. Taking the line of centres as the axis of x , and any perpendicular line as the axis of y , the equations of the three circles may be written

$$x^2 + y^2 - 2a_1x + c_1 = 0, \quad x^2 + y^2 - 2a_2x + c_2 = 0, \quad x^2 + y^2 - 2a_3x + c_3 = 0.$$

The radii are $R_1 = \sqrt{a_1^2 - c_1}$, $R_2 = \sqrt{a_2^2 - c_2}$, $R_3 = \sqrt{a_3^2 - c_3}$.

The distance between the centres of the first and second circles is $d_{12} = a_1 - a_2$, and similarly for the others.

The angles of intersection are given by

$$\cos \theta_{23} = \frac{(a_2 - a_3)^2 - R_2^2 - R_3^2}{2R_2R_3}, \quad \&c.$$

The condition that the three circles should cut each other at given angles may be expressed as follows. Let A, B, C be the angles at which the circles cut each other; then we have the system of equations:

1. The condition that the centres are in a line gives at once a simple relation between the coordinates.

2. For the angles, we have

$$\cos A = \frac{d_{23}^2 - R_2^2 - R_3^2}{2R_2R_3}, \quad \cos B = \frac{d_{31}^2 - R_3^2 - R_1^2}{2R_3R_1}, \quad \cos C = \frac{d_{12}^2 - R_1^2 - R_2^2}{2R_1R_2}.$$

3. The analytical solution may be obtained as follows. Let the angles of the triangle formed by the centres be α, β, γ ; then since the centres are in a line, we have $\alpha + \beta + \gamma = \pi$, and the angles at which the circles cut each other are connected with these angles by the relations

$$\theta_{23} = A, \quad \theta_{31} = B, \quad \theta_{12} = C.$$

The case where the three circles cut each other at right angles is particularly simple. We then have $\cos A = \cos B = \cos C = 0$, and the conditions become

$$d_{23}^2 = R_2^2 + R_3^2, \quad d_{31}^2 = R_3^2 + R_1^2, \quad d_{12}^2 = R_1^2 + R_2^2.$$

In the general case, the condition for the existence of such a system of circles is that the angles A, B, C should satisfy certain inequalities. Obviously, points on the line of centres which are centres of circles cutting two given circles at given angles lie on a conic; and the condition that three such conics should have a common point gives the required condition.

But the analytical solution may be obtained more directly. Writing the equations of the circles in the form

$$x^2 + y^2 - 2a_r x + c_r = 0, \quad (r = 1, 2, 3),$$

and expressing that the angle between the r -th and s -th circles has a given value, we obtain three equations which, together with the conditions that the centres are in a line, determine the four ratios of the five quantities $a_1, a_2, a_3, c_1, c_2, c_3$ (one of which may be taken as unity).

In these equations, y, x should be regarded as the current coordinates; and the conditions for the intersection at given angles are expressed by means of the formulae

$$\cos \theta_{rs} = \frac{(a_r - a_s)^2 - (a_r^2 - c_r) - (a_s^2 - c_s)}{2\sqrt{a_r^2 - c_r} \sqrt{a_s^2 - c_s}}.$$

0.18 876. On Systems of Rays.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 76–80.]

IN the theory of systems of rays as considered by Hamilton, a system of rays is defined by means of a characteristic function V , which is a function of the coordinates of the points A and B through which the ray passes. The differential coefficients of V with respect to these coordinates give the direction-cosines of the ray at the points A and B .

For a system of rays in a medium of variable refractive index μ , the characteristic function satisfies the partial differential equation

$$\left(\frac{\partial V}{\partial x}\right)^2 + \left(\frac{\partial V}{\partial y}\right)^2 + \left(\frac{\partial V}{\partial z}\right)^2 = \mu^2.$$

In the case of a system of rays in a plane, the equation becomes

$$\left(\frac{\partial V}{\partial x}\right)^2 + \left(\frac{\partial V}{\partial y}\right)^2 = \mu^2.$$

The theory of the characteristic function may be extended to the case of a system of rays reflected or refracted at a surface. If V_1 and V_2 are the characteristic functions for the incident and refracted systems, then at the surface of separation we have

$$\frac{\partial V_1}{\partial x} = \frac{\partial V_2}{\partial x}, \quad \frac{\partial V_1}{\partial y} = \frac{\partial V_2}{\partial y}, \quad \mu_2 \frac{\partial V_1}{\partial z} = \mu_1 \frac{\partial V_2}{\partial z}.$$

These are the equations of refraction; and for reflection we have similar equations with $\mu_1 = \mu_2$.

The caustic surfaces of a system of rays are the envelopes of the rays; they may be determined by means of the characteristic function as follows. The equations of a ray are

$$\frac{x - x_0}{\partial V / \partial x_0} = \frac{y - y_0}{\partial V / \partial y_0} = \frac{z - z_0}{\partial V / \partial z_0},$$

and the caustic is the locus of the intersections of consecutive rays. The condition for the intersection of consecutive rays gives an equation which, combined with the equations of the ray, determines the caustic surface.

For a system of rays in a medium of constant refractive index μ , the characteristic function is the distance between the points A and B multiplied by μ ; that is,

$$V = \mu \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}.$$

The direction-cosines of the ray at the point (x, y, z) are then

$$\frac{\partial V}{\partial x} = \mu \frac{x - x'}{r}, \quad \frac{\partial V}{\partial y} = \mu \frac{y - y'}{r}, \quad \frac{\partial V}{\partial z} = \mu \frac{z - z'}{r},$$

where $r = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$.

In the case of a system of rays issuing from a point, the characteristic function is simply $V = \mu r$, where r is the distance from the point. The wave-surfaces are spheres with the point as centre; and the rays are the normals to these spheres.

The theory of systems of rays was developed by Hamilton in his papers on "Theory of Systems of Rays," published in the *Transactions of the Royal Irish Academy* (1828, 1830, 1832). The characteristic function is a fundamental concept in this theory; it determines the properties of the system of rays completely.

0.19 877. Note on the Two Relations Connecting the Distances of Four Points on a Circle.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 84–86.]

THE distances of four points on a circle are connected by two relations, which may be obtained as follows. Let the four points be A, B, C, D , and let their distances from each other be $AB = c, AC = b, AD = d, BC = a, BD = f, CD = e$. The area of the quadrilateral $ABCD$ may be expressed in two ways: first, as the sum of the areas of the triangles ABC and ADC ; and second, as the sum of the areas of the triangles ABD and BCD .

The area of a triangle with sides a, b, c is given by Hero's formula

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{1}{2}(a+b+c).$$

For the four points to be on a circle, the sum of the opposite angles must be equal to π ; and this condition gives the two relations between the six distances.

The distances of four points on a circle satisfy the two equations

$$\begin{vmatrix} 0 & c^2 & b^2 & d^2 & 1 \\ c^2 & 0 & a^2 & f^2 & 1 \\ b^2 & a^2 & 0 & e^2 & 1 \\ d^2 & f^2 & e^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0,$$

and a second equation obtained by a similar determinant condition. The first of these equations may be written

$$\frac{1}{\sqrt{(ac+bd+ef)(ac+bd-ef)(ac-bd+ef)(-ac+bd+ef)}} = 0,$$

which is the condition that the quadrilateral formed by the four points should be cyclic.

The two relations may be expressed in the symmetric form

$$\frac{a^2 + e^2 - b^2 - f^2}{ae + bf} + \frac{c^2 + e^2 - d^2 - a^2}{ce + da} = 0,$$

and

$$\frac{a^2 + e^2 - b^2 - f^2}{ae + bf} \cdot \frac{c^2 + e^2 - d^2 - a^2}{ce + da} = \frac{4(ac + bd + ef)}{(ae + bf)(ce + da)}.$$

0.20 878. Note on the Anharmonic Ratio Equation.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 87, 88.]

THE anharmonic ratio of four points A, B, C, D on a line is defined as

$$\{ABCD\} = \frac{AC}{CB} \bigg/ \frac{AD}{DB} = \frac{AC \cdot DB}{CB \cdot AD}.$$

If the four points are represented by their coordinates x_1, x_2, x_3, x_4 , then the anharmonic ratio is

$$\{ABCD\} = \frac{(x_1 - x_3)(x_2 - x_4)}{(x_3 - x_2)(x_4 - x_1)}.$$

The anharmonic ratio equation is the equation connecting the anharmonic ratios of four points taken in different orders. If $\lambda = \{ABCD\}$, then the six possible values of the anharmonic ratio are

$$\lambda, \quad \frac{1}{\lambda}, \quad 1 - \lambda, \quad \frac{1}{1 - \lambda}, \quad \frac{\lambda}{\lambda - 1}, \quad \frac{\lambda - 1}{\lambda}.$$

These six values are the roots of the equation

$$(\lambda^2 - \lambda + 1)^3 - j \lambda^2 (\lambda - 1)^2 = 0,$$

where j is the absolute invariant of the binary quartic.

The equation may be written in the form

$$\frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2 (\lambda - 1)^2} = j,$$

and thus the anharmonic ratio λ is determined by the invariant j .

For the equianharmonic case, $\lambda = -\omega$ or $-\omega^2$, where ω is an imaginary cube root of unity; and $j = 0$. For the harmonic case, $\lambda = -1, \frac{1}{2}$, or 2 ; and $j = 1$.

0.21 879. Note on the Differential Equation

$$\frac{dx}{\sqrt{X}} = \frac{dy}{\sqrt{Y}}.$$

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 90, 91.]

WE have

$$\frac{\sin u \cos v - \sin v \cos u}{\sin(u+v)} = \frac{\sin(u-v)}{\sin(u+v)},$$

and writing

$$\frac{1 + \cos(u+v)}{1 - \cos(u+v)} = \cot^2 \frac{1}{2}(u+v),$$

we obtain the differential equation

$$\frac{du}{\sqrt{1 - k^2 \sin^2 u}} = \frac{dv}{\sqrt{1 - k^2 \sin^2 v}}.$$

This is the standard form of the differential equation for elliptic functions. If we write $x = \sin u$, $y = \sin v$, then $X = 1 - x^2$, $Y = 1 - y^2$, and the equation becomes

$$\frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = \frac{dy}{\sqrt{(1-y^2)(1-k^2y^2)}}.$$

The general integral of this equation is

$$y = \frac{x \Delta(\alpha) + \sin \alpha \Delta(x)}{1 - k^2 x^2 \sin^2 \alpha},$$

where α is an arbitrary constant, and $\Delta(\alpha) = \sqrt{1 - k^2 \sin^2 \alpha}$.

0.22 880. Note on the Relation Between the Distance of Five Points in Space.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 100–102.]

IN Lagrange's paper "Solutions analytiques de quelques problèmes sur les pyramides triangulaires," the relation between the distances of five points in space is considered. Let the five points be A, B, C, D, E , and let the squared distances between them be denoted by $a_{rs}^2 = (x_r - x_s)^2 + (y_r - y_s)^2 + (z_r - z_s)^2$.

The relation is expressed by means of the determinant

$$\begin{vmatrix} 0 & a_{12}^2 & a_{13}^2 & a_{14}^2 & a_{15}^2 & 1 \\ a_{21}^2 & 0 & a_{23}^2 & a_{24}^2 & a_{25}^2 & 1 \\ a_{31}^2 & a_{32}^2 & 0 & a_{34}^2 & a_{35}^2 & 1 \\ a_{41}^2 & a_{42}^2 & a_{43}^2 & 0 & a_{45}^2 & 1 \\ a_{51}^2 & a_{52}^2 & a_{53}^2 & a_{54}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0.$$

After some reductions, the value of $4\Delta^2$, in terms of these squared distances, is found to be

$$4\Delta^2 = \sum \pm a_{12}^2 a_{23}^2 a_{34}^2 a_{45}^2 a_{51}^2,$$

where the summation extends over all permutations of the indices with appropriate signs.

The condition that five points should lie on a sphere may be expressed in terms of their mutual distances; it is a relation of degree 4 in the squared distances.

For five points in space, the mutual distances are connected by a single relation, which may be written in the determinant form above. This relation is of great importance in the theory of the configuration of points in space.

0.23 881. On Hermite's q -Product Theorem.

[From the *Messenger of Mathematics*, vol. xvii. (1888), pp. 95–99.]

HERMITE'S q -product theorem relates to the multiplication of theta-functions. If we write

$$\vartheta(x, q) = \sum_{n=-\infty}^{\infty} q^{n^2} e^{2nix},$$

then the theorem asserts that

$$\vartheta(x, q) \vartheta(y, q) = \vartheta(x+y, q^2) \vartheta(x-y, q^2) + \vartheta(x+y+\frac{1}{2}\pi, q^2) \vartheta(x-y+\frac{1}{2}\pi, q^2).$$

To show this, we observe that the product $\vartheta(x, q) \vartheta(y, q)$ may be written as a double series

$$\sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} q^{m^2+n^2} e^{2i(mx+ny)}.$$

Writing $m+n = 2p$, $m-n = 2q$, or $m+n = 2p+1$, $m-n = 2q+1$, according as $m+n$ is even or odd, we have $m^2+n^2 = 2(p^2+q^2)$ or $2(p^2+q^2+p+q)+1$; and $mx+ny = p(x+y) + q(x-y)$ or $(p+\frac{1}{2})(x+y) + (q+\frac{1}{2})(x-y)$.

The number of terms in the expansion is infinite, but the series converges absolutely for $|q| < 1$. The theorem may be verified by direct multiplication of the series.

The q -product theorem has important applications in the theory of elliptic functions. It leads to the addition-theorems for the theta-functions, and thence to the addition-theorems for the elliptic functions themselves.

0.24 882. A Correspondence of Confocal Cartesians with the Right Lines of a Hyperboloid.

[From the *Messenger of Mathematics*, vol. xviii. (1889), pp. 1-5.]

THE confocal Cartesians are the surfaces defined by the equation

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1,$$

where λ is a parameter. For different values of λ , these surfaces form a system of confocal quadrics.

The right lines of a hyperboloid of one sheet are the two systems of straight lines which lie entirely on the surface. The equation of the hyperboloid may be written

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1,$$

and the right lines are given by the equations

$$\frac{x}{a} + \frac{z}{c} = \lambda \left(1 + \frac{y}{b}\right), \quad \frac{x}{a} - \frac{z}{c} = \frac{1}{\lambda} \left(1 - \frac{y}{b}\right),$$

and

$$\frac{x}{a} + \frac{z}{c} = \mu \left(1 - \frac{y}{b}\right), \quad \frac{x}{a} - \frac{z}{c} = \frac{1}{\mu} \left(1 + \frac{y}{b}\right),$$

where λ and μ are parameters.

But the equations of the right lines may also be written in the form

$$\frac{x}{a \cos \theta} + \frac{y}{b \sin \theta} = 1, \quad z = c \tan \phi,$$

where θ and ϕ are parameters; and eliminating these parameters, we obtain the equation of the hyperboloid.

The correspondence between the confocal Cartesians and the right lines of a hyperboloid may be established as follows. The three values of λ for which the confocal passes through a given point are the roots of a cubic equation; and these roots correspond to the three parameters of the right lines which pass through the point.

The three confocals which pass through a given point intersect each other at right angles. The tangent planes to the three confocals at

the point are mutually perpendicular; and the lines of intersection of these tangent planes are parallel to the axes of the confocals.

The right lines of the hyperboloid correspond to the curves of curvature on the confocals; and the asymptotic lines on the confocals correspond to the right lines on the hyperboloid. This correspondence is a remarkable example of the analogy between the properties of surfaces of the second degree.

0.25 883. Analytical Formulæ in Regard to an Octad of Points.

[From the *Messenger of Mathematics*, vol. xviii. (1889), pp. 149–152.]

THE octad of points is a system of eight points in space, connected by certain geometrical relations. Let the eight points be denoted by $P_1, P_2, \dots, P_8$, and let their coordinates be (x_r, y_r, z_r) for $r = 1, 2, \dots, 8$.

The condition that the eight points should lie on a quadric surface may be expressed by means of the determinant

$$\begin{vmatrix} x_1^2 & y_1^2 & z_1^2 & x_1y_1 & y_1z_1 & z_1x_1 & x_1 & y_1 & z_1 & 1 \\ x_2^2 & y_2^2 & z_2^2 & x_2y_2 & y_2z_2 & z_2x_2 & x_2 & y_2 & z_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_8^2 & y_8^2 & z_8^2 & x_8y_8 & y_8z_8 & z_8x_8 & x_8 & y_8 & z_8 & 1 \end{vmatrix} = 0.$$

This determinant expresses the condition that the eight points should lie on a quadric surface; and it is the generalization of the condition that six points should lie on a conic.

The foregoing condition may be expressed in a different form by means of the coordinates of the eight points. If the points are connected by the relations

$$\sum_{r=1}^8 \lambda_r x_r = 0, \quad \sum_{r=1}^8 \lambda_r y_r = 0, \quad \sum_{r=1}^8 \lambda_r z_r = 0, \quad \sum_{r=1}^8 \lambda_r = 0,$$

then the condition that they should lie on a quadric is that the coefficients λ_r should satisfy certain quadratic relations.

The octad of points has many interesting properties. The eight points may be divided into two sets of four, each set lying on a plane; and the line of intersection of these planes is the polar line of the point of intersection of the lines joining the corresponding points of the two sets.

The analytical formulae for the octad may be simplified by the introduction of homogeneous coordinates. If we write $x = X/W, y = Y/W, z = Z/W$, then the condition that the eight points should lie on a quadric is expressed by a determinant of order 10, which reduces to the determinant of order 8 given above when the coordinates are referred to a particular system of axes.

0.26 884. Note sur les Surfaces Minima et le Théorème de Joachimsthal.

[From the *Messenger of Mathematics*, vol. xviii. (1889), pp. 153–156.]

UNE surface minima est une surface dont l'aire est un minimum par rapport à toutes les surfaces ayant même contour. La condition pour qu'une surface soit minima s'exprime par l'équation

$$(1 + q^2)r - 2pqs + (1 + p^2)t = 0,$$

où $p = \partial z / \partial x$, $q = \partial z / \partial y$, $r = \partial^2 z / \partial x^2$, $s = \partial^2 z / \partial x \partial y$, $t = \partial^2 z / \partial y^2$.

Le théorème de Joachimsthal s'énonce comme suit: Si deux surfaces se coupent sous un angle constant, et si leur ligne d'intersection est une ligne de courbure sur l'une d'elles, elle est aussi une ligne de courbure sur l'autre.

Pour démontrer ce théorème, on considère les normales aux deux surfaces le long de leur ligne d'intersection. Si l'angle entre les deux surfaces est constant, les normales font entre elles un angle constant; et si la ligne d'intersection est une ligne de courbure sur la première surface, les normales à cette surface le long de la ligne forment un développable. Par conséquent, les normales à la seconde surface forment aussi un développable, et la ligne d'intersection est une ligne de courbure sur la seconde surface.

Les surfaces minima ont la propriété remarquable que leurs lignes asymptotiques sont des droites. Cela résulte de ce que l'équation des surfaces minima peut être mise sous la forme

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0,$$

lorsque la surface est représentée sous la forme $z = f(x, y)$, et que les coefficients satisfont à certaines conditions.

Le théorème de Joachimsthal a des applications importantes dans la théorie des surfaces. Il sert à déterminer les lignes de courbure sur les surfaces de révolution, sur les surfaces de translation, et sur d'autres classes de surfaces.

La démonstration analytique du théorème repose sur les formules fondamentales de la théorie des surfaces. Soient E , F , G les coefficients

de la première forme fondamentale, et L , M , N ceux de la seconde; alors les lignes de courbure sont déterminées par l'équation différentielle

$$(EM - FL) dx^2 + (EN - GL) dx dy + (FN - GM) dy^2 = 0.$$

Si deux surfaces se coupent sous un angle constant, et si leurs coefficients satisfont aux conditions du théorème, alors l'équation ci-dessus se décompose en deux équations, chacune correspondant à une ligne de courbure sur l'une des surfaces.

0.27 885. On the Diophantine Relation, $y^2 + y'^2 = \text{Square}$.

[From the *Proceedings of the London Mathematical Society*, vol. xix. (1888), pp. 527–534.]

THE Diophantine relation to be considered is the equation

$$y^2 + y'^2 = \text{a perfect square.}$$

This may be written $y^2 + y'^2 = z^2$, where y, y', z are integers. The general solution of this equation in integers is well known; it is

$$y = m^2 - n^2, \quad y' = 2mn, \quad z = m^2 + n^2,$$

or

$$y = 2mn, \quad y' = m^2 - n^2, \quad z = m^2 + n^2,$$

where m and n are any integers.

The relation may be generalized as follows. Let y and y' be functions of a variable x ; then the condition that $y^2 + y'^2$ should be a perfect square may be expressed as a differential equation. If we write $y' = y \tan \phi$, then the condition becomes $y^2 \sec^2 \phi = \text{square}$, which is satisfied if ϕ is such that $\sec \phi$ is rational.

The case where y and y' are polynomials in x is particularly interesting. Let

$$y = (x - a_1)^{\alpha_1} (x - a_2)^{\alpha_2} \cdots (x - a_n)^{\alpha_n},$$

and similarly for y' ; then the condition that $y^2 + y'^2$ should be a perfect square imposes certain conditions on the exponents α_r and the coefficients a_r .

For the case where y and y' are linear functions of x , we have

$$y = ax + b, \quad y' = cx + d,$$

and the condition that $(ax+b)^2 + (cx+d)^2$ should be a perfect square gives

$$(a^2 + c^2)x^2 + 2(ab + cd)x + (b^2 + d^2) = (Ax + B)^2,$$

which requires $a^2 + c^2 = A^2$, $ab + cd = AB$, $b^2 + d^2 = B^2$. These conditions are satisfied if (a, c, A) and (b, d, B) are Pythagorean triples with the same ratio.

The complete solution for the case where y and y' are rational functions of x may be obtained by means of the theory of elliptic functions. If we write

$$y = \sqrt{(1-x^2)(1-k^2x^2)},$$

then $y^2 + x^2$ is not in general a perfect square; but for particular values of k it may become so.

And thence

$$2(\alpha + \beta) = a + b - c - d, \quad 2(\gamma + \delta) = -(c + d - a - b),$$

that is, $\alpha + \beta = \gamma + \delta$; viz. there are, in this case also, two diptychs. If, as before, the order is taken to be $= 2\nu$, then

$$\alpha + \beta = \nu, \quad \gamma + \delta = \nu;$$

supposing that ν is a given positive integer, and that $\alpha, \beta, \gamma, \delta$ are positive integers satisfying these equations $\alpha + \beta = \nu, \gamma + \delta = \nu$, then the last-mentioned four equations between $\alpha, \beta, \gamma, \delta$ and a, b, c, d are equivalent to three relations serving to determine the differences of a, b, c, d (say $a - d, b - d, c - d$) in terms of $\alpha, \beta, \gamma, \delta$.

And we then further have

$$\begin{aligned} (a-l)(a-p) &= -(1-i)\beta(b-a), \\ (a-m)(a-q) &= -(1+i)\beta(b-a), \\ (b-l)(b-p) &= -(1-i)\alpha(a-b), \\ (b-m)(b-q) &= -(1+i)\alpha(a-b), \\ (c-l)(c-p) &= -(1+i)\gamma(d-c), \\ (c-m)(c-q) &= -(1-i)\gamma(c-d), \\ (d-l)(d-p) &= -(1+i)\delta(c-d), \\ (d-m)(d-q) &= -(1-i)\delta(d-c), \end{aligned}$$

each set equivalent to two equations; or, as these may be written,

$$\begin{aligned} 2(l+p) &= a+b+c+d+i(a+b-c-d), \\ 2lp &= ab+cd+i(ab-cd), \\ 2(m+q) &= a+b+c+d-i(a+b-c-d), \\ 2mq &= ab+cd-i(ab-cd), \end{aligned}$$

serving to determine l, p, m, q in terms of a, b, c, d .

Observe also that, u being arbitrary, we have

$$\begin{aligned} 2(u-l)(u-p) &= (1+i)(u-a)(u-b) + (1-i)(u-c)(u-d), \\ 2(u-m)(u-q) &= (1-i)(u-a)(u-b) + (1+i)(u-c)(u-d), \end{aligned}$$

(which equations, writing therein $u = a, b, c$, or d , in fact reproduce the two systems of four equations).

We have also

$$\begin{aligned} l+p+m+q &= a+b+c+d, \\ l+p-m-q &= i(a+b-c-d), \\ lp+mq &= ab+cd, \\ lp-mq &= i(ab-cd); \end{aligned}$$

and moreover

$$\begin{aligned} 4(l-p)^2 &= 2i\{(a-b)^2 - (c-d)^2\} + 4(a+b)(c+d) - 8(ab+cd), \\ 4(m-q)^2 &= -2i\{(a-b)^2 - (c-d)^2\} + 4(a+b)(c+d) - 8(ab+cd), \end{aligned}$$

which equations, combined with the foregoing values of $2(l+p)$ and $2(m+q)$, give the values of l, p, m, q .

We have thus the complete solution for the case class = 4, order = 2ν ; say

$$y = (x+a)^\alpha (x+b)^\beta (x+c)^\gamma (x+d)^\delta;$$

$$\alpha + \beta = \gamma + \delta = \nu,$$

$$y^2 + y'^2 = (x+a)^{2\alpha-2\beta} (x+b)^{2\beta-2\alpha} (x+c)^{2\gamma-2\delta} (x+d)^{2\delta-2\gamma}.$$

880.

NOTE ON THE RELATION BETWEEN THE DISTANCE OF FIVE POINTS IN SPACE.

[From the *Messenger of Mathematics*, vol. XVIII. (1889), pp.
100–102.]

In Lagrange's paper "Solutions analytiques de quelques problemes sur les pyramides triangulaires" (*Berlin Memoirs*, 1773; *Œuvres*, t. III., see p. 677), there is contained a formula for the relation between the distances from each other of five points in space; viz. this is

$$\begin{aligned}
 4\Delta^2 f = & \alpha(a+f-g)^2 + \alpha'(a'+f-g')^2 + \alpha''(a''+f-g'')^2 \\
 & + 2\beta(a'+f-g')(a''+f-g'') + 2\beta'(a+f-g)(a''+f-g'') \\
 & + 2\beta''(a+f-g)(a'+f-g'),
 \end{aligned}$$

or, in a slightly altered notation, say

$$\Pi = -4\Delta^2 f + (\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3 \setminus a_1+f-g_1, a_2+f-g_2, a_3+f-g_3)^2 = 0,$$

where, if the points are called 1, 2, 3, 4, 5, then

c_1, c_2, c_3	are the squared distances	23, 31, 12,
a_1, a_2, a_3	, , ,	41, 42, 43,
g_1, g_2, g_3	, , ,	51, 52, 53,

and

$$f \quad \text{is the squared distance} \quad 45.$$

The values of $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3$, in terms of these squared distances, are

$$\begin{aligned}
 \alpha_1 &= -(a_2 - a_3)^2 + 2c_1(a_2 + a_3) - c_1^2, \\
 \alpha_2 &= -(a_3 - a_1)^2 + 2c_2(a_3 + a_1) - c_2^2, \\
 \alpha_3 &= -(a_1 - a_2)^2 + 2c_3(a_1 + a_2) - c_3^2, \\
 \beta_1 &= -(a_3 - a_1)(a_1 - a_2) + 2c_1a_1 - c_2(a_1 + a_2) - c_3(a_3 + a_1) + c_2c_3, \\
 \beta_2 &= -(a_1 - a_2)(a_2 - a_3) + 2c_2a_2 - c_3(a_2 + a_3) - c_1(a_1 + a_2) + c_3c_1, \\
 \beta_3 &= -(a_2 - a_3)(a_3 - a_1) + 2c_3a_3 - c_1(a_3 + a_1) - c_2(a_2 + a_3) + c_1c_2.
 \end{aligned}$$

After some reductions, the value of $4\Delta^2$, in terms of these squared distances, is found to be

$$\begin{aligned} 4\Delta^2 = & c_1(a_3 - a_1)(a_1 - a_2) + a_1(-c_1^2 + c_1c_2 + c_1c_3) - c_1c_2c_3 \\ & + c_2(a_1 - a_2)(a_2 - a_3) + a_2(-c_2^2 + c_2c_3 + c_2c_1) \\ & + c_3(a_2 - a_3)(a_3 - a_1) + a_3(-c_3^2 + c_3c_1 + c_3c_2), \end{aligned}$$

which is, in fact,

$$8\Delta^2 = \begin{vmatrix} . & 1 & 1 & 1 & 1 \\ 1 & . & c_3 & c_2 & a_1 \\ 1 & c_3 & . & c_1 & a_2 \\ 1 & c_2 & c_1 & . & a_3 \\ 1 & a_1 & a_2 & a_3 & . \end{vmatrix},$$

where observe that each term of the determinant contains the factor 2.

By the formula in my paper "On a theorem in the Geometry of Position," *Camb. Math. Jour.*, t. II. (1841), pp. 267–271, [1], (introducing into it the present notation), the relation between the distances of the five points is given in the form

$$\Omega = \begin{vmatrix} . & 1 & 1 & 1 & 1 & 1 \\ 1 & . & c_3 & c_2 & a_1 & g_1 \\ 1 & c_3 & . & c_1 & a_2 & g_2 \\ 1 & c_2 & c_1 & . & a_3 & g_3 \\ 1 & a_1 & a_2 & a_3 & . & f \\ 1 & g_1 & g_2 & g_3 & f & . \end{vmatrix} = 0.$$

The equations $\Pi = 0$, $\Omega = 0$ should therefore agree with each other; we have in Ω the term

$$-f^2 \begin{vmatrix} 1 & c_3 & . & c_1 \\ 1 & c_2 & c_1 & . \\ . & 1 & 1 & 1 \\ 1 & . & c_3 & c_2 \end{vmatrix},$$

which is

$$= -f^2(c_1^2 + c_2^2 + c_3^2 - 2c_2c_3 - 2c_3c_1 - 2c_1c_2);$$

and similarly in Π we have the term

$$f^2(\alpha_1 + \alpha_2 + \alpha_3 + 2\beta_1 + 2\beta_2 + 2\beta_3),$$

which is easily shown to be

$$= -f^2(c_1^2 + c_2^2 + c_3^2 - 2c_2c_3 - 2c_3c_1 - 2c_1c_2);$$

and it thus appears that we have identically $\Pi = \Omega$.

It is to be remarked that, in Lagrange's form, the points 4 and 5 are regarded as determined each of them by means of its squared distances from the vertices of the triangle 123, and that the formula gives (by a quadratic equation) the squared distance 45; but that nevertheless the two points 4 and 5 do not present themselves symmetrically in the formula; in fact, Δ^2 and the coefficients $(\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3)$ of the formula relate all of them to the tetrahedron 4123; as noticed in the paper, $\Delta = 6 \times$ volume of tetrahedron; $\sqrt{(\alpha_1)}$, $\sqrt{(\alpha_2)}$, $\sqrt{(\alpha_3)}$ are the doubled areas of the faces 423, 431, 412 respectively, and the doubled area of the face 123 is

$$= \sqrt{(\alpha_1 + \alpha_2 + \alpha_3 + 2\beta_1 + 2\beta_2 + 2\beta_3)};$$

it may be added that

$$\beta_1 \div \sqrt{(\alpha_2\alpha_3)}, \quad \beta_2 \div \sqrt{(\alpha_3\alpha_1)}, \quad \beta_3 \div \sqrt{(\alpha_1\alpha_2)},$$

are equal to the cosines of the dihedral angles at the edges 41, 42, 43 respectively.

881.

ON HERMITE'S H -PRODUCT THEOREM.

[From the *Messenger of Mathematics*, vol. XVIII. (1889), pp. 104–107.]

I give this name to a theorem relating to the product of an even number of Eta-functions, established by M. Hermite in his “Note sur le calcul differential et le calcul integral,” forming an appendix to the sixth edition of Lacroix’s *Differential and Integral Calculus*, and separately printed, 8vo. Paris, 1862. It is the theorem stated p. 65, in the form

$$\phi(x) = F(z^2) + \frac{dz}{dx} z F_1(z^2),$$

where

$$\phi(x) = \frac{A H(x - \alpha_1) H(x - \alpha_2) \dots H(x - \alpha_{2n})}{\Theta^{2n}(x)},$$

where $\alpha_1 + \alpha_2 + \dots + \alpha_{2n} = 0$, and $z = \operatorname{sn} x$, $\operatorname{cn} x$ or $\operatorname{dn} x$ at pleasure; $F(z^2)$, $F_1(z^2)$ denote rational and integral functions of z^2 of the degrees n and $n - 2$ respectively; A is a constant, which we may if we please so determine that in $F(z^2)$ the coefficient of the highest power z^{2n} shall be $= 1$.

If, for shortness, we write s , c , d for $\operatorname{sn} x$, $\operatorname{cn} x$, $\operatorname{dn} x$ respectively; and to fix the ideas, assume $z = \operatorname{sn} x$, $= s$, then the theorem is

$$\frac{A H(x - \alpha_1) H(x - \alpha_2) \dots H(x - \alpha_{2n})}{\Theta^{2n}(x)} = F(s^2) + s c d F_1(s^2);$$

viz. the theorem is that the product of the $2n$ H -functions ($\alpha_1 + \alpha_2 + \dots + \alpha_{2n} = 0$ as above), divided by $\Theta^{2n}(x)$, is a function of the elliptic functions sn , cn , dn , of the form in question.

Hermite uses the theorem for the demonstration of Abel’s theorem, as applied to the elliptic functions; or as I would rather express it, he uses the theorem for the determination of the sn , cn , and dn of $\alpha_1 + \alpha_2 + \dots + \alpha_{2n-1}$.

To show how this is, observe that $F(s^2)$, *qua* rational and integral function of s^2 of the degree n , with its first coefficient $= 1$, contains n arbitrary coefficients; and $F_1(s^2)$, *qua* rational and integral function of s^2 of the degree $n - 2$, contains $n - 1$ arbitrary coefficients:

hence $F(s^2) + scdF_1(s^2)$ contains $2n - 1$ arbitrary coefficients; and considering $\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$ as given, the function in question must vanish for each of the values $x = \alpha_1, \alpha_2, \dots, \alpha_{2n-1}$; and we have therefore $2n - 1$ equations for obtaining the $2n - 1$ coefficients, which are thus completely determined: in particular, the constant term, say L , of $F_1(s^2)$ is a given function of $\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$, that is, of the sn , cn , and dn of these quantities; and the theorem shows that the function thus determined vanishes also for $x = \alpha_{2n}$, that is, $= -(\alpha_1 + \alpha_2 + \dots + \alpha_{2n-1})$.

Now writing $-x$ for x in the formula, and recollecting that H is an odd function, Θ an even function, we find

$$\frac{A H(x + \alpha_1) H(x + \alpha_2) \dots H(x + \alpha_{2n})}{\Theta^{2n}(x)} = F(s^2) - scdF_1(s^2);$$

and multiplying together the two sides of these equations respectively,

$$A^2 \frac{H(x - \alpha_1) H(x + \alpha_1)}{\Theta^2(x)} \dots \frac{H(x - \alpha_{2n}) H(x + \alpha_{2n})}{\Theta^2(x)} = \{F(s^2)\}^2 - s^2 c^2 d^2 \{F_1(s^2)\}^2$$

where the right-hand side is a rational and integral function of s^2 of the degree $2n$, and the coefficient of the highest term s^{4n} is $= 1$; in fact, this term arises only from the square of $F(s^2)$, which has its highest term $= s^{2n}$.

Now $\frac{H(x - \alpha_1) H(x + \alpha_1)}{\Theta^2(x)}$ is a mere constant multiple of $\text{sn}^2 x - \text{sn}^2 \alpha_1$, or say of $s^2 - \text{sn}^2 \alpha_1$; (this well-known theorem is, in fact, the particular case $n = 2$ of Hermite's theorem); and similarly for the other terms: we must clearly have A^2 , multiplied by the product of the factors thus introduced, $= 1$; and thus the theorem becomes

$$(s^2 - \text{sn}^2 \alpha_1)(s^2 - \text{sn}^2 \alpha_2) \dots (s^2 - \text{sn}^2 \alpha_{2n}) = \{F(s^2)\}^2 - s^2 c^2 d^2 \{F_1(s^2)\}^2.$$

And putting herein $s = 0$, and writing as before L for the constant term of $F(s^2)$, we have

$$\text{sn}^2 \alpha_1 \text{sn}^2 \alpha_2 \dots \text{sn}^2 \alpha_{2n} = L^2,$$

or, the sign $\pm$ being properly determined, say

$$\text{sn} \alpha_1 \text{sn} \alpha_2 \dots \text{sn} \alpha_{2n} = \pm L,$$

where, by what precedes, L is a given function of the sn , cn , and dn of $\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$. Hence we have $\text{sn} \alpha_{2n}$, that is, $-\text{sn}(\alpha_1 + \alpha_2 +$

$\dots + \alpha_{2n-1}$) as a given function of the sn, cn, and dn of $\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$.

Similarly writing $z = \operatorname{cn} x = c$, and $z = \operatorname{dn} x = d$, we have $\operatorname{cn}(\alpha_1 + \alpha_2 + \dots + \alpha_{2n-1})$ and $\operatorname{dn}(\alpha_1 + \alpha_2 + \dots + \alpha_{2n-1})$ each of them as a given function of the sn, cn, and dn of $\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$.

It is hardly necessary to remark that $F(z^2) + z \frac{dz}{dx} F_1(z^2)$ is a function of the same form, whether we have $z = s, c$ or d ; in fact, the functions F and F_1 are rational in s^2, c^2 , or d^2 , and we have $z \frac{dz}{dx} = scd, -scd$, and $-k^2 scd$ for the three values respectively.

The number of terms $\alpha_1, \alpha_2, \dots, \alpha_{2n-1}$ has been odd, but by taking one of them $= 0$, the formulae give the values of the sn, cn, and dn for the sum of an even number of terms.

It has been seen that Hermite's H -product theorem gives, say Abel's theorem, in the form

$$\Pi(s^2 - \operatorname{sn}^2 \alpha) = \{F(s^2)\}^2 - s^2 c^2 d^2 \{F_1(s^2)\}^2,$$

each side of this relation being the product of two factors, viz. for the left-hand side the factors are

$$A \Pi \frac{H(x - \alpha)}{\Theta(x)}, \quad A \Pi \frac{H(x + \alpha)}{\Theta(x)},$$

and for the right-hand side they are the rational functions of s^2 ,

$$F(s^2) + scd F_1(s^2), \quad F(s^2) - scd F_1(s^2);$$

these factors are by Hermite's theorem equal each to each; viz. this is the relation in which Hermite's stands to Abel's theorem.

The H -product theorem is given as one out of a group of four theorems; the other three may be called the H -product, H_1 -product and Θ_1 -product, odd theorems respectively,

$$\{\Theta_1(x) = \Theta(x + K), \quad H_1(x) = H(x + K)\},$$

viz. these are

$$\left\{ \begin{array}{l} \frac{A_1 H(x - \alpha_1) H(x - \alpha_2) \dots H(x - \alpha_{2n+1})}{\Theta^{2n+1}(x)} = sF(s^2) + cd\phi(s^2), \\ \frac{A_2 H_1(x - \alpha_1) H_1(x - \alpha_2) \dots H_1(x - \alpha_{2n+1})}{\Theta^{2n+1}(x)} = cF(c^2) - sd\phi(c^2), \\ \frac{A_3 \Theta_1(x - \alpha_1) \Theta_1(x - \alpha_2) \dots \Theta_1(x - \alpha_{2n+1})}{\Theta^{2n+1}(x)} = dF(d^2) - k^2 sc\phi(d^2), \end{array} \right.$$

where F, ϕ are rational and integral functions of the degrees n and $n - 1$, having their proper values in the three equations respectively, and in each case

$$\alpha_1 + \alpha_2 + \dots + \alpha_{2n+1} = 0.$$

It was seen above that, for $n = 1$, the H -product theorem became

$$\frac{A H(x - \alpha) H(x + \alpha)}{\Theta^2(x)} = \operatorname{sn}^2 x - \operatorname{sn}^2 \alpha,$$

which is the most simple case; for the odd theorems, the most simple case is $n = 0$, viz. we then have

$$\frac{A_1 H(x)}{\Theta(x)} = \operatorname{sn} x, \quad \frac{A_2 H_1(x)}{\Theta(x)} = \operatorname{cn} x, \quad \frac{A_3 \Theta_1(x)}{\Theta(x)} = \operatorname{dn} x;$$

to complete the formulae observe that the values of the constants are

$$A = \frac{2k'K}{k\pi\Theta^2(\alpha)}, \quad A_1 = \frac{1}{\sqrt{(k)}}, \quad A_2 = \sqrt{\left(\frac{k'}{k}\right)}, \quad A_3 = \sqrt{(k')}.$$

The three theorems may be used, in like manner with the H -product theorem, to give the values of the sn , cn , and dn respectively of the sum $\alpha_1 + \alpha_2 + \dots + \alpha_{2n}$.

882.

A CORRESPONDENCE OF CONFOCAL CARTESIANS WITH THE RIGHT LINES OF A HYPERBOLOID.

[From the *Messenger of Mathematics*, vol. XVIII. (1889), pp.
128–130.]

TAKE α, β, γ arbitrary, $A, B, C = \beta - \gamma, \gamma - \alpha, \alpha - \beta$ (so that $A + B + C = 0$), and writing ρ, σ, τ for rectangular coordinates, consider the hyperboloid

$$A\rho^2 + B\sigma^2 + C\tau^2 + ABC = 0.$$

Let ρ_0, σ_0, τ_0 be the coordinates of a point on the surface ($A\rho_0^2 + B\sigma_0^2 + C\tau_0^2 + ABC = 0$). The equations of a line through this point are $\rho = \rho_0 + f\Omega, \sigma = \sigma_0 + g\Omega, \tau = \tau_0 + h\Omega$ (Ω indeterminate); and if this lies on the surface, we have

$$\begin{aligned} A\rho_0 f + B\sigma_0 g + C\tau_0 h &= 0, \\ Af^2 + Bg^2 + Ch^2 &= 0, \end{aligned}$$

which equations determine the ratios $f : g : h$; the equations give

$$(A\rho_0 f + B\sigma_0 g)^2 = C\tau_0^2 \cdot Ch^2, \quad = -C\tau_0^2 (Af^2 + Bg^2);$$

that is,

$$(A^2\rho_0^2 + AC\tau_0^2)f^2 + 2AB\rho_0\sigma_0fg + (B^2\sigma_0^2 + BC\tau_0^2)g^2 = 0,$$

whence

$$\begin{aligned} &\{(B^2\sigma_0^2 + BC\tau_0^2)g + AB\rho_0\sigma_0f\}^2 \\ &= \{A^2B^2\rho_0^2\sigma_0^2 - (A^2\rho_0^2 + AC\tau_0^2)(B^2\sigma_0^2 + BC\tau_0^2)\}f^2, \\ &= -ABC(A\rho_0^2 + B\sigma_0^2 + C\tau_0^2)\tau_0^2f^2, \\ &= A^2B^2C^2\tau_0^2f^2; \end{aligned}$$

that is,

$$\{(B\sigma_0^2 + C\tau_0^2)g + A\rho_0\sigma_0f\}^2 = A^2C^2\tau_0^2f^2,$$

or say

$$(B\sigma_0^2 + C\tau_0^2)g + A(\rho_0\sigma_0 \pm C\tau_0)f = 0,$$

which equation, together with $A\rho_0f + B\sigma_0g + C\tau_0h = 0$, determines the ratios $f : g : h$. We have thus the two lines through the point $(\rho_0, \sigma_0, \tau_0)$.

But the equations of the line may be conveniently represented in a different form; writing the equation first obtained in the form

$$\sigma_0 (B\sigma_0g + A\rho_0f) + C\tau_0^2g \pm AC\tau_0f = 0,$$

this is

$$-\sigma_0C\tau_0h + C\tau_0^2g \pm AC\tau_0f = 0,$$

viz.

$$-h\sigma_0 + g\tau_0 \pm Af = 0;$$

and we have the like equations

$$-f\tau_0 + h\rho_0 \pm Bg = 0,$$

$$-g\rho_0 + f\sigma_0 \pm Ch = 0,$$

where the sign is the same in each of the three equations.

The equations of the line on the surface may be written

$$\begin{array}{lll} \cdot & h\sigma - g\tau - h\sigma_0 + g\tau_0 & = 0, \\ -h\rho & \cdot & + f\tau - f\tau_0 + h\rho_0 & = 0, \\ g\rho & -f\sigma & \cdot -g\rho_0 + f\sigma_0 & = 0, \\ (h\sigma_0 - g\tau_0)\rho + (f\tau_0 - h\rho_0)\sigma + (g\rho_0 - f\sigma_0)\tau & \cdot & = 0; \end{array}$$

and hence from the foregoing three equations, taking the sign $-$, we have

$$\begin{array}{lll} \cdot & h\sigma - g\tau + Af & = 0, \\ -h\rho & \cdot & + f\tau + Bg & = 0, \\ g\rho & -f\sigma & \cdot + Ch & = 0, \\ -Af\rho & -Bg\sigma - Ch\tau & \cdot & = 0, \end{array}$$

where $Af^2 + Bg^2 + Ch^2 = 0$, for the equations of a line on the surface.

In like manner, taking the sign $+$, and for f, g, h writing new values f', g', h' , we have

$$\begin{array}{lll} \cdot & h'\sigma - g'\tau - Af' & = 0, \\ -h'\rho & \cdot & + f'\tau - Bg' & = 0, \\ g'\rho & -f'\sigma & \cdot -Ch' & = 0, \\ Af'\rho & +Bg'\sigma + Ch'\tau & \cdot & = 0, \end{array}$$

where $Af'^2 + Bg'^2 + Ch'^2 = 0$, for the equations of a line on the surface.

The two systems of equations evidently belong to the lines of the two different kinds respectively. Writing for shortness $P, Q, R = gh' + g'h, hf' + h'f, fg' + f'g$, the two lines in fact intersect in a point, the coordinates say $(\rho_0, \sigma_0, \tau_0)$ whereof are $= \Theta QR, \Theta RP, \Theta PQ$, where

$$\Theta = \frac{A}{g^2h'^2 - g'^2h^2} = \frac{B}{h^2f'^2 - h'^2f^2} = \frac{C}{f^2g'^2 - f'^2g^2},$$

the three expressions for Θ being equal to each other in virtue of the equations

$$Af^2 + Bg^2 + Ch^2 = 0, \quad Af'^2 + Bg'^2 + Ch'^2 = 0.$$

Take now, in a plane, P, Q, R points on any line, say the axis of x , at distances α, β, γ from the origin, then for a point of the plane, coordinates (x, y) , if ρ, σ, τ be the distances of the point from these three points, or say foci, we have

$$\begin{aligned} \rho^2 &= (x - \alpha)^2 + y^2, \\ \sigma^2 &= (x - \beta)^2 + y^2, \\ \tau^2 &= (x - \gamma)^2 + y^2; \end{aligned}$$

and if as before $A, B, C = \beta - \gamma, \gamma - \alpha, \alpha - \beta$, we thence have

$$A\rho^2 + B\sigma^2 + C\tau^2 + ABC = 0.$$

A point, coordinates (ρ, σ, τ) , of the hyperboloid thus corresponds to a point in the plane, distances ρ, σ, τ from the three foci R, S, T respectively; and to any line

$$\begin{aligned} & \cdot \quad h\sigma - g\tau + Af & & = 0, \\ -h\rho & \cdot & + f\tau + Bg & = 0, \\ g\rho & -f\sigma & \cdot + Ch & = 0, \\ -Af\rho & -Bg\sigma - Ch\tau & \cdot & = 0, \end{aligned}$$

corresponds the Cartesian represented by these linear equations. Similarly, to the line represented by the other system of equations

$$\begin{aligned} & \cdot \quad h'\sigma - g'\tau - Af' & & = 0, \\ -h'\rho & \cdot & + f'\tau - Bg' & = 0, \\ g'\rho & -f'\sigma & \cdot - Ch' & = 0, \\ Af'\rho & +Bg'\sigma + Ch'\tau & \cdot & = 0, \end{aligned}$$

corresponds the Cartesian represented by these equations; the two curves intersect in the point $\rho_0, \sigma_0, \tau_0 = \Theta QR, \Theta RP, \Theta PQ$, corresponding to the intersection of the lines on the hyperboloid; and moreover, *qua* confocal Cartesians, they intersect at right angles.

883.

ANALYTICAL FORMULAE IN REGARD TO AN
OCTAD OF POINTS.

[From the *Messenger of Mathematics*, vol. XVIII. (1889), pp.
149–152.]

THE term “tetrad” is used in two distinct but not inconsistent senses, viz. a tetrad denotes any four points; and it also denotes the four vertices of a self-conjugate tetrahedron in regard to a quadric surface. In fact, given any four points, there exists a triply infinite series of quadric surfaces such that, in regard to any one of them, the four points form a self-conjugate tetrahedron.

Two or more tetrads, in regard to one and the same quadric surface, are called similar tetrads.

The eight points of intersection of any three quadric surfaces are an octad; and we have the theorem that any seven points determine the octad, viz. the quadric surfaces which pass through any seven given points pass also through a uniquely determinate eighth point.

We have the further theorem that any two similar tetrads form an octad.

In particular, the vertices of the tetrahedron

$$(x = 0, y = 0, z = 0, w = 0),$$

or, say the points 1000, 0100, 0010, 0001 are a tetrad in regard to the quadric surface $x^2 + y^2 + z^2 + w^2 = 0$. The points (x_1, y_1, z_1, w_1) , (x_2, y_2, z_2, w_2) , (x_3, y_3, z_3, w_3) , (x_4, y_4, z_4, w_4) , or say the points 1, 2, 3, 4, will be a tetrad in regard to the same quadric surface, if only

$$x_1x_2 + y_1y_2 + z_1z_2 + w_1w_2 = 0, \text{ \&c., (six equations).}$$

Hence, these equations being satisfied, the two tetrads form an octad; the equation of a general quadric surface through the points of the first tetrad is

$$fyz + gzx + hxy + lxw + myw + nzw = 0,$$

and we can from three such equations eliminate any two terms; the two terms eliminated may be such as xy , xz , (12 forms) or such as

yz, xw , (3 forms); and the equation may thus be presented either in the form

$$fyz + (lx + my + nz)w = 0,$$

or in the form

$$gzx + hxy + myw + nzw = 0.$$

The absolute magnitudes of the coordinates

$$(x_1, y_1, z_1, w_1), \text{ \&c.},$$

of these points are properly indeterminate; but we may if we please fix the absolute magnitudes by assuming

$$x_1^2 + y_1^2 + z_1^2 + w_1^2 = M, \text{ \&c.},$$

(four equations, the same quantity M in each of them), and we have as before

$$x_1x_2 + y_1y_2 + z_1z_2 + w_1w_2 = 0, \text{ \&c.}, \text{ (six equations),}$$

viz. the coordinates are taken to satisfy these 10 equations; and this being so, it is to be shown that there exist quadric surfaces as above. The 10 equations may be expressed in a correlative form in like manner with the ordinary six equations for the transformation of the rectangular coordinates (x, y, z) , viz. the new form is

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = M, \text{ \&c.}, \text{ (four equations),}$$

and

$$x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4 = 0, \text{ \&c.}, \text{ (six equations),}$$

and the coordinates thus also satisfy these 10 equations.

Thus the equation $fyz + (lx + my + nz)w = 0$, if we determine the coefficients in such wise that the surface passes through the points 1, 2, 3, becomes

$$\begin{vmatrix} yz & xw & yw & zw \\ y_1z_1 & x_1w_1 & y_1w_1 & z_1w_1 \\ y_2z_2 & x_2w_2 & y_2w_2 & z_2w_2 \\ y_3z_3 & x_3w_3 & y_3w_3 & z_3w_3 \end{vmatrix} = 0,$$

and this equation must therefore be satisfied on writing therein (x_4, y_4, z_4, w_4) for (x, y, z, w) ; viz. the equation thus obtained, det.

$(1, 2, 3, 4) = 0$, must be satisfied in virtue of the equations which connect the coordinates of the four points. But, instead of effecting this verification, it is better to exhibit the equation in (x, y, z, w) in a form wherein the coordinates of the four points enter symmetrically; and the new equation, *qua* transformation of the original form, is satisfied for $(x, y, z, w) = (x_1, y_1, z_1, w_1)$: and then of course, by reason of the symmetry, it is also satisfied for

$$(x, y, z, w) = (x_4, y_4, z_4, w_4).$$

The transformation may be effected directly; but I prefer to write down the final result, and afterwards verify it.

The form is

$$\begin{aligned} yz \cdot Mw_1w_2w_3w_4 + xw[-x_1y_1z_1 \cdot w_2w_3w_4 - x_2y_2z_2 \cdot w_3w_4w_1 - x_3y_3z_3 \cdot w_4w_1w_2 - x_4y_4z_4 \cdot w_1w_2w_3] \\ + yw[x_1^2w_1 \cdot z_2z_3z_4 + x_2^2w_2 \cdot z_3z_4z_1 + x_3^2w_3 \cdot z_4z_1z_2 + x_4^2w_4 \cdot z_1z_2z_3] \\ + zw[x_1^2w_1 \cdot y_2y_3y_4 + x_2^2w_2 \cdot y_3y_4y_1 + x_3^2w_3 \cdot y_4y_1y_2 + x_4^2w_4 \cdot y_1y_2y_3] \end{aligned}$$

We have to show that the equation is satisfied by writing therein

$$(x, y, z, w) = (x_1, y_1, z_1, w_1).$$

Calling the resulting value Ω_1 , we have

$$\begin{aligned} \Omega_1 \div w_1 = My_1z_1w_2w_3w_4 - x_1^2y_1z_1w_2w_3w_4 \\ - x_1(x_2y_2z_2w_3w_4w_1 + x_3y_3z_3w_4w_1w_2 + x_4y_4z_4w_1w_2w_3) \\ + x_1^2w_1(y_1z_2z_3z_4 + z_1y_2y_3y_4) \\ + y_1z_1[x_2^2w_2(z_3z_4 + y_3y_4) + x_3^2w_3(z_4z_2 + y_4y_2) \\ + x_4^2w_4(z_2z_3 + y_2y_3)], \end{aligned}$$

which is

$$\begin{aligned} = y_1z_1w_2w_3w_4(x_2^2 + x_3^2 + x_4^2) \\ - x_1(x_2y_2z_2w_3w_4w_1 + x_3y_3z_3w_4w_1w_2 + x_4y_4z_4w_1w_2w_3) \\ + x_1^2w_1(y_1z_2z_3z_4 + z_1y_2y_3y_4) \\ - y_1z_1[x_2^2w_2(x_3x_4 + w_3w_4) + x_3^2w_3(x_2x_4 + w_2w_4) \\ + x_4^2w_4(x_2x_3 + w_2w_3)]. \end{aligned}$$

The expression contains terms in $w_2w_3w_4$ which destroy each other; omitting them, we have

$$\begin{aligned}\Omega_1 \div w_1 = & -x_1(x_2y_2z_2w_3w_4w_1 + x_3y_3z_3w_4w_1w_2 + x_4y_4z_4w_1w_2w_3) \\ & + x_1^2w_1(y_1z_2z_3z_4 + z_1y_2y_3y_4) \\ & - x_2x_3x_4y_1z_1(x_2w_2 + x_3w_3 + x_4w_4),\end{aligned}$$

where the last line is

$$= x_2x_3x_4y_1z_1.x_1w_1.$$

Hence the whole divides by x_1w_1 , or we have

$$\begin{aligned}\Omega_1 \div x_1w_1^2 = & -x_2y_2z_2.w_3w_4 - x_3y_3z_3.w_4w_2 - x_4y_4z_4.w_2w_3 \\ & + x_1y_1z_2z_3z_4 + x_1z_1y_2y_3y_4 + y_1z_1x_2x_3x_4,\end{aligned}$$

viz. this is

$$\begin{aligned}= & x_2x_3x_4y_1z_1 + y_2y_3y_4z_1x_1 + z_2z_3z_4x_1y_1 \\ & + x_2y_2z_2(x_3x_4 + y_3y_4 + z_3z_4) \\ & + x_3y_3z_3(x_2x_4 + y_2y_4 + z_2z_4) \\ & + x_4y_4z_4(x_2x_3 + y_2y_3 + z_2z_3);\end{aligned}$$

or, finally, it is

$$\begin{aligned}= & x_2x_3x_4(y_1z_1 + y_2z_2 + y_3z_3 + y_4z_4) \\ & + y_2y_3y_4(z_1x_1 + z_2x_2 + z_3x_3 + z_4x_4) \\ & + z_2z_3z_4(x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4),\end{aligned}$$

which is $= 0$; that is, we have $\Omega_1 = 0$, the required result.

The foregoing equation of the quadric surface can be by mere interchanges of the letters exhibited in 12 different forms.

We may, in like manner, first establish and then verify the equation

$$\begin{aligned}& zx(-z_1^2w_1y_2y_3y_4 - z_2^2w_2y_3y_4y_1 - z_3^2w_3y_4y_1y_2 - z_4^2w_4y_1y_2y_3) \\ & + xy(-y_1^2w_1z_2z_3z_4 + y_2^2w_2z_3z_4z_1 + y_3^2w_3z_4z_1z_2 + y_4^2w_4z_1z_2z_3) \\ & + yw(-y_1^2x_1z_2z_3z_4 - y_2^2x_2z_3z_4z_1 - y_3^2x_3z_4z_1z_2 - y_4^2x_4z_1z_2z_3) \\ & + zw(-z_1^2x_1y_2y_3y_4 + z_2^2x_2y_3y_4y_1 + z_3^2x_3y_4y_1y_2 + z_4^2x_4y_1y_2y_3) = 0;\end{aligned}$$

this can be by a cyclical interchange of the letters (x, y, z) exhibited in 3 different forms.

Of course, the fifteen equations belong each of them to a quadric surface through the 8 points. Any three of the fifteen equations, say $U = 0$, $V = 0$, $W = 0$, may be used for determining the octad; the equation of any other quadric through the octad is then $\alpha U + \beta V + \gamma W = 0$.

884.

**NOTE SUR LES SURFACES MINIMA ET LE
THEOREME DE JOACHIMSTHAL.**

[From the *Comptes Rendus*, t. CVI. (1888), pp. 995, 996.]

I. LA seule surface minima reglee est l'helicoide a plan directeur: ce beau theoreme de M. Catalan se demontre par des considerations geometriques tres simples.

Soit un systeme de droites $\dots, P, Q, R, \dots$, tel que, pour trois droites successives P, Q, R quelconques, tout plan Π' perpendiculaire a Q rencontre les trois droites en des points p', q', r' , situes sur une droite. En particulier, si qr est la distance la plus petite des droites Q et R , le plan perpendiculaire Π , qui passe par le point q , passera par le point r et rencontrera la droite P en un point p , tel que les points p, q, r seront sur une meme droite Λ , perpendiculaire a chacune des droites P, Q, R ; c'est-a-dire, chacune de ces droites rencontre perpendiculairement une seule et meme droite Λ . Cela etant, pour que les points de rencontre p', q', r' avec un autre plan Π' soient sur la meme droite, il faut encore une condition; et, en supposant (ce que l'on peut faire sans perte de generalite) que les distances pq, qr soient egales, cette condition sera: que l'inclinaison des plans $(P\Lambda, Q\Lambda)$ est egale a l'inclinaison des plans $(Q\Lambda, R\Lambda)$.

En considerant de meme les droites Q, R, S , on demontre d'abord que la droite S rencontre perpendiculairement la droite Λ ; puis, en supposant que les distances qr, rs soient egales, on obtient la condition que l'inclinaison des plans $(Q\Lambda, R\Lambda)$ est egale a celle des plans $(R\Lambda, S\Lambda)$, et ainsi de suite: savoir, on obtient une serie de droites $\dots, P, Q, R, \dots$, qui rencontrent perpendiculairement la droite Λ et sont telles qu'en supposant que les distances $\dots, pq, qr, rs, \dots$ soient egales, les inclinaisons $\dots, (P\Lambda, Q\Lambda), (Q\Lambda, R\Lambda), (R\Lambda, S\Lambda), \dots$ sont egales: c'est-a-dire, en considerant des droites $\dots, P, Q, R, S, \dots$ consecutives, on a les generatrices d'une helicoide a plan directeur; et l'on voit ainsi que cette surface est la seule surface reglee, qui est telle que tout plan perpendiculaire a une generatrice quelconque rencontre la surface selon une courbe qui a, au point de rencontre avec la generatrice, une inflexion (ou, ce qui est la meme chose, un rayon infini de courbure). Mais, comme l'avait remarque M. Catalan, c'est la la condition pour que les deux rayons principaux

de courbure soient egaux et opposes, ou enfin pour que la surface soit une surface minima.

II. Le theoreme de Joachimsthal et aussi le theoreme plus general de Bonnet et Serret, par rapport aux lignes de courbure planes ou spheriques, se deduisent immediatement de ce theoreme elementaire de Geometrie: En considerant, dans des plans differents, deux triangles isocèles $PP'O$ et $PP'N$ avec une base commune PP' ($OP = OP'$ et $NP = NP'$), les angles OPN et $OP'N$ seront egaux. En effet, si, pour une surface quelconque, PP' est l'element d'une courbe de courbure spherique, les normales a la surface aux points P et P' se rencontrent dans un point N , et les rayons de la sphere aux memes points se rencontrent dans un point O ; on a ainsi les deux triangles isocèles $PP'O$ et $PP'N$, et de la les angles egaux OPN et $OP'N$, c'est-a-dire que, pour deux points consecutifs P et P' de la ligne de courbure spherique, l'inclinaison de la normale de la surface au rayon de la sphere a la meme valeur, et cette inclinaison a ainsi la meme valeur pour tous les points de la ligne de courbure. En prenant le point O a l'infini, on obtient le theoreme pour une ligne de courbure plane, ou, si l'on veut, le theoreme pour ce cas se deduit directement de celui-ci: Une droite quelconque PO , perpendiculaire a la base PP' d'un triangle isocèle $PP'N$, est egalemeut inclinee sur les deux droites PN et $P'N$.

885.

**ON THE DIOPHANTINE RELATION,
 $y^2 + y'^2 = \text{SQUARE.}$**

[From the *Proceedings of the London Mathematical Society*, vol. XX.
 (1889), pp. 122–127.]

THE diophantine relation $y^2 + y'^2 = \text{Square}$, where y is a function of x , and y' denotes $\frac{dy}{dx}$, is considered by Prof. Sylvester in his paper “On Reducible Cyclodes,” *Proc. Lond. Math. Soc.*, t. I. (1865–66), pp. 137–160. It is at once seen that there exists a solution

$$y = (x + a)^\alpha (x + b)^\beta (x + c)^\gamma (x + d)^\delta \dots,$$

where the roots $a, b, c, d, \dots$ are essentially unequal, and the number of simple factors $x+a, x+b, x+c, x+d, \dots$ is even; the exponents $\alpha, \beta, \gamma, \delta, \dots$ are taken to be positive integer numbers. Sylvester assumes, and it will be shown, that the factors must separate themselves into two sets, or, as he calls them, diptychs, each containing the same number of simple factors and such that the sum of the exponents for the one diptych is equal to the sum of the exponents for the other diptych; viz. the form is $y = UU_1$, where

$$U = (x + a)^\alpha (x + b)^\beta \dots, \quad U_1 = (x + a_1)^{\alpha_1} (x + b_1)^{\beta_1} \dots,$$

with the same number of simple factors, and with the relation $\alpha + \beta + \dots = \alpha_1 + \beta_1 + \dots$ between the exponents. Hence, if the number of simple factors be called the class and the sum of the exponents be called the order, the class and the order are each of them even; or, what is the same thing, the semi-class (say μ) and the semi-order (say ν) are each of them integral.

The separation of the factors into two diptychs is a remarkable theorem. I consider the analytical theory; for greater simplicity, first in the case, class = 2, and secondly in the case, class = 4; but it is easy to see that the like process is applicable to the case of any even value whatever of the class.

I write as usual $i = \sqrt{-1}$; the equation $y^2 + y'^2 = \text{square}$, implies $y + iy' = \text{square}$, and $y - iy' = \text{square}$; at least this is so, save as to a common denominator, as will appear.

First, if the class is $= 2$; we have

$$y = (x + a)^\alpha (x + b)^\beta;$$

hence

$$y + iy' = y \left(1 + \frac{i\alpha}{x + a} + \frac{i\beta}{x + b} \right),$$

$$y - iy' = y \left(1 - \frac{i\alpha}{x + a} - \frac{i\beta}{x + b} \right),$$

say these are $= \frac{(x + l)^2 y}{x + a \cdot x + b}$ and $\frac{(x + m)^2 y}{x + a \cdot x + b}$ respectively; and, this being so, we have

$$y^2 + y'^2 = \frac{y^2 (x + l)^2 (x + m)^2}{(x + a)^2 (x + b)^2}, = (x + a)^{2\alpha - 2} (x + b)^{2\beta - 2} (x + l)^2 (x + m)^2.$$

It is to be shown that the assumed relations lead to $\alpha = \beta$. Resolving the last-mentioned expressions for $y + iy'$, $y - iy'$ each into simple fractions, we have

$$\begin{aligned} i\alpha (b - a) &= (l - b)^2, & -i\alpha (b - a) &= (m - b)^2, \\ i\beta (a - b) &= (l - a)^2, & -i\beta (a - b) &= (m - a)^2. \end{aligned}$$

Hence

$$(l - b)^2 + (m - b)^2 = 0, \quad (l - a)^2 + (m - a)^2 = 0;$$

these cannot give

$$(l - b) + i(m - b) = 0, \quad (l - a) + i(m - a) = 0,$$

with the same sign for i in the two equations; for we should then have

$$(1 + i)(b - a) = 0,$$

but $1 + i$ is not $= 0$, and a, b are essentially unequal. Hence, taking (as we may do) $+i$ in the first equation, we must have $-i$ in the second equation, and the two equations are

$$\begin{aligned} l - b + i(m - b) &= 0, \text{ that is, } l + im = (1 + i)b, \\ l - a - i(m - a) &= 0, \quad \text{,,} \quad l - im = (1 - i)a, \end{aligned}$$

and thence

$$\begin{aligned} 2l &= (1+i)(b-ia), \\ 2m &= (1+i)(b+ia). \end{aligned}$$

Hence also

$$\begin{aligned} 2(l-b) &= (1-i)(a-b), & 2i(m-b) &= (1-i)(b-a), \\ 2(l-a) &= -(1+i)(a-b), & 2i(m-a) &= (1+i)(b-a); \end{aligned}$$

consequently

$$\begin{aligned} 2(l-b)^2 &= -i(a-b)^2, = -2i\alpha(a-b), \\ 2(l-a)^2 &= i(a-b)^2, = 2i\beta(a-b). \end{aligned}$$

Hence $\alpha = \beta = \frac{1}{2}(a-b)$, and the solution thus is

$$\begin{aligned} y &= (x+a)^\alpha(x+b)^\alpha, & \alpha &= \frac{1}{2}(a-b), \\ l &= \frac{1}{2}\{a+b+i(a-b)\}, \\ m &= \frac{1}{2}\{a+b-i(a-b)\}, \\ y^2 + y'^2 &= (x+a)^{2\alpha-2}(x+b)^{2\alpha-2}(x+l)^2(x+m)^2. \end{aligned}$$

The class is here $= 2$, and the order is $= 2\alpha$; considering the order as given, say it is $= 2\nu$, we have $\alpha = \nu$, and the equation $\nu = \frac{1}{2}(a-b)$ then shows that one of the roots a, b is arbitrary. Taking it to be a , we have $b = a - 2\nu$, or the solution, class 2 and order 2ν , is

$$\begin{aligned} y &= (x+a)^\nu(x+a-2\nu)^\nu, \\ l &= a-\nu+i\nu, & m &= a-\nu-i\nu, \\ y^2 + y'^2 &= (x+a)^{2\nu-2}(x+a-2\nu)^{2\nu-2}(x+l)^2(x+m)^2. \end{aligned}$$

Considering next for the case, class $= 4$, the solution

$$y = (x+a)^\alpha(x+b)^\beta(x+c)^\gamma(x+d)^\delta,$$

we have

$$\begin{aligned} y + iy' &= y \left(1 + \frac{i\alpha}{x+a} + \frac{i\beta}{x+b} + \frac{i\gamma}{x+c} + \frac{i\delta}{x+d} \right), \\ y - iy' &= y \left(1 - \frac{i\alpha}{x+a} - \frac{i\beta}{x+b} - \frac{i\gamma}{x+c} - \frac{i\delta}{x+d} \right); \end{aligned}$$

or, putting these

$$= \frac{(x+l)^2(x+p)^2 y}{x+a \cdot x+b \cdot x+c \cdot x+d} \text{ and } \frac{(x+m)^2(x+q)^2 y}{x+a \cdot x+b \cdot x+c \cdot x+d}$$

respectively, we have

$$\begin{aligned} y^2 + y'^2 &= \frac{y^2(x+l)^2(x+p)^2(x+m)^2(x+q)^2}{(x+a)^2(x+b)^2(x+c)^2(x+d)^2}, \\ &= (x+a)^{2\alpha-2}(x+b)^{2\beta-2}(x+c)^{2\gamma-2}(x+d)^{2\delta-2}(x+l)^2(x+p)^2(x+m)^2(x+q)^2 \end{aligned}$$

Also, by decomposing the expressions for $y+iy'$, $y-iy'$ into simple fractions and comparing with the original values, we find

$$\begin{aligned} i\alpha(b-a)(c-a)(d-a) &= (a-l)^2(a-p)^2, & -i\alpha(b-a)(c-a)(d-a) &= (a-m)^2(a-q)^2, \\ i\beta(a-b)(c-b)(d-b) &= (b-l)^2(b-p)^2, & -i\beta(a-b)(c-b)(d-b) &= (b-m)^2(b-q)^2, \\ i\gamma(a-c)(b-c)(d-c) &= (c-l)^2(c-p)^2, & -i\gamma(a-c)(b-c)(d-c) &= (c-m)^2(c-q)^2, \\ i\delta(a-d)(b-d)(c-d) &= (d-l)^2(d-p)^2, & -i\delta(a-d)(b-d)(c-d) &= (d-m)^2(d-q)^2. \end{aligned}$$

Hence

$$\begin{aligned} (a-l)^2(a-p)^2 + (a-m)^2(a-q)^2 &= 0, \\ (b-l)^2(b-p)^2 + (b-m)^2(b-q)^2 &= 0, \\ (c-l)^2(c-p)^2 + (c-m)^2(c-q)^2 &= 0, \\ (d-l)^2(d-p)^2 + (d-m)^2(d-q)^2 &= 0; \end{aligned}$$

we cannot from these obtain *three* equations

$$\begin{aligned} (a-m)(a-q) - i(a-l)(a-p) &= 0, \\ (b-m)(b-q) - i(b-l)(b-p) &= 0, \\ (c-m)(c-q) - i(c-l)(c-p) &= 0, \end{aligned}$$

with the same sign for i ; in fact these would give

$$(1+i)(b-c)(c-a)(a-b) = 0,$$

but $1+i$ is not $= 0$, and the a, b, c are essentially unequal. Hence we must have equations such as

$$\begin{aligned} (a-m)(a-q) - i(a-l)(a-p) &= 0; & (c-m)(c-q) + i(c-l)(c-p) &= 0, \\ (b-m)(b-q) - i(b-l)(b-p) &= 0; & (d-m)(d-q) + i(d-l)(d-p) &= 0, \end{aligned}$$

two of them with $-i$, and two of them with $+i$; viz. the a, b, c, d divide themselves into pairs which are taken to be a, b and c, d .

We hence easily obtain

$$\begin{aligned} a + b - m - q - i(a + b - l - p) &= 0, & ab - mq - i(ab - lp) &= 0, \\ c + d - m - q - i(c + d - l - p) &= 0, & cd - mq - i(cd - lp) &= 0, \end{aligned}$$

and thence

$$\begin{aligned} a + b - c - d &= i(a + b + c + d) - 2i(l + p), \\ ab - cd &= i(ab + cd) - 2ilp. \end{aligned}$$

Forming from these values of $l + p, lp$ the expression for $2i(a - l)(a - p)$, we find $2i(a - l)(a - p) = (i + 1)(a - c)(a - d)$; and we have thus the set of equations

$$\begin{aligned} 2i(a - l)(a - p) &= (i + 1)(a - c)(a - d), \\ 2i(b - l)(b - p) &= (i + 1)(b - c)(b - d), \\ 2i(c - l)(c - p) &= (i - 1)(c - a)(c - b), \\ 2i(d - l)(d - p) &= (i - 1)(d - a)(d - b). \end{aligned}$$

Hence also

$$\begin{aligned} 2(a - l)^2(a - p)^2 &= -i(a - c)^2(a - d)^2, \\ 2(b - l)^2(b - p)^2 &= -i(b - c)^2(b - d)^2, \\ 2(c - l)^2(c - p)^2 &= i(c - a)^2(c - b)^2, \\ 2(d - l)^2(d - p)^2 &= i(d - a)^2(d - b)^2; \end{aligned}$$

and, substituting these values in a former set of equations, we obtain

$$\begin{aligned} 2\alpha(b - a) &= -(a - c)(a - d), \\ 2\beta(a - b) &= -(b - c)(b - d), \\ 2\gamma(d - c) &= (c - a)(c - b), \\ 2\delta(c - d) &= (d - a)(d - b); \end{aligned}$$

and thence

$$\begin{aligned} 2(\alpha + \beta) &= a + b - c - d, \\ 2(\gamma + \delta) &= -(c + d - a - b), \end{aligned}$$

that is, $\alpha + \beta = \gamma + \delta$; viz. there are, in this case also, two diptychs.

If, as before, the order is taken to be $= 2\nu$, then $\alpha + \beta = \nu$, $\gamma + \delta = \nu$; supposing that ν is a given positive integer, and that $\alpha, \beta, \gamma, \delta$ are positive integers satisfying these equations $\alpha + \beta = \nu$, $\gamma + \delta = \nu$, then the last-mentioned four equations between $\alpha, \beta, \gamma, \delta$ and a, b, c, d are equivalent to three relations serving to determine the differences of a, b, c, d (say $a - d, b - d, c - d$) in terms of $\alpha, \beta, \gamma, \delta$. And we then further have

$$\begin{aligned}(a-l)(a-p) &= -(1-i)\alpha(b-a), & (a-m)(a-q) &= -(1+i)\alpha(b-a), \\ (b-l)(b-p) &= -(1-i)\beta(a-b), & (b-m)(b-q) &= -(1+i)\beta(a-b), \\ (c-l)(c-p) &= (1+i)\gamma(d-c), & (c-m)(c-q) &= (1-i)\gamma(c-d), \\ (d-l)(d-p) &= (1+i)\delta(c-d), & (d-m)(d-q) &= (1-i)\delta(d-c),\end{aligned}$$

each set equivalent to two equations; or, as these may be written,

$$\begin{aligned}2(l-p) &= a+b+c+d+i(a+b-c-d), \\ 2lp &= ab+cd+i(ab-cd), \\ 2(m+q) &= a+b+c+d-i(a+b-c-d), \\ 2mq &= ab+cd-i(ab-cd),\end{aligned}$$

serving to determine l, p, m, q in terms of a, b, c, d .

Observe also that, u being arbitrary, we have

$$\begin{aligned}2(u-l)(u-p) &= (1+i)(u-a)(u-b) + (1-i)(u-c)(u-d), \\ 2(u-m)(u-q) &= (1-i)(u-a)(u-b) + (1+i)(u-c)(u-d),\end{aligned}$$

(which equations, writing therein $u = a, b, c$, or d , in fact reproduce the two systems of four equations).

We have also

$$\begin{aligned}l+p+m+q &= a+b+c+d, & l+p-m-q &= i(a+b-c-d), \\ lp+mq &= ab+cd, & lp-mq &= i(ab-cd);\end{aligned}$$

and moreover

$$\begin{aligned}4(l-p)^2 &= 2i\{(a-b)^2 - (c-d)^2\} + 4(a+b)(c+d) - 8(ab+cd), \\ 4(m-q)^2 &= -2i\{(a-b)^2 - (c-d)^2\} + 4(a+b)(c+d) - 8(ab+cd),\end{aligned}$$

which equations, combined with the foregoing values of $2(l+p)$ and $2(m+q)$, give the values of l, p, m, q . We have thus the complete solution for the case class = 4, order = 2ν ; say

$$y = (x+a)^\alpha(x+b)^\beta \cdot (x+c)^\gamma(x+d)^\delta; \quad \alpha + \beta = \gamma + \delta = \nu,$$

$$y^2 + y'^2 = (x+a)^{2\alpha-2}(x+b)^{2\beta-2}(x+c)^{2\gamma-2}(x+d)^{2\delta-2}(x+l)^2(x+p)^2(x+m)^2(x$$

with the foregoing relations between the constants.

886.

ON THE SURFACES WITH PLANE OR SPHERICAL CURVES OF CURVATURE.

[From the *American Journal of Mathematics*, vol. XI. (1889), pp. 71-98; pp. 293-306.]

I CONSIDER the general question of the surfaces which have their curves of curvature plane or spherical. In the memoir "Sur les surfaces dont les rayons de courbure sont égaux mais de signes contraires," *Liouville*, t. XII., the question is considered from a different point of view, but the principal results were obtained; the present paper is an independent investigation.

The condition that the two systems of curves of curvature may be plane curves is, as is known, that the surface be a *Weingarten* surface. If in the partial differential equation $W(a, b, c, f, g, h) = 0$ of the surface, the coefficients satisfy a certain condition, then the surface has the property in question.

I recall that if the equation of the surface be $z - f(x, y) = 0$, and we write p, q, r, s, t for the first and second differential coefficients of z , then the differential equation of the curves of curvature is

$$(r - t) dx dy + s(dx^2 - dy^2) = 0,$$

or, what is the same thing,

$$s(dx^2 - dy^2) - (r - t) dx dy = 0.$$

Writing herein $dy = m dx$, the equation becomes

$$s(1 - m^2) - (r - t)m = 0,$$

that is,

$$sm^2 + (r - t)m - s = 0,$$

which determines the two values of m corresponding to the two systems of curves of curvature.

For the curves of curvature to be plane curves, the condition is that the torsion of each curve of curvature shall vanish. Now the condition for a curve on the surface to be a line of curvature is that at each point the normal section of the surface in the direction of the

curve shall coincide with the osculating plane of the curve; or, what is the same thing, the normal to the surface shall lie in the normal plane of the curve.

If the curves of curvature are plane curves, their planes envelope a surface which is the locus of the centres of principal curvature. The two sheets of the surface of centres are thus developable upon each other.

Consider a surface whose lines of curvature in one system are plane curves. The condition may be analytically expressed as follows. Let the position of a point on the surface be given as a function of the parameters u, v , which are the parameters of the lines of curvature; so that the surface is given by

$$x = x(u, v), \quad y = y(u, v), \quad z = z(u, v).$$

Then the condition is that for each of the lines of curvature $u = \text{const.}$ (say), the torsion shall be zero; this gives a relation between the fundamental quantities of the surface.

I consider in particular the case where the curves of curvature are circles. The sphere and the cyclide of Dupin are well-known examples. The Dupin cyclide is a quartic surface which is the envelope of a sphere varying in such manner that it touches three fixed spheres; or, what is the same thing, it is the inverse of a torus. The curves of curvature of a cyclide are all circles.

The condition for a surface to have its lines of curvature circles may be analytically investigated. If the lines of curvature $u = \text{const.}$ and $v = \text{const.}$ are both circles, then the surface is a cyclide. The determination depends on the following considerations.

For a line of curvature to be a circle, it is necessary and sufficient that the centres of the osculating circles of the curve be the same for all points of the curve; that is, that the radius of curvature of the curve be constant along the curve, and the curve be plane.

Now for a line of curvature $u = \text{const.}$ on the surface, the geodesic curvature and the normal curvature are connected by the relation of Euler; and the condition that the curve be a circle gives a relation between the first and second fundamental forms of the surface.

If both systems of lines of curvature are circles, then the surface belongs to the class of cyclides. It may be shown that the cyclide is the most general surface having this property. The proof depends on the fact that if both systems of lines of curvature are circles, then the

two sheets of the surface of centres degenerate into curves; and the surface is the envelope of a family of spheres with centres on these curves and radii satisfying the proper relations.

In the case where the lines of curvature in one system only are circles, we obtain a more general class of surfaces. The condition may be expressed as a partial differential equation of the second order for the function $z = f(x, y)$ which defines the surface.

I consider also the case where the curves of curvature are spherical curves. The condition that a curve be spherical is that its torsion be connected with its curvature by a certain relation; for a line of curvature to be a spherical curve, this condition must be satisfied. The investigation leads to partial differential equations which characterize the surfaces in question.

The general theory of Weingarten surfaces includes these various cases. A Weingarten surface is defined as one for which there exists a functional relation between the two principal curvatures. If the principal radii of curvature be R_1, R_2 , then the condition is

$$f(R_1, R_2) = 0.$$

For such surfaces, the lines of curvature possess remarkable properties, and the particular cases where these curves are plane or spherical are included in the general theory.

The detailed investigation of the partial differential equations, and the reduction of the problem to quadratures in particular cases, would occupy too much space to be here given. I hope to return to the subject on a future occasion.

887.

ON THE THEORY OF GROUPS.

[From the *American Journal of Mathematics*, vol. XI. (1889), pp. 139–157.]

THE present memoir is supplementary to my “Memoir on the Theory of Groups,” *American Journal of Mathematics*, vol. IX. (1887), pp. 33–44, vol. X. (1888), pp. 47–52.

I use the word group to denote a set of substitutions such that the product of any two of them is a substitution belonging to the set; the number of substitutions is the *order* of the group. The substitutions are for the most part substitutions of n letters, but I consider also linear substitutions (in particular, binary and ternary linear substitutions), and groups of such linear substitutions.

I consider first the groups of a given order. A group of prime order is necessarily cyclic, that is, of the form $\{1, P, P^2, \dots, P^{p-1}\}$. For the orders which are the products of two primes, the groups have been completely enumerated by Netto and others.

For the order p^2 , there are two types of groups: the cyclic group and the Abelian group of type $(1, 1)$, generated by two independent substitutions of order p .

For the order pq , where p and q are distinct primes, there is always a cyclic group; and if $p \equiv 1 \pmod{q}$, there is also a non-Abelian group generated by two substitutions P, Q of orders p and q respectively, with the relation $Q^{-1}PQ = P^a$, where a is a primitive q th root of unity modulo p .

For the order p^3 , there are five types of groups: three Abelian and two non-Abelian. The Abelian types are (3) , $(2, 1)$, and $(1, 1, 1)$; the two non-Abelian types are distinguished by the circumstance that in one of them the subgroup of order p^2 is cyclic, and in the other it is Abelian of type $(1, 1)$.

I consider also the transitive and intransitive groups. A group is transitive when, for any two letters of the set operated upon, there is a substitution of the group which changes the first into the second; otherwise it is intransitive. The order of a transitive group of n letters is divisible by n ; and conversely, if the order be divisible by n , the group is transitive.

For the primitive groups, the theory is more difficult. A transitive group is imprimitive when the letters can be divided into systems such

that every substitution of the group permutes these systems among themselves; otherwise the group is primitive. I have investigated the primitive groups of certain small degrees; the enumeration becomes very laborious as the degree increases.

I consider the linear groups, in particular the groups of binary and ternary linear substitutions. The general linear homogeneous group of binary substitutions has for its order $(p^2 - 1)(p^2 - p)$ when the coefficients are taken modulo p , a prime. The corresponding fractional group has order $\frac{1}{2}(p^2 - 1)p(p - 1)$ or $(p^2 - 1)p(p - 1)$, according as we do or do not identify a substitution with its negative.

The simple groups which occur as linear fractional groups have been the subject of much investigation. For $p = 2$, the group is of order 6; for $p = 3$, of order 12; for $p = 5$, of order 60; and for $p = 7$, of order 168. The group of order 168 is the well-known simple group of the modular equation of the eighth degree.

I give tables of the groups of certain orders. The following table exhibits the number of distinct groups of each order from 1 to 30:

Order	No. of groups	Order	No. of groups	Order	No. of groups
1	1	11	1	21	2
2	1	12	5	22	2
3	1	13	1	23	1
4	2	14	2	24	15
5	1	15	1	25	2
6	2	16	14	26	2
7	1	17	1	27	5
8	5	18	5	28	4
9	2	19	1	29	1
10	2	20	5	30	4

The orders which are the products of two primes have each either 1 or 2 groups. The order 12 has 5 groups: the cyclic group, the alternating group of 4 letters, the dihedral group of order 12, the dicyclic group, and a fifth group which is the direct product of a group of order 3 and the quaternion group of order 4.

For the order 16, there are 14 groups, which have been completely enumerated. The order 24 has 15 groups, which include important examples of both Abelian and non-Abelian types.

I give also a table of the transitive groups of small degrees. For degree 4, there are 4 transitive groups; for degree 5, there are 5; for

degree 6, there are 16; and for degree 7, there are 7. The symmetric and alternating groups occur in each case.

The investigation of the simple groups is continued. I prove that the alternating group of n letters is simple for $n \geq 5$, and I consider the simple groups which occur as linear fractional groups. The complete enumeration of all simple groups of a given order remains one of the most difficult problems of the theory.

I conclude with some remarks on the application of the theory of groups to the solution of algebraic equations. The group of an equation is the group of substitutions of its roots which leave the coefficients unaltered. The equation is solvable by radicals when its group is solvable; and the theory of groups thus furnishes the complete criterion for the solvability of any given equation.